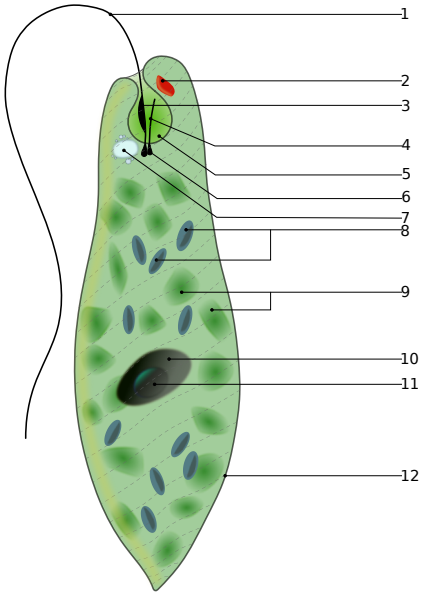

Figure 41.3*A*.1: **Contractile vacuole of Euglena**: Structure of Euglena: 1 – Flagellum; 2 – Eye spot / Pigment spot / Stigma; 3 – Photoreceptor; 4 – Short second flagellum; 5 – Reservoir; 6 – Basal body; 7 – Contractile vacuole; 8 – Paramylon granule; 9 – Chloroplasts; 10 – Nucleus; 11 – Nucleolus; 12 – Pellicle

The CV's phases of collecting water (expansion) and expelling water (contraction) are periodical. One cycle takes several seconds, depending on the species and the environment's osmolarity. The stage in which water flows into the CV is called diastole. The contraction of the CV and the expulsion of water from the cell is called systole. Water always flows from outside the cell into the cytoplasm; and only then from the cytoplasm into the CV. Species that possess a CV always use it, even in very hypertonic (high concentration of solutes) environments, since the cell tends to adjust its cytoplasm to become even more hyperosmotic (hypertonic) than the environment. The amount of water expelled from the cell and the rate of contraction are related to the osmolarity of the environment. In hyperosmotic environments, less water will be expelled and the contraction cycle will be longer.

The number of CVs per cell varies, depending on the species. Amoeba have one; *Dictyostelium discoideum*, *Paramecium aurelia*, and *Chlamydomonas reinhardtii* have two; and giant amoeba, such as *Chaos carolinensis*, have many. In some unicellular eukaryotic organisms (e.g., amoeba), cellular wastes, such as ammonia and excess water, are excreted by exocytosis as the contractile vacuoles merge with the cell membrane, expelling wastes into the environment. In *Paramecium*, which, presumably, has the most-complex and highly-evolved CV, the vacuole is surrounded by several canals, which absorb water by osmosis from the cytoplasm. After the canals fill with water, it is pumped into the vacuole. When the vacuole is full, it expels the water through a pore in the cytoplasm that can be opened and closed.

41.3B: Flame Cells of Planaria and Nephridia of Worms

LEARNING OBJECTIVES

- Compare and contrast the way in which planaria and annelids handle waste products

As multi-cellular systems evolved to have organ systems that divided the metabolic needs of the body, individual organs evolved to perform the excretory function. Excretory cells known as flame cells developed in flatworms, while nephridia developed as excretory cells in annelids.

Flame Cells of Planaria

Planaria are flatworms that live in fresh water. Their excretory system consists of two tubules connected to a highly-branched duct system that leads to pores located all along the sides of the body. The filtrate is secreted through these pores. The cells in the tubules are called flame cells (or protonephridia) because they have a cluster of cilia that looks like a flickering flame when viewed under the microscope. Flame cells function like a kidney, removing waste materials through filtration. The cilia propel waste matter down the tubules and out of the body through excretory pores that open on the body surface; cilia also draw water from the interstitial fluid, allowing for filtration. After excretion, any useful metabolites are reabsorbed by the cell. Flame cells are found in freshwater invertebrates, such as flatworms, including parasitic tapeworms and free-living planaria.

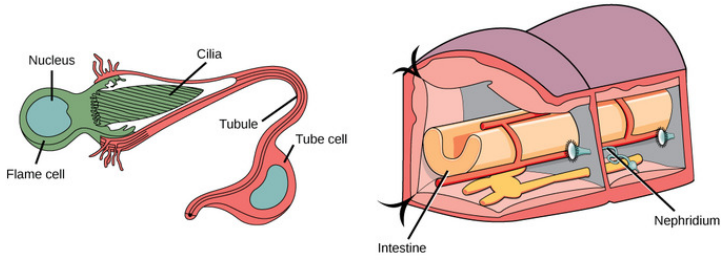

Figure $41.3B.1$: **Flame cells and nephridia**: In the excretory system of the (a) planaria, cilia of flame cells propel waste through a tubule formed by a tube cell. In (b) annelids, nephridia filter fluid from the body cavity.

Nephridia of Worms

Earthworms (annelids) and some other invertebrates, such as arthropods and mollusks, have slightly-more-evolved excretory structures called nephridia. A pair of nephridia is present on each segment of the earthworm. They are similar to flame cells in that they have tubules with cilia and function like a kidney to remove wastes, but they often open to the exterior of the organism. The ciliated tubules filter fluid from the body cavity and carry waste, including excess ions, through openings called nephrostomes. From the nephrostomes, excretion occurs through a pore called the nephridiopore. A nephridium is more evolved than a flame cell in that it has a system for reabsorption of some useful waste products, such as metabolites and ions, by a capillary network before excretion (unlike planaria that can only reabsorb useful metabolites after excretion).

Key Points

- Nephridia are more evolved than flame cells because they can reabsorb useful metabolites before excretion of waste.
- Both nephridia and flame cells are ciliated tubules that filter fluids in the cell to remove waste.
- Flame cells are connected to a duct system of pores to expel wastes, while nephridia often are open to the exterior of the organism.

Key Terms

- **flame cell**: a specialized excretory cell found in the simplest freshwater invertebrates
- **nephridium**: a tubular excretory organ in some invertebrates
- **nephridiopore**: the external opening of a nephridium, where waste is excreted from the cell
- **nephrostome**: the funnel-shaped opening of a nephridium into the body cavity

41.3C: Malpighian Tubules of Insects

LEARNING OBJECTIVES

- Explain how insects use malpighian tubules to excrete wastes and maintain osmotic balance

Malpighian tubules line the gut of some species of arthropods, such as bees. They are usually found in pairs in the posterior regions of arthropod alimentary canals; the number of tubules varies with the species of insect. The system of malpighian tubules consists of branching tubules, which increase their surface area, near the hemolymph (a mixture of blood and interstitial fluid that is found in insects, other arthropods, and most mollusks) and fat tissues. They are lined with microvilli for reabsorption and maintenance of osmotic balance. They contain actin for support.

Malpighian tubules work cooperatively with specialized glands in the wall of the rectum. Body fluids are not filtered, as in the case of nephridia. Instead, urine is produced by tubular secretion mechanisms by the cells lining the malpighian tubules that are bathed in hemolymph. Metabolic wastes, such as urea and amino acids, freely diffuse into the tubules, while ions are transported through active pump mechanisms. There are exchange pumps lining the tubules which actively transport H^+ ions into the cell and K^+ or Na^+ ions out; water passively follows to form urine. The secretion of ions alters the osmotic pressure, which draws water, electrolytes, and nitrogenous waste (uric acid) into the tubules. Water and electrolytes are reabsorbed when these organisms are faced with low-water environments and uric acid is precipitated and excreted as a thick paste or powder. By not dissolving wastes in water, these organisms are able to conserve water; this is especially important for life in dry environments.

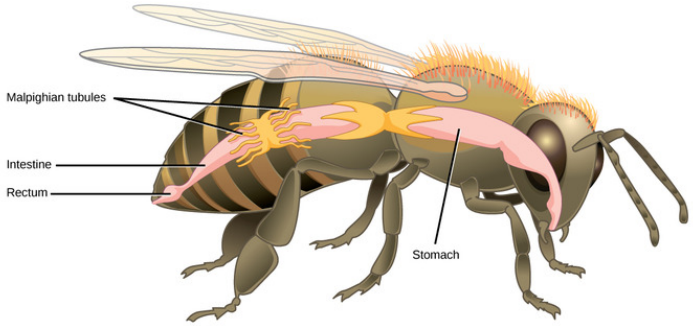

Figure $41.3C.1$: **Malpighian tubules in bees**: Malpighian tubules of insects and other terrestrial arthropods remove nitrogenous wastes and other solutes from the hemolymph.

Key Points

- Malpighian tubules are found in the posterior regions of insects, where they work with glands in the rectum to excrete waste and maintain osmotic balance.
- Ions are transported through active pumps found in the malpighian tubules; as the ions are secreted, water and waste are drawn to the tubules due to the change in osmotic pressure.
- Nitrogenous wastes, such as uric acid, are precipitated as thick pastes or powder to be excreted.

Key Terms

- **malpighian tubule**: a tubule that extends from the alimentary canal to the exterior of the organism, excreting water and wastes in the form of solid nitrogenous compounds
- **uric acid**: a bicyclic heterocyclic phenolic compound, formed in the body by the metabolism of protein and excreted in the urine
- **hemolymph**: a circulating fluid in the bodies of some invertebrates that is the equivalent of blood

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41.4: Human Osmoregulatory and Excretory Systems

Topic hierarchy

- 41.4A: Kidney Structure
- 41.4B: Nephron: The Functional Unit of the Kidney
- 41.4C: Kidney Function and Physiology

41.4A: Kidney Structure

The kidneys regulate the body's osmotic pressure in mammals.

Learning Objectives

- Explain how the kidneys serve as the main osmoregulatory organs in mammalian systems, using the functional properties of nephrons

Key Points

- Kidneys regulate the osmotic pressure of a mammal's blood through extensive filtration and purification, in a process known as osmoregulation.
- Kidneys filter the blood; urine is the filtrate that eliminates waste from the body via the ureter into the bladder.
- The kidneys are surrounded by three layers: renal fascia, perirenal fat capsule, and the renal capsule.

Key Terms

- **renal**: pertaining to the kidneys

Kidneys: The Main Osmoregulatory Organ

The kidneys are a pair of bean-shaped structures that are located just below and posterior to the liver in the peritoneal cavity. Adrenal glands, also called suprarenal glands, sit on top of each kidney. Kidneys regulate the osmotic pressure of a mammal's blood through extensive filtration and purification in a process known as osmoregulation. All the blood in the human body is filtered many times a day by the kidneys. These organs use almost 25 percent of the oxygen absorbed through the lungs to perform this function. Oxygen allows the kidney cells to efficiently manufacture chemical energy in the form of ATP through aerobic respiration. Kidneys eliminate wastes from the body; urine is the filtrate that exits the kidneys.

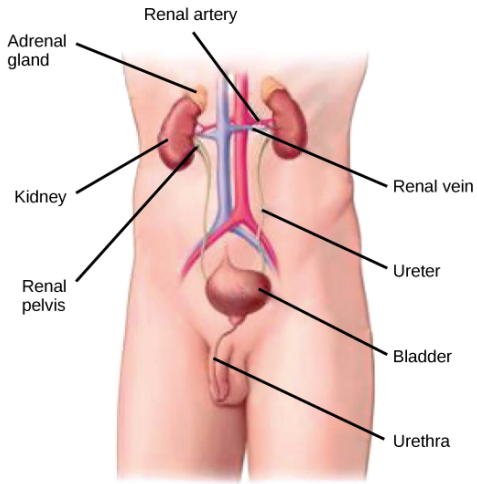

Figure 41.4*A*.1: **Kidneys' location and function**: Kidneys filter the blood, producing urine that is stored in the bladder prior to elimination through the urethra. They are located in the peritoneal cavity.

Externally, the kidneys are surrounded by three layers. The outermost layer, the renal fascia, is a tough connective tissue layer. The second layer, the perirenal fat capsule, helps anchor the kidneys in place. The third and innermost layer is the renal capsule. Internally, the kidney has three regions: an outer cortex, a medulla in the middle, and the renal pelvis in the region called the hilum of the kidney. The hilum is the concave part of the bean-shape where blood vessels and nerves enter and exit the kidney; it is also the point of exit for the ureters.

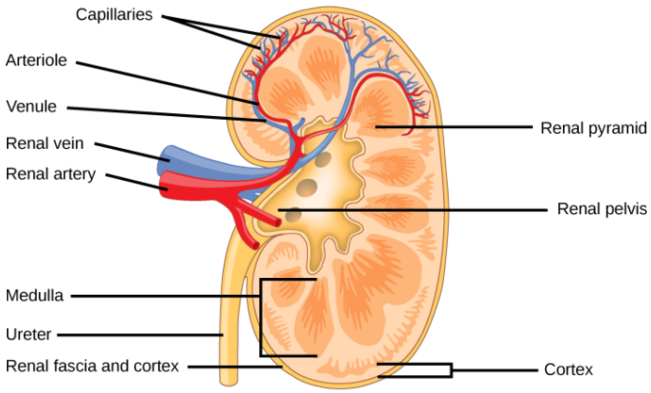

Figure 41.4A.1: **Structure of the kidney**: Externally, the kidney is surrounded by the renal fascia, the perirenal fat capsule, and the renal capsule. Internally, the kidney is most importantly filled with nephrons that filter blood and generate urine.

Because the kidney filters blood, its network of blood vessels is an important component of its structure and function. The arteries, veins, and nerves that supply the kidney enter and exit at the renal hilum. Renal blood supply starts with the branching of the aorta into the renal arteries (which are each named based on the region of the kidney they pass through) and ends with the exiting of the renal veins to join the inferior vena cava. The renal arteries split into several segmental arteries upon entering the kidneys. Each segmental artery splits further into several interlobar arteries that enter the renal columns, which supply the renal lobes. The interlobar arteries split at the junction of the renal cortex and medulla to form the arcuate arteries. The arcuate, "bow shaped" arteries form arcs along the base of the medullary pyramids. Cortical radiate arteries, as the name suggests, radiate out from the arcuate arteries, branch into numerous afferent arterioles, and then enter the capillaries supplying the nephrons.

41.4B: Nephron: The Functional Unit of the Kidney

The functional unit of the kidney, the nephron, removes waste from the body.

Learning Objectives

- Explain the role of the nephron as the functional unit of the kidney

Key Points

- Kidneys contain two types of nephrons, each located in different parts of the renal cortex: cortical nephrons and juxtamedullary nephrons.
- A nephron comprises a renal corpuscle, a renal tubule, and the associated capillary network.
- Internally, kidneys are mainly composed of over one million nephrons and an extensive network of blood vessels and capillaries.

Key Terms

- **glomerulus**: A small intertwined group of capillaries within a kidney's nephron that filters the blood to make urine.
- **loop of Henle**: A structure in a kidney's nephron that connects the proximal convoluted tubule to the distal convoluted tubule.

The nephron, the functional unit of the kidney, is responsible for removing waste from the body. Each kidney is composed of over one million nephrons that dot the renal cortex, giving it a granular appearance when sectioned sagittally (from front to rear). Eighty-five percent of nephrons are cortical nephrons, deep in the renal cortex; the remaining 15 percent are juxtamedullary nephrons, which lie in the renal cortex close to the renal medulla.

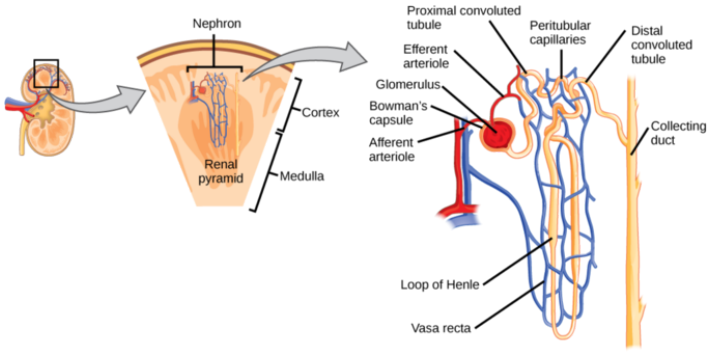

Diagram of a nephron

The nephron is the functional unit of the kidney. The glomerulus and convoluted tubules of the nephron are located in the cortex of the kidney, while the collecting ducts are located in the pyramids of the kidney's medulla.

A nephron consists of three parts: a renal corpuscle, a renal tubule, and the associated capillary network, which originates from the cortical radiate arteries. The renal corpuscle, located in the renal cortex, is composed of a network of capillaries known as the glomerulus, as well as a cup-shaped chamber that surrounds it: the glomerular or Bowman's capsule.

The renal tubule is a long, convoluted structure that emerges from the glomerulus. It can be divided into three parts based on function. The first part is called the proximal convoluted tubule (PCT), due to its proximity to the glomerulus. The second part is called the loop of Henle, or nephritic loop, because it forms a loop (with descending and ascending limbs) that goes through the renal medulla. The third part of the renal tubule is called the distal convoluted tubule (DCT); this part is also restricted to the renal cortex. This last part of the nephron connects with and empties its filtrate into collecting ducts that line the medullary pyramids. The collecting ducts amass contents from multiple nephrons, fusing together as they enter the papillae of the renal medulla.

As urine travels down the collecting duct system, it passes by the medullary interstitium, which has a high sodium concentration as a result of the loop of Henle's countercurrent multiplier system. Urine leaves the medullary collecting

ducts through the renal papillae, emptying into the renal calyces, the renal pelvis, and finally into the bladder via the ureter.

41.4C: Kidney Function and Physiology

Urine is a byproduct of the osmoregulatory function of kidneys, which filter blood, reabsorb water and nutrients, and secrete wastes.

Learning Objectives

- Outline the process by which kidneys filter blood, reabsorb nutrients and water, and produce urine

Key Points

- Glomerular filtration, tubular reabsorption, and tubular secretion are the three primary steps in which kidneys filter blood and maintain proper electrolyte balance.
- Glomerular filtration removes solutes from the blood; it is the first step of urine formation.
- In tubular reabsoption, the second step of urine formation, almost all nutrients are reabsorbed in the renal tubule by active or passive transport.
- Tubular secretion is the last step of urine formation, where solutes and waste are secreted into the collecting ducts, ultimately flowing to the bladder in the form of urine.

Key Terms

- **arteriole**: one of the small branches of an artery, especially one that connects with capillaries
- **countercurrent**: a current that flows against the prevailing one
- **electrolyte**: any of the various ions (such as sodium or chloride) that regulate the electric charge on cells and the flow of water across their membranes

Blood Filtration and Nutrient and Water Reabsorption

Kidneys filter blood in a three-step process. First, the nephrons filter blood that runs through the capillary network in the glomerulus. Almost all solutes, except for proteins, are filtered out into the glomerulus by a process called glomerular filtration. Second, the renal tubules collect the filtrate. Most of the solutes are reabsorbed in the PCT by a process called tubular reabsorption. In the loop of Henle, the filtrate continues to exchange solutes and water with the renal medulla and the peritubular capillary network.

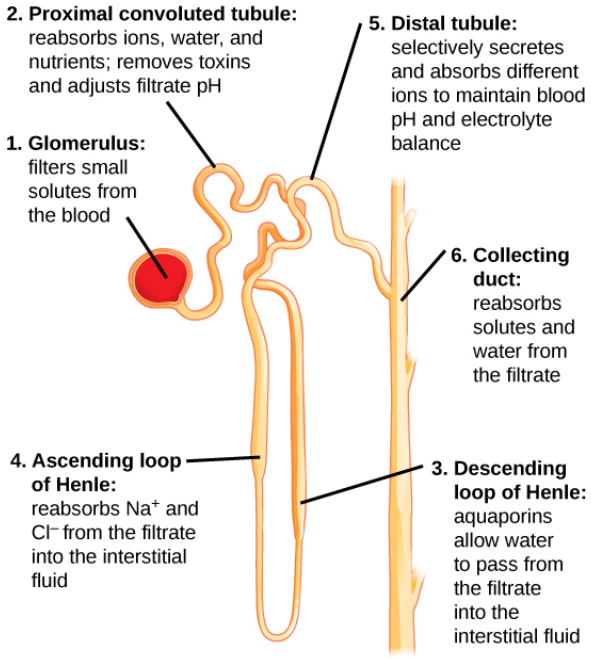

Figure 41.4C.1: **Nephron structure**: Each part of the nephron performs a different function in filtering waste and maintaining homeostatic balance.

Finally, some substances, such as electrolytes and drugs, are removed from blood through the peritubular capillary network into the distal convoluted tubule or collecting duct. Urine is a collection of substances that have not been reabsorbed during glomerular filtration or tubular reabsorbtion.

Glomerular Filtration

The formation of urine occurs through three steps: glomerular filtration, tubular reabsorption, and tubular secretion. The process of glomerular filtration filters out most of the solutes due to the high blood pressure and specialized membranes in the afferent arteriole. The blood pressure in the glomerulus is maintained independent of factors that affect systemic blood pressure. The "leaky" connections between the endothelial cells of the glomerular capillary network allow solutes to pass through easily. All solutes in the glomerular capillaries, including sodium ions and negatively and positively charged ions, pass through by passive diffusion; the only exception is macromolecules such as proteins. There is no energy requirement at this stage of the filtration process. Glomerular filtration rate (GFR) is the volume of glomerular filtrate formed per minute by the kidneys. GFR is regulated by multiple mechanisms and is an important indicator of kidney function.

Tubular Reabsorption and Secretion

Tubular reabsorption occurs in the PCT part of the renal tubule. Almost all nutrients are reabsorbed; this occurs either by passive or active transport. Reabsorption of water and key electrolytes are regulated and influenced by hormones. Sodium (Na^+) is the most abundant ion; most of it is reabsorbed by active transport and then transported to the peritubular capillaries. Because Na^+ is actively transported out of the tubule, water follows to even out the osmotic pressure. Water is also independently reabsorbed into the peritubular capillaries due to the presence of aquaporins, or water channels, in the PCT. This occurs due to the low blood pressure and high osmotic pressure in the peritubular capillaries. Every solute, however, has a transport maximum; the excess solute is not reabsorbed. Kidneys' osmolarity of body fluids is maintained at 300 milliosmole (mOsm).

In the loop of Henle, the permeability of the membrane changes. The descending limb is permeable to water, not solutes; the opposite is true for the ascending limb. Additionally, the loop of Henle invades the renal medulla, which is naturally high in salt concentration. It tends to absorb water from the renal tubule and concentrate the filtrate. The osmotic gradient increases as it moves deeper into the medulla. Because two sides of the loop of Henle perform opposing functions, it acts as a countercurrent multiplier. The vasa recta around the loop of Henle acts as the countercurrent exchanger.

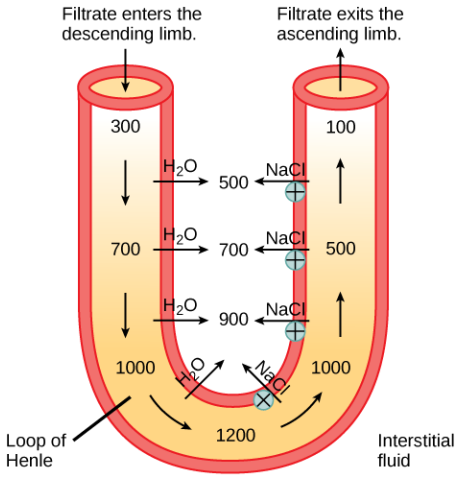

Figure 41.4C.1: **Loop of Henle**: The loop of Henle acts as a countercurrent multiplier that uses energy to create concentration gradients. The descending limb is water permeable. Water flows from the filtrate to the interstitial fluid, so osmolality inside the limb increases as it descends into the renal medulla. At the bottom, the osmolality is higher inside the loop than in the interstitial fluid. Thus, as filtrate enters the ascending limb, Na+ and Cl- ions exit through ion channels present in the cell membrane. Further up, Na+ is actively transported out of the filtrate and Cl- follows. Osmolarity is given in units of milliosmoles per liter (mOsm/L).

Additional solutes and wastes are secreted into the kidney tubules during tubular secretion, which is the opposite process to tubular reabsorption. The collecting ducts collect filtrate coming from the nephrons and fuse in the medullary papillae. From here, the papillae deliver the filtrate, now called urine, into the minor calyces that eventually connect to the ureters through the renal pelvis.

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41.5: Hormonal Control of Osmoregulatory Functions

Topic hierarchy

- 41.5A: Epinephrine and Norepinephrine
- 41.5B: Other Hormonal Controls for Osmoregulation

41.5A: Epinephrine and Norepinephrine

Epinephrine and norepinephrine are released during the flight/fight response, causing vasoconstriction of blood vessels in the kidney.

Learning Objectives

- Describe hormonal control by epinephrine and norepinephrine of osmoregulatory functions

Key Points

- Epinephrine, produced by the adrenal medulla, causes either smooth muscle relaxation in the airways or contraction of the smooth muscle in arterioles, which results in blood vessel constriction in the kidneys, decreasing or inhibiting blood flow to the nephrons.
- Norepinephrine, produced by the adrenal medulla, is a stress hormone that increases blood pressure, heart rate, and glucose from energy stores; in the kidneys, it will cause constriction of the smooth muscles, resulting in decreased or inhibited flow to the nephrons.
- Together, epinephrine and norepinephrine cause constriction of the blood vessels associated with the kidneys to inhibit flow to the nephrons.

Key Terms

- **epinephrine**: (adrenaline) an amino acid-derived hormone secreted by the adrenal gland in response to stress
- **norepinephrine**: a neurotransmitter found in the locus coeruleus which is synthesized from dopamine
- **catecholamine**: any of a class of aromatic amines derived from pyrocatechol that are hormones produced by the adrenal gland
- **adrenergic**: containing or releasing adrenaline

Epinephrine and Norepinephrine

Epinephrine

As a hormone and neurotransmitter, epinephrine acts on nearly all body tissues. Its actions vary by tissue type and tissue expression of adrenergic receptors. For example, high levels of epinephrine cause smooth muscle relaxation in the airways, but cause contraction of the smooth muscle that lines most arterioles. Epinephrine acts by binding to a variety of adrenergic receptors. Epinephrine is a nonselective agonist of all adrenergic receptors, including the major subtypes α1, α2, β1, β2, and β3. Epinephrine's binding to these receptors triggers a number of metabolic changes. Binding to α-adrenergic receptors inhibits insulin secretion by the pancreas, stimulates glycogenolysis (the breakdown of glycogen) in the liver and muscle, and stimulates glycolysis (the metabolic pathway that converts glucose into pyruvate) in muscle. β-Adrenergic receptor binding triggers glucagon secretion in the pancreas, increased adrenocorticotropic hormone (ACTH) secretion by the pituitary gland, and increased lipolysis by adipose tissue. Together, these effects lead to increased blood glucose and fatty acids, providing substrates for energy production within cells throughout the body.

Norepinephrine

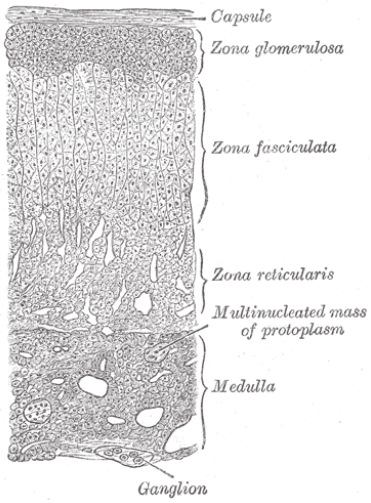

Figure 41.5A.1: **Adrenal gland**: The adrenal medulla, located toward the bottom of this image, is responsible for the release of epinephrine and norepinephrine.

Norepinephrine is a catecholamine with multiple roles. It is the hormone and neurotransmitter most responsible for vigilant concentration in contrast to its most-chemically-similar hormone, dopamine, which is most responsible for cognitive alertness. Areas of the body that produce or are affected by norepinephrine are described as noradrenergic. One of the most important functions of norepinephrine is its role as the neurotransmitter released from the sympathetic neurons to affect the heart. An increase in norepinephrine from the sympathetic nervous system increases the rate of contractions in the heart. Norepinephrine also underlies the fight-or-flight response, along with epinephrine, directly increasing heart rate, triggering the release of glucose from energy stores, and increasing blood flow to skeletal muscle. When norepinephrine acts as a drug, it increases blood pressure by increasing vascular tone through α-adrenergic receptor activation. Norepinephrine is synthesized from dopamine by dopamine β-hydroxylase in the secretory granules of the medullary chromaffin cells and is released from the adrenal medulla into the blood as a hormone. It is also a neurotransmitter in the central nervous system and sympathetic nervous system, where it is released from noradrenergic neurons in the locus coeruleus. The actions of norepinephrine are carried out via the binding to adrenergic receptors.

Role of Epinephrine and Norepinephrine in Kidney Function

Epinephrine and norepinephrine are released by the adrenal medulla and nervous system respectively. They are the flight/fight hormones that are released when the body is under extreme stress. During stress, much of the body's energy is used to combat imminent danger. Kidney function is halted temporarily by epinephrine and norepinephrine. These hormones function by acting directly on the smooth muscles of blood vessels to constrict them. Once the afferent arterioles are constricted, blood flow into the nephrons of the kidneys stops. These hormones go one step further and trigger the renin-angiotensin-aldosterone system, the hormone system that regulates blood pressure and water (fluid) imbalance.

41.5B: Other Hormonal Controls for Osmoregulation

The renin-angiotensin-aldosterone system (RAAS) stabilizes blood pressure and volume via the kidneys, liver, and adrenal cortex.

Learning Objectives

- Describe hormonal control by the renin-angiotensin-aldosterone system

Key Points

- Renin, a hormone produced by the juxtaglomerular apparatus in the kidneys, converts angiotensinogen (which is made in the liver) to angiotensin I.
- Angiotensin I is then converted to angiotensin II by the angiotensin converting enzyme (ACE), increasing blood pressure by causing vasoconstriction of the blood vessels.
- Angiotensin II causes the release of aldosterone which is produced by the adrenal cortex; it functions to maintain both sodium and water levels (osmotic balance) in the blood.
- Angiotensin II also causes the release of antidiuretic hormone (ADH) which functions to conserve water in the body when volume is low; it does this by inserting aquaporins in the collecting duct of the nephron to promote water reabsorption.
- The atrial natriuretic peptide (ANP) is another hormone that is produced to function as a vasodilator and lower blood pressure by preventing sodium reabsorption.

Key Terms

- **renin**: a circulating enzyme released by mammalian kidneys that converts angiotensinogen to angiotensin-I that plays a role in maintaining blood pressure
- **aquaporin**: any of a class of proteins that form pores in the membrane of biological cells
- **angiotensin**: any of several polypeptides that narrow blood vessels and thus regulate arterial pressure

Renin-Angiotensin-Aldosterone

The renin-angiotensin-aldosterone system (RAAS) is a hormone system that regulates blood pressure and water (fluid) balance. This system proceeds through several steps to produce angiotensin II, which acts to stabilize blood pressure and volume. Renin is secreted by a part of the juxtaglomerular complex and produced by the granular cells of the afferent and efferent arterioles. Renin is a circulating enzyme that acts on angiotensinogen, which is made in the liver, converting it to angiotensin I. Defective renin production can cause a continued decrease in blood pressure and cardiac output. After renin facilitates the production of angiotensis I, angiotensin converting enzyme (ACE) then converts angiotensin I to angiotensin II. Angiotensin II raises blood pressure by constricting blood vessels and also triggers the release of the mineralocorticoid aldosterone from the adrenal cortex. This, in turn, stimulates the renal tubules to reabsorb more sodium. Angiotensin II also triggers the release of anti-diuretic hormone (ADH) from the hypothalamus, leading to water retention in the kidneys. It acts directly on the nephrons, decreasing glomerular filtration rate. Thus, via the RAAS, the kidneys control blood pressure and volume directly. Medically, blood pressure can be controlled by drugs that inhibit ACE (called ACE inhibitors).

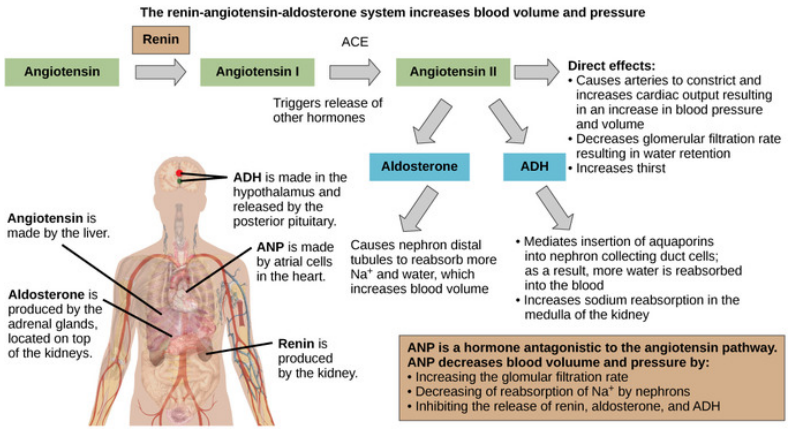

Figure 41.5B.1: **Renin-angiotensin-aldosterone system**: The renin-angiotensin-aldosterone system increases blood pressure and volume. The hormone ANP has antagonistic effects.

Mineralocorticoids

Mineralocorticoids are hormones synthesized by the adrenal cortex that affect osmotic balance. One type of mineralocorticoid, known as aldosterone, regulates sodium levels in the blood. Almost all of the sodium in the blood is reclaimed by the renal tubules under the influence of aldosterone. As sodium is always reabsorbed by active transport and water follows sodium to maintain osmotic balance, aldosterone manages not only sodium levels, but also the water levels in body fluids. Aldosterone also stimulates potassium secretion concurrently with sodium reabsorption. By contrast, absence of aldosterone means that no sodium is reabsorbed in the renal tubules; all of it is excreted in the urine. In addition, the daily dietary potassium load is not secreted; retention of potassium ions (K+) can cause a dangerous increase in plasma K+ concentration. Patients who have Addison's disease have a failing adrenal cortex and cannot produce aldosterone. They constantly lose sodium in their urine; if the supply is not replenished, the consequences can be fatal.

Antidiurectic Hormone

Antidiuretic hormone or ADH (also called vasopressin) helps the body conserve water when body fluid volume, especially that of blood, is low. It is formed by the hypothalamus, but is stored and released from the posterior pituitary gland. It acts by inserting aquaporins in the collecting ducts, promoting reabsorption of water. ADH also acts as a vasoconstrictor (constricting blood vessels) and increases blood pressure during hemorrhaging.

Atrial Natriuretic Peptide Hormone

The atrial natriuretic peptide (ANP) hormone lowers blood pressure by acting as a vasodilator (dilating or widening blood vessels). It is released by cells in the atrium of the heart in response to high blood pressure and in patients with sleep apnea. ANP affects salt release; because water passively follows salt to maintain osmotic balance, it also has a diuretic effect. ANP also prevents sodium reabsorption by the renal tubules, decreasing water reabsorption (thus acting as a diuretic) and lowering blood pressure. Its actions suppress the actions of aldosterone, ADH, and renin.

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CHAPTER OVERVIEW

42: THE IMMUNE SYSTEM

42.1: INNATE IMMUNE RESPONSE

42.1A: INNATE IMMUNE RESPONSE
42.1B: PHYSICAL AND CHEMICAL BARRIERS
42.1C: PATHOGEN RECOGNITION
42.1D: NATURAL KILLER CELLS
42.1E: THE COMPLEMENT SYSTEM
42.2: ADAPTIVE IMMUNE RESPONSE

42.2A: ANTIGEN-PRESENTING CELLS: B AND T CELLS
42.2B: HUMORAL IMMUNE RESPONSE
42.2C: CELL-MEDIATED IMMUNITY
42.2D: CYTOTOXIC T LYMPHOCYTES AND MUCOSAL SURFACES
42.2E: IMMUNOLOGICAL MEMORY
42.2F: REGULATING IMMUNE TOLERANCE
42.3: ANTIBODIES

42.3A: ANTIBODY STRUCTURE
42.3B: ANTIBODY FUNCTIONS
42.4: DISRUPTIONS IN THE IMMUNE SYSTEM

42.4A: IMMUNODEFICIENCY
42.4B: HYPERSENSITIVITIES

42.1: Innate Immune Response

Topic hierarchy

- 42.1A: Innate Immune Response
- 42.1B: Physical and Chemical Barriers
- 42.1C: Pathogen Recognition
- 42.1D: Natural Killer Cells
- 42.1E: The Complement System

42.1A: Innate Immune Response

The immune system serves to defend against pathogens: microorganisms that attempt to invade and cause disease in a host.

Learning Objectives

- Explain the purpose of the immune system

Key Points

- The mammalian immune system has evolved an extremely-diverse array of specialized cells and soluble molecules that allow it to defend against a variety of pathogens, including bacteria, protists, fungi, and other infectious organisms.
- The innate immune response serves as a general defense against all pathogens, but has no capacity to adapt or learn when new pathogens attack.
- The adaptive immune response has a “memory” about previously encountered pathogens and is able to mount pathogen-specific defenses based on this memory.

Key Terms

- **pathogen**: any organism or substance, especially a microorganism, capable of causing disease, such as bacteria, viruses, protozoa, or fungi
- **immune system**: the system that differentiates self from non-self and protects the body from foreign substances and pathogenic organisms by producing an immune response

The environment surrounding all of us consists of numerous pathogens: agents (usually microorganisms) that cause disease(s) in their hosts. A host is the organism that is invaded and often harmed by a pathogen. Pathogens, which include bacteria, protists, fungi, and other infectious organisms, can be found in food and water, on surfaces, and in the air. Concern over pathogens is one of the main reasons that we wash our hands after going to the bathroom or touching raw meat.

Mammalian immune systems evolved for protection from such pathogens. They are composed of an extremely-diverse array of specialized cells and soluble molecules that coordinate a rapid, flexible defense system capable of providing protection from a majority of these disease agents. Central to this goal, the immune system must be capable of recognizing “self” from “other” so that when it destroys cells, it destroys pathogen cells and not host cells.

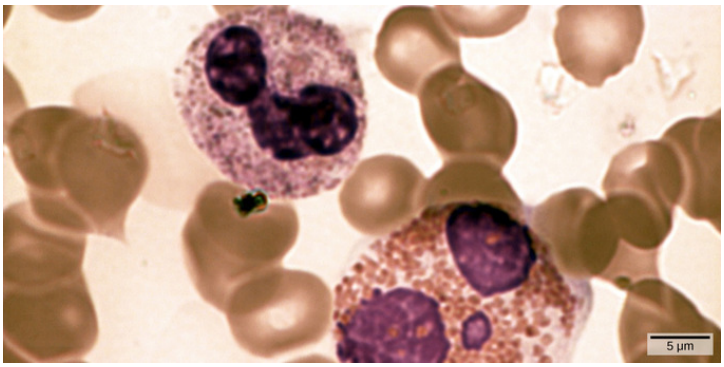

Figure 42.1*A*. 1: **Neutrophils and eosinophils**: In this compound light micrograph, purple-stained neutrophil (upper left) and eosinophil (lower right) are white blood cells that float among red blood cells in this blood smear. Neutrophils provide an early, rapid, and nonspecific defense against invading pathogens. Eosinophils play a variety of roles in the immune response.

The immune response that defends against pathogens can be classified as either innate or active. The innate immune response is present in its final state from birth and attempts to defend against all pathogens. Conversely, the adaptive immune response stores information about past infections and mounts pathogen-specific defenses. It expands over time, gaining more information about past targets so that it can respond quickly to future pathogens. The adaptive immune response functions throughout the body to combat specific pathogens that it has encountered before (a process known as reactivation). However, we are born with only innate immunity, developing our adaptive immune response after birth.

Components of both immune systems constantly search the body for signs of pathogens. When pathogens are found, immune factors are mobilized to the site of an infection. The immune factors identify the nature of the pathogen, strengthen the corresponding cells and molecules to combat it efficiently, and then halt the immune response after the infection is cleared to avoid unnecessary host cell damage. Features of the immune system (e.g., pathogen identification, specific response, amplification, retreat, and remembrance) are essential for survival against pathogens.

42.1B: Physical and Chemical Barriers

The innate immune response has physical and chemical barriers that exist as the first line of defense against infectious pathogens.

Learning Objectives

- Describe physical and chemical barriers in the innate immune response

Key Points

- The skin, or epithelial surface, serves as the primary barrier to microbial entry into the body; skin peeling, drying out, and the skin's acidity all serve to dislodge or kill foreign pathogens.
- Orifices such as the eyes and mouth, which are not covered by skin, have other mechanisms by which they prevent entry; tears wash away microbes, while cilia in the nasal passages and respiratory tract push mucus (which traps pathogens) out of the body.
- Many chemical barriers also exist once pathogens make it past the outer physical barriers; the acidity of the stomach ensures that few organisms arriving with food survive the digestive system.

Key Terms

- **cilium**: a hairlike organelle projecting from a eukaryotic cell (such as unicellular organism or one cell of a multicelled organism), which serves either for locomotion by moving or as sensors
- **microbicidal**: functioning to reduce the infectivity of microbes

Physical and chemical barriers

The immune system comprises both innate and adaptive immune responses. Innate immunity occurs naturally due to genetic factors or physiology. It is not induced by infection or vaccination, but is constantly available to reduce the workload for the adaptive immune response. The adaptive immune response expands over time, storing information about past infections and mounting pathogen-specific defenses. Both the innate and adaptive levels of the immune response involve secreted proteins, receptor-mediated signaling, and intricate cell -to-cell communication. From an historical perspective, the innate immune system developed early in animal evolution, roughly a billion years ago, as an essential response to infection. In the innate immune response, any pathogenic threat triggers a consistent sequence of events that can identify the type of pathogen and either clear the infection independently or mobilize a highly-specialized adaptive immune response.

Before any immune factors are triggered, the skin (also known as the epithelial surface) functions as a continuous, impassable barrier to potentially-infectious pathogens. The skin is considered the first defense of the innate immune system; it is the first of the nonspecific barrier defenses. Pathogens are killed or inactivated on the skin by desiccation (drying out) and by the skin's acidity. In addition, beneficial microorganisms that coexist on the skin compete with invading pathogens, preventing infection. Desquamation, or peeling skin, also serves to dislodge organisms that have adhered to the surface of the body and are awaiting entry. Regions of the body that are not protected by skin (such as the eyes and mucous membranes) have alternative methods of defense. These include tears in the eyes; mucous membranes that provide partial protection despite having to allow absorption and secretion; mucus secretions that trap and rinse away pathogens; and cilia (singular cilium) in the nasal passages and respiratory tract that push the mucus with the pathogens out of the body. Furthermore, tears and mucus secretions contain microbicidal factors that prevent many infections from entering via these routes.

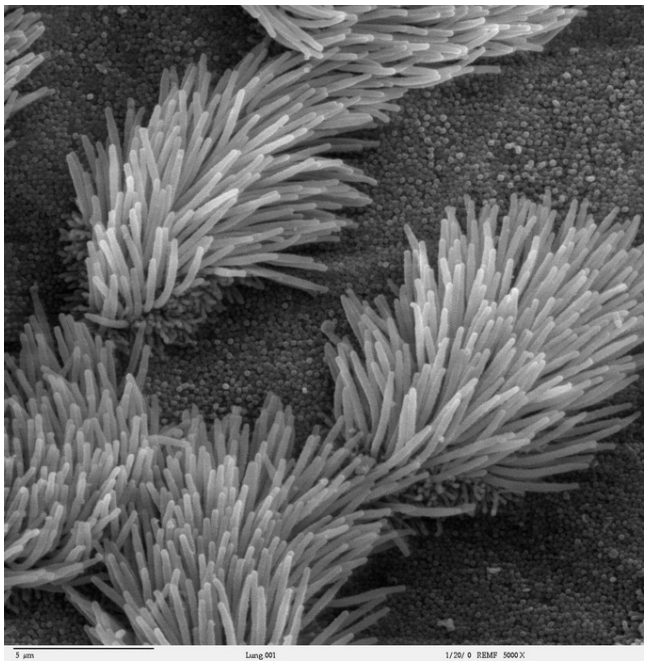

Figure 42.1B.1: **Cilia up close**: Cilia are a type of organelle found in eukaryotic cells. In the innate immune system, they serve to move pathogens out of the respiratory system via a concerted sweeping motion.

Despite these barriers, pathogens may enter the body through skin abrasions or punctures, or by collecting on mucosal surfaces in large numbers that overcome the mucus or cilia. Some pathogens have evolved specific mechanisms that allow them to overcome physical and chemical barriers.

Once inside, the body still has many other defenses, including chemical barriers. Some of these include the low pH of the stomach, which inhibits the growth of pathogens; blood proteins that bind and disrupt bacterial cell membranes; and the process of urination, which flushes pathogens from the urinary tract. The blood-brain barrier also protects the nervous system from pathogens that have already entered the blood stream, but would do significantly more damage if they entered the central nervous system.

42.1C: Pathogen Recognition

Upon pathogen entry to the body, the innate immune system uses several mechanisms to destroy the pathogen and any cells it has infected.

Learning Objectives

- Describe the role of PAMPs and PRRs, interferons, and other cytokines in innate immunity

Key Points

- Pathogens are recognized by a variety of immune cells, such as macrophages and dendritic cells, via pathogen-associated molecular patterns (PAMPs) on the pathogen surface, which interact with complementary pattern-recognition receptors (PRRs) on the immune cells' surfaces.
- Upon binding of PRRs with PAMPs (pathogen recognition), immune cells release cytokines to tell other cells to start fighting back.
- One class of cytokines, interferons, warn nearby uninfected cells of impending infection, cause cells to start cleaving RNA and reduce protein synthesis, and signal nearby infected cells to undergo apoptosis.
- Another class of cytokines, called inerleukins, mediate interactions between white blood cells (leukocytes) and help bridge the innate and adaptive immune responses.
- Inflammation (hot, red, swollen, painful tissue associated with infection) is encouraged by cytokines that are produced immediately upon pathogen recognition; the increase in blood flow associated with inflammation allows more leukocytes (a type of innate immune cell) to reach the infected area.

Key Terms

- **macrophage**: a white blood cell that phagocytizes necrotic cell debris and foreign material, including viruses, bacteria, and tattoo ink; part of the innate immune system
- **phagocytosis**: the process where a cell incorporates a particle by extending pseudopodia and drawing the particle into a vacuole of its cytoplasm
- **cytokine**: any of various small regulatory proteins that regulate the cells of the immune system; they are released upon binding of PRRs to PAMPS

Pathogen recognition

When a pathogen enters the body, cells in the blood and lymph detect the specific pathogen-associated molecular patterns (PAMPs) on the pathogen's surface. PAMPs are carbohydrate, polypeptide, and nucleic acid "signatures" that are expressed by viruses, bacteria, and parasites, but which differ from molecules on host cells. These PAMPs allow the immune system to recognize "self" from "other" so as not to destroy the host.

The immune system has specific cells with receptors that recognize these PAMPs. A macrophage is a large, phagocytic cell that engulfs foreign particles and pathogens. Macrophages recognize PAMPs via complementary pattern recognition receptors (PRRs). PRRs are molecules on macrophages and dendritic cells which are in contact with the external environment and can thus recognize PAMPs when present. A monocyte, a type of leukocyte (white blood cell) that circulates in the blood and lymph, differentiates into macrophages after it moves into infected tissue. Dendritic cells bind molecular signatures of pathogens, promoting pathogen engulfment and destruction.

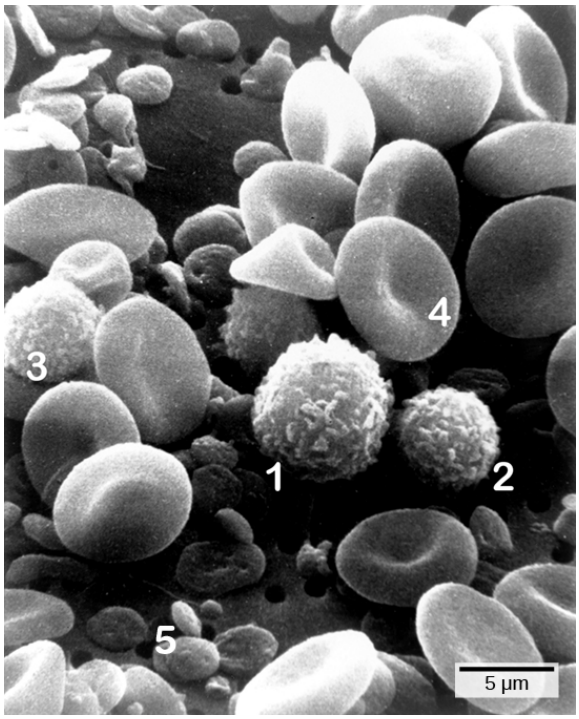

Figure 42.1C.1: **Blood cells related to the innate immune response**: Cells of the blood include (1) monocytes, (2) lymphocytes, (3) neutrophils, (4) red blood cells, and (5) platelets. Leukocytes (1, 2, 3) are white blood cells that play an important role in the body's immune system.

Cell type	Characteristics	Location	Image
Mast cell	Dilates blood vessels and induces inflammation through release of histamines and heparin. Recruits macrophages and neutrophils. Involved in wound healing and defense against pathogens but can also be responsible for allergic reactions.	Connective tissues, mucous membranes	
Macrophage	Phagocytic cell that consumes foreign pathogens and cancer cells. Stimulates response of other immune cells.	Migrates from blood vessels into tissues.	
Natural killer cell	Kills tumor cells and virus-infected cells.	Circulates in blood and migrates into tissues.	
Dendritic cell	Presents antigens on its surface, thereby triggering adaptive immunity.	Present in epithelial tissue, including skin, lung and tissues of the digestive tract. Migrates to lymph nodes upon activation.	
Monocyte	Differentiates into macrophages and dendritic cells in response to inflammation.	Stored in spleen, moves through blood vessels to infected tissues.	
Neutrophil	First responders at the site of infection or trauma, this abundant phagocytic cell represents 50-60 percent of all leukocytes. Releases toxins that kill or inhibit bacteria and fungi and recruits other immune cells to the site of infection.	Migrates from blood vessels into tissues.	
Basophil	Responsible for defense against parasites. Releases histamines that cause inflammation and may be responsible for allergic reactions.	Circulates in blood and migrates to tissues.	
Eosinophil	Releases toxins that kill bacteria and parasites but also causes tissue damage.	Circulates in blood and migrates to tissues.	

Figure 42.1C.1: **Cells involved in the innate immune system**: The immune system has specific cells whose job is to recognize pathogen-associated molecular patterns. The characteristics and location of cells involved in the innate immune system are described in this chart.

Once a pathogen is detected, the immune system must also track whether it is replicating intracellularly (inside the cell, as with most viruses and some bacteria) or extracellularly (outside of the cell, as with other bacteria, but not viruses). The innate immune system must respond accordingly by identifying the extracellular pathogen and/or by identifying host cells that have already been infected.

Cytokine release affect

The binding of PRRs with PAMPs triggers the release of cytokines, which signal that a pathogen is present and needs to be destroyed along with any infected cells. A cytokine is a chemical messenger that regulates cell differentiation (form and function), proliferation (production), and gene expression to affect immune responses. At least 40 types of cytokines exist in humans that differ in terms of the cell type that produces them, the cell type that responds to them, and the changes they produce.

One subclass of cytokines is the interleukin (IL), which mediates interactions between leukocytes (white blood cells). Interleukins are involved in bridging the innate and adaptive immune responses. In addition to being released from cells after PAMP recognition, cytokines are released by the infected cells which bind to nearby uninfected cells, inducing those cells to release cytokines, resulting in a cytokine burst.

A second class of cytokines is interferons, which are released by infected cells as a warning to nearby uninfected cells. A function an interferons is to inhibit viral replication, making them particularly effective against viruses. They also have other important functions, such as tumor surveillance. Interferons work by signaling neighboring uninfected cells to destroy RNA (often a very important biomolecule for viruses) and reduce protein synthesis; signaling neighboring infected cells to undergo apoptosis (programmed cell death); and activating immune cells.

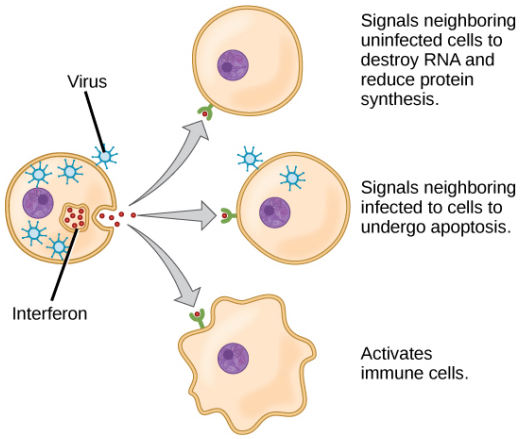

Figure 42.1C.1: **Interferon release**: Interferons are cytokines that are released by a cell infected with a virus. The response of neighboring cells to interferons helps stem the infection.

Cytokines also send feedback to cells of the nervous system to bring about the overall symptoms of feeling sick, which include lethargy, muscle pain, and nausea. These effects may have evolved because the symptoms encourage the individual to rest, preventing them from spreading the infection to others. Cytokines also increase the core body temperature, causing a fever, which causes the liver to withhold iron from the blood. Without iron, certain pathogens (such as some bacteria) are unable to replicate; this is called nutritional immunity.

Phagocytosis and inflammation

The first cytokines to be produced are pro-inflammatory; that is, they encourage inflammation, or the localized redness, swelling (edema), heat, loss of function, and pain that result from the movement of leukocytes and fluid through increasingly-permeable capillaries to a site of infection. The population of leukocytes that arrives at an infection site depends on the nature of the infecting pathogen. Both macrophages and dendritic cells engulf pathogens and cellular debris through phagocytosis. A neutrophil is also a phagocytic leukocyte that engulfs and digests pathogens. Neutrophils, the most-abundant leukocytes of the immune system, have a nucleus with two to five lobes and contain organelles (lysosomes) that digest engulfed pathogens. An eosinophil is a leukocyte that works with other eosinophils to surround a parasite. It is involved in the allergic response and in protection against helminthes (parasitic worms).

Neutrophils and eosinophils are particularly important leukocytes that engulf large pathogens, such as bacteria and fungi. A mast cell is a leukocyte that produces inflammatory molecules, such as histamine, in response to large pathogens. A basophil is a leukocyte that, like a neutrophil, releases chemicals to stimulate the inflammatory response. Basophils are also involved in allergy and hypersensitivity responses and induce specific types of inflammatory responses. Eosinophils and basophils produce additional inflammatory mediators to recruit more leukocytes. A hypersensitive immune response to harmless antigens, such as in pollen, often involves the release of histamine by basophils and mast cells; this is why many anti-allergy medications are anti-histamines.

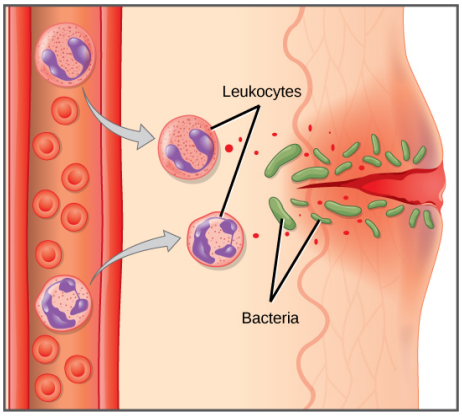

Figure 42.1C.1: **Innate immune response to cuts**: In response to a cut, mast cells secrete histamines that cause nearby capillaries to dilate. Neutrophils and monocytes leave the capillaries. Monocytes mature into macrophages. Neutrophils, dendritic cells, and macrophages release chemicals to stimulate the inflammatory response. Neutrophils and macrophages also consume invading bacteria by phagocytosis.

42.1D: Natural Killer Cells

Natural killer cells are part of the innate immune response that recognize abnormal MHC I molecules on infected/tumor cells and kill them.

Learning Objectives

- Describe the role of natural killer cells in the immune response

Key Points

- Natural killer (NK) cells are lymphocytes (a subclass of white blood cells) that recognize infected or tumorogenic cells and kill them.
- Unlike the related T cells, NK cells do not recognize fragments of the infecting particle, but rather the incorrect display of major histocompatibility complex (MHC) I molecules.
- NK cells are always active, but will not perform their killing function on cells with intact MHC I molecules.
- When NK cells detect an infected or tumor cell, they secrete granules that contain perforin, creating a pore in the target cell; granzymes then pass through these pores, degrading cellular proteins, causing cells to undergo apoptosis.

Key Terms

- **lymphocyte**: a type of white blood cell or leukocyte that is divided into two principal groups and a null group: B-cells, T-cells, and natural killer (NK) cells
- **major histocompatibility complex**: a protein present on the extracellular surface of the cell that displays portions of the proteins that are degraded inside the cell
- **T cell**: a lymphocyte, from the thymus, that can recognize specific antigens and can activate or deactivate other immune cells

Natural killer cells

Lymphocytes are leukocytes (white blood cells) that are histologically identifiable by their large, darkly-staining nuclei; they are small cells with very little cytoplasm. After a pathogen enters the body, infected cells are identified and destroyed by natural killer (NK) cells, which are a type of lymphocyte that can kill cells infected with viruses or tumor cells (abnormal cells that uncontrollably divide and invade other tissue). While NK cells are part of the innate immune response, they are best understood relative to their counterparts in the adaptive immune response,T cells, which are also classified as lymphocytes.

T cells are lymphocytes that mature in the thymus gland and identify intracellular infections, especially from viruses, by the altered expression of major histocompatibility class (MHC) I molecules on the surface of infected cells. MHC I molecules are proteins on the surfaces of all nucleated cells which help the immune system distinguish between “self” and “non-self.” If the cell is infected, the MHC I molecules display fragments of proteins from the infectious agents to T-cells. Healthy cells do not display any proteins and will be ignored by the immune system, while the cells identified as “non-self” by foreign proteins will be attacked by the immune system.

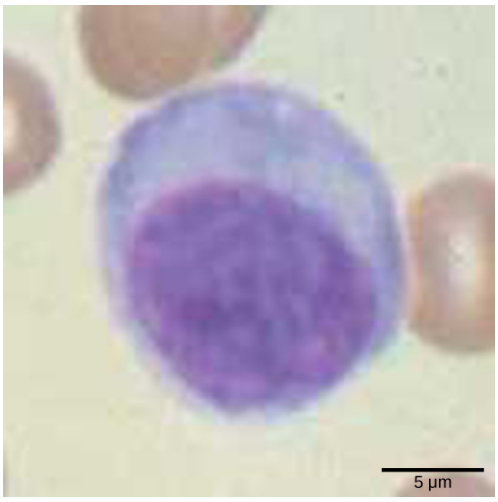

Figure 42.1D.1: **Lymphocytes**: Lymphocytes, such as NK cells, are characterized by their large nuclei that actively absorb Wright stain and, therefore, appear dark colored under a microscope.

An infected cell (or a tumor cell) is often incapable of synthesizing and displaying MHC I molecules appropriately. The metabolic resources of cells infected by some viruses produce proteins that interfere with MHC I processing and/or trafficking to the cell surface. The reduced MHC I on host cells varies from virus to virus and results from active inhibitors being produced by the viruses. This process can deplete host MHC I molecules on the cell surface, which prevents T-cells from recognizing them, but which NK cells detect as “unhealthy” or “abnormal” while searching for cellular MHC I molecules. As such, NK cells offer a complementary check for unhealthy cells, relative to T cells. Similarly, the dramatically-altered gene expression of tumor cells leads to expression of extremely- deformed or absent MHC I molecules that also signal “unhealthy” or “abnormal.”

NK cells are always active; an interaction with normal, intact MHC I molecules on a healthy cell disables the killing sequence, causing the NK cell to move on. After the NK cell detects an infected or tumor cell, its cytoplasm secretes granules comprised of perforin: a destructive protein that creates a pore in the target cell. Granzymes are released along with the perforin in the immunological synapse. A granzyme, a protease that digests cellular proteins, induces the target cell to undergo programmed cell death, or apoptosis. Phagocytic cells then digest the cell debris left behind. NK cells are constantly patrolling the body. They are an effective mechanism for controlling potential infections and preventing cancer progression.

42.1E: The Complement System

Around 20 soluble proteins comprise the complement system, which helps destroy extracellular microorganisms that have invaded the body.

Learning Objectives

- Explain how the complement system aids antibody response

Key Points

- The complement system is so named because it is complementary to the antibody response of the adaptive immune system.
- The complement system proteins are produced continuously by the liver and macrophages, are abundant in the blood serum, and are capable of immediate response to infecting microorganisms.
- The complement system works by first having several proteins bind to a target; this binding event then begins a series of highly-specific and regulated sequences wherein successive proteins are activated by cleavage and/or structural changes of the preceding proteins.
- The complement system serves as a marker to indicate targets for phagocytic cells; complement proteins can also combine to form attack complexes capable of opening pores in microbial cell membranes.

Key Terms

- **opsonization**: the process by which a pathogen is marked for ingestion and destruction by a phagocyte
- **complement system**: an aspect of the innate immune system that supplements the actions of the antibodies and phagocytic cells in clearing out pathogens from an organism

Complement

The innate immune system serves as a first responder to pathogenic threats that bypass natural physical and chemical barriers of the body. Using a combination of cellular and molecular attacks, the innate immune system identifies the nature of a pathogen and responds with inflammation, phagocytosis (where a cell engulfs a foreign particle), cytokine release, destruction by NK cells, and/or a complement system. In this concept, we will discuss the complement system.

An array of approximately 20 types of soluble proteins, called a complement system, functions to destroy extracellular pathogens. Cells of the liver and macrophages synthesize complement proteins continuously. These proteins are abundant in the blood serum and are capable of responding immediately to infecting microorganisms. The complement system is so named because it is complementary to the antibody response of the adaptive immune system. Complement proteins bind to the surfaces of microorganisms and are particularly attracted to pathogens that are already bound by antibodies. Binding of complement proteins occurs in a specific and highly-regulated sequence, with each successive protein being activated by cleavage and/or structural changes induced upon binding of the preceding protein(s). After the first few complement proteins bind, a cascade of sequential binding events follows in which the pathogen rapidly becomes coated in complement proteins.

Complement proteins perform several functions. They serve as a marker to indicate the presence of a pathogen to phagocytic cells, such as macrophages and B cells, to enhance engulfment. This process is called opsonization. Certain complement proteins can combine to form attack complexes that open pores in microbial cell membranes. These structures destroy pathogens by causing their contents to leak. When innate mechanisms are insufficient to clear an infection, the adaptive immune response is informed and mobilized.

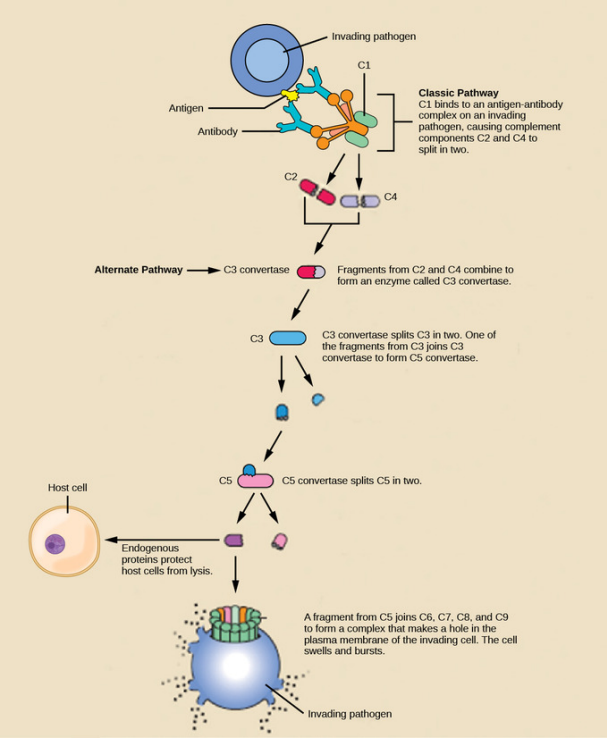

Figure 42.1E.1: **Complement cascade in the innate immune response**: The classic pathway for the complement cascade involves the attachment of several initial complement proteins to an antibody-bound pathogen, followed by rapid activation and binding of many more complement proteins and the creation of destructive pores in the microbial cell envelope and cell wall. The alternate pathway does not involve antibody activation. Rather, C3 convertase spontaneously breaks down C3. Endogenous regulatory proteins prevent the complement complex from binding to host cells. Pathogens lacking these regulatory proteins are lysed.

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42.2: Adaptive Immune Response

Topic hierarchy

- 42.2A: Antigen-presenting Cells: B and T cells
- 42.2B: Humoral Immune Response
- 42.2C: Cell-Mediated Immunity
- 42.2D: Cytotoxic T Lymphocytes and Mucosal Surfaces
- 42.2E: Immunological Memory
- 42.2F: Regulating Immune Tolerance

42.2A: Antigen-presenting Cells: B and T cells

B and T cells, parts of the adaptive immune response, contain receptors that can identify antigens derived from pathogens.

Learning Objectives

- Explain the role played by B and T cells in the adaptive immune system

Key Points

- The adaptive immune response is slower to develop than the innate immune response, but it can act much more powerfully and quickly than the innate immune response against pathogens that it has seen before.
- B and T cells are lymphocytes, or white blood cells, which are able to recognize antigens that distinguish “self” from “other” in the body.
- B and T cells that recognize “self” antigens are destroyed before they can mature; this helps to prevent the immune system from attacking its own body.

Key Terms

- **B cell**: a lymphocyte, developed in the bursa of birds and the bone marrow of other animals, that produces antibodies and is responsible for the immune system
- **T cell**: a lymphocyte, from the thymus, that can recognize specific antigens and can activate or deactivate other immune cells
- **antigen**: a substance that binds to a specific antibody; may cause an immune response

Introduction

The adaptive, or acquired, immune response to an initial infection takes days or even weeks to become established, much longer than the innate response. However, adaptive immunity is more specific to an invading pathogen and can fight back much more quickly than the innate response if it has seen the pathogen before. Adaptive immunity occurs after exposure to an antigen either from a pathogen or a vaccination. An antigen is a molecule that binds to a specific antibody, often stimulating a response in the immune system as a result.

The adaptive immune response activates when the innate immune response insufficiently controls an infection. In fact, without information from the innate immune system, the adaptive response could not be mobilized. There are two types of adaptive responses: the cell-mediated immune response, which is controlled by activated T cells, and the humoral immune response, which is controlled by activated B cells and antibodies. Upon infection, activated T and B cells that have surface binding sites with specificity to the molecules on the pathogen greatly increase in number and attack the invading pathogen. Their attack can kill pathogens directly or they can secrete antibodies that enhance the phagocytosis of pathogens and disrupt the infection. Adaptive immunity also involves a memory, which gives the host long-term protection from reinfection by the same type of pathogen; upon re-exposure, this host memory will facilitate a rapid and powerful response.

B and T Cells

Lymphocytes, which are white blood cells, are formed with other blood cells in the red bone marrow found in many flat bones, such as the shoulder or pelvic bones. The two types of lymphocytes of the adaptive immune response are B and T cells. Whether an immature lymphocyte becomes a B cell or T cell depends on where in the body it matures. The B cells remain in the bone marrow to mature (hence the name “B” for “bone marrow”), while T cells migrate to the thymus, where they mature (hence the name “T” for “thymus”).

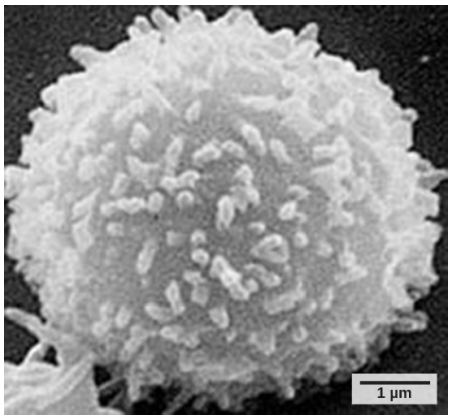

Figure 42.2A.1: **T cell by SEM**: This scanning electron micrograph shows a T lymphocyte. T and B cells are indistinguishable by light microscopy, but can be differentiated experimentally by probing their surface receptors.

B Cell Receptors

The maturation of a B or T cell involves becoming immunocompetent, meaning that it can recognize and bind to a specific molecule or antigen. This recognition, which is central to the functioning of the adaptive immune response, results from the presence of highly specific receptors on the surfaces of B and T cells. On B cells, these receptors contain antibodies, which are responsible for antigen binding. An antibody is specific for one particular antigen; typically, it will not bind to anything else. Upon antigen binding to a B cell receptor, a signal is sent into the B cell to turn on an immune response.

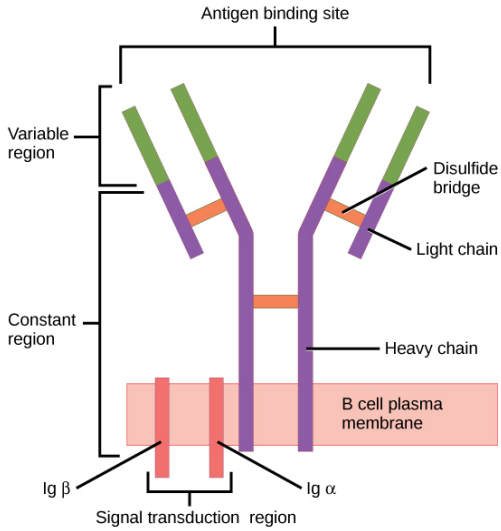

Figure 42.2A.1: **B cell receptors**: B cell receptors are embedded in the membranes of B cells and bind a variety of antigens through their variable regions, or antibodies. The signal transduction region transfers the signal into the cell.

T Cell Receptors

Meanwhile, T cell receptors are responsible for the recognition of pathogenic antigens by T cells. Unlike B cells, T cells do not directly recognize antigens. Instead, they recognize antigens presented on major histocompatibility complexes (MHCs) that cells use to display which proteins are inside of them. If a cell is infected, it will present antigenic portions of the infecting pathogen on its MHC for recognition by T cells, which will then mount an appropriate immune response. Unlike antibodies, which can typically bind one and only one antigen, T cell receptors have more flexibility in their capacity to recognize antigens presented by MHCs.

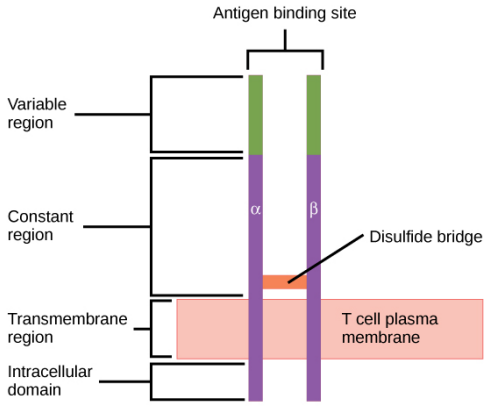

Figure 42.2*A*.1: **T cell receptors (TCRs)**: A T cell receptor spans the membrane and projects variable binding regions into the extracellular space to bind processed antigens via MHC molecules on APCs.

It is the specific pathogen recognition (via binding antigens) of B and T cells that allows the adaptive immune response to adapt. During the maturation process, B and T cells that bind too strongly to the body's own cells' antigens are eliminated in order to minimize an immune response against the body's own tissues. Only those cells that react weakly to the body's own cells will remain. This process occurs during fetal development and continues throughout life. Once they are immunocompetent, the T and B cells migrate to the spleen and lymph nodes where they remain until they are called on during an infection. B cells are involved in the humoral immune response, which targets pathogens loose in blood and lymph, while T cells are involved in the cell-mediated immune response, which targets infected cells.

42.2B: Humoral Immune Response

The humoral immune response defends against pathogens that are free in the blood by using antibodies against pathogen-specific antigens.

Learning Objectives

- Summarize the humoral immune response

Key Points

- Antigens are proteins and other macromolecules that bind to a specific antibody and are used by the immune system to recognize pathogens.
- B cells express receptors (BCRs) on their membrane which contain antibodies; these antibodies allow B cells to detect pathogens and release further antibodies to fight the infection.
- Antibodies fight infections in three ways: they mark pathogens for destruction by phagocytic cells in a process known as opsonization, they coat key sites on pathogens necessary for infection, and they induce the complement cascade to occur against antibody-bound pathogens.
- Once the adaptive immune response has encountered an antigen, B cells will divide to produce plasma cells, which rapidly secrete antibodies to that antigen in a process called active immunity.

Key Terms

- **antibody**: a protein produced by B-lymphocytes that binds to a specific antigen
- **opsonize**: to make (bacteria or other cells) more susceptible to the action of phagocytes by use of opsonins
- **MHC**: an acronym for major histocompatibility complex; these extracellular protein receptors display antigens derived from extracellular (class I) or intracellular (class II) proteins and other biomolecules

The humoral immune response fights pathogens that are free in the bodily fluids, or “humours”. It relies on antigens (which are also often free in the humours) to detect these pathogens. An antigen is a biomolecule, such as a protein or sugar, that binds to a specific antibody. An antibody/antigen interaction may stimulate an immune response. Not every biomolecule is antigenic and not all antigens produce an immune response. B cells are the major cell type involved in the humoral immune response. When a foreign antigen (one coming from a pathogen, for example) is detected, B cells in the body that recognize that antigen will begin to produce antibodies as a means of fighting off the foreign invader.

B cell maturation

During maturation, B cells gain antigen receptor molecules, termed B cell receptors (BCRs), which are displayed in large numbers, extracellularly on their membrane. These membrane-bound protein complexes contain antibodies, which enable specific antigen recognition. Each B cell initially produced has only one kind of antibody (antigen receptor), which makes every B cell unique. It is the immense number of B cells in the body, each of which produces a unique antibody, that allows the immune system to detect such a wide variety of pathogenic antigens. B cells containing antibodies that recognize “self” antigens are destroyed before they can mature, preventing the immune system from attacking the host. Once B cells mature in the bone marrow, they migrate to lymph nodes or other lymphatic organs, where they may begin to encounter pathogens.

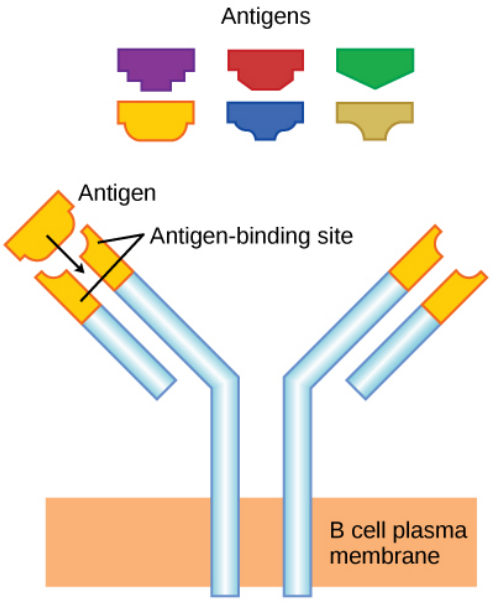

Figure 42.2B.1: **B cell receptors**: B cell receptors, containing antibodies (termed antigen-binding site in the picture) are embedded in the membranes of B cells and bind a variety of antigens through their variable regions.

B cell activation

When a B cell encounters the antigen that binds to its receptor, the antigen molecule is brought into the cell by endocytosis, reappearing on the surface of the cell bound to an MHC class II molecule. When this process is complete, the B cell is sensitized. In most cases, the sensitized B cell must then encounter a specific kind of T cell, called a helper T cell, before it is activated. This activation of the helper T cell occurs when a dendritic cell presents an antigen on its MHC II molecule, allowing the T cell to recognize it and mature.

The helper T cell binds to the antigen-MHC class II complex and is induced to release cytokines that induce the B cell to divide rapidly, making thousands of identical (clonal) cells. These daughter cells become either plasma cells or memory B cells. The memory B cells remain inactive at this point. A later encounter with the antigen, caused by a reinfection by the same bacteria or virus, will result in them dividing into a new population of plasma cells. The plasma cells, on the other hand, produce and secrete large quantities, up to 100 million molecules per hour, of antibody molecules. An antibody, also known as an immunoglobulin (Ig), is a protein that is produced by plasma cells after stimulation by an antigen. Antibodies are the agents of humoral immunity. Antibodies occur in the blood, in gastric and mucus secretions, and in breast milk. Antibodies in these bodily fluids can bind pathogens and mark them for destruction by phagocytes before they are able to infect cells.

Antibodies

These antibodies circulate in the blood stream and lymphatic system, binding with the antigen whenever it is encountered. The binding can fight infection in several ways. Antibodies can bind to viruses or bacteria, which interferes with the chemical interactions required for them to infect or bind to other cells. The antibodies may create bridges between different particles containing antigenic sites, clumping them all together and preventing their proper functioning. Antibody neutralization can prevent pathogens from entering and infecting host cells. The neutralized antibody-coated pathogens can then be filtered by the spleen to be eliminated in urine or feces. The antigen-antibody complex stimulates the complement system described previously, destroying the cell bearing the antigen. Antibodies also opsonize pathogen cells, wherein they mark them for destruction by phagocytic cells, such as macrophages or neutrophils. Additionally, antibodies stimulate inflammation, while their presence in mucus and on the skin prevents pathogen attack.

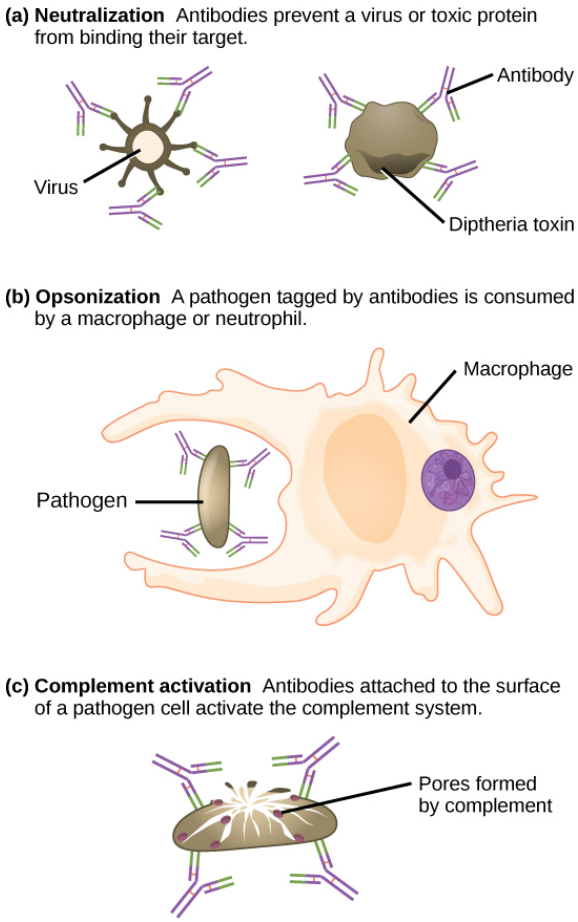

Figure 42.2B.1: **Methods by which antibodies inhibit infection**: Antibodies may inhibit infection by (a) preventing the antigen from binding its target, (b) tagging a pathogen for destruction by macrophages or neutrophils, or (c) activating the complement cascade.

The production of antibodies by plasma cells in response to an antigen is called active immunity. This describes the host's active response of the immune system to an infection or to a vaccination. There is also a passive immune response wherein antibodies are introduced into the host from an outside source, instead of the individual's own plasma cells. For example, antibodies circulating in a pregnant woman's body move across the placenta into the developing fetus. The child benefits from the presence of these antibodies for up to several months after birth. In addition, a passive immune response is possible by injecting antibodies into an individual in the form of an antivenom to a snake-bite toxin or antibodies in blood serum to help fight a hepatitis infection, giving immediate relief.

42.2C: Cell-Mediated Immunity

Cell-mediated immunity involves cytotoxic T cells recognizing infected cells and bringing about their destruction.

Learning Objectives

- Summarize the cell-mediated immune response

Key Points

- Once a pathogen enters a cell, it can no longer be detected by the humoral immune response; instead, the cell-mediated immune response must take over to kill the infected cell before it can allow the virus or bacteria to replicate and spread.
- T cells recognize infected cells by interacting with antigen present on their MHC II molecules; before a T cell can do so, it must be activated via interaction with an antigen presenting cell, or APC.
- Once a cytotoxic T cell (T_C) is activated, it will clone itself, producing many T_C cells with the correct receptors; some portion of the cells are active and will help destroy infected cells, while others are inactive memory cells that will create more active T_C cells if the infection returns.
- Helper T cells (T_H cells) also aid in cell-mediated immunity by releasing signaling molecules known as cytokines which can recruit natural killer cells and phagocytes to destroy infected cells and further activate T_C cells; they do not directly destroy pathogens.

Key Terms

- **cytotoxic T cell**: a subgroup of lymphocytes (white blood cells) that are capable of inducing death to infected somatic or tumor cells; part of cell-mediated immunity
- **cytokine**: any of various small regulatory proteins that regulate the cells of the immune system; they are released upon binding of PRRs to PAMPS

T cells

Just as the humoral immune response has B cells which mediate its response, the cellular immune response has T cells, which recognize infected cells and destroy them before the pathogen inside can replicate and spread to infect other cells. Unlike B cells, T lymphocytes (T cells) are unable to recognize pathogens without assistance. First, an antigen-presenting cell (APC, such as a dendritic cell or a macrophage) detects, engulfs (via phagocytosis in the case of macrophages or by entry of the pathogen of its own accord in the case of dendritic cells), and digests pathogens into hundreds or thousands of antigen fragments. These fragments are then transported to the surface of the APC, where they are presented on proteins known as Major Histocompatibility Complexes class II (MHC II, see). T cells become activated towards a certain antigen once they encounter it displayed on an MHC II. After a virus or bacteria enters a cell, it can no longer be detected by the humoral immune response. Instead, the cellular immune response must take over. To do so, a T cell will become activated by interacting with an antigen of the infecting cell or virus presented on the MHC II of an APC.

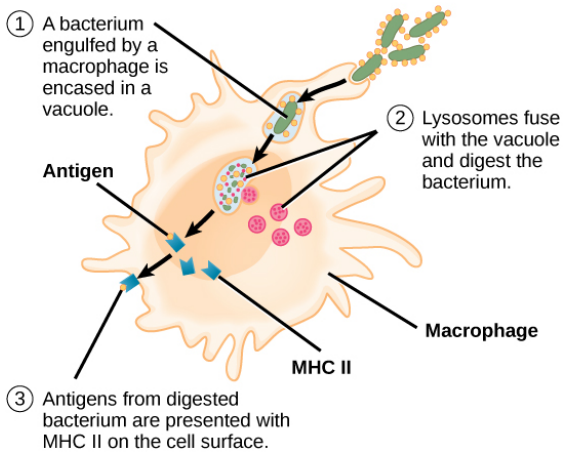

Figure 42.2C.1: **APCs, MHCs and lymphocytes**: An antigen-presenting cell (APC), such as a macrophage, engulfs a foreign antigen, partially digests it in a lysosome, and then embeds it in an MHC class II molecule for presentation at the cell surface. Lymphocytes of the adaptive immune response must interact with antigen-embedded MHC class II molecules to mature into functional immune cells.

Cytotoxic T cells mediate one arm of the cellular immune response

There are two main types of T cells: helper T lymphocytes (T_H) and the cytotoxic T lymphocytes (T_C). The T_H lymphocytes function indirectly to tell other immune cells about potential pathogens, while cytotoxic T cells (T_C) are the key component of the cell-mediated part of the adaptive immune system which attacks and destroys infected cells. T_C cells are particularly important in protecting against viral infections because viruses replicate within cells where they are shielded from extracellular contact with circulating antibodies. Once activated, the T_C creates a large clone of cells with one specific set of cell-surface receptors, similar to the proliferation of activated B cells. As with B cells, the clone includes active T_C cells and inactive memory T_C cells. The resulting active T_C cells then identify infected host cells.

T_C cells attempt to identify and destroy infected cells by triggering apoptosis (programmed cell death) before the pathogen can replicate and escape, thereby halting the progression of intracellular infections. To recognize which cells to pursue, T_C recognize antigens presented on MHC I complexes, which are present on all nucleated cells. MHC I complexes display a current readout of intracellular proteins inside a cell and will present pathogen antigens if the pathogen is present in the cell. T_C cells also support NK lymphocytes to destroy early cancers.

Cytokines released by T_H cells recruit NK cells and phagocytes

Cytokines are signaling molecules secreted by a T_H cell in response to a pathogen-infected cell; they stimulate natural killer cells and phagocytes such as macrophages. Phagocytes will then engulf infected cells and destroy them. Cytokines are also involved in stimulating T_C cells, enhancing their ability to identify and destroy infected cells and tumors. A summary of how the humoral and cell-mediated immune responses are activated appears in. B plasma cells and T_C cells are collectively called effector cells because they are involved in “effecting” (bringing about) the immune response of killing pathogens and infected host cells.

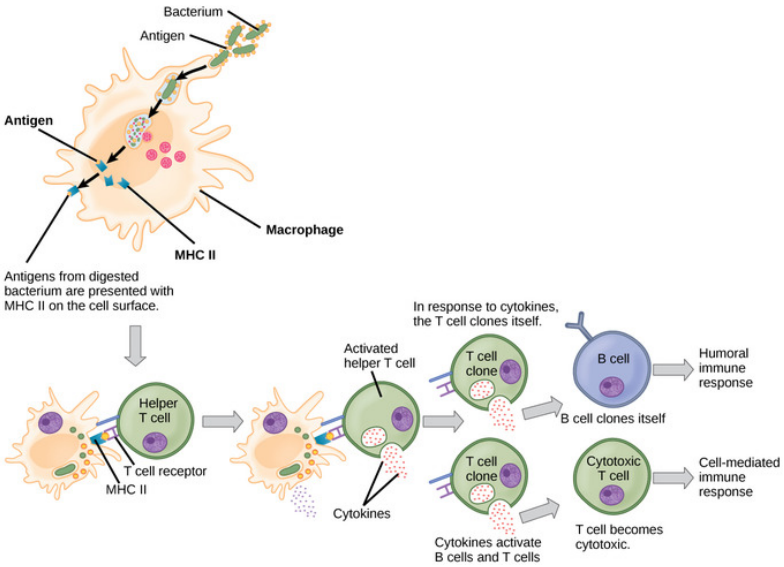

Figure 42.2C.1: **Helper T cells in the immune response**: A helper T cell becomes activated by binding to an antigen presented by an APC via the MHCII receptor, causing it to release cytokines. Depending on the cytokines released, this activates either the humoral or the cell-mediated immune response.

42.2D: Cytotoxic T Lymphocytes and Mucosal Surfaces

The lymphatic system houses large populations of immune cells which are released upon detection of a pathogen.

Learning Objectives

- Describe the features of the lymphatic system as they relate to the immune response

Key Points

- The lymphatic system contains lymph: a fluid that bathes tissues and organs and contains white blood cells (not red blood cells).
- Once B and T cells mature, the majority of them enter the lymphatic system, where they are stored in lymph nodes until needed.
- Lymph nodes also store dendritic cells and macrophages; as antigens are filtered through the lymphatic system, these cells collect them so as to present them to B and T cells.
- The spleen, which is to blood what lymph nodes are to lymph, filters foreign substances and antibody -complexed pathogens from the blood.

Key Terms

- **lymph**: a colorless, watery, bodily fluid carried by the lymphatic system, consisting mainly of white blood cells

Lymphatic system

Lymph, the watery fluid that bathes tissues and organs, contains protective white blood cells, but does not contain erythrocytes (red blood cells). Lymph moves about the body through the lymphatic system, which is made up of vessels, lymph ducts, lymph glands, and organs such as tonsils, adenoids, thymus, and spleen. Although the immune system is characterized by circulating cells throughout the body, the regulation, maturation, and intercommunication of immune factors occur at specific sites that are known as lymph nodes.

The blood circulates immune cells, proteins, and other factors through the body. Approximately 0.1 percent of all cells in the blood are leukocytes, which include monocytes (the precursor of macrophages) and lymphocytes. Most cells in the blood are red blood cells. Cells of the immune system can travel between the distinct lymphatic and blood circulatory systems, which are separated by interstitial space, by a process called extravasation (passing through to surrounding tissue).

Recall that cells of the immune system originate from stem cells in the bone marrow. B cell maturation occurs in the bone marrow, whereas progenitor cells migrate from the bone marrow and develop and mature into naïve T cells in the organ called the thymus. On maturation, T and B lymphocytes circulate to various destinations. Lymph nodes scattered throughout the body house large populations of T and B cells, dendritic cells, and macrophages. Lymph gathers antigens as it drains from tissues. These antigens are filtered through lymph nodes before the lymph is returned to circulation. Antigen-presenting cells (APCs) in the lymph nodes capture and process antigens, informing nearby lymphocytes about potential pathogens.

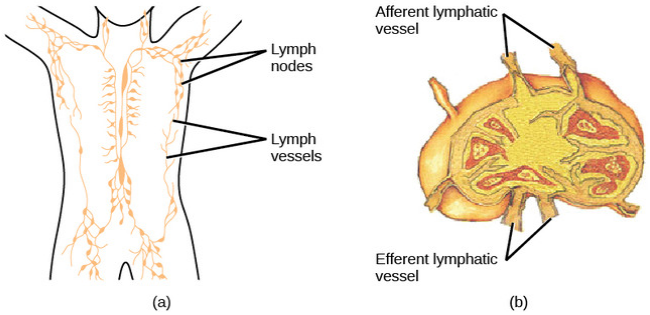

Figure 42.2D.1: **Lymphatic system**: (a) Lymphatic vessels carry a clear fluid called lymph throughout the body. The liquid passes through (b) lymph nodes that filter the lymph that enters the node through afferent vessels, leaving through efferent vessels. Lymph nodes are filled with lymphocytes that purge infecting cells.

The spleen houses B and T cells, macrophages, dendritic cells, and NK cells. The spleen is also the site where APCs that have trapped foreign particles in the blood can communicate with lymphocytes. Antibodies are synthesized and secreted by activated plasma cells in the spleen, which filters foreign substances and antibody-complexed pathogens from the blood. Functionally, the spleen is to the blood as lymph nodes are to the lymph.

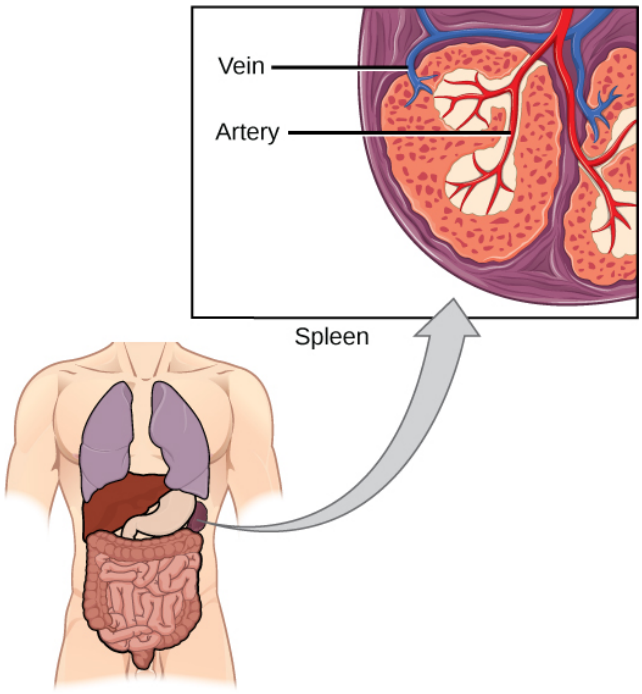

Figure $42.2D.1$: **Spleen in the lymphatic system**: The spleen functions to immunologically filter the blood and allow for communication between cells corresponding to the innate and adaptive immune responses.

42.2E: Immunological Memory

Immunological memory allows the adaptive immune system to very rapidly clear infections that it has encountered before.

Learning Objectives

- Describe the role of memory B and T cells in immulogical memory

Key Points

- During the adaptive immune response to a pathogen that the body has not previously encountered (the primary immune response), the level of plasma cells that secrete antibodies to the new pathogen and T cells that recognize the pathogen will increase at a modest pace.
- If a pathogen is re-encountered, memory B and T cells can immediately differentiate into plasma cells and cytotoxic T cells without input from APCs or T_H cells; this secondary immune response occurs much more rapidly than the primary immune response.
- Vaccination can be used to generate a mild, primary immune response against an inactivated pathogen, which will allow the secondary immune response to function the first time the immune system encounters the actual pathogen.

Key Terms

- **vaccination**: inoculation in order to protect against a particular disease or strain of disease; causes a primary immune response without illness, allowing the secondary response to destroy subsequent infection
- **effector cell**: a plasma B cell or cytotoxic T cell, which are the main types of cells responsible for the humoral and cellular immune responses, respectively

Immunological Memory

The adaptive immune system has a memory component that allows for a rapid and large response upon re-invasion of the same pathogen. During the adaptive immune response to a pathogen that has not been encountered before, known as the primary immune response, plasma cells secreting antibodies and differentiated T cells increase, then plateau over time. As B cells and T cells mature into effector cells, a subset of the naïve populations differentiates into B and T memory cells with the same antigen specificities. A memory cell is an antigen-specific B or T lymphocyte that does not differentiate into an effector cell during the primary immune response, but that can immediately become an effector cell on re-exposure to the same pathogen. As the infection is cleared and pathogenic stimuli subside, the effector cells are no longer needed; they undergo apoptosis. In contrast, the memory cells persist in circulation.

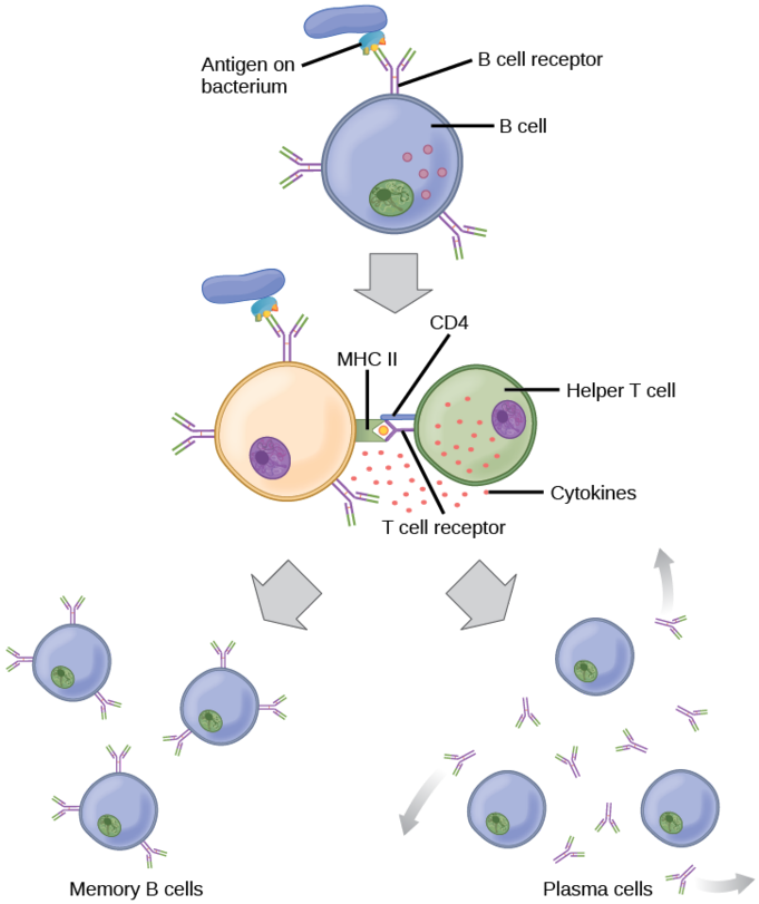

Figure 42.2E. 1: **Formation of memory B cells and plasma cells**: After initially binding an antigen to the B cell receptor (BCR), a B cell internalizes the antigen and presents it on MHC II. A helper T cell recognizes the MHC II–antigen complex and activates the B cell. As a result, memory B cells and plasma cells are made.

If the pathogen is not encountered again during the individual's lifetime, B and T memory cells will circulate for a few years or even several decades, gradually dying off, having never functioned as effector cells. However, if the host is re-exposed to the same pathogen type, circulating memory cells will immediately differentiate into plasma cells and T_C cells without input from APCs or T_H cells. This is known as the secondary immune response. One reason why the adaptive immune response is delayed is that it takes time for naïve B and T cells with the appropriate antigen specificities to be identified, activated, and proliferate. On reinfection, this step is skipped. The result is a more rapid production of immune defenses. Memory B cells that differentiate into plasma cells output ten to hundred-fold greater antibody amounts than were secreted during the primary response. This rapid and dramatic antibody response may stop the infection before it can even become established. Individuals may not realize they had been exposed.

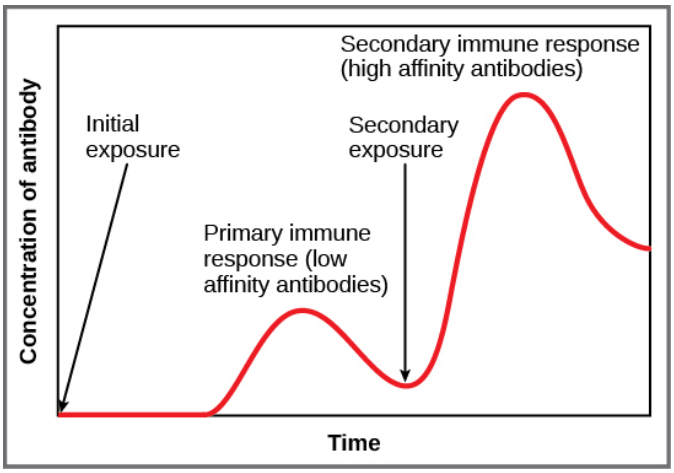

Figure 42.2E.1: **Primary versus secondary immune response**: In the primary response to infection, antibodies are secreted first from plasma cells. Upon re-exposure to the same pathogen, memory cells differentiate into antibody-secreting plasma cells that output a greater amount of antibody for a longer period of time.

Vaccination is based on the knowledge that exposure to noninfectious antigens, derived from known pathogens, generates a mild primary immune response. The immune response to vaccination may not be perceived by the host as illness, but still confers immune memory. When exposed to the corresponding pathogen to which an individual was vaccinated, the reaction is similar to a secondary exposure. Because each reinfection generates more memory cells and increased resistance to the pathogen, some vaccine courses involve one or more booster vaccinations to mimic repeat exposures.

Figure 42.2E.1: **Vaccination**: Vaccines, often delivered by injection into the arm, result in a secondary immune response if the vaccinated individual is later exposed to that pathogen.

42.2F: Regulating Immune Tolerance

Immune tolerance of self and harmless antigens occurs by deleting B and T cells that recognize those antigens, often near mucosal surfaces.

Learning Objectives

- Describe the process by which immune tolerance is acquired

Key Points

- Mucosal surfaces are those that are in contact with air, but, unlike the skin, allow fluid to flow in and out of them.
- Due to the location of mucosal surfaces, they are often the first sites to encounter new antigens.
- The immune system has to be regulated to prevent wasteful, unnecessary responses to harmless substances and, more importantly, so that it does not attack “self”.
- Mucosal-associated lymphoid tissue, or MALT, is involved in immune tolerance to harmless, foreign antigens.

Key Terms

- **self-antigen**: antigens (substances that bind to antibodies) that are usually well tolerated by the immune system, which has been educated to non-reactivity against the structures
- **mucous membrane**: a membrane which secretes mucus; it forms the lining of body passages that contact the air, such as the respiratory and genitourinary tracts including the mouth, nasal passages, vagina and urethra

Mucosal surfaces

The innate and adaptive immune responses discussed thus far comprise the systemic immune system (affecting the whole body), which is distinct from the mucosal immune system. Mucosa are another name for mucous membranes. Mucosal immunity is formed by mucosa-associated lymphoid tissue, or MALT, which functions independently of the systemic immune system; it has its own innate and adaptive components. MALT is a collection of lymphatic tissue that combines with epithelial tissue lining the mucosa throughout the body. This tissue functions as the immune barrier and immune response in areas of the body in direct contact to the external environment. The systemic and mucosal immune systems use many of the same cell types. Foreign particles that make their way to MALT are taken up by absorptive epithelial cells called M cells and delivered to APCs (antigen-presenting cells) located directly below the mucosal tissue. M cells are located in the Peyer’s patch, which is a lymphoid nodule. APCs of the mucosal immune system are primarily dendritic cells, with B cells and macrophages playing minor roles. Processed antigens displayed on APCs are detected by T cells in the MALT and at various mucosal induction sites, such as the tonsils, adenoids, appendix, or the mesenteric lymph nodes of the intestine. Activated T cells then migrate through the lymphatic system and into the circulatory system to mucosal sites of infection.

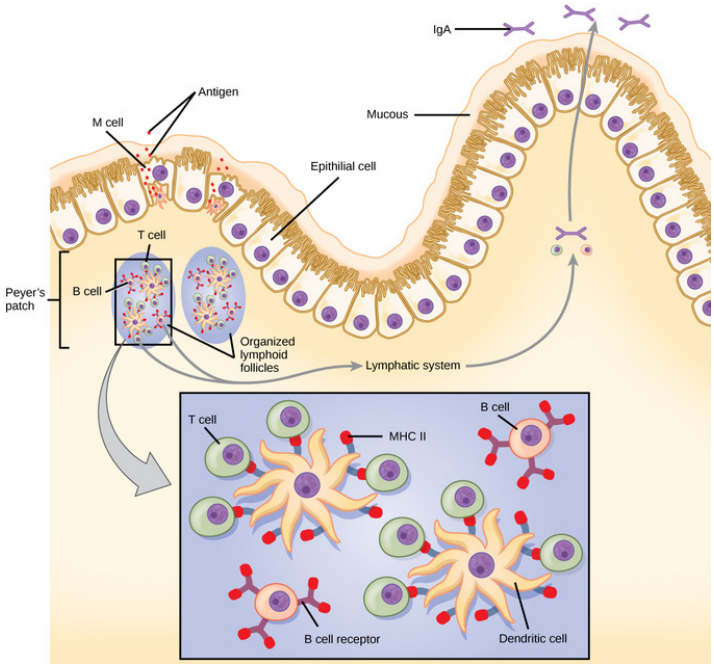

Figure 42.2F.1: **MALT tissue**: The topology and function of intestinal MALT is shown. Pathogens are taken up by M cells in the intestinal epithelium and excreted into a pocket formed by the inner surface of the cell. The pocket contains antigen-presenting cells, such as dendritic cells, which engulf the antigens, then present them with MHC II molecules on the cell surface. The dendritic cells migrate to an underlying tissue called a Peyer's patch. Antigen-presenting cells, T cells, and B cells aggregate within the Peyer's patch, forming organized lymphoid follicles. There, some T cells and B cells are activated. Other antigen-loaded dendritic cells migrate through the lymphatic system where they activate B cells, T cells, and plasma cells in the lymph nodes. The activated cells then return to MALT tissue effector sites. IgA and other antibodies are secreted into the intestinal lumen.

MALT is a crucial component of a functional immune system because mucosal surfaces, such as the nasal passages, are the first tissues onto which inhaled or ingested pathogens are deposited. This allows the immune system to detect and deal with pathogens very quickly after they enter the body through various mucous membranes. The mucosal tissue includes the mouth, pharynx, and esophagus, along with the gastrointestinal, respiratory, and urogenital tracts.

Immune tolerance

The immune system has to be regulated to prevent wasteful, unnecessary responses to harmless substances and, more importantly, so that it does not attack "self." The acquired ability to prevent an unnecessary or harmful immune response to a detected foreign substance known not to cause disease or to self-antigens is described as immune tolerance. The primary mechanism for developing immune tolerance to self-antigens occurs during the selection for weakly, self-binding cells during T and B lymphocyte maturation. Any T or B lymphocytes that recognize harmless foreign or "self" antigens are deleted before they can fully mature into immunocompetent cells.

There are populations of T cells that suppress the immune response to self-antigens. They also suppress the immune response after the infection has cleared to minimize host cell damage induced by inflammation and cell lysis. Immune tolerance is especially well developed in the mucosa of the upper digestive system because of the tremendous number of foreign substances (such as food proteins) that APCs of the oral cavity, pharynx, and gastrointestinal mucosa encounter. Immune tolerance is brought about by specialized APCs in the liver, lymph nodes, small intestine, and lung that present harmless antigens to a diverse population of regulatory T (T_{reg}) cells: specialized lymphocytes that suppress local inflammation and inhibit the secretion of stimulatory immune factors. The combined result of T_{reg} cells is to prevent immunologic activation and inflammation in undesired tissue compartments, allowing the immune system to focus on hazardous pathogens instead.

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42.3: Antibodies

Topic hierarchy

- 42.3A: Antibody Structure
- 42.3B: Antibody Functions

42.3A: Antibody Structure

Variations in antibody structure allow great diversity of antigen recognition among different antibodies.

Learning Objectives

- Differentiate among the classes of antibodies

Key Points

- Antibodies contain four polypeptides: two identical (to each other) heavy chains in a “Y” formation and two idenitical (to each other) light chains on the outside of the top of the “Y” portion.
- Each antibody has a unique variable region, which is responsible for antigen detection and specificity.
- There are five classes of antibodies, each utilized by the body under different conditions, including IgM, IgG, IgA, IgD, and IgE; Ig stands for immunoglobulin.
- IgAs, secreted in the milk, tears and mucous, are the most numerous antibodies produced; inside of the body, circulating IgGs are the most abundant.

Key Terms

- **immunoglobulin**: any of the glycoproteins in blood serum that respond to invasion by foreign antigens and that protect the host by removing pathogens; also known as an antibody
- **antigen**: a substance that binds to a specific antibody; may cause an immune response
- **B cell**: a lymphocyte, developed in the bursa of birds and the bone marrow of other animals, that produces antibodies and is responsible for the immune system
- **epitope**: that part of a biomolecule (such as a protein) that is the target of an immune response; the part of the antigen recognized by the immune system

Antibody Structure

An antibody is a molecule that recognizes a specific antigen; this recognition is a vital component of the adaptive immune response. Antibodies are composed of four polypeptides: two identical heavy chains (large peptide units) that are partially bound to each other in a “Y” formation, which are flanked by two identical light chains (small peptide units). The area where the antigen is recognized on the antibody is known as the variable domain or variable region. This is why there are numerous antibodies that can each recognize a different antigen. The antibody base is known as the constant domain or constant region. The portion of an antigen that is recognized by the antibody is known as the epitope.

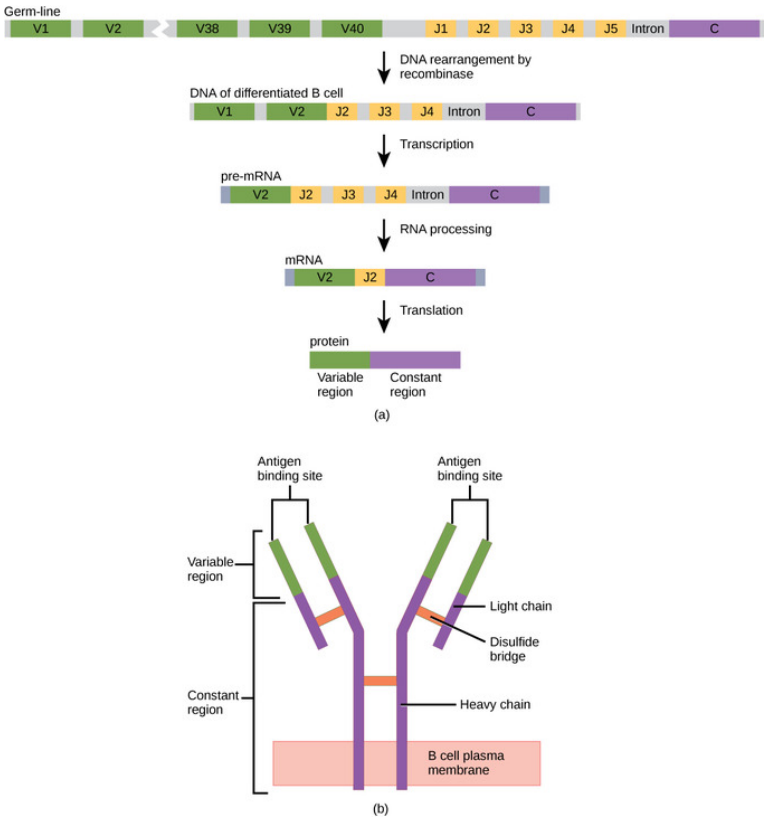


Figure 42.3*A*.1: **Antibodies**: (a) As a germ-line B cell matures, an enzyme called DNA recombinase randomly excises V and J segments from the light chain gene. Splicing at the mRNA level results in further gene rearrangement. As a result, (b) each antibody has a unique variable region capable of binding a different antigen.

Antibody variation

In B cells, the variable region of the light chain gene has 40 variable (V) and five joining (J) segments. An enzyme called DNA recombinase randomly excises most of these segments out of the gene, splicing one V segment to one J segment. During RNA processing, all but one V and J segment are spliced out. Recombination and splicing may result in over 10^6 possible VJ combinations. As a result, each differentiated B cell in the human body typically has a unique variable chain. The constant domain, which does not bind to an antibody, is the same for all antibodies. The large diversity of antibody structure translates into the large diversity of antigens that antibodies can bind and recognize.

Similar to TCRs (T cell receptors) and BCRs (B cell receptors), antibody diversity is produced by the mutation and recombination of approximately 300 different gene segments encoding the light and heavy chain variable domains in precursor cells that are destined to become B cells. The variable domains from the heavy and light chains interact to form the binding site through which an antibody can bind a specific epitope on an antigen. The numbers of repeated constant domains in Ig classes (discussed below) are the same for all antibodies corresponding to a specific class. Antibodies are structurally similar to the extracellular component of the BCRs. The maturation of B cells into plasma cells occurs when the cells gain the ability to secrete the antibody portion of its BCR in large quantities.

Antibody Classes

Antibodies can be divided into five classes (IgM, IgG, IgA, IgD, and IgE) based on their physiochemical, structural, and immunological properties. Ig stands for immunoglobulin, another term for an antibody. IgGs, which make up about 80 percent of all antibodies in circulation, have heavy chains that consist of one variable domain and three identical constant domains. IgA and IgD also have three constant domains per heavy chain, whereas IgM and IgE each have four constant domains per heavy chain. The variable domain determines binding specificity, while the constant domain of the heavy chain determines the immunological mechanism of action of the corresponding antibody class. It is possible for two antibodies to have the same binding specificities, but be in different classes and, therefore, to be involved in different functions.

In an adaptive defense is produced against a pathogen, typically plasma cells first secrete IgM into the blood. BCRs on B cells are of the IgM class and, occasionally, the IgD class. IgM molecules comprise approximately ten percent of all antibodies. Prior to antibody secretion, plasma cells assemble IgM molecules into pentamers (five individual antibodies) linked by a joining (J) chain. The pentamer arrangement means that these macromolecules can bind ten identical antigens. However, IgM molecules released early in the adaptive immune response do not bind to antigens as stably as do IgGs, which are one of the possible types of antibodies secreted in large quantities upon re-exposure to the same pathogen. The properties of immunoglobulins and their basic structures are shown in the table.

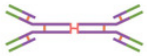
Name	Properties	Structure
IgA	Found in mucous, saliva, tears, and breast milk. Protects against pathogens.	
IgD	Part of the B cell receptor. Activates basophils and mast cells.	
IgE	Protects against parasitic worms. Responsible for allergic reactions.	
IgG	Secreted by plasma cells in the blood. Able to cross the placenta into the fetus.	
IgM	May be attached to the surface of a B cell or secreted into the blood. Responsible for early stages of immunity.	

Figure 42.3A.1: **Classes of antibodies**: Immunoglobulins (antibody classes) have different functions, but all are composed of light and heavy chains that form a Y-shaped structure.

IgAs populate the saliva, tears, breast milk, and mucus secretions of the gastrointestinal, respiratory, and genitourinary tracts. Collectively, these bodily fluids coat and protect the extensive mucosa (4000 square feet in humans). The total number of IgA molecules in these bodily secretions is greater than the number of IgG molecules in the blood serum. A small amount of IgA is also secreted into the serum in monomeric form. Conversely, some IgM is secreted into bodily fluids of the mucosa. Similarly to IgM, IgA molecules are secreted as polymeric structures linked with a J chain. However, IgAs are secreted mostly as dimeric molecules, not pentamers.

IgE is present in the serum in small quantities and is best characterized in its role as an allergy mediator. IgD is also present in small quantities. Similarly to IgM, BCRs containing the IgD class of antibodies are found on the surface of naïve B cells. This class supports antigen recognition and subsequent maturation of B cells to plasma cells.

42.3B: Antibody Functions

Antibodies, part of the humoral immune response, are involved in pathogen detection and neutralization.

Learning Objectives

- Differentiate among affinity, avidity, and cross-reactivity in antibodies

Key Points

- Antibodies are produced by plasma cells, but, once secreted, can act independently against extracellular pathogen and toxins.
- Antibodies bind to specific antigens on pathogens; this binding can inhibit pathogen infectivity by blocking key extracellular sites, such as receptors involved in host cell entry.
- Antibodies can also induce the innate immune response to destroy a pathogen, by activating phagocytes such as macrophages or neutrophils, which are attracted to antibody-bound cells.
- Affinity describes how strongly a single antibody binds a given antigen, while avidity describes the binding of a multimeric antibody to multiple antigens.
- A multimeric antibody may have individual arms with low affinity, but have high overall avidity due to synergistic effects between binding sites.
- Cross reactivity occurs when an antibody binds to a different-but-similar antigen than the one for which it was raised; this can increase pathogen resistance or result in an autoimmune reaction.

Key Terms

- **avidity**: the measure of the synergism of the strength individual interactions between proteins
- **affinity**: the attraction between an antibody and an antigen

Antibody Functions

Differentiated plasma cells are crucial players in the humoral immunity response. The antibodies they secrete are particularly significant against extracellular pathogens and toxins. Once secreted, antibodies circulate freely and act independently of plasma cells. Sometimes, antibodies can be transferred from one individual to another. For instance, a person who has recently produced a successful immune response against a particular disease agent can donate blood to a non-immune recipient, confering temporary immunity through antibodies in the donor's blood serum. This phenomenon, called passive immunity, also occurs naturally during breastfeeding, which makes breastfed infants highly resistant to infections during the first few months of life.

Antibodies coat extracellular pathogens and neutralize them by blocking key sites on the pathogen that enhance their infectivity, such as receptors that "dock" pathogens on host cells. Antibody neutralization can prevent pathogens from entering and infecting host cells, as opposed to the cytotoxic T-cell-mediated approach of killing cells that are already infected to prevent progression of an established infection. The neutralized antibody-coated pathogens can then be filtered by the spleen and eliminated in urine or feces.

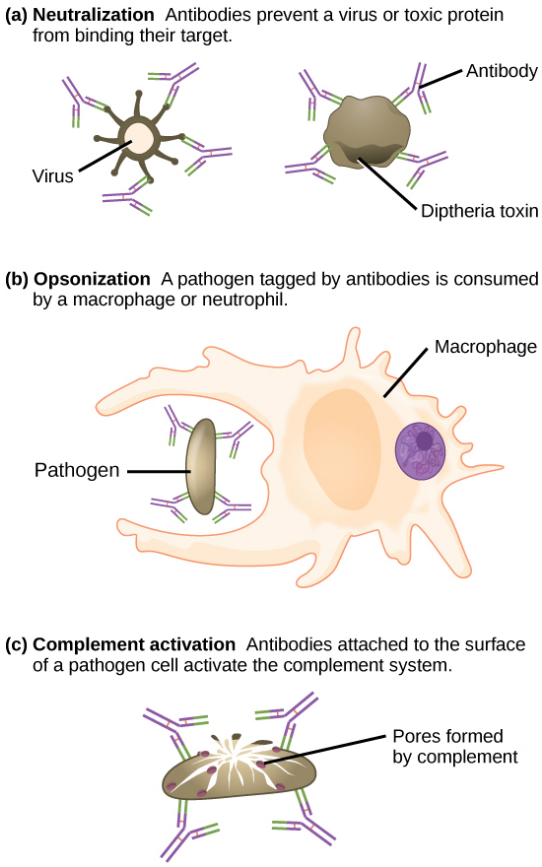


Figure $42.3B.1$: **Mechanisms of antibody action**: Antibodies may inhibit infection by (a) preventing the antigen from binding to its target, (b) tagging a pathogen for destruction by macrophages or neutrophils, or (c) activating the complement cascade.

Antibodies also mark pathogens for destruction by phagocytic cells, such as macrophages or neutrophils, because they are highly attracted to macromolecules complexed with antibodies. Phagocytic enhancement by antibodies is called opsonization. In another process, complement fixation, IgM and IgG in serum bind to antigens, providing docking sites onto which sequential complement proteins can bind. The combination of antibodies and complement enhances opsonization even further, promoting rapid clearing of pathogens.

Affinity, avidity, and cross reactivity

Not all antibodies bind with the same strength, specificity, and stability. In fact, antibodies exhibit different affinities (attraction) depending on the molecular complementarity between antigen and antibody molecules. An antibody with a higher affinity for a particular antigen would bind more strongly and stably. It would be expected to present a more challenging defense against the pathogen corresponding to the specific antigen.

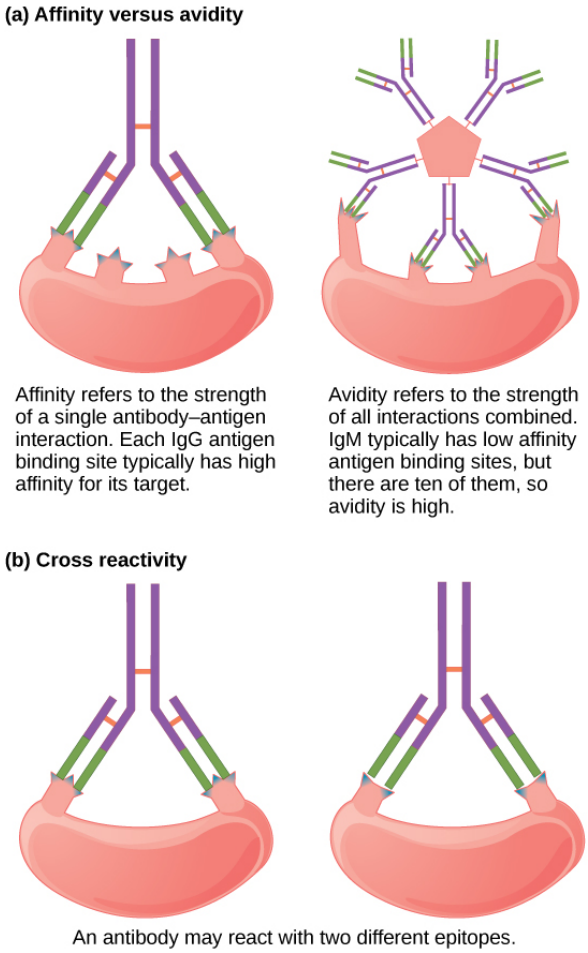


Figure $42.3B.1$: **Antibody affinity, avidity, and cross reactivity**: (a) Affinity refers to the strength of single interactions between antigen and antibody, while avidity refers to the strength of all interactions combined. (b) An antibody may cross-react with different epitopes.

The term avidity describes binding by antibody classes that are secreted as joined, multivalent structures (such as IgM and IgA). Although avidity measures the strength of binding, just as affinity does, the avidity is not simply the sum of the affinities of the antibodies in a multimeric structure. The avidity depends on the number of identical binding sites on the antigen being detected, as well as other physical and chemical factors. Typically, multimeric antibodies, such as pentameric IgM, are classified as having lower affinity than monomeric antibodies, but high avidity. Essentially, the fact that multimeric antibodies can bind many antigens simultaneously balances their slightly-lower-binding strength for each antibody/antigen interaction.

Antibodies secreted after binding to one epitope on an antigen may exhibit cross reactivity for the same or similar epitopes on different antigens. Cross reactivity occurs when an antibody binds not to the antigen that elicited its synthesis and secretion, but to a different antigen. Because an epitope corresponds to such a small region (the surface area of about four to six amino acids), it is possible for different macromolecules to exhibit the same molecular identities and orientations over short regions.

Cross reactivity can be beneficial if an individual develops immunity to several related pathogens despite having been exposed to or vaccinated against only one of them. For instance, antibody cross reactivity may occur against the similar surface structures of various Gram-negative bacteria. Conversely, antibodies raised against pathogenic molecular components that resemble self molecules may incorrectly mark host cells for destruction, causing autoimmune damage. Patients who develop systemic lupus erythematosus (SLE) commonly exhibit antibodies that react with their own DNA. These antibodies may have been initially raised against the nucleic acid of microorganisms, but later cross-reacted with self-antigens. This phenomenon is also called molecular mimicry.

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Topic hierarchy

- 42.4A: Immunodeficiency
- 42.4B: Hypersensitivities

42.4A: Immunodeficiency

Immunodeficiency occurs when the immune system cannot appropriately respond to infections.

Learning Objectives

- Explain the problems associated with immunodeficiency

Key Points

- If a pathogen is allowed to proliferate to certain levels, the immune system can become overwhelmed; immunodeficiency occurs when the immune system fails to respond sufficiently to a pathogen.
- Immunodeficiency can be caused by many factors, including certain pathogens, malnutrition, chemical exposure, radiation exposure, or even extreme stress.
- HIV is a virus that causes immunodeficiency by infecting helper T cells, causing cytotoxic T cells to destroy them.

Key Terms

- **phagocyte**: a cell of the immune system, such as a neutrophil, macrophage or dendritic cell, that engulfs and destroys viruses, bacteria, and waste materials
- **lysis**: the disintegration or destruction of cells
- **immunodeficiency**: a depletion in the body's natural immune system, or in some component of it

Immunodeficiency

Failures, insufficiencies, or delays at any level of the immune response can allow pathogens or tumor cells to gain a foothold to replicate or proliferate to high enough levels that the immune system becomes overwhelmed, leading to immunodeficiency; it may be acquired or inherited. Immunodeficiency can be acquired as a result of infection with certain pathogens (such as HIV), chemical exposure (including certain medical treatments), malnutrition, or, possibly, by extreme stress. For instance, radiation exposure can destroy populations of lymphocytes, elevating an individual's susceptibility to infections and cancer. Dozens of genetic disorders result in immunodeficiencies, including Severe Combined Immunodeficiency (SCID), bare lymphocyte syndrome, and MHC II deficiencies. Rarely, primary immunodeficiencies that are present from birth may occur. Neutropenia is one form in which the immune system produces a below-average number of neutrophils, the body's most abundant phagocytes. As a result, bacterial infections may go unrestricted in the blood, causing serious complications.

HIV/AIDS

Human immunodeficiency virus infection / acquired immunodeficiency syndrome (HIV/AIDS), is a disease of the human immune system caused by infection with human immunodeficiency virus (HIV). During the initial infection, a person may experience a brief period of influenza-like illness. This is typically followed by a prolonged period without symptoms. As the illness progresses, it interferes more and more with the immune system. The person has a high probability of becoming infected, including from opportunistic infections and tumors that do not usually affect people who have working immune systems.

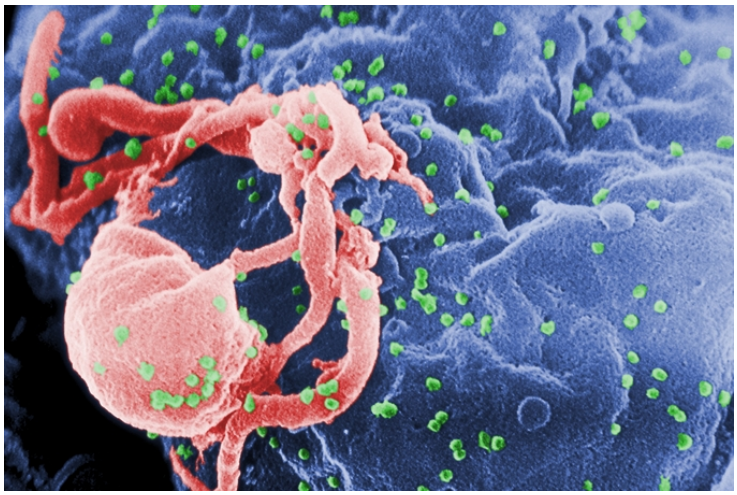

Figure 42.4A.1: **Image of HIV: scanning electron micrograph of HIV-1 budding (in green, color added) from cultured lymphocyte**: Multiple round bumps on cell surface represent sites of assembly and budding of HIV. During primary infection, the level of HIV may reach several million virus particles per milliliter of blood.

After the virus enters the body, there is a period of rapid viral replication, leading to an abundance of virus in the peripheral blood. During primary infection, the level of HIV may reach several million virus particles per milliliter of blood. This response is accompanied by a marked drop in the number of circulating CD4+ T cells, cells that are or will become helper T cells. The acute viremia, or spreading of the virus, is almost invariably associated with activation of CD8+ T cells (which kill HIV-infected cells) and, subsequently, with antibody production. The CD8+ T cell response is thought to be important in controlling virus levels, which peak and then decline, as the CD4+ T cell counts recover.

Ultimately, HIV causes AIDS by depleting CD4+ T cells (helper T cells). This weakens the immune system, allowing opportunistic infections. T cells are essential to the immune response; without them, the body cannot fight infections or kill cancerous cells. The mechanism of CD4+ T cell depletion differs in the acute and chronic phases. During the acute phase, HIV-induced cell lysis and killing of infected cells by cytotoxic T cells accounts for CD4+ T cell depletion, although apoptosis (programmed cell death) may also be a factor. During the chronic phase, the consequences of generalized immune activation coupled with the gradual loss of the ability of the immune system to generate new T cells appear to account for the slow decline in CD4+ T cell numbers.

42.4B: Hypersensitivities

Hypersensitivities are maladaptive immune reactions against harmless antigens (allergies) or against self antigens (autoimmunity).

Learning Objectives

- Distinguish between the disruptions to the immune system caused by allergies and autoimmunity

Key Points

- While the immune system is normally very tightly controlled to disregard "self" and harmless antigens, a condition known as hypersensitivity can override this control, causing illness and injury to an individual.
- Allergies result when the immune system recognizes harmless antigens, such as pollen or dust; they are characterized by red, swollen eyes, sneezing and itching, and can also encompass more serious symptoms such as anaphylactic shock.
- Autoimmunity occurs when the immune system recognizes "self" antigens and begins attacking them; antibodies that recognize self are termed autoantibodies.

Key Terms

- **autoantibody**: an antibody formed in response to an agent (autoantigen) produced by the organism itself
- **histamine**: an amine that causes dilatation of capillaries, contraction of smooth muscle, and stimulation of gastric acid secretion; it is released during allergic reactions

Hypersensitivities

Maladaptive immune responses toward harmless foreign substances or self antigens that occur after tissue sensitization are termed hypersensitivities. This can potentially be very dangerous for an individual, as the immune response can be very powerful; it can destroy host tissue if not kept in check. The types of hypersensitivities include immediate, delayed, and autoimmunity hypersensitivities. A large proportion of the population is affected by one or more types of hypersensitivity.

Allergies

The immune reaction that results from immediate hypersensitivities, in which an antibody-mediated immune response occurs within minutes of exposure to a harmless antigen, is called an allergy. In the United States, 20 percent of the population exhibits symptoms of allergy or asthma, whereas 55 percent test positive against one or more allergens. Upon initial exposure to a potential allergen, an allergic individual synthesizes antibodies of the IgE class; this class of antibodies also mediates the immune response to parasitic worms. The constant domain of the IgE molecules interacts with mast cells embedded in connective tissues. This process primes, or sensitizes, the tissue. Upon subsequent exposure to the same allergen, IgE molecules on mast cells bind the antigen via their variable domains, stimulating the mast cell to release the modified amino acids histamine and serotonin. These chemical mediators then recruit eosinophils which mediate allergic responses. The effects of an allergic reaction range from mild symptoms such as sneezing and itchy, watery eyes, to more severe or even life-threatening reactions involving intensely-itchy welts known as hives, airway contraction with severe respiratory distress, and plummeting blood pressure. This extreme reaction is known as anaphylactic shock. If not treated with epinephrine to counter the blood pressure and breathing effects, this condition can be fatal.

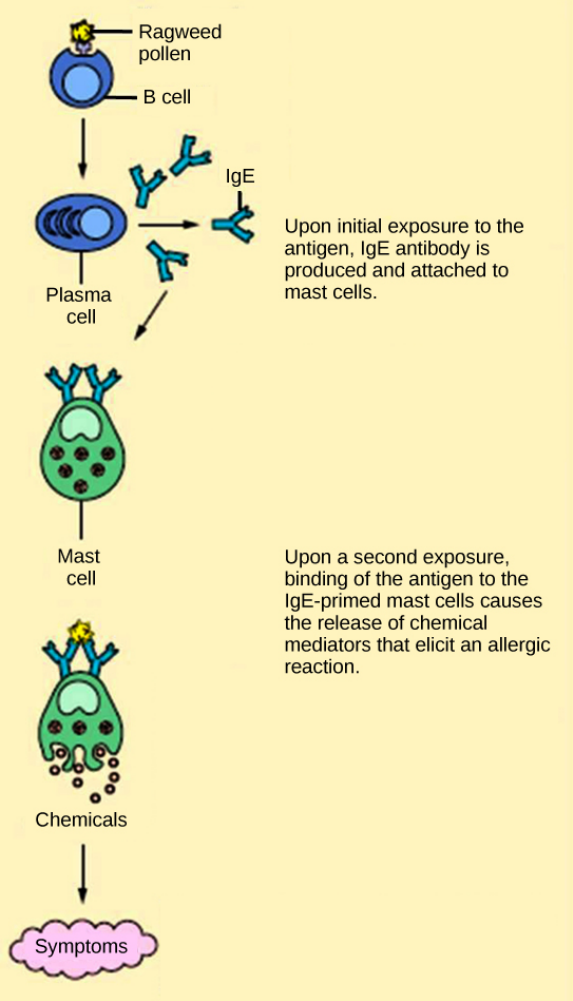

Figure 42.4B.1: **Allergens**: This image shows an example of an allergic response to ragweed pollen. On first exposure to an allergen, an IgE antibody is synthesized by plasma cells in response to a harmless antigen. The IgE molecules bind to mast cells. On secondary exposure, the mast cells release histamines and other modulators that affect the symptoms of allergy.

Delayed hypersensitivity is a cell-mediated immune response that takes approximately one to two days after secondary exposure for a maximal reaction to be observed. This type of hypersensitivity involves the T_H1 cytokine -mediated inflammatory response. It may manifest as local tissue lesions or contact dermatitis (rash or skin irritation). Delayed hypersensitivity occurs in some individuals in response to contact with certain types of jewelry or cosmetics. It also facilitates the immune response to poison ivy and is the reason why the skin test for tuberculosis results in a small region of inflammation on individuals who were previously exposed to *Mycobacterium tuberculosis*. Cortisone is typically used to treat such responses as it inhibits cytokine production.

Autoimmunity

Autoimmunity is a type of hypersensitivity to self antigens that affects approximately five percent of the population. Most types of autoimmunity involve the humoral immune response. Antibodies that inappropriately mark self components as foreign are termed autoantibodies. In patients with the autoimmune disease myasthenia gravis, muscle cell receptors that induce contraction in response to acetylcholine are targeted by antibodies. The result is muscle weakness that may include marked difficultly with fine and/or gross motor functions. In systemic lupus erythematosus, a diffuse autoantibody response to the individual's own DNA and proteins results in various systemic diseases. Systemic lupus erythematosus may affect the heart, joints, lungs, skin, kidneys, central nervous system, or other tissues, causing tissue damage via antibody binding, complement recruitment, lysis, and inflammation.

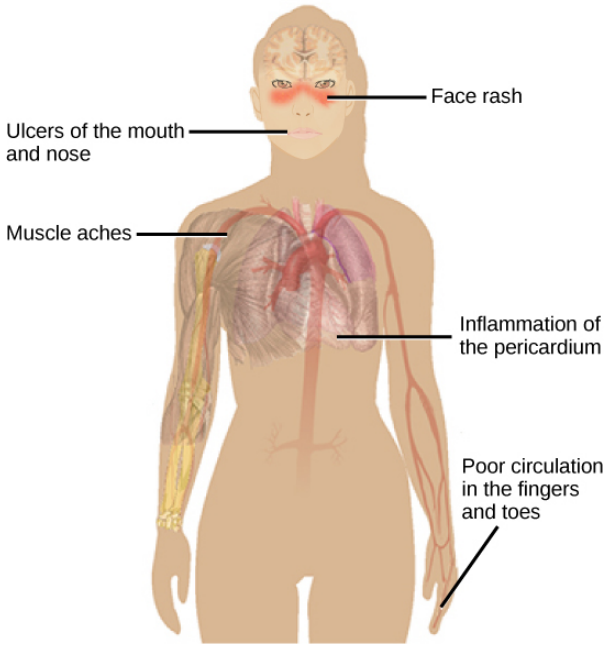

Figure 42.4B.1: **SLE and autoimmunity**: Systemic lupus erythematosus is characterized by autoimmunity to the individual's own DNA and/or proteins, which leads to varied dysfunction of the organs.

Autoimmunity can develop with time; its causes may be rooted in molecular mimicry. Antibodies and TCRs may bind self antigens that are structurally similar to pathogen antigens, which the immune receptors first raised. As an example, infection with *Streptococcus pyogenes* (bacterium that causes strep throat) may generate antibodies or T cells that react with heart muscle, which has a similar structure to the surface of *S. pyogenes*. These antibodies can damage heart muscle with autoimmune attacks, leading to rheumatic fever. Insulin -dependent (Type 1) diabetes mellitus arises from a destructive inflammatory T_H1 response against insulin-producing cells of the pancreas. Patients with this autoimmunity must be injected with insulin that originates from other sources.

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CHAPTER OVERVIEW

43: ANIMAL REPRODUCTION AND DEVELOPMENT

43.1: REPRODUCTION METHODS

43.1A: METHODS OF REPRODUCING
43.1B: TYPES OF SEXUAL AND ASEXUAL REPRODUCTION
43.1C: SEX DETERMINATION
43.2: FERTILIZATION

43.2A: EXTERNAL AND INTERNAL FERTILIZATION
43.2B: THE EVOLUTION OF REPRODUCTION
43.3: HUMAN REPRODUCTIVE ANATOMY AND GAMETOGENESIS

43.3A: MALE REPRODUCTIVE ANATOMY
43.3B: FEMALE REPRODUCTIVE ANATOMY
43.3C: GAMETOGENESIS (SPERMATOGENESIS AND OOGENESIS)
43.4: HORMONAL CONTROL OF HUMAN REPRODUCTION

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43.5A: FERTILIZATION
43.5B: CLEAVAGE, THE BLASTULA STAGE, AND GASTRULATION
43.6: ORGANOGENESIS AND VERTEBRATE FORMATION

43.6A: ORGANOGENESIS
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43.7: HUMAN PREGNANCY AND BIRTH

43.7A: HUMAN GESTATION
43.7B: LABOR AND BIRTH
43.7C: CONTRACEPTION AND BIRTH CONTROL
43.7D: INFERTILITY

Infertility, or the ability to conceive, can be treated by assisted reproductive technologies which can aid conception.

43.1: Reproduction Methods

Topic hierarchy

- 43.1A: Methods of Reproducing
- 43.1B: Types of Sexual and Asexual Reproduction
- 43.1C: Sex Determination

43.1A: Methods of Reproducing

Animal reproduction is essential to the survival of a species; it can occur through either asexual or sexual means.

Learning Objectives

- Describe reproduction in animals

Key Points

- Reproduction (or procreation) is the biological process by which new "offspring" are produced from their "parents".
- Asexual reproduction yields genetically-identical organisms because an individual reproduces without another.
- In sexual reproduction, the genetic material of two individuals from the same species combines to produce genetically-different offspring; this ensures mixing of the gene pool of the species.
- Organisms that reproduce through asexual reproduction tend to grow exponentially and rely on mutations for DNA variation, while those that reproduce sexually yield a smaller number of offspring, but have larger genetic variation.

Key Terms

- **reproduction**: the act of producing new individuals biologically
- **clone**: a living organism produced asexually from a single ancestor, to which it is genetically identical

Animal Reproduction

Reproduction (or procreation) is the biological process by which new "offspring" (individual organisms) are produced from their "parents. " It is a fundamental feature of all known life that each individual organism exists as the result of reproduction. Most importantly, reproduction is necessary for the survival of a species. The known methods of reproduction are broadly grouped into two main types: sexual and asexual.

In asexual reproduction, an individual can reproduce without involvement with another individual of that species. The division of a bacterial cell into two daughter cells is an example of asexual reproduction. This type of reproduction produces genetically-identical organisms (clones), whereas in sexual reproduction, the genetic material of two individuals combines to produce offspring that are genetically different from their parents.

During sexual reproduction, the male gamete (sperm) may be placed inside the female's body for internal fertilization, or the sperm and eggs may be released into the environment for external fertilization. Humans provide an example of the former, while seahorses provide an example of the latter. Following a mating dance, the female seahorse lays eggs in the male seahorse's abdominal brood pouch where they are fertilized. The eggs hatch and the offspring develop in the pouch for several weeks.

Figure 43.1A.1: **Sexual reproduction in seahorses**: Female seahorses produce eggs for reproduction that are then fertilized by the male. Unlike almost all other animals, the male seahorse then gestates the young until birth.

Asexual versus Sexual Reproduction

Organisms that reproduce through asexual reproduction tend to grow in number exponentially. However, because they rely on mutation for variations in their DNA, all members of the species have similar vulnerabilities. Organisms that reproduce sexually yield a smaller number of offspring, but the large amount of variation in their genes makes them less susceptible to disease.

Many organisms can reproduce sexually as well as asexually. Aphids, slime molds, sea anemones, and some species of starfish are examples of animal species with this ability. When environmental factors are favorable, asexual reproduction is employed to exploit suitable conditions for survival, such as an abundant food supply, adequate shelter, favorable climate, disease, optimum pH, or a proper mix of other lifestyle requirements. Populations of these organisms increase exponentially via asexual reproductive strategies to take full advantage of the rich supply resources. When food sources have been depleted, the climate becomes hostile, or individual survival is jeopardized by some other adverse change in living conditions, these organisms switch to sexual forms of reproduction.

Sexual reproduction ensures a mixing of the gene pool of the species. The variations found in offspring of sexual reproduction allow some individuals to be better suited for survival and provide a mechanism for selective adaptation to occur. In addition, sexual reproduction usually results in the formation of a life stage that is able to endure the conditions that threaten the offspring of an asexual parent. Thus, seeds, spores, eggs, pupae, cysts, or other “over-wintering” stages of sexual reproduction ensure the survival during unfavorable times as the organism can “wait out” adverse situations until a swing back to suitability occurs.

43.1B: Types of Sexual and Asexual Reproduction

Asexual and sexual reproduction, two methods of reproduction among animals, produce offspring that are clones or genetically unique.

Learning Objectives

- Discuss sexual and asexual reproduction methods

Key Points

- Asexual reproduction includes fission, budding, fragmentation, and parthenogenesis, while sexual reproduction is achieved through the combination of reproductive cells from two individuals.
- The ability of a species to reproduce through fragmentation depends on the size of part that breaks off, while in binary fission, an individual splits off and forms two individuals of the same size.
- Budding may lead to the production of a completely new adult that forms away from the original body or may remain attached to the original body.
- Observed in invertebrates and some vertebrates, parthenogenesis produce offspring that may be either haploid or diploid.
- Sexual reproduction, the production of an offspring with a new combination of genes, may also involve hermaphroditism in which an organism can self-fertilize or mate with another individual of the same species.

Key Terms

- **binary fission**: the process whereby a cell divides asexually to produce two daughter cells
- **hermaphroditism**: having sexual organs of both sexes
- **parthenogenesis**: a form of asexual reproduction where growth and development of embryos occur without fertilization

Methods of Reproduction: Asexual & Sexual

Asexual Reproduction

Asexual reproduction produces offspring that are genetically identical to the parent because the offspring are all clones of the original parent. This type of reproduction occurs in prokaryotic microorganisms (bacteria) and in some eukaryotic single-celled and multi-celled organisms. Animals may reproduce asexually through fission, budding, fragmentation, or parthenogenesis.

Fission

Fission, also called binary fission, occurs in prokaryotic microorganisms and in some invertebrate, multi-celled organisms. After a period of growth, an organism splits into two separate organisms. Some unicellular eukaryotic organisms undergo binary fission by mitosis. In other organisms, part of the individual separates, forming a second individual. This process occurs, for example, in many asteroid echinoderms through splitting of the central disk. Some sea anemones and some coral polyps also reproduce through fission.

Figure 43.1B.1: **Fission**: Coral polyps reproduce asexually by fission, where an organism splits into two separate organisms.

Budding

Budding is a form of asexual reproduction that results from the outgrowth of a part of a cell or body region leading to a separation from the original organism into two individuals. Budding occurs commonly in some invertebrate animals such as corals and hydras. In hydras, a bud forms that develops into an adult, which breaks away from the main body; whereas in coral budding, the bud does not detach and multiplies as part of a new colony.

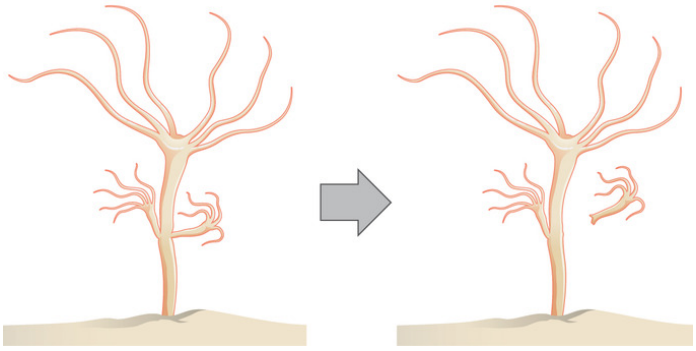

Figure 43.1B.1: **Budding**: Hydra reproduce asexually through budding, where a bud forms that develops into an adult and breaks away from the main body.

Fragmentation

Fragmentation is the breaking of the body into two parts with subsequent regeneration. If the animal is capable of fragmentation, and the part is big enough, a separate individual will regrow.

Many sea stars reproduce asexually by fragmentation. For example, if the arm of an individual sea star is broken off it will regenerate a new sea star. Fishery workers have been known to try to kill the sea stars that eat their clam or oyster beds by cutting them in half and throwing them back into the ocean. Unfortunately for the workers, the two parts can each regenerate a new half, resulting in twice as many sea stars to prey upon the oysters and clams. Fragmentation also occurs in annelid worms, turbellarians, and poriferans.

Figure 43.1*B*. 1 : **Fragmentation**: Sea stars can reproduce through fragmentation. The large arm, a fragment from another sea star, is developing into a new individual.

Note that in fragmentation, there is generally a noticeable difference in the size of the individuals, whereas in fission, two individuals of approximately the same size are formed.

Parthenogenesis

Parthenogenesis is a form of asexual reproduction where an egg develops into a complete individual without being fertilized. The resulting offspring can be either haploid or diploid, depending on the process and the species. Parthenogenesis occurs in invertebrates such as water fleas, rotifers, aphids, stick insects, some ants, wasps, and bees. Bees use parthenogenesis to produce haploid males (drones) and diploid females (workers). If an egg is fertilized, a queen is produced. The queen bee controls the reproduction of the hive bees to regulate the type of bee produced.

Some vertebrate animals, such as certain reptiles, amphibians, and fish, also reproduce through parthenogenesis. Although more common in plants, parthenogenesis has been observed in animal species that were segregated by sex in terrestrial or marine zoos. Two Komodo dragons, a bonnethead shark, and a blacktip shark have produced parthenogenic young when the females have been isolated from males.

Sexual Reproduction

Sexual reproduction is the combination of (usually haploid, or having a single set of unpaired chromosomes) reproductive cells from two individuals to form a third (usually diploid, or having a pair of each type of chromosome) unique offspring. Sexual reproduction produces offspring with novel combinations of genes. This can be an adaptive advantage in unstable or unpredictable environments. As humans, we are used to thinking of animals as having two separate sexes, male and female, determined at conception. However, in the animal kingdom, there are many variations on this theme.

Hermaphroditism

Hermaphroditism occurs in animals where one individual has both male and female reproductive parts. Invertebrates, such as earthworms, slugs, tapeworms and snails, are often hermaphroditic. Hermaphrodites may self-fertilize or may mate with another of their species, fertilizing each other and both producing offspring. Self fertilization is common in animals that have limited mobility or are not motile, such as barnacles and clams.

43.1C: Sex Determination

Sex determination in animals may be regulated by the presence of chromosomes or through the impact of an environmental factor.

Learning Objectives

- Differentiate among the various ways animals determine the sex of offspring

Key Points

- Mammals, birds, and some other animal species depend on heterozygous or homozygous chromosome combinations for sex determination.
- Cool or warm temperatures affect sex determination in species such as crocodiles and turtles.
- Some species, such as oysters, have the capability of alternating their sex several times within their life span.

Key Terms

- **protandry**: the condition in which an organism begins life as a male and then changes into a female
- **protogyny**: the condition in which an organism begins life as a female and then changes into a male
- **homozygous**: of an organism in which both copies of a given gene have the same allele
- **heterozygous**: of an organism which has two different alleles of a given gene

Sex Determination

Mammalian sex is determined genetically by the presence of X and Y chromosomes. Individuals homozygous for X (XX) are female, while heterozygous individuals (XY) are male. The presence of a Y chromosome causes the development of male characteristics, while its absence results in female characteristics. The XY system is also found in some insects and plants.

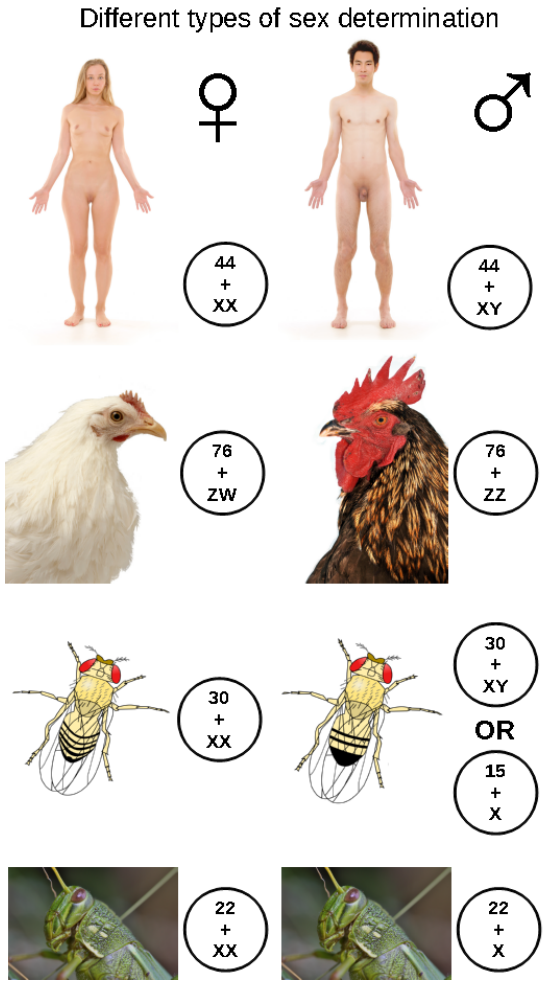

Figure 43.1C.1: **Sex determination**: The presence of X and Y chromosomes are one of the factors responsible for sex determination in mammals, with males being the heterozygous sex. In birds, Z and W chromosomes determine sex, with females being the heterozygous sex.

Avian sex determination is dependent on the presence of Z and W chromosomes. Homozygous for Z (ZZ) results in a male, while heterozygous (ZW) results in a female. The W appears to be essential in determining the sex of the individual, similar to the Y chromosome in mammals. Some fish, crustaceans, insects (such as butterflies and moths), and reptiles use this system.

The sex of some species is not determined by genetics, but by some aspect of the environment. Sex determination in some crocodiles and turtles, for example, is often dependent on the temperature during critical periods of egg development. This is referred to as environmental sex determination or, more specifically, as temperature-dependent sex determination. In many turtles, cooler temperatures during egg incubation produce males, while warm temperatures produce females. In some crocodiles, moderate temperatures produce males, while both warm and cool temperatures produce females. In some species, sex is both genetic- and temperature-dependent.

Individuals of some species change their sex during their lives, alternating between male and female. If the individual is female first, it is termed protogyny or “first female;” if it is male first, it is termed protandry or “first male.” Oysters, for example, are born male, grow, become female, and lay eggs; some oyster species change sex multiple times.

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: Fertilization

Topic hierarchy

- 43.2A: External and Internal Fertilization
- 43.2B: The Evolution of Reproduction

43.2A: External and Internal Fertilization

External and internal fertilization are forms of reproduction that vary in method and embryo development.

Learning Objectives

- Compare and contrast external and internal methods of fertilization

Key Points

- External fertilization is characterized by the release of both sperm and eggs into an external environment; sperm will fertilize the egg outside of the organism, as seen in spawning.
- Internal fertilization is characterized by sperm fertilizing the egg within the female; the three methods include: oviparity (egg laid outside female body), ovoviparity (egg held within female), and viviparity (development within female followed by live birth).
- Internal fertilization protects the fertilized egg or embryo from predation and harsh environments, which results in higher survival rates than can occur with external fertilization.
- Ovoviparity is characterized by an organism retaining a fertilized egg inside the body where development occurs and nourishment is received from the yolk.
- Viparity is characterized by an organism which has its young develop within the female and nourishment is received directly from the mother via a placenta.

Key Terms

- **oviparous**: egg-laying; depositing eggs that develop and hatch outside the body as a reproductive strategy
- **viviparous**: being born alive, as are most mammals, some reptiles, and a few fish (as opposed to being laid as an egg)
- **ovoviparity**: eggs are retained in the female, but the embryo obtains its nourishment from the egg's yolk

External and Internal Fertilization

External Fertilization

External fertilization usually occurs in aquatic environments where both eggs and sperm are released into the water. After the sperm reaches the egg, fertilization can then take place. Most external fertilization happens during the process of spawning where one or several females release their eggs and the male(s) release sperm in the same area, at the same time. The release of the reproductive material may be triggered by water temperature or the length of daylight. Nearly all fish spawn, as do crustaceans (such as crabs and shrimp), mollusks (such as oysters), squid, and echinoderms (such as sea urchins and sea cucumbers). Pairs of fish that are not broadcast spawners may exhibit courtship behavior. This allows the female to select a particular male. The trigger for egg and sperm release (spawning) causes the egg and sperm to be placed in a small area, enhancing the possibility of fertilization.

Figure 43.2A.1: **Anemone fish protecting its spawn**: An image of an Anemone fish protecting its spawn. The anemone fish utilizes a form of external fertilization.

al fertilization in an aquatic environment protects the eggs from drying out. Broadcast spawning can result in a mixture of the genes within a group, leading to higher genetic diversity and a greater chance of species survival in a hostile environment. For sessile aquatic organisms such as sponges, broadcast spawning is the only mechanism for fertilization and colonization of new environments. The presence of the fertilized eggs and developing young in the water provides opportunities for predation, resulting in a loss of offspring. Therefore, millions of eggs must be produced by individuals. The offspring produced through this method must mature rapidly. The survival rate of eggs produced through broadcast spawning is low.

Internal Fertilization

Internal fertilization occurs most often in land-based animals, although some aquatic animals also use this method. There are three ways that offspring are produced following internal fertilization: oviparity, ovoviparity, and viviparity.

In oviparity, fertilized eggs are laid outside the female's body and develop there, receiving nourishment from the yolk that is a part of the egg. This occurs in most bony fish, many reptiles, some cartilaginous fish, most amphibians, two mammals, and all birds. Reptiles and insects produce leathery eggs, while birds and turtles produce eggs with high concentrations of calcium carbonate in the shell, making them hard. These animals are classified as oviparous.

In ovoviparity, fertilized eggs are retained in the female, but the embryo obtains its nourishment from the egg's yolk; the young are fully developed when they are hatched. This occurs in some bony fish (such as the guppy, *Lebistes reticulatus*), some sharks, some lizards, some snakes (such as the garter snake, *Thamnophis sirtalis*), some vipers, and some invertebrate animals (such as the Madagascar hissing cockroach, *Gromphadorhina portentosa*).

In viviparity, the young develop within the female, receiving nourishment from the mother's blood through a placenta. The offspring develops in the female and is born alive. This occurs in most mammals, some cartilaginous fish, and a few reptiles, making these animals viviparous.

Internal fertilization has the advantage of protecting the fertilized egg from dehydration on land. The embryo is isolated within the female, which limits predation on the young. Internal fertilization also enhances the fertilization of eggs by a specific male. Even though fewer offspring are produced through this method, their survival rate is higher than that for external fertilization.

43.2B: The Evolution of Reproduction

Sexually-reproducing organisms have evolved specialized gonads, along with a variety of ways to transfer sperm during reproduction.

Learning Objectives

- Differentiate among the types of reproductive systems that have evolved

Key Points

- Annelids undergo sexual reproduction by producing sperm or eggs within the coelom and storing them within the cavity until they are ready to be released through an excretory opening.
- Insects have developed complete reproductive systems for the separate sexes and will often have a specialized sac for sperm called the spermatheca.
- Non-mammals will utilize a common body opening called the cloaca to transfer sperm between animals.
- The means by which sperm is transferred varies and can include releasing sperm into the environment as well as direct delivery to the vagina.

Key Terms

- **coelom**: a fluid-filled cavity within the body of an animal; the digestive system is suspended within the cavity, which is lined by a tissue called the peritoneum
- **gonad**: a sex organ that produces gametes; specifically, a testicle or ovary
- **cloaca**: the common duct in fish, reptiles, birds, and some primitive mammals that serves as the anus as well as the genital opening
- **spermatheca**: a small sac within the reproductive tract of some female invertebrates, such as insects, which stores sperm until it is used to fertilize the ova

The Evolution of Reproduction

Several competing scientific hypotheses have been proposed to explain the evolution of sexual reproduction. All sexually-reproducing eukaryotic organisms derive from a common ancestor that was a single-celled eukaryotic species. Many protists reproduce sexually, as do multicellular plants, animals, and fungi. However, there are a few species which have secondarily lost this feature. The evolution of sex contains two related, yet distinct, themes: its origin and its maintenance. However, since the hypotheses for the origins of sex are difficult to test experimentally, most current work has been focused on the maintenance of sexual reproduction.

Once multicellular organisms evolved and developed specialized cells, some also developed tissues and organs with specialized functions. The evolution of reproductive organs arrived with the development of gonads that produced sperm and eggs. These cells develop through meiosis, an adaption of mitosis, which reduced the number of chromosomes in each reproductive cell by half, while increasing the number of cells through cell division. The development of specialized gonads to produce sperm and egg was a major step in the evolutionary process.

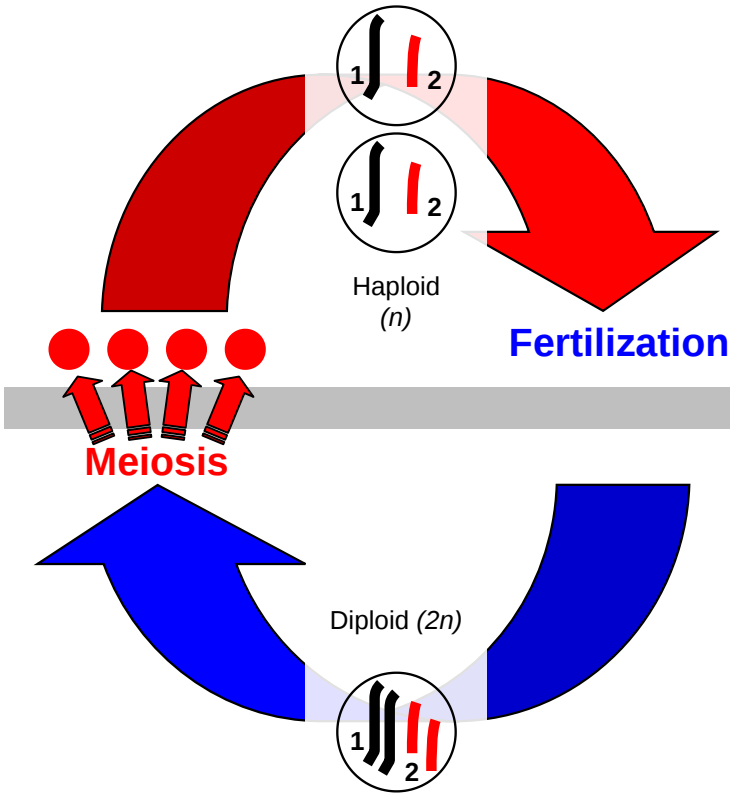

Figure 43.2B.1: **Meiosis**: The development of sperm and eggs occurs via the process of meiosis which reduces the number of chromosomes in each cell by half.

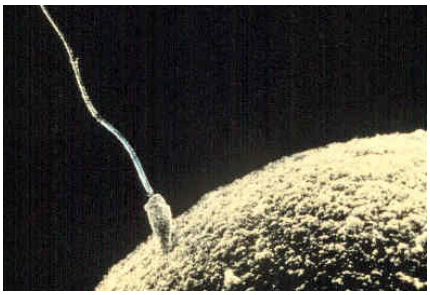

Figure 43.2B.1: **Sperm fertilizing an egg**: The evolution of sex-specific gonads led to the development of sperm (male) and eggs (female). When fertilized, if the conditions are favorable, the egg may develop into a new organism.

An early development in reproduction occurred in the Annelids. These organisms produce sperm and eggs from undifferentiated cells in their coelom, storing them in that cavity. When the coelom becomes filled, the cells are released through an excretory opening or by the body splitting open. Further evolution of reproductive systems resulted in the development of reproductive systems that are sex specific. In these more advanced systems, sperm is made in the testes and then travels through coiled tubes to the epididymis for storage. Additionally, in these more advanced systems, eggs are matured in the ovary; when released, they travel to the uterine tubes for fertilization. These types of reproductive systems developed in insects (compared to annelids which have a coelom for storage). Specifically, in the insect reproductive system, a specialized sac developed, called a spermatheca, which is used to store sperm for later use, sometimes up to a year. This was a key development since fertilization in insects can be timed with environmental or food conditions that are optimal for offspring survival.

Vertebrates have similar structures (i.e., gonads that specialize in sex cell production) with a few differences in their reproductive systems. Non-mammals, such as birds and reptiles, have a common body opening, called a cloaca, for the digestive, excretory, and reproductive systems. Coupling between birds usually involves positioning the cloaca openings opposite each other for transfer of sperm. In mammals, there are separate openings for the systems in the female and a

uterus for support of developing offspring. Depending on the type of species, there are differences in the uterus. In species that produce large numbers of offspring, the uterus has two chambers. In other species that produce one offspring, such as in primates, there is a single uterus.

Another development in the evolution of reproduction is the means by which sperm is transferred. During reproduction, sperm transfer from the male to the female ranges from releasing the sperm into the watery environment for external fertilization, to the joining of cloaca in birds, to the development of a penis for direct delivery into the female's vagina in mammals. All of these methods of sperm transfer represent the varying ways reproduction has evolved and become specialized to specific organisms.

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43.3: Human Reproductive Anatomy and Gametogenesis

Topic hierarchy

- 43.3A: Male Reproductive Anatomy
- 43.3B: Female Reproductive Anatomy
- 43.3C: Gametogenesis (Spermatogenesis and Oogenesis)

43.3A: Male Reproductive Anatomy

The male reproductive system is a series of organs and glands that produce sperm, fluids, hormones, and enable the ejaculation of sperm.

Learning Objectives

- Diagram the structures of human male reproductive anatomy

Key Points

- The male gonads, or testes, produce sperm within the seminiferous tubules; the sperm are housed here until they are nearly mature, at which point they enter the epididymis for full maturation.
- The testes are housed in the scrotum, an external sac that keeps the sperm at a temperature lower than that of the body.
- At ejaculation, sperm leave the epididymis and enter the vas deferens, a duct which carries the sperm out of the body through the urethra, along with the fluids of various glands of the male reproductive system.
- The seminal vesicles produce a thick fluid that is alkaline in order to protect sperm from the acidic nature of the female vagina; it also contains sugars to nourish the sperm.
- The prostate gland produces the force necessary to push the sperm out of the epididymis at ejaculation, while the bulbourethral gland emits a fluid just prior to ejaculation that neutralizes acid from any urine left over in the urethra.
- During sexual arousal, the spongy tissue inside the penis (the corpus spongiosum) fills with blood, causing the penis to become erect and hard; after ejaculation, the blood flows back out of the penis, leaving it flaccid.

Key Terms

- **epididymis**: a narrow, tightly-coiled tube connecting the efferent ducts from the rear of each testicle to its vas deferens, where sperm are stored during maturation
- **prostate gland**: a gland in male mammals surrounding the urethra just below the urinary bladder that controls the release of urine from the bladder and produces a secretion that is the fluid part of semen
- **seminiferous tubule**: any of many threadlike structures, located in the testes, that are the specialized areas of sperm production

Human Reproductive Anatomy

The reproductive tissues of male and female humans develop similarly *in utero* until a low level of the hormone testosterone is released from male gonads. Testosterone causes the undeveloped tissues to differentiate into male sexual organs. Primitive gonads become testes; other tissues produce a penis and scrotum in males.

Male Reproductive Anatomy

In the male reproductive system, the scrotum houses the testicles or testes, providing passage for blood vessels, nerves, and muscles related to testicular function. The testes, a pair of male reproductive organs, produce sperm and male sex hormones, including the steroid testosterone. Coiled in each testis are seminiferous tubules that produce sperm.

Sperm

Sperm are immobile at body temperature; therefore, the scrotum and penis are external to the body so that a proper temperature is maintained for motility. In land mammals, the pair of testes must be suspended outside the body at about 2° C lower than body temperature to produce viable sperm.

Sperm develop in the seminiferous tubules that are coiled inside the testes. The walls of the seminiferous tubules are composed of the developing sperm cells, with the least-developed sperm at the periphery of the tubule and the fully-developed sperm in the lumen. The sperm cells are mixed with “nursemaid” cells called Sertoli cells which protect the germ cells and promote their development. Other cells mixed in the wall of the tubules are the interstitial cells of Leydig; these cells produce high levels of testosterone once the male reaches adolescence.

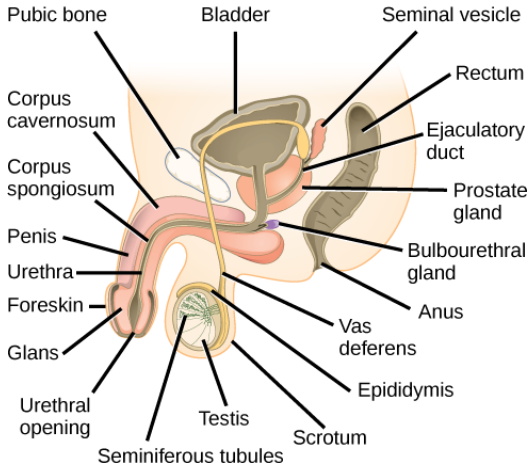

Figure 43.3A.1: **Male reproductive system**: The reproductive structures of the human male are shown. Sperm are produced in the seminiferous tubules, mature in the epididymis, and then are forced out into the vas deferens and out of the body through the urethra during ejaculation.

Sperm consist of a flagellum (as a tail), a neck that contains the cell's energy-producing mitochondria, and a head that contains the genetic material. When the sperm have developed flagella, or lash-like appendages that protrude from the cell body, and are nearly mature, they leave the testicles and enter the epididymis. This structure lies along the top and posterior portion of the testes; it is the site of sperm maturation. The sperm leave the epididymis and enter the vas deferens, which is the duct in the testicle that carries sperm from the epididymis to the ejaculatory duct.

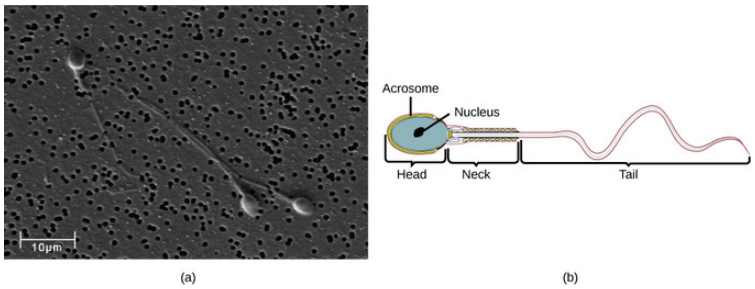

Figure 43.3A.1: **Structure of a human sperm**: Human sperm, visualized using scanning electron microscopy, have a flagellum, neck, and head.

Semen is a mixture of sperm and spermatic duct secretions (about 10 percent of the total), along with fluids from accessory glands, that contribute most of the semen's volume. An ejaculate will contain from two to five milliliters of fluid with from 50–120 million sperm per milliliter. The bulk of the semen comes from the accessory glands associated with the male reproductive system, including the seminal vesicles, the prostate gland, and the bulbourethral gland.

Seminal vesicles, penis, prostate, and bulbourethral gland

The seminal vesicles are a pair of glands that lie along the posterior border of the urinary bladder. The glands make a solution that is thick, yellowish, and alkaline. As sperm are only motile in an alkaline environment, a basic pH is important to reverse the acidity of the vaginal environment. The solution also contains mucus, fructose (a sperm mitochondrial nutrient), a coagulating enzyme, ascorbic acid, and local-acting hormones called prostaglandins.

The penis is an organ that drains urine from the renal bladder and functions as a copulatory organ during intercourse. The penis contains three tubes of erectile tissue running through the length of the organ. These consist of a pair of tubes on the dorsal side, called the corpus cavernosum, and a single tube of tissue on the ventral side, called the corpus spongiosum. This tissue, when engorged with blood, becomes erect and hard, in preparation for intercourse. The organ is inserted into the vagina, culminating with an ejaculation, which is the forcible ejection of semen from the urethra. An orgasm is a two-stage process: first, glands and accessory organs connected to the testes contract; second, semen (containing sperm) is expelled through the urethra during ejaculation. After intercourse, the blood drains from the erectile tissue and the penis becomes flaccid.

The walnut-shaped prostate gland surrounds the urethra, the connection to the urinary bladder. It has a series of short ducts that directly connect to the urethra. The gland is a mixture of smooth muscle and glandular tissue. The muscle provides much of the force needed for ejaculation to occur.

The bulbourethral gland, or Cowper's gland, is an exocrine gland which secretes a clear fluid known as pre-ejaculate that is generated upon sexual arousal. This gland releases its secretion prior to the release of the bulk of the semen. It neutralizes any acid residue in the urethra left over from urine. This usually accounts for a couple of drops of fluid in the total ejaculate and may contain a few sperm. Withdrawal of the penis from the vagina before ejaculation to prevent pregnancy may not work if sperm are present in the bulbourethral gland secretions.

43.3B: Female Reproductive Anatomy

The female reproductive structures produce eggs, support a growing embryo, and provide a birth canal to the fetus.

Learning Objectives

- Diagram the structures of human female reproductive anatomy

Key Points

- The external anatomy of the female reproductive system is referred to as the vulva; it includes the labia minora, which protects the vagina and urethra, and the labia majora, which surrounds it.
- Internal female reproductive structures include ovaries, oviducts, the uterus, and the vagina.
- The eggs develop in structures called follicles, which are located on the surface of the ovaries; at maturity, one of the follicles will rupture and release the egg, which is captured by the fimbrae of the oviduct.
- If fertilization occurs, it generally does so in the oviduct; the fertilized egg then travels down the oviduct and enters the uterus, where it will implant in the lining of the uterus, known as the endometrium.
- If fertilization does not occur, the endometrium of the uterus will slough off at the end of the menstrual period and is shed through the vagina, which is also the opening through which the penis enters during intercourse and through which the baby will exit during birth.

Key Terms

- **clitoris**: a small sensitive elongated erectile organ at the anterior part of the vulva in female mammals, homologous with the penis
- **ovary**: a female reproductive organ, often paired, that produces ova and in mammals secretes the hormones estrogen and progesterone
- **vagina**: the passage leading from the opening of the vulva to the cervix of the uterus for copulation and childbirth in female mammals
- **uterus**: an organ of the female reproductive system in which the young are conceived and develop until birth; the womb
- **vulva**: the external female sexual organs, collectively

Female reproductive anatomy

Female reproductive anatomy includes both external and internal structures. Among the external structures are the vulva, which consists of the mons pubis, clitoris, labia majora, labia minora, and the vestibular glands. The vulva is an area associated with the vestibule that includes the structures found in the inguinal (groin) area of women. The mons pubis is a round, fatty area that overlies the pubic symphysis. The clitoris is a structure with erectile tissue that contains a large number of sensory nerves and serves as a source of stimulation during intercourse. The labia majora are a pair of elongated folds of tissue that run posterior from the mons pubis and enclose the other components of the vulva. The labia majora derive from the same tissue that produces the scrotum in a male. The labia minora are thin folds of tissue centrally located within the labia majora. These labia protect the openings to the vagina and urethra. The mons pubis and the anterior portion of the labia majora become covered with hair during adolescence; the labia minora is hairless. The greater vestibular glands are found at the sides of the vaginal opening and provide lubrication during intercourse.

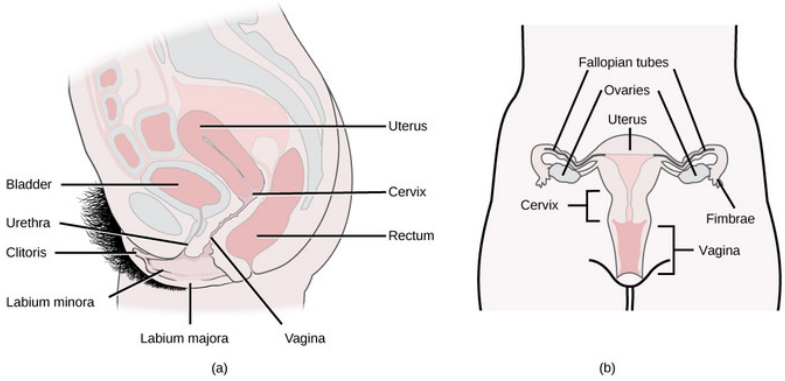

Figure 43.3B.1: **Female reproductive anatomy**: The reproductive structures of the human female are shown. Several structures are on the outside of the body, such as the labia, vulva, and clitoris, while others, such as the ovaries and uterus, are internal.

Internal female reproductive structures include ovaries, oviducts, the uterus, and the vagina. An ovary consists of a medulla and cortex: the medulla contains nerves and blood vessels to supply the cortex with nutrients and remove waste. The outer layers of cells of the cortex are the functional parts of the ovaries. The cortex is made up of follicular cells that surround eggs. During the menstrual cycle, a batch of follicular cells develops, preparing the eggs for release. At ovulation, one follicle ruptures and one egg is released.

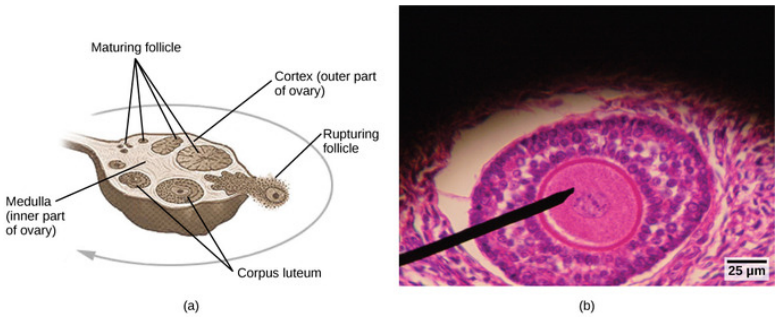

Figure 43.3B.1: **Oocyte development**: Oocytes develop in (a) follicles, located in the ovary. At the beginning of the menstrual cycle, the follicle matures. At ovulation, the follicle ruptures, releasing the egg. The follicle becomes a corpus luteum, which eventually degenerates. The (b) follicle in this light micrograph has an oocyte at its center.

The oviducts, or fallopian tubes, extend from the uterus in the lower abdominal cavity to the ovaries, but they are not in contact with the ovaries. The lateral ends of the oviducts flare out into a trumpet-like structure and have a fringe of finger-like projections called fimbriae. When an egg is released at ovulation, the fimbrae help the non-motile egg enter into the tube, the passage to the uterus. The walls of the oviducts are ciliated (covered in cilia) and are primarily smooth muscle. The cilia beat toward the middle, while the smooth muscle contracts in the same direction, moving the egg toward the uterus. Fertilization usually takes place within the oviducts. The embryo is moved toward the uterus for further development. It usually takes the egg or embryo a week to travel through the oviduct.

The uterus, a structure about the size of a woman's fist, is lined with an endometrium that is rich in blood vessels and mucus glands. The uterus supports the developing embryo and fetus during gestation. The thickest portion of the wall of the uterus is made of smooth muscle. Contractions of the smooth muscle in the uterus aid in passing the baby through the vagina during labor. A portion of the lining of the uterus sloughs off during each menstrual period if an egg has not been fertilized; it builds up again in preparation for an implantation. Part of the uterus, called the cervix, protrudes into the top of the vagina, which functions as the birth canal.

The vagina is a muscular tube that serves several purposes. It allows menstrual flow to leave the body, is the receptacle for the penis during intercourse, and serves as the vessel for the delivery of offspring. It is lined by stratified squamous epithelial cells to protect the underlying tissue.

43.3C: Gametogenesis (Spermatogenesis and Oogenesis)

Spermatogenesis and oogenesis are both forms of gametogenesis, in which a diploid gamete cell produces haploid sperm and egg cells, respectively.

Learning Objectives

- Distinguish between spermatogenesis and oogenesis

Key Points

- Gametogenesis, the production of sperm (spermatogenesis) and eggs (oogenesis), takes place through the process of meiosis.
- In oogenesis, diploid oogonium go through mitosis until one develops into a primary oocyte, which will begin the first meiotic division, but then arrest; it will finish this division as it develops in the follicle, giving rise to a haploid secondary oocyte and a smaller polar body.
- The secondary oocyte begins the second meiotic division and then arrests again; it will not finish this division unless it is fertilized by a sperm; if this occurs, a mature ovum and another polar body is produced.
- In spermatogenesis, diploid spermatogonia go through mitosis until they begin to develop into gametes; eventually, one develops into a primary spermatocyte that will go through the first meiotic division to form two haploid secondary spermatocytes.
- The secondary spermatocytes will go through a second meiotic division to each produce two spermatids; these cells will eventually develop flagella and become mature sperm.

Key Terms

- **spermatocyte**: a male gametocyte, from which a spermatozoon develops
- **oocyte**: a cell that develops into an egg or ovum; a female gametocyte
- **polar body**: one of the small cells that are by-products of the meiosis that forms an egg
- **mitosis**: the division of a cell nucleus in which the genome is copied and separated into two identical halves. It is normally followed by cell division
- **meiosis**: cell division of a diploid cell into four haploid cells, which develop to produce gametes

Gametogenesis (Spermatogenesis and Oogenesis)

Gametogenesis, the production of sperm and eggs, takes place through the process of meiosis. During meiosis, two cell divisions separate the paired chromosomes in the nucleus and then separate the chromatids that were made during an earlier stage of the cell's life cycle, resulting in gametes that each contain half the number of chromosomes as the parent. The production of sperm is called spermatogenesis and the production of eggs is called oogenesis.

Oogenesis

Oogenesis occurs in the outermost layers of the ovaries. As with sperm production, oogenesis starts with a germ cell, called an oogonium (plural: oogonia), but this cell undergoes mitosis to increase in number, eventually resulting in up to one to two million cells in the embryo.

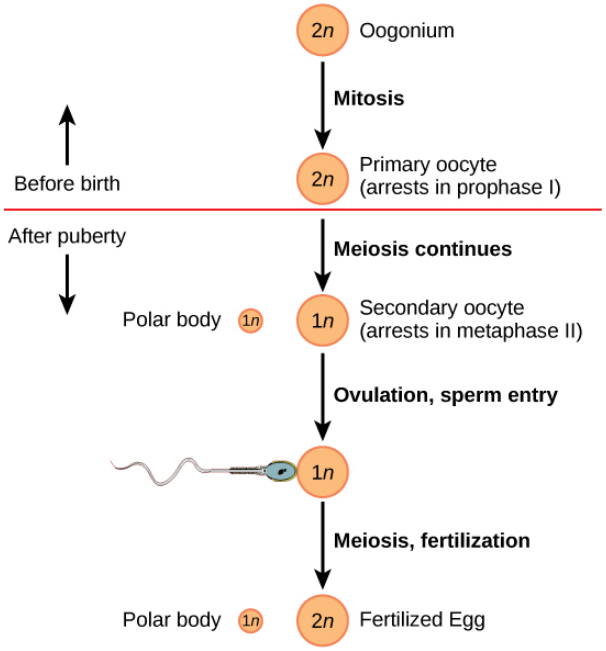

Figure 43.3C.1: **Oogenesis**: The process of oogenesis occurs in the ovary's outermost layer. A primary oocyte begins the first meiotic division, but then arrests until later in life when it will finish this division in a developing follicle. This results in a secondary oocyte, which will complete meiosis if it is fertilized.

The cell starting meiosis is called a primary oocyte. This cell will begin the first meiotic division, but be arrested in its progress in the first prophase stage. At the time of birth, all future eggs are in the prophase stage. At adolescence, anterior pituitary hormones cause the development of a number of follicles in an ovary. This results in the primary oocyte finishing the first meiotic division. The cell divides unequally, with most of the cellular material and organelles going to one cell, called a secondary oocyte, and only one set of chromosomes and a small amount of cytoplasm going to the other cell. This second cell is called a polar body and usually dies. A secondary meiotic arrest occurs, this time at the metaphase II stage. At ovulation, this secondary oocyte will be released and travel toward the uterus through the oviduct. If the secondary oocyte is fertilized, the cell continues through the meiosis II, completing meiosis, producing a second polar body and a fertilized egg containing all 46 chromosomes of a human being, half of them coming from the sperm.

Spermatogenesis

Spermatogenesis occurs in the wall of the seminiferous tubules, with stem cells at the periphery of the tube and the spermatozoa at the lumen of the tube. Immediately under the capsule of the tubule are diploid, undifferentiated cells. These stem cells, called spermatogonia (singular: spermatagonium), go through mitosis with one offspring going on to differentiate into a sperm cell, while the other gives rise to the next generation of sperm.

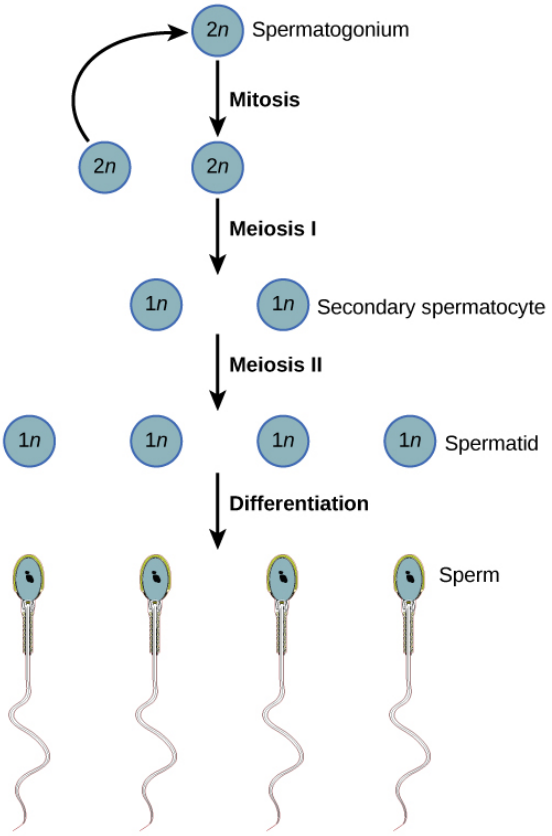

Figure 43.3C.1: **Spermatogenesis**: During spermatogenesis, four sperm result from each primary spermatocyte, which divides into two haploid secondary spermatocytes; these cells will go through a second meiotic division to produce four spermatids.

Meiosis begins with a cell called a primary spermatocyte. At the end of the first meiotic division, a haploid cell is produced called a secondary spermatocyte. This haploid cell must go through another meiotic cell division. The cell produced at the end of meiosis is called a spermatid. When it reaches the lumen of the tubule and grows a flagellum (or "tail"), it is called a sperm cell. Four sperm result from each primary spermatocyte that goes through meiosis.

Stem cells are deposited during gestation and are present at birth through the beginning of adolescence, but in an inactive state. During adolescence, gonadotropic hormones from the anterior pituitary cause the activation of these cells and the production of viable sperm. This continues into old age.

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43.4: Hormonal Control of Human Reproduction

Topic hierarchy

- 43.4A: Male Hormones
- 43.4B: Female Hormones

43.4A: Male Hormones

The onset of puberty is controlled by two major hormones: FSH initiates spermatogenesis and LH signals the release of testosterone.

Learning Objectives

- Explain the function of male hormones in reproduction

Key Points

- The onset of puberty is signaled by high pulses of GnRH secreted by the hypothalamus; this in turn signals the release of FSH and LH from the pituitary gland.
- FSH causes the Sertoli cells of the testes (which help nurse developing sperm cells) to begin the process of spermatogenesis in the testes.
- LH triggers the production of testosterone from the Leydig cells of the testis; testosterone causes the development of secondary sex characteristics in the male.
- As spermatogenesis and testosterone production increase, the Sertoli cells produce inhibin, which, together with rising levels of testosterone, inhibit the release of FSH and LH from the pituitary gland.

Key Terms

- **puberty**: the age at which a person is first capable of sexual reproduction
- **Sertoli cell**: a kind of sustentacular cell which serves as a “nurse” cell of the testes and which is part of a seminiferous tubule
- **Leydig cell**: one of the interstitial cells, located next to the seminiferous tubules inside the testicle, that produce testosterone
- **follicle stimulating hormone**: a gonadotropic glycoprotein hormone, secreted in the anterior pituitary, that stimulates the growth of ovarian follicles in female mammals, and induces spermatogenesis in male mammals
- **luteinizing hormone**: a hormone, produced by part of the pituitary gland, that stimulates ovulation and the development of the corpus luteum in female mammals, and the production of androgens by male mammals
- **inhibin**: a peptide hormone, secreted by the gonads, which inhibits the secretion of follicle-stimulating hormone
- **testosterone**: steroid hormone produced primarily in the testes of the male; it is responsible for the development of secondary sex characteristics in the male

Male Hormones

Puberty is a period of several years in which rapid physical growth and psychological changes occur, culminating in sexual maturity. The average onset of puberty is age 11 or 12 for boys. Some of the most significant parts of pubertal development involve distinctive physiological changes in individuals’ height, weight, body composition, and circulatory and respiratory systems. These changes are largely influenced by hormonal activity. Hormones play an organizational role, priming the body to behave in a certain way once puberty begins, and an activational role, referring to changes in hormones during adolescence that trigger behavioral and physical changes.

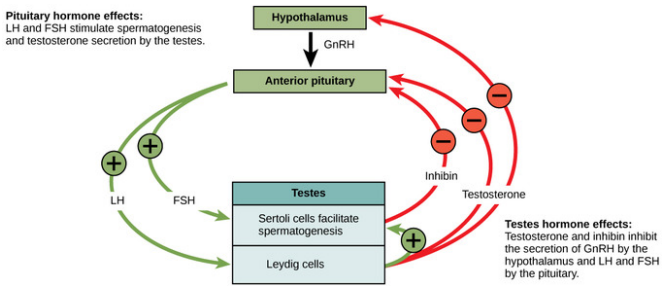

Figure 43.4A.1: **Hormonal regulation of the male reproductive system**: GnRH stimulates the production of FSH and LH, which act on the testes to begin spermatogenesis and to develop secondary sex characteristics in the male. In turn, the testes production of testosterone and the hormone inhibin inhibit the release of GnRH, FSH, and LH in a negative feedback loop.

onset of puberty, the hypothalamus begins secreting high pulses of GnRH, or gonadotropin-releasing hormone. In response, the pituitary gland releases follicle stimulating hormone (FSH) and luteinizing hormone (LH) into the male system for the first time. FSH enters the testes, stimulating the Sertoli cells, which help to nourish the sperm cells that the testes produce, to begin facilitating spermatogenesis. LH also enters the testes, stimulating the interstitial cells, called Leydig cells, to make and release testosterone into the testes and the blood.

Testosterone, the hormone responsible for the secondary sexual characteristics that develop in the male during adolescence, stimulates spermatogenesis, or the process of sperm production in the testes. Secondary sex characteristics include a deepening of the voice, the growth of facial, axillary, and pubic hair, and the beginnings of the sex drive.

A negative feedback system occurs in the male with rising levels of testosterone acting on the hypothalamus and anterior pituitary to inhibit the release of GnRH, FSH, and LH. The Sertoli cells produce the hormone inhibin, which is released into the blood when the sperm count is too high. This inhibits the release of GnRH and FSH, which will cause spermatogenesis to slow down. If the sperm count reaches 20 million/ml, the Sertoli cells cease the release of inhibin, allowing the sperm count to increase.

B: Female Hormones

The stages of the ovarian cycle in the female are regulated by hormones secreted by the hypothalamus, pituitary, and the ovaries.

Learning Objectives

- Explain the function of female hormones in reproduction

Key Points

- As in males, GnRH secreted by the hypothalamus triggers the release of FSH and LH from the pituitary; however, in females, this signals the ovaries to produce estradiol and progesterone.
- FSH stimulates the growth and maturation of follicles on the ovaries, which house and nourish the developing eggs; the follicle, in turn, releases inhibin, which inhibits the production of FSH.
- Progesterone stimulates the growth of the endometrial lining of the uterus in order to prepare it for pregnancy; a strong surge of LH at around day 14 of the cycle triggers ovulation of an egg from the most mature follicle.
- After ovulation, the ruptured follicle becomes a corpus luteum, which secretes progesterone to either regrow the uterine lining or to support the pregnancy if it occurs.
- During middle age, a woman's ovaries become less sensitive to FSH and LH and, therefore, cease to mature follicles and undergo ovulation; this is known as menopause.

Key Terms

- **corpus luteum**: a yellow mass of cells that forms from an ovarian follicle during the luteal phase of the menstrual cycle in mammals; it secretes steroid hormones
- **menopause**: the ending of menstruation; the time in a woman's life when this happens
- **endometrium**: the mucous membrane that lines the uterus in mammals and in which fertilized eggs are implanted
- **estradiol**: a potent estrogenic hormone produced in the ovaries of all vertebrates; the synthetic compound is used medicinally to treat estrogen deficiency and breast cancer
- **menstruation**: the periodic discharging of the menses, the flow of blood and cells from the lining of the uterus in females of humans and other primates

Female Hormones

The control of reproduction in females is more complex than that of the male. As with the male, the hypothalamic hormone GnRH (gonadotropin-releasing hormone) causes the release of the hormones FSH (follicle stimulating hormone) and LH (luteinizing hormone) from the anterior pituitary. In addition, estrogens and progesterone are released from the developing follicles, which are structures on the ovaries that contain the maturing eggs.

In females, FSH stimulates the development of egg cells, called ova, which develop in structures called follicles. Follicle cells produce the hormone inhibin, which inhibits FSH production. LH also plays a role in the development of ova, as well as in the induction of ovulation and stimulation of estradiol and progesterone production by the ovaries. Estradiol and progesterone are steroid hormones that prepare the body for pregnancy. Estradiol is the reproductive hormone in females that assists in endometrial regrowth, ovulation, and calcium absorption; it is also responsible for the secondary sexual characteristics of females. These include breast development, flaring of the hips, and a shorter period necessary for bone maturation. Progesterone assists in endometrial re-growth and inhibition of FSH and LH release.

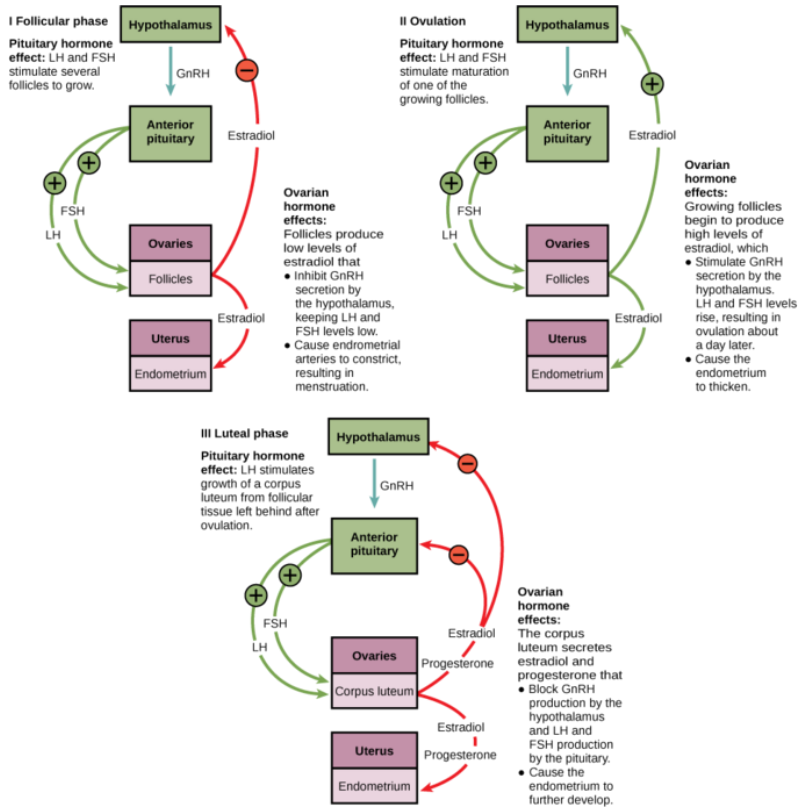

Figure 43.4B.1: **Hormonal control of the female reproductive cycle**: The ovarian and menstrual cycles of female reproduction are regulated by hormones produced by the hypothalamus, pituitary, and ovaries. The pattern of activation and inhibition of these hormones varies between phases of the reproductive cycle.

The Ovarian Cycle and the Menstrual Cycle

The ovarian cycle governs the preparation of endocrine tissues and release of eggs, while the menstrual cycle governs the preparation and maintenance of the uterine lining. These cycles occur concurrently and are coordinated over a 22–32 day cycle, with an average length of 28 days.

The first half of the ovarian cycle is the follicular phase. Slowly-rising levels of FSH and LH cause the growth of follicles on the surface of the ovary, which prepares the egg for ovulation. As the follicles grow, they begin releasing estrogens and a low level of progesterone. Progesterone maintains the endometrium, the lining of the uterus, to help ensure pregnancy. Just prior to the middle of the cycle (approximately day 14), the high level of estrogen causes FSH and, especially, LH to rise rapidly and then fall. The spike in LH causes ovulation: the most mature follicle ruptures and releases its egg. The follicles that did not rupture degenerate and their eggs are lost. The level of estrogen decreases when the extra follicles degenerate.

Figure 43.4B.1: **Follicle**: This mature egg follicle may rupture and release an egg in response to a surge of LH.

If pregnancy implantation does not occur, the lining of the uterus is sloughed off, a process known as menstruation. After about five days, estrogen levels rise and the menstrual cycle enters the proliferative phase. The endometrium begins to regrow, replacing the blood vessels and glands that deteriorated during the end of the last cycle.

Following ovulation, the ovarian cycle enters its luteal phase and the menstrual cycle enters its secretory phase, both of which run from about day 15 to 28. The luteal and secretory phases refer to changes in the ruptured follicle. The cells in the follicle undergo physical changes, producing a structure called a corpus luteum, which produces estrogen and progesterone. The progesterone facilitates the regrowth of the uterine lining and inhibits the release of further FSH and LH. The uterus is again being prepared to accept a fertilized egg, should it occur during this cycle. The inhibition of FSH and LH prevents any further eggs and follicles from developing. The level of estrogen produced by the corpus luteum increases to a steady level for the next few days.

If no fertilized egg is implanted into the uterus, the corpus luteum degenerates and the levels of estrogen and progesterone decrease. The endometrium begins to degenerate as the progesterone levels drop, initiating the next menstrual cycle. The decrease in progesterone also allows the hypothalamus to send GnRH to the anterior pituitary, releasing FSH and LH to start the cycles again.

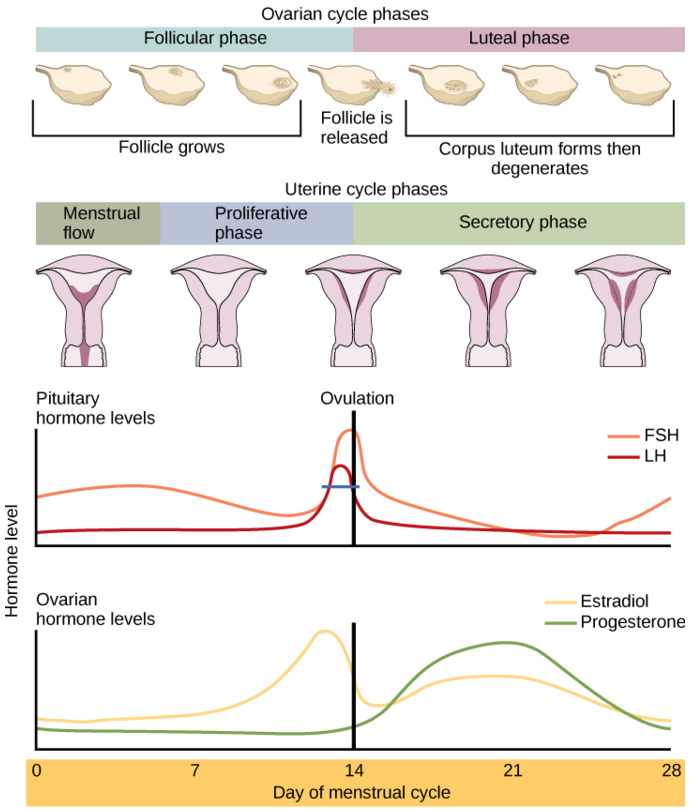

Figure 43.4B.1: **Stages of the menstrual cycle**: Rising and falling hormone levels result in progression of the ovarian and menstrual cycles.

Menopause

As women approach their mid-40s to mid-50s, their ovaries begin to lose their sensitivity to FSH and LH. Menstrual periods become less frequent and finally cease; this process is known as menopause. There are still eggs and potential follicles on the ovaries, but without the stimulation of FSH and LH, they will not produce a viable egg to be released. The outcome of this is the inability to have children.

Various symptoms are associated with menopause, including hot flashes, heavy sweating, headaches, some hair loss, muscle pain, vaginal dryness, insomnia, depression, weight gain, and mood swings. Estrogen is involved in calcium metabolism and, without it, blood levels of calcium decrease. To replenish the blood, calcium is lost from bone, which may decrease the bone density and lead to osteoporosis. Supplementation of estrogen in the form of hormone replacement therapy (HRT) can prevent bone loss, but the therapy can have negative side effects, such as an increased risk of stroke or heart attack, blood clots, breast cancer, ovarian cancer, endometrial cancer, gall bladder disease, and, possibly, dementia.

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: Fertilization and Early Embryonic Development

Topic hierarchy

- 43.5A: Fertilization
- 43.5B: Cleavage, the Blastula Stage, and Gastrulation

43.5A: Fertilization

In fertilization, the sperm binds to the egg, allowing their membranes to fuse and the sperm to transfer its nucleus into the egg.

Learning Objectives

- Describe the process of fertilization

Key Points

- A mammalian egg is covered by a layer of glycoproteins called the zona pellucida, which the sperm must penetrate in order to fertilize the egg.
- Upon binding with the egg, the sperm initiates the acrosome reaction, in which it releases digestive enzymes that degrade the zona pellucida, allowing the plasma membrane of the sperm to fuse with that of the egg.
- Upon fusion of the two plasma membranes, the sperm's nucleus enters the egg and fuses with the nucleus of the egg.
- Both the sperm and the egg each contain one half the normal number of chromosomes, so when they fuse the resulting zygote is a diploid organism with a complete set of chromosomes.
- When the egg is successfully fertilized, it releases proteins that prevent it from being fertilized by another sperm, a condition known as polyspermy.

Key Terms

- **fertilization**: the act of fecundating or impregnating animal or vegetable gametes
- **zona pellucida**: a glycoprotein membrane surrounding the plasma membrane of an oocyte
- **acrosome**: a structure forming the end of the head of a spermatozoon
- **polyspermy**: the penetration of an ovum by more than one sperm

Fertilization

Fertilization is the process in which gametes (an egg and sperm) fuse to form a zygote. The egg and sperm are haploid, which means they each contain one set of chromosomes; upon fertilization, they will combine their genetic material to form a zygote that is diploid, having two sets of chromosomes. A zygote that has more than two sets of chromosomes will not be viable; therefore, to ensure that the offspring has only two sets of chromosomes, only one sperm must fuse with one egg.

In mammals, the egg is protected by a layer of extracellular matrix consisting mainly of glycoproteins called the zona pellucida. When a sperm binds to the zona pellucida, a series of biochemical events, called the acrosomal reaction, take place. In placental mammals, the acrosome contains digestive enzymes that initiate the degradation of the glycoprotein matrix protecting the egg and allowing the sperm plasma membrane to fuse with the egg plasma membrane. The fusion of these two membranes creates an opening through which the sperm nucleus is transferred into the ovum. Fusion between the oocyte plasma membrane and sperm follows and allows the sperm nucleus, centriole, and flagellum, but not the mitochondria, to enter the oocyte. The nuclear membranes of the egg and sperm break down and the two haploid genomes condense to form a diploid genome. This process ultimately leads to the formation of a diploid cell called a zygote. The zygote divides to form a blastocyst and, upon entering the uterus, implants in the endometrium, beginning pregnancy.

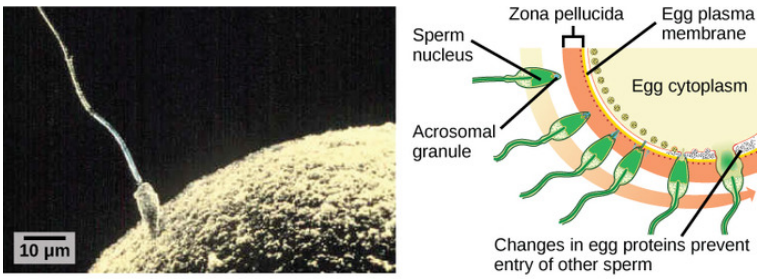

Figure 43.5A.1:

ocess of fertilization: (a) Fertilization is the process in which sperm and egg fuse to form a zygote. (b) Acrosomal tions help the sperm degrade the glycoprotein matrix protecting the egg and allow the sperm to transfer its nucleus.

To ensure that no more than one sperm fertilizes the egg, once the acrosomal reactions take place at one location of the egg membrane, the egg releases proteins in other locations to prevent other sperm from fusing with the egg. If this mechanism fails, multiple sperm can fuse with the egg, resulting in polyspermy. The resulting embryo is not genetically viable and dies within a few days.

43.5B: Cleavage, the Blastula Stage, and Gastrulation

A zygote undergoes rapid cell divisions (cleavage) to form a spherical ball of cells: the blastula; this will further develop into a blastocyst.

Learning Objectives

- Describe the events that occur from the formation of a zygote to gastrulation

Key Points

- A single-celled zygote will undergo multiple rounds of cleavage, or cell division, in order to produced a ball of cells, called a blastula, with a fluid-filled cavity in its center, called a blastocoel.
- In animals with little yolk in the egg, the zygote undergoes holoblastic cleavage, in which the entire zygote is cleaved repeatedly; in animals with a lot of yolk in the egg, the zygote undergoes meroblastic cleavage, in which only part of the zygote is cleaved.
- The blastula eventually organizes itself into two layers: the inner cell mass (which will become the embryo) and the outer layer or trophoblast (which will become the placenta); the structure is now called a blastocyst.
- During gastrulation, the blastula folds in on itself to form three germ layers, the ectoderm, the mesoderm, and the endoderm, that will give rise to the internal structures of the organism.

Key Terms

- **blastula**: a 6-32-celled hollow structure that is formed after a zygote undergoes cell division
- **blastomere**: any cell that results from division of a fertilized egg
- **meroblastic**: undergoing only partial cleavage
- **holoblastic**: cleaving, and separating into separate blastomeres
- **inner cell mass**: a mass of cells within a primordial embryo that will eventually develop into the distinct form of a fetus in most eutherian mammals
- **gastrulation**: the stage of embryo development at which a gastrula is formed from the blastula by the inward migration of cells
- **trophoblast**: the membrane of cells that forms the wall of a blastocyst during early pregnancy, providing nutrients to the embryo and later developing into part of the placenta

Cleavage and Blastula Stage

The development of multi-cellular organisms begins from a single-celled zygote, which undergoes rapid cell division to form the blastula. The rapid, multiple rounds of cell division are termed cleavage. After the cleavage has produced over 100 cells, the embryo is called a blastula. The blastula is usually a spherical layer of cells (the blastoderm) surrounding a fluid-filled or yolk-filled cavity (the blastocoel). Mammals at this stage form a structure called the blastocyst, characterized by an inner cell mass that is distinct from the surrounding blastula. During cleavage, the cells divide without an increase in mass; that is, one large single-celled zygote divides into multiple smaller cells. Each cell within the blastula is called a blastomere.

Cleavage can take place in two ways: holoblastic (total) cleavage or meroblastic (partial) cleavage. The type of cleavage depends on the amount of yolk in the eggs. In placental mammals (including humans) where nourishment is provided by the mother's body, the eggs have a very small amount of yolk and undergo holoblastic cleavage. Other species, such as birds, with a lot of yolk in the egg to nourish the embryo during development, undergo meroblastic cleavage.

In mammals, the blastula forms the blastocyst in the next stage of development. Here the cells in the blastula arrange themselves in two layers: the inner cell mass and an outer layer called the trophoblast. The inner cell mass is also known as the embryoblast; this mass of cells will go on to form the embryo. At this stage of development, the inner cell mass consists of embryonic stem cells that will differentiate into the different cell types needed by the organism. The trophoblast will contribute to the placenta and nourish the embryo.

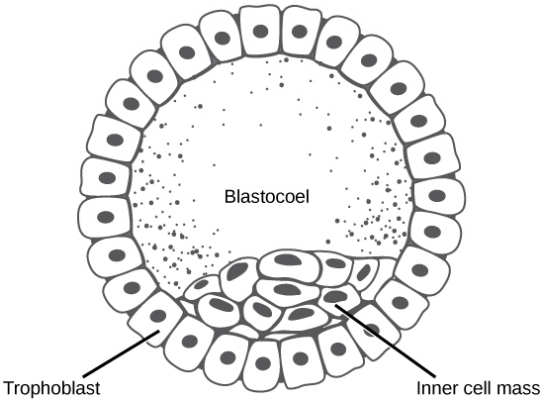

Figure 43.5B.1: **Formation of the blastocyst**: The rearrangement of the cells in the mammalian blastula to two layers, the inner cell mass and the trophoblast, results in the formation of the blastocyst.

Gastrulation

The typical blastula is a ball of cells. The next stage in embryonic development is the formation of the body plan. The cells in the blastula rearrange themselves spatially to form three layers of cells in a process known as gastrulation. During gastrulation, the blastula folds upon itself to form the three layers of cells. Each of these layers is called a germ layer, which differentiate into different organ systems.

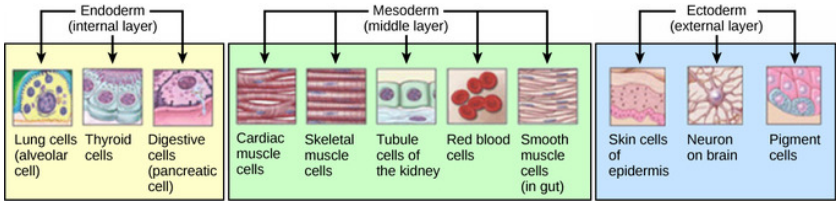

Figure 43.5B.1:

Differentiation of germ layers: The three germ layers give rise to different cell types in the animal body: the ectoderm forms the nervous system and the outer layer of skin, the mesoderm gives rise to muscles and connective tissues, and the endoderm gives rise to the lining of the digestive system and other internal organs.

The three germs layers are the endoderm, the ectoderm, and the mesoderm. The ectoderm gives rise to the nervous system and the epidermis; the mesoderm gives rise to the muscle cells and connective tissue in the body; and the endoderm gives rise to columnar cells found in the digestive system and many internal organs.

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43.6: Organogenesis and Vertebrate Formation

Topic hierarchy

- 43.6A: Organogenesis
- 43.6B: Vertebrate Axis Formation

43.6A: Organogenesis

During organogenesis, the three germ layers of the embryo differentiate and further specialize to form the various organs of the body.

Learning Objectives

- Describe the process of organogenesis in vertebrates

Key Points

- Cells in the ectoderm are signaled by molecules called growth factors to form the neural plate, which rolls up to form a structure called the neural tube; the neural tube will eventually develop into the brain and spinal cord.
- The differing expression of various genes controls the differentiation of the mesoderm into connective tissue, as well as the ribs, spine, skeletal muscle, and lungs.
- The endoderm forms the lining of the digestive tract, as well as the linings of all the glands that will empty into the digestive tract; it also forms a wide variety of internal organs.

Key Terms

- **organogenesis**: the formation and development of the organs of an organism from embryonic cells
- **ectoderm**: outermost of the three tissue layers in the embryo of a metazoan animal, which will produce the epidermis (skin) and nervous system of the adult
- **mesoderm**: one of the three tissue layers in the embryo of a metazoan animal, which will produce many internal organs of the adult such as the muscles, spine and circulatory system
- **endoderm**: one of the three tissue layers in the embryo of a metazoan animal, which will produce the digestive system and other internal organs of the adult
- **neural plate**: a thick, flat bundle of ectoderm formed in vertebrate embryos after induction by the notochord

Organogenesis

Organogenesis is the process by which the three germ tissue layers of the embryo, which are the ectoderm, endoderm, and mesoderm, develop into the internal organs of the organism. Organs form from the germ layers through the differentiation: the process by which a less-specialized cell becomes a more-specialized cell type. This must occur many times as a zygote becomes a fully-developed organism. During differentiation, the embryonic stem cells express specific sets of genes which will determine their ultimate cell type. For example, some cells in the ectoderm will express the genes specific to skin cells. As a result, these cells will differentiate into epidermal cells. Therefore, the process of differentiation is regulated by cellular signaling cascades.

In vertebrates, one of the primary steps during organogenesis is the formation of the neural system. The ectoderm forms epithelial cells and tissues, as well as neuronal tissues. During the formation of the neural system, special signaling molecules called growth factors signal some cells at the edge of the ectoderm to become epidermis cells. The remaining cells in the center form the neural plate. If the signaling by growth factors were disrupted, then the entire ectoderm would differentiate into neural tissue. The neural plate undergoes a series of cell movements where it rolls up and forms a tube called the neural tube. In further development, the neural tube will give rise to the brain and the spinal cord.

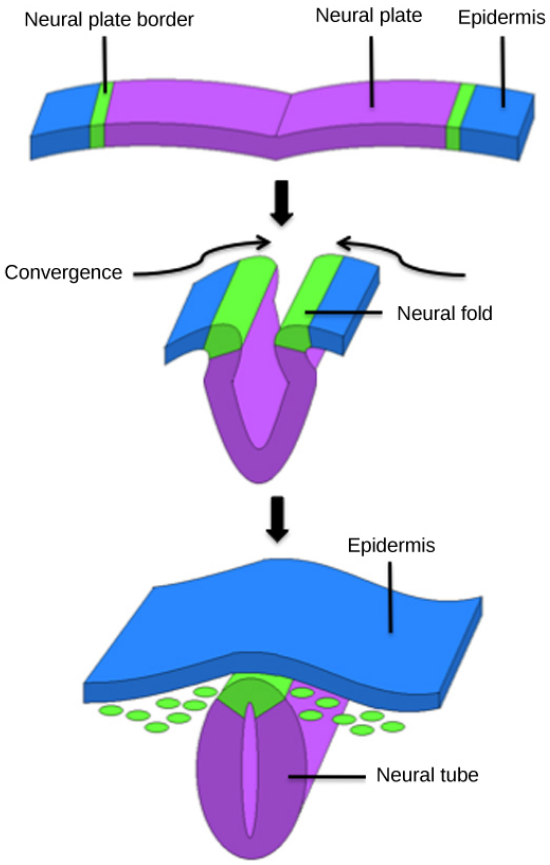

Figure 43.6A.1: **Neural tube formation**: The central region of the ectoderm forms the neural tube, which gives rise to the brain and the spinal cord.

The mesoderm that lies on either side of the vertebrate neural tube will develop into the various connective tissues of the animal body. A spatial pattern of gene expression reorganizes the mesoderm into groups of cells called somites, with spaces between them. The somites will further develop into the ribs, lungs, and segmental (spine) muscle. The mesoderm also forms a structure called the notochord, which is rod-shaped and forms the central axis of the animal body.

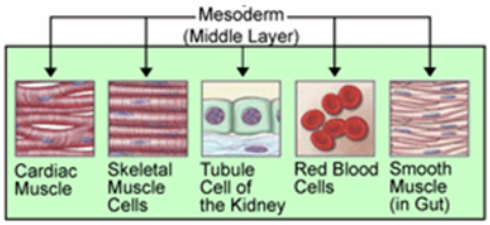

Figure 43.6A.1: **Mesoderm**: The mesoderm aids in the production of cardiac muscles, skeletal muscle, smooth muscle, tissues within the kidneys, and red blood cells.

The endoderm consists, at first, of flattened cells, which subsequently become columnar. It forms the epithelial lining of the whole of the digestive tube (except part of the mouth and pharynx) and the terminal part of the rectum (which is lined by involutions of the ectoderm). It also forms the lining cells of all the glands which open into the digestive tube, including those of the liver and pancreas; the epithelium of the auditory tube and tympanic cavity; the trachea, bronchi, and air cells of the lungs; the urinary bladder and part of the urethra; and the follicle lining of the thyroid gland and thymus. Additionally, the endoderm forms internal organs including the stomach, the colon, the liver, the pancreas, the urinary bladder, the epithelial parts of trachea, the lungs, the pharynx, the thyroid, the parathyroid, and the intestines.

43.6B: Vertebrate Axis Formation

Through the expression patterns of different genes, the three axes of the body are established, aiding in tissue and organ development.

Learning Objectives

- Describe the formation of body axes in vertebrates

Key Points

- As an animal develops, it must organize its internal and external structures such that the anterior/posterior (forward/backward), dorsal / ventral (back/belly), and lateral/medial (side/middle) axes are correctly determined.
- Proteins that are part of the Wnt signaling pathway help determine the anterior/posterior axis by guiding the axons of the spinal cord in an anterior/posterior direction.
- Together with the sonic hedgehog (Shh) protein, Wnt determines the dorsal/ventral axis; Wnt levels are highest in the dorsal region and lessen toward the ventral region, while Shh levels are highest in the ventral region and lessen toward the dorsal region.

Key Terms

- **dorsal**: with respect to, or concerning the side in which the backbone is located, or the analogous side of an invertebrate
- **ventral**: on the front side of the human body, or the corresponding surface of an animal, usually the lower surface
- **notochord**: a flexible rodlike structure that forms the main support of the body in the lowest chordates; a primitive spine
- **Wnt signaling pathway**: a group of signal transduction pathways made of proteins that pass signals from outside of a cell through cell surface receptors to the inside of the cell

Vertebrate Axis Formation

Even as the germ layers form, the ball of cells still retains its spherical shape. However, animal bodies have lateral-medial (toward the side-toward the midline), dorsal-ventral (toward the back-toward the belly), and anterior-posterior (toward the front-toward the back) axes. As the body forms, it must develop in such a way that cells, tissues, and organs are organized correctly along these axes.

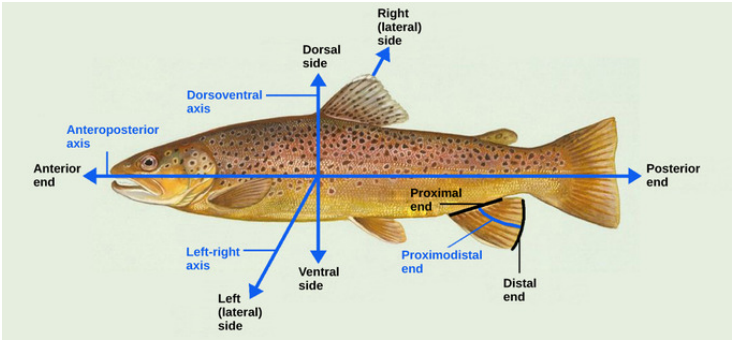

Figure 43.6B.1: **Body axes**: Animal bodies have three axes for symmetry: anterior/posterior (front/behind), dorsal/ventral (back/belly), and lateral/medial (side/middle).

How are these established? In one of the most seminal experiments ever to be carried out in developmental biology, Spemann and Mangold took dorsal cells from one embryo and transplanted them into the belly region of another embryo. They found that the transplanted embryo now had two notochords: one at the dorsal site from the original cells and another at the transplanted site. This suggested that the dorsal cells were genetically programmed to form the notochord and define the dorsal-ventral axis. Since then, researchers have identified many genes that are responsible for axis formation. Mutations in these genes leads to the loss of symmetry required for organism development. Many of these genes are involved in the Wnt signaling pathway.

y embryonic development, the formation of the primary body axes is a crucial step in establishing the overall body f each particular organism. Wnt signaling can be implicated in the formation of the anteroposterior and dorsoventral axes. Wnt signaling activity in anterior-posterior development can be seen in several organisms including mammals, fish, and frogs. Wnt signaling is also involved in the axis formation of specific body parts and organ systems that are a part of later development. In vertebrates, sonic hedgehog (Shh) and Wnt morphogenetic signaling gradients establish the dorsoventral axis of the central nervous system during neural tube axial patterning. High Wnt signaling establishes the dorsal region while high Shh signaling indicates in the ventral region. Wnt is also involved in the dorsal-ventral formation of the central nervous system through its involvement in axon guidance. Wnt proteins guide the axons of the spinal cord in an anterior-posterior direction. Wnt is also involved in the formation of the limb dorsal-ventral axis. Specifically, Wnt7a helps produce the dorsal patterning of the developing limb.

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43.7: Human Pregnancy and Birth

Topic hierarchy

- 43.7A: Human Gestation
- 43.7B: Labor and Birth
- 43.7C: Contraception and Birth Control
- 43.7D: Infertility

Infertility, or the ability to conceive, can be treated by assisted reproductive technologies which can aid conception.

43.7A: Human Gestation

Once the zygote implants in the uterine wall, embryonic and fetal development continue through three trimesters to birth.

Learning Objectives

- Describe the development of the human fetus from fertilization through the third trimester

Key Points

- After fertilization, the zygote implants itself in the uterine wall; its outer layer grows into the endometrium, where it begins to produce human chorionic gonadotropin.
- During the first trimester, the placenta forms along with the internal organs and structures; however, not all of the internal organs function at this point.
- During the second trimester, internal organs continue to develop and the fetus becomes active.
- The third trimester is one of rapid growth, in which the fetus reaches its full size; pregnancy often becomes uncomfortable for the mother.

Key Terms

- **zygote**: a diploid fertilized egg cell
- **chorion**: allows exchange of oxygen and carbon dioxide between the embryo and the egg's external environment
- **human chorionic gonadotropin**: a peptide hormone, produced during pregnancy, that prevents the breakdown of the corpus luteum and maintains progesterone production
- **placenta**: a vascular organ in mammals that supplies food and oxygen from the mother to the fetus, while passing back waste; it is implanted in the wall of the uterus

Human gestation

Twenty-four hours before fertilization, the egg has finished meiosis and become a mature oocyte. When fertilized (at conception), the egg, now known as a zygote, travels through the oviduct to the uterus. The developing embryo must implant into the wall of the uterus within seven days or it will deteriorate and die. The outer layers of the zygote (blastocyst) grow into the endometrium by digesting the endometrial cells. Wound healing of the endometrium closes up the blastocyst into the tissue. Another layer of the blastocyst, the chorion, begins releasing a hormone called human chorionic gonadotropin (hCG) which makes its way to the corpus luteum, keeping it active. This ensures adequate levels of progesterone that will maintain the endometrium of the uterus for the support of the developing embryo. Pregnancy tests determine the level of hCG in urine or serum: if the hormone is present, the test is positive.

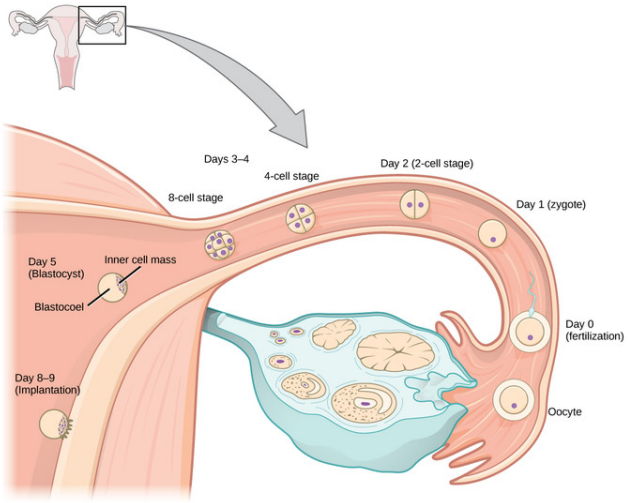

Figure 43.7A.1: **Development of the embryo**: In humans, fertilization occurs soon after the oocyte leaves the ovary. Implantation occurs eight or nine days later. The embryo divides several times as it travels.

First trimester

The gestation period is divided into three equal periods or trimesters. During the first two to four weeks of the first trimester, nutrition and waste are handled by the endometrial lining through diffusion. As the trimester progresses, the outer layer of the embryo begins to merge with the endometrium and the placenta forms. This organ takes over the nutrient and waste requirements of the embryo and fetus, with the mother's blood passing nutrients to the placenta and removing waste from it. Chemicals from the fetus, such as bilirubin, are processed by the mother's liver for elimination. Some of the mother's immunoglobulins will pass through the placenta, providing passive immunity against some potential infections.

Internal organs and body structures begin to develop during the first trimester. By five weeks, limb buds, eyes, the heart, and liver have been basically formed. By eight weeks, the term fetus applies; the body is essentially formed. The individual is about five centimeters (two inches) in length and many of the organs, such as the lungs and liver, are not yet functioning. Exposure to any toxins is especially dangerous during the first trimester, as all of the body's organs and structures are going through initial development. Anything that affects that development can have a severe effect on the fetus' survival.

Figure 43.7A.1: **First trimester**: Fetal development is shown at nine weeks gestation. At this stage, the body is essentially formed; however many of the organs are not yet functioning.

Second trimester

During the second trimester, the fetus grows to about 30 cm (12 inches). As it becomes active, the mother usually feels the first movements. All organs and structures continue to develop. The placenta has taken over the functions of nutrition and waste, along with the production of estrogen and progesterone from the corpus luteum, which has degenerated. The placenta will continue functioning up through the delivery of the fetus.

Figure 43.7*A*. 1: **Second trimester**: This fetus is just entering the second trimester, when the placenta takes over more of the functions performed as the baby develops.

Third trimester

During the third trimester, the fetus grows to 3 to 4 kg (6 ½ -8 ½ lbs.) and about 50 cm (19-20 inches) long. This is the period of the most rapid growth during the pregnancy. Organ development continues to birth (and some systems, such as the nervous system and liver, continue to develop after birth). The mother will be at her most uncomfortable during this trimester. She may urinate frequently due to pressure on the bladder from the fetus. There may also be intestinal blockage and circulatory problems, especially in her legs. Clots may form in her legs due to pressure from the fetus on returning veins as they enter the abdominal cavity.

43.7B: Labor and Birth

Labor and birth are divided into three stages: the dilation of the cervix, the delivery of the baby, and the expulsion of the placenta.

Learning Objectives

- Describe the process of labor and birth in humans

Key Points

- At the end of gestation, estrogen receptors on the uterine wall bind oxytocin, which causes the uterine muscles to contract; as the muscles contract, they signal for the release of more oxytocin in a positive feedback loop.
- During the first stage of labor, the cervix thins and dilates to allow passage of the baby into the birth canal; typically over the course of several hours, the cervix will dilate to its full width of 10 centimeters.
- During the second stage of labor, contractions become very strong and, aided by the pushing of the mother, the baby is expelled from the uterus.
- During the third stage of labor, the placenta, amniotic sac, and the remainder of the umbilical cord are expelled from the uterus, usually within a few minutes after birth.
- When the baby begins suckling at the breast after delivery, prolactin signals the release of milk from the mammary glands, providing nutrition and immunity against disease to the infant.

Key Terms

- **prolactin**: a peptide gonadotrophic hormone secreted by the pituitary gland; it stimulates growth of the mammary glands and lactation in females
- **parturition**: the act of giving birth; childbirth
- **oxytocin**: a hormone that stimulates contractions during labor, and then the production of milk

Labor and Birth

Labor is the physical effort of expulsion of the fetus and the placenta from the uterus during birth (parturition). The total gestation period from fertilization to birth is about 38 weeks (birth usually occurring 40 weeks after the last menstrual period). Toward the end of the third trimester, estrogen causes receptors on the uterine wall to develop and bind the hormone oxytocin. At this time, the baby reorients, facing forward and down with the back or crown of the head engaging the cervix (uterine opening). This causes the cervix to stretch, sending nerve impulses to the hypothalamus, which signals for the release of oxytocin from the posterior pituitary. The oxytocin causes the smooth muscle in the uterine wall to contract. At the same time, the placenta releases prostaglandins into the uterus, increasing the contractions. A positive feedback relay occurs between the uterus, hypothalamus, and the posterior pituitary to assure an adequate supply of oxytocin. As more smooth muscle cells are recruited, the contractions increase in intensity and force.

There are three stages to labor. During stage one, the cervix thins and dilates. This is necessary for the baby and placenta to be expelled during birth. The cervix will eventually dilate to about 10 cm, a process that may take many hours, especially in a woman bearing her first child. At some point, the amniotic sac bursts and the amniotic fluid escapes. During stage two, the baby is expelled from the uterus with the umbilical cord still attached. The uterus contracts and the mother pushes as she compresses her abdominal muscles to aid the delivery. The last stage is the passage of the placenta after the baby has been born and the organ has completely disengaged from the uterine wall, usually within a few minutes. If labor should stop before stage two is reached, synthetic oxytocin, known as Pitocin, can be administered to restart and maintain labor.

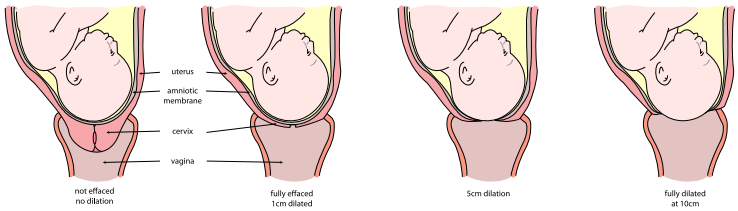

Figure $43.7B.1$: **Cervix dilation**: During the first stage of labor, the cervix, which is normally closed, must open and widen to accommodate the passage of the baby. A cervix is considered fully dilated at 10 centimeters.

The mother's mammary glands go through changes during the third trimester to prepare for lactation and breastfeeding. When the baby begins suckling at the breast, signals are sent to the hypothalamus causing the release of prolactin from the anterior pituitary, which signals the mammary glands to produce milk. Oxytocin is also released, promoting the release of the milk. The milk contains nutrients for the baby's development and growth as well as immunoglobulins to protect the child from bacterial and viral infections.

C: Contraception and Birth Control

Contraception, also known as birth control, is methods used to prevent pregnancy; some of these methods are more successful than others.

Learning Objectives

- Describe the various methods of birth control and their associated failure rates

Key Points

- Barrier methods include those that prevent the sperm from reaching the egg (condoms, diaphragms, cervical caps, sponges, and spermicides); these methods have a failure rate of 15-24%.
- Hormonal methods include those that prevent a woman from ovulating (oral birth control pills, hormone injections, and hormone implants); these have a failure rate of 8%.
- Sterilization, such as a tubal ligation in a woman or a vasectomy in a man, is an extremely effective (although permanent) method of contraception, with a failure rate of 1%.
- More “natural” methods of birth control include avoiding intercourse when a woman is ovulating, or withdrawing the penis from the vagina before ejaculation; these methods are much less successful, with a failure rate of around 25%.

Key Terms

- **condom**: a flexible sleeve made of latex or other impermeable material such as sheepskin, worn over an erect penis during intercourse as a contraceptive or as a way to prevent the spread of STDs
- **intrauterine device**: a contraceptive device consisting of a spiral or similar shape of plastic or metal inserted through the vagina into the uterus in order to prevent the implantation of a fertilized egg
- **vasectomy**: the surgical removal of all or part of the vas deferens, usually as a means of male sterilization
- **contraception**: the use of a device or procedure to prevent conception as a result of sexual activity
- **spermicide**: a substance used for killing sperm

Contraception and Birth Control

Contraception, also known as birth control, is methods or devices used to prevent pregnancy. Strictly speaking, contraception aims to prevent the sperm and egg from joining, while birth control can refer to methods used to prevent a fertilized egg from developing into a fetus. Both terms are, however, frequently used interchangeably.

There are many methods of birth control, including barriers to sperm, hormones that prevent ovulation, sterilization procedures, and “natural” methods. Each method has an associated “failure rate”, which is the number of pregnancies resulting from the method’s use over a twelve-month period.

Barrier methods include those that prevent the sperm from reaching the egg. These include condoms, diaphragms, cervical caps, sponges, and spermicides; they have a failure rate of 15-24%. Barrier methods such as condoms, cervical caps, and diaphragms serve to block sperm from entering the uterus, thereby preventing fertilization. Chemicals such as spermicides, which are designed to kill sperm, are often used in conjunction; sponges, for example, are saturated with spermicides and are placed in the vagina at the cervical opening. Combinations of spermicidal chemicals and barrier methods achieve lower failure rates than do the methods when used separately.

Hormonal methods use synthetic progesterone (sometimes in combination with estrogen) to inhibit the hypothalamus from releasing FSH or LH, preventing an egg from being available for fertilization. The method of administering the hormone affects failure rate, although, in general, hormonal methods have a failure rate of 8%. The most reliable method, with a failure rate of less than 1 percent, is the implantation of the hormone under the skin. The same rate can be achieved through the sterilization procedures of vasectomy in the man or of tubal ligation in the woman, or by using an intrauterine device (IUD). IUDs are inserted into the uterus where they establish an inflammatory condition that prevents fertilized eggs from implanting into the uterine wall.

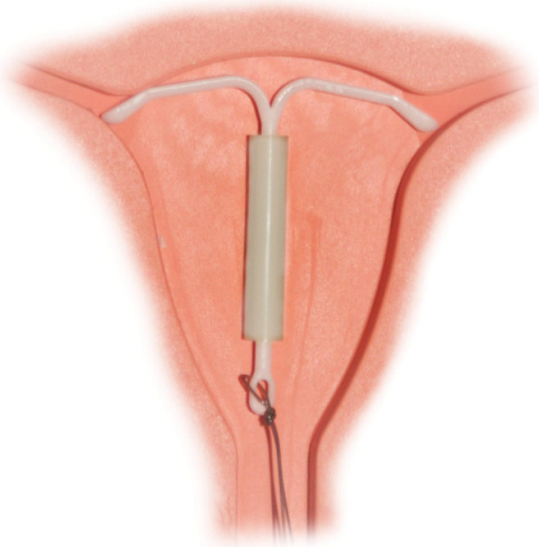

Figure 43.7C.1: **Intrauterine device**: An intrauterine device, or IUD, is implanted inside a woman's uterus, causing the uterus to respond with slight inflammation. This inflammation prevents a fertilized egg from implanting in the uterine wall.

Sterilization is a one-time, permanent, surgical procedure. In a vasectomy, the vasa deferentia of a male are severed and then tied/sealed in a manner that prevents sperm from entering into the seminal stream (ejaculate). Tubal ligation or tubectomy is a surgical procedure for sterilization in which a woman's fallopian tubes are clamped and blocked, or severed and sealed; either method prevents eggs from reaching the uterus for fertilization. Both of these procedures have a less than 1% failure rate.

Natural methods include avoiding intercourse when ovulation is occurring ("natural family planning) or withdrawing the penis from the vagina before ejaculation. Nearly 25% of the couples using natural family planning or withdrawal can expect a failure of the method. Natural family planning is based on the monitoring of the menstrual cycle and having intercourse only during times when the egg is not available. A woman's body temperature may rise a degree Celsius at ovulation and the cervical mucus may increase in volume, becoming more pliable. These changes give a general indication of when intercourse is more or less likely to result in fertilization. Withdrawal involves the removal of the penis from the vagina during intercourse, before ejaculation occurs; it has a failure rate of 27%. This is, therefore, a risky method. The high failure rate is due to the possible presence of sperm in the bulbourethral gland's secretion, which may enter the vagina prior to removing the penis.

Termination of an existing pregnancy can be spontaneous or voluntary. Spontaneous termination in the first trimester is referred to as spontaneous abortion, while in the second and third trimesters is referred to as a miscarriage. Spontaneous abortion usually occurs very early in the pregnancy (typically within the first few weeks). Pregnancy loss occurs when the embryo/fetus cannot develop properly so the gestation is naturally terminated. Voluntary termination of a pregnancy is called an elective abortion. Laws regulating elective abortion vary between states; the laws tend to view fetal viability as the criteria for allowing or preventing the procedure.

43.7D: Infertility

LEARNING OBJECTIVES

- Define infertility and discuss ways in which it can be treated

Infertility is the inability to conceive or carry a fetus to birth. About 75% of causes of infertility can be identified. These include diseases (such as sexually-transmitted diseases that can cause scarring of the reproductive tubes in either men or women) or developmental problems frequently related to abnormal hormone levels in one of the individuals. Inadequate nutrition, especially starvation, can delay menstruation. Stress can also lead to infertility. Short-term stress can affect hormone levels, while long-term stress can delay puberty, causing less-frequent menstrual cycles. Other factors that affect fertility include toxins (such as cadmium), tobacco smoking, marijuana use, gonadal injuries, and aging.

If the cause of infertility is identified, several assisted reproductive technologies (ART) are available to aid conception. A common type of assisted reproductive technology is *in vitro* fertilization (IVF) where an egg and sperm are combined outside the body and then placed in the uterus. Eggs are obtained from the woman after extensive hormonal treatments that prepare mature eggs for fertilization and prepare the uterus for implantation of the fertilized egg. Sperm are obtained from the male and combined with the eggs, which are then supported through several cell divisions to ensure viability of the zygotes. When the embryos have reached the eight-cell stage, one or more is implanted into the woman's uterus. If fertilization is not accomplished by simple IVF, a procedure known as intracytoplasmic sperm injection (ICSI) can be used to inject the sperm into an egg. IVF procedures produce a surplus of fertilized eggs and embryos that can be frozen and stored for future use; the procedures can also result in multiple births.

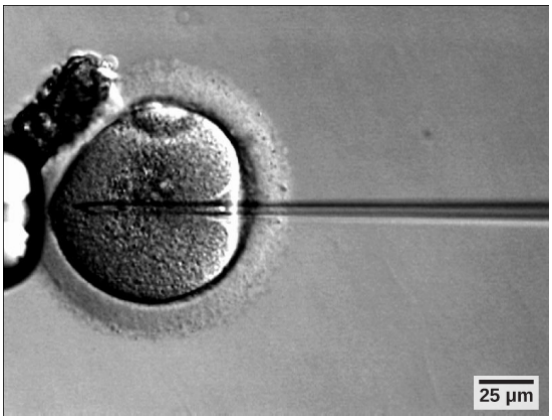

Figure $43.7D.1$: **Intracytoplasmic sperm injection**: A sperm is inserted into an egg for fertilization during intracytoplasmic sperm injection (ICSI).

Key Points

- Infertility can have many causes, such as sexually transmitted diseases, toxins, malnutrition, drug use, stress, and age.
- Infertility can be treated by assisted reproductive technologies in which science intervenes, attempting to aid conception through artificial means.
- *In vitro* fertilization is a type of assisted reproductive technology in which an egg is fertilized by a sperm outside the body, grows to the eight- cell stage, and is then implanted in the uterus.

Key Terms

- **infertility**: the inability to conceive children
- **assisted reproductive technology**: the technological means of inducing pregnancy (by means of artificial insemination or other techniques)

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CHAPTER OVERVIEW

44: ECOLOGY AND THE BIOSPHERE

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44.1C: COMMUNITY ECOLOGY AND ECOSYSTEM ECOLOGY
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44.5B: CAUSES OF GLOBAL CLIMATE CHANGE
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44.1: The Scope of Ecology

Topic hierarchy

- 44.1A: Introduction to Ecology
- 44.1B: Organismal Ecology and Population Ecology
- 44.1C: Community Ecology and Ecosystem Ecology

44.1A: Introduction to Ecology

Ecology is the study of organisms, populations, and communities as they relate to one another and interact in the ecosystems they comprise.

Learning Objectives

- Describe the study of ecology

Key Points

- In ecology, ecosystems are composed of organisms, the communities they comprise, and the non-living aspects of their environment.
- The four main levels of study in ecology are the organism, population, community, and ecosystem.
- Ecosystem processes are those that sustain and regulate the environment.
- Ecological areas of study include topics ranging from the interactions and adaptations of organisms within an ecosystem to the abiotic processes that drive the development of those ecosystems.

Key Terms

- **ecology**: the branch of biology dealing with the relationships of organisms with their environment and with each other
- **ecosystem**: a system formed by an ecological community and its environment that functions as a unit
- **ecophysiology**: the study of the relationships between, and adaptation of, the physiology of an organism and its environment

An Introduction to Ecology

Ecology is the study of the interactions of living organisms with their environment. Within the discipline of ecology, researchers work at four specific levels, sometimes discretely and sometimes with overlap. These levels are organism, population, community, and ecosystem. In ecology, ecosystems are composed of dynamically-interacting parts, which include organisms, the communities they comprise, and the non-living (abiotic) components of their environment. Ecosystem processes, such as primary production, pedogenesis (the formation of soil), nutrient cycling, and various niche construction activities, regulate the flux of energy and matter through an environment. These processes are sustained by organisms with specific life-history traits. The variety of organisms, called biodiversity, which refer to the differing species, genes, and ecosystems, enhances certain ecosystem services.

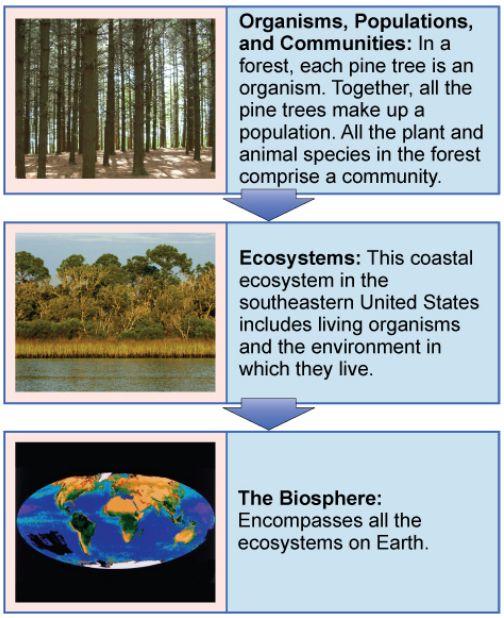

Figure 44.1A.1: **Levels of ecological study**: Ecologists study within several biological levels of organization, which include organism, population, community, and ecosystem.

In essence, ecologists seek to explain:

- life processes
- interactions, interrelationships, behaviors, and adaptations of organisms
- the movement of materials and energy through living communities
- the successional development of ecosystems
- the abundance and distribution of organisms and biodiversity in the context of the environment

There are many practical applications of ecology in conservation biology, wetland management, natural resource management (agroecology, agriculture, forestry, agroforestry, fisheries), city planning (urban ecology), community health, economics, basic and applied science, and human social interaction (human ecology). Organisms and resources comprise ecosystems which, in turn, maintain biophysical feedback mechanisms that moderate processes acting on living (biotic) and nonliving (abiotic) components of the planet. Ecosystems sustain life-supporting functions and produce natural capital, such as biomass production (food, fuel, fiber and medicine), the regulation of climate, global biogeochemical cycles, water filtration, soil formation, erosion control, flood protection, and many other natural features of scientific, historical, economic, or intrinsic value.

There are also many subcategories of ecology, such as ecosystem ecology, animal ecology, and plant ecology, which look at the differences and similarities of various plants in various climates and habitats. In addition, physiological ecology, or ecophysiology, studies the responses of the individual organism to the environment, while population ecology looks at the similarities and dissimilarities of populations and how they replace each other over time.

Finally, it is important to note that ecology is not synonymous with environment, environmentalism, natural history, or environmental science. It is also different from, though closely related to, the studies of evolutionary biology, genetics, and ethology.

44.1B: Organismal Ecology and Population Ecology

Organismal and population ecology study the adaptations that allow organisms to live in a habitat and organisms' relationships to one another.

Learning Objectives

- Describe populations as studied in population ecology and organisms as studied in organismal ecology

Key Points

- Organismal ecology focuses on the morphological, physiological, and behavioral adaptations that let an organism survive in a specific habitat.
- Population ecology studies the number of individuals in an area, as well as how and why their population size changes over time.
- The Karner blue butterfly, an endangered species, makes a good model for both organismal and population ecology since it is dependent, as a population, on a specific plant that grows within specific areas, which, thus, influences butterfly distribution and numbers.

Key Terms

- **conspecific**: an organism belonging to the same species as another
- **population**: a collection of organisms of a particular species, sharing a particular characteristic of interest, most often that of living in a given area
- **oviposit**: to lay eggs

Organismal Ecology

Figure 44.1B.1: **Karner blue butterfly**: The Karner blue butterfly (Lycaeides melissa samuelis) is a rare butterfly that lives only in open areas with few trees or shrubs, such as pine barrens and oak savannas. It can only lay its eggs on lupine plants.

Researchers studying ecology at the organismal level are interested in the adaptations that enable individuals to live in specific habitats. These adaptations can be morphological (pertaining to the study of form or structure), physiological, and behavioral. For instance, the Karner blue butterfly (*Lycaeides melissa samuelis*) is considered a specialist because the females preferentially oviposit (that is, lay eggs) on wild lupine. This preferential adaptation means that the Karner blue butterfly is highly dependent on the presence of wild lupine plants for its continued survival.

Figure 44.1B.1: **Wild lupine**: The wild lupine (Lupinus perennis) is the host plant for the Karner blue butterfly.

After hatching, the larval caterpillars emerge to spend four to six weeks feeding solely on wild lupine. The caterpillars pupate (undergo metamorphosis), emerging as butterflies after about four weeks. The adult butterflies feed on the nectar of flowers of wild lupine and other plant species. A researcher interested in studying Karner blue butterflies at the organismal level might, in addition to asking questions about egg laying, ask questions about the butterflies' preferred temperature (a physiological question) or the behavior of the caterpillars when they are at different larval stages (a behavioral question).

Population Ecology

A population is a group of interbreeding organisms that are members of the same species living in the same area at the same time. Organisms that are all members of the same species, a population, are called conspecifics. A population is identified, in part, by where it lives; its area of population may have natural or artificial boundaries. Natural boundaries might be rivers, mountains, or deserts, while examples of artificial boundaries include mowed grass or manmade structures such as roads. The study of population ecology focuses on the number of individuals in an area and how and why population size changes over time. Population ecologists are particularly interested in counting the Karner blue butterfly, for example, because it is classified as federally endangered. However, the distribution and density of this species is highly influenced by the distribution and abundance of wild lupine. Researchers might ask questions about the factors leading to the decline of wild lupine and how these affect Karner blue butterflies. For example, ecologists know that wild lupine thrives in open areas where trees and shrubs are largely absent. In natural settings, intermittent wildfires regularly remove trees and shrubs, helping to maintain the open areas that wild lupine requires. Mathematical models can be used to understand how wildfire suppression by humans has led to the decline of this important plant for the Karner blue butterfly.

44.1C: Community Ecology and Ecosystem Ecology

Community ecology studies interactions between different species; abiotic and biotic factors affect these on an ecosystem level.

Learning Objectives

- Distinguish between community ecology and ecosystem ecology

Key Points

- Community ecology focuses on the processes driving interactions between differing species and their overall consequences.
- Ecosystem ecology studies all organismal, population, and community components of an area, as well as the non-living counterparts.
- The mutualistic relationship between the Karner blue butterfly and ants are of interest to community ecology studies since both species interact within an area and affect each other's survival rate; in turn, they are both affected by nutrient -poor soils, which are part of the ecosystem ecology.

Key Terms

- **community**: a group of interdependent organisms inhabiting the same region and interacting with each other
- **conspecific**: an organism belonging to the same species as another
- **heterospecific**: an organism belonging to a different species to another

Community Ecology

A biological community consists of the different species within an area, typically a three-dimensional space, and the interactions within and among these species. Community ecologists are interested in the processes driving these interactions and their consequences. Questions about conspecific interactions often focus on competition among members of the same species for a limited resource. Ecologists also study interactions among various species; members of different species are called heterospecifics. Examples of heterospecific interactions include predation, parasitism, herbivory, competition, and pollination. These interactions can have regulating effects on population sizes and can impact ecological and evolutionary processes affecting diversity.

For example, the larvae of the Karner blue butterfly form mutualistic relationships with ants. Mutualism is a form of a long-term relationship that has coevolved between two species and from which each species benefits. For mutualism to exist between individual organisms, each species must receive some benefit from the other as a consequence of the relationship. Researchers have shown that there is an increase in the probability of survival when Karner blue butterfly larvae (caterpillars) are tended by ants. This might be because the larvae spend less time in each life stage when tended by ants, which provides an advantage for the larvae. Meanwhile, the Karner blue butterfly larvae secrete a carbohydrate-rich substance that is an important energy source for the ants. Both the Karner blue larvae and the ants benefit from their interaction.

Figure 44.1C.1: **Karner blue butterfly caterpillar**: Karner blue butterfly caterpillars form beneficial interactions with ants. This mutualistic relationship is an example of a community ecological study, which aims to examine the interactions between different species living in an area.

Ecosystem Ecology

Ecosystem ecology is an extension of organismal, population, and community ecology. The ecosystem is composed of all the biotic components (living things) in an area along with that area's abiotic components (non-living things). Some of the abiotic components include air, water, and soil. Ecosystem biologists ask questions about how nutrients and energy are stored, along with how they move among organisms and the surrounding atmosphere, soil, and water.

The Karner blue butterflies and the wild lupine live in an oak-pine barren habitat. This habitat is characterized by natural disturbance and nutrient-poor soils that are low in nitrogen. The availability of nutrients is an important factor in the distribution of the plants that live in this habitat. Researchers interested in ecosystem ecology could ask questions about the importance of limited resources and the movement of resources, such as nutrients, though the biotic and abiotic portions of the ecosystem.

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44.2: Biogeography

Topic hierarchy

- 44.2A: Biogeography
- 44.2B: Energy Sources
- 44.2C: Temperature and Water
- 44.2D: Inorganic Nutrients and Other Factors
- 44.2E: Abiotic Factors Influencing Plant Growth

44.2A: Biogeography

Biogeography is an ecological field of interest that focuses on the distribution of organisms and the abiotic factors that affect them.

Learning Objectives

- Explain the role of biogeography in the analysis of species distribution

Key Points

- The composition of plant and animal communities change as abiotic factors, which include temperature and altitude, start to vary.
- Some species exist only in specific geographical areas while others can thrive in a variety of areas; however, no single species can be found everywhere in the world.
- Studying an area where a species is not found is also of importance to ecologists in determining unique patterns of species distribution.
- As with animals, plant species can also be either endemic, usually found in isolated land masses, or generalists, found in many regions.

Key Terms

- **biogeography**: the study of the geographical distribution of living things
- **generalist**: species which can thrive in a wide variety of environmental conditions
- **endemic**: unique to a particular area or region; not found in other places

Biogeography

Biogeography is the study of the geographic distribution of living things and the abiotic (non-living) factors that affect their distribution. Abiotic factors can include temperature, moisture, nutrients, oxygen, and energy availability, as well as disturbances from events such as wind and fire. Differences in temperature and rainfall are primarily based on latitude and elevation. As these abiotic factors change, the composition of plant and animal communities also changes. For example, if you were to begin a journey at the equator and walk north, you would notice gradual changes in plant communities. At the beginning of your journey, you would see tropical wet forests with broad-leaved evergreen trees, which are characteristic of plant communities found near the equator. As you continued to travel north, you would see these broad-leaved evergreen plants eventually give rise to seasonally-dry forests with scattered trees. You would also begin to notice changes in temperature and moisture. At about 30 degrees north, these forests would give way to deserts, which are characterized by low precipitation.

Moving farther north, you would see that deserts are replaced by grasslands or prairies. Eventually, grasslands are replaced by deciduous temperate forests. These deciduous forests give way to the boreal forests found in the subarctic, the area south of the Arctic Circle. Finally, you would reach the Arctic tundra, which is found at the most northern latitudes. This trek north reveals gradual changes in both climate and the types of organisms that have adapted to environmental factors associated with ecosystems found at different latitudes. However, different ecosystems exist at the same latitude due in part to abiotic factors such as jet streams, the Gulf Stream, and ocean currents. If you were to hike up a mountain, the changes you would see in the vegetation would parallel those as you move to higher latitudes.

Ecologists who study biogeography examine patterns of species distribution. No species exists everywhere. For example, the Venus flytrap is endemic to a small area in North and South Carolina. An endemic species is one which is naturally found only in a specific geographic area that is usually restricted in size. Other species are generalists, living in a wide variety of geographic areas. The raccoon, for example, is native to most of North and Central America.

Species distribution patterns are based on biotic and abiotic factors and the influences these factors have had during the very long periods of time required for species evolution. Therefore, early studies of biogeography were closely linked to the emergence of evolutionary thinking in the eighteenth century. Some of the most distinctive assemblages of plants and animals occur in regions that have been physically separated for millions of years by geographic barriers. Biologists

estimate that Australia, for example, has between 600,000 and 700,000 species of plants and animals. Approximately 3/4 of living plant and mammal species are endemic species found solely in Australia.

Figure 44.2A.1: **Endemic species of Australia**: Australia is home to many endemic species. The (a) wallaby (Wallabia bicolor), a medium-sized member of the kangaroo family, is a pouched mammal, or marsupial. The (b) echidna (Tachyglossus aculeatus) is an egg-laying mammal.

Sometimes ecologists discover unique patterns of species distribution by determining where species are not found. Hawaii, for example, has no native land species of reptiles or amphibians and has only one native terrestrial mammal, the hoary bat. Most of New Guinea, as another example, lacks placental mammals.

Plants can be endemic or generalists. Endemic plants are found only in specific regions of the earth, while generalists are found in many regions. Isolated land masses, such as Australia, Hawaii, and Madagascar, often have large numbers of endemic plant species. Some of these plants are endangered due to human activity. The forest gardenia (*Gardenia brighamii*), for instance, is endemic to Hawaii; only an estimated 15–20 trees are thought to exist.

Figure 44.2A.1: **Endangered forest gardenia**: Listed as federally endangered, the forest gardenia is a small tree with distinctive flowers. It is found only in five of the Hawaiian Islands in small populations consisting of a few individual specimens.

44.2B: Energy Sources

The availability of energy and nutrient sources affects species distribution and their adaptation to land or aquatic habitats.

Learning Objectives

- Assess how energy availability affects species distribution within an ecosystem

Key Points

- In land habitats, plant adaptations include life cycles that are dependent on the availability of light; for example, species will flower or grow at varying times to ensure they capture enough available light suitable to their needs.
- In aquatic ecosystems, species growth and distribution are adapted to deal with the sometimes-limited availability of light due to its absorption by water, plants, suspended particles, microorganisms, and water depth.
- Ocean upwelling and spring and fall turnovers are important processes regulating the distribution of nutrients in an aquatic ecosystems.
- Nutrient availability is connected to the energy needs of organisms in aquatic ecosystems since sequestered energy is reused by living organisms from dead ones.

Key Terms

- **ephemeral**: lasting for a short period of time
- **upwelling**: the oceanographic phenomenon that occurs when strong, usually seasonal, winds push water away from the coast, bringing cold, nutrient-rich deep waters up to the surface
- **thermocline**: a layer within a body of water or air where the temperature changes rapidly with depth

Energy Availability & Nutrient Cycling as Abiotic Factors

Energy Sources

Energy from the sun is captured by green plants, algae, cyanobacteria, and photosynthetic protists. These organisms convert solar energy into the chemical energy needed by all living things. Light availability can be an important abiotic force directly affecting the evolution of adaptations in photosynthesizers. For instance, plants in the understory of a temperate forest are shaded when the trees above them in the canopy completely leaf out in the late spring. Not surprisingly, understory plants have adaptations to successfully capture available light. One such adaptation is the rapid growth of spring ephemeral plants, such as the spring beauty. These spring flowers achieve much of their growth and finish their life cycle (reproduce) early in the season before the trees in the canopy develop leaves.

Figure 44.2B.1: **Ephemeral plant**: The spring beauty is an ephemeral spring plant that flowers early in the spring to avoid competing with larger forest trees for sunlight.

In aquatic ecosystems, the availability of light may be limited because sunlight is absorbed by water, plants, suspended particles, and resident microorganisms. Toward the bottom of a lake, pond, or ocean, there is a zone that light cannot reach. Photosynthesis cannot take place there and, as a result, a number of adaptations have evolved that enable living things to survive without light. For instance, aquatic plants have photosynthetic tissue near the surface of the water. The broad, floating leaves of a water lily cannot survive without light. In environments such as hydrothermal vents, some bacteria extract energy from inorganic chemicals because there is no light for photosynthesis.

Nutrient Cycling

The availability of nutrients in aquatic systems is also an important aspect of energy or photosynthesis. Many organisms sink to the bottom of the ocean when they die in the open water. When this occurs, the energy found in that organism is sequestered for some time unless ocean upwelling occurs. Ocean upwelling is the rising of deep ocean waters that occurs when prevailing winds blow along surface waters near a coastline. As the wind pushes ocean waters offshore, water from the bottom of the ocean moves up to replace this water. As a result, the nutrients once contained in dead organisms become available for reuse by other living organisms.

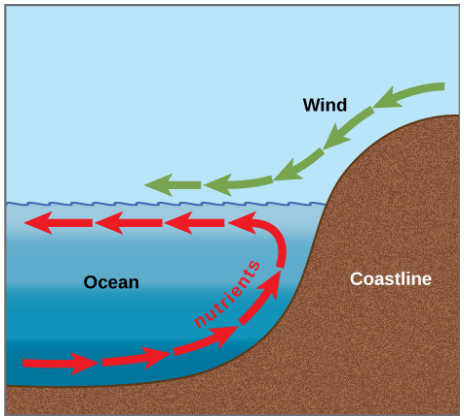

Figure 44.2B.1: **Upwelling**: Ocean upwelling is an important process that recycles nutrients and energy in the ocean. As wind (green arrows) pushes offshore, it causes water from the ocean bottom (red arrows) to move to the surface, bringing up nutrients from the ocean depths.

In freshwater systems, the recycling of nutrients occurs in response to air temperature changes. The nutrients at the bottom of lakes are recycled twice each year: in the spring and fall turnover, which recycles nutrients and oxygen from the bottom of a freshwater ecosystem to the top of a body of water. These turnovers are caused by the formation of a thermocline: a layer of water with a temperature that is significantly different from that of the surrounding layers. In wintertime, the surface of lakes found in many northern regions is frozen. However, the water under the ice is slightly warmer, while the water at the bottom of the lake is warmer yet at 4 °C to 5 °C (39.2 °F to 41 °F). Water is densest at 4 °C; therefore, the deepest water is also the densest. The deepest water is oxygen poor because the decomposition of organic material at the bottom of the lake uses up available oxygen that cannot be replaced by means of oxygen diffusion into the water due to the surface ice layer.

In springtime, air temperatures increase and surface ice melts. When the temperature of the surface water begins to reach 4 °C, the water becomes heavier and sinks to the bottom. The water at the bottom of the lake, displaced by the heavier surface water, rises to the top. As it rises, the sediments and nutrients from the lake bottom are brought along with it. During the summer months, the lake water stratifies, or forms layers, with the warmest water at the lake surface.

As air temperatures drop in the fall, the temperature of the lake water cools to 4 °C; this causes fall turnover as the heavy cold water sinks and displaces the water at the bottom. The oxygen-rich water at the surface of the lake then moves to the bottom of the lake, while the nutrients at the bottom of the lake rise to the surface (). During the winter, the oxygen at the bottom of the lake is used by decomposers and other organisms requiring oxygen, such as fish.

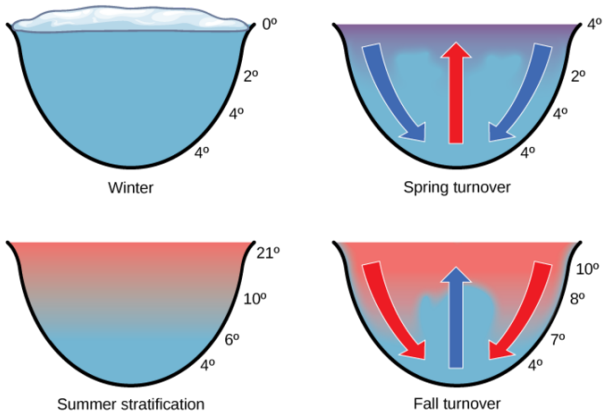

Figure 44.2B.1: **Nurient recycling in freshwater systems**: The spring and fall turnovers are important processes in freshwater lakes that act to move the nutrients and oxygen at the bottom of deep lakes to the top. Turnover occurs because water has a maximum density at 4 °C. Surface water temperature changes as the seasons progress, causing denser water to sink.

C: Temperature and Water

Temperature and water are important abiotic factors that affect species distribution.

Learning Objectives

- Describe species adaptations to temperature fluctuations and the availability of water

Key Points

- Temperature is a factor that influences species distribution because organisms must either maintain a specific internal temperature or inhabit an environment that will keep the body within a temperature range that supports their metabolism.
- Many species have developed adaptations, such as migration, hibernation, and estivation, to deal with temperature fluctuations in the environments in which they live.
- Water retention is vital to all living beings; adaptations have evolved within both terrestrial and aquatic species to minimize water loss.

Key Terms

- **osmosis**: The net movement of solvent molecules from a region of high solvent potential to a region of lower solvent potential through a partially permeable membrane
- **estivate**: to go into stasis or torpor in the summer months
- **extremophile**: an organism that lives under extreme conditions of temperature, salinity, etc; commercially important as a source of enzymes that operate under similar conditions
- **torpor**: a state of being inactive or stuporous
- **hibernation**: a state of inactivity and metabolic depression in animals during winter

Temperature & Water as Abiotic Influences

Temperature

Temperature affects the physiology of living things as well as the density and state of water. It exerts an important influence on living organisms because few can survive at temperatures below 0 °C (32 °F) due to metabolic constraints. It is also rare for them to survive at temperatures exceeding 45 °C (113 °F). This is a reflection of evolutionary response to typical temperatures. Enzymes are most efficient within a narrow and specific range of temperatures; enzyme degradation can occur at higher temperatures. Therefore, organisms must either maintain an internal temperature or inhabit an environment that will keep the body within a temperature range that supports metabolism. Some animals have adapted to enable their bodies to survive significant temperature fluctuations, as seen in hibernation or reptilian torpor. Similarly, some bacteria have adapted to survive in extremely-hot temperatures found in places such as geysers. Such bacteria are examples of extremophiles: organisms that thrive in extreme environments.

Temperature can limit the distribution of living things. Animals faced with temperature fluctuations may respond with adaptations, such as migration, in order to survive. Migration, the movement from one place to another, is common in animals, including many that inhabit seasonally-cold climates. Migration solves problems related to temperature, locating food, and finding a mate. In migration, for instance, the arctic tern (*Sterna paradisaea*) makes a 40,000 km (24,000 mi) round trip flight each year between its feeding grounds in the southern hemisphere and its breeding grounds in the Arctic Ocean. Monarch butterflies (*Danaus plexippus*) live in the eastern United States in the warmer months, but migrate to Mexico and the southern United States in the wintertime. Some species of mammals also make migratory forays: reindeer (*Rangifer tarandus*) travel about 5,000 km (3,100 mi) each year to find food. Amphibians and reptiles are more limited in their distribution because they lack migratory ability. Not all animals that can migrate do so as migration carries risk and comes at a high energy cost.

Figure 44.2C. 1: **Arctic tern**: The arctic tern is an example of a species that must migrate yearly to deal with temperature fluctuations that exist in the regions where it is found.

Some animals hibernate or estivate to survive hostile temperatures. Hibernation enables animals to survive cold conditions, while estivation allows animals to survive the hostile conditions of a hot, dry climate. Animals that hibernate or estivate enter a state known as torpor, a condition in which their metabolic rate is significantly lowered. This enables the animal to wait until its environment better supports its survival. Some amphibians, such as the wood frog (*Rana sylvatica*), have an antifreeze-like chemical in their cells, which retains the cells' integrity and prevents them from bursting.

Figure 44.2C. 1: **Wood frog**: The wood frog, like all other amphibians and reptiles, cannot migrate; as a result, the species survives extreme temperature changes through the antifreeze-like chemical found in their cells.

Water

Water is required by all living things because it is critical for cellular processes. Since terrestrial organisms lose water to the environment by simple diffusion, they have evolved many adaptations to retain water.

Examples of adaptations used by terrestrial and aquatic species include the following:

- Plants have a number of interesting features on their leaves, such as leaf hairs and a waxy cuticle, that serve to decrease the rate of water loss via transpiration.
- Freshwater organisms, surrounded by water, are constantly in danger of having water rush into their cells because of osmosis. Many adaptations of organisms living in freshwater environments have evolved to ensure that solute concentrations in their bodies remain within appropriate levels. One such adaptation is the excretion of dilute urine.
- Marine organisms are surrounded by water with a higher solute concentration than the organism and, thus, are in danger of losing water to the environment because of osmosis. These organisms have morphological and physiological adaptations to retain water and release solutes into the environment. For example, marine iguanas (*Amblyrhynchus*

status) sneeze out water vapor that is high in salt in order to maintain solute concentrations within an acceptable ge while swimming in the ocean and eating marine plants.

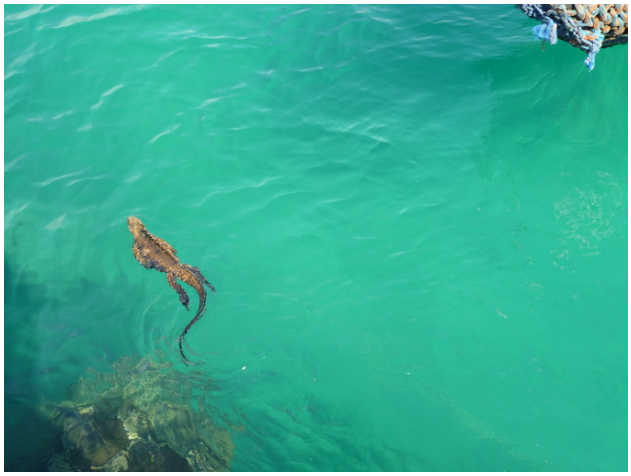

Figure 44.2C.1: **Marine iguanas**: Marine iguanas have a special, salt-secretion adaption that allows them to minimize bodily water loss.

44.2D: Inorganic Nutrients and Other Factors

Soil structure, oxygen availability, wind, and fire are abiotic factors that have influences on species distribution and quantity.

Learning Objectives

- Identify further abiotic factors that affect species distribution and abundance

Key Points

- Soil structure, pH, and its nutrient content affect the distribution of plants, which in turn influences the distribution of the animals that feed on them.
- Oxygen availability is an important abiotic factor affecting species in both aquatic and terrestrial environments.
- Wind and fire impose physical disturbances that species must be adapted to in order to live in affected areas.

Key Terms

- **inorganic**: relating to a compound that does not contain carbon
- **transpiration**: the loss of water by evaporation in terrestrial plants, especially through the stomata; accompanied by a corresponding uptake from the roots

Other Important Abiotic Factors

Inorganic nutrients, soil structure, and aquatic oxygen availability are further abiotic factors that affect species distribution in an ecosystem. The same is true for terrestrial factors, such as wind and fire, which can impact the types of species that inhabit regions exposed to these types of disturbances.

Inorganic Nutrients and Soil

Inorganic nutrients, such as nitrogen and phosphorus, are important in the distribution and the abundance of living things. Plants obtain these inorganic nutrients from the soil when water moves into the plant through the roots. Therefore, soil structure (the particle size of soil components), soil pH, and soil nutrient content play an important role in the distribution of plants. Animals obtain inorganic nutrients from the food they consume. Therefore, animal distributions are related to the distribution of what they eat. In some cases, animals will follow their food resource as it moves through the environment.

Oxygen Availability

Figure $44.2D.1$: **Jack pine cones**: The mature cones of the jack pine (Pinus banksiana) open only when exposed to high temperatures, such as during a forest fire. A fire will probably kill most vegetation, so a seedling that germinates after a fire is more likely to receive ample sunlight than one that germinates under normal conditions.

Some abiotic factors, such as oxygen, are important in aquatic ecosystems as well as terrestrial environments. Terrestrial animals obtain oxygen from the air they breathe. Oxygen availability can be an issue for organisms living at very high elevations, where there are fewer molecules of oxygen in the air. In aquatic systems, the concentration of dissolved oxygen is related to water temperature and the speed at which the water moves. Cold water has more dissolved oxygen than warmer water. In addition, salinity, water current, and tide can be important abiotic factors in aquatic ecosystems.

Other Terrestrial Factors

Wind can be an important abiotic factor because it influences the rate of evaporation and transpiration. The physical force of wind is also important because it can move soil, water, or other abiotic factors, as well as an ecosystem's organisms.

Fire is another terrestrial factor that can be an important agent of disturbance in terrestrial ecosystems. Some organisms are adapted to fire and, thus, require the high heat associated with fire to complete a part of their life cycle. For example, the jack pine, a coniferous tree, requires heat from fire for its seed cones to open. Through the burning of pine needles, fire adds nitrogen to the soil and limits competition by destroying undergrowth.

E: Abiotic Factors Influencing Plant Growth

The two most important abiotic factors affecting plant primary productivity in an ecosystem are temperature and moisture.

Learning Objectives

- Identify the abiotic factors that affect plant growth

Key Points

- Primary production, on which almost all of life on earth is dependent, occurs through either photosynthesis or chemosynthesis.
- Annual biomass production, used to estimate net primary productivity by plants in an area, is directly influenced by an environment's abiotic factors, which include temperature and moisture.
- Warm and wet climates have the greatest amount of plant biomass because they offer conditions in which photosynthesis, plant growth, and the resulting net primary productivity are highest.

Key Terms

- **biomass**: the total mass of all living things within a specific area, habitat, etc.
- **eco-region**: a region, smaller than an ecozone, that contains a distinct biodiversity of flora and fauna
- **chemosynthesis**: the production of carbohydrates and other compounds using the oxidation of chemical nutrients as a source of energy rather than sunlight; it is limited to certain bacteria and fungi

Abiotic Factors Influencing Plant Growth

Temperature and moisture are important influences on plant production (primary productivity) and the amount of organic matter available as food (net primary productivity). Primary production is the synthesis of organic compounds from atmospheric or aqueous carbon dioxide. It principally occurs through the process of photosynthesis, which uses light as its source of energy, but it also occurs through chemosynthesis, which uses the oxidation or reduction of chemical compounds as its source of energy. Almost all life on earth is directly or indirectly reliant on primary production. The organisms responsible for primary production, known as primary producers or autotrophs, form the base of the food chain. In terrestrial eco-regions, these are mainly plants, while in aquatic eco-regions, they are mainly algae.

Net primary productivity is an estimation of all of the organic matter available as food. It is calculated as the total amount of carbon fixed per year minus the amount that is oxidized during cellular respiration. In terrestrial environments, net primary productivity is estimated by measuring the aboveground biomass per unit area, which is the total mass of living plants, excluding roots. This means that a large percentage of plant biomass which exists underground is not included in this measurement. Net primary productivity is an important variable when considering differences in biomes. Very productive biomes have a high level of aboveground biomass.

Annual biomass production is directly related to the abiotic components of the environment. Environments with the greatest amount of biomass have conditions in which photosynthesis, plant growth, and the resulting net primary productivity are optimized. The climate of these areas is warm and wet. Photosynthesis can proceed at a high rate, enzymes can work most efficiently, and stomata can remain open without the risk of excessive transpiration. Together, these factors lead to the maximal amount of carbon dioxide (CO_2) moving into the plant, resulting in high biomass production. The aboveground biomass produces several important resources for other living things, including habitat and food. Conversely, dry and cold environments have lower photosynthetic rates and, therefore, less biomass. The animal communities living there will also be affected by the decrease in available food.

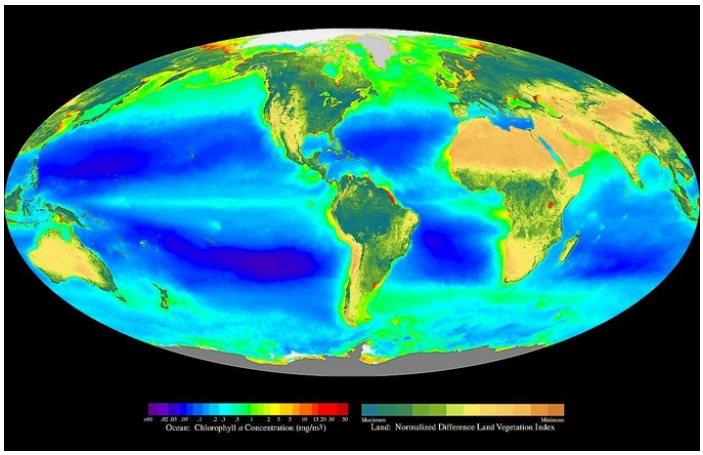

Figure 44.2E.1: **Primary productivity and biomass production**: The magnitude and distribution of global primary production varies between biomes. However, warm and wet climates have the greatest amount of annual biomass production.

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44.3: Terrestrial Biomes

Topic hierarchy

- 44.3A: What constitutes a biome?
- 44.3B: Tropical Wet Forest and Savannas
- 44.3C: Subtropical Deserts and Chaparral
- 44.3D: Temperate Grasslands
- 44.3E: Temperate Forests
- 44.3F: Boreal Forests and Arctic Tundra

44.3A: What constitutes a biome?

A biome consists of all the habitats of a community that make up similar ecosystems in a particular region.

Learning Objectives

- Differentiate biomes from other levels of ecological classification, including habitat

Key Points

- The climate, including precipitation and temperature, and the geography control the type of biome found in a region.
- There are two major classifications of biomes, which are terrestial and aquatic, and these include the types of biomes known as deserts, forests, grasslands, savannas, tundra, and freshwater environments.
- A habitat is the location where a group of one type of organism (a population) lives, while a biome is a community made of all the habitats in a given region and climate.
- Different organisms inhabit different types of biomes.
- Each type of biome can be found in multiple locations on Earth depending on its climate, geography, and organisms.

Key Terms

- **biome**: any major regional biological community such as that of forest or desert
- **ecotone**: a transition area between two adjacent ecosystems
- **habitat**: a specific place or natural conditions in which a plant or animal lives
- **population**: a collection of organisms of a particular species, sharing a particular characteristic of interest, most often that of living in a given area
- **ecosystem**: a system formed by an ecological community and its environment that functions as a unit

What Constitutes a Biome?

A group of living organisms of the same kind that live in the same place simultaneously is known as a population. Populations live together in habitats, which together make up a community. An ecosystem is a community of living organisms interacting with the non-living components of that environment.

A biome is a community on a global scale, where habitats flank each other, and is usually defined by the temperature, precipitation, and types of plants and animals that inhabit it. The Earth's biomes are categorized into two major groups: terrestrial and aquatic. Terrestrial biomes are based on land, while aquatic biomes include both ocean and freshwater biomes. The major types of biomes include: aquatic, desert, forest, grassland, savannas, and tundra.

Biome Attributes

Generally, biome classification is determined by the climate and geography of an area. Each biome consists of communities that have adapted to the different climate and environment inside the biome. Specifically, there are special vegetation adaptations as well as physical and behavioral adaptions made by animals in order to accommodate the environment. The eight major terrestrial biomes on Earth are each distinguished by characteristic temperatures and amount of precipitation. Comparing the annual totals of precipitation and fluctuations in precipitation from one biome to another provides clues as to the importance of abiotic factors in the distribution of biomes. Temperature variation on a daily and seasonal basis is also important for predicting the geographic distribution of the biome and the vegetation type in the biome.

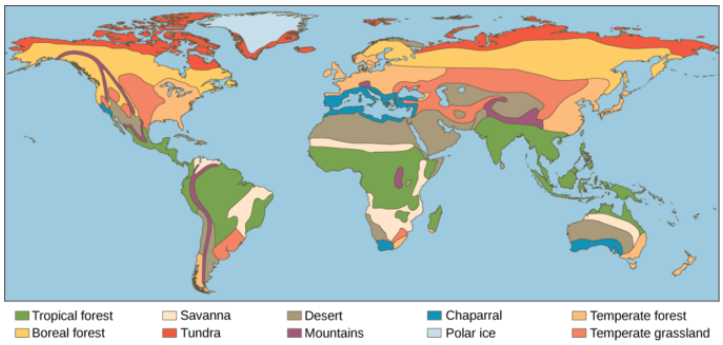

Figure 44.3*A*. 1: **Earth's Major Biomes**: Each of the world's major biomes is distinguished by characteristic temperatures and amounts of precipitation. Each type of biome can be found in multiple places.

The distribution of these biomes shows that the same biome can occur in geographically distinct areas with similar climates. Biomes have no distinct boundaries. Instead, there is a transition zone called an ecotone, which contains a variety of plants and animals. For example, an ecotone might be a transition region between a grassland and a desert, with species from both.

44.3B: Tropical Wet Forest and Savannas

Tropical wet forests are characterized by high precipitation and humidity, while savannas have scattered trees and an extensive dry season.

Learning Objectives

- Recognize the distinguishing characteristics of tropical wet forests and savannas

Key Points

- Tropical wet forests, located near the equator, have temperatures that range from 20°C – 34°C (68°F – 93°F), with little variation in seasonal temperatures.
- The lack of seasonality, constant daily sunlight, ideal temperatures, and high rate of precipitation in tropical wet forests lead to increased plant growth and high species diversity.
- The annual rainfall in tropical wet forests ranges from 125 to 660 cm (50–200 in); there is a high rate of precipitation even in the dry months.
- Savannas, grasslands with scattered trees, are located in Africa, South America, and northern Australia.
- Temperatures in savannas range from 24°C – 29°C (75°F – 84°F), with annual rainfall ranges from 10–40 cm (3.9–15.7 in).
- Because savannas are very dry, trees do not grow as well as they do in other forest biomes and diversity is minimal.

Key Terms

- **deciduous**: of or pertaining to trees which lose their leaves in winter or the dry season
- **biome**: any major regional biological community such as that of forest or desert
- **understory**: the layer of plants that grow in the shade of the canopy of a forest

Tropical Wet Forest

Tropical wet forests, also referred to as tropical rainforests, are found in equatorial regions. The vegetation is characterized by plants with broad leaves that fall off throughout the year. Unlike the trees of deciduous forests, the trees in this biome do not have a seasonal loss of leaves associated with variations in temperature and sunlight; these forests are “evergreen” year-round.

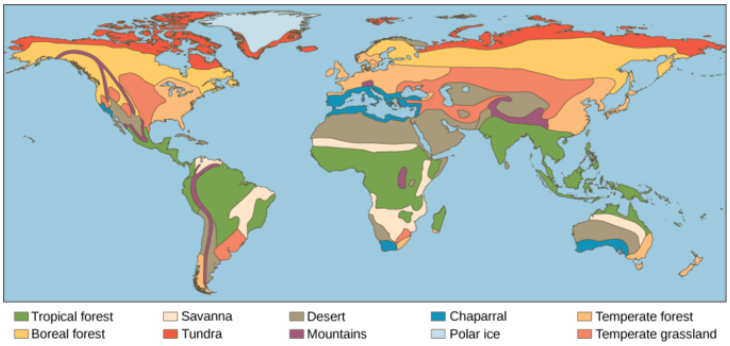

Figure 44.3B.1: **Terrestrial biomes**: Each of the world’s major biomes is distinguished by characteristic temperatures and amounts of precipitation. Polar ice and mountains are also shown. Tropical wet forests are depicted in green and are usually found at equatorial regions. Savannas are depicted in light pink and are usually located in Africa, South America, and Northern Australia.

The temperature and sunlight profiles of tropical wet forests are very stable in comparison to that of other terrestrial biomes, with the temperatures ranging from 20°C – 34°C (68°F – 93°F). Compared to other forest biomes, tropical wet forests have little variation in seasonal temperatures. This lack of seasonality leads to year-round plant growth, rather than the seasonal (spring, summer, and fall) growth seen in other biomes. In contrast to other ecosystems, tropical ecosystems do not have long days and short days during the yearly cycle. Instead, a constant daily amount of sunlight (11–12 hrs per day) provides more solar radiation and, thereby, a longer period of time for plant growth.

The annual rainfall in tropical wet forests ranges from 125-660 cm (50–200 in), with some monthly variation. While sunlight and temperature remain fairly consistent, annual rainfall is highly variable. Tropical wet forests have wet months in which there can be more than 30 cm (11–12 in) of precipitation, as well as dry months in which there are fewer than 10 cm (3.5 in) of rainfall. However, the driest month of a tropical wet forest still exceeds the annual rainfall of some other biomes, such as deserts.

Tropical wet forests have high net primary productivity because the annual temperatures and precipitation values in these areas are ideal for plant growth. Therefore, the extensive biomass present in the tropical wet forest leads to plant communities with very high species diversity. Tropical wet forests have more species of trees than any other biome. On average, between 100 and 300 species of trees are present in a single hectare (2.5 acres) of South America. One way to visualize this is to compare the distinctive horizontal layers within the tropical wet forest biome. On the forest floor is a sparse layer of plants and decaying plant matter. Above that is an understory of short shrubby foliage. A layer of trees rising above this understory is topped by a closed upper canopy: the uppermost overhead layer of branches and leaves. Some additional trees emerge through this closed upper canopy. These layers provide diverse and complex habitats for the variety of plants, fungi, animals, and other organisms within the tropical wet forests. For instance, epiphytes are plants that grow on other plants. Host plants are typically unharmed. Epiphytes are found throughout tropical wet forest biomes. Many species of animals use the variety of plants and the complex structure of the tropical wet forests for food and shelter. Some organisms live several meters above ground, having adapted to this arboreal lifestyle.

Figure 44.3B.1: **Tropical wet forests**: Tropical wet forests, such as the forests of Madre de Dios near the Amazon River in Peru, have high species diversity.

Savannas

Savannas are grasslands with scattered trees located in Africa, South America, and northern Australia. Savannas are hot, tropical areas with temperatures averaging from 24°C – 29°C (75°F – 84°F) and an annual rainfall of 10–40 cm (3.9–15.7 in). They have an extensive dry season. For this reason, forest trees do not grow as well as they do in the tropical wet forest or other forest biomes. As a result, there are relatively few trees within the grasses and forbs (herbaceous flowering plants) that dominate the savanna. Since fire is an important source of disturbance in this biome, plants have evolved well-developed root systems that allow them to quickly re-sprout after a fire.

Figure 44.3B.1: **Savannas**: Savannas, such as this one in Taita Hills Wildlife Sanctuary in Kenya, are dominated by grasses.

Animals commonly found in savannas in Africa include the African elephant, lions, gazelles, giraffes, ostriches, and many other mammals, birds, plants and invertebrates. The northern Australian savannas also have many types of plants, animals, insects, and reptiles, including marsupials (kangaroos and wallabies), bats, and rodents. In addition to the native animals such as foxes and Patagonian maras (rabbit-like rodents), savannas in South America are commonly used for grazing domestic livestock, such as sheep, goats, and cattle because of their open grasslands and herbaceous layer of plants.

44.3C: Subtropical Deserts and Chaparral

Subtropical deserts are characterized by their dry environments, while chaparrals are characterized by the presence of shrubs.

Learning Objectives

- Recognize the distinguishing characteristics of subtropical deserts and chaparrals

Key Points

- Subtropical deserts are centered on the Tropics of Cancer and Capricorn.
- Subtropical deserts can be hot or cold, but they are all very dry,having very low annual precipitation.
- Because precipitation is so low in subtropical deserts, most plants are annuals which utilize adaptations to conserve water.
- Chaparrals (scrub forests) are found in California, along the Mediterranean Sea, and along the southern coast of Australia.
- Chaparrals are very wet in the winter, but very dry in the summer months; most chaparral plants stay dormant during the summer.
- Most chaparral plants are shrubs adapted to fires; some seeds only germinate after a fire.

Key Terms

- **chaparral**: a region of shrubs, typically dry in the summer and rainy in the winter
- **subtropical desert**: dry region centered on the Tropics of Cancer and Capricorn where evaporation exceeds precipitation

Subtropical Deserts

Subtropical deserts, which exist between 15° and 30° north and south latitude, are centered on the Tropics of Cancer and Capricorn. In some years, evaporation exceeds precipitation in this very dry biome. Subtropical hot deserts may have daytime soil surface temperatures above 60°C (140°F) and nighttime temperatures approaching 0°C (32°F). In cold deserts, temperatures may be as high as 25°C (77°F) and may drop below -30°C (-22°F). Subtropical deserts are characterized by low annual precipitation of fewer than 30 cm (12 in), with little monthly variation and lack of predictability in rainfall. In some cases, the annual rainfall can be as low as 2 cm (0.8 in), such as in central Australia (“the Outback”) and northern Africa.

Types of Deserts

There are several types of deserts including high-pressure deserts, mid-continent deserts, rain-shadow deserts, and upwelling deserts. In high-pressure deserts, the high atmospheric pressure enables the air to retain more moisture and there is little rainfall. High-pressure deserts include the Sahara, Arabian, Thar, and Kalahari deserts, and the desert regions within the Arctic and Antarctic circles. Areas in the middle of a continent can receive little rainfall because moisture tends to condense before it reaches the middle of a large continent. Modern examples of mid-continent deserts are the Turkmenistan, Gobi, and Great Australian deserts. Third, rain-shadow deserts are created when moisture from the ocean condenses on one side of a mountain range. These mountain ranges usually have a rainforest on one side and a desert on the other. Examples of rain-shadow deserts include the Mojave desert in the rain-shadow of the Sierra Nevada, the Patagonian desert in the rain-shadow of the Andes, and the Iranian desert in the rain-shadow of the Zagros mountains. Finally, upwelling deserts exist adjacent to areas where cold currents rise to the ocean surface, reducing evaporation. Examples include the Atacama desert, the Western Sahara, and the Namib desert.

Adaptations for Deserts

The type of vegetation and limited animal diversity of this biome are closely related to the low and unpredictable precipitation. Very dry deserts lack perennial vegetation that lives from one year to the next. Instead, many plants are annuals that grow quickly, reproduce when rainfall does occur, and then die. Many other plants in these areas are characterized by having a number of adaptations that conserve water, such as deep roots, reduced foliage, and water-

stems. Seed plants in the desert produce seeds that can remain in dormancy for extended periods between rains. To water loss and conserve energy, many desert animals like the fennec fox are nocturnal and burrow during the day.

Figure 44.3C.1: **Fennec fox**: Fennec foxes live in the deserts of Northern Africa and have large ears for heat dissipation during the day. They stay in burrows during the day and are mostly active at night.

Figure 44.3C.1: **Desert plants**: To reduce water loss, many desert plants have tiny leaves or no leaves at all. The leaves of ocotillo (*Fouquieria splendens*), shown here in the Sonora Desert near Gila Bend, Arizona, appear only after rainfall and then are shed.

Chaparral

The chaparral, also called the scrub forest, is found in California, along the Mediterranean Sea, and along the southern coast of Australia. The annual rainfall in this biome ranges from 65 cm to 75 cm (25.6–29.5 in), with the majority of rain falling in the winter. Due to the very dry summers, many chaparral plants are dormant during that season. The chaparral vegetation is dominated by shrubs and is adapted to periodic fires, with some plants producing seeds that only germinate after a hot fire. The ashes left behind after a fire are rich in nutrients, such as nitrogen, that fertilize the soil and promote plant regrowth.

Figure 44.3C.1: **Chaparrals**: The chaparral, or scrub forest, is dominated by shrubs adapted to periodic fires. Some plants produce seeds that only germinate after a hot fire.

44.3D: Temperate Grasslands

Temperate grasslands are areas with low annual precipitation, fluctuating seasonal temperatures, and few trees.

Learning Objectives

- Recognize the distinguishing characteristics of temperate grasslands

Key Points

- Temperate grasslands are found throughout central North America, where they are also known as prairies; they are also found in Eurasia, where they are known as steppes.
- Temperate grasslands have hot summers and cold winters; the growing season occurs during the spring, summer, and fall.
- Because of the low annual precipitation, temperate grasslands have very few trees.
- Grasses are the dominant vegetation; their roots and rhizomes provide increased fertility to the soil.
- Fires caused by lightening occur often in grasslands; without fires grasslands are converted to scrub forests.

Key Terms

- **steppe**: the grasslands of Eastern Europe and Asia
- **prairie**: an extensive area of relatively flat grassland with few, if any, trees, especially in North America

Temperate Grasslands

Temperate grasslands are found throughout central North America, where they are also known as prairies, and within Eurasia, where they are known as steppes. Temperate grasslands have pronounced annual fluctuations in temperature, with hot summers and cold winters. The annual temperature variation produces specific growing seasons for plants. Plant growth is possible when temperatures are warm enough and when ample water is available to sustain it, which typically occurs in the spring, summer, and fall. During much of the winter, temperatures are low and water, which is stored in the form of ice, is not available for plant growth.

Figure 44.3D.1: **Temperate grasslands in New Zealand**: Temperate tussock grasslands form expansive and distinctive landscapes in the South Island and to a lesser extent in the central plateau region of the North Island of New Zealand.

Annual precipitation ranges from 25 cm to 75 cm (9.8–29.5 in). Because of relatively-lower annual precipitation in temperate grasslands, there are few trees, except for those found growing along rivers or streams. The dominant vegetation tends to consist of grasses; some prairies sustain populations of grazing animals. The vegetation is very dense and the soils

are fertile because the subsurface of the soil is packed with the roots and rhizomes (underground stems) of these grasses, which anchor plants into the ground and replenish the organic material (humus) in the soil when they die and decay.

Figure 44.3D.1: **Przewalski's horses on the Asian steppes**: The Przewalski's horse is a rare and endangered subspecies of wild horse. By the end of the 1950s, only 12 individual Przewalski's horses were left in the world and several reintroduction programs have released populations back into the grasslands in Mongolia and Ukraine.

Fires, mainly caused by lightning, are a natural disturbance in temperate grasslands. When fire is suppressed, the vegetation eventually converts to scrub and dense forests. Often, the restoration or management of temperate grasslands requires the use of controlled burns to suppress the growth of trees and maintain the grasses. Burning causes new grass to grow, which brings back the grazing animals.

Organisms Found in Temperate Grasslands

Mites, insect larvae, nematodes and earthworms inhabit deep soil, which can reach 6 meters (20 ft) underground in undisturbed grasslands on the richest soils of the world. These invertebrates, along with symbiotic fungi, extend the root systems, break apart hard soil, enrich it with urea and other natural fertilizers, trap minerals and water, and promote growth. Some types of fungi make the plants more resistant to insect and microbial attacks. Grasslands also are home to a vast variety of mammals, reptiles, birds, and insects. Typical large mammals include the Giant Anteater and populations of grazing animals, such as the Blue Wildebeest, Przewalski's Horse, and the American Bison.

Figure 44.3D.1: **Grazing animals**: The American bison (*Bison bison*), more commonly called the buffalo, is a grazing mammal that once populated American prairies in huge numbers.

44.3E: Temperate Forests

Temperate forests are characterized by fluctuating seasonal temperatures and constant-but-moderate rainfall.

Learning Objectives

- Recognize the distingushing chracteristics of temperate forests

Key Points

- Temperate forests are the most common biome in eastern North America, Western Europe, Eastern Asia, Chile, and New Zealand.
- Temperatures in temperate forests fluctuate; there are defined growing seasons during spring, summer, and early fall.
- Because temperate forests have moderate annual precipitation, the dominant plants are the deciduous trees.
- Deciduous trees experience a dormant period in the winter, which is why temperate forests have less net productivity than tropical forests.
- Temperate forests are more open than tropical wet forests since their trees do not grow as tall.
- The soils of the temperate forests are rich in inorganic and organic nutrients; this is due to the thick layer of leaf litter on forest floors, which returns nutrients to the soil.

Key Terms

- **deciduous**: of or pertaining to trees which lose their leaves in winter or the dry season
- **temperate forest**: forest concentrations formed in the northern and southern hemisphere, or in temperate regions; main characteristics include wide leaves, large and tall trees, and seasonal vegetation

Temperate Forests

Temperate forests are the most common biome in eastern North America, Western Europe, Eastern Asia, Chile, and New Zealand. This biome is found throughout mid-latitude regions. Temperatures ranging between -30°C – 30°C (-22°F – 86°F) drop below freezing on an annual basis, resulting in defined growing seasons during the spring, summer, and early fall. Precipitation is relatively constant throughout the year, ranging between 75 cm and 150 cm (29.5–59 in).

Because of the moderate, annual rainfall and temperatures, deciduous trees are the dominant plant in this biome. Deciduous trees lose their leaves each fall, remaining leafless in the winter; thus, no photosynthesis occurs during the dormant winter period. Each spring, new leaves appear as the temperature increases. Because of the dormant period, the net primary productivity of temperate forests is less than that of tropical wet forests. In addition, temperate forests show less diversity of tree species than do tropical wet forest biomes.

Figure 44.3E.1: **Deciduous trees**: Deciduous trees, those that lose their leaves seasonally, are the dominant plant in the temperate forest.

The trees of the temperate forests leaf out and shade much of the ground; however, this biome is more open than tropical wet forests because trees in the temperate forests do not grow as tall as the trees in tropical wet forests. The soils of the temperate forests are rich in inorganic and organic nutrients due to the thick layer of leaf litter on forest floors. As this leaf litter decays, nutrients are returned to the soil. The leaf litter also protects soil from erosion, insulates the ground, and provides habitats for invertebrates (such as the pill bug or roly-poly, *Armadillidium vulgare*) and their predators, such as the red-backed salamander (*Plethodon cinereus*).

Organisms Found in Temperate Deciduous Forests

The leaf litter is home to invertebrates (such as the pill bug or roly-poly, *Armadillidium vulgare*) and their predators, including the red-backed salamander (*Plethodon cinereus*). Pileated woodpeckers depend upon dead or dying trees as a source of food and for constructing their nests, and many migratory birds, such as the spring warblers, time their arrival to coincide with the opening of the tree canopy, which provides the insects that are their principal food sources for raising young. Many well-known animals are found in temperate deciduous forests including squirrels, deer, and bears. The top predators in deciduous forest were once wolves and cougars, but their populations have been in decline.

Figure $44.3E.1$: **White-tailed deer**: Deer such as the white-tailed deer of North America are common inhabitants of temperate forests.

44.3F: Boreal Forests and Arctic Tundra

The boreal forest is characterized by coniferous trees, while the arctic tundra is characterized by permanently frozen soils.

Learning Objectives

- Recognize the distinguishing characteristics of boreal forests and arctic tundra

Key Points

- The boreal forest is found across most of Canada, Alaska, Russia, and northern Europe; the arctic tundra lies north of the boreal forest.
- The boreal forest has cold, dry winters and short, cool, wet summers with precipitation that takes the form of snow; due to this environment, evergreen coniferous trees are the dominant plants.
- The soil in boreal forest regions is usually acidic and contains little available nitrogen.
- Boreal forests have lower productivity than tropical or temperate forests; they also have less diversity, with only a tree layer and ground layer.
- Temperatures in the arctic tundra are cold year-round and precipitation is very low.
- Plants in the arctic tundra have a very short growing season of approximately 10–12 weeks, but during this time, growth is rapid; plants are low to the ground and the soil is permanently frozen.

Key Terms

- **permafrost**: permanently frozen ground
- **arctic tundra**: a biome found in the far Northern Hemisphere, north of the boreal forests, where the subsoil is permanently frozen
- **boreal forest**: a biome found in the Northern Hemisphere and characterized by coniferous forests consisting mostly of pines, spruces, and larches

Boreal Forests

The boreal forest, also known as taiga or coniferous forest, the world's largest terrestrial biome, is found south of the Arctic Circle and across most of Canada, Alaska, Russia, and northern Europe. This biome has cold, dry winters and short, cool, wet summers. Temperatures vary from −54°C – 30°C (-65°F – 86°F) throughout the whole year. The summers, while short, are generally warm and humid. In much of the taiga, -20°C (-4°F) would be a typical winter day temperature, while 18°C (64°F) would be an average summer day.The annual precipitation, from 40 cm -100 cm (15.7–39 in), usually takes the form of snow. Little evaporation occurs because of the cold temperatures.

The long and cold winters in the boreal forest have led to the predominance of cold-tolerant, cone-bearing plants. These are evergreen, coniferous trees such as pines, spruces, and firs, which retain their needle-shaped leaves year-round. Evergreen trees can photosynthesize earlier in the spring than can deciduous trees because less energy from the sun is required to warm a needle-like leaf than a broad leaf. This benefits evergreen trees, which grow faster than deciduous trees in the boreal forest. In addition, soils in boreal forest regions tend to be acidic, with little available nitrogen. Leaves are a nitrogen-rich structure that deciduous trees must produce yearly. Therefore, coniferous trees that retain nitrogen-rich needles may have a competitive advantage over the broad-leafed deciduous trees.

The net primary productivity of boreal forests is lower than that of temperate forests and tropical wet forests. The aboveground biomass of boreal forests is high because these slow-growing tree species are long-lived, accumulating standing biomass over time. Plant species diversity is less than that seen in temperate forests and tropical wet forests. Boreal forests lack the pronounced elements of the layered forest structure seen in tropical wet forests. The structure of a boreal forest is often only a tree layer and a ground layer. When conifer needles are dropped, they decompose more slowly than do broad leaves; therefore, fewer nutrients are returned to the soil to fuel plant growth.

Figure 44.3F.1: **Plants in the boreal forest**: The boreal forest (taiga) has low-lying plants and conifer trees; it often consists of only a tree layer and a ground layer.

Arctic Tundra

The Arctic tundra, lying north of the subarctic boreal forest, is located throughout the Arctic regions of the northern hemisphere. The average winter temperature is -34°C (-29.2°F), while the average summer temperature is from 3°C – 12°C (37°F – 52°F). Plants in the arctic tundra have a very short growing season of approximately 10–12 weeks. However, during this time, there are almost 24 hours of daylight, so plant growth is rapid. The annual precipitation of the Arctic tundra is very low (about 15-25 cm), with little annual variation in precipitation. As in the boreal forests, there is little evaporation due to the cold temperatures.

Plants in the Arctic tundra are generally low to the ground. There is little species diversity, low net primary productivity, and low aboveground biomass. The soils of the Arctic tundra may remain in a perennially frozen state referred to as permafrost. The permafrost makes it impossible for roots to penetrate deep into the soil and slows the decay of organic matter, which inhibits the release of nutrients from organic matter. During the growing season, the ground of the Arctic tundra can be completely covered with plants or lichens.

Figure 44.3F.1: **Plants in the Arctic tundra**: Low-growing plants such as shrub willow dominate the tundra landscape, shown here in the Arctic National Wildlife Refuge.

The biodiversity of the tundras is low: there are 1,700 species of vascular plants and only 48 species of land mammals. Notable animals in the Arctic tundra include caribou (reindeer), musk ox, arctic hare, arctic fox, snowy owl, lemmings, and polar bears. Due to the harsh climate, tundra regions have seen little human activity, even though they are sometimes rich in natural resources such as oil and uranium.

Figure 44.3F.1: **Caribou (reindeer) in the tundra**: Some populations of the North American caribou migrate the farthest of any terrestrial mammal, traveling up to 5,000 km (3,100 mi) a year.

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44.4: Aquatic Biomes

Topic hierarchy

- 44.4A: Abiotic Factors Influencing Aquatic Biomes
- 44.4B: Marine Biomes
- 44.4C: Estuaries: Where the Ocean Meets Fresh Water
- 44.4D: Freshwater Biomes

44.4A: Abiotic Factors Influencing Aquatic Biomes

Abiotic factors that influence aquatic biomes include light availability, depth, stratification, temperature, currents, and tides.

Learning Objectives

- Differentiate among the abiotic factors that affect aquatic biomes

Key Points

- In aquatic biomes, light is an important factor that influences the communities of organisms found in both freshwater and marine ecosystems.
- In freshwater biomes, stratification, a major abiotic factor, is related to the energy aspects of light.
- Marine systems are influenced by the physical water movements, such as currents and tides, along with the thermal properties of water.
- Oceans zones can be categorized into photic or aphotic zones, depending on the presence or absence of light and photosynthesis.

Key Terms

- **stratification**: the process leading to the formation or deposition of layers
- **photic**: of, related to, or irradiated by light; especially describing that part of the near-surface ocean is which photosynthesis is possible
- **aphotic**: describing that part of the deep oceans and lakes where photosynthesis is not possible

Abiotic Factors Influencing Aquatic Biomes

As with terrestrial biomes, aquatic biomes are influenced by a series of abiotic factors. However, these factors differ since water has different physical and chemical properties than does air. Even if the water in a pond or other body of water is perfectly clear (there are no suspended particles), water, on its own, absorbs light. As one descends into a deep body of water, there will eventually be a depth which the sunlight cannot reach. While there are some abiotic and biotic factors in a terrestrial ecosystem that might obscure light (such as fog, dust, or insect swarms), usually these are not permanent features of the environment. The importance of light in aquatic biomes is central to the communities of organisms found in both freshwater and marine ecosystems. In freshwater systems, stratification due to differences in density is perhaps the most critical abiotic factor and is related to the energy aspects of light. The thermal properties of water (rates of heating and cooling) are significant to the function of marine systems and have major impacts on global climate and weather patterns. Marine systems are also influenced by large-scale physical water movements, such as currents; these are less important in most freshwater lakes.

The ocean is categorized by several areas or zones. All of the ocean's open water is referred to as the pelagic realm (or zone). The benthic realm (or zone) extends along the ocean bottom from the shoreline to the deepest parts of the ocean floor. Within the pelagic realm is the photic zone, which is the portion of the ocean that light can penetrate (approximately 200 m or 650 ft). At depths greater than 200 m, light cannot penetrate; thus, this is referred to as the aphotic zone. The majority of the ocean is aphotic, lacking sufficient light for photosynthesis. The deepest part of the ocean, the Challenger Deep (in the Mariana Trench, located in the western Pacific Ocean), is about 11,000 m (about 6.8 mi) deep. To give some perspective on the depth of this trench, the ocean is, on average, 4267 m or 14,000 ft deep. These realms and zones are relevant to freshwater lakes as well, as they determine the types of organisms that will inhabit each region.

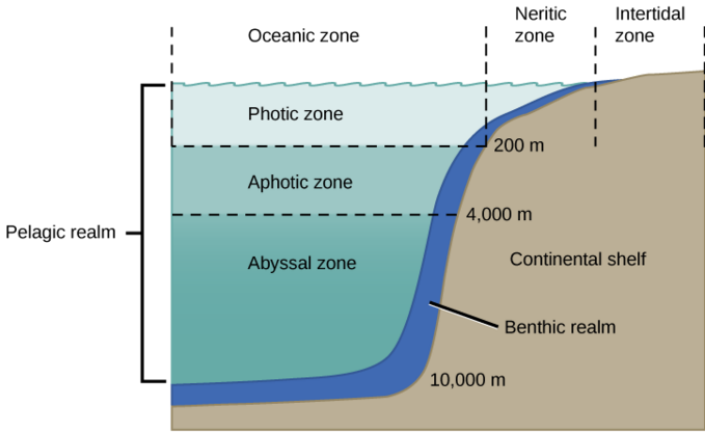

Figure 44.4A.1: **Oceanic zones**: The ocean is divided into different zones based on water depth, light availability, and distance from the shoreline.

44.4B: Marine Biomes

The ocean and coral reefs make up two types of marine biomes where organisms are influenced by depth and light availability.

Learning Objectives

- Describe coral reefs and the various zones in the ocean and the types of organisms living in each

Key Points

- The ocean is divided into different zones with groups of species adapted to deal with the differences in light level, as well as other biotic and abiotic conditions particular to the zones.
- The intertidal zone is characterized by its high and low tides, as well as wave action; as the zone closest to land, it can be a sandy, rocky, or muddy beach.
- The neritic zone is silted, well-oxygenated, low in pressure, and stable in temperature; it is adjacent to the oceanic zone, in which warm and cold waters mix due to currents in the open ocean.
- The benthic zone is nutrient rich because of the sand, silt, and dead organisms that comprise the bottom of the region; as water depth increases, temperature within this deepwater region decreases.
- The abyssal zone is the deepest part of the ocean and, as such, it is cold, has very high pressure, high oxygen content, and low nutrient content.
- Coral reefs are ocean ridges formed by a mutualistic relationship between cnidarians and photosythetic algae; climate change and run-off are just two reasons why these important organisms are now in decline.

Key Terms

- **desiccation**: the state or process of drying-out
- **zooxanthellae**: animals of the genus Symbiodinium, a yellow dinoflagellate, notably found in coral reefs
- **photic**: of, related to, or irradiated by light; especially describing that part of the near-surface ocean is which photosynthesis is possible

Marine Biomes

The ocean is the largest marine biome. It is a continuous body of salt water that is relatively uniform in chemical composition; it is a weak solution of mineral salts and decayed biological matter. Within the ocean, coral reefs are a second kind of marine biome.

Ocean

Physical diversity has a significant influence on the ocean, which is categorized into different zones based on how far light reaches into the water. Each zone has a distinct group of species adapted to the biotic and abiotic conditions particular to that zone.

The intertidal zone (between high and low tide) is the oceanic region that is closest to land. It includes sandy beaches, but can also be rocky or muddy. This zone is an extremely variable environment because of tides. Organisms, exposed to air and sunlight at low tide, are underwater most of the time, especially during high tide. Therefore, living things that thrive in the intertidal zone are adapted to being dry for long periods of time. Since the shore of the intertidal zone is repeatedly struck by waves, the organisms found there are adapted to withstand damage from the pounding action of the waves. The exoskeletons of shoreline crustaceans are tough, protecting them from desiccation and wave damage.

Figure 44.4B.1: **Intertidal zone species**: Sea urchins, mussel shells, and starfish, often found in the intertidal zone, are adapted to this rugged environment.

The neritic zone extends from the intertidal zone to depths of about 200 m (or 650 ft) at the edge of the continental shelf. Since light can penetrate this depth, photosynthesis can occur in the neritic zone. The water here contains silt, is well-oxygenated, low in pressure, and stable in temperature. Phytoplankton and floating *Sargassum*, a marine seaweed, provide a habitat for some sea life found in the neritic zone, including zooplankton, protists, small fishes, and shrimp, which are the base of the food chain for most of the world's fisheries.

Beyond the neritic zone is the open ocean area known as the oceanic zone. Within the oceanic zone there is thermal stratification where warm and cold waters mix because of ocean currents. Abundant plankton serve as the base of the food chain for larger animals. Nutrients are scarce in this less-productive part of the marine biome. When photosynthetic organisms and the protists and animals that feed on them die, their bodies fall to the bottom of the ocean where they remain. In contrast to freshwater lakes, the open ocean lacks a process for bringing the organic nutrients back up to the surface. The majority of organisms in the aphotic zone include sea cucumbers and other organisms that survive on the nutrients contained in the dead bodies of organisms in the photic zone.

A lower layer is the benthic realm, the deepwater region beyond the continental shelf. The bottom of the benthic realm comprises sand, silt, and dead organisms. Temperature decreases, remaining above freezing, as water depth increases. Due to the dead organisms that fall from the upper layers of the ocean, this nutrient-rich portion of the ocean allows a diversity of life to exist, including fungi, sponges, sea anemones, marine worms, sea stars, fishes, and bacteria.

Figure 44.4B.1: **Coral reefs**: Coral reefs are formed by the calcium carbonate skeletons of coral organisms, which are marine invertebrates in the phylum Cnidaria.

The deepest part of the ocean, the abyssal zone, at depths of 4000 m or greater, is very cold and has very high pressure, high oxygen content, and low nutrient content. There are a variety of invertebrates and fishes found in this zone, but the

abyssal zone does not have plants due to the lack of light.

Coral Reefs

Coral reefs are ocean ridges formed by marine invertebrates living in warm, shallow waters of the ocean. They are found in the photic zone and are important in shore protection. Other coral reef systems are fringing islands, which are directly adjacent to land, or atolls, which are circular reef systems surrounding a former landmass that is now underwater. The coral organisms are colonies of cnidarian polyps that secrete a calcium carbonate skeleton, which slowly accumulates, forming the underwater reef. Corals found in shallower waters have a mutually symbiotic relationship with photosynthetic unicellular algae, which provides corals with the majority of the nutrition and the energy they require. The waters are nutritionally poor; therefore, without this mutualism, it would not be possible for large corals to grow. In contrast, corals living in deeper and colder water attain energy and nutrients by capturing prey using stinging cells on their tentacles.

Global Decline of Coral Reefs

Climate change and human activity pose dual threats to the long-term survival of the world's coral reefs. Corals evolved to survive at the upper limit of ocean water temperature. Excessive warmth induced by climate change causes reefs to expel their symbiotic, food-producing algae, resulting in a phenomenon known as bleaching. When bleaching occurs, the reefs lose much of their characteristic color as the algae and the coral animals die if loss of the symbiotic zooxanthellae is prolonged.

44.4C: Estuaries: Where the Ocean Meets Fresh Water

Estuaries, composed of a mix of fresh and salt water and their living communities, are influenced by salinity and the changing tides.

Learning Objectives

- Explain the ecology of estuaries

Key Points

- Estuaries acts as nursery grounds for crustaceans, mollusks, and fish.
- Salinity, regulated by the influx of seawater and outflow of freshwater once or twice each day, is a determining factor in the types of organisms that can live there.
- To deal with the short-term and rapid variation in salinity, estuary species have developed specialized adaptations that enable them to live with the salty conditions; as a result, most plant species found in estuaries are halophytes.

Key Terms

- **brackish**: salty or slightly salty, as a mixture of fresh and sea water, such as that found in estuaries
- **halophyte**: any plant that tolerates an environment having a high salt content
- **estuary**: coastal water body where ocean tides and river water merge

Estuaries: Where the Ocean Meets Fresh Water

Estuaries form a unique marine biome that occurs where a source of fresh water, such as a river, meets the ocean. Therefore, both fresh water and salt water are found in the same vicinity. Mixing results in a diluted (brackish) saltwater. Estuaries form protected areas where many of the young offspring of crustaceans, mollusks, and fish begin their lives. Salinity of estuaries is a very important factor that influences the organisms found there and their adaptations. The salinity, which varies, is based on the rate of flow of its freshwater sources. Once or twice a day, high tides bring salt water into the estuary. Low tides, occurring at the same frequency, reverse the current of salt water.

Figure 44.4C.1: **Low tide**: The salinity levels within an estuary are dependent on the tides; during low tide, salt water influx levels are reduced.

The short-term and rapid variation in salinity due to the mixing of fresh water and salt water is a difficult physiological challenge for the plants and animals that inhabit estuaries. Many estuarine plant species are halophytes: plants that can tolerate salty water on their roots or sea spray. In some halophytes, filters in the roots remove the salt from the water that the plant absorbs. Other plants are able to pump oxygen into their roots. Animals, such as mussels and clams, have developed behavioral adaptations that expend a lot of energy to function in this rapidly-changing environment. When these animals are exposed to low salinity, they stop feeding, close their shells, and switch from aerobic respiration (in which they use gills) to anaerobic respiration (a process that does not require oxygen). When high tide returns to the estuary, the

salinity and oxygen content of the water increases, causing these animals to open their shells, begin feeding, and to return to aerobic respiration.

44.4D: Freshwater Biomes

Lakes, ponds, rivers, streams, and wetlands are all freshwater biomes, which differ in depth, water movement, and other abiotic factors.

Learning Objectives

- Differentiate among the freshwater biomes of lakes and ponds, rivers and streams, and wetlands

Key Points

- Temperature, as well as the availability of nitrogen and phosphorus, are factors that affect living things in lakes and ponds.
- When available in large amounts, nitrogen and phosphorus cause potentially-detrimental algal blooms in lakes; nitrogen is also a limiting factor for plant growth in bogs.
- The continuous movement of rivers and streams are their defining characteristic; these bodies of water carry large amounts of water from the source to a lake or ocean.
- Wetlands are shallow bodies of water with soil that is either permanently or periodically saturated with water; every type of wetland has three shared characteristics: their hydrology, hydrophytic vegetation, and hydric soils.

Key Terms

- **algal bloom**: a dense spread of algae on the surface of water
- **percolation**: the seepage or filtration of a liquid through a porous substance
- **hydrology**: the science of the properties, distribution, and effects of water on a planet's surface, in the soil and underlying rocks, and in the atmosphere

Freshwater Biomes

Freshwater biomes occur throughout the world's terrestrial biomes. They include lakes and ponds, rivers and streams, and wetlands.

Lakes and Ponds

Lakes and ponds can range in area from a few square meters to thousands of square kilometers. Temperature is an important abiotic factor affecting organisms found there. In the summer, thermal stratification of lakes and ponds occurs when the upper layer of water is warmed by the sun, but does not mix with deeper, cooler water. Light can penetrate within the photic zone of the lake or pond. Phytoplankton found here carry out photosynthesis, providing the base of the food web. At the bottom of lakes and ponds, bacteria in the aphotic zone break down dead organisms that sink to the bottom.

Nitrogen and phosphorus are important limiting nutrients. Because of this, they are determining factors in the amount of phytoplankton growth that occurs in lakes and ponds. When there is a large input of nitrogen and phosphorus (from sewage and run-off from fertilized lawns and farms, for example), the growth of algae skyrockets, resulting in a large accumulation called an algal bloom. These blooms can become so extensive that they reduce light penetration in water. As a result, the lake or pond becomes aphotic: photosynthetic plants cannot survive. When the algae die and decompose, severe oxygen depletion of the water occurs. Fish and other organisms that require oxygen are more likely to die. The resulting dead zones are found across the globe.

Figure 44.4D.1: **Algal blooms**: The uncontrolled growth of algae in this lake has resulted in an algal bloom.

Rivers and Streams

Rivers and streams are continuously moving bodies of water that carry large amounts of water from the source, or headwater, to a lake or ocean. Abiotic features of a river or stream vary along its length. The origin point of streams (source water) is usually cold, low in nutrients, and clear.

Because the source channel is narrow, the current is often faster here than at any other point of the river or stream. This fast-moving water results in minimal silt accumulation at the bottom of the river or stream, resulting in clear water. Photosynthesis occurs primarily in algae growing on rocks since the swift current in channels inhibits the growth of phytoplankton. An additional input of energy can come from leaves or other organic material that falls into the river or stream from trees and other plants that border the water. When the leaves decompose, the organic material and nutrients in the leaves are returned to the water. Plants and animals have adapted to this fast-moving water. For instance, leeches have elongated bodies and suckers on both ends that attach to the substrate, keeping the leech anchored in place.

As the river or stream flows away from the source, the width of the channel gradually widens and the current slows. This slow-moving water, caused by the gradient decrease and the volume increase as tributaries unite, has more sedimentation. The water is as clear as it is near the source since phytoplankton can be suspended in slow-moving water. The water is also warmer. Worms and insects can be found burrowing into the mud. The higher order predator vertebrates, which include waterfowl, frogs, and fishes, often depend on taste or chemical cues to find prey due to the murkiness of the water.

Wetlands

Wetlands are environments in which the soil is either permanently or periodically saturated with water. They differ from lakes in that they are shallow bodies of water. Emergent vegetation consists of wetland plants that are rooted in the soil, but have portions of leaves, stems, and flowers extending above the water's surface. Types of wetlands include marshes, swamps, bogs, mudflats, and salt marshes. The three shared characteristics among these types are their hydrology, hydrophytic vegetation, and hydric soils.

Figure 44.4D.1: **Wetlands**: Located in southern Florida, Everglades National Park is vast array of wetland environments, including sawgrass marshes, cypress swamps, and estuarine mangrove forests. Here, a great egret walks among cypress trees.

Freshwater marshes and swamps are characterized by slow and steady water flow. Bogs develop in depressions where water flow is low or non-existent. Bogs usually occur in areas where there is a clay bottom with poor percolation: the movement of water through the pores in the soil or rocks. The water found in a bog is stagnant and oxygen-depleted because the oxygen that is used during the decomposition of organic matter is not replaced, resulting in a slowing of decomposition. This leads to organic acids and other acids building up, which lower the pH of the water. At a lower pH, nitrogen becomes unavailable to plants, creating a challenge for them. Some types of bog plants capture insects and extract the nitrogen from their bodies.

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44.5: Climate and the Effects of Global Climate Change

Topic hierarchy

- 44.5A: Climate and Weather
- 44.5B: Causes of Global Climate Change
- 44.5C: Evidence of Global Climate Change
- 44.5D: Past and Present Effects of Climate Change

44.5A: Climate and Weather

Climate refers to long-term, predictable atmospheric conditions, while weather refers to atmospheric conditions during a short period of time.

Learning Objectives

- Distinguish between climate and weather

Key Points

- Climate refers to the long-term, predictable atmospheric conditions of a specific area; it does not address the amount of rain that fell on one particular day or the colder-than-average temperatures on a given day in a biome.
- Weather refers to the conditions of the atmosphere during a short period of time; weather forecasts are usually made for 48-hour cycles.
- Specific, one-off weather occurrences are not necessarily indicators of climate change.

Key Terms

- **biome**: any major regional biological community such as that of forest or desert
- **climate**: long-term manifestations of weather and other atmospheric conditions in a given area or country in a period long enough to ensure that representative values are obtained (generally 30 years)
- **weather**: the short term state of the atmosphere at a specific time and place, including the temperature, humidity, cloud cover, precipitation, wind, etc.

Climate and Weather

A common misconception about global climate change is that a specific weather event occurring in a particular region (for example, a very cool week in June in central Indiana) is evidence of global climate change. However, a cold week in June is a weather-related event and not a climate-related one. These misconceptions often arise because of confusion over the terms climate and weather.

Climate refers to the long-term, predictable atmospheric conditions of a specific area. The climate of a biome is characterized by having consistent temperature and annual rainfall ranges. Climate does not address the amount of rain that fell on one particular day in a biome or the colder-than-average temperatures that occurred on one day. In contrast, weather refers to the conditions of the atmosphere during a short period of time. Weather forecasts are usually made for 48-hour cycles; while long-range weather forecasts are available, they can be unreliable.

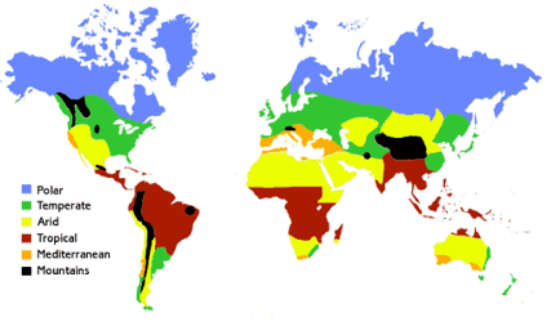

Figure 44.5*A*.1: **Climate**: Climate refers to long-term, predictable atmospheric conditions of a specific area. Weather, in contrast, refers to the conditions of the atmosphere during a short period of time. This map illustrates the various climate conditions around the world.

To better understand the difference between climate and weather, imagine that you are planning an outdoor event in northern Wisconsin. You would be thinking about climate when you plan the event in the summer rather than the winter because you have long-term knowledge that any given Saturday in the months of May to August would be a better choice for an outdoor event in Wisconsin than any given Saturday in January. However, you cannot determine the specific day that the event should be held because it is difficult to accurately predict the weather on a specific day. Climate can be considered "average" weather.

44.5B: Causes of Global Climate Change

Global climate change is cyclical and happens naturally; however, modern human society's impact has had unprecedented negative effects.

Learning Objectives

- Explain the drivers of climate change, past and present

Key Points

- Data show a correlation between the timing of temperature changes and drivers of climate change.
- Prior to the Industrial Era (pre-1780), there were three drivers of climate change that were not related to human activity: Milankovitch cycles, sun intensity, and volcanic eruptions.
- Greenhouse gases, probably the most significant drivers of the climate, include carbon dioxide, methane, water vapor, nitrous oxide, and ozone.
- Human activity, such as the burning of fossil fuels, releases carbon dioxide and methane, two of the most important greenhouse gases, into the atmosphere.
- Deforestation, cement manufacture, animal agriculture, the clearing of land, and the burning of forests are other human activities that release carbon dioxide.

Key Terms

- **Milankovitch cycle**: any of the three cyclic variations in the earth's orbit around the sun, respectively the obliquity of its axis, the precession of the equinoxes, and the eccentricity of its orbit
- **greenhouse gas**: any gas, such as carbon dioxide, that contributes to the greenhouse effect (continued warming) when released into the atmosphere
- **greenhouse effect**: the process by which a planet is warmed by its atmosphere

Current and Past Drivers of Global Climate Change

Since it is not possible to go back in time to directly observe and measure climate, scientists use indirect evidence to determine the drivers, or factors, that may be responsible for climate change. The indirect evidence includes data collected using ice cores, boreholes (narrow shafts bored into the ground), tree rings, glacier lengths, pollen remains, and ocean sediments. The data shows a correlation between the timing of temperature changes and drivers of climate change. Before the Industrial Era (pre-1780), there were three drivers of climate change that were not related to human activity or atmospheric gases: the Milankovitch cycles, solar intensity, and volcanic eruptions.

Natural Causes of Climate Change

The Milankovitch cycles describe how slight changes in the earth's orbit affect the earth's climate. The length of the Milankovitch cycles ranges between 19,000 and 100,000 years. In other words, one could expect to see some predictable changes in the earth's climate associated with changes in the earth's orbit at a minimum of every 19,000 years.

The variation in the sun's intensity is the second natural factor responsible for climate change. Solar intensity is the amount of solar power or energy the sun emits in a given length of time. There is a direct relationship between solar intensity and temperature: as solar intensity increases (or decreases), the earth's temperature correspondingly increases (or decreases). Changes in solar intensity have been proposed as one of several possible explanations for the Little Ice Age.

Finally, volcanic eruptions are a third natural driver of climate change. Volcanic eruptions can last a few days, but the solids and gases released during an eruption can influence the climate over a period of a few years, causing short-term climate changes. The gases and solids released by volcanic eruptions can include carbon dioxide, water vapor, sulfur dioxide, hydrogen sulfide, hydrogen, and carbon monoxide.

Human Activity-Related Causes of Climate Change

Greenhouse gases are probably the most significant drivers of the climate. When heat energy from the sun strikes the earth, gases known as greenhouse gases trap the heat in the atmosphere, similar to how the glass panes of a greenhouse keep heat

from escaping. The greenhouse gases that affect earth include carbon dioxide, methane, water vapor, nitrous oxide, and ozone. Approximately half of the radiation from the sun passes through these gases in the atmosphere, striking the earth. This radiation is converted into thermal radiation on the earth's surface; a portion of that energy is re-radiated into the atmosphere. Greenhouse gases, however, reflect much of the thermal energy back to the earth's surface. The more greenhouse gases there are in the atmosphere, the more thermal energy is reflected back to the earth's surface. Greenhouse gases, as they absorb and emit radiation, are an important factor in the greenhouse effect, or the warming of earth due to carbon dioxide and other greenhouse gases in the atmosphere.

Figure 44.5B.1: **Greenhouse gasses**: The burning of fossil fuels in industry and by vehicles releases carbon dioxide and other greenhouse gases into the atmosphere.

Beginning recently, atmospheric carbon dioxide concentrations have increased beyond the historical maximum of 300 ppm. The current increases in atmospheric carbon dioxide have happened very quickly: in a matter of hundreds of years rather than thousands of years. A key factor that must be recognized when comparing the historical data and the current data is the presence of modern human society. No other driver of climate change has yielded changes in atmospheric carbon dioxide levels at this rate or to this magnitude.

Human activity releases carbon dioxide and methane, two of the most important greenhouse gases, into the atmosphere in several ways. The primary mechanism that releases carbon dioxide is the burning of fossil fuels, such as gasoline, coal, and natural gas. Deforestation, cement manufacture, animal agriculture, the clearing of land, and the burning of forests are other human activities that release carbon dioxide. Methane (CH_4) is produced when bacteria break down organic matter under anaerobic conditions (i.e., without oxygen), which can happen when organic matter is trapped underwater, as in rice paddies, or in the intestines of herbivores. Methane can also be released from natural gas fields and the decomposition that occurs in landfills. Another source of methane is the melting of clathrates: frozen chunks of ice and methane found at the bottom of the ocean. When water warms, these chunks of ice melt, releasing methane. As the ocean's water temperature increases, the rate at which clathrates melt is increasing, releasing even more methane. This leads to increased levels of methane in the atmosphere, which further accelerates the rate of global warming. This is an example of the positive feedback loop that is leading to the rapid rate of increase of global temperatures.

44.5C: Evidence of Global Climate Change

Global climate change can be understood by analyzing past historical climate data, such as atmospheric CO_2 concentrations in ice cores.

Learning Objectives

- Evaluate the evidence for global climate change

Key Points

- Climate change can be understood by approaching three areas of study: (1) current and past global climate change, (2) causes of past and present-day global climate change, and (3) ancient and current results of climate change.
- Since we cannot go back in time to directly measure climatic variables, such as average temperature and precipitation, we must rely on historical evidence of earth's past climate, such as Antarctic ice cores.
- Three significant temperature anomalies, or irregularities, have occurred in the last 2000 years: the Medieval Climate Anomaly (or the Medieval Warm Period), the Little Ice Age, and the Industrial Era.
- With the beginning of the Industrial Era, atmospheric carbon dioxide began to rise.

Key Terms

- **fossil fuel**: any fuel derived from hydrocarbon deposits such as coal, petroleum, natural gas, and, to some extent, peat; these fuels are irreplaceable; their burning generates the greenhouse gas carbon dioxide

Global Climate Change

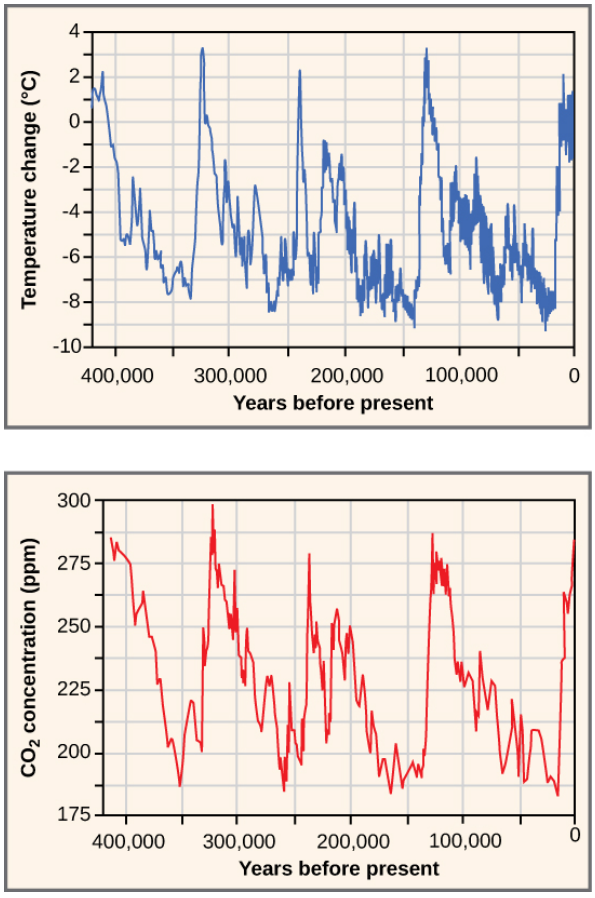

Figure 44.5C.1: **Measuring historical climate change**: By measuring the amount of CO2 trapped in ice, scientists can determine past atmospheric CO2 concentrations. Temperatures relative to the present are determined from the amount of deuterium (an isotope of hydrogen) present.

Climate change can be understood by approaching three areas of study: (1) evidence of current and past global climate change, (2) causes of past and present-day global climate change, and (3) ancient and current results of climate change.

It is helpful to keep these three different aspects of climate change clearly separated when consuming media reports about global climate change. It is common for reports and discussions about global climate change to confuse the data showing that earth's climate is changing with the factors that drive this climate change.

Evidence for Global Climate Change

Since scientists cannot go back in time to directly measure climatic variables, such as average temperature and precipitation, they must, instead, indirectly measure temperature. To do this, scientists rely on historical evidence of earth's past climates.

Antarctic ice cores are a key example of such evidence. These ice cores are samples of polar ice obtained by means of drills that reach thousands of meters into ice sheets or high mountain glaciers. Viewing the ice cores is like traveling backwards through time; the deeper the sample, the earlier the time period. Trapped within the ice are bubbles of air and other biological evidence that can reveal temperature and carbon dioxide data. Antarctic ice cores have been collected and analyzed to indirectly estimate the temperature of the earth over the past 400,000 years.

Before the late 1800s, the earth had been as much as 9°C cooler and about 3°C warmer. Atmospheric concentration of carbon dioxide also rose and fell in periodic cycles; note the relationship between carbon dioxide concentration and temperature. Carbon dioxide levels in the atmosphere have historically cycled between 180 and 300 parts per million (ppm) by volume.

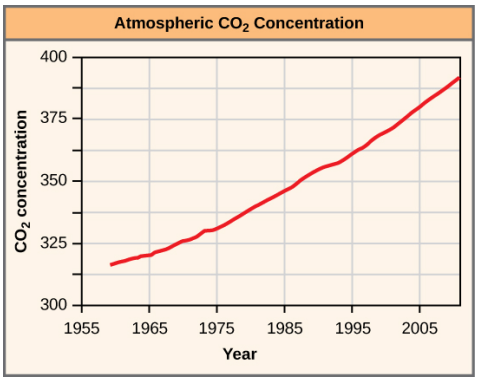

Figure 44.5C.1: **Climate change and the Industrial Era**: The atmospheric concentration of CO2 has risen steadily since the beginning of industrialization.The Industrial Revolution, which began around 1750, was characterized by changes in much of human society. Advances in agriculture increased the food supply, which improved the standard of living for people in Europe and the United States. New technologies were invented, providing jobs and cheaper goods. These new technologies were powered using fossil fuels, especially coal. The Industrial Revolution in the early nineteenth century ushered in the beginning of the Industrial Era, a period when atmospheric carbon dioxide began to rise. When a fossil fuel is burned, carbon dioxide is released.

Two significant temperature anomalies, or irregularities, have occurred in the last 2000 years. These are the Medieval Climate Anomaly (or the Medieval Warm Period) and the Little Ice Age. A third temperature anomaly aligns with the Industrial Era. The Medieval Climate Anomaly occurred between 900 and 1300 AD. During this time period, many climate scientists think that slightly-warmer conditions prevailed in many parts of the world; the higher-than-average temperature changes varied between 0.10 °C and 0.20 °C above the norm. Although 0.10 °C does not seem large enough to produce any noticeable change, it did free seas of ice. Because of this warming, the Vikings were able to colonize Greenland.

The Little Ice Age was a cold period that occurred between 1550 AD and 1850 AD. During this time, a slight cooling of a little less than 1 °C was observed in North America, Europe, and possibly other areas of the world. This 1 °C change is a seemingly-small deviation in temperature (as was observed during the Medieval Climate Anomaly); however, it also resulted in noticeable changes. Historical accounts reveal a time of exceptionally-harsh winters with much snow and frost.

44.5D: Past and Present Effects of Climate Change

Results of climate change, past and present, have been documented and include species extinction, rising sea levels, and effects on organisms.

Learning Objectives

- Describe the effects of current and geological climate change

Key Points

- Global warming has been associated with at least one planet-wide extinction event during the geological past; scientists estimate that during the Permian period, approximately 70 percent of the terrestrial plant and animal species along with 84 percent of marine species became extinct.
- Glacier recession and melting ice caps are direct effects of current global climate change, ultimately leading to higher global sea levels; as glaciers and polar ice caps melt, there is a significant contribution of liquid water that was previously frozen.
- Changes in temperature and precipitation are causing plants to flower earlier, before their insect pollinators have emerged; mismatched timing of plants and pollinators could result in injurious ecosystem effects.
- This mismatched timing of plants and pollinators could result in injurious ecosystem effects because, for continued survival, insect-pollinated plants must flower when their pollinators are present.

Key Terms

- **phenology**: the study of the effect of climate on periodic biological phenomena
- **Permian**: of a geologic period within the Paleozoic era; comprises the Cisuralian, Guadalupian, and Lopingian epochs from about 280 to 248 million years ago

Documented results of climate change: past and present

Scientists have geological evidence of the consequences of long-ago climate change. Modern-day phenomena, such as retreating glaciers and melting polar ice, cause a continual rise in sea level. Changes in climate can negatively affect organisms.

Geological climate change effects

Global warming has been associated with at least one planet-wide extinction event during the geological past. The Permian extinction event occurred about 251 million years ago toward the end of the roughly 50-million-year-long geological time span known as the Permian period. This geologic time period was one of the three warmest periods in earth's geologic history. Scientists estimate that approximately 70 percent of the terrestrial plant and animal species and 84 percent of marine species became extinct, vanishing forever near the end of the Permian period. Organisms that had adapted to wet and warm climatic conditions, such as annual rainfall of 300–400 cm (118–157 in) and 20 °C–30 °C (68 °F–86 °F) in the tropical wet forest, may not have been able to survive the Permian climate change.

Present climate change effects

A number of global events have occurred that may be attributed to recent climate change during our lifetimes. Glacier National Park in Montana, among others, is undergoing the retreat of many of its glaciers, a phenomenon known as glacier recession. In 1850, the area contained approximately 150 glaciers. By 2010, however, the park contained only about 24 glaciers greater than 25 acres in size. One of these glaciers is the Grinnell Glacier at Mount Gould. Between 1966 and 2005, the size of Grinnell Glacier shrank by 40 percent. Similarly, the mass of the ice sheets in Greenland and the Antarctic is decreasing: Greenland lost 150–250 km^3 of ice per year between 2002 and 2006. In addition, the size and thickness of the Arctic sea ice is decreasing.

Figure 44.5D.1: **Glacial recession**: The effect of global warming can be seen in the continuing retreat of Grinnel Glacier. The loss of a glacier results in the loss of summer meltwaters, sharply reducing seasonal water supplies and severely affecting local ecosystems.

This loss of ice is leading to rises in the global sea level. On average, the sea is rising at a rate of 1.8 mm per year. However, between 1993 and 2010, the rate of sea-level increase ranged between 2.9 and 3.4 mm per year. A variety of factors affect the volume of water in the ocean, including the temperature of the water (the density of water is related to its temperature) and the amount of water found in rivers, lakes, glaciers, polar ice caps, and sea ice. As glaciers and polar ice caps melt, there is a significant contribution of liquid water that was previously frozen.

In addition to some abiotic conditions changing in response to climate change, many organisms are also being affected by the changes in temperature. Temperature and precipitation play key roles in determining the geographic distribution and phenology of plants and animals. Phenology is the study of the effects of climatic conditions on the timing of periodic lifecycle events, such as flowering in plants or migration in birds. Researchers have shown that 385 plant species in Great Britain are flowering 4.5 days sooner than was recorded during the previous 40 years. In addition, insect-pollinated species were more likely to flower earlier than wind-pollinated species. The impact of changes in flowering date would be mitigated if the insect pollinators emerged earlier. This mismatched timing of plants and pollinators could result in injurious ecosystem effects because, for continued survival, insect-pollinated plants must flower when their pollinators are present.

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CHAPTER OVERVIEW

45: POPULATION AND COMMUNITY ECOLOGY

45.1: POPULATION DEMOGRAPHY

45.1A: POPULATION DEMOGRAPHY
45.1B: POPULATION SIZE AND DENSITY
45.1C: SPECIES DISTRIBUTION
45.1D: THE STUDY OF POPULATION DYNAMICS
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Logistic growth of a population size occurs when resources are limited, thereby setting a maximum number an environment can support.

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45.7A: SIMPLE LEARNED BEHAVIORS
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Cognitive learning relies on cognitive processes such as reasoning and abstract thinking; it is much more efficient than conditioning.

45.1: Population Demography

Topic hierarchy

- 45.1A: Population Demography
- 45.1B: Population Size and Density
- 45.1C: Species Distribution
- 45.1D: The Study of Population Dynamics

45.1A: Population Demography

Demography is the study of population dynamics, using statistical and mathematical tools.

Learning Objectives

- Explain the importance and function of population demography

Key Points

- Demographic studies help scientists understand the population dynamics of species, such as invasive species like the Asian carp.
- Population fluctuations depend on the weather, food availability, natural disasters such as forest fires or volcanic eruptions, predation, and biological competition.
- Researchers originally designed demographic tools to study human populations, but demographic approaches can be applied to all living populations.

Key Terms

- **demography**: the study of populations and how they change
- **population dynamics**: the variation of populations due to birth and death rates, by immigration and emigration, and concerning topics such as aging populations or population decline
- **statistics**: a systematic collection of data on measurements or observations, often related to demographic information such as population counts, incomes, population counts at different ages, etc.

Introduction

Imagine someone sailing down a river in a small motorboat on a warm day. She is enjoying the warm sunshine and cool breeze when suddenly a 20-pound silver carp hits her in the head. The presence of Asian carp in U.S. waterways make this risk very real on rivers and canal systems, particularly in Illinois and Missouri.

Figure 45.1*A*.1: **Asian carp jump out of the water in response to electrofishing**: The Asian carp in the inset photograph were harvested from the Little Calumet River in Illinois in May, 2010, using rotenone, also used as an insecticide.

The several types of Asian carp include the silver, black, grass, and big head carp. These have been farmed and eaten in China for over 1,000 years; it is among the top aquaculture foods worldwide. In the United States, however, Asian carp is considered to be an invasive species. It disrupts the structure and composition of native fish communities to the point of threatening native aquatic species.

Biologists are working to understand the biology and ecology of Asian carp. The aim is to develop methods of controlling the species without damaging native fish. Understanding the population dynamics of the carp will help biologists develop and implement measures that reduce its population, allowing scientists to model the statistics of carp populations.

Population demography

Populations are dynamic entities, consisting of all of the species living within a specific area. They fluctuate based on a number of factors: seasonal and yearly changes in the environment, natural disasters such as forest fires and volcanic eruptions, and competition for resources between and within species.

Demography, the statistical study of population dynamics, uses mathematical tools to investigate how populations respond to changes in their biotic and abiotic environments. Researchers originally designed demographic tools, such as life tables, to study human populations. To determine insurance rates, life insurance companies developed methods to analyze life expectancies of individuals in a population. The term “demographics” is often used in discussions of human populations, but demographic approaches can be applied to all living populations.

45.1B: Population Size and Density

Scientists study population size and density using a variety of field sampling methods, including quadrats and mark-recapture.

Learning Objectives

- Choose the appropriate method to sample a population, given features of the organisms in that population

Key Points

- A population ‘s size refers to the number of individuals (N) it comprises.
- Population density is the number of individuals within a given area or volume.
- Scientists usually study populations by sampling, which involves counting individuals within a certain area or volume that is part of the population’s habitat.
- The quadrat method is used to sample sessile organisms, using a square within which all individuals are counted; extrapolation of the data to the entire habitat results in a population size estimate.
- The mark and recapture technique is used for mobile organisms; it involves marking a sample of individuals and then estimating population size from the number of marked individuals in subsequent samples.

Key Terms

- **population density**: the average number of a population’s individuals that inhabit a unit area or volume
- **quadrat**: a square area, marked with boundaries for studying the population size and density of plants and sessile animals
- **mark and recapture**: a sample technique is used for study of the populations of mobile organisms, estimating population size from the number of marked individuals in samples

Population size and density

Population size and density are the two most important statistics scientists use to describe and understand populations. A population’s size refers to the number of individuals (N) it comprises. Its density is the number of individuals within a given area or volume. These data allow scientists to model the fluctuations of a population over time. For example, a larger population may be more stable than a smaller population. With less genetic variation, a smaller population will have reduced capacity to adapt to environmental changes. Individuals in a low-density population are thinly dispersed; hence, they may have more difficulty finding a mate compared to individuals in a higher-density population. On the other hand, high-density populations often result in increased competition for food. Many factors influence density, but, as a rule-of-thumb, smaller organisms have higher population densities than do larger organisms.

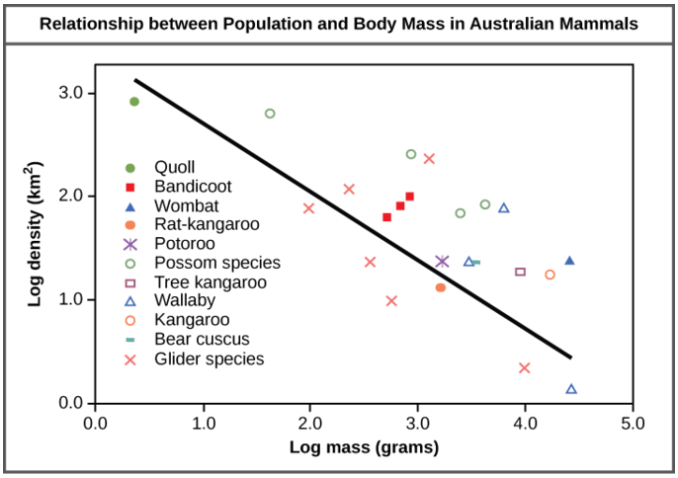

Figure 45.1B.1: **Population density is negatively correlated with body size**: Australian mammals show a typical inverse relationship between population density and body size.

Population research methods

Counting all individuals in a population is the most accurate way to determine its size. However, this approach is not usually feasible, especially for large populations or extensive habitats. Instead, scientists study populations by sampling, which involves counting individuals within a certain area or volume that is part of the population's habitat. Analyses of sample data enable scientists to infer population size and population density about the entire population.

A variety of methods can be used to sample populations. Scientists usually estimate the populations of sessile or slow-moving organisms with the quadrat method. A quadrat is a square that encloses an area within a habitat. The area may be defined by staking it out with sticks and string, or using a square made of wood, plastic, or metal placed on the ground.

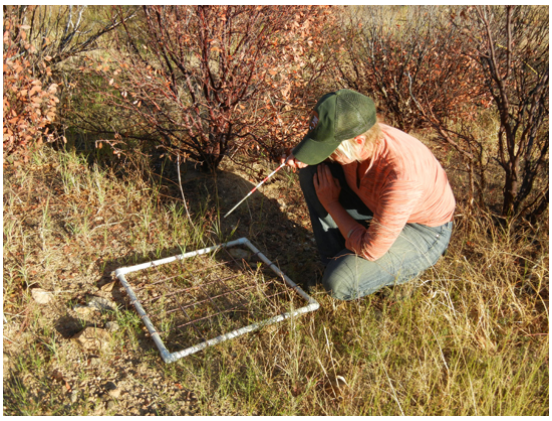

Figure 45.1B.1: **Scientist uses a quadrat to measure plant population size and density**: A quadrat is a square frame of known area in which species of interest can be easily counted and measured.

A field study usually includes several quadrat samples at random locations or along a transect in representative habitat. After they place the quadrats, researchers count the number of individuals that lie within the quadrat boundaries. The researcher decides the quadrat size and number of samples from the type of organism, its spatial distribution, and other factors. For sampling daffodils, a 1 m^2 quadrat could be appropriate. Giant redwoods are larger and live further apart from each other, so a larger quadrat, such as 100 m^2, would be necessary. The correct quadrat size ensures counts of enough individuals to get a sample representative of the entire habitat.

Scientists typically use the mark and recapture technique for mobile organisms such as mammals, birds, or fish. With this method, researchers capture animals and mark them with tags, bands, paint, body markings, or some other sign. The marked animals are then released back into their environment where they mix with the rest of the population. Later, a new sample is collected, including some individuals that are marked (recaptures) and some individuals that are unmarked.

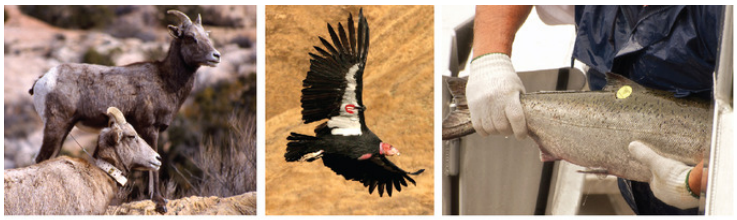

Figure 45.1B.1: **Mark and recapture is used to measure the population size of mobile animals.**: With the mark and recapture method, researchers capture animals and mark them with tags, bands, paint, body markings or some other sign.

The ratio of marked to unmarked individuals allows scientists to calculate how many individuals are in the population as an estimate of total population size. This method assumes that the larger the population, the lower the percentage of tagged organisms that will be recaptured since they will have mixed with more untagged individuals. For example, if 80 deer are captured, tagged, and released into the forest, and later 100 deer are captured with 20 of them are already marked, we can determine the population size (N) using the following equation:

numbermarkedinfirstcatch×totalnumberofsecondcatchnumberofmarkedrecapturesinsecondcatchnumbermarkedinfirstcatch ×totalnumberofsecondcatchnumberofmarkedrecapturesinsecondcatch

Plugging the example data into the equation, the calculation gives an estimated total population size of 400.

80×10020=40080×10020=400

Using the example data, if only 10 already-marked deer had been recaptured, the calculated total population size would be 800.

The mark and recapture method has limitations. Some animals from the first catch may learn to avoid capture in the second round. Such behavior would cause inflated population estimates. Alternatively, animals may preferentially be retrapped (especially if a food reward is offered), resulting in an underestimate of population size. Also, some species may be harmed by the marking technique, reducing their survival. A variety of other techniques have been developed, including the electronic tracking of animals tagged with radio transmitters and the use of data from commercial fishing and trapping operations to estimate the size and health of populations and communities.

45.1C: Species Distribution

Scientists gain insight into a species' biology and ecology from studying spatial distribution of individuals.

Learning Objectives

- Differentiate among the ways in which species distribute themselves in space

Key Points

- Dispersion or distribution patterns show the spatial relationship between members of a population within a habitat.
- Individuals of a population can be distributed in one of three basic patterns: uniform, random, or clumped.
- In a uniform distribution, individuals are equally spaced apart, as seen in negative allelopathy where chemicals kill off plants surrounding sages.
- In a random distribution, individuals are spaced at unpredictable distances from each other, as seen among plants that have wind-dispersed seeds.
- In a clumped distribution, individuals are grouped together, as seen among elephants at a watering hole.

Key Terms

- **dispersion pattern**: the spatial relationship between members of a population within a habitat, often characteristic of a particular species
- **allelopathy**: the release by a plant of a toxin to suppress growth of nearby competing plants, often resulting in a uniform dispersion pattern
- **habitat heterogeneity**: variation in physical environmental features within an area, such as topography, soil chemistry, temperature, moisture, and biological factors

Species distribution

Density and size are useful measures for characterizing populations. Scientists gain additional insight into a species' biology and ecology from studying how individuals are spatially distributed. Dispersion or distribution patterns show the spatial relationship between members of a population within a habitat. Patterns are often characteristic of a particular species; they depend on local environmental conditions and the species' growth characteristics (as for plants) or behavior (as for animals).

Individuals of a population can be distributed in one of three basic patterns: they can be more or less equally spaced apart (uniform dispersion), dispersed randomly with no predictable pattern (random dispersion), or clustered in groups (clumped dispersion).

Figure 45.1C.1: **Three patterns of distribution in populations of organisms**: A population may have a uniform, random, or clumped distribution. Territorial birds, such as penguins, tend to have uniform distribution. Plants with wind-dispersed seeds, such as dandelions, are usually distributed randomly. Animals, such as elephants, that travel in groups exhibit clumped distribution.

Uniform dispersion is observed in plant species that inhibit the growth of nearby individuals. For example, the sage plant, *Salvia leucophylla*, secretes toxins, a phenomenon called negative allelopathy. The chemicals kill off surrounding plants in a circle around the individual sage plants, leading to a uniform distance between each plant. Animals that maintain defined territories, such as nesting penguins, also exhibit uniform dispersion.

Random dispersion occurs with dandelion and other plants that have wind-dispersed seeds that germinate wherever they happen to fall in a favorable environment. Clumped dispersion is seen in plants that drop their seeds straight to the ground, such as oak trees, or animals that live in groups, such as schools of fish or herds of elephants. Clumped dispersions may also result from habitat heterogeneity. If favorable conditions are localized, organisms will tend to clump around those, such as lions around a watering hole.

In this way, the dispersion pattern of the individuals within a population provides more information about how they interact with each other and their environment than does a simple density measurement. Just as lower density species might have more difficulty finding a mate, solitary species with a random distribution might have a similar difficulty when compared to social species clumped together in groups.

45.1D: The Study of Population Dynamics

Demography, or the study of population dynamics, is studied using tools such as life tables and survivorship curves.

Learning Objectives

- Distinguish between life tables and survivorship curves as used in demography

Key Points

- The key statistics used in demography are birth rates, death rates, and life expectancies which may be influenced by population characteristics and biological factors.
- Birth rates, death rates, and life expectancies are the basic determinants of how a population changes over time.
- Life tables are demographic tools which shows a population's life expectancy and mortality within age groups.
- A survivorship curve is a graph of the number of individuals surviving at each age interval plotted versus time.
- The characteristics and behavior of a species, such as number of offspring produced, its percentage of surviving offspring, and degree of parental care, determine the shape of its survivorship curve.

Key Terms

- **life table**: a demographic tool which shows a population's life expectancy and mortality within age groups
- **survivorship curve**: a graph of the number of individuals surviving at each age interval plotted versus time
- **mortality rate**: the number of deaths per given unit of population over a given period of time

Demography

Population size, density, and distribution patterns describe a population at a fixed point in time. To study how a population changes over time, scientists must use the tools of demography: the statistical study of population changes over time. The key statistics demographers use are birth rates, death rates, and life expectancies; although, in practice, scientists also study immigration and emigration rates, which also affect populations.

These measures, especially birth rates, may be related to the population characteristics described in prior sections. For example, a large population would have a relatively-high birth rate if it has more reproductive individuals. Alternatively, a large population may also have a high death rate because of competition, disease, or waste accumulation. A high population density may lead to more reproductive encounters between individuals, as would a clumped distribution pattern. Such conditions would increase the birth rate.

Biological features of the population also affect population changes over time. Birth rates will be higher in a population with the ratio of males to females biased towards females, or in a population composed of relatively more individuals of reproductive age.

The demographic characteristics of a population are the basic determinants of how the population changes over time. If birth and death rates are equal, the population remains stable. The population will increase if birth rates exceed death rates, but will decrease if birth rates are lower than death rates. Life expectancy, another important factor, is the length of time individuals remain in the population. It is impacted by local resources, reproduction, and the overall health of the population. These demographic characteristics are often displayed in the form of a life table.

Life tables

Life tables, which provide important information about the life history of an organism, divide the population into age groups and often sexes; they show how long a member of that group will probably live. The tables are modeled after actuarial tables used by the insurance industry for estimating human life expectancy. Life tables may include:

- the probability of individuals dying before their next birthday (i.e., mortality rate)
- the percentage of surviving individuals at a particular age interval
- the life expectancy at each interval

The life table shown is from a study of Dall mountain sheep, a species native to northwestern North America. The population is divided into age intervals, as seen in the leftmost column. The mortality rate per 1,000 individuals is

calculated by dividing the number of individuals dying during an age interval by the number of individuals surviving at the beginning of the interval, multiplied by 1,000.

Life Table of Dall Mountain Sheep[1]				
Age interval (years)	Number dying in age interval out of 1000 born	Number surviving at beginning of age interval out of 1000 born	Mortality rate per 1000 alive at beginning of age interval	Life expectancy or mean lifetime remaining to those attaining age interval
0-0.5	54	1000	54.0	7.06
0.5-1	145	946	153.3	--
1-2	12	801	15.0	7.7
2-3	13	789	16.5	6.8
3-4	12	776	15.5	5.9
4-5	30	764	39.3	5.0
5-6	46	734	62.7	4.2
6-7	48	688	69.8	3.4
7-8	69	640	107.8	2.6
8-9	132	571	231.2	1.9
9-10	187	439	426.0	1.3
10-11	156	252	619.0	0.9
11-12	90	96	937.5	0.6
12-13	3	6	500.0	1.2
13-14	3	3	1000	0.7

Table 1: This life table of *Ovis dalli* shows the number of deaths, number of survivors, mortality rate, and life expectancy at each age interval for the Dall mountain sheep.

Figure 45.1*D*. 1: **Life table of Dall mountain sheep**: This life table of *Ovis dalli* shows the number of deaths, number of survivors, mortality rate, and life expectancy at each age interval for the Dall mountain sheep.

For example, between ages three and four, 12 individuals die out of the 776 that were remaining from the original 1,000 sheep. This number is then multiplied by 1,000 to get the mortality rate per thousand.

As can be seen from the mortality rate data (column D), a high death rate occurred when the sheep were between 6 and 12 months old, which then increased even more from 8 to 12 years old, after which there were few survivors. The data indicate that if a sheep in this population were to survive to age one, it could be expected to live another 7.7 years on average, as shown by the life expectancy numbers in column E.

Survivorship curves

Another tool used by population ecologists is a survivorship curve, which is a graph of the number of individuals surviving at each age interval plotted versus time (usually with data compiled from a life table). These curves allow comparison of life histories of different populations.

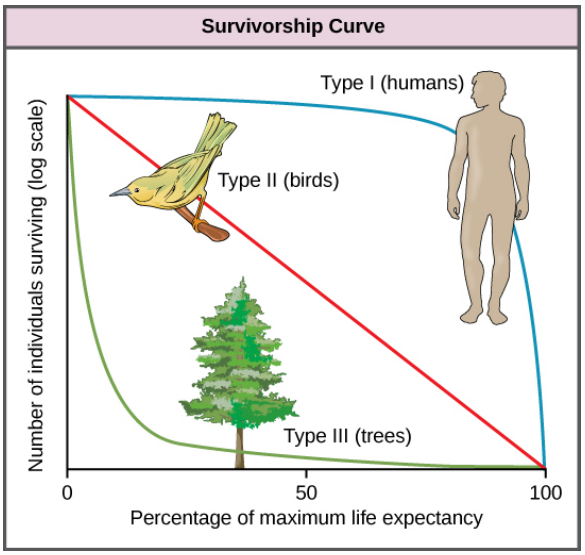

Figure 45.1D.1: **Survivorship curves show the distribution of individuals in a population according to age**: Humans and most mammals have a Type I survivorship curve because death primarily occurs in the older years. Birds have a Type II survivorship curve, as death at any age is equally probable. Trees have a Type III survivorship curve because very few survive the younger years, but after a certain age, individuals are much more likely to survive.

Humans and most primates exhibit a Type I survivorship curve because a high percentage of offspring survive early and middle years; death occurs predominantly in older individuals. These species have few offspring as they invest in parental care to increase survival.

Birds show the Type II survivorship curve because equal numbers of birds tend to die at each age interval. These species may also have relatively-few offspring and provide significant parental care.

Trees, marine invertebrates, and most fishes exhibit a Type III survivorship curve. Very few individuals survive the younger years; however, those that live to old age are likely to survive for a relatively-long period. Organisms in this category usually have large numbers of offspring and provide little parental care. Such offspring are “on their own” and suffer high mortality due to predation or starvation; however, their abundance ensures that enough individuals survive to the next generation, perpetuating the population.

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45.2: Environmental Limits to Population Growth

Topic hierarchy

- 45.2A: Exponential Population Growth
- 45.2B: Logistic Population Growth

Logistic growth of a population size occurs when resources are limited, thereby setting a maximum number an environment can support.

- 45.2C: Density-Dependent and Density-Independent Population Regulation

45.2A: Exponential Population Growth

When resources are unlimited, a population can experience exponential growth, where its size increases at a greater and greater rate.

Learning Objectives

- Describe exponential growth of a population size

Key Points

- To get an accurate growth rate of a population, the number that died in the time period (death rate) must be removed from the number born during the same time period (birth rate).
- When the birth rate and death rate are expressed in a per capita manner, they must be multiplied by the population to determine the number of births and deaths.
- Ecologists are usually interested in the changes in a population at either a particular point in time or over a small time interval.
- The intrinsic rate of increase is the difference between birth and death rates; it can be positive, indicating a growing population; negative, indicating a shrinking population; or zero, indicting no change in the population.
- Different species have a different intrinsic rate of increase which, when under ideal conditions, represents the biotic potential or maximal growth rate for a species.

Key Terms

- **fission**: the process by which a bacterium splits to form two daughter cells
- **per capita**: per person or individual

Exponential growth

In his theory of natural selection, Charles Darwin was greatly influenced by the English clergyman Thomas Malthus. Malthus published a book in 1798 stating that populations with unlimited natural resources grow very rapidly, after which population growth decreases as resources become depleted. This accelerating pattern of increasing population size is called exponential growth.

The best example of exponential growth is seen in bacteria. Bacteria are prokaryotes that reproduce by prokaryotic fission. This division takes about an hour for many bacterial species. If 1000 bacteria are placed in a large flask with an unlimited supply of nutrients (so the nutrients will not become depleted), after an hour there will be one round of division (with each organism dividing once), resulting in 2000 organisms. In another hour, each of the 2000 organisms will double, producing 4000; after the third hour, there should be 8000 bacteria in the flask; and so on. The important concept of exponential growth is that the population growth rate, the number of organisms added in each reproductive generation, is accelerating; that is, it is increasing at a greater and greater rate. After 1 day and 24 of these cycles, the population would have increased from 1000 to more than 16 billion. When the population size, N, is plotted over time, a J-shaped growth curve is produced.

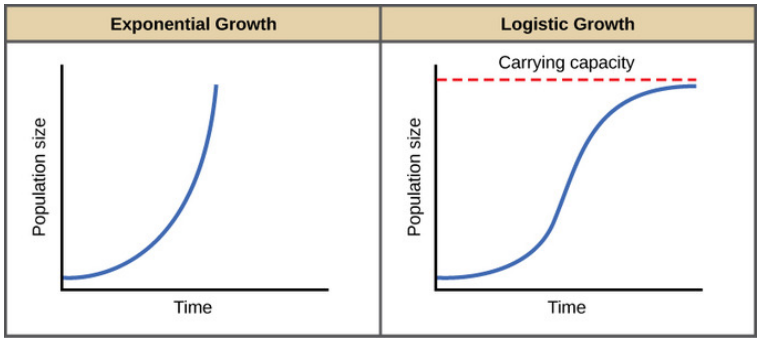

Figure 45.2A.1: **Exponential population growth**: When resources are unlimited, populations exhibit exponential growth, resulting in a J-shaped curve. When resources are limited, populations exhibit logistic growth. In logistic growth, population expansion decreases as resources become scarce. It levels off when the carrying capacity of the environment is reached, resulting in an S-shaped curve.

The bacteria example is not representative of the real world where resources are limited. Furthermore, some bacteria will die during the experiment and, thus, not reproduce, lowering the growth rate. Therefore, when calculating the growth rate of a population, the death rate (D; the number organisms that die during a particular time interval) is subtracted from the birth rate (B; the number organisms that are born during that interval). This is shown in the following formula:

ΔN/ΔT=B−DΔN/ΔT=B−D

where ΔNΔN = change in number, ΔTΔT = change in time, BB = birth rate, and DD = death rate. The birth rate is usually expressed on a per capita (for each individual) basis. Thus, B (birth rate) = bN (the per capita birth rate "b" multiplied by the number of individuals "N") and D (death rate) = dN (the per capita death rate "d" multiplied by the number of individuals "N"). Additionally, ecologists are interested in the population at a particular point in time: an infinitely small time interval. For this reason, the terminology of differential calculus is used to obtain the "instantaneous" growth rate, replacing the change in number and time with an instant-specific measurement of number and time.

dN/dT=BN DN=(BD)NdN/dT=BN DN=(BD)N

Notice that the "d" associated with the first term refers to the derivative (as the term is used in calculus) and is different from the death rate, also called "d." The difference between birth and death rates is further simplified by substituting the term "r" (intrinsic rate of increase) for the relationship between birth and death rates:

dN/dT=rNdN/dT=rN

The value "r" can be positive, meaning the population is increasing in size; negative, meaning the population is decreasing in size; or zero, where the population's size is unchanging, a condition known as zero population growth. A further refinement of the formula recognizes that different species have inherent differences in their intrinsic rate of increase (often thought of as the potential for reproduction), even under ideal conditions. Obviously, a bacterium can reproduce more rapidly and have a higher intrinsic rate of growth than a human. The maximal growth rate for a species is its biotic potential, or r_{max}, thus changing the equation to:

dN/dT=rmaxN

45.2B: Logistic Population Growth

LEARNING OBJECTIVES

- Describe logistic growth of a population size

Exponential growth is possible only when infinite natural resources are available; this is not the case in the real world. Charles Darwin recognized this fact in his description of the “struggle for existence,” which states that individuals will compete (with members of their own or other species) for limited resources. The successful ones will survive to pass on their own characteristics and traits (which we know now are transferred by genes) to the next generation at a greater rate: a process known as natural selection. To model the reality of limited resources, population ecologists developed the logistic growth model.

Carrying Capacity and the Logistic Model

In the real world, with its limited resources, exponential growth cannot continue indefinitely. Exponential growth may occur in environments where there are few individuals and plentiful resources, but when the number of individuals becomes large enough, resources will be depleted, slowing the growth rate. Eventually, the growth rate will plateau or level off. This population size, which represents the maximum population size that a particular environment can support, is called the **carrying capacity**, or K.

The formula we use to calculate logistic growth adds the carrying capacity as a moderating force in the growth rate. The expression “K – N” is indicative of how many individuals may be added to a population at a given stage, and “K – N” divided by “K” is the fraction of the carrying capacity available for further growth. Thus, the exponential growth model is restricted by this factor to generate the logistic growth equation:

$$\begin{aligned}\frac{dN}{dT} &= r_{max}\frac{dN}{dT} \\ &= r_{max} \times N \times (\frac{K-N}{K})\frac{dN}{dT} \\ &= rmax * (dN/dT) = rmax * N * ((KN)/K)\end{aligned}$$

Notice that when N is very small, (K-N)/K becomes close to K/K or 1; the right side of the equation reduces to $r_{max}N$, which means the population is growing exponentially and is not influenced by carrying capacity. On the other hand, when N is large, $(K-N)/K$ come close to zero, which means that population growth will be slowed greatly or even stopped. Thus, population growth is greatly slowed in large populations by the carrying capacity K. This model also allows for negative population growth or a population decline. This occurs when the number of individuals in the population exceeds the carrying capacity (because the value of (K-N)/K is negative).

A graph of this equation yields an S-shaped curve; it is a more-realistic model of population growth than exponential growth. There are three different sections to an S-shaped curve. Initially, growth is exponential because there are few individuals and ample resources available. Then, as resources begin to become limited, the growth rate decreases. Finally, growth levels off at the carrying capacity of the environment, with little change in population size over time.

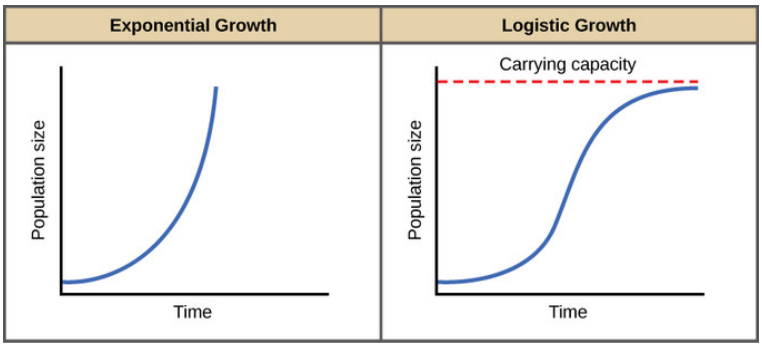

Figure $45.2B.1$: **Exponential and logistical population growth**: When resources are unlimited, populations exhibit exponential growth, resulting in a J-shaped curve. When resources are limited, populations exhibit logistic growth. In logistic growth, population expansion decreases as resources become scarce, leveling off when the carrying capacity of the environment is reached, resulting in an S-shaped curve.

Role of Intraspecific Competition

The logistic model assumes that every individual within a population will have equal access to resources and, thus, an equal chance for survival. For plants, the amount of water, sunlight, nutrients, and the space to grow are the important resources, whereas in animals, important resources include food, water, shelter, nesting space, and mates.

In the real world, the variation of phenotypes among individuals within a population means that some individuals will be better adapted to their environment than others. The resulting competition between population members of the same species for resources is termed intraspecific competition (intra- = “within”; -specific = “species”). Intraspecific competition for resources may not affect populations that are well below their carrying capacity as resources are plentiful and all individuals can obtain what they need. However, as population size increases, this competition intensifies. In addition, the accumulation of waste products can reduce an environment’s carrying capacity.

Examples of Logistic Growth

Yeast, a microscopic fungus used to make bread and alcoholic beverages, exhibits the classical S-shaped curve when grown in a test tube (a). Its growth levels off as the population depletes the nutrients that are necessary for its growth. In the real world, however, there are variations to this idealized curve. Examples in wild populations include sheep and harbor seals (b). In both examples, the population size exceeds the carrying capacity for short periods of time and then falls below the carrying capacity afterwards. This fluctuation in population size continues to occur as the population oscillates around its carrying capacity. Still, even with this oscillation, the logistic model is confirmed.

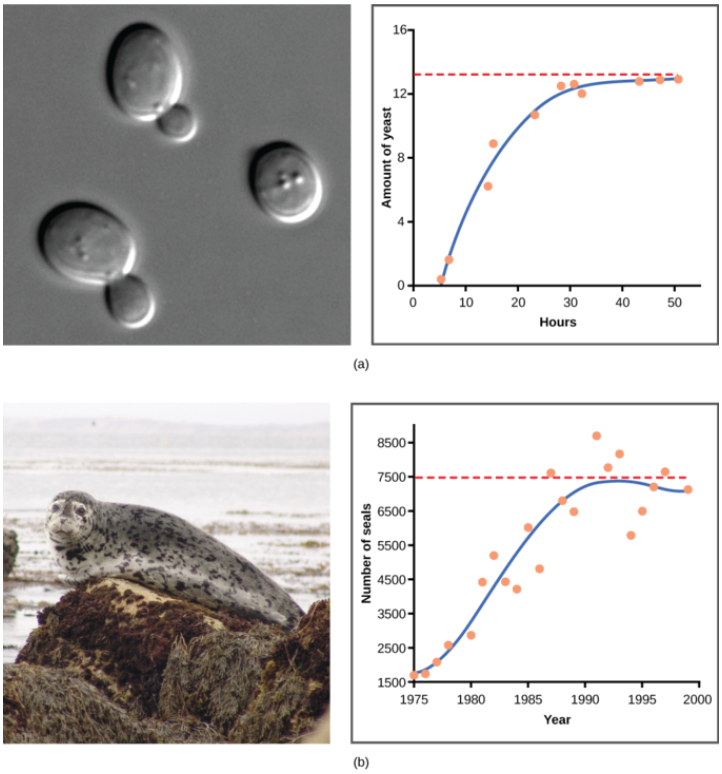

Figure $45.2B.1$: **Logistic population growth**: (a) Yeast grown in ideal conditions in a test tube show a classical S-shaped logistic growth curve, whereas (b) a natural population of seals shows real-world fluctuation.

Key Points

- The carrying capacity of a particular environment is the maximum population size that it can support.
- The carrying capacity acts as a moderating force in the growth rate by slowing it when resources become limited and stopping growth once it has been reached.
- As population size increases and resources become more limited, intraspecific competition occurs: individuals within a population who are more or less better adapted for the environment compete for survival.

Key Terms

- **phenotype**: the appearance of an organism based on a multifactorial combination of genetic traits and environmental factors, especially used in pedigrees
- **carrying capacity**: the number of individuals of a particular species that an environment can support; indicated by the letter “K”

45.2C: Density-Dependent and Density-Independent Population Regulation

Population regulation is a density-dependent process, meaning that population growth rates are regulated by the density of a population.

Learning Objectives

- Differentiate between density-dependent and density-independent population regulation.

Key Points

- The density of a population can be regulated by various factors, including biotic and abiotic factors and population size.
- Density-dependent regulation can be affected by factors that affect birth and death rates such as competition and predation.
- Density-independent regulation can be affected by factors that affect birth and death rates such as abiotic factors and environmental factors, i.e. severe weather and conditions such as fire.
- New models of life history incorporate ecological concepts that are typically included in r- and K-selection theory in combination with population age structures and mortality factors.

Key Terms

- **interspecific**: existing or occurring between different species
- **intraspecific**: occurring among members of the same species
- **fecundity**: number, rate, or capacity of offspring production

Density-dependent regulation

In population ecology, density-dependent processes occur when population growth rates are regulated by the density of a population. Most density-dependent factors, which are biological in nature (biotic), include predation, inter- and intraspecific competition, accumulation of waste, and diseases such as those caused by parasites. Usually, the denser a population is, the greater its mortality rate. For example, during intra- and interspecific competition, the reproductive rates of the individuals will usually be lower, reducing their population's rate of growth. In addition, low prey density increases the mortality of its predator because it has more difficulty locating its food source.

An example of density-dependent regulation is shown with results from a study focusing on the giant intestinal roundworm (*Ascaris lumbricoides*), a parasite of humans and other mammals. The data shows that denser populations of the parasite exhibit lower fecundity: they contained fewer eggs. One possible explanation for this phenomenon was that females would be smaller in more dense populations due to limited resources so they would have fewer eggs. This hypothesis was tested and disproved in a 2009 study which showed that female weight had no influence. The actual cause of the density-dependence of fecundity in this organism is still unclear and awaiting further investigation.

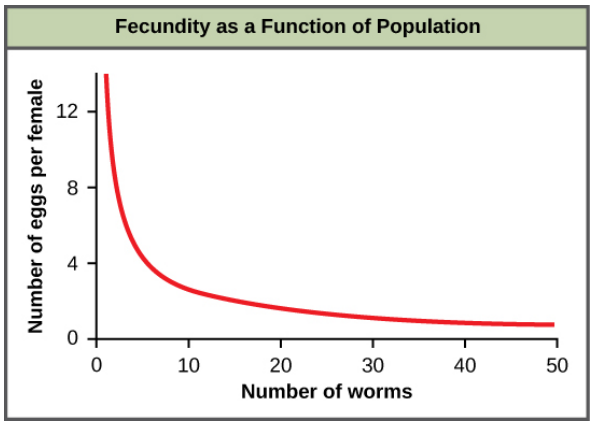

Figure $45.2C.1$: **Effect of population density on fecundity**: In this population of roundworms, fecundity (number of eggs) decreases with population density.

Density-independent regulation and interaction with density-dependent factors

Many factors, typically physical or chemical in nature (abiotic), influence the mortality of a population regardless of its density. They include weather, natural disasters, and pollution. An individual deer may be killed in a forest fire regardless of how many deer happen to be in that area. Its chances of survival are the same whether the population density is high or low.

In real-life situations, population regulation is very complicated and density-dependent and independent factors can interact. A dense population that is reduced in a density-independent manner by some environmental factor(s) will be able to recover differently than would a sparse population. For example, a population of deer affected by a harsh winter will recover faster if there are more deer remaining to reproduce.

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45.3: Life History Patterns

Topic hierarchy

- 45.3A: Life History Patterns and Energy Budgets
- 45.3B: Theories of Life History

45.3A: Life History Patterns and Energy Budgets

Energy budgets and life history strategies determine the type of reproductive capacity displayed by a population.

Learning Objectives

- Describe the energy budgets of, and the life history strategies used in, reproduction

Key Points

- The amount of parental care given to an individual offspring is inversely related to the reproductive capacity of an animal species.
- Animal species that produce many small, vulnerable offspring tend to provide little or no care for them due to their energy budget constraints; just enough offspring survive to maintain the species.
- Animal species that have few offspring expend large amounts of their energy budgets on caring for helpless offspring that need to develop before being on their own.
- Plants with low fecundity produce few energy-rich seeds with high germination rates, while plants with high fecundity usually have many small, energy-poor seeds with poor survival rates.
- Species that reproduce early ensure a greater chance of having surviving offspring than do those that must survive to a later reproductive age.
- Semelparous species use all of their reproductive budgets on one single reproductive event, while iteroparous species spend it on multiple mating seasons.

Key Terms

- **iteroparous**: reproducing more than once in a lifetime
- **semelparous**: reproducing only once in a lifetime
- **fecundity**: number, rate, or capacity of offspring production

Life history patterns and energy budgets

Energy is required by all living organisms for their growth, maintenance, and reproduction. At the same time, energy is often a major limiting factor in determining an organism's survival. Plants, for example, acquire energy from the sun via photosynthesis, but must expend this energy to grow, maintain health, and produce energy-rich seeds to produce the next generation. Animals also have the additional burden of using some of their energy reserves to acquire food. In addition, some animals must expend energy caring for their offspring. Thus, all species have an energy budget in which they must balance energy intake with their use of energy for metabolism, reproduction, parental care, and energy storage, as when bears build up body fat for winter hibernation.

Parental care and fecundity

Fecundity is the potential reproductive capacity of an individual within a population. In other words, it describes how many offspring could ideally be produced if an individual has as many offspring as possible, repeating the reproductive cycle as soon as possible after the birth of the offspring. In animals, fecundity is inversely related to the amount of parental care given to an individual offspring. Species that produce a large number of offspring, such as many marine invertebrates, usually provide little if any care for those offspring, as they would not have the energy or the ability to do so. Most of their energy budget is used to produce many tiny offspring. Animals with this strategy are often self-sufficient at a very early age. This is because of the energy trade-off these organisms have made to maximize their evolutionary fitness. Since their energy is used for producing offspring instead of parental care, it makes sense that these offspring have some ability to be able to move within their environment to find food and perhaps shelter. Even with these abilities, their small size makes them extremely vulnerable to predation, so the production of many offspring allows enough of them to survive to maintain the species.

Animal species that have few offspring during a reproductive event usually give extensive parental care, devoting much of their energy budget to these activities, sometimes at the expense of their own health. This is the case with many mammals,

such as humans, kangaroos, and pandas. The offspring of these species are relatively helpless at birth, needing to develop before they achieve self-sufficiency.

Plants with low fecundity produce few energy-rich seeds (such as coconuts and chestnuts) that have a good chance to germinate into a new organism. Plants with high fecundity usually have many small, energy-poor seeds (as do orchids) that have a relatively-poor chance of surviving. Although it may seem that coconuts and chestnuts have a better chance of surviving, the energy trade-off of the orchid is also very effective. It is a matter of where the energy is used: for large numbers of seeds or for fewer seeds with more energy.

Early versus late reproduction

Figure 45.3*A*.1: **Semelparous species**: Chinook salmon are an example of a population that uses its energy budget in one major reproductive event, dying shortly thereafter.

The timing of reproduction in a life history also affects species survival. Organisms that reproduce at an early age have a greater chance of producing offspring, but this is usually at the expense of their growth and the maintenance of their health. Conversely, organisms that start reproducing later in life often have greater fecundity or are better able to provide parental care, but they risk not surviving to reproductive age. Examples of this can be seen in fish. Small fish, such as guppies, use their energy to reproduce rapidly, but never attain the size that would give them defense against some predators. Larger fish, such as bluefin tuna and mako sharks, use their energy to attain a large size, but do so with the risk that they will die before they can reproduce or reproduce to their maximum. These different energy strategies and trade-offs are key to understanding the evolution of each species as it maximizes its fitness and fills its niche.

Single versus multiple reproductive events

Some life history traits, such as fecundity, timing of reproduction, and parental care, can be grouped together into general strategies that are used by multiple species. Semelparous species are those that only reproduce once during their lifetime and then die. Such species use most of their resource budget during a single reproductive event, sacrificing their health to the point that they do not survive. Examples of semelparity are bamboo, which flowers once and then dies, and the Chinook salmon, which uses most of its energy reserves to migrate from the ocean to its freshwater nesting area, where it reproduces and then dies. In contrast, iteroparous species reproduce repeatedly during their lives. Some animals are able to mate only once per year, but survive multiple mating seasons. Primates, including humans and chimpanzees, are examples of animals that display iteroparity.

45.3B: Theories of Life History

Modern theories of life history incorporate life and survivorship factors with ecological concepts associated with r- and K-selection theories.

Learning Objectives

- Distinguish between K- and r-selected species

Key Points

- K-selection species are defined as those present in stable and predictable environments that produce fewer offspring, have longer gestation periods, and provide long-term care after birth.
- r-selected species are defined as those present in fluctuating environments that have large numbers of offspring and do not provide long-term care after birth.
- Based on evidence that shows not all species follow solely the K- or r-selection theories, new models of life history are being developed which incorporate K- and r-selection theories with additional factors that affect life and survivorship such as population age structure and mortality factors.

Key Terms

- **carrying capacity**: the number of individuals of a particular species that an environment can support; indicated by the letter "K"
- **intraspecific**: occurring among members of the same species

Early theories about life history: *K*-selected and *r*-selected species

While reproductive strategies play a key role in life histories, they do not account for important factors such as limited resources and competition. The regulation of population growth by these factors can be used to introduce a classical concept in population biology: that of *K*-selected versus *r*-selected species.

By the second half of the twentieth century, the concept of K- and r-selected species was used extensively and successfully to study populations. The concept relates not only to reproductive strategies, but also to a species' habitat and behavior. This includes the way they obtain resources and care for their young, as well as length of life and survivorship factors. For this analysis, population biologists have grouped species into the two large categories, *K*-selected and *r*-selected, although they are really two ends of a continuum. The first variable is *K* (the carrying capacity of a population; density dependent), and the second variable is *r* (the intrinsic rate of natural increase in population size, density independent).

K-selected species

K-selected species are those in stable, predictable environments. Populations of *K*-selected species tend to exist close to their carrying capacity (hence the term *K*-selected) where intraspecific competition is high. These species produce few offspring, have a long gestation period, and often give long-term care to their offspring. While larger in size when born, the offspring are relatively helpless and immature at birth. By the time they reach adulthood, they must develop skills to compete for natural resources. Examples of *K*-selected species are primates including humans, other mammals such as elephants, and plants such as oak trees.

(a) K-selected species

(b) r-selected species

Figure 45.3B.1: **K- and r- selected species**: (A) Elephants are considered K-selected species as they live long, mature late, and provide long-term parental care to few offspring. Oak trees produce many offspring that do not receive parental care, but are considered K-selected species based on longevity and late maturation. (B) Dandelions and jellyfish are both considered r-selected species as they mature early, have short lifespans, and produce many offspring that receive no parental care

In plants, scientists think of parental care more broadly: how long fruit takes to develop or how long it remains on the plant are determining factors in the time to the next reproductive event. Oak trees grow very slowly and take, on average, 20 years to produce their first seeds, known as acorns. As many as 50,000 acorns can be produced by an individual tree, but the germination rate is low as many of these rot or are eaten by animals such as squirrels. In some years, oaks may produce an exceptionally large number of acorns; these years may be on a two- or three-year cycle depending on the species of oak (r-selection).

As oak trees grow to a large size (and for many years before they begin to produce acorns) they devote a large percentage of their energy budget to growth and maintenance. The tree's height and size allow it to dominate other plants in the competition for sunlight, the oak's primary energy resource. Furthermore, when it does reproduce, the oak produces large, energy-rich seeds that use their energy reserve to become quickly established (*K*-selection).

r-selected species

In contrast to K-selected species, *r*-selected species have a large number of small offspring (hence their *r* designation). This strategy is often employed in unpredictable or changing environments. Animals that are *r*-selected do not give long-term parental care and the offspring are relatively mature and self-sufficient at birth. Examples of *r*-selected species are marine invertebrates, such as jellyfish, and plants, such as the dandelion. Dandelions have small seeds that are dispersed long distances by wind; many seeds are produced simultaneously to ensure that at least some of them reach a hospitable environment. Seeds that land in inhospitable environments have little chance for survival since the seeds are low in energy content. Note that survival is not necessarily a function of energy stored in the seed itself.

Modern theories of life history

The r- and K-selection theory, although accepted for decades and used for much groundbreaking research, has now been reconsidered. Many population biologists have abandoned or modified it. Over the years, several studies attempted to confirm the theory, but these attempts have largely failed. Many species were identified that did not follow the theory's predictions. Furthermore, the theory ignored the age-specific mortality of the populations which scientists now know is very important. New demographic-based models of life history evolution have been developed which incorporate many ecological concepts included in r- and K-selection theory, as well as population age structure and mortality factors.

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45.4: Human Population Growth

Topic hierarchy

- 45.4A: Human Population Growth
- 45.4B: Overcoming Density-Dependent Regulation
- 45.4C: Age Structure, Population Growth, and Economic Development

45.4A: Human Population Growth

The exponential growth of the human population could lead to food shortages, global warming, and other issues of resource scarcity.

Learning Objectives

- Predict the long-term consequences of exponential human population growth

Key Points

- Global human population growth is around 75 million annually, or 1.1% per year. The global population has grown from 1 billion in 1800 to 7 billion in 2012.
- Although the direst consequences of human population growth have not yet been realized, exponential growth cannot continue indefinitely.
- In the late 1970s, China's "one-child" policy tried to control population growth, but restrictions were relaxed in the early 2000s.
- One of the major consequences of population growth is the potential for widespread food shortages.
- Most scientists agree that humans and human population growth are causing climate change by emission of the greenhouse gas carbon dioxide (CO_2).
- International treaties to limit greenhouse gas emissions have not been ratified by every country due to economic and political concerns.

Key Terms

- **greenhouse gas**: Any gas, such as carbon dioxide, that contributes to the greenhouse effect (continued warming) when released into the atmosphere.
- **climate change**: Changes in the earth's climate, especially those said to be produced by global warming.

Human Population Growth

Global human population growth is around 75 million annually, or 1.1% per year. The global population has grown from 1 billion in 1800 to 7 billion in 2012. It is expected to keep growing, though predictions differ as to when and if this growth will plateau.

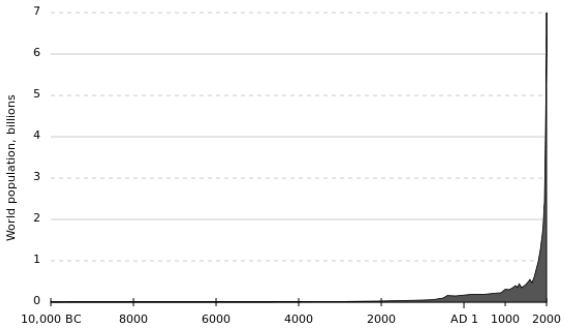


Figure 45.4A.1: **The estimated growth of the human population from 10,000 BCE–2000 CE.**: The human population has grown most sharply in the past 200 years.

The "population growth rate" is the rate at which the number of individuals in a population increases in a given time period as a fraction of the initial population. Specifically, population growth rate refers to the change in population over a time period, often expressed as a percentage of the number of individuals in the population at the beginning of that period. This can be written as the formula:

population growth rate=P(t2)−P(t1)P(t1)population growth rate=P(t2)−P(t1)P(t1)

Globally, the growth rate of the human population has been declining since 1962 and 1963, when it was 2.20% per annum. In 2009, the estimated annual growth rate was 1.1%. The CIA World Factbook gives the world annual birthrate, mortality

nd growth rate as 1.89%, 0.79%, and 1.096% respectively. The last 100 years have seen a rapid increase in tion due to medical advances and massive increase in agricultural productivity.

Each region of the globe has seen reductions in growth rate in recent decades, though growth rates remain above 2% in some countries of the Middle East and Sub-Saharan Africa, and also in South Asia, Southeast Asia, and Latin America. This does not mean that the population is declining;
rather, it means the population is growing more slowly. However, some countries do experience negative population growth, mainly due to low fertility rates, high death rates and emigration.

According to the UN's 2010 revision to its population projections, world population will peak at 10.1 billion in 2100 compared to 7 billion in 2011. However, some experts dispute the UN's forecast and have argued that birthrates will fall below replacement rates (the number of births needed to maintain a stable population) in the 2020s. According to these forecasters, population growth will be only sustained until the 2040s by rising longevity, but will peak below 9 billion by 2050, followed by a long decline.

Figure 45.4*A*. 1: **Growing Population Rate and Resource Scarcity**: Greater Los Angeles lies on a coastal Mediterranean Savannah with a small watershed that is able to support at most one million people on its own water; as of 2015, the area has a population of over 18 million. Researchers predict that similar cases of resource scarcity will grow more common as the world population increases.

Long-term Consequences of Population Growth

The "population explosion" seen in the last century has led to dire predictions. In 1968, biologist Paul Ehrlich wrote, "The battle to feed all of humanity is over. In the 1970s, hundreds of millions of people will starve to death in spite of any crash programs embarked upon now. At this late date, nothing can prevent a substantial increase in the world death rate. " Although many critics view Ehrlich's view as an exaggeration, the human population continues to grow exponentially. The laws of nature dictate that exponential growth cannot continue indefinitely.

Despite efforts to curb population growth, such as the "one-child policy" in China (introduced in 1979 but relaxed in the early 2000s), the human population continues to grow. A primary concern regarding this growth is that the demand for ever-more food will lead to widespread shortages, as forecast by Ehrlich.

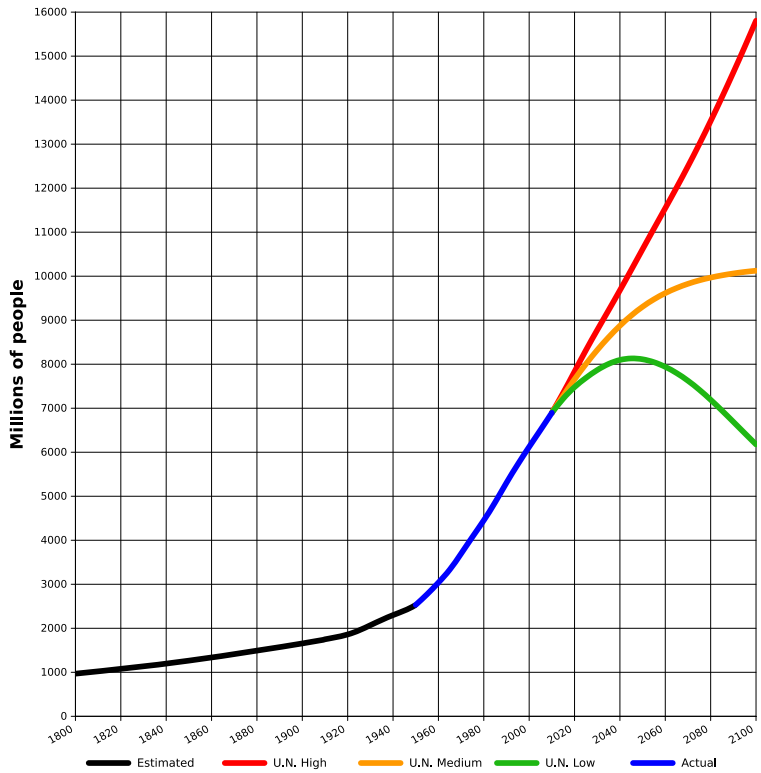


Figure 45.4*A*.1: **World population growth from 1800 to 2100**: United Nations projections in 2010 give "high" (red line), "medium" (orange) and "low" (green) scenarios for world population growth. The highest estimate projects the world population may rise to 16 billion by 2100 or it may decline to 6 billion, according to the lowest estimate.

In addition to the threat of food shortages, human population growth is damaging to the environment in potentially permanent ways. Most scientists agree that climate change caused by the emission of the greenhouse gas carbon dioxide (CO_2) is a significant consequence of human activities. In a series of treaties in the late 20th century, many countries committed to reducing their CO_2emissions to prevent continuous global warming; however these treaties have not been ratified by every country, largely due to economic and political concerns. The role of human activity in climate change is hotly debated in some circles. The future holds considerable uncertainty for curbing human population growth and protecting the environment.

45.4B: Overcoming Density-Dependent Regulation

Humans have exceeded density-dependent limits on population by enacting various environmental changes to accommodate our needs for hygiene, shelter, and food.

Learning Objectives

- Describe ways in which humans overcome density-dependent regulation of population size

Key Points

- Humans' ability to alter their environment is an underlying reason for human population growth, enabling people to overcome density-dependent limits on growth, in contrast with all other organisms.
- Abilities, such as construction of shelter, food cultivation, and the sharing of technology, have helped humans overcome factors that would have otherwise limited their population growth.
- Originating from Africa, human migration to nearly every inhabitable area of the globe has enabled colonization of areas where people were previously absent.
- Advances in medicine, notably vaccines and antibiotics, as well as improvements in nutrition and vector control, have significantly curbed mortality from disease.

Key Terms

- **density-dependent**: Processes that occur when population growth rates are regulated by the size of a population in a given amount of resources such as food or habitat area.
- **vaccine**: A substance given to stimulate the body's production of antibodies and provide immunity against a disease, prepared from the agent that causes the disease, or a synthetic substitute.
- **infectious disease**: Illness caused by introduction of a pathogen or parasite into the body via contact with a transmitting agent such as vector organism or an infected person.

Humans are uniquely able to consciously alter their environment to increase its carrying capacity. This capability is an underlying reason for human population growth as humans are able to overcome density-dependent limits on population growth, in contrast with all other organisms.

Human intelligence, society, and communication have enabled this capacity. For instance, people can construct shelters to protect them from the elements; food supply has increased because of agriculture and domestication of animals; and humans use language to pass on technology to new generations, allowing continual improvement upon previous accomplishments. Migration has also contributed to human population growth. Originating from Africa, humans have migrated to nearly every inhabitable area on the planet.

Public health, sanitation, and the use of antibiotics and vaccines have lessened the impact of infectious disease on human populations. In the fourteenth century, the bubonic plague killed as many as 100 million people: between 30 to 60 percent of Europe's population. Today, however, the plague and other infectious diseases have much less of an impact. Through vaccination programs, better nutrition, and vector control (carriers of disease), international agencies have significantly reduced the global infectious disease burden. For example, reported cases of measles in the United States dropped from around 700,000 a year in the 1950s to practically zero by the late 1990s. Globally, measles fell 60 percent from an estimated 873,000 deaths in 1999 to 164,000 in 2008. This advance is attributed entirely to a comprehensive vaccination program.

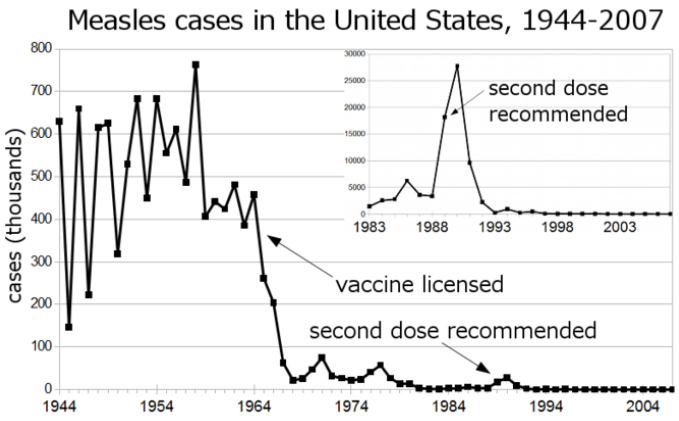


Figure 45.4B.1: **Measles cases reported in the United States, 1944-2007**: Measles cases reported in the United States, represented as thousands of cases per year, declined sharply after the measles vaccine was introduced, in 1964.

Developing countries have also made advances in curbing mortality from infectious disease. For example, deaths from infectious and parasitic diseases in Brazil fell from second place as the most important causes of death in 1977 to fifth place in 1984. The improvement is attributed in part to increased access to essential goods and services, reflecting the country's rising prosperity. Through changes in economic status, as in Brazil, as well as global disease control efforts, human population growth today is less limited by infectious disease than has been the case historically.

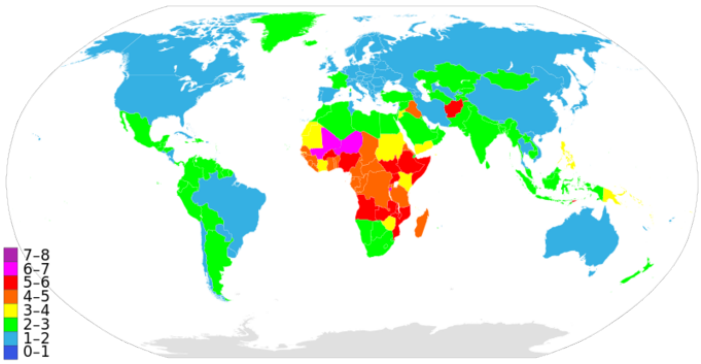


Figure 45.4B.1: **Countries by Fertility Rate Comparison**: The advent of modern medicine is very closely tied to childhood mortality, as well as the number of children per mother (Fertility Rate). As modern medicine decrease child mortality, the birth rate decreases.

C: Age Structure, Population Growth, and Economic Development

A population's growth is strongly influenced by the proportions of individuals in different age brackets, which in turn is influenced by economic development.

Learning Objectives

- Explain how age structure in a population is associated with population growth and economic development

Key Points

- Population dynamics are influenced by age structure, which is characteristic for populations growing at different rates.
- Age structure varies according to the age distribution of individuals within a population.
- Fast-growing populations with a high proportion of young people have a triangle-shaped age structure, representing younger ages at the bottom and older ages at the top.
- Slow-growing populations with a smaller proportion of young people have a column-shaped age structure, representing a relatively even distribution of ages.
- Improvements in health care have led to the population explosion in underdeveloped countries, causing a "youth bulge" which is associated with social unrest.

Key Terms

- **population dynamics**: Variation among populations due to birth and death rates, by immigration and emigration, and concerning topics such as aging populations or population decline.
- **youth bulge**: Age structure typical of fast-growing populations in which a majority of the population are relatively young.
- **age structure**: The composition of a population in terms of the proportions of individuals of different ages; represented as a bar graph with younger ages at the bottom and males and females on either side.

The variation of populations over time, also known as population dynamics, depends on biological and environmental processes that determine population changes. A population's growth rate is strongly influenced by the proportions of individuals of particular ages. With knowledge of this age structure, population growth can be more accurately predicted. Age structure data allow the rate of growth (or decline) to be associated with a population's level of economic development.

For example, the population of a country with rapid growth has a triangle-shaped age structure with a greater proportion of younger individuals who are at or close to reproductive age. This pattern typically occurs where fewer people live to old age because of sub-optimal living standards, such as occurs in underdeveloped countries.

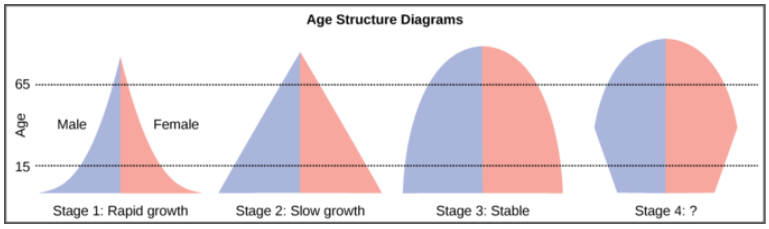
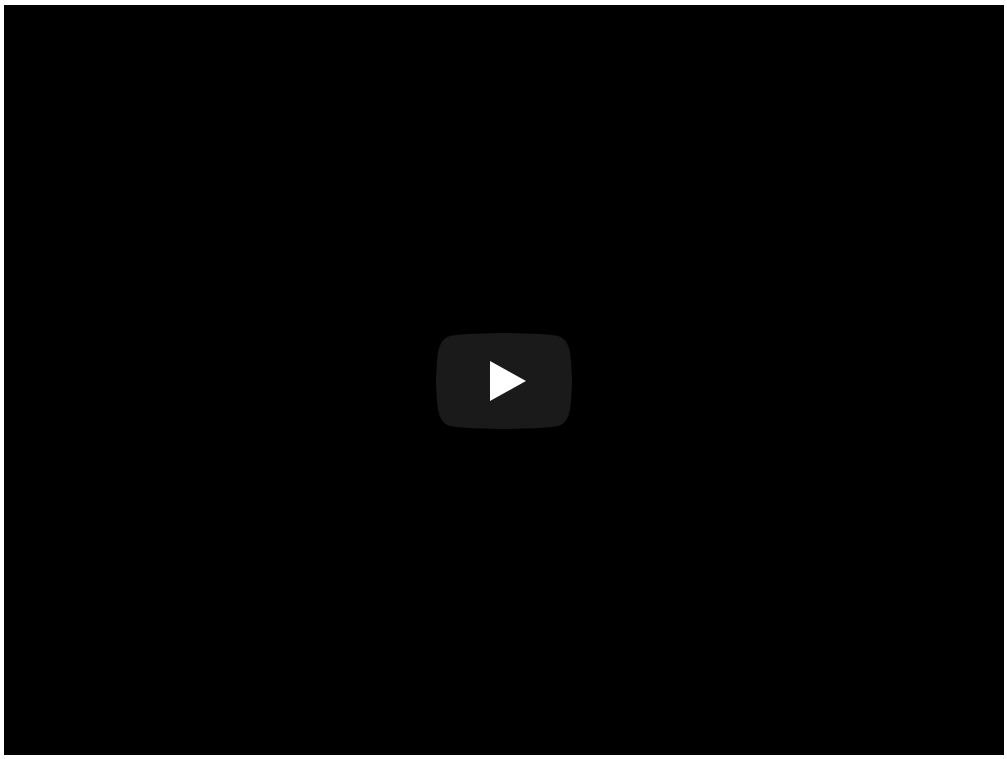
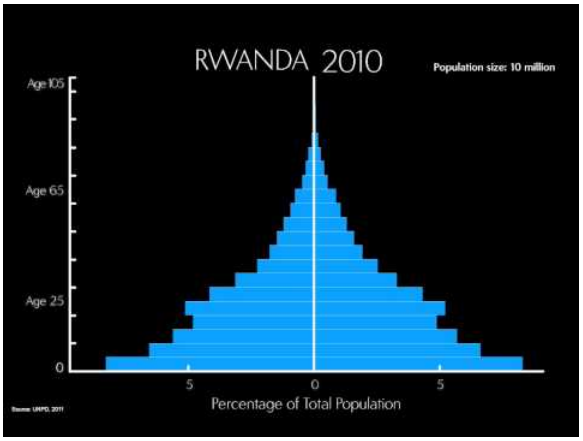


Figure 45.4C.1: **Population bar graphs for stages of demographic change from expansion to contraction**: The leftmost diagram (representing the age structure of a rapidly-growing population) indicates that the number of individuals decreases rapidly with age. The slow-growth model shows that the proportion of individuals decreases steadily with age. The stable population diagram is rounded on top; the older part of the population is a larger proportion of the population than in the other age diagrams. The rightmost diagram represents a population that may be stable or even declining. The relatively few young people may not be making up for the mortality among the older age groups.





Changing Population Age Structure: This 3:28 minute movie discusses age structures and gives examples.

Some developed countries, including the United States, have a slowly-growing population. This results in a column-shaped age structure diagram with steeper sides. In these cases, the population has fewer young and reproductive-aged individuals, with a greater proportion of older individuals. Some developed countries, such as Italy, have zero population growth. Countries with declining populations, such as Japan, have a bulge in the middle of their age structure diagram. The bulge indicates relatively-few young individuals, and a higher proportion of middle-aged and older individuals.

Globally, less-economically developed countries in Africa and Asia have the highest growth rates, leading to populations consisting mostly of younger people. Improved health care, beginning in the 1960s, is one of the leading causes of the increased growth rates that created the population explosion. For example, in the Middle East and North Africa, around 65 percent of the population is under the age of 30. These high growth rates lead to the so-called “youth bulge,” which some experts believe is a cause of social unrest and economic problems such as high unemployment.

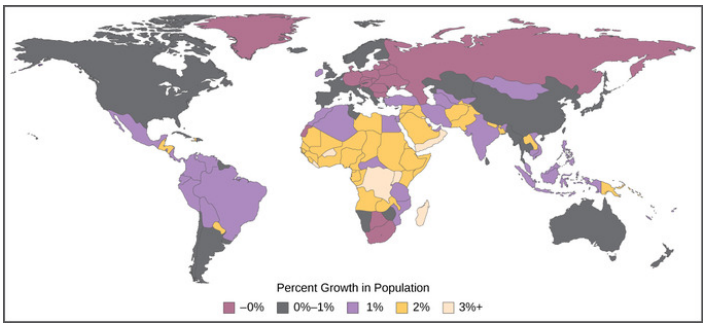


Figure 45.4C.1: **Global percent growth rate of population**: The percent growth rate of population in different countries shows that the highest growth rates are in underdeveloped regions of Africa and Asia.

All of the factors above also have an impact on the average life expectancy. As economic development and quality of health care increase, the life expectancy also increases.

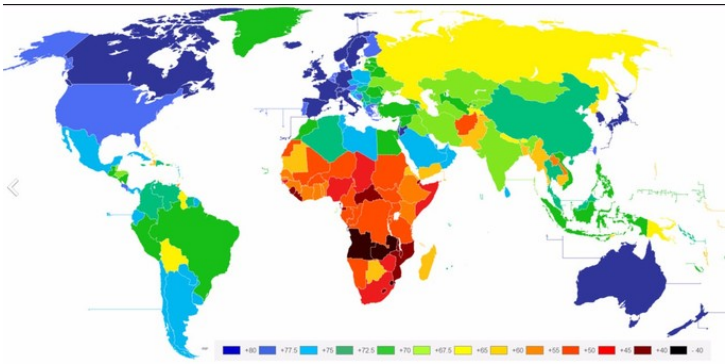


Figure 45.4C.1: **Worldwide Life Expectancy by Country**: 2012 map showing average life expectancy by country in years. In 2012, the World Health Organization estimated the average global life expectancy as 70.5 years.

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Topic hierarchy

- 45.5A: The Role of Species within Communities
- 45.5B: Predation, Herbivory, and the Competitive Exclusion Principle
- 45.5C: Symbiosis
- 45.5D: Ecological Succession

45.5A: The Role of Species within Communities

Communities are shaped by foundation species and keystone species, while invasive species disrupt the natural balance of an area.

Learning Objectives

- Distinguish between foundation, keystone, and invasive species

Key Points

- A community is defined by the structure of different species that occupy it and how those structures change over time.
- Foundation species change the environment where other species live, modifying it to benefit the organisms that live there.
- Keystone species maintain biodiversity; their removal can greatly alter the dynamics within the community.
- Invasive species are non-native organisms introduced into an area that may be better competitors and reproduce faster than native species; they tend to upset the natural balance.

Key Terms

- **invasive species**: any species that has been introduced to an environment where it is not native and has since become a nuisance through rapid spread and increase in numbers, often to the detriment of native species
- **community**: a group of interdependent organisms inhabiting the same region and interacting with each other
- **keystone species**: a species that exerts a large, stabilizing influence throughout an ecological community, despite its relatively small numerical abundance

Characteristics of Communities

Communities are complex entities that can be characterized by their structure (the types and numbers of species present) and dynamics (how communities change over time). Understanding community structure and dynamics enables community ecologists to manage ecosystems more effectively. There are three main types of species that serve as the basis for a community. These include the foundation species, keystone species, and invasive species. Each of these has a specific role in how communities are formed.

Foundation Species

Foundation species are considered the “base” or “bedrock” of a community, having the greatest influence on its overall structure. They are usually the primary producers: organisms that bring most of the energy into the community. Kelp, a brown algae, is a foundation species that forms the basis of the kelp forests off the coast of California.

Foundation species may physically modify the environment to produce and maintain habitats that benefit the other organisms that use them. An example is the photosynthetic corals of the coral reef. Corals themselves are not photosynthetic, but harbor symbionts within their body tissues (dinoflagellates called zooxanthellae) that perform photosynthesis; this is another example of a mutualism. The exoskeletons of living and dead coral make up most of the reef structure, which protects many other species from waves and ocean currents.

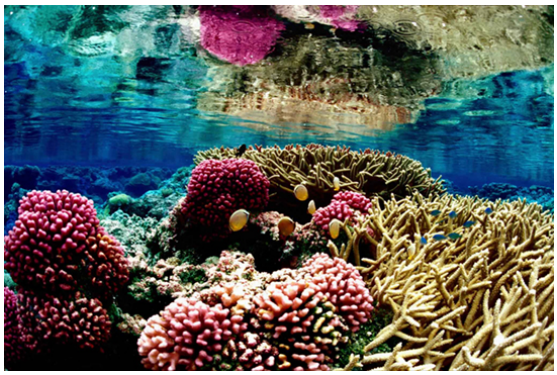

Figure 45.5A.1: **Foundation species**: Coral is the foundation species of coral reef ecosystems. The photosynthetic algae within the corals provides energy for them so that they can build the reefs.

Keystone Species

A keystone species is one whose presence is key to maintaining biodiversity within an ecosystem and to upholding an ecological community's structure. The intertidal sea star, *Pisaster ochraceus*, of the northwestern United States is a keystone species. Studies have shown that when this organism is removed from communities, populations of their natural prey (mussels) increase, completely altering the species composition and reducing biodiversity. Another keystone species is the banded tetra, a fish in tropical streams, which supplies nearly all of the phosphorus, a necessary inorganic nutrient, to the rest of the community. If these fish were to become extinct, the community would be greatly affected.

Figure 45.5A.1: **Keystone species maintain biodiveristy**: The *Pisaster ochraceus* sea star is a keystone species. When this animal is removed from certain areas, its prey species greatly alters the dynamics of the ecosystem, reducing biodiversity.

Invasive Species

Invasive species are foreign species whose introduction can cause harm to the economy and the environment. These species have many ways of entering foreign environments, including through ship's ballast water: when planes take off, organisms can sometimes become stuck in the cargo area. When the plane arrives in its destination, the organisms are now in a foreign environment. Travelers sometimes illegally smuggle items, such as fruits, plants, or even animals as pets, from one state or country to another..

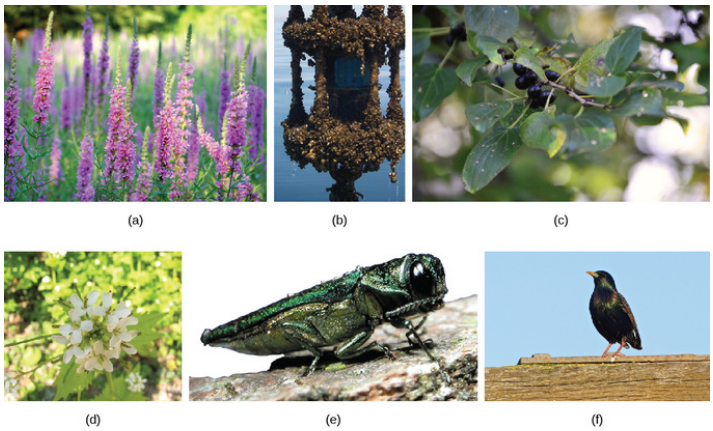

Figure 45.5A.1: **Invasive species threaten ecosystems**: In the United States, invasive species such as (a) purple loosestrife (*Lythrum salicaria*) and the (b) zebra mussel (*Dreissena polymorpha*) threaten certain ecosystems. Some forests are threatened by the spread of (c) common buckthorn (*Rhamnus cathartica*), (d) garlic mustard (*Alliaria petiolata*), and (e) the emerald ash borer (*Agrilus planipennis*). The (f) European starling (*Sturnus vulgaris*) may compete with native bird species for nest holes.

Invasive species are often better competitors than native species, resulting in population explosions. These new species usually overtake the native populations, driving them to localized extinctions.

One of the many recent proliferations of an invasive species concerns the growth of Asian carp populations. Asian carp were introduced to the United States in the 1970s by fisheries and sewage treatment facilities that used the fish's excellent filter feeding capabilities to clean their ponds of excess plankton. Some of the fish escaped, however, and by the 1980s, they had colonized many waterways of the Mississippi River basin, including the Illinois and Missouri Rivers.

Voracious eaters and rapid reproducers, Asian carp may outcompete native species for food, potentially leading to native species extinctions. For example, black carp are voracious eaters of native mussels and snails, limiting this food source for native fish species. Silver carp eat plankton that native mussels and snails feed upon, reducing this food source by a different alteration of the food web. In some areas of the Mississippi River, Asian carp species have become predominant, effectively outcompeting native fish for habitat. In some parts of the Illinois River, Asian carp constitute 95 percent of the community's biomass. Although edible, the fish is bony and not a desirable food in the United States. Moreover, their presence threatens the native fish and fisheries of the Great Lakes, which are important to local economies and recreational anglers. Asian carp have even injured humans. The fish, frightened by the sound of approaching motorboats, thrust themselves into the air, often landing in the boat or directly hitting the boaters.

One infested waterway of particular importance is the Chicago Sanitary and Ship Channel, the major supply waterway linking the Great Lakes to the Mississippi River. To prevent the Asian carp from leaving the canal, a series of electric barriers have been successfully used to discourage their migration; however, the threat is significant enough that several states and Canada have sued to have the Chicago channel permanently cut off from Lake Michigan. Local and national politicians have weighed in on how to solve the problem, but no one knows whether the Asian carp will ultimately be considered a nuisance, like other invasive species, such as the water hyacinth and zebra mussel, or whether it will be the destroyer of the largest freshwater fishery of the world.

45.5B: Predation, Herbivory, and the Competitive Exclusion Principle

Predation and herbivory are two methods animals use to obtain energy; many species have developed defenses against them.

Learning Objectives

- Distinguish between predation and herbivory and describe defense mechanisms against each

Key Points

- Predation, the hunting and consuming of animals by other animals, often shows cyclical patterns of predator/prey population sizes; predators increase in numbers when prey species are plentiful.
- Herbivory is the eating of plant material for energy and can assist the plants with seed distribution.
- Plants have evolved spines and toxins to defend against being eaten by herbivores.
- Animals use bright colors to advertise that they are toxic; mimicry to hide from predators; or have spines, shells, and scales to protect themselves.
- Batesian mimicry is when a non-toxic species looks similar to a poisonous one, which deters predator attacks.

Key Terms

- **camouflage**: resemblance of an organism to its surroundings for avoiding detection
- **herbivory**: the consumption of living plant tissue by animals
- **Batesian mimicry**: the resemblance of one or more non-poisonous species to a poisonous species, for example, the scarlet king snake and the coral snake

Predation and Herbivory

Most animals fall into one of two major categories when it comes to obtaining the energy they need to survive in the environment: predation or herbivory. An animal that hunts, kills, and eats other animals is called a predator. Examples of predators include tigers, snakes, and hawks. Herbivory, on the other hand, refers to animals that eat plant matter. Deer, mice, and most song birds are examples. To protect themselves against these feeding mechanisms, many organisms have developed methods that keep them from being eaten.

Predation is the hunting of prey by its predator. Populations of predators and prey in a community are not constant over time; in most cases, they vary in cycles that appear to be related. The most-often-cited example of predator-prey dynamics is seen in the cycling of the lynx (predator) and the snowshoe hare (prey), which is based on nearly 200-year-old trapping data from North American forests. This cycle of predator and prey lasts approximately 10 years, with the predator population lagging 1–2 years behind that of the prey population. As the hare numbers increase, there is more food available for the lynx, allowing the lynx population to increase as well. When the lynx population grows to a threshold level, they kill so many hares that the hare population begins to decline. This is followed by a decline in the lynx population because of scarcity of food. When the lynx population is low, the hare population size begins to increase due, at least in part, to low predation pressure, starting the cycle anew.

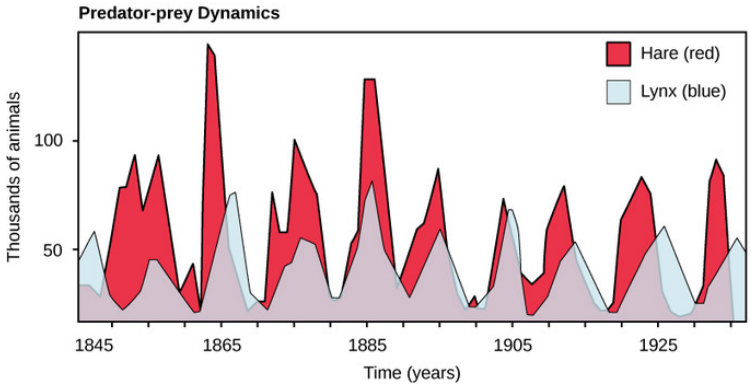

Figure $45.5B.1$: **Predator and prey population cycling**: The cycling of lynx and snowshoe hare populations in Northern Ontario is an example of predator-prey dynamics. As hare populations increase,the lynx populations also increase due to increased food supplies. Many lynx eating many hares causes a decline in the hare population size. This results in a decrease in the lynx population. Then the cycle begins again.

Herbivory describes the consumption of plants by insects and other animals. Unlike animals, plants cannot outrun predators or use mimicry to hide from hungry animals. Some plants have developed mechanisms to defend against herbivory. Other species have developed mutualistic relationships; for example, herbivory provides a mechanism of seed distribution that aids in plant reproduction.

Defense Mechanisms against Predation and Herbivory

The study of communities must consider evolutionary forces that act on the members of the various populations contained within it. Species are not static, but slowly changing and adapting to their environment by natural selection and other evolutionary forces. Species have evolved numerous mechanisms to escape predation and herbivory. These defenses may be mechanical, chemical, physical, or behavioral.

Mechanical defenses, such as the presence of thorns on plants or the hard shell on turtles, discourage animal predation and herbivory by causing physical pain to the predator or by physically preventing the predator from being able to eat the prey. Chemical defenses are produced by many animals as well as plants, such as the foxglove which is extremely toxic when eaten.

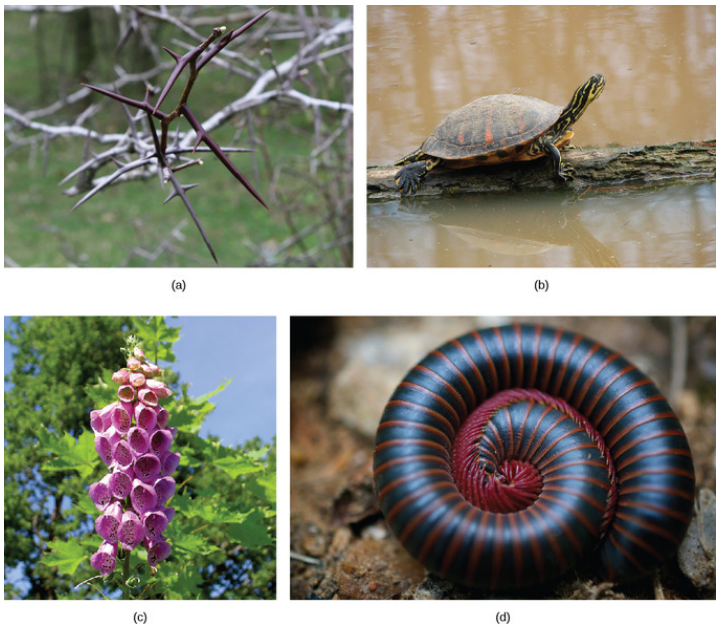

Figure 45.5B.1: **Defenses against predation and herbivory**: The (a) honey locust tree (*Gleditsia triacanthos*) uses thorns, a mechanical defense, against herbivores, while the (b) Florida red-bellied turtle (*Pseudemys nelsoni*) uses its shell as a mechanical defense against predators. (c) Foxglove (*Digitalis* sp.) uses a chemical defense: toxins produced by the plant can cause nausea, vomiting, hallucinations, convulsions, or death when consumed. (d) The North American millipede (*Narceus americanus*) uses both mechanical and chemical defenses: when threatened, the millipede curls into a defensive ball, producing a noxious substance that irritates eyes and skin.

Many species use their body shape and coloration to avoid being detected by predators. The tropical walking stick is an insect with the coloration and body shape of a twig, which makes it very hard to see when stationary against a background of real twigs. In another example, the chameleon can change its color to match its surroundings. Both of these are examples of camouflage: avoiding detection by blending in with the background.

Some species use coloration as a way of warning predators they are not good to eat. For example, the cinnabar moth caterpillar, the fire-bellied toad, and many species of beetle have bright colors that warn of a foul taste, the presence of toxic chemical, and/or the ability to sting or bite, respectively. Predators that ignore this coloration and eat the organisms will experience their unpleasant taste or presence of toxic chemicals and learn not to eat them in the future. This type of defensive mechanism is called aposematic coloration, or warning coloration.

Figure 45.5B.1: **Examples of aposematic coloration**: (a) The strawberry poison dart frog (*Oophaga pumilio*) uses aposematic coloration to warn predators that it is toxic, while the (b) striped skunk (*Mephitis mephitis)*uses aposematic coloration to warn predators of the unpleasant odor it produces.

While some predators learn to avoid eating certain potential prey because of their coloration, other species have evolved mechanisms to mimic this coloration to avoid being eaten, even though they themselves may not be unpleasant to eat or contain toxic chemicals. In Batesian mimicry, a harmless species imitates the warning coloration of a harmful one. Assuming they share the same predators, this coloration then protects the harmless ones, even though they do not have the same level of physical or chemical defenses against predation as the organism they mimic. Many insect species mimic the coloration of wasps or bees, which are stinging, venomous insects, thereby discouraging predation.

45.5C: Symbiosis

Commensalism, mutualism, and parasitism are three symbiotic ways organisms interact with each other with differing degrees of benefit.

Learning Objectives

- Differentiate among the types of symbiosis: commensalism, mutualism, and parasitism

Key Points

- Commensalism is when two organisms share the same environment, where one benefits and the other is unharmed.
- Trees and birds have a commensalistic relationship; the birds benefit from having a place to build their nests, while the trees are unharmed and not impacted by the bird's presence.
- Mutualism is when two species sharing the same environment both benefit from their interactions.
- The protozoans living within the intestines of termites create a mutualistic relationship with them; the protozoans get a safe place to live while the termites get help digesting the cellulose in their diet.
- Parasitism occurs when two organisms interact, but while one benefits, the other experiences harm.
- Parasites harm their hosts, as with the tapeworm attaching itself to the intestine of a cow; the tapeworm absorbs the nutrients from the cow's diet, preventing them from being absorbed by the cow.

Key Terms

- **mutualism**: Any interaction between two species that benefits both.
- **commensalism**: A sharing of the same environment by two organisms where one species benefits and the other is unaffected; e.g., barnacles on whales.
- **parasitism**: Interaction between two organisms, in which one organism (the parasite) benefits and the other (the host) is harmed.

Symbiotic relationships, or symbioses (plural), are close interactions between individuals of different species over an extended period of time which impact the abundance and distribution of the associating populations. Most scientists accept this definition, but some restrict the term to only those species that are mutualistic, where both individuals benefit from the interaction.

Commensalism

A commensalistic relationship occurs when one species benefits from the close, prolonged interaction, while the other neither benefits nor is harmed. Birds nesting in trees provide an example of a commensal relationship. The tree is not harmed by the presence of the nest among its branches. The nests are light and produce little strain on the structural integrity of the branch. Most of the leaves, which the tree uses to obtain energy by photosynthesis, are above the nest, so they are unaffected. The bird, on the other hand, benefits greatly. If the bird had to nest in the open, its eggs and young would be vulnerable to predators.

Figure 45.5C.1: **Commensalistic relationship**: The southern masked-weaver bird is starting to make a nest in a tree in Zambezi Valley, Zambia. This is an example of a commensal relationship, in which one species (the bird) benefits, while the other (the tree) neither benefits nor is harmed.

Mutualism

A second type of symbiotic relationship, mutualism, is where two species both benefit from their interaction. Some scientists believe that these are the only true examples of symbiosis. For example, termites have a mutualistic relationship with protozoa that live in the insect's gut. The termite benefits from the ability of bacterial symbionts within the protozoa to digest cellulose. The termite itself cannot do this; without the protozoa, it would not be able to obtain energy from its food (cellulose from the wood it chews and eats). The protozoa and the bacterial symbionts benefit by having a protective environment and a constant supply of food from the wood-chewing actions of the termite.

Parasitism

A parasite is an organism that lives in or on another living organism, deriving nutrients from it. In this relationship the parasite benefits, but the organism being fed upon, the host, is harmed. The host is usually weakened by the parasite as it siphons resources the host would normally use to maintain itself. The parasite, however, is unlikely to kill the host. This is because the parasite needs the host to complete its reproductive cycle by spreading to another host.

The reproductive cycles of parasites are often very complex, sometimes requiring more than one host species. A tapeworm is a parasite that causes disease in humans when contaminated, undercooked meat such as pork, fish, or beef is consumed. The tapeworm can live inside the intestine of the host for several years, benefiting from the food the host is bringing into its gut by eating; it may grow to be over 50 ft long by adding segments. The parasite moves from species to species as it requires two hosts to complete its life cycle.

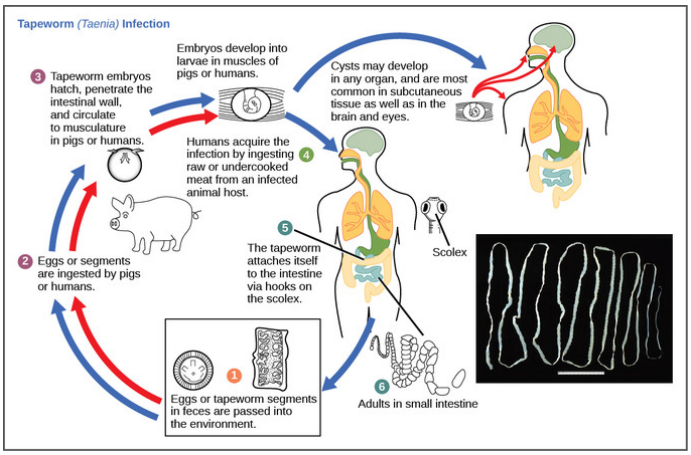

Figure 45.5C.1: **Lifecycle of a parasitic tapeworm**: This diagram shows the life cycle of a pork tapeworm (*Taenia solium*), a human worm parasite. The eggs of the tapeworm are ingested by the host. When they hatch, the worms travel through the wall of the intestine and begin to grow. Here, the parasite will absorb the nutrition from the host and continue to grow.

45.5D: Ecological Succession

When disturbances occur, succession allows for communities to become re-established over periods of time.

Learning Objectives

- Distinguish between primary and secondary succession following disturbances to communities

Key Points

- After an environmental disturbance such as a volcanic eruption or forest fire, communities are able to replace lost species through the process of succession.
- Primary succession occurs after a volcanic eruption or earthquake; it involves the breakdown of rocks by lichens to create new, nutrient -rich soils.
- The first species to colonize an area after a major disturbance are called pioneer species; they help to form the new environment.
- Secondary succession occurs after a disturbance such as a forest fire, where there is still some organic matter to allow new plants to grow.
- Both types of succession take place over long periods of time and result in the communities reaching a state of equilibrium.

Key Terms

- **succession**: an act of following in sequence
- **equilibrium**: the condition of a system in which competing influences are balanced, resulting in no net change

Community dynamics

Community dynamics are the changes in community structure and composition over time. Sometimes these changes are induced by environmental disturbances such as volcanoes, earthquakes, storms, fires, and climate change. Communities with a stable structure are said to be at equilibrium. Following a disturbance, the community may or may not return to the equilibrium state.

Succession describes the sequential appearance and disappearance of species in a community over time. In primary succession, newly-exposed or newly-formed land is colonized by living things. In secondary succession, part of an ecosystem is disturbed, but remnants of the previous community remain.

Primary succession and pioneer species

Primary succession occurs when new land is formed or rock is exposed; for example, following the eruption of volcanoes, such as those on the Big Island of Hawaii. As lava flows into the ocean, new land is continually being formed. On the Big Island, approximately 32 acres of land are added each year. First, weathering and other natural forces break down the substrate enough for the establishment of certain hearty plants and lichens with few soil requirements, known as pioneer species. These species help to further break down the mineral-rich lava into soil where other, less-hardy species will grow, eventually replacing the pioneer species. In addition, as these early species grow and die, they add to an ever-growing layer of decomposing organic material, contributing to soil formation. Over time, the area will reach an equilibrium state with a set of organisms quite different from the pioneer species.

Figure 45.5D.1: **Pioneer species are the first to colonize**: During primary succession in lava on Maui, Hawaii, succulent plants are the pioneer species. After weathering breaks down the lava into soil, the plants are able to take root. These plants will help to further change the soil by adding nutrients to it.

Secondary succession

A classic example of secondary succession occurs in oak and hickory forests cleared by wildfire. Wildfires will burn most vegetation and kill those animals unable to flee the area. Their nutrients, however, are returned to the ground in the form of ash. Thus, even when areas are devoid of life due to severe fires, they will soon be ready for new life to take hold.

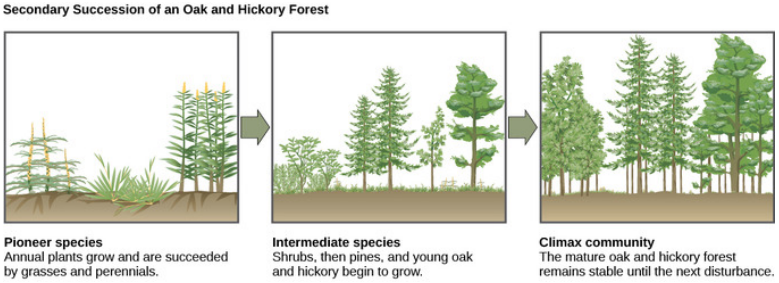

Figure 45.5D.1: **Secondary succession in the forest**: Secondary succession is shown in an oak and hickory forest after a forest fire. Those organisms that could not escape are killed, but their bodies decompose, adding nutrients to the soil. These nutrients provide the basis for new plants to grow.

Before a wildfire, vegetation is often dominated by tall trees with access to the major plant energy resource: sunlight. Their height gives them access to sunlight while also shading the ground and other low-lying species. After the fire, however, these trees are no longer dominant. Thus, the first plants to grow back are usually annual plants followed, within a few years, by quickly-growing and spreading grasses along with other pioneer species. Due to changes in the environment brought on by the growth of the grasses and other species, over many years, shrubs will emerge along with small pine, oak, and hickory trees. These organisms are called intermediate species. Eventually, over 150 years, the forest will reach its equilibrium point where species composition is no longer changing and resembles the community before the fire. This equilibrium state is referred to as the climax community, which will remain stable until the next disturbance.

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45.6: Innate Animal Behavior

Topic hierarchy

- 45.6A: Introduction to Animal Behavior
- 45.6B: Movement and Migration
- 45.6C: Animal Communication and Living in Groups
- 45.6D: Altruism and Populations
- 45.6E: Mating Systems and Sexual Selection

45.6A: Introduction to Animal Behavior

Behavior is the change in activity of an organism in response to a stimulus and can be grouped as innate or learned.

Learning Objectives

- Distinguish between innate and learned behaviors

Key Points

- Behavioral biology is the study of the biological and evolutionary bases for changes in activity in response to a stimulus.
- Comparative psychology is an extension of work done in human and behavioral psychology. Ethology is an extension of genetics, evolution, anatomy, physiology, and other biological disciplines.
- Innate behaviors have a strong genetic component and are largely independent of environmental influences; they are “hard wired.”
- Learned behaviors result from environmental conditioning; they allow an organism to adapt to changes in the environment and are modified by previous experiences..

Key Terms

- **behavioral biology**: A systematic approach to the understanding of human and animal behavior assuming that the behavior of a human or animal is a consequence of that individual’s history.
- **comparative psychology**: The scientific study of the behavior and mental processes of non-human animals, especially as these relate to the phylogenetic history, adaptive significance, and development of behavior.

Behavior is the change in activity of an organism in response to a stimulus. Behavioral biology is the study of the biological and evolutionary bases for such changes. The idea that behaviors evolved as a result of the pressures of natural selection is not new.

Animal behavior has been studied for decades, by biologists in the science of ethology, by psychologists in the science of comparative psychology, and by scientists of many disciplines in the study of neurobiology. Although there is overlap between these disciplines, scientists in these behavioral fields take different approaches. Comparative psychology is an extension of work done in human and behavioral psychology. Ethology is an extension of genetics, evolution, anatomy, physiology, and other biological disciplines. One cannot study behavioral biology without touching on both comparative psychology and ethology.

Figure 45.6A.1: **A Range of Animal Behaviors**: Ethology has many aspects ranging from animal communication, emotions, culture, learning and sexuality.

One goal of behavioral biology is to distinguish the innate behaviors, which have a strong genetic component and are largely independent of environmental influences, from the learned behaviors, which result from environmental conditioning.

Innate behavior, or instinct, is important because there is no risk of an incorrect behavior being learned. These behaviors are “hard wired” into the system. In contrast, learned behaviors are flexible, dynamic, and can be altered relative to changes in the environment. Learned behaviors, even though they may have instinctive components, allow an organism to adapt to changes in the environment and are modified by previous experiences. Simple learned behaviors include habituation and imprinting—both are important to the maturation process of young animals.

45.6B: Movement and Migration

Innate behaviors, such as kinesis, taxis, and migration, are instinctual responses to external stimuli.

Learning Objectives

- Distinguish between kinesis, taxis, and migration in response to stimuli

Key Points

- Innate behaviors are instinctual, relying on responses to stimuli.
- Kinesis is the undirected movement in response to a stimulus, which can include orthokinesis (related to speed) or klinokinesis (related to turning).
- Taxis is the directed movement towards or away from a stimulus, which can be in response to light (phototaxis), chemical signals (chemotaxis), or gravity (geotaxis).
- Migration is an innate behavior characterized as the long-range seasonal movement of animals; it is an evolved, adapted response to variation in resource availability.
- Migration is a variable innate behavior as some migrating species always migrate (obligate migration) while in other animals, only a portion of the population migrates (incomplete migration).

Key Terms

- **orthokinesis**: the speed of movement of the individual is dependent upon the intensity of the stimulus
- **taxis**: the movement of an organism in response to a stimulus; similar to kinesis, but more direct
- **kinesis**: the undirected movement of an organism in response to an external stimulus

Innate behaviors: movement and migration

Innate or instinctual behaviors rely on response to stimuli. The simplest example of this is a reflex action: an involuntary and rapid response to stimulus. To test the “knee-jerk” reflex, a doctor taps the patellar tendon below the kneecap with a rubber hammer. The stimulation of the nerves there leads to the reflex of extending the leg at the knee. This is similar to the reaction of someone who touches a hot stove and instinctually pulls his or her hand away. Even humans, with our great capacity to learn, still exhibit a variety of innate behaviors.

Kinesis and taxis

Another activity or movement of innate behavior is kinesis: undirected movement in response to a stimulus. Orthokinesis is the increased or decreased speed of movement of an organism in response to a stimulus. Woodlice, for example, increase their speed of movement when exposed to high or low temperatures. This movement, although random, increases the probability that the insect spends less time in the unfavorable environment. Another example is klinokinesis, an increase in turning behaviors. It is exhibited by bacteria such as *E. coli* which, in association with orthokinesis, helps the organisms randomly find a more hospitable environment.

A similar, but more-directed version of kinesis is taxis: the directed movement towards or away from a stimulus. This movement can be in response to light (phototaxis), chemical signals (chemotaxis), or gravity (geotaxis). It can be directed toward (positive) or away (negative) from the source of the stimulus. An example of a positive chemotaxis is exhibited by the unicellular protozoan *Tetrahymena thermophila*. This organism swims using its cilia, at times moving in a straight line and at other times making turns. The attracting chemotactic agent alters the frequency of turning as the organism moves directly toward the source, following the increasing concentration gradient.

Migration as innate behavior

Migration is the long-range seasonal movement of animals. An evolved, adapted response to variation in resource availability, it is a common phenomenon found in all major groups of animals. Birds fly south for the winter to get to warmer climates with sufficient food, while salmon migrate to their spawning grounds. The popular 2005 documentary *March of the Penguins* followed the 62-mile migration of emperor penguins through Antarctica to bring food back to their breeding site and to their young. Wildebeests migrate over 1800 miles each year in search of new grasslands.

Figure 45.6B.1: **Migration**: Wildebeests migrate in a clockwise fashion over 1800 miles each year in search of rain-ripened grass.

Although migration is thought of as an innate behavior, only some migrating species always migrate (obligate migration). Animals that exhibit facultative migration can choose to migrate or not. Additionally, in some animals, only a portion of the population migrates, whereas the rest does not migrate (incomplete migration). For example, owls that live in the tundra may migrate in years when their food source, small rodents, is relatively scarce, but not migrate during the years when rodents are plentiful.

45.6C: Animal Communication and Living in Groups

Animals communicate using signals, which can be chemical (pheromones), aural (sound), visual (courtship displays), or tactile (touch).

Learning Objectives

- Differentiate among the ways in which animals communicate

Key Points

- Animals need to communicate with one another in order to successfully mate, which usually involves one animal signaling another; the energy-intensive behaviors or displays associated with mating are called mating rituals.
- Animal signaling is not the same as the communication we associate with language, which has been observed only in humans, but may also occur in some non-human primates and cetaceans.
- Animal communication by stimuli known as signals may be instinctual, learned, or a combination of both.

Key Terms

- **pheromone**: a chemical secreted by an animal that affects the development or behavior of other members of the same species, functioning often as a means of attracting a member of the opposite sex

Innate behaviors: living in groups

Not all animals live in groups, but even those that live relatively-solitary lives (with the exception of those that can reproduce asexually) must mate. Mating usually involves one animal signaling another so as to communicate the desire to mate. There are several types of energy-intensive behaviors or displays associated with mating called mating rituals. Other behaviors found in populations that live in groups are described in terms of which animal benefits from the behavior. In selfish behavior, only the animal in question benefits; in altruistic behavior, one animal's actions benefit another animal; cooperative behavior occurs when both animals benefit. All of these behaviors involve some sort of communication between population members.

Communication within a species

Animals communicate with each other using stimuli known as signals. These signals are chemical (pheromones), aural (sound), visual (courtship and aggressive displays), or tactile (touch). These types of communication may be instinctual, learned, or a combination of both. These are not the same as the communication we associate with language, which has been observed only in humans and, perhaps, in some species of primates and cetaceans.

A pheromone is a secreted, chemical signal used to obtain a response from another individual of the same species. The purpose of pheromones is to elicit a specific behavior from the receiving individual. Pheromones are especially common among social insects, but they are used by many animal species to attract the opposite sex, to sound alarms, to mark food trails, and to elicit other, more-complex behaviors. Even humans are thought to respond to certain pheromones called axillary steroids. These chemicals influence human perception of other people. In one study, they were responsible for a group of women synchronizing their menstrual cycles. The role of pheromones in human-to-human communication is still somewhat controversial and continues to be researched.

Songs are an example of an aural signal: one that needs to be heard by the recipient. Perhaps the best known of these are songs of birds, which identify the species and are used to attract mates. Other well-known songs are those of whales, which are of such low frequency that they can travel long distances underwater. Dolphins communicate with each other using a wide variety of vocalizations. Male crickets make chirping sounds using a specialized organ to attract a mate, repel other males, and to announce a successful mating.

Courtship displays are a series of ritualized visual behaviors (signals) designed to attract and convince a member of the opposite sex to mate. These displays are ubiquitous in the animal kingdom. They often involve a series of steps, including an initial display by one member followed by a response from the other. If at any point the display is performed incorrectly or a proper response is not given, the mating ritual is abandoned and the mating attempt will be unsuccessful.

Figure 45.6C.1: **Courtship displays**: A male peacock's extravagant, eye-spotted tail is used in courtship displays to attract a mate.

Aggressive displays are also common in the animal kingdom. As, for example, when a dog bares its teeth to get another dog to back down. Presumably, these displays communicate not only the willingness of the animal to fight, but also its fighting ability. Although these displays do signal aggression on the part of the sender, it is thought that they are actually a mechanism to reduce the amount of fighting that occurs between members of the same species: they allow individuals to assess the fighting ability of their opponent and thus decide whether it is "worth the fight."

Distraction displays are seen in birds and some fish. They are designed to attract a predator away from the nest that contains their young. This is an example of an altruistic behavior: it benefits the young more than the individual performing the display, which is putting itself at risk by doing so.

Many animals, especially primates, communicate with other members in the group through touch. Activities such as grooming, touching the shoulder or root of the tail, embracing, lip contact, and greeting ceremonies have all been observed in the Indian langur, an Old World monkey. Similar behaviors are found in other primates, especially in the great apes.

45.6D: Altruism and Populations

Altruistic behaviors may be explained by the natural instinct to improve the chances of passing on one's genes.

Learning Objectives

- Explain how altruistic behaviors can benefit populations

Key Points

- Behaviors that lower the fitness of the individual, but increase the fitness of another individual are termed altruistic; why altruistic behaviors exist has been the topic of some debate.
- One explanation for altruistic-type behaviors is found in the genetics of natural selection and the "selfish gene " theory: although a gene cannot be selfish in the human sense, it may appear that way if the sacrifice of an individual benefits related individuals that share genes that are identical.
- Even less-related individuals, those with less genetic identity than that shared by parent and offspring, benefit from seemingly-altruistic behavior, such as sterile worker bees protecting the queen.
- Unrelated individuals may also act altruistically to each other, which seems to defy the "selfish gene" explanation; however, this altruism is typically reciprocal, in that both benefit from the interaction.
- Most of the behaviors described when speaking of altruism do not seem to satisfy the definition of "pure" altruism; some evolutionary game theorists suggest that we get rid of the terms "altruistic" and "selfish" altogether since they describe human behavior.

Key Terms

- **kin selection**: an evolutionary mechanism by which an organism's behavior benefits the reproductive success of its relatives, including at the expense of its own survival or reproduction
- **altruism**: devotion to the interests of others; brotherly kindness; opposed to egoism or selfishness
- **game theory**: a branch of applied mathematics that studies strategic situations in which individuals or organizations choose various actions in an attempt to maximize their returns

Altruistic Behaviors

Behaviors that lower the fitness of the individual engaging in the behavior, but increase the fitness of another individual, are termed altruistic. Examples of such behaviors are seen widely across the animal kingdom. Social insects, such as worker bees, have no ability to reproduce, yet they maintain the queen so she can populate the hive with her offspring. Meerkats keep a sentry standing guard to warn the rest of the colony about intruders, even though the sentry is putting itself at risk. Wolves and wild dogs bring meat to pack members not present during a hunt. Lemurs take care of infants unrelated to them. Although on the surface these behaviors appear to be altruistic, it may not be so simple.

Why Does Altruism Exist?

There has been much discussion over why altruistic behaviors exist. Do these behaviors lead to overall evolutionary advantages for their species ? Do they help the altruistic individual pass on its own genes? And what about such activities between unrelated individuals? One explanation for altruistic-type behaviors is found in the genetics of natural selection. In the 1976 book, *The Selfish Gene*, scientist Richard Dawkins attempted to explain many seemingly-altruistic behaviors from the viewpoint of the gene itself. Although a gene obviously cannot be selfish in the human sense, it may appear that way if the sacrifice of an individual benefits related individuals that share genes that are identical by descent (present in relatives because of common lineage). Mammal parents make this sacrifice to take care of their offspring. Emperor penguins migrate miles in harsh conditions to bring food back for their young. Selfish gene theory has been controversial over the years and is still discussed among scientists in related fields.

Figure 45.6D.1: **Altruistic actions**: Emperor penguins migrate miles in harsh conditions to bring back food for their young. Are extreme, altruistic actions such as these motivated by the selfish need to pass on genes?

Even less-related individuals (those with less genetic identity than that shared by parent and offspring) benefit from seemingly altruistic behavior. The activities of social insects such as bees, wasps, ants, and termites are good examples. Sterile workers in these societies take care of the queen because they are closely related to it; as the queen has offspring, she is passing on genes from the workers indirectly. Thus, it is of fitness benefit for the worker to maintain the queen without having any direct chance of passing on its genes due to its sterility. The lowering of individual fitness to enhance the reproductive fitness of a relative and, thus, one's inclusive fitness evolves through kin selection. This phenomenon can explain many superficially-altruistic behaviors seen in animals. However, these behaviors may not be truly defined as altruism in these cases because the actor is actually increasing its own fitness either directly (through its own offspring) or indirectly (through the inclusive fitness it gains through relatives that share genes with it).

Unrelated individuals may also act altruistically to each other; this seems to defy the "selfish gene" explanation. An example of this is observed in many monkey species where a monkey will present its back to an unrelated monkey to have that individual pick the parasites from its fur. After a certain amount of time, the roles are reversed and the first monkey now grooms the second monkey. Thus, there is reciprocity in the behavior. Both benefit from the interaction and their fitness is raised more than if neither cooperated or if one cooperated and the other did not. This behavior is still not necessarily altruism, as the "giving" behavior of the actor is based on the expectation that it will be the "receiver" of the behavior in the future; a concept termed reciprocal altruism. Reciprocal altruism requires that individuals repeatedly encounter each other, often the result of living in the same social group, and that cheaters (those that never "give back") are punished.

Evolutionary Game Theory and Altruism

According to evolutionary game theory, a modification of classical game theory in mathematics, many of these so-called "altruistic behaviors" are not altruistic at all. The definition of "pure" altruism, based on human behavior, is an action that benefits another without any direct benefit to oneself. Most of the behaviors previously described do not seem to satisfy this definition; game theorists are good at finding "selfish" components in them. Others have argued that the terms "selfish" and "altruistic" should be dropped completely when discussing animal behavior, as they describe human behavior and may not be directly applicable to instinctual animal activity. What is clear, though, is that heritable behaviors that improve the chances of passing on one's genes or a portion of one's genes are favored by natural selection and will be retained in future generations as long as those behaviors convey a fitness advantage.

45.6E: Mating Systems and Sexual Selection

In mating, there are two types of selection (intersexual, intrasexual) and three mating systems (monogamous, polygynous, polyandrous).

Learning Objectives

- Differentiate among monogamous, polygynous, and polyandrous mating systems, and distinguish between intersexual and intrasexual mate selection

Key Points

- Two types of mate selection occur: intersexual selection (the choice of a mate where individuals of one sex choose mates of the other sex) and intrasexual selection (the competition for mates between species members of the same sex).
- Three general mating systems, all involving innate as opposed to learned behaviors, are seen in animal populations: monogamous (monogamy), polygynous (polygyny), and polyandrous (polyandry).
- In monogamous systems, one male and one female are paired for at least one breeding season; although in some animals, these partnerships can last even longer, sometimes an entire lifetime; males provide substantial parental care.
- Polygynous mating refers to one male mating with multiple females; in these situations, the female must be responsible for most of the parental care as the single male is not capable of providing care to that many offspring.
- In polyandrous mating systems, one female mates with many males; these types of systems are much rarer than monogamous and polygynous mating systems.

Key Terms

- **polyandry**: the mating pattern whereby a female copulates with several males
- **polygyny**: the mating patterns whereby a male copulates with several females
- **monogamy**: a form of sexual bonding involving an exclusive pair bond between two individuals

Finding Sexual Partners

Not all animals reproduce sexually, but many that do have the same challenge: they need to find a suitable mate and often have to compete with other individuals to obtain one. Significant energy is spent in the process of locating, attracting, and mating with a sex partner.

Types of Mate Selection

Two types of selection that occur during the process of choosing a mate may be involved in the evolution of reproductive traits called secondary sexual characteristics. These types are: intersexual selection (the choice of a mate where individuals of one sex choose mates of the other sex) and intrasexual selection (the competition for mates between species members of the same sex). Intersexual selection is often complex because choosing a mate may be based on a variety of visual, aural, tactile, and chemical cues. An example of intersexual selection is when female peacocks choose to mate with the male with the brightest plumage. This type of selection often leads to traits in the chosen sex that do not enhance survival, but are those traits most attractive to the opposite sex (often at the expense of survival). Intrasexual selection involves mating displays and aggressive mating rituals such as rams butting heads; the winner of these battles is the one that is able to mate. Many of these rituals use up considerable energy, but result in the selection of the healthiest, strongest, and/or most dominant individuals for mating.

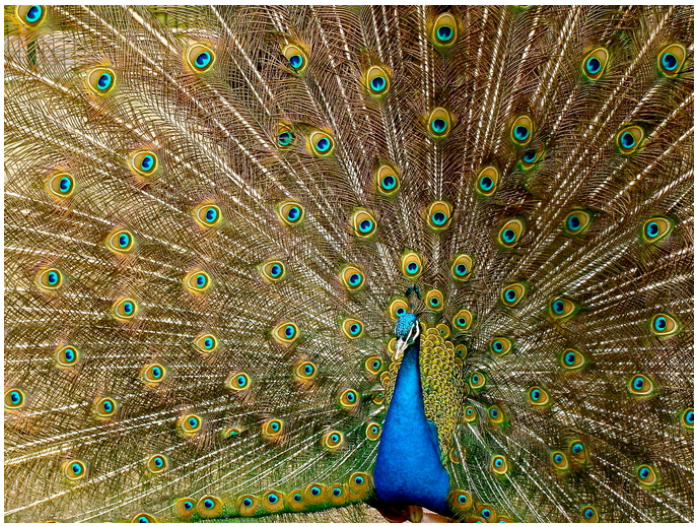

Figure 45.6E. 1: **Courtship display of the male peacock**: This male peacock's courtship display—his intricate, colorful tail feathers—is designed to attract potential mates.

Mating Systems

Three general mating systems, all involving innate as opposed to learned behaviors, are seen in animal populations: monogamous (monogamy), polygynous (polygyny), and polyandrous (polyandry).

In monogamous systems, one male and one female are paired for at least one breeding season. In some animals, such as the gray wolf, these associations can last much longer, even a lifetime. Several explanations have been proposed for this type of mating system. The "mate-guarding hypothesis" states that males stay with the female to prevent other males from mating with her. This behavior is advantageous in such situations where mates are scarce and difficult to find. Another explanation is the "male-assistance hypothesis," where males that remain with a female to help guard and rear their young will have more and healthier offspring. Monogamy is observed in many bird populations where, in addition to the parental care from the female, the male is also a major provider of parental care for the chicks. A third explanation for the evolutionary advantages of monogamy is the "female-enforcement hypothesis." In this scenario, the female ensures that the male does not have other offspring that might compete with her own, so she actively interferes with the male's signaling to attract other mates.

Polygynous mating refers to one male mating with multiple females. In these situations, the female must be responsible for most of the parental care as the single male is not capable of providing care to that many offspring. In resourced-based polygyny, males compete for territories with the best resources. They then mate with females that enter the territory, drawn to its resource richness. The female benefits by mating with a dominant, genetically-fit male; however, it is at the cost of having no male help in caring for the offspring. An example is seen in the yellow-rumped honeyguide, a bird whose males defend beehives because the females feed on the wax. As the females approach, the male defending the nest will mate with them. Harem mating structures are a type of polygynous system where certain males dominate mating while controlling a territory with resources. Elephant seals, where the alpha male dominates the mating within the group, are an example. A third type of polygyny is a lek system. Here there is a communal courting area where several males perform elaborate displays for females; the females choose their mate from this group. This behavior is observed in several bird species.

In polyandrous mating systems, one female mates with many males. These types of systems are much rarer than monogamous and polygynous mating systems. In pipefishes and seahorses, males receive the eggs from the female, fertilize them, protect them within a pouch, and give birth to the offspring. Therefore, the female is able to provide eggs to several males without the burden of carrying the fertilized eggs.

Figure 45.6E. 1: **Polyandry**: Seahorses are a good example of a polyandrous mating system, in which one female mates with several males. In seahorse reproduction, the male receives the eggs from the female, fertilizes them, protects them within a pouch, and gives birth to the offspring.

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45.7: Learned Animal Behavior

Topic hierarchy

- 45.7A: Simple Learned Behaviors
- 45.7B: Conditioned Behavior
- 45.7C: Cognitive Learning and Sociobiology

Cognitive learning relies on cognitive processes such as reasoning and abstract thinking; it is much more efficient than conditioning.

45.7A: Simple Learned Behaviors

Simple learned behaviors include habituation and imprinting, both of which are important to the maturation process of young animals.

Learning Objectives

- Distinguish between the simple learned behaviors of habituation and imprinting

Key Points

- Learned behaviors stand in opposition to innate behaviors: while learned behaviors may have an innate component, they allow the organism to modify its behavior according to environmental factors or previous experiences.
- Habituation is a simple form of learning in which an animal stops responding to a stimulus after a period of repeated exposure; it is a form of non-associative learning, as the stimulus is not associated with any punishment or reward.
- Imprinting is a type of learning that occurs at a particular age or a life stage that is rapid and independent of the species involved.

Key Terms

- **imprinting**: any kind of phase-sensitive learning (learning occurring at a particular age or a particular life stage) that is rapid and apparently independent of the consequences of behavior
- **habituation**: a learned behavior involving modifying behavior according to the environment or previous expriences
- **innate**: inborn; native; natural

Simple Learned Behaviors

The majority of the behaviors discussed in previous sections are innate or at least have an innate component. In other words, variations on the innate behaviors may be learned. Innate behaviors are inherited and do not change in response to signals from the environment. Conversely, learned behaviors, even though they may have instinctive components, allow an organism to adapt to changes in the environment and are modified by previous experiences. Simple learned behaviors include habituation and imprinting, both of which are important to the maturation process of young animals.

Habituation

Habituation is a simple form of learning in which an animal stops responding to a stimulus after a period of repeated exposure. This is a form of non-associative learning as the stimulus is not associated with any punishment or reward. Prairie dogs typically sound an alarm call when threatened by a predator, but they become habituated to the sound of human footsteps when no harm is associated with this sound; therefore, they no longer respond to them with an alarm call. In this example, habituation is specific to the sound of human footsteps, as the animals still respond to the sounds of potential predators.

Imprinting

Imprinting is a type of learning that occurs at a particular age or a life stage that is rapid and independent of the species involved. Hatchling ducks recognize the first adult they see, their mother, and make a bond with her. A familiar sight is ducklings walking or swimming after their mothers. This type of non-associative learning is very important in the maturation process of these animals as it encourages them to stay near their mother in order to be be protected, greatly increasing their chances of survival. However, if newborn ducks see a human before they see their mother, they will imprint on the human and follow it in just the same manner as they would follow their real mother.

Figure 45.7A.1: **Imprinting**: The attachment of ducklings to their mother is an example of imprinting.

45.7B: Conditioned Behavior

In classical conditioning, a behavior is paired with an unrelated stimulus; in operant conditioning, behaviors are modified by consequences.

Learning Objectives

- Distinguish between classical and operant conditioning techniques

Key Points

- In classical conditioning, a response called the conditioned response is associated with a stimulus that it had previously not been associated with, the conditioned stimulus; the response to the original, unconditioned stimulus is called the unconditioned response.
- Classical conditioning is a major tenet of behaviorism, a branch of psychological philosophy that proposes that all actions, thoughts, and emotions of living things are behaviors that can be treated by behavior modification and changes in the environment.
- In operant conditioning, the conditioned behavior is gradually modified by its consequences as the animal responds to the stimulus.
- Operant conditioning relies on the use of reinforcement (i.e. a reward) and/or punishment to modify a conditioned behavior; in this way, the animal is conditioned to associate a type of behavior with the punishment or reward.

Key Terms

- **classical conditioning**: the use of a neutral stimulus, originally paired with one that invokes a response, to generate a conditioned response
- **operant conditioning**: a technique of behavior modification through positive and negative reinforcement and positive and negative punishment

Conditioned Behavior

Conditioned behaviors are types of associative learning where a stimulus becomes associated with a consequence. Two types of conditioning techniques include classical and operant conditioning.

Classical Conditioning

In classical conditioning, a response called the conditioned response is associated with a stimulus that it had previously not been associated with, the conditioned stimulus. The response to the original, unconditioned stimulus is called the unconditioned response. The most cited example of classical conditioning is Ivan Pavlov's experiments with dogs. In Pavlov's experiments, the unconditioned response was the salivation of dogs in response to the unconditioned stimulus of seeing or smelling their food. The conditioning stimulus that researchers associated with the unconditioned response was the ringing of a bell. During conditioning, every time the animal was given food, the bell was rung. This was repeated during several trials. After some time, the dog learned to associate the ringing of the bell with food and to respond by salivating. After the conditioning period was finished, the dog would respond by salivating when the bell was rung, even when the unconditioned stimulus (the food) was absent. Thus, the ringing of the bell became the conditioned stimulus and the salivation became the conditioned response. Although it is thought by some scientists that the unconditioned and conditioned responses are identical, Pavlov discovered that the saliva in the conditioned dogs had characteristic differences when compared to the unconditioned dog.

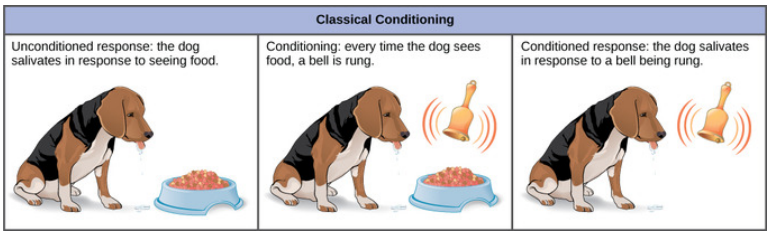


Figure 45.7B.1: **Classical conditioning**: In the classic Pavlovian response, the dog becomes conditioned to associate the ringing of the bell with food.

Some believe that this type of conditioning requires multiple exposures to the paired stimulus and response, but it is now known that this is not necessary in all cases; some conditioning can be learned in a single pairing experiment. Classical conditioning is a major tenet of behaviorism, a branch of psychological philosophy that proposes that all actions, thoughts, and emotions of living things are behaviors that can be treated by behavior modification and changes in the environment.

Operant Conditioning

In operant conditioning, the conditioned behavior is gradually modified by its consequences as the animal responds to the stimulus. A major proponent of such conditioning was psychologist B.F. Skinner, the inventor of the Skinner box. Skinner put rats in his boxes that contained a lever that would dispense food to the rat when depressed. While initially the rat would push the lever a few times by accident, it eventually associated pushing the lever with getting the food. This type of learning is an example of operant conditioning. Operant learning is the basis of most animal training: the conditioned behavior is continually modified by positive or negative reinforcement (such as being given a reward or having a negative stimulus removed) or by positive or negative punishment (such as being given a punishment or having a pleasing stimulus removed). In this way, the animal is conditioned to associate a type of behavior with the punishment or reward. Over time, the animal can be induced to perform behaviors that they would not have done in the wild, such as the “tricks” dolphins perform at marine amusement park shows.

Figure 45.7B.1: **Operant conditioning**: The training of dolphins by rewarding them with food is an example of positive reinforcement operant conditioning.

45.7C: Cognitive Learning and Sociobiology

Learning Objectives

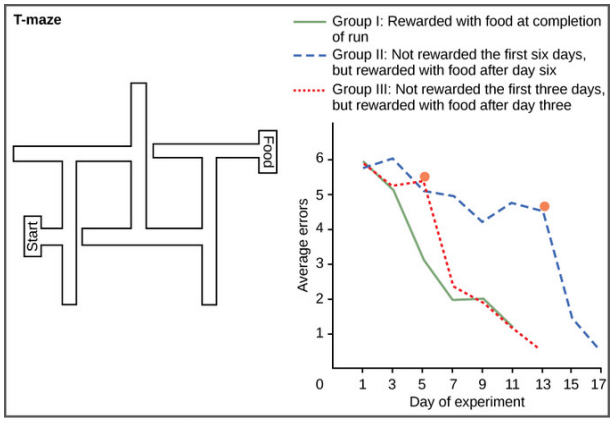
- Describe research models that indicate the presence of cognitive learning in animals

Cognitive Learning

Classical and operant conditioning are inefficient ways for humans and other intelligent animals to learn. Some primates, including humans, are able to learn by imitating the behavior of others and by taking instructions. The development of complex language by humans has made cognitive learning, or the manipulation of information using the mind, the most prominent method of human learning. In fact, that is how you are learning right now by reading this information. As students read, they can make mental images of objects or organisms, imagining changes to them or behaviors by them as they anticipate the consequences. In addition to visual processing, cognitive learning is also enhanced by remembering past experiences, touching physical objects, hearing sounds, tasting food, and a variety of other sensory-based inputs. Cognitive learning is so powerful that it can be used to understand conditioning (discussed in the previous concept) in detail. In the reverse scenario, conditioning cannot help someone learn about cognition.

Classic work on cognitive learning was done by Wolfgang Köhler with chimpanzees. He demonstrated that these animals were capable of abstract thought by showing that they could learn how to solve a puzzle. When a banana was hung in their cage too high for them to reach along with several boxes placed randomly on the floor, some of the chimps were able to stack the boxes one on top of the other, climb on top of them, and get the banana. This implies that they could visualize the result of stacking the boxes even before they had performed the action. This type of learning is much more powerful and versatile than conditioning.

Cognitive learning is not limited to primates, although they are the most efficient in using it. Maze-running experiments done with rats in the 1920s were the first to show cognitive skills in a simple mammal. The motivation for the animals to work their way through the maze was the presence of a piece of food at its end. In these studies, the animals in Group I were run in one trial per day and had food available to them each day on completion of the run. Group II rats were not fed in the maze for the first six days and then subsequent runs were done with food for several days after. Group III rats had food available on the third day and every day thereafter. The results were that the control rats, Group I, learned quickly, figuring out how to run the maze in seven days. Group III did not learn much during the three days without food, but rapidly caught up to the control group when given the food reward. Group II learned very slowly for the six days with no reward to motivate them. They did not begin to catch up to the control group until the day food was given; it then took two days longer to learn the maze.



Redrawn after H. C. Blodgett, The effect of the introduction of reward upon the maze performance of rats. Univ. Calif. Publ. Psychol., 1929, 4, No. 8, pages 117 and 120.

Figure 45.7C.1: **Cognitive learning**: Group I (the green solid line) found food at the end of each trial; group II (the blue dashed line) did not find food for the first 6 days; and group III (the red dotted line) did not find food during runs on the first three days. Notice that rats given food earlier learned faster and eventually caught up to the control group. The orange dots on the group II and III lines show the days when food rewards were added to the mazes.

It may not be immediately obvious that this type of learning is different from conditioning. Although one might be tempted to believe that the rats simply learned how to find their way through a conditioned series of right and left turns, E.C. Tolman proved a decade later that the rats were making a representation of the maze in their minds, which he called a "cognitive map." This was an early demonstration of the power of cognitive learning and how these abilities were not limited just to humans.

Sociobiology

Sociobiology is an interdisciplinary science originally popularized by social insect researcher E.O. Wilson in the 1970s. Wilson defined the science as "the extension of population biology and evolutionary theory to social organization." The main thrust of sociobiology is that animal and human behavior, including aggressiveness and other social interactions, can be explained almost solely in terms of genetics and natural selection. This science is controversial; some have criticized the approach for ignoring the environmental effects on behavior. This is another example of the "nature versus nurture" debate of the role of genetics versus the role of environment in determining an organism's characteristics.

Sociobiology also links genes with behaviors and has been associated with "biological determinism," the belief that all behaviors are hardwired into our genes. No one disputes that certain behaviors can be inherited and that natural selection plays a role retaining them. It is the application of such principles to human behavior that sparks this controversy, which remains active today.

Key Points

- Cognitive learning involves the manipulation of information using the mind; it is a great deal more powerful than either operant or classical conditioning.
- The development of complex language by humans has made cognitive learning the most prominent method of human learning.
- Cognitive learning is not limited to primates; rats have demonstrated the ability to build cognitive maps, as well, which are mental representations used to acquire, code, store, recall, and decode information about the environment.
- Sociobiology argues that all animal and human behavior, including aggressiveness and other social interactions, can be explained almost solely in terms of genetics and natural selection.
- Sociobiology is controversial: some have criticized the approach for ignoring the environmental effects on behavior and for being similar to "biological determinism," or the belief that all behaviors are hardwired into our genes.

Key Terms

- **cognitive map**: a mental epresentation which serves an organism to acquire, code, store, recall, and decode information about the relative locations and attributes of phenomena in their everyday environment
- **cognitive learning**: the process by which one acquires knowledge or skill in cognitive processes, which include reasoning, abstract thinking, and problem solving
- **sociobiology**: the science that applies the principles of evolutionary biology to the study of social behaviour in both humans and animals

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CHAPTER OVERVIEW

46: ECOSYSTEMS

46.1: ECOLOGY OF ECOSYSTEMS

46.1A: ECOSYSTEM DYNAMICS
46.1B: FOOD CHAINS AND FOOD WEBS
46.1C: STUDYING ECOSYSTEM DYNAMICS
46.1D: MODELING ECOSYSTEM DYNAMICS
46.2: ENERGY FLOW THROUGH ECOSYSTEMS

46.2A: STRATEGIES FOR ACQUIRING ENERGY
46.2B: PRODUCTIVITY WITHIN TROPHIC LEVELS
46.2C: TRANSFER OF ENERGY BETWEEN TROPHIC LEVELS
46.2D: ECOLOGICAL PYRAMIDS
Ecological pyramids, which can be inverted or upright, depict biomass, energy, and the number of organisms in each trophic level.

46.2E: BIOLOGICAL MAGNIFICATION
When toxic substances are introduced into the environment, organisms at the highest trophic levels suffer the most damage.

46.3: BIOGEOCHEMICAL CYCLES

46.3A: BIOGEOCHEMICAL CYCLES
46.3B: THE WATER (HYDROLOGIC) CYCLE
46.3C: THE CARBON CYCLE
46.3D: THE NITROGEN CYCLE
46.3E: THE PHOSPHORUS CYCLE
46.3E: THE SULFUR CYCLE

46.1: Ecology of Ecosystems

Topic hierarchy

- 46.1A: Ecosystem Dynamics
- 46.1B: Food Chains and Food Webs
- 46.1C: Studying Ecosystem Dynamics
- 46.1D: Modeling Ecosystem Dynamics

46.1A: Ecosystem Dynamics

Ecosystems are controlled both by external and internal factors; they can be both resistant or resilient to ecosystem disturbances.

Learning Objectives

- Explain ecosystem dynamics

Key Points

- Biotic and abiotic factors interact through nutrient cycles and energy flows.
- External factors control resource inputs and are not influenced by the ecosystem itself.
- Internal factors are processes that exist within the ecosystem, such as decomposition, succession, and the types of species present.
- While in equilibrium, an ecosystem can recover from small changes through negative feedback, returning to its original state.
- Resistance describes an ecosystem's ability to resist disturbances to the ecosystems dynamics.
- Human disturbances to ecosystems can overwhelm the ecosystem's resilience, crippling its ability to return to equilibrium.

Key Terms

- **resilience**: the speed with which an ecosystem returns to its initial state after a disturbance
- **equilibrium**: the condition of a system in which competing influences are balanced, resulting in no net change
- **resistance**: the tendency of a system to remain close to its equilibrium state, despite disturbances

Ecosystem Dynamics

An ecosystem is a community of living organisms (plants, animals, and microbes) existing in conjunction with the nonliving components of their environment (air, water, and mineral soil), interacting as a system. These biotic and abiotic components are linked together through nutrient cycles and energy flows. As ecosystems are defined by the network of interactions among organisms, or between organisms and their environment, they can be of any size, but usually encompass specific, limited spaces.

Internal and External Factors

Ecosystems are dynamic entities controlled both by external and internal factors. External factors, such as climate and the parent material that forms the soil, control the overall structure of an ecosystem and the way things work within it, but are not themselves influenced by the ecosystem. While the resource inputs are generally controlled by external processes, the availability of these resources within the ecosystem is controlled by internal factors such as decomposition, root competition, or shading. Other internal factors include disturbance, succession, and the types of species present. From one year to another, ecosystems experience variation in their biotic and abiotic environments. A drought, an especially cold winter, and a pest outbreak all constitute short-term variability in environmental conditions. Animal populations vary from year to year, building up during resource-rich periods, but crashing as the food supply becomes scarce.

Equilibrium is the steady state of an ecosystem where all organisms are in balance with their environment and with each other. In equilibrium, any small changes to the system will be balanced by negative feedback, allowing the system to return to its original state.

Resistance and Resilience

In ecology, two parameters are used to measure changes in ecosystems: resistance and resilience. Resistance is the ability of an ecosystem to remain at equilibrium despite disturbances. Resilience is the speed at which an ecosystem recovers to equilibrium after being disturbed. Humans may impact the nature of an ecosystem to such a degree that the ecosystem can lose its resilience entirely. In these cases, external human influences can lead to the complete destruction or irreversible altering of the ecosystem equilibrium.

Figure 46.1*A*.1: **Human intervention in ecosystem equilibrium**: The Australian Aboriginal practice of "Fire-stick farming" has fundamentally modified Australian ecosystems. The legacy of this practice over long periods has resulted in forests being converted to grasslands. In this example, the forests became less and less resilient over time until the fundamental system equilibrium had changed.

The Sin Nombre Virus: Ecosystem Dynamics in a Human Population

In 1993, a change in ecosystem dynamics caused a disease outbreak in a human population. In May of 1993, an unexplained pulmonary illness struck inhabitants of the southwestern United States in an area shared by Arizona, New Mexico, Colorado and Utah known as "The Four Corners. " A young, physically fit Navajo man suffering from shortness of breath was rushed to a hospital in New Mexico and died rapidly. After further investigation, state officials located another five young, healthy people who had all died after acute respiratory failure.

When laboratory tests failed to identify the disease causing the deaths, New Mexico state health officials notified the Centers for Disease Control (CDC), the United States government agency responsible for managing potential epidemics. As additional cases of the disease were reported in the following weeks, physicians and scientists worked intensively to narrow down the list of possible causes. Virologists at the CDC linked the pulmonary syndrome with a virus – a previously unknown type of hantavirus. The hantavirus became known as *Sin Nombre*, the virus "with no name. "

Figure 46.1*A*.1: **Sin Nombre hantavirus**: After a series of sudden deaths in 1993, scientists in the Four Corners area of the Southwestern United States rushed to determine the cause. They isolated a previously unknown hantavirus that caused pulmonary failure or Hantavirus Pulmonary Syndrome (HPS). The new virus was named Sin Nombre, or virus with "no name. "

Although they identified the virus as the cause of the disease, researchers did not understand how it was transmitted. The researchers trapped and examined rodents that lived in and around the homes of the victims, and found that almost 30% of the deer mice were infected with the Sin Nombre hantavirus. The virus had been transmitted to humans via aerosolized mouse droppings, and a dramatic increase in the deer mouse population increased human infection rates.

The Four Corners area had experienced a drought until early 1993, when there were heavy snows and rainfall. The end of the drought caused an increase in vegetation, and particularly pinon nut production. With the sudden increase in food supply, the local deer mice population exploded and reproduced so rapidly that there were ten times more mice in May 1993 than there had been in May of 1992. The higher population of deer mice meant more mouse droppings and more opportunities to transmit hantavirus to humans.

Figure 46.1*A*.1: **Ecosystem dynamics can affect human populations**: The Four Corners area had been in a drought for several years. In early 1993, the rainfall caused an increase in vegetation, which caused an increase the local deer mice population. Hantavirus infected the high deer mouse population and was quickly transmitted to humans via aerosolized mouse droppings.

As part of the effort to locate the source of the virus, researchers located and examined stored samples of lung tissue from people who had died of unexplained lung disease. Some of these samples showed evidence of previous infection with Sin Nombre virus, indicating that the earlier cases of the disease had not been recognized. The Navajo Native Americans recognize a similar disease in their medical traditions, and associate its occurrence with mice.

46.1B: Food Chains and Food Webs

A food web describes the flow of energy and nutrients through an ecosystem, while a food chain is a linear path through a food web.

Learning Objectives

- Distinguish between food chains and food webs as models of energy flow in ecosystems

Key Points

- Organisms can be organized into trophic levels: primary producer, primary consumer, secondary consumer, and tertiary or higher-order consumer.
- Energy decreases in each successive trophic level, preventing more than four or five levels in a food chain.
- An ecosystem usually has two different types of food webs: a grazing food web based on photosynthetic plants or algae, along with a detrital food web based on decomposers (such as fungi).
- There are different types of food webs including grazing food webs based on photosynthetic plants (such as algae) or detrital food webs based on decomposers (such as fungi).

Key Terms

- **detritivore**: an organism that feeds on detritus; a decomposer
- **food chain**: the feeding relationships between species in a biotic community; a linear path through a food web
- **trophic level**: a particular position occupied by a group of organisms in a food chain (primary producer, primary consumer, secondary consumer, or tertiary consumer)

Food Chains and Food Webs

In ecology, a food web describes the feeding connections between organisms in a biotic community. Both energy and nutrients flow through a food web, moving through organisms as they are consumed by an organism above them in the food web. A single path of energy through a food web is called a food chain.

Trophic Levels

Each organism within a food web can be classified by trophic level according to their position within the web. Depending on an organism's location in a food web, it may be grouped into more than one of these categories. Energy and nutrients move up trophic levels in the following order:

1. Primary producers
2. Primary consumers
3. Secondary consumers
4. Tertiary and other high-level consumers

In both food webs and food chains, arrows point from an organism that is consumed to the organism that consumes it. In many ecosystems, the bottom of the food chain consists of photosynthetic organisms, such as plants or phytoplankton, known as primary producers. The organisms that consume the primary producers are herbivores: the primary consumers. Secondary consumers are usually carnivores that eat the primary consumers, while tertiary consumers are carnivores that eat other carnivores. Higher-level consumers feed on the next lower trophic levels, and so on, up to the organisms at the top of the food chain, which are called the apex consumers. Some lines within a food web may point to more than one organism; those organisms may occupy different trophic levels depending on their position in each food chain within the web.

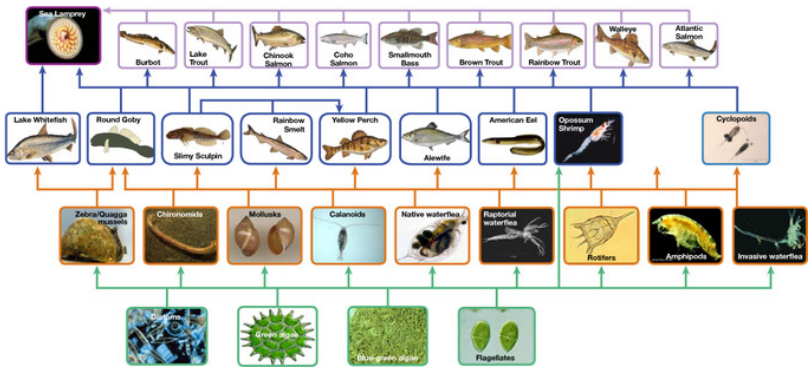

Figure 46.1B.1: **Food web**: This food web shows the interactions between organisms across trophic levels in the Lake Ontario ecosystem. Primary producers are outlined in green, primary consumers in orange, secondary consumers in blue, and tertiary (apex) consumers in purple. The opossum shrimp eats both primary producers and primary consumers; it is, therefore, both a primary consumer and a secondary consumer.

The Loss of Energy in Tropic Levels

It is rare to find food chains that have more than four or five links because the loss of energy limits the length of food chains. At each trophic level, most of the energy is lost through biological processes such as respiration or finding food. Only the energy that is directly assimilated into an animal's consumable mass will be transferred to the next level when that animal is eaten. Therefore, after a limited number of trophic energy transfers, the amount of energy remaining in the food chain cannot support a higher trophic level. Although energy is lost, nutrients are recycled through waste or decomposition.

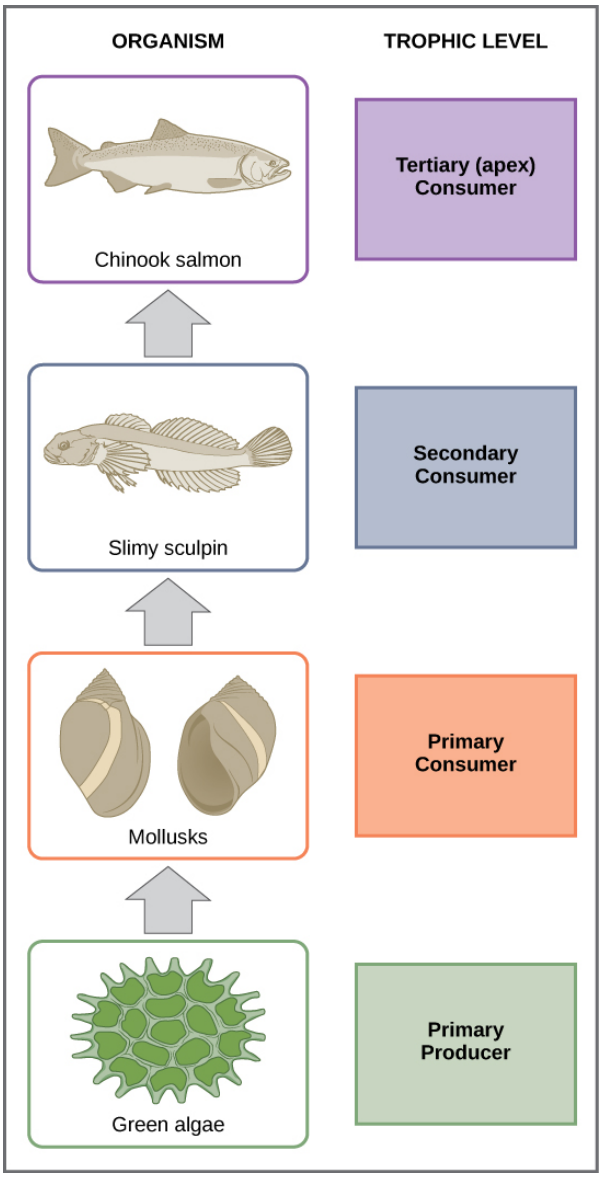

Figure 46.1B.1: **Food chain**: These are the trophic levels of a food chain in Lake Ontario. Energy and nutrients flow from photosynthetic green algae at the bottom to the salmon at the top of the food chain. There are only four links in this chain because significant energy is lost between each successive trophic level.

A scientist named Howard T. Odum demonstrated the loss of energy in each trophic level in the Silver Springs, Florida, ecosystem in the 1940s. He found that the primary producers generated 20,819 kcal/m^2/yr (kilocalories per square meter per year), the primary consumers generated 3368 kcal/m^2/yr, the secondary consumers generated 383 kcal/m^2/yr, and the tertiary consumers only generated 21 kcal/m^2/yr. In each successive trophic level, the energy available to the next level decreased significantly.

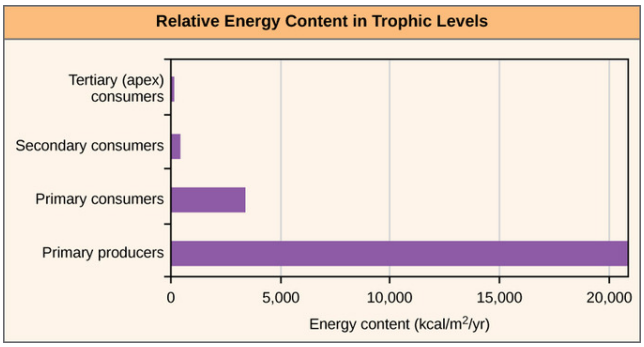

Figure 46.1*B*.1: **Energy decreases per trophic level**: The relative energy in trophic levels in a Silver Springs, Florida, ecosystem is shown. Each trophic level has less energy available and supports fewer organisms at the next level.

Types of Food Webs

Two general types of food webs are often shown interacting within a single ecosystem. As an example, a grazing food web has plants or other photosynthetic organisms at its base, followed by herbivores and various carnivores. A detrital food web consists of a base of organisms that feed on decaying organic matter (dead organisms), called decomposers or detritivores. These organisms are usually bacteria or fungi that recycle organic material back into the biotic part of the ecosystem as they themselves are consumed by other organisms. As all ecosystems require a method to recycle material from dead organisms, most grazing food webs have an associated detrital food web. For example, in a meadow ecosystem, plants may support a grazing food web of different organisms, primary and other levels of consumers, while at the same time supporting a detrital food web of bacteria, fungi, and detrivorous invertebrates feeding off dead plants and animals.

46.1C: Studying Ecosystem Dynamics

Many different models are used to study ecosystem dynamics, including holistic, experimental, conceptual, analytical, and simulation models.

Learning Objectives

- Differentiate between conceptual, analytical, and simulation models of ecosystem dynamics, and mesocosm and microcosm research studies

Key Points

- A holistic ecosystem model quantifies the dynamics of an entire ecosystem.
- Scientists can use experimental systems, such as a microcosms or mesocosms, to study ecosystems under controlled laboratory conditions.
- A conceptual model uses flow charts to show the interactions between living and nonliving components of the ecosystem.
- An analytical model uses simple mathematical formulas to predict the effects of environmental disturbances on an ecosystem's structure and dynamics.
- A simulation model predicts the effects of environmental disturbances using complex computer algorithms; they are usually fairly-reliable predictors.

Key Terms

- **mesocosm**: a small portion of the natural environment that is brought under controlled conditions for experimental purposes
- **microcosm**: an artificial, simplified ecosystem that is used to simulate and predict the behaviour of natural ecosystems under controlled conditions

Research into Ecosystem Dynamics: Ecosystem Experimentation and Modeling

Ecosystem dynamics is the study of the changes in ecosystem structure caused by environmental disturbances or by internal forces. Various research methodologies measure ecosystem dynamics. Some ecologists study ecosystems using controlled experimental systems, while some study entire ecosystems in their natural state; others use both approaches.

Holistic Ecosystem Model

A holistic ecosystem model attempts to quantify the composition, interaction, and dynamics of entire ecosystems. A food web is an example of a holistic ecosystem model, which is the most representative of the ecosystem in its natural state. However, this type of study is limited by time and expense, as well as its limited feasibility to conduct experiments on large natural ecosystems.

Experimental Systems

For these reasons, scientists study ecosystems under more controlled conditions. Experimental systems usually involve either partitioning a part of a natural ecosystem that can be used for experiments, termed a mesocosm, or by re-creating an ecosystem entirely in an indoor or outdoor laboratory environment, which is referred to as a microcosm. A major limitation to these approaches is that removing individual organisms from their natural ecosystem or altering a natural ecosystem through partitioning may change the dynamics of the ecosystem. These changes are often due to differences in species numbers and diversity, but also to environment alterations caused by partitioning (mesocosm) or re-creating (microcosm) the natural habitat. Thus, these types of experiments are not totally predictive of changes that would occur in the ecosystem from which they were gathered.

Figure 46.1C.1: **Mesocosm**: Greenhouses contribute to mesocosm studies because they allow us to control the environment and, thus, the experiment. The mesocosms in this example, tomato plants, have been placed in a greenhouse to control the air, temperature, water, and light distribution in order to observe the effects when exposed to different amounts of each factor.

As both of these approaches have their limitations, some ecologists suggest that results from these experimental systems should be used only in conjunction with holistic ecosystem studies to obtain the most representative data about ecosystem structure, function, and dynamics.

Ecosystem Models

Scientists use the data generated by these experimental studies to develop ecosystem models that demonstrate the structure and dynamics of ecosystems. Three basic types of ecosystem modeling are routinely used in research and ecosystem management: conceptual models, analytical models, and simulation models.

A conceptual model consists of flow charts to show interactions of different compartments of the living and nonliving components of the ecosystem. A conceptual model describes ecosystem structure and dynamics and shows how environmental disturbances affect the ecosystem, although its ability to predict the effects of these disturbances is limited.

Analytical and simulation models are mathematical methods of describing ecosystems that are capable of predicting the effects of potential environmental changes without direct experimentation, although with limitations in accuracy. An analytical model is created using simple mathematical formulas to predict the effects of environmental disturbances on ecosystem structure and dynamics.

A simulation model is created using complex computer algorithms to holistically model ecosystems and to predict the effects of environmental disturbances on ecosystem structure and dynamics. Ideally, these models are accurate enough to determine which components of the ecosystem are particularly sensitive to disturbances. They can serve as a guide to ecosystem managers (such as conservation ecologists or fisheries biologists) in the practical maintenance of ecosystem health.

46.1D: Modeling Ecosystem Dynamics

Conceptual models describe ecosystem structure, while analytical and simulation models use algorithms to predict ecosystem dynamics.

Learning Objectives

- Compare and contrast conceptual, analytical, and simulation models of ecosystem dynamics

Key Points

- Conceptual models are often flow charts that demonstrate the relationships between different organisms in a community and their environment, including the transfer of energy and nutrients.
- Analytical models use mathematical equations to predict and describe simple, linear components of ecosystems, such as food chains.
- Simulation models use computer algorithms to predict ecosystem dynamics; they are considered the most ecologically-realistic and accurate.

Key Terms

- **conceptual model**: a model that is represented by conceptual representations of the relationships between different organisms in a community and their environment
- **analytical model**: a model that works best when dealing with relatively simple (often linear) systems, specifically those that can be accurately described by a set of mathematical equations whose behavior is well known
- **simulation model**: a model that utilizes mathematical algorithms to predict complex responses in ecosystem dynamics

Conceptual models

Conceptual models are useful for describing ecosystem structure and dynamics and for demonstrating the relationships between different organisms in a community and their environment. Conceptual models are usually depicted graphically as flow charts. The organisms and their resources are grouped into specific compartments with arrows showing the relationship and transfer of energy or nutrients between them. These diagrams are sometimes called compartment models.

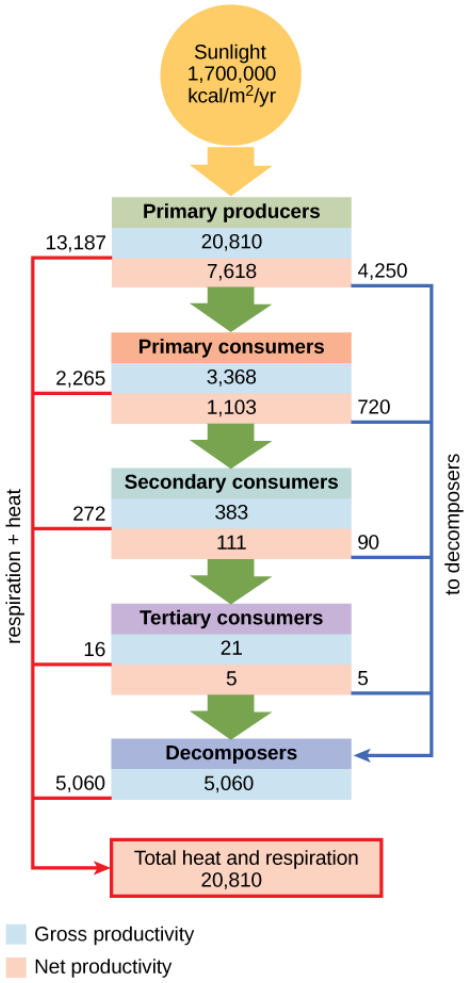

Figure 46.1D.1: **Conceptual model of energy**: This conceptual model shows the flow of energy through a spring ecosystem in Silver Springs, Florida. Notice that the energy decreases with each increase in trophic level. Conceptual models are useful for describing ecosystem structure, but are limited by their poor prediction of ecosystem changes.

To model the cycling of mineral nutrients, organic and inorganic nutrients are subdivided into those that are bioavailable (ready to be incorporated into biological macromolecules) and those that are not. For example, in a terrestrial ecosystem near a deposit of coal, carbon will be available to the plants of this ecosystem as carbon dioxide gas in a short-term period, not from the carbon-rich coal itself. However, over a longer period, microorganisms capable of digesting coal will incorporate its carbon or release it as natural gas (methane, CH_4), changing this unavailable organic source into an available one.

Human combustion of fossil fuels accelerates this conversion by releasing large amounts of carbon dioxide into the atmosphere, which may be a large contributor to the rise of the atmospheric carbon dioxide levels in the industrial age. The carbon dioxide released from burning fossil fuels is produced faster than photosynthetic organisms can use it, while the number of photosynthetic trees have decreased because of worldwide deforestation. Most scientists agree that high atmospheric carbon dioxide is a major cause of global climate change.

Analytical and simulation models

Conceptual models are limited; they poorly predict the consequences of changes in ecosystem species and/or environment. Ecosystems are dynamic entities that are subject to a variety of abiotic and biotic disturbances. In these cases, scientists often use analytical or simulation models. These models predict how ecosystems recover from disturbances, returning to a state of equilibrium. As most ecosystems are subject to periodic disturbances and are often in a state of change, they are usually either moving toward or away from multiple equilibrium states. Since human impact can greatly and rapidly alter

the species content and habitat of an ecosystem, it is crucial for scientists to develop models that predict how ecosystems respond to these changes.

Analytical models

Analytical models generally work best when dealing with relatively-simple, linear systems; specifically, those that can be accurately described by a set of mathematical equations whose behavior is well known. They are mathematically complex models that are good at predicting components of ecosystems such as food chains. However, their accuracy is limited by their simplification of complex ecosystems.

Simulation models

Like analytical models, simulation models use complex algorithms to predict ecosystem dynamics. However, sophisticated computer programs have enabled simulation models to predict responses in complex ecosystems. Simulation models use numerical techniques to solve problems for which analytic solutions are impractical or impossible. These kinds of models tend to be more widely used. They are generally considered more ecologically realistic, while analytic models are valued for their mathematical elegance and explanatory power. These simulations are considered to be the most accurate and predictive of ecosystem dynamics.

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46.2: Energy Flow through Ecosystems

Topic hierarchy

46.2A: Strategies for Acquiring Energy

46.2B: Productivity within Trophic Levels

46.2C: Transfer of Energy between Trophic Levels

46.2D: Ecological Pyramids

Ecological pyramids, which can be inverted or upright, depict biomass, energy, and the number of organisms in each trophic level.

46.2E: Biological Magnification

When toxic substances are introduced into the environment, organisms at the highest trophic levels suffer the most damage.

46.2A: Strategies for Acquiring Energy

Autotrophs (producers) synthesize their own energy, creating organic materials that are utilized as fuel by heterotrophs (consumers).

Learning Objectives

- Distinguish between photoautotrophs and chemoautotrophs and the ways in which they acquire energy

Key Points

- Food webs illustrate how energy flows through ecosystems, including how efficiently organisms acquire and use it.
- Autotrophs, producers in food webs, can be photosynthetic or chemosynthetic.
- Photoautotrophs use light energy to synthesize their own food, while chemoautotrophs use inorganic molecules.
- Chemoautotrophs are usually bacteria that live in ecosystems where sunlight is unavailable.
- Heterotrophs cannot synthesize their own energy, but must obtain it from autotrophs or other heterotrophs; they act as consumers in food webs.

Key Terms

- **photoautotroph**: an organism that can synthesize its own food by using light as a source of energy
- **chemoautotroph**: a simple organism, such as a protozoan, that derives its energy from chemical processes rather than photosynthesis
- **heterotroph**: an organism that requires an external supply of energy in the form of food, as it cannot synthesize its own

How Organisms Acquire Energy in a Food Web

All living things require energy in one form or another since energy is required by most, complex, metabolic pathways (often in the form of ATP); life itself is an energy-driven process. Living organisms would not be able to assemble macromolecules (proteins, lipids, nucleic acids, and complex carbohydrates) from their monomeric subunits without a constant energy input.

It is important to understand how organisms acquire energy and how that energy is passed from one organism to another through food webs and their constituent food chains. Food webs illustrate how energy flows directionally through ecosystems, including how efficiently organisms acquire it, use it, and how much remains for use by other organisms of the food web. Energy is acquired by living things in three ways: photosynthesis, chemosynthesis, and the consumption and digestion of other living or previously-living organisms by heterotrophs.

Photosynthetic and chemosynthetic organisms are grouped into a category known as autotrophs: organisms capable of synthesizing their own food (more specifically, capable of using inorganic carbon as a carbon source). Photosynthetic autotrophs (photoautotrophs) use sunlight as an energy source, whereas chemosynthetic autotrophs (chemoautotrophs) use inorganic molecules as an energy source. Autotrophs act as producers and are critical for all ecosystems. Without these organisms, energy would not be available to other living organisms and life itself would not be possible.

Photoautotrophs, such as plants, algae, and photosynthetic bacteria, serve as the energy source for a majority of the world's ecosystems. These ecosystems are often described by grazing food webs. Photoautotrophs harness the solar energy of the sun by converting it to chemical energy in the form of ATP (and NADP). The energy stored in ATP is used to synthesize complex organic molecules, such as glucose.

Chemoautotrophs are primarily bacteria that are found in rare ecosystems where sunlight is not available, such as in those associated with dark caves or hydrothermal vents at the bottom of the ocean. Many chemoautotrophs in hydrothermal vents use hydrogen sulfide (H_2S), which is released from the vents, as a source of chemical energy. This allows chemoautotrophs to synthesize complex organic molecules, such as glucose, for their own energy and in turn supplies energy to the rest of the ecosystem.

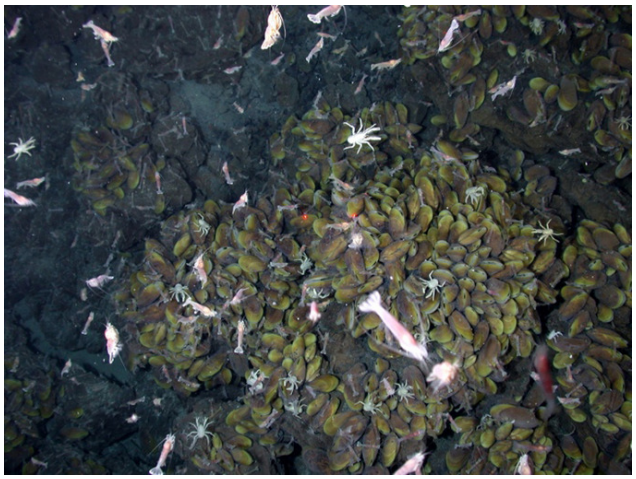

Figure 46.2A.1: **Chemoautotrophs**: Swimming shrimp, a few squat lobsters, and hundreds of vent mussels are seen at a hydrothermal vent at the bottom of the ocean. As no sunlight penetrates to this depth, the ecosystem is supported by chemoautotrophic bacteria and organic material that sinks from the ocean's surface.

Heterotrophs function as consumers in the food chain; they obtain energy in the form of organic carbon by eating autotrophs or other heterotrophs. They break down complex organic compounds produced by autotrophs into simpler compounds, releasing energy by oxidizing carbon and hydrogen atoms into carbon dioxide and water, respectively. Unlike autotrophs, heterotrophs are unable to synthesize their own food. If they cannot eat other organisms, they will die.

46.2B: Productivity within Trophic Levels

Productivity, measured by gross and net primary productivity, is defined as the amount of energy that is incorporated into a biomass.

Learning Objectives

- Explain the concept of primary production and distinguish between gross primary production and net primary production

Key Points

- A biomass is the total mass of living and previously-living organisms within a trophic level; ecosystems have characteristic amounts of biomass at each trophic level.
- The productivity of the primary producers (gross primary productivity) is important to ecosystems because these organisms bring energy to other living organisms.
- Net primary productivity (energy that remains in the primary producers after accounting for respiration and heat loss) is available to the primary consumers at the next trophic level.

Key Terms

- **biomass**: the total mass of all living things within a specific area, habitat, etc.
- **gross primary productivity**: rate at which photosynthetic primary producers incorporate energy from the sun
- **net primary productivity**: energy that remains in the primary producers after accounting for the organisms' respiration and heat loss
- **trophic level**: a particular position occupied by a group of organisms in a food chain (primary producer, primary consumer, secondary consumer, or tertiary consumer)

Productivity within trophic levels

Productivity within an ecosystem can be defined as the percentage of energy entering the ecosystem incorporated into biomass in a particular trophic level. Biomass is the total mass in a unit area (at the time of measurement) of living or previously-living organisms within a trophic level. Ecosystems have characteristic amounts of biomass at each trophic level. For example, in the English Channel ecosystem, the primary producers account for a biomass of 4 g/m^2 (grams per meter squared), while the primary consumers exhibit a biomass of 21 g/m^2.

The productivity of the primary producers is especially important in any ecosystem because these organisms bring energy to other living organisms by photoautotrophy or chemoautotrophy. Photoautotrophy is the process by which an organism (such as a green plant) synthesizes its own food from inorganic material using light as a source of energy; chemoautotrophy, on the other hand, is the process by which simple organisms (such as bacteria or archaea) derive energy from chemical processes rather than photosynthesis. The rate at which photosynthetic primary producers incorporate energy from the sun is called gross primary productivity. An example of gross primary productivity is the compartment diagram of energy flow within the Silver Springs aquatic ecosystem. In this ecosystem, the total energy accumulated by the primary producers was shown to be 20,810 $kcal/m^2/yr$.

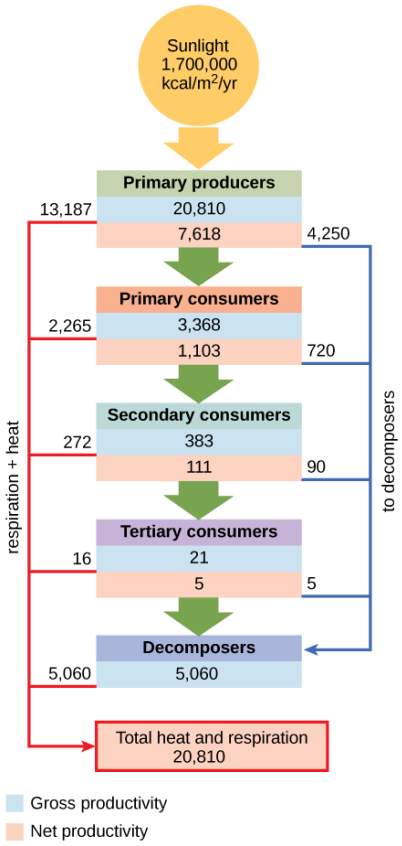

Figure 46.2B.1: **Energy flow in Silver Springs**: This conceptual model shows the flow of energy through a spring ecosystem in Silver Springs, Florida. Notice that the energy decreases with each increase in trophic level.

Because all organisms need to use some of this energy for their own functions (such as respiration and resulting metabolic heat loss), scientists often refer to the net primary productivity of an ecosystem. Net primary productivity is the energy that remains in the primary producers after accounting for the organisms' respiration and heat loss. The net productivity is then available to the primary consumers at the next trophic level. In the Silver Spring example, 13,187 of the 20,810 kcal/m^2/yr were used for respiration or were lost as heat, leaving 7,632 kcal/m^2/yr of energy for use by the primary consumers.

46.2C: Transfer of Energy between Trophic Levels

Energy is lost as it is transferred between trophic levels; the efficiency of this energy transfer is measured by NPE and TLTE.

Learning Objectives

- Illustrate the transfer of energy between trophic levels

Key Points

- Energy decreases as it moves up trophic levels because energy is lost as metabolic heat when the organisms from one trophic level are consumed by organisms from the next level.
- Trophic level transfer efficiency (TLTE) measures the amount of energy that is transferred between trophic levels.
- A food chain can usually sustain no more than six energy transfers before all the energy is used up.
- Net production efficiency (NPE) measures how efficiently each trophic level uses and incorporates the energy from its food into biomass to fuel the next trophic level.
- Endotherms have a low NPE and use more energy for heat and respiration than ectotherms, so most endotherms have to eat more often than ectotherms to get the energy they need for survival.
- Since cattle and other livestock have low NPEs, it is more costly to produce energy content in the form of meat and other animal products than in the form of corn, soybeans, and other crops.

Key Terms

- **assimilation**: the biomass of the present trophic level after accounting for the energy lost due to incomplete ingestion of food, energy used for respiration, and energy lost as waste
- **net consumer productivity**: energy content available to the organisms of the next trophic level
- **net production efficiency (NPE)**: measure of the ability of a trophic level to convert the energy it receives from the previous trophic level into biomass
- **trophic level transfer efficiency (TLTE)**: energy transfer efficiency between two successive trophic levels

Ecological efficiency: the transfer of energy between trophic levels

Large amounts of energy are lost from the ecosystem between one trophic level and the next level as energy flows from the primary producers through the various trophic levels of consumers and decomposers. The main reason for this loss is the second law of thermodynamics, which states that whenever energy is converted from one form to another, there is a tendency toward disorder (entropy) in the system. In biologic systems, this means a great deal of energy is lost as metabolic heat when the organisms from one trophic level are consumed by the next level. The measurement of energy transfer efficiency between two successive trophic levels is termed the trophic level transfer efficiency (TLTE) and is defined by the formula:

TLTE=productionatpresenttrophiclevelproductionatprevioustrophiclevelx100TLTE=productionatpresenttrophiclevelproductionatprevioustrophiclevelx100

In Silver Springs, the TLTE between the first two trophic levels was approximately 14.8 percent. The low efficiency of energy transfer between trophic levels is usually the major factor that limits the length of food chains observed in a food web. The fact is, after four to six energy transfers, there is not enough energy left to support another trophic level. In the Lake Ontario ecosystem food web, only three energy transfers occurred between the primary producer (green algae) and the tertiary, or apex, consumer (Chinook salmon).

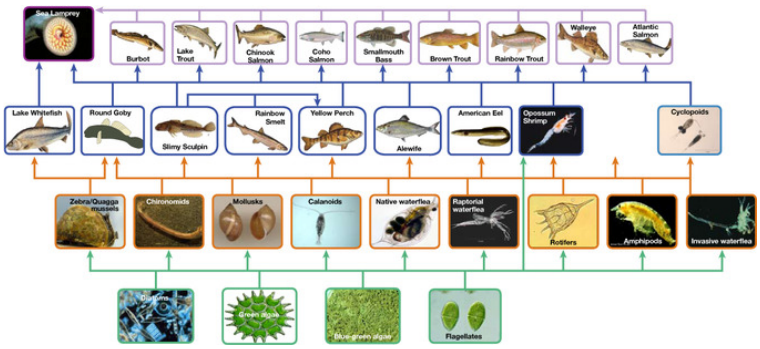

Figure 46.2C.1: **Food web of Lake Ontario**: This food web shows the interactions between organisms across trophic levels in the Lake Ontario ecosystem. Primary producers are outlined in green, primary consumers in orange, secondary consumers in blue, and tertiary (apex) consumers in purple. Arrows point from an organism that is consumed to the organism that consumes it. Notice how some lines point to more than one trophic level. For example, the opossum shrimp eats both primary producers and primary consumers.

Ecologists have many different methods of measuring energy transfers within ecosystems. Some transfers are easier or more difficult to measure depending on the complexity of the ecosystem and how much access scientists have to observe the ecosystem. In other words, some ecosystems are more difficult to study than others; sometimes the quantification of energy transfers has to be estimated.

Net production efficiency

Another main parameter that is important in characterizing energy flow within an ecosystem is the net production efficiency. Net production efficiency (NPE) allows ecologists to quantify how efficiently organisms of a particular trophic level incorporate the energy they receive into biomass. It is calculated using the following formula:

NPE=netconsumerproductivityassimilationx100NPE=netconsumerproductivityassimilationx100

Net consumer productivity is the energy content available to the organisms of the next trophic level. Assimilation is the biomass (energy content generated per unit area) of the present trophic level after accounting for the energy lost due to incomplete ingestion of food, energy used for respiration, and energy lost as waste. Incomplete ingestion refers to the fact that some consumers eat only a part of their food. For example, when a lion kills an antelope, it will eat everything except the hide and bones. The lion is missing the energy-rich bone marrow inside the bone, so the lion does not make use of all the calories its prey could provide.

Thus, NPE measures how efficiently each trophic level uses and incorporates the energy from its food into biomass to fuel the next trophic level. In general, cold-blooded animals (ectotherms), such as invertebrates, fish, amphibians, and reptiles, use less of the energy they obtain for respiration and heat than warm-blooded animals (endotherms), such as birds and mammals. The extra heat generated in endotherms, although an advantage in terms of the activity of these organisms in colder environments, is a major disadvantage in terms of NPE. Therefore, many endotherms have to eat more often than ectotherms to obtain the energy they need for survival. In general, NPE for ectotherms is an order of magnitude (10x) higher than for endotherms. For example, the NPE for a caterpillar eating leaves has been measured at 18 percent, whereas the NPE for a squirrel eating acorns may be as low as 1.6 percent.

The inefficiency of energy use by warm-blooded animals has broad implications for the world's food supply. It is widely accepted that the meat industry uses large amounts of crops to feed livestock. Because the NPE is low, much of the energy from animal feed is lost. For example, it costs about $0.01 to produce 1000 dietary calories (kcal) of corn or soybeans, but approximately $0.19 to produce a similar number of calories growing cattle for beef consumption. The same energy content of milk from cattle is also costly, at approximately $0.16 per 1000 kcal. Much of this difference is due to the low NPE of cattle. Thus, there has been a growing movement worldwide to promote the consumption of non-meat and non-dairy foods so that less energy is wasted feeding animals for the meat industry.

46.2D: Ecological Pyramids

Learning Objectives

- Explain the shape and structure of the ecological pyramid

Modeling ecosystems energy flow: ecological pyramids

The structure of ecosystems can be visualized with ecological pyramids, which were first described by the pioneering studies of Charles Elton in the 1920s. Ecological pyramids show the relative amounts of various parameters (such as number of organisms, energy, and biomass) across trophic levels. Ecological pyramids can also be called trophic pyramids or energy pyramids.

Pyramids of numbers can be either upright or inverted, depending on the ecosystem. A typical grassland during the summer has an upright shape since it has a base of many plants, with the numbers of organisms decreasing at each trophic level. However, during the summer in a temperate forest, the base of the pyramid consists of few trees compared with the number of primary consumers, mostly insects. Because trees are large, they have great photosynthetic capability and dominate other plants in this ecosystem to obtain sunlight. Even in smaller numbers, primary producers in forests are still capable of supporting other trophic levels.

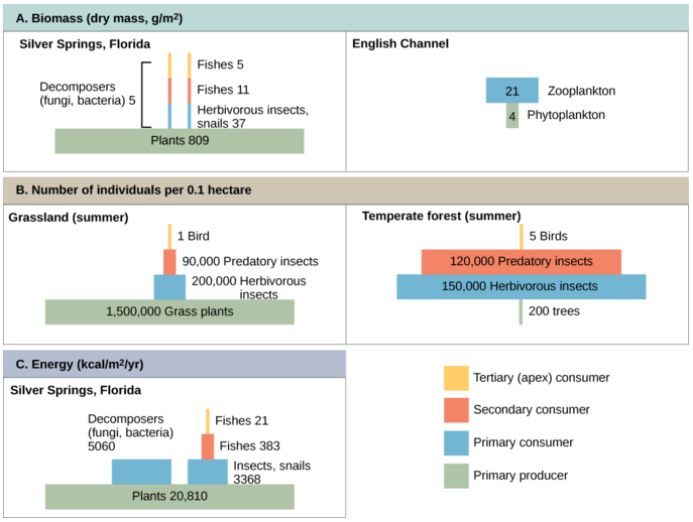

Figure 46.2D.1: **Ecological pyramids**: Ecological pyramids depict the (a) biomass, (b) number of organisms, and (c) energy in each trophic level.

Another way to visualize ecosystem structure is with pyramids of biomass. This pyramid measures the amount of energy converted into living tissue at the different trophic levels. Using the Silver Springs ecosystem example, this data exhibits an upright biomass pyramid, whereas the pyramid from the English Channel example is inverted. The plants (primary producers) of the Silver Springs ecosystem make up a large percentage of the biomass found there. However, the phytoplankton in the English Channel example make up less biomass than the primary consumers, the zooplankton. As with inverted pyramids of numbers, the inverted biomass pyramid is not due to a lack of productivity from the primary producers, but results from the high turnover rate of the phytoplankton. The phytoplankton are consumed rapidly by the primary consumers, which minimizes their biomass at any particular point in time. However, since phytoplankton reproduce quickly, they are able to support the rest of the ecosystem.

Pyramid ecosystem modeling can also be used to show energy flow through the trophic levels. Pyramids of energy are always upright, since energy is lost at each trophic level; an ecosystem without sufficient primary productivity cannot be supported. All types of ecological pyramids are useful for characterizing ecosystem structure. However, in the study of energy flow through the ecosystem, pyramids of energy are the most consistent and representative models of ecosystem structure.

Key Points

- Pyramids of numbers can be either upright or inverted, depending on the ecosystem.
- Pyramids of biomass measure the amount of energy converted into living tissue at the different trophic levels.
- The English Channel ecosystem exhibits an inverted biomass pyramid since the primary producers make up less biomass than the primary consumers.
- Pyramid ecosystem modeling can also be used to show energy flow through the trophic levels; pyramids of energy are always upright since energy decreases at each trophic level.
- All types of ecological pyramids are useful for characterizing ecosystem structure; however, in the study of energy flow through the ecosystem, pyramids of energy are the most consistent and representative models of ecosystem structure.

Key Terms

- **ecological pyramid**: diagram that shows the relative amounts of energy or matter or numbers of organisms within each trophic level in a food chain or food web

46.2E: Biological Magnification

Learning Objectives

- Describe the consequences of biomagnification between trophic levels

Consequences of Food Webs: Biological Magnification

One of the most important environmental consequences of ecosystem dynamics is biomagnification: the increasing concentration of persistent, toxic substances in organisms at each trophic level, from the primary producers to the apex consumers. Many substances have been shown to bioaccumulate, including classical studies with the pesticide dichlorodiphenyltrichloroethane (DDT), which was published in the 1960s bestseller, *Silent Spring*, by Rachel Carson. DDT was a commonly-used pesticide before its dangers became known. In some aquatic ecosystems, organisms from each trophic level consumed many organisms of the lower level, which caused DDT to increase in birds (apex consumers) that ate fish. Thus, the birds accumulated sufficient amounts of DDT to cause fragility in their eggshells. This effect increased egg breakage during nesting, which was shown to have adverse effects on these bird populations. The use of DDT was banned in the United States in the 1970s.

Other substances that biomagnify are polychlorinated biphenyls (PCBs), which were used in coolant liquids in the United States until their use was banned in 1979, and heavy metals, such as mercury, lead, and cadmium. These substances were best studied in aquatic ecosystems where fish species at different trophic levels accumulate toxic substances brought through the ecosystem by the primary producers. In a study performed by the National Oceanic and Atmospheric Administration (NOAA) in the Saginaw Bay of Lake Huron, PCB concentrations increased from the ecosystem's primary producers (phytoplankton) through the different trophic levels of fish species. The apex consumer (walleye) had more than four times the amount of PCBs compared to phytoplankton. Also, based on results from other studies, birds that eat these fish may have PCB levels at least one order of magnitude higher than those found in the lake fish.

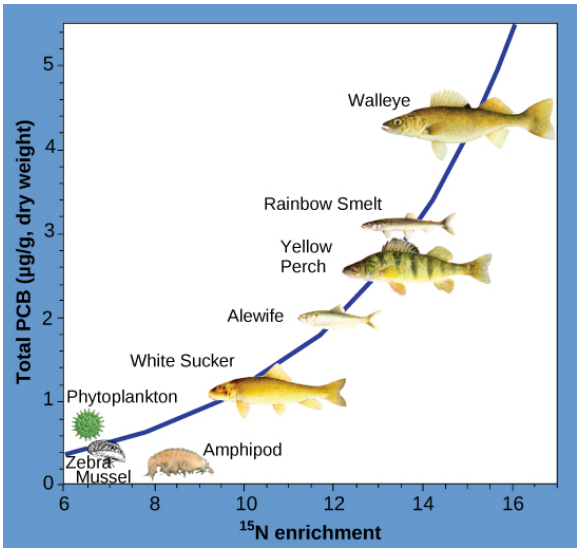

Figure 46.2*E*.1: **PCB concentration in Lake Huron**: This chart shows the PCB concentrations found at the various trophic levels in the Saginaw Bay ecosystem of Lake Huron. Numbers on the x-axis reflect enrichment with heavy isotopes of nitrogen (15N), which is a marker for increasing trophic levels. Notice that the fish in the higher trophic levels accumulate more PCBs than those in lower trophic levels.

Other concerns have been raised by the accumulation of heavy metals, such as mercury and cadmium, in certain types of seafood. The United States Environmental Protection Agency (EPA) recommends that pregnant women and young children should not consume any swordfish, shark, king mackerel, or tilefish because of their high mercury content. These individuals are advised to eat fish low in mercury: salmon, sardines, tilapia, shrimp, pollock, and catfish. Biomagnification is a good example of how ecosystem dynamics can affect our everyday lives, even influencing the food we eat.

Key Points

- Biomagnification increases the concentration of toxic substances in organisms at higher trophic levels.
- DDT is an example of a substance that biomagnifies; birds accumulate sufficient amounts of DDT from eating fish to cause adverse effects on bird populations.
- The presence of polychlorinated biphenyls (PCB) in phytoplankton causes increased PCB concentrations in walleyes and birds.
- Heavy metals, such as mercury and cadmium, found in certain types of seafood can also biomagnify.

Key Terms

- **biomagnification**: the process, in an ecosystem, in which a higher concentration of a substance in an organism is obtained higher up the food chain
- **dichlorodiphenyltrichloroethane**: a chlorinated hydrocarbon which is mainly used as an insecticide (DDT)
- **apex consumer**: consumers with few to no predators of their own, residing at the top of their food chain

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46.3: Biogeochemical Cycles

Topic hierarchy

- 46.3A: Biogeochemical Cycles
- 46.3B: The Water (Hydrologic) Cycle
- 46.3C: The Carbon Cycle
- 46.3D: The Nitrogen Cycle
- 46.3E: The Phosphorus Cycle
- 46.3E: The Sulfur Cycle

46.3A: Biogeochemical Cycles

The elemental components of organic matter are cycled through the biosphere in an interconnected process called the biogeochemical cycle.

Learning Objectives

- Summarize the concept of biogeochemical cycles

Key Points

- Carbon, nitrogen, hydrogen, oxygen, phosphorus, and sulfur are conserved and recycled in the atmosphere, on land, in water, or beneath the earth's surface.
- Materials are recycled via erosion, weathering, water drainage, and the movement of tectonic plates.
- Water is essential to all living processes, while carbon is found in all organic macromolecules.
- Nitrogen and phosphorus are major components of nucleic acids and play major roles in agriculture.
- Sulfur plays a role in the three-dimensional folding of proteins and is released into the atmosphere by the burning of fossil fuels.

Key Terms

- **hydrosphere**: combined mass of water found on, under, and over the surface of a planet
- **biogeochemical cycle**: cycling of mineral nutrients through ecosystems and through the non-living world

Biogeochemocal Cycles: Introduction

Energy flows directionally through ecosystems, entering as sunlight (or inorganic molecules for chemoautotrophs) and leaving as heat during the many transfers between trophic levels. However, the matter that makes up living organisms is conserved and recycled. The six most common elements associated with organic molecules (carbon, nitrogen, hydrogen, oxygen, phosphorus, and sulfur) take a variety of chemical forms and may exist for long periods in the atmosphere, on land, in water, or beneath the earth's surface. Geologic processes, such as weathering, erosion, water drainage, and the movement of the continental plates, all play a role in this recycling of materials. Because geology and chemistry have major roles in the study of this process, the recycling of inorganic matter between living organisms and their environment is called a biogeochemical cycle.

The components of organic molecules are constantly being stored and recycled as part of their biogeochemical cycle. Water, which contains hydrogen and oxygen, is essential to all living processes. The hydrosphere is the area of the earth where water movement and storage occurs. Water can be liquid on the surface and beneath the surface or frozen (rivers, lakes, oceans, groundwater, polar ice caps, and glaciers) or exist as water vapor in the atmosphere. Carbon, found in all organic macromolecules, is an important constituent of fossil fuels. Nitrogen, a major component of our nucleic acids and proteins, is critical to human agriculture. Phosphorus, a major component of nucleic acid (along with nitrogen), is one of the main ingredients in artificial fertilizers used in agriculture and their associated environmental impacts on our surface water. Sulfur, critical to the 3–D folding of proteins (as in disulfide binding), is released into the atmosphere by the burning of fossil fuels, such as coal.

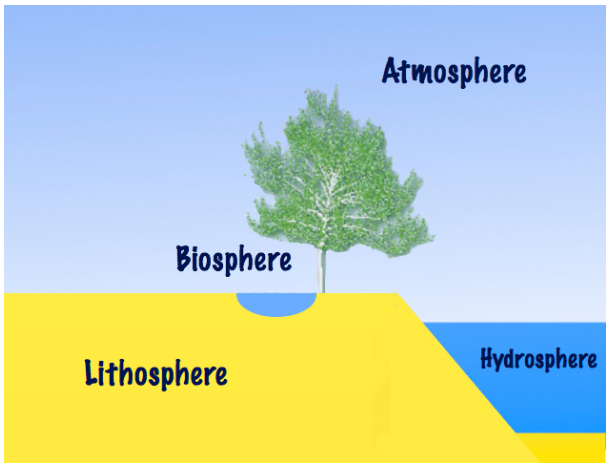

Figure 46.3A.1: **Importance of the hydrosphere**: Earth has a hydrosphere, where water movement and storage occurs. It is important for leaching certain components of organic matter into rivers, lakes, and oceans, and is a reservoir for carbon.

The cycling of all of these elements is interconnected. For example, the movement of water is critical for the leaching of nitrogen and phosphate into rivers, lakes, and oceans. Furthermore, the ocean itself is a major reservoir for carbon. Thus, mineral nutrients are cycled, either rapidly or slowly, through the entire biosphere, from one living organism to another, and between the biotic and abiotic world.

46.3B: The Water (Hydrologic) Cycle

Water has a large effect on climate, ecosystems, and living organisms and is continuously cycled through the environment.

Learning Objectives

- Explain the path of the hydrologic cycle and its importance

Key Points

- Water cycling affects the climate, transports minerals, purifies water, and replenishes the land with fresh water.
- Water with a longer residence time, such as water in oceans and glaciers, is not available for short-term cycling, which occurs via evaporation.
- Surface water evaporates (water to water vapor) or sublimates (ice to water vapor), which deposits large amounts of water vapor into the atmosphere.
- Water vapor in the atmosphere condenses into clouds and is eventually followed by precipitation, which returns water to the earth's surface.
- Rain percolates into the ground, where it may evaporate or enter bodies of water.
- Surface runoff enters oceans directly or via streams and lakes.

Key Terms

- **residence time**: the average time a particular molecule of water will remain in a body of water
- **condensation**: the conversion of a gas to a liquid; the condensate so formed
- **surface runoff**: overland flow of excess water (with or without accumulated contaminants) that cannot be absorbed by the ground as infiltration
- **Evaporation**: the process of a liquid converting to the gaseous state
- **sublimation**: the transition of a substance from the solid phase directly to the vapor state such that it does not pass through the intermediate liquid phase

The Water (Hydrologic) Cycle

Water is the basis of all living processes. More than half of the human body is made up of water, while human cells are more than 70 percent water. Thus, most land animals need a supply of fresh water to survive. However, when examining the stores of water on earth, 97.5 percent of it is non-potable salt water. Of the remaining water, 99 percent is locked underground as water or as ice. Thus, less than 1 percent of fresh water is easily accessible from lakes and rivers. Many living things, such as plants, animals, and fungi, are dependent on the small amount of fresh surface water supply, a lack of which can have massive effects on ecosystem dynamics. Humans, of course, have developed technologies to increase water availability, such as digging wells to harvest groundwater, storing rainwater, and using desalination to obtain drinkable water from the ocean. Although this pursuit of drinkable water has been ongoing throughout human history, the supply of fresh water is still a major issue in modern times.

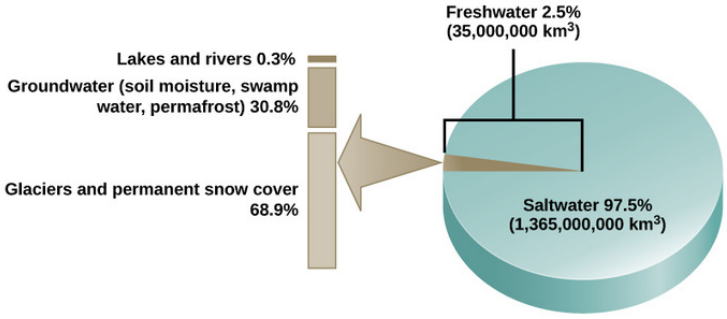

Figure 46.3B.1: **Water availability**: Only 2.5 percent of water on earth is fresh water. Less than 1 percent of fresh water is easily accessible to living things.

Water cycling is extremely important to ecosystem dynamics as it has a major influence on climate and, thus, on the environments of ecosystems. For example, when water evaporates, it takes up energy from its surroundings, cooling the

environment. When it condenses, it releases energy, warming the environment. The evaporation phase of the cycle purifies water, which then replenishes the land with fresh water. The flow of liquid water and ice transports minerals across the globe. It is also involved in reshaping the geological features of the earth through processes including erosion and sedimentation. The water cycle is also essential for the maintenance of most life and ecosystems on the planet. Most of the water on earth is stored for long periods in the oceans, underground, and as ice. Residence time is a measure of the average time an individual water molecule stays in a particular reservoir. A large amount of the earth's water is locked in place in these reservoirs as ice, beneath the ground, and in the ocean, and, thus, is unavailable for short-term cycling (only surface water can evaporate).

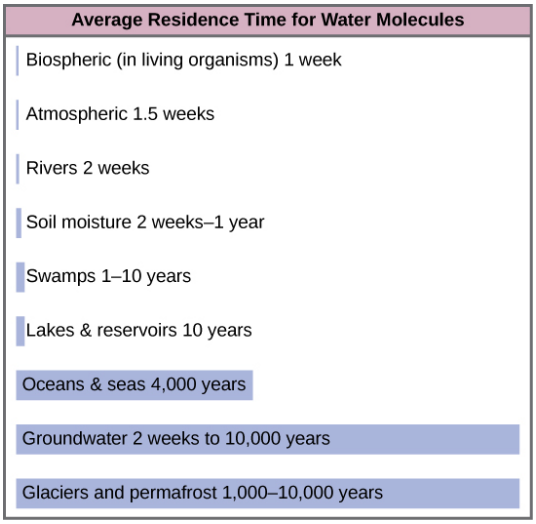

Figure $46.3B.1$: **Residence time of water**: This graph shows the average residence time for water molecules in the earth's water reservoirs. Most of the earth's water is unavailable for use because it is locked in ice, beneath the ground, or in the oceans.

There are various processes that occur during the cycling of water, which include the following:

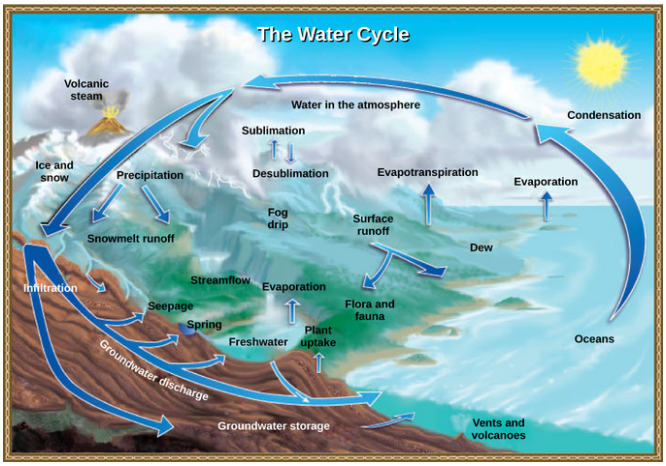

Figure $46.3B.1$: **Cycling of water**: Water from the land and oceans enters the atmosphere by evaporation or sublimation, where it condenses into clouds and falls as rain or snow. Precipitated water may enter freshwater bodies or infiltrate the soil. The cycle is complete when surface or groundwater reenters the ocean.

- evaporation/ sublimation
- condensation/precipitation
- subsurface water flow
- surface runoff/snowmelt
- streamflow

The water cycle is driven by the sun's energy as it warms the oceans and other surface waters. This leads to the evaporation (water to water vapor) of liquid surface water and the sublimation (ice to water vapor) of frozen water, which

deposits large amounts of water vapor into the atmosphere. Over time, this water vapor condenses into clouds as liquid or frozen droplets, which is eventually followed by precipitation (rain or snow), returning water to the earth's surface. Rain eventually percolates into the ground, where it may evaporate again (if it is near the surface), flow beneath the surface, or be stored for long periods. More easily observed is surface runoff: the flow of fresh water either from rain or melting ice. Runoff can then make its way through streams and lakes to the oceans or flow directly to the oceans themselves. Rain and surface runoff are major ways in which minerals, including carbon, nitrogen, phosphorus, and sulfur, are cycled from land to water.

46.3C: The Carbon Cycle

Carbon enters the atmosphere in the form of carbon dioxide via the carbon cycle and returns to organic carbon via photosynthesis.

Learning Objectives

- Distinguish between the biological and biogeochemical cycles of carbon

Key Points

- Carbon is present in all organic molecules; carbon compounds contain large amounts of energy, which humans use as fuel.
- The biological carbon cycle is the rapid exchange of carbon among living things; autotrophs use carbon dioxide produced by heterotrophs to produce glucose and oxygen, which are then utilized by heterotrophs.
- The biogeochemical cycle occurs at a much slower rate than the biological cycle since carbon is stored in carbon reservoirs for long periods of time.
- Carbon dioxide from the atmosphere dissolves in water, combining with water molecules to form carbonic acid, which then ionizes to carbonate and bicarbonate ions.
- Most of the carbon in the ocean is in the form of bicarbonate ions, which can combine with seawater calcium to form calcium carbonate ($CaCO3$), a major component of marine organism shells.
- Carbon can enter the soil as a result of the decomposition of living organisms, the weathering of rocks, the eruption of volcanoes, and other geothermal systems.

Key Terms

- **subduction**: movement of one tectonic plate beneath another
- **non-renewable resource**: resource, such as fossil fuel, that is either regenerated very slowly or not at all
- **autotroph**: Any organism that can synthesize its food from inorganic substances, using heat or light as a source of energy
- **heterotroph**: an organism that requires an external supply of energy in the form of food, as it cannot synthesize its own

The Carbon Cycle

Carbon, the second most abundant element in living organisms, is present in all organic molecules. Its role in the structure of macromolecules is of primary importance to living organisms. Carbon compounds contain especially- high forms of energy, which humans use as fuel. Since the 1800s (the beginning of the Industrial Revolution), the number of countries using massive amounts of fossil fuels increased, which raised the levels of carbon dioxide in the atmosphere. This increase in carbon dioxide has been associated with climate change and other disturbances of the earth’s ecosystems. It is a major environmental concern worldwide.

The carbon cycle is most easily studied as two interconnected sub-cycles: one dealing with rapid carbon exchange among living organisms and the other dealing with the long-term cycling of carbon through geologic processes.

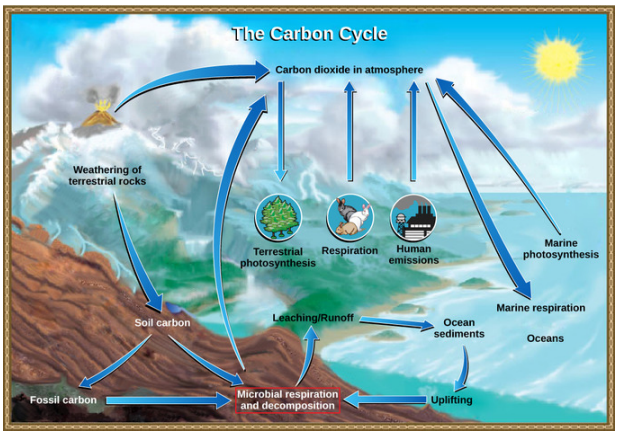

Figure 46.3C.1: **Carbon cycle**: Carbon dioxide gas exists in the atmosphere and is dissolved in water. Photosynthesis converts carbon dioxide gas to organic carbon, while respiration cycles the organic carbon back into carbon dioxide gas. Long-term storage of organic carbon occurs when matter from living organisms is buried deep underground and becomes fossilized. Volcanic activity and human emissions bring this stored carbon back into the carbon cycle.

The Biological Carbon Cycle

Living organisms are connected in many ways, even between ecosystems. A good example of this connection is the exchange of carbon between autotrophs and heterotrophs. Carbon dioxide is the basic building block that most autotrophs use to build multi-carbon, high-energy compounds, such as glucose. The energy harnessed from the sun is used by these organisms to form the covalent bonds that link carbon atoms together. These chemical bonds store this energy for later use in the process of respiration. Most terrestrial autotrophs obtain their carbon dioxide directly from the atmosphere, while marine autotrophs acquire it in the dissolved form (carbonic acid, $H_2CO_3^-$). However carbon dioxide is acquired, a by-product of the process is oxygen. The photosynthetic organisms are responsible for depositing approximately 21 percent of the oxygen content in the atmosphere that we observe today.

Heterotrophs acquire the high-energy carbon compounds from the autotrophs by consuming them and breaking them down by respiration to obtain cellular energy, such as ATP. The most efficient type of respiration, aerobic respiration, requires oxygen obtained from the atmosphere or dissolved in water. Thus, there is a constant exchange of oxygen and carbon dioxide between the autotrophs (which need the carbon) and the heterotrophs (which need the oxygen). Gas exchange through the atmosphere and water is one way that the carbon cycle connects all living organisms on Earth.

The Biogeochemical Carbon Cycle

The movement of carbon through the land, water, and air is complex and, in many cases, it occurs much more slowly than the biological carbon cycle. Carbon is stored for long periods in what are known as carbon reservoirs, which include the atmosphere, bodies of liquid water (mostly oceans), ocean sediment, soil, land sediments (including fossil fuels), and the earth's interior.

As stated, the atmosphere, a major reservoir of carbon in the form of carbon dioxide, is essential to the process of photosynthesis. The level of carbon dioxide in the atmosphere is greatly influenced by the reservoir of carbon in the oceans. The exchange of carbon between the atmosphere and water reservoirs influences how much carbon is found in each location; each affects the other reciprocally. Carbon dioxide (CO_2) from the atmosphere dissolves in water, combining with water molecules to form carbonic acid. It then ionizes to carbonate and bicarbonate ions.

Step 1: CO_2 (atmospheric) $\rightleftharpoons$ CO_2 (dissolved)
Step 2: CO_2 (dissolved) + H_2O $\rightleftharpoons$ H_2CO_3 (carbonic acid)
Step 3: H_2CO_3 $\rightleftharpoons$ + H^+ + HCO_3^- (biocarbonate ion)
Step 4: HCO_3^- $\rightleftharpoons$ H^+ + CO_3^{2-} (carbonate ion)

Figure 46.3C.1: **Formation of bicarbonate**: Carbon dioxide reacts with water to form bicarbonate and carbonate ions.

More than 90 percent of the carbon in the ocean is found as bicarbonate ions. Some of these ions combine with seawater calcium to form calcium carbonate ($CaCO_3$), a major component of marine organism shells. These organisms eventually form sediments on the ocean floor. Over geologic time, the calcium carbonate forms limestone, which comprises the largest carbon reservoir on earth.

On land, carbon is stored in soil as a result of the decomposition of living organisms or the weathering of terrestrial rock and minerals. This carbon can be leached into the water reservoirs by surface runoff. Deeper underground, on land and at sea, are fossil fuels: the anaerobically-decomposed remains of plants that take millions of years to form. Fossil fuels are considered a non-renewable resource because their use far exceeds their rate of formation. A non-renewable resource is either regenerated very slowly or not at all. Another way for carbon to enter the atmosphere is from land by the eruption of volcanoes and other geothermal systems. Carbon sediments from the ocean floor are taken deep within the earth by the process of subduction: the movement of one tectonic plate beneath another. Carbon is released as carbon dioxide when a volcano erupts or from volcanic hydrothermal vents.

Carbon dioxide is also added to the atmosphere by the breeding and raising of livestock. The large numbers of land animals raised to feed the earth's growing population results in increased carbon dioxide levels in the atmosphere due to farming practices, respiration, and methane production. This is another example of how human activity indirectly affects biogeochemical cycles in a significant way. Although much of the debate about the future effects of increasing atmospheric carbon on climate change focuses on fossils fuels, scientists take natural processes, such as volcanoes and respiration, into account as they model and predict the future impact of this increase.

46.3D: The Nitrogen Cycle

Nitrogen, the most abundant gas in the atmosphore, is cycled through the biosphere via the multi-step process of nitrogen fixation, which is carried out by bacteria.

Learning Objectives

- Describe the nitrogen cycle

Key Points

- Bacteria, such as cyanobacteria, convert nitrogen into nitrogen gas via nitrogen fixation.
- Nitrogen fixation occurs in three steps: ammonification, nitrification, and denitrification.
- Human activity can release nitrogen into the environment by the combustion of fossil fuels and by the use of artificial fertilizers in agriculture.
- Atmospheric nitrogen is responsible for acid rain, the release of greenhouse gasses, and eutrophication.
- Nitrogen fixation can be performed by marine bacteria; nitrogen falls to the ocean floor as sediment and is then moved to land, becoming incorporated into terrestrial rock.

Key Terms

- **denitrification**: process of converting nitrates into nitrogen gas, especially by the action of bacteria
- **nitrification**: the conversion of ammonium into nitrites (NO2−) by nitrifying bacteria
- **ammonification**: the formation of ammonia or its compounds from nitrogenous compounds, especially as a result of bacterial decomposition

The Nitrogen Cycle

Getting nitrogen into the living world is difficult. Plants and phytoplankton are not equipped to incorporate nitrogen from the atmosphere (which exists as tightly-bonded, triple-covalent N_2), even though this molecule comprises approximately 78 percent of the atmosphere. Nitrogen enters the living world via free-living and symbiotic bacteria, which incorporate nitrogen into their macromolecules through nitrogen fixation (conversion of N_2). Cyanobacteria live in most aquatic ecosystems where sunlight is present; they play a key role in nitrogen fixation. Cyanobacteria are able to use inorganic sources of nitrogen to “fix” nitrogen. *Rhizobium*bacteria live symbiotically in the root nodules of legumes (such as peas, beans, and peanuts), providing them with the organic nitrogen they need. Free-living bacteria, such as *Azotobacter*, are also important nitrogen fixers.

Organic nitrogen is especially important to the study of ecosystem dynamics as many ecosystem processes, such as primary production and decomposition, are limited by the available supply of nitrogen. The nitrogen that enters living systems by nitrogen fixation is successively converted from organic nitrogen back into nitrogen gas by bacteria. This process occurs in three steps in terrestrial systems: ammonification, nitrification, and denitrification. First, the ammonification process converts nitrogenous waste from living animals or from the remains of dead animals into ammonium (NH_4^+) by certain bacteria and fungi. Second, the ammonium is converted to nitrites (NO_2^-) by nitrifying bacteria, such as *Nitrosomonas*, through nitrification. Subsequently, nitrites are converted to nitrates (NO_3^-) by similar organisms. Third, the process of denitrification occurs, whereby bacteria, such as *Pseudomonas* and *Clostridium*, convert the nitrates into nitrogen gas, allowing it to re-enter the atmosphere.

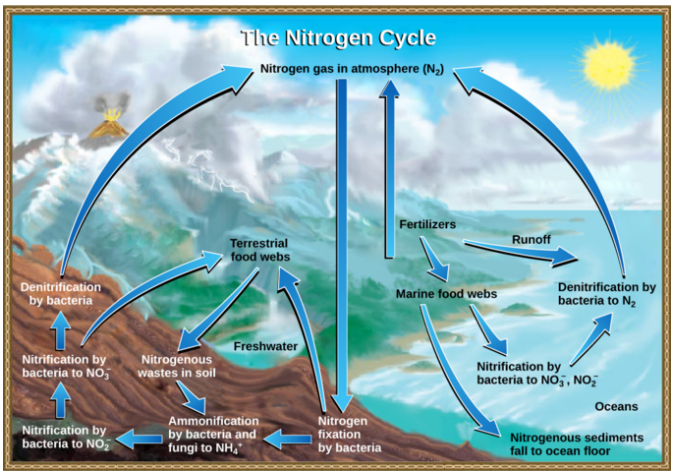

Figure $46.3D.1$: **Nitrogen fixation**: Nitrogen enters the living world from the atmosphere via nitrogen-fixing bacteria. This nitrogen and nitrogenous waste from animals is then processed back into gaseous nitrogen by soil bacteria, which also supply terrestrial food webs with the organic nitrogen they need.

Human activity can release nitrogen into the environment by two primary means: the combustion of fossil fuels, which releases different nitrogen oxides, and the use of artificial fertilizers in agriculture, which are then washed into lakes, streams, and rivers by surface runoff. Atmospheric nitrogen is associated with several effects on earth's ecosystems, including the production of acid rain (as nitric acid, HNO_3) and greenhouse gas (as nitrous oxide, N_2O), potentially causing climate change. A major effect from fertilizer runoff is saltwater and freshwater eutrophication: a process whereby nutrient runoff causes the excess growth of microorganisms, depleting dissolved oxygen levels and killing ecosystem fauna.

A similar process occurs in the marine nitrogen cycle, where the ammonification, nitrification, and denitrification processes are performed by marine bacteria. Some of this nitrogen falls to the ocean floor as sediment, which can then be moved to land in geologic time by uplift of the earth's surface, becoming incorporated into terrestrial rock. Although the movement of nitrogen from rock directly into living systems has been traditionally seen as insignificant compared with nitrogen fixed from the atmosphere, a recent study showed that this process may indeed be significant and should be included in any study of the global nitrogen cycle.

46.3E: The Phosphorus Cycle

Phosphorus is an essential element of living things, but, in excess, it can cause damage to ecosystems.

Learning Objectives

- Describe the phosphorus cycle and the effects of phosphorus on the environment

Key Points

- Phosphorus, a major component of nucleic acid and phospholipids, also makes up the supportive components of our bones; it is often necessary for growth in aquatic ecosystems.
- Phosphates (PO_4^{3-}) are sent into rivers, lakes, and the ocean by leaching and natural surface runoff.
- Phosphate-containing ocean sediments slowly move to land by the uplifting of areas of the earth's surface.
- Excess phosphorus and nitrogen in the ecosystem leads to the death of many organisms, causing dead zones.
- Dead zones are caused by by eutrophication, oil spills, dumping of toxic chemicals, and other human activities.

Key Terms

- **eutrophication**: process whereby excess levels of nitrogen or phosphorus cause excessive growth of microorganisms, depleting dissolved oxygen levels and kill ecosystem fauna
- **dead zone**: an area within a freshwater or marine ecosystem where large areas are depleted of their normal flora and fauna; caused by excessive nutrient pollution

The Phosphorus Cycle

Phosphorus is an essential nutrient for living processes. It is a major component of nucleic acid, both DNA and RNA; of phospholipids, the major component of cell membranes; and, as calcium phosphate, makes up the supportive components of our bones. Phosphorus is often the limiting nutrient (necessary for growth) in aquatic ecosystems.

Phosphorus occurs in nature as the phosphate ion (PO_4^{3-}). In addition to phosphate runoff as a result of human activity, natural surface runoff occurs when it is leached from phosphate-containing rock by weathering, thus sending phosphates into rivers, lakes, and the ocean. This rock has its origins in the ocean. Phosphate-containing ocean sediments form primarily from the bodies of ocean organisms and from their excretions. However, in remote regions, volcanic ash, aerosols, and mineral dust may also be significant phosphate sources. This sediment then is moved to land over geologic time by the uplifting of areas of the earth's surface.

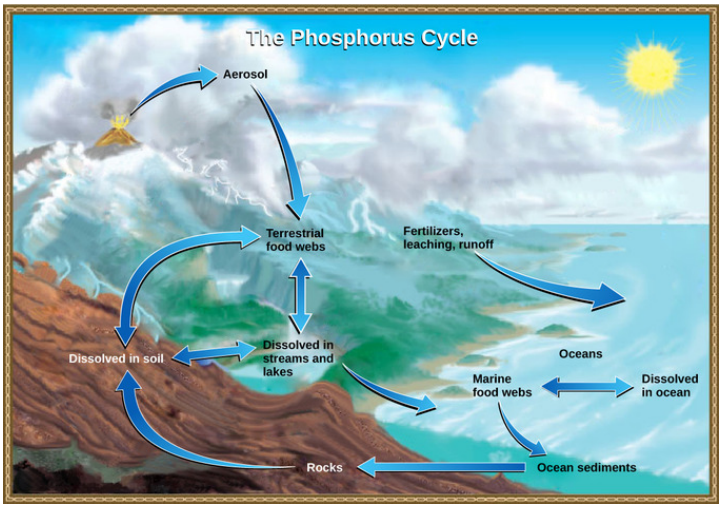

Figure 46.3E.1: **Phosphorus cycle**: In nature, phosphorus exists as the phosphate ion (PO43−). Weathering of rocks and volcanic activity releases phosphate into the soil, water, and air, where it becomes available to terrestrial food webs. Phosphate enters the oceans via surface runoff, groundwater flow, and river flow. Phosphate dissolved in ocean water cycles into marine food webs. Some phosphate from the marine food webs falls to the ocean floor, where it forms sediment.

Phosphorus is also reciprocally exchanged between phosphate dissolved in the ocean and marine ecosystems. The movement of phosphate from the ocean to the land and through the soil is extremely slow, with the average phosphate ion having an oceanic residence time between 20,000 and 100,000 years.

Excess phosphorus and nitrogen that enters these ecosystems from fertilizer runoff and from sewage causes excessive growth of microorganisms and depletes the dissolved oxygen, which leads to the death of many ecosystem fauna, such as shellfish and finfish. This process is responsible for dead zones in lakes and at the mouths of many major rivers.

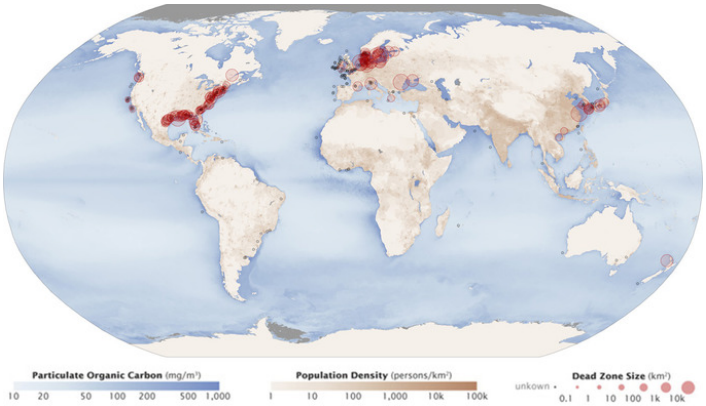

Figure 46.3E.1: **Dead zones**: Dead zones occur when phosphorus and nitrogen from fertilizers cause excessive growth of microorganisms, which depletes oxygen, killing flora and fauna. Worldwide, large dead zones are found in coastal areas of high population density.

A dead zone is an area within a freshwater or marine ecosystem where large areas are depleted of their normal flora and fauna. These zones can be caused by eutrophication, oil spills, dumping of toxic chemicals, and other human activities. The number of dead zones has been increasing for several years; more than 400 of these zones were present as of 2008. One of the worst dead zones is off the coast of the United States in the Gulf of Mexico, where fertilizer runoff from the Mississippi River basin has created a dead zone of over 8,463 square miles. Phosphate and nitrate runoff from fertilizers also negatively affect several lake and bay ecosystems, including the Chesapeake Bay in the eastern United States, which was one of the first ecosystems to have identified dead zones.

46.3E: The Sulfur Cycle

Sulfur is deposited on land as precipitation, fallout, and rock weathering, and reintroduced when organisms decompose.

Learning Objectives

- Describe the sulfur cycle

Key Points

- Sulfur is an essential element for the macromolecules of living things since it determines the 3-D folding patterns of proteins.
- On land, sulfur enters the atmosphere via acid rain, fallout, the weathering of rocks, decompostion of organic materials, and geothermal vents.
- Sulfur enters the ocean via runoff, fallout, and underwater geothermal vents; some marine ecosystems also rely on chemoautotrophs as a sulfur source.
- The burning of fossil fuels increases the amount of sulfide in the atmosphere and causes acid rain.
- Acid rain is corrosive rain that causes damage to aquatic ecosystems by lowering the pH of lakes, killing many of the resident fauna; it also degrades buildings and human-made structures.

Key Terms

- **chemoautotroph**: a simple organism, such as a protozoan, that derives its energy from chemical processes rather than photosynthesis
- **fallout**: direct deposit of solid minerals on land or in the ocean from the atmosphere
- **acid rain**: corrosive rain caused by rainwater falling to the ground through sulfur dioxide gas, turning it into weak sulfuric acid; can damage structures and ecosystems

The Sulfur Cycle

Sulfur is an essential element for the macromolecules of living things. As a part of the amino acid cysteine, it is involved in the formation of disulfide bonds within proteins, which help to determine their 3-D folding patterns and, hence, their functions. Sulfur cycles exist between the oceans, land, and atmosphere.

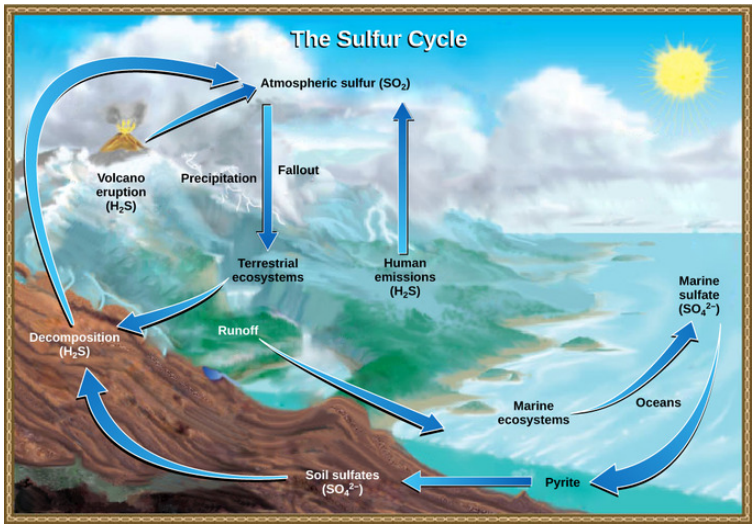

Figure 46.3E. 1: **Sulfur cycle**: Sulfur dioxide from the atmosphere becomes available to terrestrial and marine ecosystems when it is dissolved in precipitation as weak sulfuric acid or when it falls directly to the earth as fallout. Weathering of rocks also makes sulfates available to terrestrial ecosystems. Decomposition of living organisms returns sulfates to the ocean, soil, and atmosphere.

On land, sulfur is deposited in four major ways: precipitation, direct fallout from the atmosphere, rock weathering, and decomposition of organic materials. Atmospheric sulfur is found in the form of sulfur dioxide (SO_2). As rain falls through the atmosphere, sulfur is dissolved in the form of weak sulfuric acid (H_2SO_4), creating acid rain. Sulfur can also fall

directly from the atmosphere in a process called fallout. The weathering of sulfur-containing rocks also releases sulfur into the soil. These rocks originate from ocean sediments that are moved to land by the geologic uplift. Terrestrial ecosystems can then make use of these soil sulfates (SO_4^{2-}). Upon the death and decomposition of these organisms, sulfur is released back into the atmosphere as hydrogen sulfide (H_2S) gas. Sulfur may also enter the atmosphere through geothermal vents.

Figure $46.3E.1$: **Sulfur vents**: At this sulfur vent in Lassen Volcanic National Park in northeastern California, the yellowish sulfur deposits are visible near the mouth of the vent.

Sulfur enters the ocean via runoff from land, fallout, and underwater geothermal vents. Some marine ecosystems rely on chemoautotrophs, using sulfur as a biological energy source. This sulfur then supports marine ecosystems in the form of sulfates.

Human activities have played a major role in altering the balance of the global sulfur cycle. The burning of large quantities of fossil fuels, especially from coal, releases large amounts of hydrogen sulfide gas into the atmosphere, creating acid rain. Acid rain is corrosive rain that causes damage to aquatic ecosystems and the natural environment by lowering the pH of lakes, which kills many of the resident fauna; it also affects the human-made environment through the chemical degradation of buildings. For example, many marble monuments, such as the Lincoln Memorial in Washington, DC, have suffered significant damage from acid rain over the years. These examples show the wide-ranging effects of human activities on our environment and the challenges that remain for our future.

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CHAPTER OVERVIEW

47: CONSERVATION BIOLOGY AND BIODIVERSITY

47.1: THE BIODIVERSITY CRISIS

47.1A: LOSS OF BIODIVERSITY
47.1B: TYPES OF BIODIVERSITY
47.1C: BIODIVERSITY CHANGE THROUGH GEOLOGICAL TIME
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47.4A: MEASURING BIODIVERSITY
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47.1: The Biodiversity Crisis

Topic hierarchy

- 47.1A: Loss of Biodiversity
- 47.1B: Types of Biodiversity
- 47.1C: Biodiversity Change through Geological Time
- 47.1D: The Pleistocene Extinction
- 47.1E: Present-Time Extinctions

47.1A: Loss of Biodiversity

Human activity is the driving force behind the current biodiversity crisis, which is causing great species loss in a short time period.

Learning Objectives

- Explain the biodiversity crisis

Key Points

- Biodiversity is the variety of species present in the biosphere; the main goal of conservationists is to preserve biodiversity.
- An example of biodiversity loss was the extinction of over 200 species of cichlids in Lake Victoria; this was caused by the introduction of the Nile Perch as well as increased agriculture and fishing.
- Unlike the five previous mass extinctions, the current one is a result of detrimental human activities.
- Current human-caused biodiversity loss is the result activities such as habitat destruction, the introduction of exotic invasive species, and the over-harvesting of species.

Key Terms

- **biodiversity**: the diversity (number and variety of species) of plant and animal life within a region
- **adaptive radiation**: the diversification of species into separate forms that each adapt to occupy a specific environmental niche
- **invasive species**: any species that has been introduced to an environment where it is not native and has since become a nuisance through rapid spread and increase in numbers, often to the detriment of native species
- **extinction**: disappearance of a species from earth; local extinction is the disappearance of a species from a region

The Biodiversity Crisis

Traditionally, ecologists have measured biodiversity, a general term for the variety of species present in the biosphere, by taking into account both the number of species and their commonness. Biodiversity can be estimated at a number of levels of organization of living things. These estimation indexes, which came from information theory, are most useful as a first step in quantifying biodiversity between and within ecosystems, yet they are less useful when the main concern among conservation biologists is simply the loss of biodiversity. However, biologists recognize that measures of biodiversity, in terms of species diversity, may help focus efforts to preserve the biologically or technologically important elements of biodiversity.

Cichlids in Lake Victoria

The Lake Victoria cichlids provide an example through which we can begin to understand biodiversity. The biologists studying cichlids in the 1980s discovered hundreds of cichlid species representing a variety of specializations to particular habitat types and specific feeding strategies: eating plankton floating in the water, scraping and then eating algae from rocks, eating insect larvae from the bottom, and eating the eggs of other species of cichlid. The cichlids of Lake Victoria are the product of an adaptive radiation. An adaptive radiation is a rapid (less than three million years in the case of the Lake Victoria cichlids) branching through speciation of a phylogenetic tree into many closely-related species; typically, the species “radiate” into different habitats and niches. The Galápagos finches are an example of a modest adaptive radiation with 15 species. The cichlids of Lake Victoria are an example of a spectacular adaptive radiation that includes about 500 species.

At the time biologists were making this discovery, some species began to quickly disappear. A culprit in these declines was a species of large fish that was introduced to Lake Victoria by fisheries to feed the people living around the lake. The Nile perch was introduced in 1963, but was not a problem until the 1980s when its population began to surge by consuming cichlids, driving species after species to the point of extinction (the disappearance of a species). In fact, there were several factors that played a role in the extinction of perhaps 200 cichlid species in Lake Victoria. These factors included not only the Nile perch, but also the declining lake water quality due to agriculture and land clearing on the shores of Lake Victoria,

and increased fishing pressure. Scientists had not even cataloged all of the species present, so many were lost that they were never named. The diversity is now a shadow of what it once was.

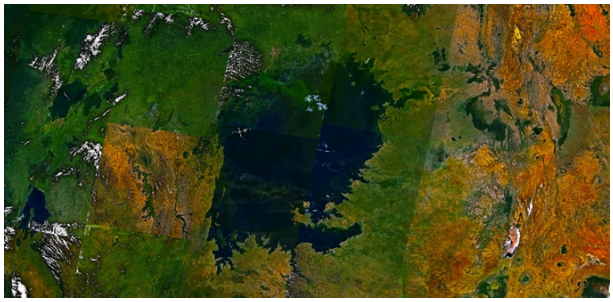


Figure 47.1*A*.1: **Lake Victoria and biodiversity loss**: Lake Victoria in Africa, shown in this satellite image, was the site of one of the most extraordinary evolutionary findings on the planet, as well as a casualty of devastating biodiversity loss.

Causes of Biodiversity

The cichlids of Lake Victoria are a thumbnail sketch of contemporary rapid species loss caused by human activity occurring all over earth. Extinction, a natural process of macroevolution, occurs at the rate of about one out of 1 million species becoming extinct per year. The fossil record reveals that there have been five periods of mass extinction in history with much higher rates of species loss. The rate of species loss today is comparable to those periods of mass extinction. However, there is a major difference between the previous mass extinctions and the current extinction we are experiencing: human activity. Specifically, three human activities have a major impact: destruction of habitat, introduction of exotic invasive species, and over-harvesting. Predictions of species loss within the next century, a tiny amount of time on geological timescales, range from 10 percent to 50 percent. The five previous extinctions on this scale were caused by cataclysmic events that changed the course of the history of life in each instance. Earth is now in one of those times.

47.1B: Types of Biodiversity

Genetic diversity, ecosystem diversity, and human-derived diversity are measures of biodiversity that currently define life on earth.

Learning Objectives

- Differentiate among the types of biodiversity

Key Points

- A genus with a high variety of species will have more genetic diversity; the most genetically-diverse species will have the greatest potential for evolution and preservation.
- The loss of ecosystem diversity results in the loss of interactions between species, unique features of co- adaptation, and biological productivity.
- Human-generated species diversity has decreased due to migration, market forces, and agriculture.
- Humans have only been able to estimate the number of species that inhabit earth; this estimate accounts for about 20 percent of predicted species on the planet.
- Humans have only been able to estimate the number of species that inhabit Earth; this estimate only accounts for 20 percent of predicted species on the planet.

Key Terms

- **genetic diversity**: variety of genes in a species or other taxonomic group or ecosystem; can refer to allelic diversity or genome-wide diversity
- **ecosystem diversity**: variety of ecosystems in a biosphere or the variety of species and ecological processes that occur in different physical settings
- **chemical diversity**: variety of metabolic compounds in an ecosystem

Types of Biodiversity

Scientists generally accept that the term biodiversity describes the number and kinds of species in a location or on the planet. Species can be difficult to define, but most biologists still feel comfortable with the concept and are able to identify and count eukaryotic species in most contexts. Biologists have also identified alternate measures of biodiversity, some of which are important for planning how to preserve biodiversity.

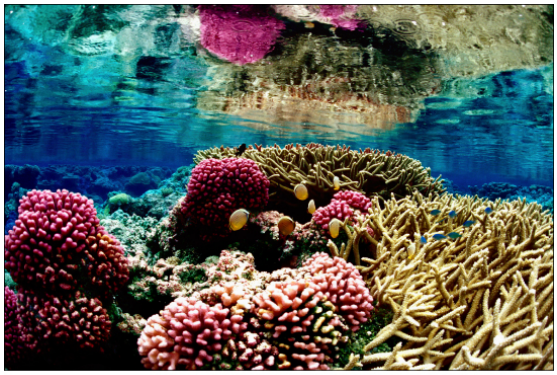
Genetic diversity is one of those alternate concepts. Genetic diversity or variation is the raw material for adaptation in a species. A species' future potential for adaptation depends on the genetic diversity held in the genomes of the individuals in populations that make up the species. The same is true for higher taxonomic categories. A genus with very different types of species will have more genetic diversity than a genus with species that look alike and have similar ecologies. If there were a choice between one of these genera of species being preserved, the one with the greatest potential for subsequent evolution is the most genetically-diverse one. It would be ideal not to have to make such choices, but, increasingly, this may be the norm.

Many genes code for proteins, which in turn carry out the metabolic processes that keep organisms alive and reproducing. Genetic diversity can be measured as chemical diversity in that different species produce a variety of chemicals in their cells, both the proteins as well as the products and by-products of metabolism. This chemical diversity has potential benefit for humans as a source of pharmaceuticals, so it provides one way to measure diversity that is important to human health and welfare.

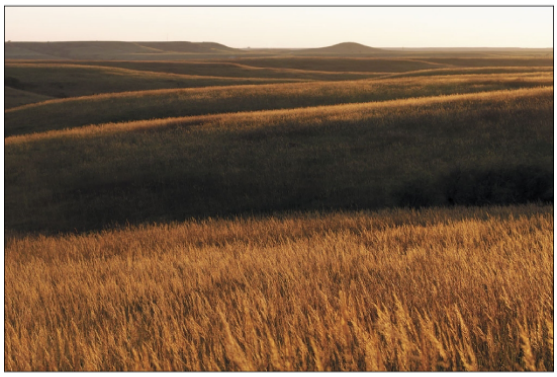
Humans have generated diversity in domestic animals, plants, and fungi. This diversity is also suffering losses because of migration, market forces, and increasing globalism in agriculture, especially in heavily-populated regions such as China, India, and Japan. The human population directly depends on this diversity as a stable food source; its decline is troubling to biologists and agricultural scientists.

It is also useful to define ecosystem diversity: the number of different ecosystems on the planet or in a given geographic area. Whole ecosystems can disappear even if some of the species might survive by adapting to other ecosystems. The loss

of an ecosystem means the loss of interactions between species, the loss of unique features of co-adaptation, and the loss of biological productivity that an ecosystem is able to create. An example of a largely-extinct ecosystem in North America is the prairie ecosystem. Prairies once spanned central North America from the boreal forest in northern Canada down into Mexico. Now, they have mostly disappeared, replaced by crop fields, pasture lands, and suburban sprawl. Many of the species survive, but the hugely-productive ecosystem that was responsible for creating the most productive agricultural soils is now gone. As a consequence, soils are disappearing or must be maintained at greater expense.



(a)



(b)

Figure 47.1B.1: **Ecosystem diversity**: The variety of ecosystems on earth, from (a) coral reef to (b) prairie, enables a great diversity of species to exist.

Current Species Diversity

Despite considerable effort, knowledge of the species that inhabit the planet is limited. A recent estimate suggests that the number of identified eukaryote species, about 1.5 million species, account for less than 20 percent of the total number of eukaryote species present on the planet (8.7 million species, by one estimate). Estimates of numbers of prokaryotic species are largely guesses, but biologists agree that science has only begun to catalog their diversity. Even with what is known, there is no central repository of names or samples of the described species; therefore, there is no way to be sure that 1.5 million is an accurate number. It is a best guess based on the opinions of experts in different taxonomic groups. Given that earth is losing species at an accelerating pace, science knows little about what is being lost.

Estimates of the Numbers of Described and Predicted Species by Taxonomic Group						
	Mora et al. 2011[1]		Chapman 2009[2]		Groombridge & Jenkins 2002[3]	
	Described	Predicted	Described	Predicted	Described	Predicted
Animalia	1,124,516	9,920,000	1,424,153	6,836,330	1,225,500	10,820,000
Chromista	17,892	34,900	25,044	200,500	—	—
Fungi	44,368	616,320	98,998	1,500,000	72,000	1,500,000
Plantae	224,244	314,600	310,129	390,800	270,000	320,000
Protozoa	16,236	72,800	28,871	1,000,000	80,000	600,000
Prokaryotes	—	—	10,307	1,000,000	10,175	—
Total	1,438,769	10,960,000	1,897,502	10,897,630	1,657,675	13,240,000

Figure 47.1B.1: **Estimates of species diversity**: Scientists have only been able to account for less than 20 percent of eukaryotic species that live on the planet. This table depicts the estimates based on taxonomic group.

There are various initiatives to catalog described species in accessible ways; the internet is facilitating that effort. Nevertheless, it has been pointed out that at the current rate of species description, which according to the State of Observed Species Report is 17,000 to 20,000 new species per year, it will take close to 500 years to finish describing life on this planet.

Naming and counting species may seem an unimportant pursuit given the other needs of humanity, but it is not as simple as counting. Describing species is a complex process by which biologists determine an organism's unique characteristics and whether or not that organism belongs to any other described species. It allows biologists to find and recognize the species after the initial discovery, which allows them to follow up on questions about its biology. In addition, the unique characteristics of each species make it potentially valuable to humans or other species on which humans depend. Understanding these characteristics is the value of finding and naming species.

47.1C: Biodiversity Change through Geological Time

Biodiversity has been affected by five mass extinction periods, which greatly influenced speciation and extinction rates.

Learning Objectives

- Describe how biodiversity has changed through geological time as a result of mass extinctions

Key Points

- Fossil records have been instrumental in differentiating between five periods that appear to show sudden and dramatic losses in biodiversity.
- Although there have been smaller periods of extinction, the five mass extinction periods demonstrate a continuous series of large extinction events.
- The Ordovician-Silurian extinction event was caused by glaciation followed by warming; it led to the extinction of 85 percent of marine species.
- The late Devonian extinction affected only marine species; the end- Permian extinction, the largest in history, may have been caused by volcanic activity, leading to a global-warming type event.
- The Triassic–Jurassic extinction event may have been caused by climate change, asteroid impact, or volcanic eruptions; however, the main cause is still unclear.
- The end- Cretaceous extinction event, in which the extinction of the dinosaurs occurred, is believed to have been caused by the impact of a large asteroid.

Key Terms

- **macroevolution**: large-scale patterns or processes in the history of life, including the origins of novel organism designs, evolutionary trends, adaptive radiations and extinctions
- **theropod**: any large, bipedal dinosaur, of the suborder Theropoda, from the Jurassic and Cretaceous periods
- **anoxic**: suffering from a reduced supply of oxygen

Biodiversity Change through Geological Time

The number of species on the planet, or in any geographical area, is the result of an equilibrium of two evolutionary processes that are ongoing: speciation and extinction. Both are natural “birth” and “death” processes of macroevolution. When speciation rates begin to outstrip extinction rates, the number of species will increase; likewise, the number of species will decrease when extinction rates begin to overtake speciation rates. Throughout earth’s history, these two processes have fluctuated, sometimes leading to dramatic changes in the number of species on earth.

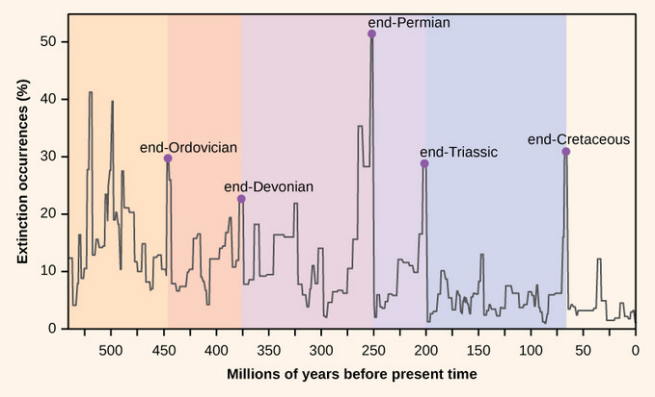


Figure 47.1C.1: **Fossil record**: Extinction occurrences, as reflected in the fossil record, have fluctuated throughout earth’s history. Sudden and dramatic losses of biodiversity, called mass extinctions, have occurred five times.

Paleontologists have identified five strata in the fossil record that appear to show sudden and dramatic losses in biodiversity known as mass extinctions. There are many lesser, yet still dramatic, extinction events, but the five mass extinctions have attracted the most research. An argument can be made that the five mass extinctions are only the five most

extreme events in a continuous series of large extinction events throughout the Phanerozoic (since 542 million years ago). In most cases, the hypothesized causes are still controversial.

The Five Mass Extinctions

The fossil record of the mass extinctions was the basis for defining periods of geological history, so they typically occur at the transition point between geological periods. The transition in fossils from one period to another reflects the dramatic loss of species and the gradual origin of new species.

Mass Extinctions		
Geological Period	**Mass Extinction Name**	**Time (millions of years ago)**
Ordovician–Silurian	end-Ordovician O–S	450–440
Late Devonian	end-Devonian	375–360
Permian–Triassic	end-Permian	251
Triassic–Jurassic	end-Triassic	205
Cretaceous–Paleogene	end-Cretaceous K–Pg (K–T)	65.5

Figure 47.1C.1: **Five mass extinctions**: The transitions between the five main mass extinctions can be seen in the rock strata. The table shows the time that elapsed between each period.

The Ordovician-Silurian extinction event is the first-recorded mass extinction and the second largest. During this period, about 85 percent of marine species (few species lived outside the oceans) became extinct. The main hypothesis for its cause was a period of glaciation followed by warming. These two extinction events, cooling and warming, were separated by about 1 million years; the climate changes affected temperatures and sea levels. Some researchers have suggested that a gamma-ray burst caused by a nearby supernova is a possible cause of the Ordovician-Silurian extinction. The gamma-ray burst would have stripped away the earth's ozone layer, causing intense ultraviolet radiation from the sun. It may account for climate changes observed at the time.

The late Devonian extinction may have occurred over a relatively long period of time. Its causes are poorly-understood and it appears to have have affected only marine species.

The end-Permian extinction was the largest in the history of life. Estimates predict that 96 percent of all marine species and 70 percent of all terrestrial species were lost.The causes for this mass extinction are not clear, but the leading suspect is extended and widespread volcanic activity that led to a runaway global-warming event. The oceans became largely anoxic, suffocating marine life. Terrestrial tetrapod diversity took 30 million years to recover after the end-Permian extinction. The Permian extinction dramatically altered earth's biodiversity composition and the course of evolution.

The causes of the Triassic–Jurassic extinction event are not clear. Hypotheses of climate change, asteroid impact, and volcanic eruptions have been argued. The extinction event occurred just before the breakup of the supercontinent Pangaea; although, recent scholarship suggests that the extinctions may have occurred more gradually throughout the Triassic.

The causes of the end-Cretaceous extinction event are the ones that are best understood. It was during this extinction event, about 65 million years ago, that the dinosaurs, the dominant vertebrate group for millions of years, disappeared from the planet (with the exception of a theropod clade that gave rise to birds). Indeed, every land animal that weighed more then 25 kg became extinct. The cause of this extinction is now understood to be the result of a cataclysmic impact of a large meteorite or asteroid off the coast of what is now the Yucatán Peninsula. This hypothesis, proposed first in 1980, was a radical explanation based on a sharp spike in the levels of iridium (which rains down from space in meteors at a fairly constant rate, but is otherwise absent on earth's surface) at the rock stratum that marks the boundary between the Cretaceous and Paleogene periods. The Cretaceous-Paleogene (K-Pg) boundary marked the disappearance of the dinosaurs in fossils, as well as many other taxa. The researchers who discovered the iridium spike interpreted it as a rapid influx of iridium from space to the atmosphere (in the form of a large asteroid), rather than a slowing in the deposition of sediments during that period. It was a radical explanation, but the report of an appropriately aged and sized impact crater in 1991 made the hypothesis more credible. Now, an abundance of geological evidence supports the hypothesis. Recovery times for biodiversity after the end-Cretaceous extinction were shorter, in geological time, than for the end-Permian extinction: on the order of 10 million years.

Figure 47.1C.1: **K-Pg mass extinction**: In 1980, Luis and Walter Alvarez, Frank Asaro, and Helen Michels discovered, across the world, a spike in the concentration of iridium within the sedimentary layer at the K–Pg boundary. These researchers hypothesized that this iridium spike was caused by an asteroid impact that resulted in the K–Pg mass extinction. In the photo, the iridium layer is the light band.

47.1D: The Pleistocene Extinction

Biodiversity loss, especially the disappearance of megafauna, during the Pleistocene Extinction has been linked to the arrival of humans.

Learning Objectives

- Describe the biodiversity loss associated with the Pleistocene extinction

Key Points

- The Pleistocene extinction in North America appears to have happened 10,000–12,000 years ago; this is several thousand years after the arrival of humans, who may have contributed to the extinction through hunting.
- Regions that experienced relatively-recent human arrivals show patterns of dramatic extinctions of megafauna hundreds to thousands of years after humans arrived; the same patterns are not observed in Eurasia and Africa as they did not experience a relatively-recent arrival of humans.
- To date, over-hunting by humans is one probable factor for the extinction of large mammals in the Pleistocene period.

Key Terms

- **Pleistocene**: of a geologic epoch within the Neogene period from about 2.6 million to 11,000 years ago; marked by the evolution of humans and the extinction of the large mammals
- **megafauna**: the large animals of a given region or time, considered as a group

The Pleistocene Extinction

The Pleistocene Extinction is one of the lesser extinctions and a relatively-recent one. It is well known that the North American, and to some degree Eurasian, megafauna disappeared toward the end of the last glaciation period. The extinction appears to have happened in a relatively-restricted time period between 10,000–12,000 years ago. In North America, the losses were quite dramatic and included the woolly mammoths (last dated about 4,000 years ago in an isolated population), mastodons, giant beavers, giant ground sloths, saber-toothed cats, and the North American camel, to name just a few. The possibility that the rapid extinction of these large animals was caused by over-hunting was first suggested in the 1900s; research into this hypothesis continues today. It seems probable that over-hunting was a factor in extinctions in many regions of the world.

Figure 47.1*D*.1: **Giant ground sloth**: Giant ground sloths, relatives of the living South American tree sloths, lived across much of North America. The giant sloths disappeared, along with the mammoths, mastodons, and many other large animals, at the end of the Pleistocene Epoch.

In general, the timing of the Pleistocene extinctions correlated with the arrival of humans and not with climate -change events, which is the main competing hypothesis for these extinctions. The extinctions began in Australia about 40,000 to 50,000 years ago, 10,000 to 20,000 years after the arrival of humans in the area. A marsupial lion, a giant one-ton wombat, and several giant kangaroo species disappeared. In North America, the extinctions of almost all of the large mammals occurred 10,000 to 12,000 years ago, several thousand years after the first evidence of humans in North America. All that are left are the smaller mammals such as bears, elk, moose, and cougars. Finally, on many remote oceanic islands, the extinctions of many species occurred with the coincidence of human arrivals. Not all of the islands had large animals, but when there were large animals, they were lost. Madagascar was colonized about 2,000 years ago; the large mammals (prosimians) that lived there became extinct. Eurasia and Africa do not show this pattern, but they also did not experience a recent arrival of humans. Humans arrived in Eurasia hundreds of thousands to over one million years ago, after the origin of the species in Africa. This topic remains an area of active research and hypothesizing. It seems clear that even if climate played a role, human hunting was an additional factor in the extinctions.

47.1E: Present-Time Extinctions

Human activities probably caused the Holocene mass extinctions; many methods have been employed to estimate these extinction rates.

Learning Objectives

- Describe the biodiversity loss during the Holocene extinction

Key Points

- The dodo was one of the first-known examples of a species that went extinct (in the 1600s) during the Holocene period.
- Steller's sea cows, passenger pigeons, Carolina parakeets, Japanese sea lions, and Caribbean monk seals are examples of species that went extinct from the 1700s-1900s.
- Major reasons for species extinctions during the Holocene period are due to overhunting, overfishing, and other human-related activities.
- Extinction rates can be estimated by comparing extinction events since the 1500s, but this method may underestimate the actual extinction rate value.
- Extinction rates can be estimated by observing species-area relationships and correlating species loss with habitat loss; however, this method may lead to overestimation.

Key Terms

- **species-area relationship**: relationship between area surveyed and number of species encountered; typically measured by incrementally increasing the area of a survey and determining the cumulative numbers of species
- **Holocene**: of a geologic epoch within the Neogene period from about 11,000 years ago to the present
- **extinction rate**: number of species becoming extinct over time, sometimes defined as extinctions per million species–years to make numbers manageable (E/MSY)

Present-Time Extinctions

The sixth, or Holocene, mass extinction appears to have begun earlier than previously believed and is mostly due to the activities of *Homo sapiens*. Since the beginning of the Holocene period, there have been numerous recent extinctions of individual species that are recorded in human writings. Most of these coincide with the expansion of the European colonies in the 1500s.

One of the earlier and popularly-known examples of extinction in this period is the dodo bird. The dodo bird lived in the forests of Mauritius, an island in the Indian Ocean, but became extinct around 1662. It was hunted for its meat by sailors as it was easy prey because the dodo, which did not evolve with humans, would approach people without fear. Introduced pigs, rats, and dogs, brought to the island by European ships, also killed dodo young and eggs.

Another example, Steller's sea cows, became extinct in 1768.The sea cow, first discovered by Europeans in 1741, was hunted for meat and oil. The last of the species was killed in 1768, which amounts to 27 years between the species' first contact with Europeans and its extinction. In addition, the last living passenger pigeon died in a zoo in Cincinnati, Ohio in 1914. This species was hunted and suffered from habitat loss through the clearing of forests for farmland. Furthermore, in 1918, the last living Carolina parakeet died in captivity. This species, once common in the eastern United States, was a victim of habitat loss and hunting as well. Adding to the extinction list, the Japanese sea lion, which inhabited a broad area around Japan and the coast of Korea, became extinct in the 1950s due to overfishing. The Caribbean monk seal, found in the Caribbean Sea, was driven to extinction through hunting by 1952.

These are only a few of the recorded extinctions in the past 500 years. The International Union for Conservation of Nature (IUCN) keeps a list of extinct and endangered species called the Red List. The list is not complete, but it describes 380 extinct species of vertebrates after 1500 AD, 86 of which were made extinct by over-hunting or overfishing.

Estimates of Present-Time Extinction Rates

Estimates of extinction rates are hampered by the fact that most extinctions are probably happening without observation since there are many organisms that are of less interest to humans and many that are undescribed.

The background extinction rate is estimated to be about one per million species per year (E/MSY). For example, assuming there are about ten million species in existence, the expectation is that ten species would become extinct each year.

One contemporary extinction rate estimate uses the extinctions in the written record since the year 1500. For birds alone, this method yields an estimate of 26 E/MSY. However, this value may be underestimated for three reasons. First, many species would not have been described until much later in the time period, so their loss would have gone unnoticed. Secondly, the number of recently-extinct species is increasing because extinct species now are being described from skeletal remains. Lastly, some species are probably already extinct even though conservationists are reluctant to name them as such. Taking these factors into account raises the estimated extinction rate closer to 100 E/MSY. The predicted rate by the end of the century is 1500 E/MSY.

A second approach to estimating present-day extinction rates is to correlate species loss with habitat loss by measuring forest-area loss and understanding species-area relationships. The species-area relationship is the rate at which new species are seen when the area surveyed is increased. Studies have shown that the number of species present increases as the size of the island increases. This phenomenon has also been shown to hold true in other habitats as well. Turning this relationship around, if the habitat area is reduced, the number of species living there will also decline. Estimates of extinction rates based on habitat loss and species-area relationships have suggested that with about 90 percent habitat loss an expected 50 percent of species would become extinct. Species-area estimates have led to species extinction rate calculations of about 1000 E/MSY and higher. In general, actual observations do not show this amount of loss, suggesting that there is a delay in extinction. Recent work has also called into question the applicability of the species-area relationship when estimating the loss of species. This work argues that the species-area relationship leads to an overestimate of extinction rates. A better relationship to use may be the endemics-area relationship. Using this method would bring estimates down to around 500 E/MSY in the coming century. Note that this value is still 500 times the background rate.

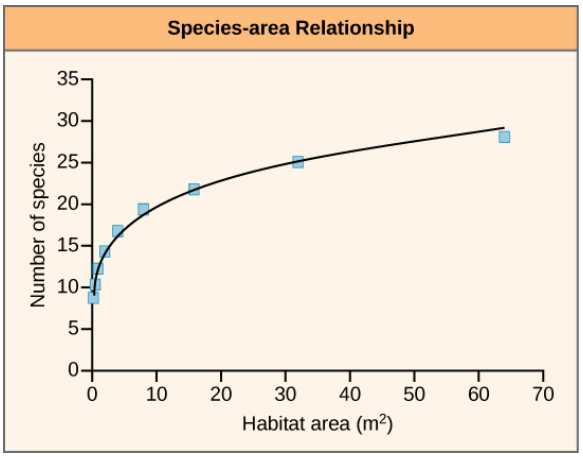


Figure 47.1E.1: **Species-area extinction estimates**: Studies have shown that the number of species that are present increases with the size of the habitat. As habitat is lost, the number of species present will decrease.

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47.2: The Importance of Biodiversity to Human Life

Topic hierarchy

- 47.2A: Human Health and Biodiversity
- 47.2B: Agricultural Diversity
- 47.2C: Managing Fisheries

47.2A: Human Health and Biodiversity

Maintaining biodiversity ultimately helps maintain of human health; many medicines are derived from plants and, recently, animal toxins.

Learning Objectives

- Describe the benefits to human health of maintaining biodiversity

Key Points

- Most plants produce secondary plant compounds, which are toxins used to protect the plant from insects and other animals that eat them, but some of which also work as medication.
- Antibiotics, which are responsible for extraordinary improvements in health and lifespans in developed countries, are compounds largely derived from fungi and bacteria.
- In recent years, animal venoms and poisons have excited intense research for their medicinal potential; at least five of these drugs have been FDA approved since 2007.

Key Terms

- **atropine**: an alkaloid extracted from the plant deadly nightshade (Atropa belladonna); used as a drug in medicine for its paralytic effects (in surgery to relax muscles, in dentistry to dry the mouth, etc.)
- **vincristine**: a particular drug used in chemotherapy
- **antibiotic**: any substance that can destroy or inhibit the growth of bacteria and similar microorganisms

Human Health

Contemporary societies that live close to the land often have a broad knowledge of the medicinal uses of plants growing in their area. Most plants produce secondary plant compounds, which are toxins used to protect the plants from insects and other animals that eat them, but some of which also work as medicines. For centuries in Europe, older knowledge about the medicinal uses of plants was compiled in herbals: books that identified plants and their uses. Humans are not the only species to use plants for medicinal reasons: the great apes (orangutans, chimpanzees, bonobos, and gorillas) have all been observed self-medicating with plants.

Modern pharmaceutical science also recognizes the importance of these plant compounds. Examples of significant medicines derived from plant compounds include aspirin, codeine, digoxin, atropine, and vincristine. Many medicines were once derived from plant extracts, but are now synthesized. It is estimated that, at one time, 25 percent of modern drugs contained at least one plant extract. That number has probably decreased to about 10 percent as natural plant ingredients are replaced by synthetic versions. Antibiotics, which are responsible for extraordinary improvements in health and lifespans in developed countries, are compounds largely derived from fungi and bacteria.

Figure 47.2A.1: **Medicinal uses of plants**: Catharanthus roseus, the Madagascar periwinkle, has various medicinal properties. Among other uses, it is a source of vincristine, a drug used in the treatment of lymphomas.

In recent years, animal venoms and poisons have excited intense research for their medicinal potential. By 2007, the FDA had approved five drugs based on animal toxins to treat diseases such as hypertension, chronic pain, and diabetes. Another

five drugs are undergoing clinical trials. At least six drugs are being used in other countries. Other toxins under investigation come from mammals, snakes, lizards, various amphibians, fish, snails, octopuses, and scorpions.

Aside from representing billions of dollars in profits, these medicines improve people's lives. Pharmaceutical companies are actively looking for new compounds synthesized by living organisms that can function as medicine. It is estimated that one-third of pharmaceutical research and development is spent on natural compounds. About 35 percent of new drugs brought to market between 1981 and 2002 were from natural compounds. The opportunities for new medications will be reduced in direct proportion to the disappearance of species. It is beneficial to humans, therefore, for medicinal purposes and many others, to maintain biodiversity.

47.2B: Agricultural Diversity

Maintaining genetic biodiversity of wild species of our crops that are related to domesticated species ensures our continued food supply.

Learning Objectives

- Assess the interactions of biodiversity with agricultural diversity

Key Points

- Agricultural diversity is driven by the demands of the topography, the limited movement of people, and the needs for crop rotation of varieties that do well in different fields.
- Resistance to disease is a chief benefit to maintaining crop biodiversity; lack of diversity in crop species risks an entire crop being wiped out by a disease to which it is susceptible.
- The ability to create new crop varieties relies on the diversity of varieties available and the accessibility of wild forms related to the crop plant that can be bred with existing varieties.
- Seed companies must continually breed new varieties to keep up with evolving pest organisms.

Key Terms

- **blight**: any of many plant diseases causing damage to, or the death of, leaves, fruit or other parts

Agricultural Diversity

Since the beginning of human agriculture more than 10,000 years ago, human groups have been breeding and selecting crop varieties. This crop diversity matched the cultural diversity of highly-subdivided populations of humans. For example, potatoes were domesticated beginning around 7,000 years ago in the central Andes of Peru and Bolivia. The potatoes grown in that region belong to seven species, while the number of varieties is probably in the thousands. Each variety has been bred to thrive at particular elevations and soil and climate conditions. The diversity is driven by the demands of the topography, the limited movement of people, and the demands created by crop rotation for different varieties that will do well in different fields and microclimates.

Potatoes are only one example of human-generated diversity. Every plant, animal, and fungus that has been cultivated by humans has been bred from original wild ancestor species into diverse varieties arising from the demands for food value, adaptation to growing conditions, and resistance to pests. The potato demonstrates a well-known example of the risks of low crop diversity. The tragic, Irish potato famine occurred when the single variety grown in Ireland became susceptible to a potato blight, wiping out the crop. The loss of the crop led to famine, death, and mass emigration. Resistance to disease is a chief benefit to maintaining crop biodiversity; lack of diversity in contemporary crop species carries similar risks. Seed companies, which are the source of most crop varieties in developed countries, must continually breed new varieties to keep up with evolving pest organisms. These same seed companies, however, have participated in the decline of the number of varieties available as they focus on selling fewer varieties in more areas of the world.

Figure 47.2*B*.1: **Svalbard Global Seed Vault**: The Svalbard Global Seed Vault is a protected storage facility in the arctic for seeds of earth's diverse crops.

The ability to create new crop varieties relies on the diversity of varieties available and the accessibility of wild forms related to the crop plant. These wild forms are often the source of new gene variants that can be bred with existing varieties to create varieties with new attributes. Loss of wild species related to a crop will mean the loss of potential in crop improvement. Maintaining the genetic diversity of wild species related to domesticated species ensures our continued food supply.

Since the 1920s, government agriculture departments have maintained seed banks of crop varieties as a way to maintain crop diversity. Sometimes, however, seed banks are lost through accidents; there is no way to replace them. In 2008, the Svalbard Global Seed Vault began storing seeds from around the world as a backup system to the regional seed banks. If a regional seed bank stores varieties in Svalbard, losses can be replaced from those stored here. The seed vault is located deep into the rock of an arctic island. Conditions within the vault are maintained at ideal temperature and humidity for seed survival, but the deep underground location of the vault in the arctic means that failure of the vault's systems will not compromise the climatic conditions inside the vault.

47.2C: Managing Fisheries

Overfishing leads to fishery extinctions, loss of a food source, and affects many other species in ways that may be impossible to predict.

Learning Objectives

- Explain the impact that the collapse of marine fisheries will have on human diet and health

Key Points

- Dramatic changes in species composition can result in an ecosystem shift, where species compositions differ from those that had been present before the depletion of the original fish stock.
- In general, the fish taken from fisheries have shifted to smaller species as larger species are fished to extinction; if such trends continue, aquatic ecosystems could become unavailable as food sources.
- Sustainable seafood is seafood from either fished or farmed sources that can maintain or increase production in the future without affecting the ecosystem in which it lives.

Key Terms

- **sustainability**: configuring society so that each can meet his own needs and greatest potential, while preserving biodiversity and natural ecosystems, and planning for future generations to maintain this potential.
- **overfishing**: fishing that reduces an aquatic population, or stock, to a level that is inadequate for the stock to replenish itself

Managing Wild Food Resources

In addition to growing crops and raising animals for food, humans obtain food resources from wild populations, primarily fish populations. For approximately 1 billion people, aquatic resources provide the main source of animal protein. But since 1990, global fish production has declined dramatically. Despite considerable effort, few fisheries on the planet are managed for sustainability.

Overfishing

Overfishing is the harvest of an aquatic population to a level that is too low for that population to replenish itself. Resource depletion, low biological growth rates, and critically low biomass levels result from overfishing. For example, overfishing of sharks has disrupted entire marine ecosystems.

Figure $47.2C.1$: **Overfishing**: When a fish stock is reduced to a level that is too low to allow that stock to replenish itself, overfishing has occurred. In addition to eliminating major food sources, overfishing is a threat to aquatic biodiversity.

The ability of a fishery to recover from overfishing depends on the species' natural ability to replenish itself as well as the ecosystem's conditions. Dramatic changes in species composition can result in an ecosystem shift, where other equilibrium energy flows involve species compositions different from those that had been present before overfishing occurred. For example, once trout have been overfished, carp might take over in a way that makes it impossible for the trout to re-establish a breeding population.

Sustainable Ecosystem Management

Fishery extinctions rarely lead to complete extinction of the harvested species, but rather to a radical restructuring of the marine ecosystem in which an abundant species is so over-harvested that it becomes a minor player, ecologically. In general, the fish taken from fisheries have shifted to smaller species as larger species are fished to extinction. In addition to humans losing the food source, these alterations affect many other species in ways that are difficult or impossible to predict. The collapse of fisheries has dramatic and long-lasting effects on local populations that work in the fishery. In addition, the loss of an inexpensive protein source to populations that cannot afford to replace it will increase the cost of living and limit societies in other ways. In general, the fish taken from fisheries have shifted to smaller species as larger species are fished to extinction. In addition to humans losing the food source, these alterations affect many other species in ways that are difficult or impossible to predict. The collapse of fisheries has dramatic and long-lasting effects on local populations that work in the fishery, not only the fishermen who depend directly on the fish stocks but all of the workers who are part of the industry, such as chefs, fish mongers, truck drivers, boat mechanics, and many more. In addition, the loss of an inexpensive protein source to populations that cannot afford to replace it can increase the cost of living and limit societies in other ways. The ultimate outcomes of overfishing and the commercial extinctions of fish stocks could include the loss of aquatic systems as food sources and economic crises for societies that depend on them.

Sustainable seafood is a movement that has gained momentum as more people become aware of overfishing and environmentally-destructive fishing methods. Sustainable seafood is seafood from either fished or farmed sources that can maintain or increase production in the future without jeopardizing the ecosystems from which it was acquired. In general, slow-growing fish that reproduce late in life, such as orange roughy, are vulnerable to overfishing and are considered unsustainable seafood. Seafood species that grow quickly and breed young, such as anchovies and sardines, are much more resistant to overfishing and are therefore labeled "sustainable" and promoted as good alternatives.

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47.3: Threats to Biodiversity

Topic hierarchy

- 47.3A: Habitat Loss and Sustainability
- 47.3B: Overharvesting
- 47.3C: Exotic Species
- 47.3D: Climate Change and Biodiversity

47.3A: Habitat Loss and Sustainability

Through increased adoption of sustainable practices, we can reduce habitat loss and its consequences.

Learning Objectives

- Describe the effects of habitat loss to biodiversity and concept of sustainability

Key Points

- Habitat destruction renders entire habitats functionally unable to support the species present; biodiversity is reduced in this process when existing organisms in the habitat are displaced or destroyed.
- Clearing areas for agricultural purposes is the main cause of habitat destruction; other principal causes include mining, logging, and urban sprawl.
- The primary cause of species extinction worldwide is habitat destruction.
- Sustainability is a term that describes how biological systems remain diverse and productive over time, creating the potential for long-term maintenance of human well-being.
- Reducing negative human impact requires three concepts: environmental management, management of human consumption of resources, and awareness of cultural and political concerns to increase sustainability.

Key Terms

- **sustainability**: Configuring society so that each person can meet their own needs and greatest potential, while preserving biodiversity and natural ecosystems, and planning for future generations to maintain this potential.
- **endemism**: The ecological state of a species being unique to a defined geographic location, such as an island, nation, country or other defined zone, or habitat type; organisms that are indigenous to a place are not endemic to it if they are also found elsewhere.
- **biodiversity**: The diversity (number and variety of species) of plant and animal life within a region.

Habitat Loss

Humans rely on technology to modify their environment and replace certain functions that were once performed by the natural ecosystem. Other species cannot do this. Elimination of their ecosystem – whether it is a forest, a desert, a grassland, a freshwater estuary, or a marine environment – will kill the individuals within most species. Remove the entire habitat within the range of a species and, unless they are one of the few species that do well in human-built environments, the species will become extinct.

Effects of Habitat Loss on Biodiversity

Habitat loss is a process of environmental change in which a natural habitat is rendered functionally unable to support the species present. This process may be natural or unnatural, and may be caused by habitat fragmentation, geological processes, climate change, or human activities such as the introduction of invasive species or ecosystem nutrient depletion. In the process of habitat destruction, the organisms that previously used the site are displaced or destroyed, reducing biodiversity.

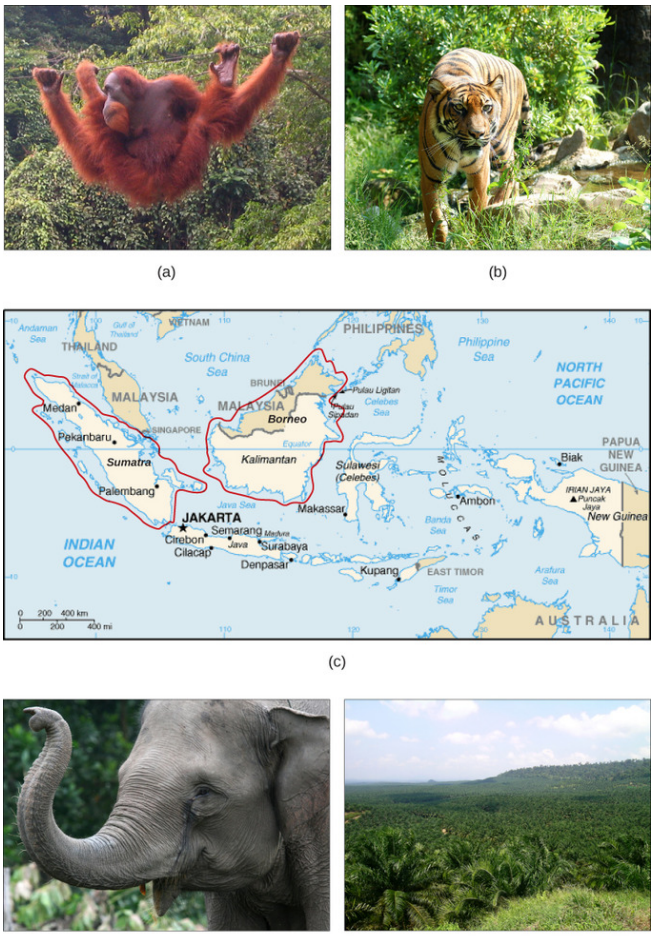

Figure 47.3*A*.1: **Biodiversity loss in Sumatra**: (a) One sub-species of orangutan is found only in the rain forests of Borneo, while the other sub-species of orangutan is found only in the rain forests of Sumatra. These animals are examples of the exceptional biodiversity of (c) the islands of Sumatra and Borneo. Other species include the (b) Sumatran tiger and the (d) Sumatran elephant, both of which are critically endangered. Rainforest habitat is being removed to make way for (e) oil palm plantations such as this one in Borneo's Sabah Province.

Human destruction of habitats has accelerated greatly in the latter half of the twentieth century. Natural habitats are often destroyed through human activity for the purpose of harvesting natural resources for industry production and urbanization. Clearing habitats for agriculture, for example, is the principal cause of habitat destruction. Other important causes of habitat destruction include mining, logging, and urban sprawl. Habitat destruction is currently ranked as the primary cause of species extinction worldwide.

Consider the exceptional biodiversity of Sumatra. It is home to one sub-species of orangutan, a species of critically endangered elephant, and the Sumatran tiger; however half of Sumatra's forest is now gone. The neighboring island of Borneo, home to the other sub-species of orangutan, has lost a similar area of forest, and forest loss continues in protected areas. The orangutan in Borneo is listed as endangered by the International Union for Conservation of Nature (IUCN), but it is simply the most visible of thousands of species that will not survive the disappearance of the forests of Borneo. The forests are being removed for their timber, and to clear space for plantations of palm oil, an oil used in Europe for many items including food products, cosmetics, and biodiesel.

A five-year estimate of global forest cover loss for the years 2000–2005 was 3.1 percent. In the humid tropics where forest loss is primarily from timber extraction, 272,000 km^2 was lost out of a global total of 11,564,000 km^2 (or 2.4 percent). In the tropics, these losses also represent the extinction of species because of high levels of endemism.

Since the Neolithic Revolution, about 47% of the world's forests have been lost to human use. Present-day forests occupy about a quarter of the world's ice-free land, with about half of these occurring in the tropics. In temperate and boreal

regions, forest area is gradually increasing (with the exception of Siberia), but deforestation in the tropics is of major concern.

Feeding more than seven billion human bodies takes a heavy toll on the earth's resources. This begins with the appropriation of about 38 percent of the earth's land surface and about 20 percent of its net primary productivity. Added to this are the resource-hungry activities of industrial agribusiness: everything from crops' need for irrigation water, synthetic fertilizers, and pesticides, to the resource costs of food packaging, transport (now a major part of global trade), and retail.

Figure 47.3A.1: **Sustainability and deforestation**: Since the Neolithic Revolution, nearly half of the world's forests have been destroyed for human use. Sustainable practices, which preserve environments for long-term maintenance and well-being, can help preserve habitats and ecosystems for greater biodiversity.

Sustainability

Sustainability is a concept that describes how biological systems remain diverse and productive over time. Long-lived and healthy wetlands and forests are examples of sustainable biological systems. For humans, sustainability is the potential for long-term maintenance of well-being, which has ecological, economic, political, and cultural dimensions. Sustainability requires the reconciliation of environmental, social, and economic demands, which are also referred to as the "three pillars" of sustainability.

Healthy ecosystems and environments are necessary for the survival and flourishing of humans and other organisms, and there are a number of ways to reduce humans' negative impact on the environment. One approach is environmental management, which is based largely on information gained from earth science, environmental science, and conservation biology. A second approach is management of human consumption of resources, which is based largely on information gained from economics. A third, more recent, approach adds cultural and political concerns into the sustainability matrix.

Loss of biodiversity stems largely from the habitat loss and fragmentation produced by human appropriation of land for development, forestry and agriculture as natural capital is progressively converted to human-made capital. At the local human scale, sustainability benefits accrue from the creation of green cities and sustainable parks and gardens. Similarly, environmental problems associated with industrial agriculture and agribusiness are now being addressed through such movements as sustainable agriculture, organic farming, and more-sustainable business practices.

47.3B: Overharvesting

Overharvesting threatens biodiversity by degrading ecosystems and eliminating species of plants, animals, and other organisms.

Learning Objectives

- Explain why overharvesting is a threat to biodiversity

Key Points

- Until recently, human populations harvested resources in limited quantities. Today, new methods of harvest and capture contribute to overharvesting and overexploitation.
- Overharvesting stems from several factors, including an exponential increase in the human population, expanding markets, increasing demand, and improved access and techniques for capture.
- Overharvesting natural resources for extended periods of time depletes these resources until they cannot recover within a short period of time; some may never recover.
- Overharvesting is one of five primary activities threatening global biodiversity; others include pollution, introduced species, habitat fragmentation, and habitat destruction.
- Aquatic species are especially threatened by overharvesting, due to a situation known as the tragedy of the commons.

Key Terms

- **overexploitation**: Excessive and damaging use of natural resources, including plants and animals.
- **trawler**: A fishing boat that uses a dragnet, or "trawl net," to catch fish.
- **apex predator**: An animal at the top of the food chain, preying on other species but not prey itself.

Overharvesting

Overharvesting, also called overexploitation, refers to harvesting a renewable resource to the point of diminishing returns. Ecologists use the term to describe populations that are harvested at a rate that is unsustainable, given their natural rates of mortality and capacities for reproduction. The term applies to natural resources such as wild medicinal plants, grazing pastures, game animals, fish stocks, forests, and water aquifers. Sustained overharvesting can lead to the destruction of the resource, and is one of the five main activities – along with pollution, introduced species, habitat fragmentation, and habitat destruction – that threaten global biodiversity today.

All living organisms require resources to survive. Overharvesting these resources for extended periods of time can deplete natural resources to the point where they are unable to recover within a short time frame. Humans have always harvested food and other resources they have needed to survive; however, human populations, historically, were small and methods of collection limited to small quantities. Exponential increase in human population, expanding markets, and increasing demand, combined with improved access and techniques for capture, are causing the exploitation of many species beyond sustainable levels.

Effects of overharvesting

As mentioned above, sustained overharvesting is one of the primary threats to biodiversity. Overharvesting can lead to resource destruction, including extinction at the population level and even extinction of whole species. Depleting the numbers or amount of certain resources can also change their quality; for example, the overharvesting of footstool palm (a wild palm tree found in Southeast Asia, the leaves of which are used for thatching and food wrapping) has resulted in its leaf size becoming smaller.

Overharvesting not only threatens the resource being harvested, but can directly impact humans as well – for example by decreasing the biodiversity necessary for medicinal resources. A significant proportion of drugs and medicines are natural products which are derived, directly or indirectly, from biological sources. However, unregulated and inappropriate harvesting could potentially lead to overexploitation, ecosystem degradation, and loss of biodiversity; further, it can negatively impact the rights of the communities and states from which the resources are taken.

Tragedy of the commons

Overharvesting is a serious threat to many species, especially aquatic ones. Common resources – or resources that are shared, such as fisheries – are subject to an economic pressure known as "the tragedy of the commons," in which essentially no harvester has a motivation to exercise restraint in harvesting from a certain area, because that area is not owned by that harvester. The natural outcome of harvesting common resources is their overexploitation.

For example, most fisheries are managed as a common resource even when the fishing territory lies within a country's territorial waters; because of this, fishers have very little motivation to limit their harvesting, and in fact technology gives fishers the ability to overfish. In a few fisheries, the biological growth of the resource is less than the potential growth of the profits made from fishing if that time and money were invested elsewhere. In these cases (for example, whales) economic forces will always drive toward fishing the population to extinction.

Figure $47.3B.1$: **Cod trawler and net**: Overharvesting fisheries is an especially salient problem because of a situation termed the tragedy of the commons. In this situation, fishers have no real incentive to practice restraint when harvesting fish because they do not own the fisheries.

Cascade Effects

Overexploitation of species can also result in cascade effects, particularly if a habitat loses its apex predator. Because of the loss of the top predator, a dramatic increase in their prey species can occur. In turn, the unchecked prey can then overexploit their own food resources until population numbers dwindle, possibly to the point of extinction.

C: Exotic Species

Exotic species introduced into foreign ecosystems can threaten native species through competition for resources, predation, and disease.

Learning Objectives

- Describe the impact of exotic and invasive species on native species

Key Points

- Exotic species introduced to new environments often reset the ecological conditions in that new habitat, threatening the species that exist there; this is the reason that they are also termed invasive species.
- Invasive species that are closely related to rare native species have the potential to hybridize with the native species; harmful effects of hybridization have led to a decline and even extinction of native species.
- Biologists studying frogs and toads may be inadvertently responsible for the worldwide spread of a fungus deadly to amphibians.

Key Terms

- **invasive species**: any species that has been introduced to an environment where it is not native and has since become a nuisance through rapid spread and increase in numbers, often to the detriment of native species

Exotic Species

Exotic species are those that have been intentionally or unintentionally introduced by humans into an ecosystem in which they did not evolve. Such introductions probably occur frequently as natural phenomena. For example, Kudzu (*Pueraria lobata*), which is native to Japan, was introduced in the United States in 1876. It was later planted for soil conservation. Problematically, it grows too well in the southeastern United States: up to one foot each day. It is now a pest species, covering over seven million acres in the southeastern United States. If an introduced species is able to survive in its new habitat, that introduction is now reflected in the observed range of the species. Human transportation of people and goods, including the intentional transport of organisms for trade, has dramatically increased the introduction of species into new ecosystems, sometimes at distances that are well beyond the capacity of the species to ever travel itself and outside the range of the species' natural predators.

Figure 47.3C.1: **Exotic threats**: The brown tree snake, Boiga irregularis, is an exotic species that has caused numerous extinctions on the island of Guam since its accidental introduction in 1950.

Most exotic species introductions probably fail because of the low number of individuals introduced or poor adaptation to the ecosystem they enter. Some species, however, possess preadaptations that can make them especially successful in a new ecosystem. These exotic species often undergo dramatic population increases in their new habitat, resetting the

ical conditions in the new environment, while threatening the species that exist there. For this reason, exotic species, lled invasive species, can threaten other species through competition for resources, predation, or disease.

Exotic Species Threaten Native Species

Invasive species can change the functions of ecosystems. For example, invasive plants can alter the fire regimen, nutrient cycling, and hydrology in native ecosystems. Invasive species that are closely related to rare native species have the potential to hybridize with the native species. Harmful effects of hybridization have led to a decline and even extinction of native species. For example, hybridization with introduced cordgrass, *Spartina alterniflora*, threatens the existence of California cordgrass in San Francisco Bay. Invasive species cause competition for native species. Four hundred of the 958 endangered species under the Endangered Species Act are at risk due to this competition.

Figure 47.3C.1: **Global decline in amphibian species**: This Limosa Harlequin Frog (Atelopus limosus), an endangered species from Panama, died from a fungal disease called chytridiomycosis. The red lesions are symptomatic of the disease.

Lakes and islands are particularly vulnerable to extinction threats from introduced species. In Lake Victoria, as mentioned earlier, the intentional introduction of the Nile perch was largely responsible for the extinction of about 200 species of cichlids. The accidental introduction of the brown tree snake via aircraft from the Solomon Islands to Guam in 1950 has led to the extinction of three species of birds and three to five species of reptiles endemic to the island. Several other species are still threatened. The brown tree snake is adept at exploiting human transportation as a means to migrate; one was even found on an aircraft arriving in Corpus Christi, Texas. Constant vigilance on the part of airport, military, and commercial aircraft personnel is required to prevent the snake from moving from Guam to other islands in the Pacific, especially Hawaii. Islands do not make up a large area of land on the globe, but they do contain a disproportionate number of endemic species because of their isolation from mainland ancestors.

It now appears that the global decline in amphibian species recognized in the 1990s is, in some part, caused by the fungus *Batrachochytrium dendrobatidis*, which causes the disease chytridiomycosis. There is evidence that the fungus, native to Africa, may have been spread throughout the world by transport of a commonly-used laboratory and pet species: the African clawed toad (*Xenopus laevis*). It may well be that biologists themselves are responsible for spreading this disease worldwide. The North American bullfrog, *Rana catesbeiana*, which has also been widely introduced as a food animal, but which easily escapes captivity, survives most infections of Batrachochytriumdendrobatidis and can act as a reservoir for the disease.

47.3D: Climate Change and Biodiversity

The global warming trend is recognized as a major biodiversity threat, especially when combined with other threats such as habitat loss.

Learning Objectives

- Evaluate climate change and its impact on biodiversity

Key Points

- This warming trend is persistently shifting colder climates further toward the north and south poles, forcing species to move with their own adapted climate norms, while also facing habitat gaps along the way.
- Climate shifts will move up mountains, resulting in the crowding of species higher in altitude and eliminating the habitat for those species adapted to those high elevations; indeed, some climates will completely disappear.
- Global warming will also raise ocean water levels due to melted water from glaciers and the greater volume of warmer water; shorelines will be flooded, affecting many species; many islands will disappear altogether.

Key Terms

- **anthropogenic**: having its origin in the influence of human activity on nature
- **biodiversity**: the diversity (number and variety of species) of plant and animal life within a region

Climate Change

Climate change, specifically, the anthropogenic (caused by humans) warming trend presently underway, is recognized as a major extinction threat, particularly when combined with other threats such as habitat loss. Scientists disagree about the probable magnitude of the effects, with extinction rate estimates ranging from 15 percent to 40 percent of species by 2050. Scientists do agree, however, that climate change will alter regional climates, including rainfall and snowfall patterns, making habitats less hospitable to the species living in them.

Impact of Climate Change on Biodiversity

Figure 47.3D.1: **Grizzly-polar bear hybrid**: Since 2008, grizzly bears (Ursus arctos horribilis) have been spotted farther north than their historic range, a possible consequence of climate change. As a result, grizzly bear habitat now overlaps polar bear (Ursus maritimus) habitat. The two kinds of bears, which are capable of mating and producing viable offspring, are considered separate species as historically they lived in different habitats and never met. However, in 2006 a hunter shot a wild grizzly-polar bear hybrid known as a grolar bear, the first wild hybrid ever found.

The warming trend will shift colder climates toward the north and south poles, forcing species to move with their adapted climate norms while facing habitat gaps along the way. The shifting ranges will impose new competitive regimes on species as they find themselves in contact with other species not present in their historic range. One such unexpected species contact is between polar bears and grizzly bears. Previously, these two species had separate ranges. Now, with their ranges are overlapping, there are documented cases of these two species mating and producing viable offspring. Changing climates also throw off species' delicate timing adaptations to seasonal food resources and breeding times. Many contemporary mismatches to shifts in resource availability and timing have recently been documented.

Range shifts are already being observed. For example, some European bird species' ranges have moved 91 km northward. The same study suggests that the optimal shift based on warming trends was double that distance, suggesting that the populations are not moving quickly enough. Range shifts have also been observed in plants, butterflies, other insects, freshwater fishes, reptiles, and mammals.

Climate gradients will also move up mountains, eventually crowding species higher in altitude and eliminating the habitat for those species adapted to the highest elevations. Some climates will completely disappear. The rate of warming appears to be accelerated in the arctic, which is recognized as a serious threat to polar bear populations that require sea ice to hunt seals during the winter months; seals are the only source of protein available to polar bears. A trend to decreasing sea ice coverage has occurred since observations began in the mid-twentieth century. The rate of decline observed in recent years is far greater than previously predicted by climate models.

Finally, global warming will raise ocean levels due to glacial melt and the greater volume of warmer water. Shorelines will be inundated, reducing island size, which will have an effect on many species; a number of islands will disappear entirely. Additionally, the gradual melting and subsequent refreezing of the poles, glaciers, and higher elevation mountains, a cycle that has provided freshwater to environments for centuries, will also be jeopardized. This could result in an overabundance of salt water and a shortage of fresh water.

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47.4: Preserving Biodiversity

Topic hierarchy

- 47.4A: Measuring Biodiversity
- 47.4B: Changing Human Behavior in Response to Biodiversity Loss
- 47.4C: Ecological Restoration

47.4A: Measuring Biodiversity

Technology has matured to the point where we can begin cataloging the planet's species in accessible ways; DNA barcoding is one such method.

Learning Objectives

- Explain how DNA barcoding aids in measuring biodiversity

Key Points

- DNA barcoding is a taxonomic method that uses a short genetic marker in an organism's DNA to identify it as belonging to a particular species.
- Barcoding allows us to classify organisms that would otherwise be difficult to identify, such as in situations where only part of an organism is available, or is too immature to identify by conventional methods.
- At the present rate of description of new species, it will take close to 500 years before the complete catalog of life is known; however, most species will be extinct before this time.
- Even with barcoding, it is difficult to know which species are threatened and to what degree they are threatened, a task carried out by the non-profit IUCN (International Union for Conservation of Nature).

Key Terms

- **barcoding**: a taxonomic method that uses a short genetic marker in an organism's DNA to identify it as belonging to a particular species
- **phylogeny**: the evolutionary history of an organism
- **taxonomy**: the science of finding, describing, classifying and naming organisms

Measuring Biodiversity

The technologies of molecular genetics, data processing, and data storage are maturing to the point where cataloging the planet's species in an accessible way is close to feasible. DNA barcoding is one molecular genetic method, which takes advantage of the rapid evolution in a mitochondrial gene present in eukaryotes, to identify species using the sequence of portions of the gene. Plants may be barcoded using a combination of chloroplast genes.

DNA Barcoding

DNA barcoding is a taxonomic method that uses a short genetic marker in an organism's DNA to identify it as belonging to a particular species. It differs from molecular phylogeny in that the main goal is not to determine patterns of relationship, but to identify an unknown sample in terms of a preexisting classification. The most commonly-used barcode region for animals, at least, is a segment of approximately 600 base pairs of the mitochondrial gene cytochrome oxidase I (COI).

Applications include, for example, identifying plant leaves (even when flowers or fruit are not available), identifying insect larvae (which may have fewer diagnostic characters than adults and are frequently less well-known), identifying the diet of an animal (based on its stomach contents or feces), and identifying products in commerce (for example, herbal supplements or wood).

Rapid, mass-sequencing machines make the molecular genetics portion of the work relatively inexpensive and quick. Computer resources store and make available the large volumes of data. Projects are currently underway to use DNA barcoding to catalog museum specimens, which have already been named and studied, as well as testing the method on less studied groups. As of mid-2012, close to 150,000 named species had been barcoded. Early studies suggest there are significant numbers of undescribed species that looked too much like sibling species to previously be recognized as different. These now can be identified with DNA barcoding.

Numerous computer databases now provide information about named species and a framework for adding new species. However, as already noted, at the present rate of description of new species, it will take close to 500 years before the complete catalog of life is known. Many, perhaps most, species on the planet do not have that much time.

There is also the problem of understanding which species known to science are threatened and to what degree they are threatened. This task is carried out by the non-profit IUCN (International Union for Conservation of Nature) which maintains the Red List: an online listing of endangered species categorized by taxonomy, type of threat, and other criteria. The Red List is supported by scientific research. In 2011, the list contained 61,000 species, all with supporting documentation.

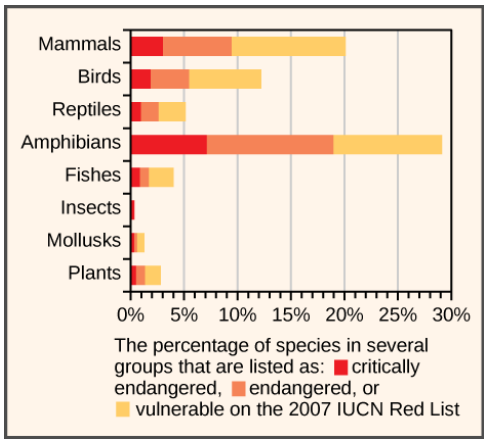

Figure 47.4*A*.1: **IUCN Red List**: This chart shows the percentage of various animal species, by group, on the IUCN Red List as of 2007. The Red List is an online listing of endangered species categorized by taxonomy, type of threat, and other criteria.

47.4B: Changing Human Behavior in Response to Biodiversity Loss

Human responses to climate change and species loss include national and international legal measures, as well as the creation of preserves.

Learning Objectives

- Detail the benefits and limitations of different human responses to climate change and species loss, such as using preserves to conserve species

Key Points

- Legislation throughout the world has been enacted to protect species; the legislation includes international treaties as well as national and state laws.
- The international response to global warming has been mixed; the Kyoto Protocol, an international agreement that came out of the United Nations climate change convention that committed countries to reducing greenhouse gas emissions, was ratified by some countries, but spurned by others.
- Many nations have laws that protect endangered species: for example, forbidding hunting, restricting land development, or creating preserves (purchased land for the explicit purpose of attempting to protect species and ecosystems).
- Preserves protect biodiversity in many ways, two of which are captive breeding and private farming.

Key Terms

- **endangered species**: a species which is in danger of becoming extinct

Changing Human Behavior

Human Responses to Climate Change and Species Loss

Legislation throughout the world has been enacted to protect species and include international treaties as well as national and state laws. The Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) treaty came into force in 1975. The treaty (and the national legislation that supports) it provides a legal framework for preventing approximately 33,000 listed species from being transported across nations' borders, thus protecting them from being caught or killed when international trade is involved. The treaty is limited in its reach because it only deals with international movement of organisms or their parts. It is also limited by various countries' ability or willingness to enforce the treaty and supporting legislation. The illegal trade in organisms and their parts is probably a market worth hundreds of millions of dollars. Illegal wildlife trade is monitored by another non-profit, the Trade Records Analysis of Flora and Fauna in Commerce (TRAFFIC).

Within many countries there are laws that protect endangered species and regulate hunting and fishing. In the United States, the Endangered Species Act (ESA) was enacted in 1973. The U.S. Fish & Wildlife Service is required by law to develop management plans that protect the species listed as at risk in the Act in order to bring them back to sustainable numbers. The Act, and others like it in other countries, is a useful tool, but it suffers because it is often difficult to get a species listed or to get an effective management plan in place once it is listed. Additionally, species may be controversially taken off the list without necessarily having had a change in their situation. More fundamentally, the approach to protecting individual species rather than entire ecosystems is inefficient as it focuses efforts on a few highly-visible and often charismatic species, perhaps at the expense of other species that go unprotected. At the same time, the Act has a critical habitat provision outlined in the recovery mechanism that may benefit species other than the one targeted for management.

The Migratory Bird Treaty Act (MBTA) is an agreement between the United States and Canada that was signed into law in 1918 in response to declines in North American bird species caused by hunting. The Act now lists over 800 protected species. It makes it illegal to disturb or kill the protected species or distribute their parts (much of the hunting of birds in the past was for their feathers).

In relation to global warming, The Kyoto Protocol, an international agreement that came out of the United Nations Framework Convention on Climate Change that committed countries to reducing greenhouse gas emissions by 2012, was

ratified by some countries, but spurned by others. Two important countries in terms of their potential impact that did not ratify the Kyoto Protocol were the United States and China. The United States rejected it as a result of a powerful fossil fuel industry, while China did so because of a concern that it would stifle the nation's growth. Some goals for reduction in greenhouse gasses were met and exceeded by individual countries, but worldwide, the effort to limit greenhouse gas production is not succeeding. The intended replacement for the Kyoto Protocol has not materialized because governments cannot agree on timelines and benchmarks. Meanwhile, climate scientists predict the resulting costs to human societies and biodiversity will be high.

Using Preserves to Conserve Species

The private, non-profit sector plays a large role in the conservation effort both in North America and around the world. The approaches range from species-specific organizations to the broadly-focused groups mentioned above.

Many nations have laws that protect endangered species: for example, forbidding hunting, restricting land development, or creating preserves. Preserves are purchased land for the explicit purpose of attempting to protect species and ecosystems. They also protect biodiversity in many ways, such as through captive breeding and private farming.

Captive breeding is the process of breeding rare or endangered species in human-controlled environments with restricted settings, such as wildlife preserves and conservation facilities. Captive breeding is meant to prevent species extinction and to stabilize the population of the species so that it will not disappear. This technique has worked for many species for some time, such as for Pere David's deer. However, captive breeding techniques are usually difficult to implement for such highly-mobile species as some migratory birds (e.g. cranes) and fish (e.g. hilsa). Additionally, if the captive-breeding population is too small, then inbreeding may occur due to a reduced gene pool, which may also reduce immunity.

Figure 47.4*B*.1: **Captive breeding**: Père David's deer is a species of deer that is currently extinct in the wild: all known specimens are found only in captivity. This species was successfully saved through captive breeding programs after almost being hunted to extinction in China.

47.4C: Ecological Restoration

The purpose of ecological restoration projects, such as wildlife and ecosystem preserves, is to return ecosystems to pre-disturbance states.

Learning Objectives

- Explain the purpose of ecological restoration projects

Key Points

- Ecological preserves, while effective in the short term, are not yet viable, long-term solutions.
- Some of the limitations on preserves as conservation tools include preserve designs, political and economic pressures, and climate change.
- Habitat restoration is a promising tool for restoring and maintaining biodiversity; restoration can improve the biodiversity of degraded ecosystems.

Key Terms

- **biodiversity**: the diversity (number and variety of species) of plant and animal life within a region
- **keystone species**: a species that exerts a large, stabilizing influence throughout an ecological community, despite its relatively small numerical abundance
- **carrion**: dead flesh; carcasses

Conservation in Preserves

Establishment of wildlife and ecosystem preserves is one of the key tools in conservation efforts. A preserve is an area of land set aside with varying degrees of protection for the organisms that exist within the boundaries of the preserve. Preserves can be effective in the short term for protecting both species and ecosystems, but they face challenges that scientists are still exploring in order to strengthen their viability as long-term solutions.

Due to the way protected lands are allocated (they tend to contain less economically-valuable resources rather than being set aside specifically for at-risk species or ecosystems) and the way biodiversity is distributed, determining a target percentage of land or marine habitat that should be protected to maintain biodiversity levels is challenging. The IUCN (International Union for Conservation of Nature) World Parks Congress estimated that 11.5 percent of earth's land surface was covered by preserves of various kinds in 2003. This area is greater than previous goals; however, it only represents 9 out of 14 recognized major biomes. Research has shown that 12 percent of all species live solely outside preserves; these percentages are much higher when only threatened species and high-quality preserves are considered.

Limitations on Preserves

Some of the limitations on preserves as conservation tools stem from preserve designs. Additionally, political and economic pressures typically make preserves smaller, never larger, so setting aside areas that are large enough is difficult. Climate change will create inevitable problems with the location of preserves. The species within them will migrate to higher latitudes as the habitat of the preserve becomes less favorable. Scientists are planning for the effects of global warming on future preserves and striving to predict the need for new preserves to accommodate anticipated changes to habitats. Some argue that conservation preserves reinforce the cultural perception that humans are separate from nature, can exist outside of it, and can only operate in ways that do damage to biodiversity. Creating preserves reduces the pressure on human activities outside the preserves to be sustainable and non-damaging to biodiversity.

Habitat Restoration

Habitat restoration holds considerable promise as a mechanism for restoring and maintaining biodiversity. Restoration ecology aims to return ecosystems to a more natural, pre-disturbance state. Of course, once a species has become extinct, its restoration is impossible. However, restoration can improve the biodiversity of degraded ecosystems.

Reintroducing wolves, a top predator, to Yellowstone National Park in 1995 led to dramatic changes in the ecosystem that increased biodiversity. The wolves, which function to suppress elk and coyote populations, provide more-abundant

resources to the guild of carrion (flesh) eaters. Reducing elk populations has allowed re-vegetation of riparian areas, which has increased the diversity of species in that habitat. Decreasing the coyote population increased the populations of species that were previously suppressed by this predator. The number of species of carrion eaters has increased because of the predatory activities of the wolves. In this habitat, the wolf is a keystone species: it is a species that is instrumental in maintaining diversity in an ecosystem. Removing a keystone species from an ecological community may cause a collapse in diversity. Similarly, restoring a keystone species can have dramatic effects. The results from the Yellowstone experiment suggest that restoring a keystone species can have the effect of restoring biodiversity in the community. Ecologists have argued for the identification of keystone species where possible and for focusing protection efforts on those species. It also makes sense to attempt to return them to their ecosystem if they have been removed.

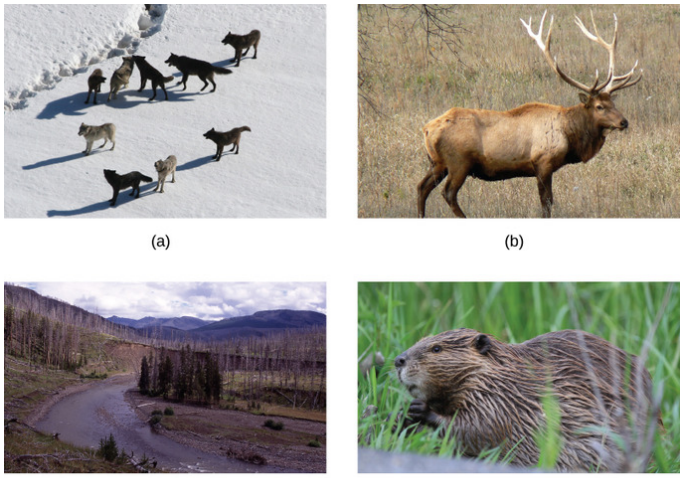

Figure 47.4C.1: **Yellowstone National Park restoration efforts**: (a) The Gibbon wolf pack in Yellowstone National Park represents a keystone species. The reintroduction of wolves into Yellowstone National Park in 1995 led to a change in the grazing behavior of (b) elk. To avoid predation, the elk no longer grazed exposed stream and riverbeds, such as (c) the Lamar Riverbed in Yellowstone. This allowed willow and cottonwood seedlings to grow. The seedlings decreased erosion and provided shading to the creek, which improved fish habitat. A new colony of (d) beaver may also have benefited from the habitat change.

Other large-scale restoration experiments underway involve dam removal. In the United States, since the mid-1980s, many aging dams are being considered for removal rather than replacement because of shifting beliefs about the ecological value of free-flowing rivers and because many dams no longer provide the benefit and functions that they did when they were first built. The measured benefits of dam removal include restoration of naturally-fluctuating water levels (the purpose of dams is frequently to reduce variation in river flows), which leads to increased fish diversity and improved water quality. In the Pacific Northwest, dam removal projects are expected to increase populations of salmon, which is considered a keystone species because it transports key nutrients to inland ecosystems during its annual spawning migrations. In other regions, such as the Atlantic coast, dam removal has allowed the return of spawning anadromous fish species (species that are born in fresh water, live most of their lives in salt water, and return to fresh water to spawn). Some of the largest dam removal projects have yet to occur or have happened too recently for the consequences to be measured. The large-scale ecological experiments that these removal projects constitute will provide valuable data for other dam projects slated either for removal or construction.

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CHAPTER OVERVIEW

5: STRUCTURE AND FUNCTION OF PLASMA MEMBRANES

5.1: COMPONENTS AND STRUCTURE

5.1A: COMPONENTS OF PLASMA MEMBRANES
5.1B: FLUID MOSAIC MODEL
The fluid mosaic model describes the plasma membrane structure as a mosaic of phospholipids, cholesterol, proteins, and carbohydrates.

5.1C: MEMBRANE FLUIDITY
5.2: PASSIVE TRANSPORT

5.2A: THE ROLE OF PASSIVE TRANSPORT
Passive transport, such as diffusion and osmosis, moves materials of small molecular weight across membranes.

5.2B: SELECTIVE PERMEABILITY
The hydrophobic and hydrophilic regions of plasma membranes aid the diffusion of some molecules and hinder the diffusion of others.

5.2C: DIFFUSION
Diffusion is a process of passive transport in which molecules move from an area of higher concentration to one of lower concentration.

5.2D: FACILITATED TRANSPORT
Facilitated diffusion is a process by which molecules are transported across the plasma membrane with the help of membrane proteins.

5.2E: OSMOSIS
Osmosis is the movement of water through a semipermeable membrane according to the concentration gradient of water across the membrane, which is inversely proportional to the concentration of solutes. Semipermeable membranes, also termed selectively permeable membranes or partially permeable membranes, allow certain molecules or ions to pass through by diffusion.

5.2F: TONICITY
Tonicity describes how an extracellular solution can change the volume of a cell by affecting osmosis. A solution’s tonicity often directly correlates with the osmolarity of the solution. Osmolarity describes the total solute concentration of the solution. A solution with low osmolarity has a greater number of water molecules relative to the number of solute particles; a solution with high osmolarity has fewer water molecules with respect to solute particles.

5.3: ACTIVE TRANSPORT

5.3A: ELECTROCHEMICAL GRADIENT
5.3B: PRIMARY ACTIVE TRANSPORT
5.3C: SECONDARY ACTIVE TRANSPORT
5.4: BULK TRANSPORT

5.4A: ENDOCYTOSIS
Endocytosis is a type of active transport that moves particles, such as large molecules, parts of cells, and even whole cells, into a cell. There are different variations of endocytosis, but all share a common characteristic: the plasma membrane of the cell invaginates, forming a pocket around the target particle. The pocket pinches off, resulting in the particle being contained in a newly-created intracellular vesicle formed from the plasma membrane.

5.4B: EXOCYTOSIS
Exocytosis is the process by which cells release particles from within the cell into the extracellular space.

5.1: Components and Structure

Topic hierarchy

- 5.1A: Components of Plasma Membranes
- 5.1B: Fluid Mosaic Model

The fluid mosaic model describes the plasma membrane structure as a mosaic of phospholipids, cholesterol, proteins, and carbohydrates.

- 5.1C: Membrane Fluidity

5.1A: Components of Plasma Membranes

The plasma membrane protects the cell from its external environment, mediates cellular transport, and transmits cellular signals.

Learning Objectives

- Describe the function and components of the plasma membrane

Key Points

- The principal components of the plasma membrane are lipids (phospholipids and cholesterol), proteins, and carbohydrates.
- The plasma membrane protects intracellular components from the extracellular environment.
- The plasma membrane mediates cellular processes by regulating the materials that enter and exit the cell.
- The plasma membrane carries markers that allow cells to recognize one another and can transmit signals to other cells via receptors.

Key Terms

- **plasma membrane**: The semipermeable barrier that surrounds the cytoplasm of a cell.
- **receptor**: A protein on a cell wall that binds with specific molecules so that they can be absorbed into the cell.

Structure of Plasma Membranes

The plasma membrane (also known as the cell membrane or cytoplasmic membrane) is a biological membrane that separates the interior of a cell from its outside environment.

The primary function of the plasma membrane is to protect the cell from its surroundings. Composed of a phospholipid bilayer with embedded proteins, the plasma membrane is selectively permeable to ions and organic molecules and regulates the movement of substances in and out of cells. Plasma membranes must be very flexible in order to allow certain cells, such as red blood cells and white blood cells, to change shape as they pass through narrow capillaries.

The plasma membrane also plays a role in anchoring the cytoskeleton to provide shape to the cell, and in attaching to the extracellular matrix and other cells to help group cells together to form tissues. The membrane also maintains the cell potential.

In short, if the cell is represented by a castle, the plasma membrane is the wall that provides structure for the buildings inside the wall, regulates which people leave and enter the castle, and conveys messages to and from neighboring castles. Just as a hole in the wall can be a disaster for the castle, a rupture in the plasma membrane causes the cell to lyse and die.

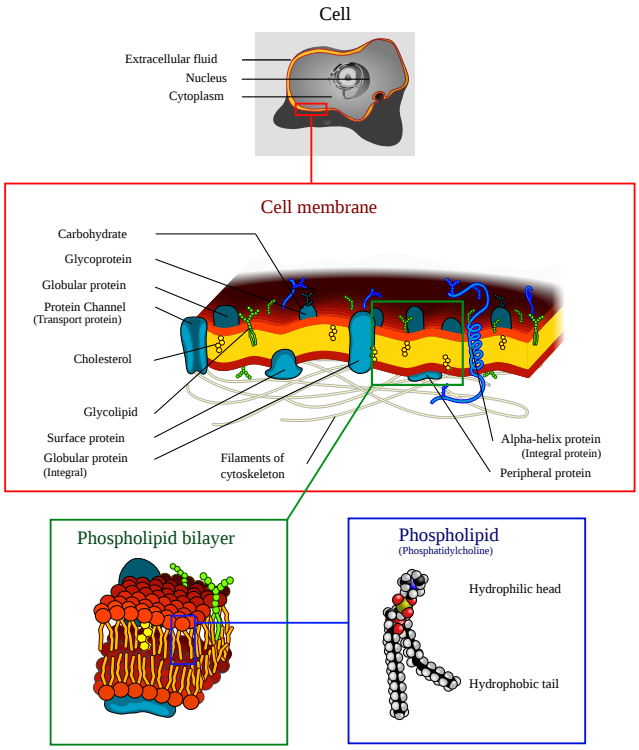

Figure 5.1*A*. 1: **The plasma membrane**: The plasma membrane is composed of phospholipids and proteins that provide a barrier between the external environment and the cell, regulate the transportation of molecules across the membrane, and communicate with other cells via protein receptors.

The Plasma Membrane and Cellular Transport

The movement of a substance across the selectively permeable plasma membrane can be either "passive"—i.e., occurring without the input of cellular energy —or "active"—i.e., its transport requires the cell to expend energy.

The cell employs a number of transport mechanisms that involve biological membranes:

1. Passive osmosis and diffusion: transports gases (such as O_2 and $CO_{2)}$ and other small molecules and ions
2. Transmembrane protein channels and transporters: transports small organic molecules such as sugars or amino acids
3. Endocytosis: transports large molecules (or even whole cells) by engulfing them
4. Exocytosis: removes or secretes substances such as hormones or enzymes

The Plasma Membrane and Cellular Signaling

Among the most sophisticated functions of the plasma membrane is its ability to transmit signals via complex proteins. These proteins can be receptors, which work as receivers of extracellular inputs and as activators of intracellular processes, or markers, which allow cells to recognize each other.

Membrane receptors provide extracellular attachment sites for effectors like hormones and growth factors, which then trigger intracellular responses. Some viruses, such as Human Immunodeficiency Virus (HIV), can hijack these receptors to gain entry into the cells, causing infections.

Membrane markers allow cells to recognize one another, which is vital for cellular signaling processes that influence tissue and organ formation during early development. This marking function also plays a later role in the "self"-versus-"non-self" distinction of the immune response. Marker proteins on human red blood cells, for example, determine blood type (A, B, AB, or O).

5.1B: Fluid Mosaic Model

LEARNING OBJECTIVES

- Describe the fluid mosaic model of cell membranes

The fluid mosaic model was first proposed by S.J. Singer and Garth L. Nicolson in 1972 to explain the structure of the plasma membrane. The model has evolved somewhat over time, but it still best accounts for the structure and functions of the plasma membrane as we now understand them. The fluid mosaic model describes the structure of the plasma membrane as a mosaic of components —including phospholipids, cholesterol, proteins, and carbohydrates—that gives the membrane a fluid character. Plasma membranes range from 5 to 10 nm in thickness. For comparison, human red blood cells, visible via light microscopy, are approximately 8 µm wide, or approximately 1,000 times wider than a plasma membrane. The proportions of proteins, lipids, and carbohydrates in the plasma membrane vary with cell type. For example, myelin contains 18% protein and 76% lipid. The mitochondrial inner membrane contains 76% protein and 24% lipid.

The Components and Functions of the Plasma Membrane	
Component	**Location**
Phospholipid	Main fabric of the membrane
Cholesterol	Attached between phospholipids and between the two phospholipid layers
Integral proteins (for example, integrins)	Embedded within the phospholipid layer(s). May or may not penetrate through both layers
Peripheral proteins	On the inner or outer surface of the phospholipid bilayer; not embedded within the phospholipids
Carbohydrates (components of glycoproteins and glycolipids)	Generally attached to proteins on the outside membrane layer

Figure 5.1B.1: **The Components and functions of the Plasma Membrane**: The principal components of a plasma membrane are lipids (phospholipids and cholesterol), proteins, and carbohydrates attached to some of the lipids and some of the proteins.

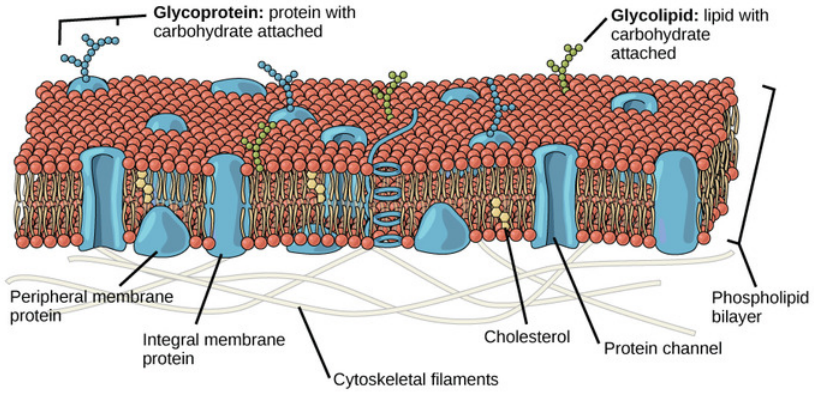

Figure 5.1B.1: **The fluid mosaic model of the plasma membrane**: The fluid mosaic model of the plasma membrane describes the plasma membrane as a fluid combination of phospholipids, cholesterol, and proteins. Carbohydrates attached to lipids (glycolipids) and to proteins (glycoproteins) extend from the outward-facing surface of the membrane.

The main fabric of the membrane is composed of amphiphilic or dual-loving, phospholipid molecules. The hydrophilic or water-loving areas of these molecules are in contact with the aqueous fluid both inside and outside the cell. Hydrophobic, or water-hating molecules, tend to be non- polar. A phospholipid molecule consists of a three-carbon glycerol backbone with two fatty acid molecules attached to carbons 1 and 2, and a phosphate-containing group attached to the third carbon. This arrangement gives the overall molecule an area described as its head (the phosphate-containing group), which has a polar character or negative charge, and an area called the tail (the fatty acids), which has no charge. They interact with other non-polar molecules in chemical reactions, but generally do not interact with polar molecules. When placed in water, hydrophobic molecules tend to form a ball or cluster. The hydrophilic regions of the phospholipids tend to form hydrogen bonds with water and other polar molecules on both the exterior and interior of the cell. Thus, the membrane surfaces that face the interior and exterior of the cell are hydrophilic. In contrast, the middle of the cell membrane is hydrophobic and will not interact with water. Therefore, phospholipids form an excellent lipid bilayer cell membrane that separates fluid within the cell from the fluid outside of the cell.

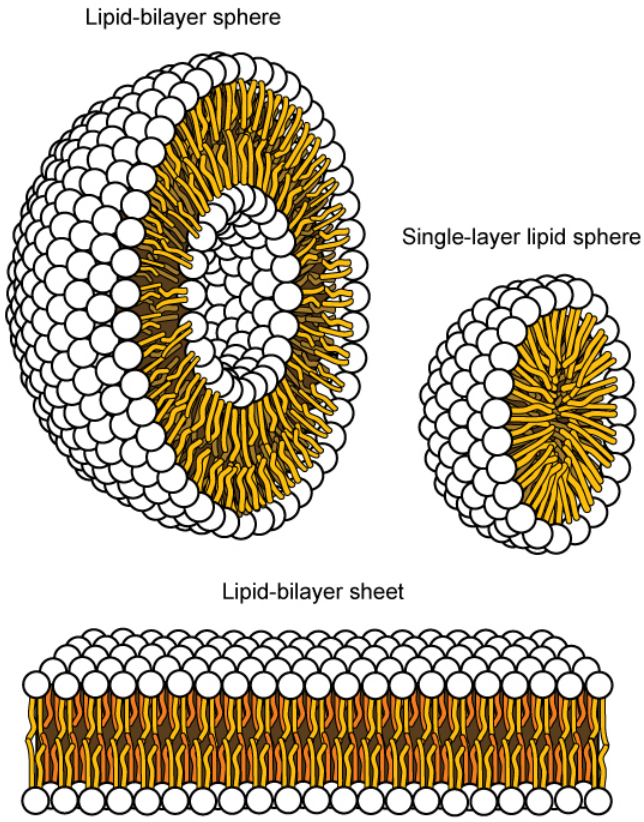

Figure 5.1B.1: **Phospholipid aggregation**: In an aqueous solution, phospholipids tend to arrange themselves with their polar heads facing outward and their hydrophobic tails facing inward.

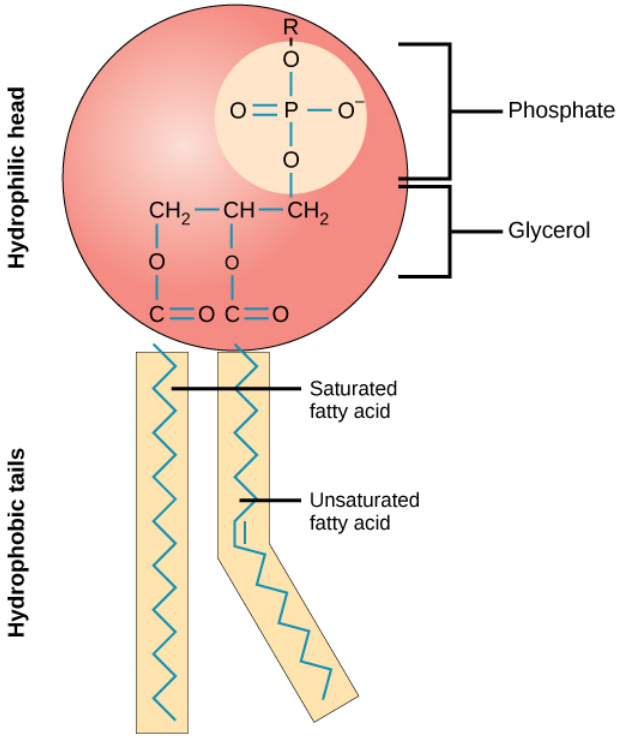

Figure 5.1B.1: **The structure of a phospholipid molecule**: This phospholipid molecule is composed of a hydrophilic head and two hydrophobic tails. The hydrophilic head group consists of a phosphate-containing group attached to a glycerol molecule. The hydrophobic tails, each containing either a saturated or an unsaturated fatty acid, are long hydrocarbon chains.

Proteins make up the second major component of plasma membranes. Integral proteins (some specialized types are called integrins) are, as their name suggests, integrated completely into the membrane structure, and their hydrophobic membrane-spanning regions interact with the hydrophobic region of the the phospholipid bilayer. Single-pass integral membrane proteins usually have a hydrophobic transmembrane segment that consists of 20–25 amino acids. Some span only part of the membrane—associating with a single layer—while others stretch from one side of the membrane to the other, and are exposed on either side. Some complex proteins are composed of up to 12 segments of a single protein, which are extensively folded and embedded in the membrane. This type of protein has a hydrophilic region or regions, and one or several mildly hydrophobic regions. This arrangement of regions of the protein tends to orient the protein alongside the phospholipids, with the hydrophobic region of the protein adjacent to the tails of the phospholipids and the hydrophilic region or regions of the protein protruding from the membrane and in contact with the cytosol or extracellular fluid.

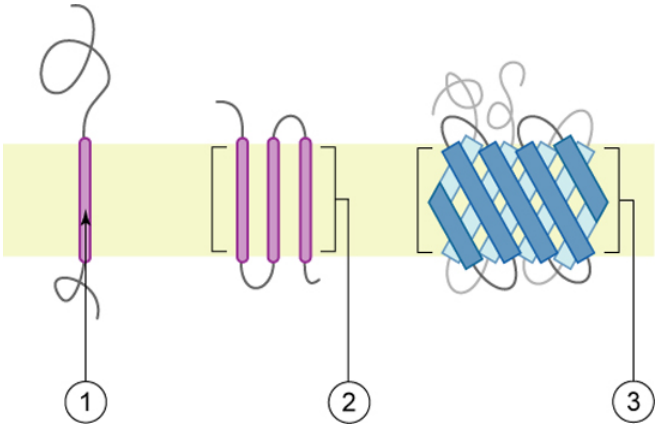

Figure 5.1B.1: **Structure of integral membrane proteins**: Integral membrane proteins may have one or more alpha-helices that span the membrane (examples 1 and 2), or they may have beta-sheets that span the membrane (example 3).

Carbohydrates are the third major component of plasma membranes. They are always found on the exterior surface of cells and are bound either to proteins (forming glycoproteins) or to lipids (forming glycolipids). These carbohydrate chains may consist of 2–60 monosaccharide units and can be either straight or branched. Along with peripheral proteins, carbohydrates form specialized sites on the cell surface that allow cells to recognize each other. This recognition function is very important to cells, as it allows the immune system to differentiate between body cells (called “self”) and foreign cells or tissues (called “non-self”). Similar types of glycoproteins and glycolipids are found on the surfaces of viruses and may change frequently, preventing immune cells from recognizing and attacking them. These carbohydrates on the exterior surface of the cell—the carbohydrate components of both glycoproteins and glycolipids—are collectively referred to as the glycocalyx (meaning “sugar coating”). The glycocalyx is highly hydrophilic and attracts large amounts of water to the surface of the cell. This aids in the interaction of the cell with its watery environment and in the cell’s ability to obtain substances dissolved in the water.

Key Points

- The main fabric of the membrane is composed of amphiphilic or dual-loving, phospholipid molecules.
- Integral proteins, the second major component of plasma membranes, are integrated completely into the membrane structure with their hydrophobic membrane-spanning regions interacting with the hydrophobic region of the phospholipid bilayer.
- Carbohydrates, the third major component of plasma membranes, are always found on the exterior surface of cells where they are bound either to proteins (forming glycoproteins) or to lipids (forming glycolipids).

Key Terms

- **amphiphilic**: Having one surface consisting of hydrophilic amino acids and the opposite surface consisting of hydrophobic (or lipophilic) ones.
- **hydrophilic**: Having an affinity for water; able to absorb, or be wetted by water, “water-loving.”
- **hydrophobic**: Lacking an affinity for water; unable to absorb, or be wetted by water, “water-fearing.”

5.1C: Membrane Fluidity

The mosaic nature of the membrane, its phospholipid chemistry, and the presence of cholesterol contribute to membrane fluidity.

Learning Objectives

- Explain the function of membrane fluidity in the structure of cells

Key Points

- The membrane is fluid but also fairly rigid and can burst if penetrated or if a cell takes in too much water.
- The mosaic nature of the plasma membrane allows a very fine needle to easily penetrate it without causing it to burst and allows it to self-seal when the needle is extracted.
- If saturated fatty acids are compressed by decreasing temperatures, they press in on each other, making a dense and fairly rigid membrane.
- If unsaturated fatty acids are compressed, the “kinks” in their tails push adjacent phospholipid molecules away, which helps maintain fluidity in the membrane.
- The ratio of saturated and unsaturated fatty acids determines the fluidity in the membrane at cold temperatures.
- Cholesterol functions as a buffer, preventing lower temperatures from inhibiting fluidity and preventing higher temperatures from increasing fluidity.

Key Terms

- **phospholipid**: Any lipid consisting of a diglyceride combined with a phosphate group and a simple organic molecule such as choline or ethanolamine; they are important constituents of biological membranes
- **fluidity**: A measure of the extent to which something is fluid. The reciprocal of its viscosity.

Membrane Fluidity

There are multiple factors that lead to membrane fluidity. First, the mosaic characteristic of the membrane helps the plasma membrane remain fluid. The integral proteins and lipids exist in the membrane as separate but loosely-attached molecules. The membrane is not like a balloon that can expand and contract; rather, it is fairly rigid and can burst if penetrated or if a cell takes in too much water. However, because of its mosaic nature, a very fine needle can easily penetrate a plasma membrane without causing it to burst; the membrane will flow and self-seal when the needle is extracted.

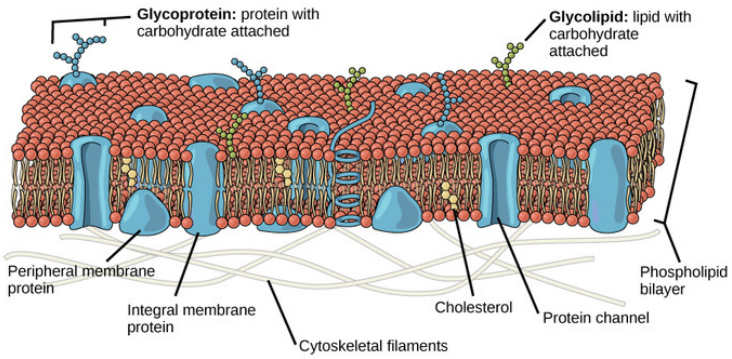

Figure 5.1C.1: **Membrane Fluidity**: The plasma membrane is a fluid combination of phospholipids, cholesterol, and proteins. Carbohydrates attached to lipids (glycolipids) and to proteins (glycoproteins) extend from the outward-facing surface of the membrane.

The second factor that leads to fluidity is the nature of the phospholipids themselves. In their saturated form, the fatty acids in phospholipid tails are saturated with bound hydrogen atoms; there are no double bonds between adjacent carbon atoms. This results in tails that are relatively straight. In contrast, unsaturated fatty acids do not contain a maximal number of hydrogen atoms, although they do contain some double bonds between adjacent carbon atoms; a double bond results in a bend of approximately 30 degrees in the string of carbons. Thus, if saturated fatty acids, with their straight tails, are compressed by decreasing temperatures, they press in on each other, making a dense and fairly rigid membrane. If

unsaturated fatty acids are compressed, the “kinks” in their tails elbow adjacent phospholipid molecules away, maintaining some space between the phospholipid molecules. This “elbow room” helps to maintain fluidity in the membrane at temperatures at which membranes with saturated fatty acid tails in their phospholipids would “freeze” or solidify. The relative fluidity of the membrane is particularly important in a cold environment. A cold environment tends to compress membranes composed largely of saturated fatty acids, making them less fluid and more susceptible to rupturing. Many organisms (fish are one example) are capable of adapting to cold environments by changing the proportion of unsaturated fatty acids in their membranes in response to the lowering of the temperature.

In animals, the third factor that keeps the membrane fluid is cholesterol. It lies alongside the phospholipids in the membrane and tends to dampen the effects of temperature on the membrane. Thus, cholesterol functions as a buffer, preventing lower temperatures from inhibiting fluidity and preventing higher temperatures from increasing fluidity too much. Cholesterol extends in both directions the range of temperature in which the membrane is appropriately fluid and, consequently, functional. Cholesterol also serves other functions, such as organizing clusters of transmembrane proteins into lipid rafts.

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Passive Transport

Topic hierarchy

5.2A: The Role of Passive Transport

Passive transport, such as diffusion and osmosis, moves materials of small molecular weight across membranes.

5.2B: Selective Permeability

The hydrophobic and hydrophilic regions of plasma membranes aid the diffusion of some molecules and hinder the diffusion of others.

5.2C: Diffusion

Diffusion is a process of passive transport in which molecules move from an area of higher concentration to one of lower concentration.

5.2D: Facilitated transport

Facilitated diffusion is a process by which molecules are transported across the plasma membrane with the help of membrane proteins.

5.2E: Osmosis

Osmosis is the movement of water through a semipermeable membrane according to the concentration gradient of water across the membrane, which is inversely proportional to the concentration of solutes. Semipermeable membranes, also termed selectively permeable membranes or partially permeable membranes, allow certain molecules or ions to pass through by diffusion.

5.2F: Tonicity

Tonicity describes how an extracellular solution can change the volume of a cell by affecting osmosis. A solution's tonicity often directly correlates with the osmolarity of the solution. Osmolarity describes the total solute concentration of the solution. A solution with low osmolarity has a greater number of water molecules relative to the number of solute particles; a solution with high osmolarity has fewer water molecules with respect to solute particles.

5.2A: The Role of Passive Transport

LEARNING OBJECTIVES

- Indicate the manner in which various materials cross the cell membrane

Plasma membranes must allow or prevent certain substances from entering or leaving a cell. In other words, plasma membranes are selectively permeable; they allow some substances to pass through, but not others. If they were to lose this selectivity, the cell would no longer be able to sustain itself, and it would be destroyed. Some cells require larger amounts of specific substances than other cells; they must have a way of obtaining these materials from extracellular fluids. This may happen passively, as certain materials move back and forth, or the cell may have special mechanisms that facilitate transport. Some materials are so important to a cell that it spends some of its energy (hydrolyzing adenosine triphosphate (ATP)) to obtain these materials. Red blood cells use some of their energy to do this. All cells spend the majority of their energy to maintain an imbalance of sodium and potassium ions between the interior and exterior of the cell.

The most direct forms of membrane transport are passive. Passive transport is a naturally-occurring phenomenon and does not require the cell to exert any of its energy to accomplish the movement. In passive transport, substances move from an area of higher concentration to an area of lower concentration. A physical space in which there is a range of concentrations of a single substance is said to have a concentration gradient.

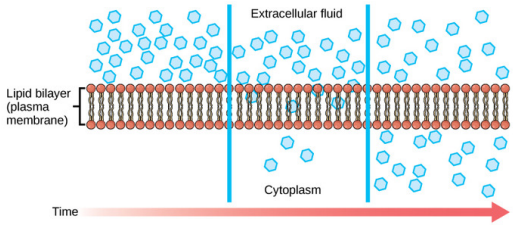


Figure 5.2A.1: **Passive Transport**: Diffusion is a type of passive transport. Diffusion through a permeable membrane moves a substance from an area of high concentration (extracellular fluid, in this case) down its concentration gradient (into the cytoplasm).

The passive forms of transport, diffusion and osmosis, move materials of small molecular weight across membranes. Substances diffuse from areas of high concentration to areas of lower concentration; this process continues until the substance is evenly distributed in a system. In solutions containing more than one substance, each type of molecule diffuses according to its own concentration gradient, independent of the diffusion of other substances. Many factors can affect the rate of diffusion, including, but not limited to, concentration gradient, size of the particles that are diffusing, and temperature of the system.

In living systems, diffusion of substances in and out of cells is mediated by the plasma membrane. Some materials diffuse readily through the membrane, but others are hindered; their passage is made possible by specialized proteins, such as channels and transporters. The chemistry of living things occurs in aqueous solutions; balancing the concentrations of those solutions is an ongoing problem. In living systems, diffusion of some substances would be slow or difficult without membrane proteins that facilitate transport.

Key Points

- Plasma membranes are selectively permeable; if they were to lose this selectivity, the cell would no longer be able to sustain itself.
- In passive transport, substances simply move from an area of higher concentration to an area of lower concentration, which does not require the input of energy.
- Concentration gradient, size of the particles that are diffusing, and temperature of the system affect the rate of diffusion.
- Some materials diffuse readily through the membrane, but others require specialized proteins, such as channels and transporters, to carry them into or out of the cell.

Key Terms

- **concentration gradient**: A concentration gradient is present when a membrane separates two different concentrations of molecules.
- **passive transport**: A movement of biochemicals and other atomic or molecular substances across membranes that does not require an input of chemical energy.
- **permeable**: Of or relating to substance, substrate, membrane or material that absorbs or allows the passage of fluids.

5.2B: Selective Permeability

The hydrophobic and hydrophilic regions of plasma membranes aid the diffusion of some molecules and hinder the diffusion of others.

Learning Objectives

- Describe how membrane permeability, concentration gradient, and molecular properties affect biological diffusion rates.

Key Points

- The interior and exterior surfaces of the plasma membrane are not identical, which adds to the selective permeability of the membrane.
- Fat soluble substances are able to pass easily to the hydrophobic interior of the plasma membrane and diffuse into the cell.
- Polar molecules and charged molecules do not diffuse easily through the lipid core of the plasma membrane and must be transported across by proteins, sugars, or amino acids.

Key Terms

- **polar**: a separation of electric charge leading to a molecule or its chemical groups having an electric dipole
- **amphiphilic**: Having one surface consisting of hydrophilic amino acids and the opposite surface consisting of hydrophobic (or lipophilic) ones.

Selective Permeability

Plasma membranes are asymmetric: the interior of the membrane is not identical to the exterior of the membrane. In fact, there is a considerable difference between the array of phospholipids and proteins between the two leaflets that form a membrane. On the interior of the membrane, some proteins serve to anchor the membrane to fibers of the cytoskeleton. There are peripheral proteins on the exterior of the membrane that bind elements of the extracellular matrix. Carbohydrates, attached to lipids or proteins, are also found on the exterior surface of the plasma membrane. These carbohydrate complexes help the cell bind substances that the cell needs in the extracellular fluid. This adds considerably to the selective nature of plasma membranes.

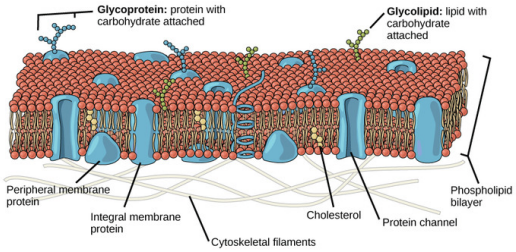
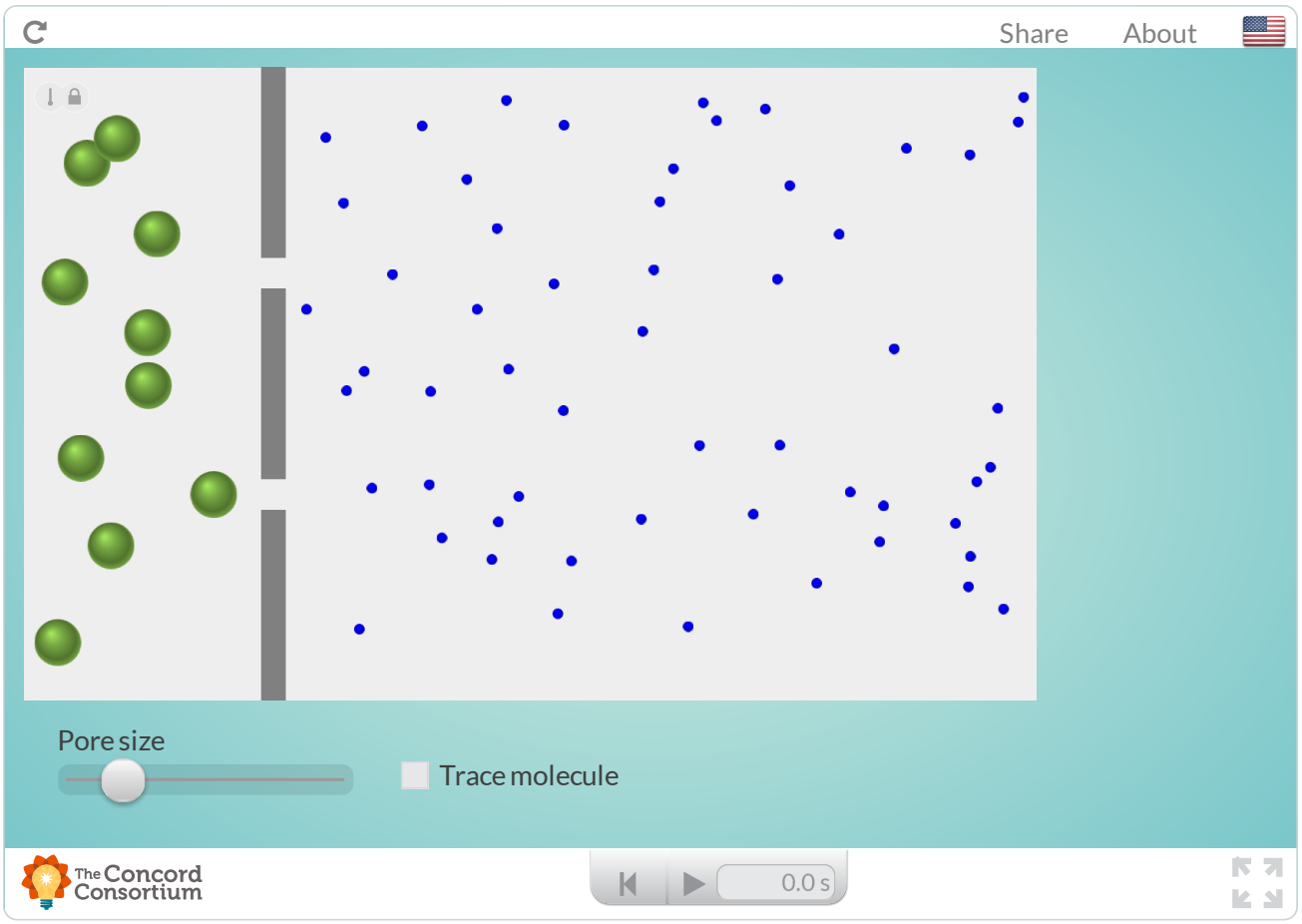


Figure 5.2*B*.1: **Asymmetry in Plasma Membranes**: The exterior surface of the plasma membrane is not identical to the interior surface of the same membrane.

Recall that plasma membranes are amphiphilic; that is, they have hydrophilic and hydrophobic regions. This characteristic helps the movement of some materials through the membrane and hinders the movement of others. Lipid-soluble material with a low molecular weight can easily slip through the hydrophobic lipid core of the membrane. Substances such as the fat-soluble vitamins A, D, E, and K readily pass through the plasma membranes in the digestive tract and other tissues. Fat-soluble drugs and hormones also gain easy entry into cells and are readily transported into the body's tissues and organs. Molecules of oxygen and carbon dioxide have no charge and so pass through membranes by simple diffusion.

Polar substances present problems for the membrane. While some polar molecules connect easily with the outside of a cell, they cannot readily pass through the lipid core of the plasma membrane. Additionally, while small ions could easily slip through the spaces in the mosaic of the membrane, their charge prevents them from doing so. Ions such as sodium, potassium, calcium, and chloride must have special means of penetrating plasma membranes. Simple sugars and amino acids also need help with transport across plasma membranes, achieved by various transmembrane proteins (channels).



Diffusion across a semipermeable membrane: This interactive shows that smaller molecules have an easier time making it across a semipermeable membrane. Adjust the pore size so the larger molecules can make it through!

5.2C: Diffusion

Diffusion is a process of passive transport in which molecules move from an area of higher concentration to one of lower concentration.

Learning Objectives

- Describe diffusion and the factors that affect how materials move across the cell membrane.

Key Points

- Substances diffuse according to their concentration gradient; within a system, different substances in the medium will each diffuse at different rates according to their individual gradients.
- After a substance has diffused completely through a space, removing its concentration gradient, molecules will still move around in the space, but there will be no net movement of the number of molecules from one area to another, a state known as dynamic equilibrium.
- Several factors affect the rate of diffusion of a solute including the mass of the solute, the temperature of the environment, the solvent density, and the distance traveled.

Key Terms

- **diffusion**: The passive movement of a solute across a permeable membrane
- **concentration gradient**: A concentration gradient is present when a membrane separates two different concentrations of molecules.

Diffusion

Diffusion is a passive process of transport. A single substance tends to move from an area of high concentration to an area of low concentration until the concentration is equal across a space. You are familiar with diffusion of substances through the air. For example, think about someone opening a bottle of ammonia in a room filled with people. The ammonia gas is at its highest concentration in the bottle; its lowest concentration is at the edges of the room. The ammonia vapor will diffuse, or spread away, from the bottle; gradually, more and more people will smell the ammonia as it spreads. Materials move within the cell ‘s cytosol by diffusion, and certain materials move through the plasma membrane by diffusion. Diffusion expends no energy. On the contrary, concentration gradients are a form of potential energy, dissipated as the gradient is eliminated.

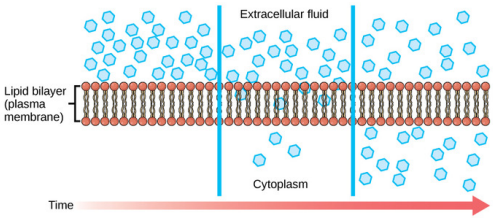


Figure 5.2C.1: **Diffusion**: Diffusion through a permeable membrane moves a substance from an area of high concentration (extracellular fluid, in this case) down its concentration gradient (into the cytoplasm).

Each separate substance in a medium, such as the extracellular fluid, has its own concentration gradient independent of the concentration gradients of other materials. In addition, each substance will diffuse according to that gradient. Within a system, there will be different rates of diffusion of the different substances in the medium.

Factors That Affect Diffusion

Molecules move constantly in a random manner at a rate that depends on their mass, their environment, and the amount of thermal energy they possess, which in turn is a function of temperature. This movement accounts for the diffusion of molecules through whatever medium in which they are localized. A substance will tend to move into any space available to it until it is evenly distributed throughout it. After a substance has diffused completely through a space removing its concentration gradient, molecules will still move around in the space, but there will be no net movement of the number of molecules from one area to another. This lack of a concentration gradient in which there is no net movement of a substance

is known as dynamic equilibrium. While diffusion will go forward in the presence of a concentration gradient of a substance, several factors affect the rate of diffusion:

- Extent of the concentration gradient: The greater the difference in concentration, the more rapid the diffusion. The closer the distribution of the material gets to equilibrium, the slower the rate of diffusion becomes.
- Mass of the molecules diffusing: Heavier molecules move more slowly; therefore, they diffuse more slowly. The reverse is true for lighter molecules.
- Temperature: Higher temperatures increase the energy and therefore the movement of the molecules, increasing the rate of diffusion. Lower temperatures decrease the energy of the molecules, thus decreasing the rate of diffusion.
- Solvent density: As the density of a solvent increases, the rate of diffusion decreases. The molecules slow down because they have a more difficult time getting through the denser medium. If the medium is less dense, diffusion increases. Because cells primarily use diffusion to move materials within the cytoplasm, any increase in the cytoplasm's density will inhibit the movement of the materials. An example of this is a person experiencing dehydration. As the body's cells lose water, the rate of diffusion decreases in the cytoplasm, and the cells' functions deteriorate. Neurons tend to be very sensitive to this effect. Dehydration frequently leads to unconsciousness and possibly coma because of the decrease in diffusion rate within the cells.
- Solubility: As discussed earlier, nonpolar or lipid-soluble materials pass through plasma membranes more easily than polar materials, allowing a faster rate of diffusion.
- Surface area and thickness of the plasma membrane: Increased surface area increases the rate of diffusion, whereas a thicker membrane reduces it.
- Distance travelled: The greater the distance that a substance must travel, the slower the rate of diffusion. This places an upper limitation on cell size. A large, spherical cell will die because nutrients or waste cannot reach or leave the center of the cell. Therefore, cells must either be small in size, as in the case of many prokaryotes, or be flattened, as with many single-celled eukaryotes.

A variation of diffusion is the process of filtration. In filtration, material moves according to its concentration gradient through a membrane; sometimes the rate of diffusion is enhanced by pressure, causing the substances to filter more rapidly. This occurs in the kidney where blood pressure forces large amounts of water and accompanying dissolved substances, or solutes, out of the blood and into the renal tubules. The rate of diffusion in this instance is almost totally dependent on pressure. One of the effects of high blood pressure is the appearance of protein in the urine, which is "squeezed through" by the abnormally high pressure.

5.2D: Facilitated transport

LEARNING OBJECTIVES

- Explain why and how passive transport occurs

Facilitated transport is a type of passive transport. Unlike simple diffusion where materials pass through a membrane without the help of proteins, in facilitated transport, also called facilitated diffusion, materials diffuse across the plasma membrane with the help of membrane proteins. A concentration gradient exists that would allow these materials to diffuse into the cell without expending cellular energy. However, these materials are ions or polar molecules that are repelled by the hydrophobic parts of the cell membrane. Facilitated transport proteins shield these materials from the repulsive force of the membrane, allowing them to diffuse into the cell.

The material being transported is first attached to protein or glycoprotein receptors on the exterior surface of the plasma membrane. This allows the material that is needed by the cell to be removed from the extracellular fluid. The substances are then passed to specific integral proteins that facilitate their passage. Some of these integral proteins are collections of beta-pleated sheets that form a channel through the phospholipid bilayer. Others are carrier proteins which bind with the substance and aid its diffusion through the membrane.

Channels

The integral proteins involved in facilitated transport are collectively referred to as transport proteins; they function as either channels for the material or carriers. In both cases, they are transmembrane proteins. Channels are specific for the substance that is being transported. Channel proteins have hydrophilic domains exposed to the intracellular and extracellular fluids; they additionally have a hydrophilic channel through their core that provides a hydrated opening through the membrane layers. Passage through the channel allows polar compounds to avoid the nonpolar central layer of the plasma membrane that would otherwise slow or prevent their entry into the cell. Aquaporins are channel proteins that allow water to pass through the membrane at a very high rate.

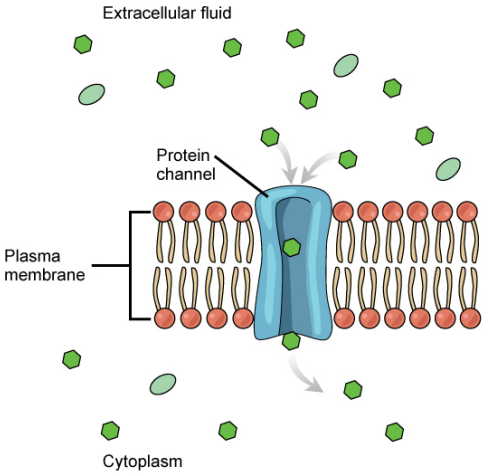


Figure $5.2D.1$: **Channel Proteins in Facilitated Transport**: Facilitated transport moves substances down their concentration gradients. They may cross the plasma membrane with the aid of channel proteins.

Channel proteins are either open at all times or they are “gated,” which controls the opening of the channel. The attachment of a particular ion to the channel protein may control the opening or other mechanisms or substances may be involved. In some tissues, sodium and chloride ions pass freely through open channels, whereas in other tissues, a gate must be opened to allow passage. An example of this occurs in the kidney, where both forms of channels are found in different parts of the renal tubules. Cells involved in the transmission of electrical impulses, such as nerve and muscle cells, have gated channels for sodium, potassium, and calcium in their membranes. Opening and closing of these channels changes the relative concentrations on opposing sides of the membrane of these ions, resulting in the facilitation of electrical transmission along membranes (in the case of nerve cells) or in muscle contraction (in the case of muscle cells).

Carrier Proteins

Another type of protein embedded in the plasma membrane is a carrier protein. This protein binds a substance and, in doing so, triggers a change of its own shape, moving the bound molecule from the outside of the cell to its interior; depending on the gradient, the material may move in the opposite direction. Carrier proteins are typically specific for a single substance. This adds to the overall selectivity of the plasma membrane. The exact mechanism for the change of shape is poorly understood. Proteins can change shape when their hydrogen bonds are affected, but this may not fully explain this mechanism. Each carrier protein is specific to one substance, and there are a finite number of these proteins in any membrane. This can cause problems in transporting enough of the material for the cell to function properly.

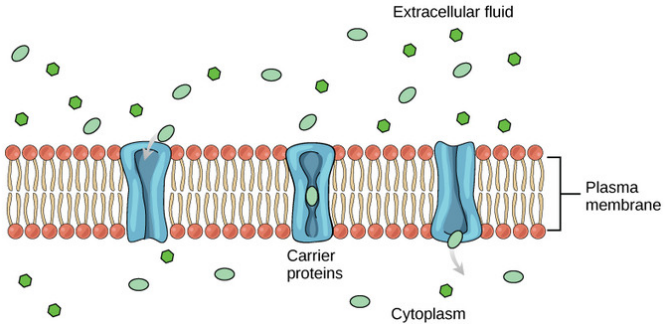


Figure 5.2D.1: **Carrier Proteins**: Some substances are able to move down their concentration gradient across the plasma membrane with the aid of carrier proteins. Carrier proteins change shape as they move molecules across the membrane.

An example of this process occurs in the kidney. Glucose, water, salts, ions, and amino acids needed by the body are filtered in one part of the kidney. This filtrate, which includes glucose, is then reabsorbed in another part of the kidney. Because there are only a finite number of carrier proteins for glucose, if more glucose is present than the proteins can handle, the excess is not transported; it is excreted from the body in the urine. In a diabetic individual, this is described as "spilling glucose into the urine." A different group of carrier proteins called glucose transport proteins, or GLUTs, are involved in transporting glucose and other hexose sugars through plasma membranes within the body.

Channel and carrier proteins transport material at different rates. Channel proteins transport much more quickly than do carrier proteins. Channel proteins facilitate diffusion at a rate of tens of millions of molecules per second, whereas carrier proteins work at a rate of a thousand to a million molecules per second.

Key Points

- A concentration gradient exists that would allow ions and polar molecules to diffuse into the cell, but these materials are repelled by the hydrophobic parts of the cell membrane.
- Facilitated diffusion uses integral membrane proteins to move polar or charged substances across the hydrophobic regions of the membrane.
- Channel proteins can aid in the facilitated diffusion of substances by forming a hydrophilic passage through the plasma membrane through which polar and charged substances can pass.
- Channel proteins can be open at all times, constantly allowing a particular substance into or out of the cell, depending on the concentration gradient; or they can be gated and can only be opened by a particular biological signal.
- Carrier proteins aid in facilitated diffusion by binding a particular substance, then altering their shape to bring that substance into or out of the cell.

Key Terms

- **facilitated diffusion**: The spontaneous passage of molecules or ions across a biological membrane passing through specific transmembrane integral proteins.
- **membrane protein**: Proteins that are attached to, or associated with the membrane of a cell or an organelle.

5.2E: Osmosis

LEARNING OBJECTIVES

- Describe the process of osmosis and explain how concentration gradient affects osmosis

Osmosis and Semipermeable Membranes

Osmosis is the movement of water through a semipermeable membrane according to the concentration gradient of water across the membrane, which is inversely proportional to the concentration of solutes. Semipermeable membranes, also termed selectively permeable membranes or partially permeable membranes, allow certain molecules or ions to pass through by diffusion.

While diffusion transports materials across membranes and within cells, osmosis transports only water across a membrane. The semipermeable membrane limits the diffusion of solutes in the water. Not surprisingly, the aquaporin proteins that facilitate water movement play a large role in osmosis, most prominently in red blood cells and the membranes of kidney tubules.

Mechanism of Osmosis

Osmosis is a special case of diffusion. Water, like other substances, moves from an area of high concentration to one of low concentration. An obvious question is what makes water move at all? Imagine a beaker with a semipermeable membrane separating the two sides or halves. On both sides of the membrane the water level is the same, but there are different concentrations of a dissolved substance, or solute, that cannot cross the membrane (otherwise the concentrations on each side would be balanced by the solute crossing the membrane). If the volume of the solution on both sides of the membrane is the same but the concentrations of solute are different, then there are different amounts of water, the solvent, on either side of the membrane. If there is more solute in one area, then there is less water; if there is less solute in one area, then there must be more water.

To illustrate this, imagine two full glasses of water. One has a single teaspoon of sugar in it, whereas the second one contains one-quarter cup of sugar. If the total volume of the solutions in both cups is the same, which cup contains more water? Because the large amount of sugar in the second cup takes up much more space than the teaspoon of sugar in the first cup, the first cup has more water in it.

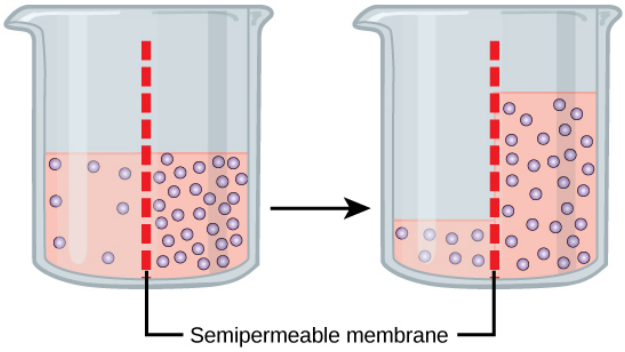


Figure $5.2E.1$: **Osmosis**: In osmosis, water always moves from an area of higher water concentration to one of lower concentration. In the diagram shown, the solute cannot pass through the selectively permeable membrane, but the water can.

Returning to the beaker example, recall that it has a mixture of solutes on either side of the membrane. A principle of diffusion is that the molecules move around and will spread evenly throughout the medium if they can. However, only the material capable of passing through the membrane will diffuse through it. In this example, the solute cannot diffuse through the membrane, but the water can. Water has a concentration gradient in this system. Thus, water will diffuse down its concentration gradient, crossing the membrane to the side where it is less concentrated. This diffusion of water through the membrane—osmosis—will continue until the concentration gradient of water goes to zero or until the hydrostatic pressure of the water balances the osmotic pressure. In the beaker example, this means that the level of fluid in the side with a higher solute concentration will go up.

Key Points

- Osmosis occurs according to the concentration gradient of water across the membrane, which is inversely proportional to the concentration of solutes.
- Osmosis occurs until the concentration gradient of water goes to zero or until the hydrostatic pressure of the water balances the osmotic pressure.
- Osmosis occurs when there is a concentration gradient of a solute within a solution, but the membrane does not allow diffusion of the solute.

Key Terms

- **solute**: Any substance that is dissolved in a liquid solvent to create a solution
- **osmosis**: The net movement of solvent molecules from a region of high solvent potential to a region of lower solvent potential through a partially permeable membrane
- **semipermeable membrane**: A type of biological membrane that will allow certain molecules or ions to pass through it by diffusion and occasionally by specialized facilitated diffusion

5.2F: Tonicity

LEARNING OBJECTIVES

- Define tonicity and describe its relevance to osmosis

Tonicity describes how an extracellular solution can change the volume of a cell by affecting osmosis. A solution's tonicity often directly correlates with the osmolarity of the solution. Osmolarity describes the total solute concentration of the solution. A solution with low osmolarity has a greater number of water molecules relative to the number of solute particles; a solution with high osmolarity has fewer water molecules with respect to solute particles. In a situation in which solutions of two different osmolarities are separated by a membrane permeable to water, though not to the solute, water will move from the side of the membrane with lower osmolarity (and more water) to the side with higher osmolarity (and less water). This effect makes sense if you remember that the solute cannot move across the membrane, and thus the only component in the system that can move—the water—moves along its own concentration gradient. An important distinction that concerns living systems is that osmolarity measures the number of particles (which may be molecules) in a solution. Therefore, a solution that is cloudy with cells may have a lower osmolarity than a solution that is clear if the second solution contains more dissolved molecules than there are cells.

Hypotonic Solutions

Three terms—hypotonic, isotonic, and hypertonic—are used to relate the osmolarity of a cell to the osmolarity of the extracellular fluid that contains the cells. In a hypotonic situation, the extracellular fluid has lower osmolarity than the fluid inside the cell, and water enters the cell. (In living systems, the point of reference is always the cytoplasm, so the prefix hypo- means that the extracellular fluid has a lower concentration of solutes, or a lower osmolarity, than the cell cytoplasm.) It also means that the extracellular fluid has a higher concentration of water in the solution than does the cell. In this situation, water will follow its concentration gradient and enter the cell, causing the cell to expand.

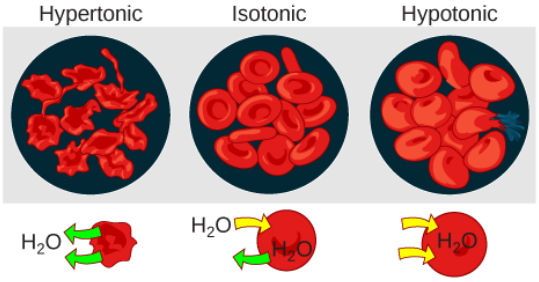


Figure $5.2F.1$: **Changes in Cell Shape Due to Dissolved Solutes**: Osmotic pressure changes the shape of red blood cells in hypertonic, isotonic, and hypotonic solutions.

Hypertonic Solutions

As for a hypertonic solution, the prefix hyper- refers to the extracellular fluid having a higher osmolarity than the cell's cytoplasm; therefore, the fluid contains less water than the cell does. Because the cell has a relatively higher concentration of water, water will leave the cell, and the cell will shrink.

Isotonic Solutions

In an isotonic solution, the extracellular fluid has the same osmolarity as the cell. If the osmolarity of the cell matches that of the extracellular fluid, there will be no net movement of water into or out of the cell, although water will still move in and out.

Blood cells and plant cells in hypertonic, isotonic, and hypotonic solutions take on characteristic appearances. Cells in an isotonic solution retain their shape. Cells in a hypotonic solution swell as water enters the cell, and may burst if the concentration gradient is large enough between the inside and outside of the cell. Cells in a hypertonic solution shrink as water exits the cell, becoming shriveled.

Key Points

- Osmolarity describes the total solute concentration of a solution; solutions with a low solute concentration have a low osmolarity, while those with a high osmolarity have a high solute concentration.
- Water moves from the side of the membrane with lower osmolarity (and more water) to the side with higher osmolarity (and less water).
- In a hypotonic solution, the extracellular fluid has a lower osmolarity than the fluid inside the cell; water enters the cell.
- In a hypertonic solution, the extracellular fluid has a higher osmolarity than the fluid inside the cell; water leaves the cell.
- In an isotonic solution, the extracellular fluid has the same osmolarity as the cell; there will be no net movement of water into or out of the cell.

Key Terms

- **osmolarity**: The osmotic concentration of a solution, normally expressed as osmoles of solute per litre of solution.
- **hypotonic**: Having a lower osmotic pressure than another; a cell in this environment causes water to enter the cell, causing it to swell.
- **hypertonic**: having a greater osmotic pressure than another
- **isotonic**: having the same osmotic pressure

5.3: Active Transport

Topic hierarchy

- 5.3A: Electrochemical Gradient
- 5.3B: Primary Active Transport
- 5.3C: Secondary Active Transport

5.3A: Electrochemical Gradient

To move substances against the membrane's electrochemical gradient, the cell utilizes active transport, which requires energy from ATP.

Learning Objectives

- Define an electrochemical gradient and describe how a cell moves substances against this gradient

Key Points

- The electrical and concentration gradients of a membrane tend to drive sodium into and potassium out of the cell, and active transport works against these gradients.
- To move substances against a concentration or electrochemical gradient, the cell must utilize energy in the form of ATP during active transport.
- Primary active transport, which is directly dependent on ATP, moves ions across a membrane and creates a difference in charge across that membrane.
- Secondary active transport, created by primary active transport, is the transport of a solute in the direction of its electrochemical gradient and does not directly require ATP.
- Carrier proteins such as uniporters, symporters, and antiporters perform primary active transport and facilitate the movement of solutes across the cell's membrane.

Key Terms

- **adenosine triphosphate**: a multifunctional nucleoside triphosphate used in cells as a coenzyme, often called the "molecular unit of energy currency" in intracellular energy transfer
- **active transport**: movement of a substance across a cell membrane against its concentration gradient (from low to high concentration) facilitated by ATP conversion
- **electrochemical gradient**: The difference in charge and chemical concentration across a membrane.

Electrochemical Gradients

Simple concentration gradients are differential concentrations of a substance across a space or a membrane, but in living systems, gradients are more complex. Because ions move into and out of cells and because cells contain proteins that do not move across the membrane and are mostly negatively charged, there is also an electrical gradient, a difference of charge, across the plasma membrane. The interior of living cells is electrically negative with respect to the extracellular fluid in which they are bathed. At the same time, cells have higher concentrations of potassium (K^+) and lower concentrations of sodium (Na^+) than does the extracellular fluid. In a living cell, the concentration gradient of Na^+ tends to drive it into the cell, and the electrical gradient of Na^+ (a positive ion) also tends to drive it inward to the negatively-charged interior. The situation is more complex, however, for other elements such as potassium. The electrical gradient of K^+, a positive ion, also tends to drive it into the cell, but the concentration gradient of K^+ tends to drive K^+ out of the cell. The combined gradient of concentration and electrical charge that affects an ion is called its electrochemical gradient.

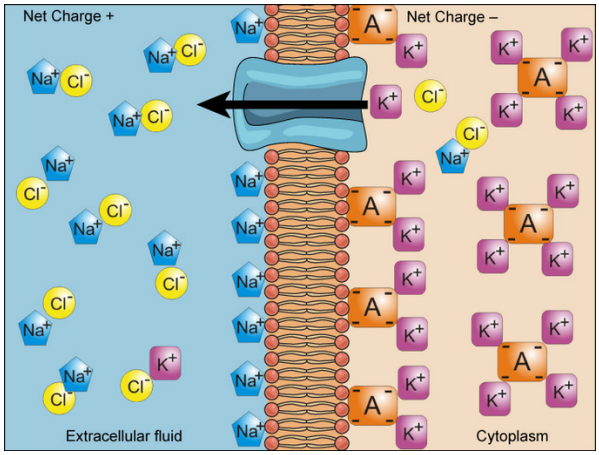


Figure 5.3A.1: **Electrochemical Gradient**: Electrochemical gradients arise from the combined effects of concentration gradients and electrical gradients.

Moving Against a Gradient

To move substances against a concentration or electrochemical gradient, the cell must use energy. This energy is harvested from adenosine triphosphate (ATP) generated through the cell's metabolism. Active transport mechanisms, collectively called pumps, work against electrochemical gradients. Small substances constantly pass through plasma membranes. Active transport maintains concentrations of ions and other substances needed by living cells in the face of these passive movements. Much of a cell's supply of metabolic energy may be spent maintaining these processes. For example, most of a red blood cell's metabolic energy is used to maintain the imbalance between exterior and interior sodium and potassium levels required by the cell. Because active transport mechanisms depend on a cell's metabolism for energy, they are sensitive to many metabolic poisons that interfere with the supply of ATP.

Two mechanisms exist for the transport of small-molecular weight material and small molecules. Primary active transport moves ions across a membrane and creates a difference in charge across that membrane, which is directly dependent on ATP. Secondary active transport describes the movement of material that is due to the electrochemical gradient established by primary active transport that does not directly require ATP.

Carrier Proteins for Active Transport

An important membrane adaption for active transport is the presence of specific carrier proteins or pumps to facilitate movement. There are three types of these proteins or transporters: uniporters, symporters, and antiporters. A uniporter carries one specific ion or molecule. A symporter carries two different ions or molecules, both in the same direction. An antiporter also carries two different ions or molecules, but in different directions. All of these transporters can also transport small, uncharged organic molecules like glucose. These three types of carrier proteins are also found in facilitated diffusion, but they do not require ATP to work in that process. Some examples of pumps for active transport are Na^+-K^+ ATPase, which carries sodium and potassium ions, and H^+-K^+ ATPase, which carries hydrogen and potassium ions. Both of these are antiporter carrier proteins. Two other carrier protein pumps are Ca^{2+} ATPase and H^+ATPase, which carry only calcium and only hydrogen ions, respectively.

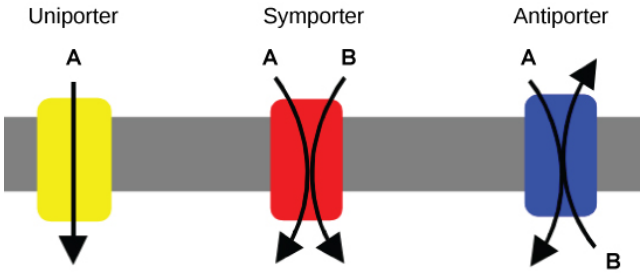


Figure 5.3A.1: **Uniporters, Symporters, and Antiporters**: A uniporter carries one molecule or ion. A symporter carries two different molecules or ions, both in the same direction. An antiporter also carries two different molecules or ions, but in different directions.

5.3B: Primary Active Transport

The sodium-potassium pump maintains the electrochemical gradient of living cells by moving sodium in and potassium out of the cell.

Learning Objectives

- Describe how a cell moves sodium and potassium out of and into the cell against its electrochemical gradient

Key Points

- The sodium-potassium pump moves K+ into the cell while moving Na+ at a ratio of three Na+ for every two K+ ions.
- When the sodium-potassium- ATPase enzyme points into the cell, it has a high affinity for sodium ions and binds three of them, hydrolyzing ATP and changing shape.
- As the enzyme changes shape, it reorients itself towards the outside of the cell, and the three sodium ions are released.
- The enzyme's new shape allows two potassium to bind and the phosphate group to detach, and the carrier protein repositions itself towards the interior of the cell.
- The enzyme changes shape again, releasing the potassium ions into the cell.
- After potassium is released into the cell, the enzyme binds three sodium ions, which starts the process over again.

Key Terms

- **electrogenic pump**: An ion pump that generates a net charge flow as a result of its activity.
- **Na+-K+ ATPase**: An enzyme located in the plasma membrane of all animal cells that pumps sodium out of cells while pumping potassium into cells.

Primary Active Transport

The primary active transport that functions with the active transport of sodium and potassium allows secondary active transport to occur. The secondary transport method is still considered active because it depends on the use of energy as does primary transport.

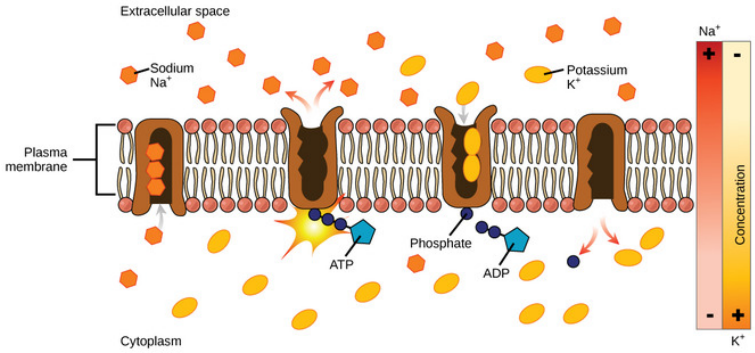


Figure 5.3B.1: **Active Transport of Sodium and Potassium**: Primary active transport moves ions across a membrane, creating an electrochemical gradient (electrogenic transport).

One of the most important pumps in animals cells is the sodium-potassium pump (Na^+-K^+ ATPase), which maintains the electrochemical gradient (and the correct concentrations of Na^+ and K^+) in living cells. The sodium-potassium pump moves two K^+ into the cell while moving three Na^+ out of the cell. The Na+-K+ ATPase exists in two forms, depending on its orientation to the interior or exterior of the cell and its affinity for either sodium or potassium ions. The process consists of the following six steps:

- With the enzyme oriented towards the interior of the cell, the carrier has a high affinity for sodium ions. Three sodium ions bind to the protein.
- ATP is hydrolyzed by the protein carrier, and a low-energy phosphate group attaches to it.
- As a result, the carrier changes shape and re-orients itself towards the exterior of the membrane. The protein's affinity for sodium decreases, and the three sodium ions leave the carrier.

- The shape change increases the carrier's affinity for potassium ions, and two such ions attach to the protein. Subsequently, the low-energy phosphate group detaches from the carrier.
- With the phosphate group removed and potassium ions attached, the carrier protein repositions itself towards the interior of the cell.
- The carrier protein, in its new configuration, has a decreased affinity for potassium, and the two ions are released into the cytoplasm. The protein now has a higher affinity for sodium ions, and the process starts again.

Several things have happened as a result of this process. At this point, there are more sodium ions outside of the cell than inside and more potassium ions inside than out. For every three ions of sodium that move out, two ions of potassium move in. This results in the interior being slightly more negative relative to the exterior. This difference in charge is important in creating the conditions necessary for the secondary process. The sodium-potassium pump is, therefore, an electrogenic pump (a pump that creates a charge imbalance), creating an electrical imbalance across the membrane and contributing to the membrane potential.

5.3C: Secondary Active Transport

In secondary active transport, a molecule is moved down its electrochemical gradient as another is moved up its concentration gradient.

Learning Objectives

- Differentiate between primary and secondary active transport

Key Points

- While secondary active transport consumes ATP to generate the gradient down which a molecule is moved, the energy is not directly used to move the molecule across the membrane.
- Both antiporters and symporters are used in secondary active transport.
- Secondary active transport brings sodium ions into the cell, and as sodium ion concentrations build outside the plasma membrane, an electrochemical gradient is created.
- If a channel protein is open via primary active transport, the ions will be pulled through the membrane along with other substances that can attach themselves to the transport protein through the membrane.
- Secondary active transport is used to store high-energy hydrogen ions in the mitochondria of plant and animal cells for the production of ATP.
- The potential energy in the hydrogen ions is translated into kinetic energy as the ions surge through the channel protein ATP synthase, and that energy is used to convert ADP into ATP.

Key Terms

- **secondary active transport**: A method of transport in which the electrochemical potential difference created by pumping ions out of the cell is used to transport molecules across a membrane.

Secondary Active Transport (Co-transport)

Unlike in primary active transport, in secondary active transport, ATP is not directly coupled to the molecule of interest. Instead, another molecule is moved up its concentration gradient, which generates an electrochemical gradient. The molecule of interest is then transported down the electrochemical gradient. While this process still consumes ATP to generate that gradient, the energy is not directly used to move the molecule across the membrane, hence it is known as secondary active transport. Both antiporters and symporters are used in secondary active transport. Co-transporters can be classified as symporters and antiporters depending on whether the substances move in the same or opposite directions across the cell membrane.

Secondary active transport brings sodium ions, and possibly other compounds, into the cell. As sodium ion concentrations build outside the plasma membrane because of the action of the primary active transport process, an electrochemical gradient is created. If a channel protein exists and is open, the sodium ions will be pulled through the membrane. This movement is used to transport other substances that can attach themselves to the transport protein through the membrane. Many amino acids, as well as glucose, enter a cell this way. This secondary process is also used to store high-energy hydrogen ions in the mitochondria of plant and animal cells for the production of ATP. The potential energy that accumulates in the stored hydrogen ions is translated into kinetic energy as the ions surge through the channel protein ATP synthase, and that energy is used to convert ADP into ATP.

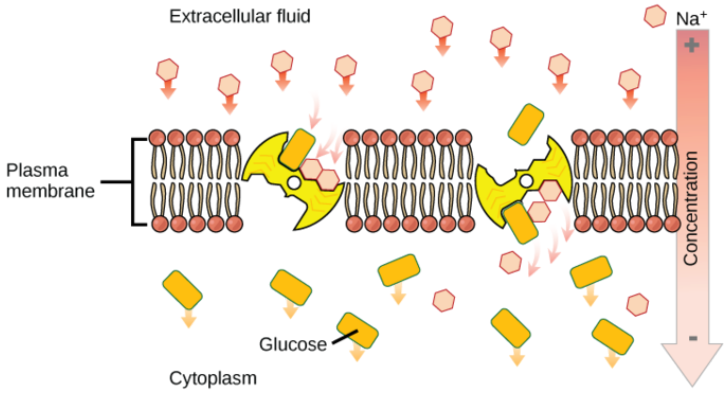


Figure 5.3C.1: **Secondary Active Transport**: An electrochemical gradient, created by primary active transport, can move other substances against their concentration gradients, a process called co-transport or secondary active transport.

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5.4: Bulk Transport

Topic hierarchy

5.4A: Endocytosis

Endocytosis is a type of active transport that moves particles, such as large molecules, parts of cells, and even whole cells, into a cell. There are different variations of endocytosis, but all share a common characteristic: the plasma membrane of the cell invaginates, forming a pocket around the target particle. The pocket pinches off, resulting in the particle being contained in a newly-created intracellular vesicle formed from the plasma membrane.

5.4B: Exocytosis

Exocytosis is the process by which cells release particles from within the cell into the extracellular space.

5.4A: Endocytosis

LEARNING OBJECTIVES

- Describe endocytosis, including phagocytosis, pinocytosis, and receptor-mediated endocytosis.

Endocytosis is a type of active transport that moves particles, such as large molecules, parts of cells, and even whole cells, into a cell. There are different variations of endocytosis, but all share a common characteristic: the plasma membrane of the cell invaginates, forming a pocket around the target particle. The pocket pinches off, resulting in the particle being contained in a newly-created intracellular vesicle formed from the plasma membrane.

Phagocytosis

Phagocytosis (the condition of "cell eating") is the process by which large particles, such as cells or relatively large particles, are taken in by a cell. For example, when microorganisms invade the human body, a type of white blood cell called a neutrophil will remove the invaders through this process, surrounding and engulfing the microorganism, which is then destroyed by the neutrophil.

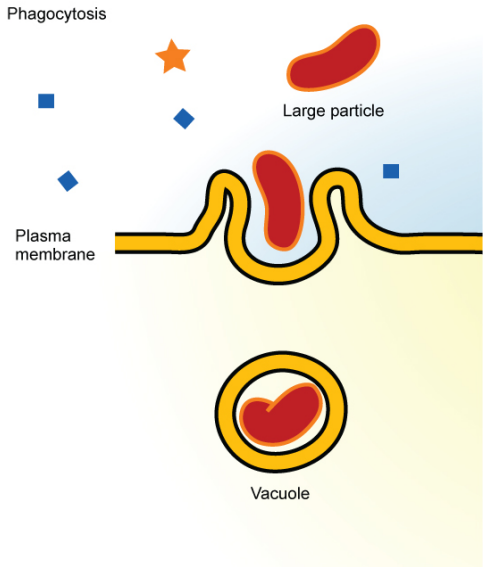

Figure 5.4A.1: **Phagocytosis**: In phagocytosis, the cell membrane surrounds the particle and engulfs it.

In preparation for phagocytosis, a portion of the inward-facing surface of the plasma membrane becomes coated with a protein called clathrin, which stabilizes this section of the membrane. The coated portion of the membrane then extends from the body of the cell and surrounds the particle, eventually enclosing it. Once the vesicle containing the particle is enclosed within the cell, the clathrin disengages from the membrane and the vesicle merges with a lysosome for the breakdown of the material in the newly-formed compartment (endosome). When accessible nutrients from the degradation of the vesicular contents have been extracted, the newly-formed endosome merges with the plasma membrane and releases its contents into the extracellular fluid. The endosomal membrane again becomes part of the plasma membrane.

Pinocytosis

A variation of endocytosis is called pinocytosis. This literally means "cell drinking" and was named at a time when the assumption was that the cell was purposefully taking in extracellular fluid. In reality, this is a process that takes in molecules, including water, which the cell needs from the extracellular fluid. Pinocytosis results in a much smaller vesicle than does phagocytosis, and the vesicle does not need to merge with a lysosome.

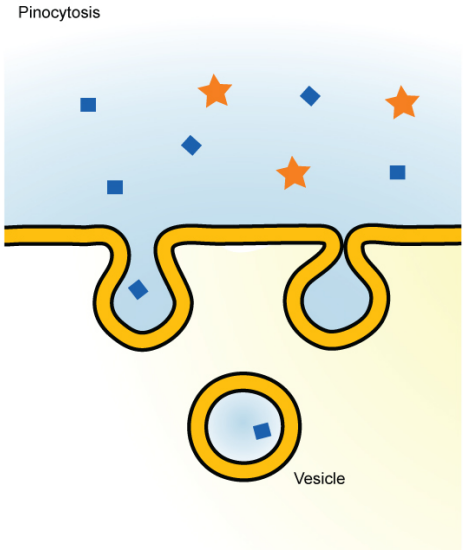

Figure 5.4A.1: **Pinocytosis**: In pinocytosis, the cell membrane invaginates, surrounds a small volume of fluid, and pinches off.

Potocytosis, a variant of pinocytosis, is a process that uses a coating protein, called caveolin, on the cytoplasmic side of the plasma membrane, which performs a similar function to clathrin. The cavities in the plasma membrane that form the vacuoles have membrane receptors and lipid rafts in addition to caveolin. The vacuoles or vesicles formed in caveolae (singular caveola) are smaller than those in pinocytosis. Potocytosis is used to bring small molecules into the cell and to transport these molecules through the cell for their release on the other side of the cell, a process called transcytosis.

Receptor-mediated Endocytosis

A targeted variation of endocytosis, known as receptor-mediated endocytosis, employs receptor proteins in the plasma membrane that have a specific binding affinity for certain substances. In receptor-mediated endocytosis, as in phagocytosis, clathrin is attached to the cytoplasmic side of the plasma membrane. If uptake of a compound is dependent on receptor-mediated endocytosis and the process is ineffective, the material will not be removed from the tissue fluids or blood. Instead, it will stay in those fluids and increase in concentration. Some human diseases are caused by the failure of receptor-mediated endocytosis. For example, the form of cholesterol termed low-density lipoprotein or LDL (also referred to as “bad” cholesterol) is removed from the blood by receptor-mediated endocytosis. In the human genetic disease familial hypercholesterolemia, the LDL receptors are defective or missing entirely. People with this condition have life-threatening levels of cholesterol in their blood, because their cells cannot clear LDL particles from their blood.

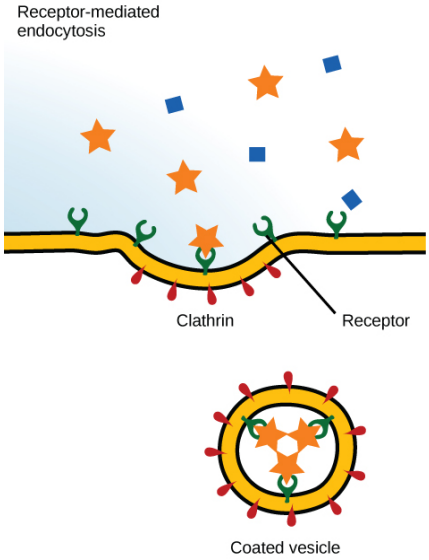

Figure 5.4A.1: **Receptor-Mediated Endocytosis**: In receptor-mediated endocytosis, uptake of substances by the cell is targeted to a single type of substance that binds to the receptor on the external surface of the cell membrane.

Although receptor-mediated endocytosis is designed to bring specific substances that are normally found in the extracellular fluid into the cell, other substances may gain entry into the cell at the same site. Flu viruses, diphtheria, and cholera toxin all have sites that cross-react with normal receptor-binding sites and gain entry into cells.

Key Points

- Endocytosis consists of phagocytosis, pinocytosis, and receptor -mediated endocytosis.
- Endocytosis takes particles into the cell that are too large to passively cross the cell membrane.
- Phagocytosis is the taking in of large food particles, while pinocytosis takes in liquid particles.
- Receptor-mediated endocytosis uses special receptor proteins to help carry large particles across the cell membrane.

Key Terms

- **endosome**: An endocytic vacuole through which molecules internalized during endocytosis pass en route to lysosomes
- **neutrophil**: A cell, especially a white blood cell that consumes foreign invaders in the blood.

5.4B: Exocytosis

LEARNING OBJECTIVES

- Describe exocytosis and the processes used to release materials from the cell.

Exocytosis' main purpose is to expel material from the cell into the extracellular fluid; this is the opposite of what occurs in endocytosis. In exocytosis, waste material is enveloped in a membrane and fuses with the interior of the plasma membrane. This fusion opens the membranous envelope on the exterior of the cell and the waste material is expelled into the extracellular space. Exocytosis is used continuously by plant and animal cells to excrete waste from the cells.

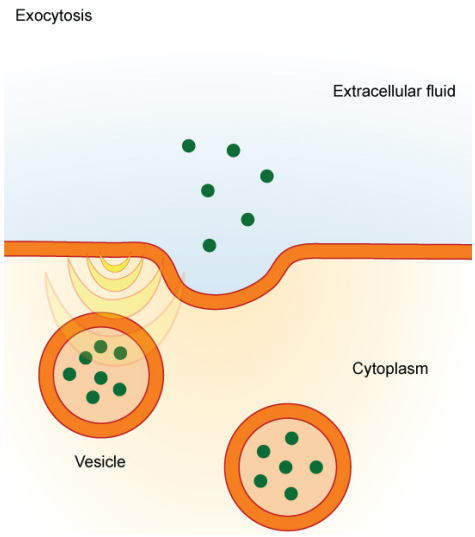

Figure 5.4B.1: **Exocytosis**: In exocytosis, vesicles containing substances fuse with the plasma membrane. The contents are then released to the exterior of the cell.

Exocytosis is composed of five main stages. The first stage is called vesicle trafficking. This involves the steps required to move, over a significant distance, the vesicle containing the material that is to be disposed. The next stage that occurs is vesicle tethering, which links the vesicle to the cell membrane by biological material at half the diameter of a vesicle. Next, the vesicle's membrane and the cell membrane connect and are held together in the vesicle docking step. This stage of exocytosis is then followed by vesicle priming, which includes all of the molecular rearrangements and protein and lipid modifications that take place after initial docking. In some cells, there is no priming. The final stage, vesicle fusion, involves the merging of the vesicle membrane with the target membrane. This results in the release of the unwanted materials into the space outside the cell.

Some examples of cells releasing molecules via exocytosis include the secretion of proteins of the extracellular matrix and secretion of neurotransmitters into the synaptic cleft by synaptic vesicles. Some examples of cells using exocytosis include: the secretion of proteins like enzymes, peptide hormones and antibodies from different cells, the flipping of the plasma membrane, the placement of integral membrane proteins(IMPs) or proteins that are attached biologically to the cell, and the recycling of plasma membrane bound receptors (molecules on the cell membrane that intercept signals).

Key Points

- Exocytosis is the opposite of endocytosis as it involves releasing materials from the cell.
- Exocytosis has five stages, each leading up to the vesicle binding with the cell membrane.
- Many bodily functions include the use of exocytosis, such as the release of neurotransmitters into the synaptic cleft and the release of enzymes into the blood.

Key Terms

- **secretion**: The act of secreting (producing and discharging) a substance, especially from a gland.
- **vesicle**: A membrane-bound compartment found in a cell.

membrane that intercept signals).

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CHAPTER OVERVIEW

6: METABOLISM

6.1: ENERGY AND METABOLISM

6.1A: THE ROLE OF ENERGY AND METABOLISM
6.1B: TYPES OF ENERGY
6.1C: METABOLIC PATHWAYS
6.1D: METABOLISM OF CARBOHYDRATES
6.2: POTENTIAL, KINETIC, FREE, AND ACTIVATION ENERGY

6.2A: FREE ENERGY
6.2B: THE FIRST LAW OF THERMODYNAMICS
6.2C: THE SECOND LAW OF THERMODYNAMICS
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6.3: ATP: ADENOSINE TRIPHOSPHATE

6.3A: ATP: ADENOSINE TRIPHOSPHATE
6.4: ENZYMES

6.4A: ENZYME ACTIVE SITE AND SUBSTRATE SPECIFICITY
6.4B: CONTROL OF METABOLISM THROUGH ENZYME REGULATION

6.1: Energy and Metabolism

Topic hierarchy

- 6.1A: The Role of Energy and Metabolism
- 6.1B: Types of Energy
- 6.1C: Metabolic Pathways
- 6.1D: Metabolism of Carbohydrates

: The Role of Energy and Metabolism

All organisms require energy to complete tasks; metabolism is the set of the chemical reactions that release energy for cellular processes.

Learning Objectives

- Explain the importance of metabolism

Key Points

- All living organisms need energy to grow and reproduce, maintain their structures, and respond to their environments; metabolism is the set of the processes that makes energy available for cellular processes.
- Metabolism is a combination of chemical reactions that are spontaneous and release energy and chemical reactions that are non-spontaneous and require energy in order to proceed.
- Living organisms must take in energy via food, nutrients, or sunlight in order to carry out cellular processes.
- The transport, synthesis, and breakdown of nutrients and molecules in a cell require the use of energy.

Key Terms

- **metabolism**: the complete set of chemical reactions that occur in living cells
- **bioenergetics**: the study of the energy transformations that take place in living organisms
- **energy**: the capacity to do work

Energy and Metabolism

All living organisms need energy to grow and reproduce, maintain their structures, and respond to their environments. Metabolism is the set of life-sustaining chemical processes that enables organisms transform the chemical energy stored in molecules into energy that can be used for cellular processes. Animals consume food to replenish energy; their metabolism breaks down the carbohydrates, lipids, proteins, and nucleic acids to provide chemical energy for these processes. Plants convert light energy from the sun into chemical energy stored in molecules during the process of photosynthesis.

Bioenergetics and Chemical Reactions

Scientists use the term bioenergetics to discuss the concept of energy flow through living systems such as cells. Cellular processes such as the building and breaking down of complex molecules occur through step-by-step chemical reactions. Some of these chemical reactions are spontaneous and release energy, whereas others require energy to proceed. All of the chemical reactions that take place inside cells, including those that use energy and those that release energy, are the cell's metabolism.

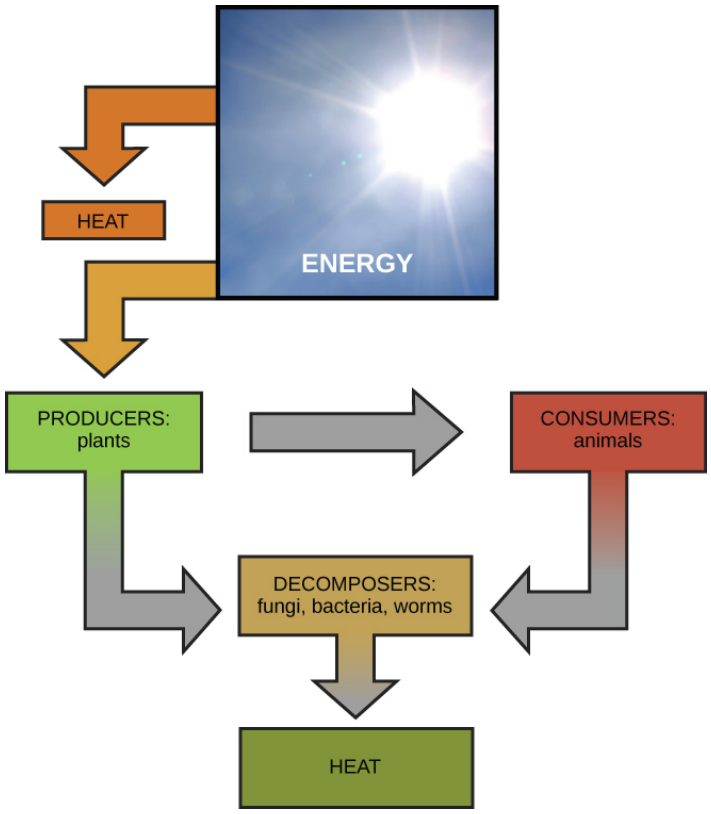

Figure 6.1.*A*. 1: **Most energy comes from the sun, either directly or indirectly**: Most life forms on earth get their energy from the sun. Plants use photosynthesis to capture sunlight, and herbivores eat those plants to obtain energy. Carnivores eat the herbivores, and decomposers digest plant and animal matter.

Cellular Metabolism

Every task performed by living organisms requires energy. Energy is needed to perform heavy labor and exercise, but humans also use a great deal of energy while thinking and even while sleeping. For every action that requires energy, many chemical reactions take place to provide chemical energy to the systems of the body, including muscles, nerves, heart, lungs, and brain.

The living cells of every organism constantly use energy to survive and grow. Cells break down complex carbohydrates into simple sugars that the cell can use for energy. Muscle cells may consumer energy to build long muscle proteins from small amino acid molecules. Molecules can be modified and transported around the cell or may be distributed to the entire organism. Just as energy is required to both build and demolish a building, energy is required for both the synthesis and breakdown of molecules.

Many cellular process require a steady supply of energy provided by the cell's metabolism. Signaling molecules such as hormones and neurotransmitters must be synthesized and then transported between cells. Pathogenic bacteria and viruses are ingested and broken down by cells. Cells must also export waste and toxins to stay healthy, and many cells must swim or move surrounding materials via the beating motion of cellular appendages like cilia and flagella.

Figure 6.1A.1: **Eating provides energy for activities like flight**: A hummingbird needs energy to maintain prolonged periods of flight. The hummingbird obtains its energy from taking in food and transforming the nutrients into energy through a series of biochemical reactions. The flight muscles in birds are extremely efficient in energy production.

6.1B: Types of Energy

The various types of energy include kinetic, potential, and chemical energy.

Learning Objectives

- Differentiate between types of energy

Key Points

- All organisms use different forms of energy to power the biological processes that allow them to grow and survive.
- Kinetic energy is the energy associated with objects in motion.
- Potential energy is the type of energy associated with an object's potential to do work.
- Chemical energy is the type of energy released from the breakdown of chemical bonds and can be harnessed for metabolic processes.

Key Terms

- **chemical energy**: The net potential energy liberated or absorbed during the course of a chemical reaction.
- **potential energy**: Energy possessed by an object because of its position (in a gravitational or electric field), or its condition (as a stretched or compressed spring, as a chemical reactant, or by having rest mass).
- **kinetic energy**: The energy possessed by an object because of its motion, equal to one half the mass of the body times the square of its velocity.

Energy is a property of objects which can be transferred to other objects or converted into different forms, but cannot be created or destroyed. Organisms use energy to survive, grow, respond to stimuli, reproduce, and for every type of biological process. The potential energy stored in molecules can be converted to chemical energy, which can ultimately be converted to kinetic energy, enabling an organism to move. Eventually, most of energy used by organisms is transformed into heat and dissipated.

Kinetic Energy

Energy associated with objects in motion is called kinetic energy. For example, when an airplane is in flight, the airplane is moving through air very quickly—doing work to enact change on its surroundings. The jet engines are converting potential energy in fuel to the kinetic energy of movement. A wrecking ball can perform a large amount of damage, even when moving slowly. However, a still wrecking ball cannot perform any work and therefore has no kinetic energy. A speeding bullet, a walking person, the rapid movement of molecules in the air that produces heat, and electromagnetic radiation, such as sunlight, all have kinetic energy.

Potential Energy

What if that same motionless wrecking ball is lifted two stories above a car with a crane? If the suspended wrecking ball is not moving, is there energy associated with it? Yes, the wrecking ball has energy because the wrecking ball has the potential to do work. This form of energy is called potential energy because it is possible for that object to do work in a given state.

Objects transfer their energy between potential and kinetic states. As the wrecking ball hangs motionlessly, it has 0%0% kinetic and 100%100%potential energy. Once the ball is released, its kinetic energy increases as the ball picks up speed. At the same time, the ball loses potential energy as it nears the ground. Other examples of potential energy include the energy of water held behind a dam or a person about to skydive out of an airplane.

Figure 6.1B.1: **Potential energy vs. kinetic energy**: Water behind a dam has potential energy. Moving water, such as in a waterfall or a rapidly flowing river, has kinetic energy.

Chemical Energy

Potential energy is not only associated with the location of matter, but also with the structure of matter. A spring on the ground has potential energy if it is compressed, as does a rubber band that is pulled taut. The same principle applies to molecules. On a chemical level, the bonds that hold the atoms of molecules together have potential energy. This type of potential energy is called chemical energy, and like all potential energy, it can be used to do work.

For example, chemical energy is contained in the gasoline molecules that are used to power cars. When gas ignites in the engine, the bonds within its molecules are broken, and the energy released is used to drive the pistons. The potential energy stored within chemical bonds can be harnessed to perform work for biological processes. Different metabolic processes break down organic molecules to release the energy for an organism to grow and survive.

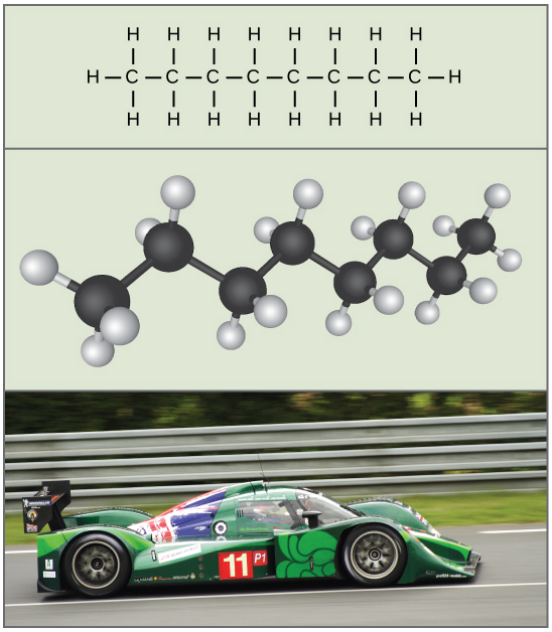

Figure 6.1B.1: **Chemical energy**: The molecules in gasoline (octane, the chemical formula shown) contain chemical energy. This energy is transformed into kinetic energy that allows a car to race on a racetrack.

: Metabolic Pathways

An anabolic pathway requires energy and builds molecules while a catabolic pathway produces energy and breaks down molecules.

Learning Objectives

- Describe the two major types of metabolic pathways

Key Points

- A metabolic pathway is a series of chemical reactions in a cell that build and breakdown molecules for cellular processes.
- Anabolic pathways synthesize molecules and require energy.
- Catabolic pathways break down molecules and produce energy.
- Because almost all metabolic reactions take place non-spontaneously, proteins called enzymes help facilitate those chemical reactions.

Key Terms

- **catabolism**: destructive metabolism, usually including the release of energy and breakdown of materials
- **enzyme**: a globular protein that catalyses a biological chemical reaction
- **anabolism**: the constructive metabolism of the body, as distinguished from catabolism

Metabolic Pathways

The processes of making and breaking down carbohydrate molecules illustrate two types of metabolic pathways. A metabolic pathway is a step-by-step series of interconnected biochemical reactions that convert a substrate molecule or molecules through a series of metabolic intermediates, eventually yielding a final product or products. For example, one metabolic pathway for carbohydrates breaks large molecules down into glucose. Another metabolic pathway might build glucose into large carbohydrate molecules for storage. The first of these processes requires energy and is referred to as anabolic. The second process produces energy and is referred to as catabolic. Consequently, metabolism is composed of these two opposite pathways:

1. Anabolism (building molecules)
2. Catabolism (breaking down molecules)

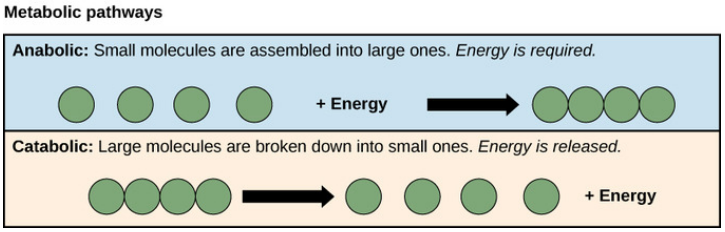

Figure 6.1C.1: **Anabolic and catabolic pathways**: Anabolic pathways are those that require energy to synthesize larger molecules. Catabolic pathways are those that generate energy by breaking down larger molecules. Both types of pathways are required for maintaining the cell's energy balance.

Anabolic Pathways

Anabolic pathways require an input of energy to synthesize complex molecules from simpler ones. One example of an anabolic pathway is the synthesis of sugar from CO_2. Other examples include the synthesis of large proteins from amino acid building blocks and the synthesis of new DNA strands from nucleic acid building blocks. These processes are critical to the life of the cell, take place constantly, and demand energy provided by ATP and other high-energy molecules like NADH (nicotinamide adenine dinucleotide) and NADPH.

bolic Pathways

blic pathways involve the degradation of complex molecules into simpler ones, releasing the chemical energy stored in the bonds of those molecules. Some catabolic pathways can capture that energy to produce ATP, the molecule used to power all cellular processes. Other energy-storing molecules, such as lipids, are also broken down through similar catabolic reactions to release energy and make ATP.

Importance of Enzymes

Chemical reactions in metabolic pathways rarely take place spontaneously. Each reaction step is facilitated, or catalyzed, by a protein called an enzyme. Enzymes are important for catalyzing all types of biological reactions: those that require energy as well as those that release energy.

6.1D: Metabolism of Carbohydrates

Organisms break down carbohydrates to produce energy for cellular processes, and photosynthetic plants produce carbohydrates.

Learning Objectives

- Analyze the importance of carbohydrate metabolism to energy production

Key Points

- The breakdown of glucose living organisms utilize to produce energy is described by the equation: $C_6H_{12}O_6+6O_2 \rightarrow 6CO_2+6H_2O+\text{energy}$.
- The photosynthetic process plants utilize to synthesize glucose is described by the equation:$6CO_2+6H_2O+\text{energy} \rightarrow C_6H_{12}O_6+6O_2$
- Glucose that is consumed is used to make energy in the form of ATP, which is used to perform work and power chemical reactions in the cell.
- During photosynthesis, plants convert light energy into chemical energy that is used to build molecules of glucose.

Key Terms

- **adenosine triphosphate**: a multifunctional nucleoside triphosphate used in cells as a coenzyme, often called the "molecular unit of energy currency" in intracellular energy transfer
- **glucose**: a simple monosaccharide (sugar) with a molecular formula of C6H12O6; it is a principal source of energy for cellular metabolism

Metabolism of Carbohydrates

Carbohydrates are one of the major forms of energy for animals and plants. Plants build carbohydrates using light energy from the sun (during the process of photosynthesis), while animals eat plants or other animals to obtain carbohydrates. Plants store carbohydrates in long polysaccharides chains called starch, while animals store carbohydrates as the molecule glycogen. These large polysaccharides contain many chemical bonds and therefore store a lot of chemical energy. When these molecules are broken down during metabolism, the energy in the chemical bonds is released and can be harnessed for cellular processes.

Figure 6.1*D*.1: **All living things use carbohydrates as a form of energy.**: Plants, like this oak tree and acorn, use energy from sunlight to make sugar and other organic molecules. Both plants and animals (like this squirrel) use cellular respiration to derive energy from the organic molecules originally produced by plants

Energy Production from Carbohydrates (Cellular Respiration)

The metabolism of any monosaccharide (simple sugar) can produce energy for the cell to use. Excess carbohydrates are stored as starch in plants and as glycogen in animals, ready for metabolism if the energy demands of the organism suddenly increase. When those energy demands increase, carbohydrates are broken down into constituent monosaccharides, which are then distributed to all the living cells of an organism. Glucose ($C_6H_{12}O_6$) is a common example of the monosaccharides used for energy production.

Inside the cell, each sugar molecule is broken down through a complex series of chemical reactions. As chemical energy is released from the bonds in the monosaccharide, it is harnessed to synthesize high-energy adenosine triphosphate (ATP) molecules. ATP is the primary energy currency of all cells. Just as the dollar is used as currency to buy goods, cells use molecules of ATP to perform immediate work and power chemical reactions.

The breakdown of glucose during metabolism is call cellular respiration can be described by the equation:

$$C_6H_{12}O_6+6O_2 \rightarrow 6CO_2+6H_2O+\text{energy}$$

Producing Carbohydrates (Photosynthesis)

Plants and some other types of organisms produce carbohydrates through the process called photosynthesis. During photosynthesis, plants convert light energy into chemical energy by building carbon dioxide gas molecules (CO_2) into sugar molecules like glucose. Because this process involves building bonds to synthesize a large molecule, it requires an input of energy (light) to proceed. The synthesis of glucose by photosynthesis is described by this equation (notice that it is the reverse of the previous equation):

$$6CO_2+6H_2O+\text{energy} \rightarrow C_6H_{12}O_6+6O_2$$

As part of plants' chemical processes, glucose molecules can be combined with and converted into other types of sugars. In plants, glucose is stored in the form of starch, which can be broken down back into glucose via cellular respiration in order to supply ATP.

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6.2: Potential, Kinetic, Free, and Activation Energy

Topic hierarchy

- 6.2A: Free Energy
- 6.2B: The First Law of Thermodynamics
- 6.2C: The Second Law of Thermodynamics
- 6.2D: Activation Energy

6.2A: Free Energy

Free energy, called Gibbs free energy (G), is usable energy or energy that is available to do work.

Learning Objectives

- Discuss the concept of free energy.

Key Points

- Every chemical reaction involves a change in free energy, called delta G (ΔG).
- To calculate ΔG, subtract the amount of energy lost to entropy (ΔS) from the total energy change of the system; this total energy change in the system is called enthalpy (ΔH): $\Delta G = \Delta H - T\Delta S$.
- Endergonic reactions require an input of energy; the ΔG for that reaction will be a positive value.
- Exergonic reactions release free energy; the ΔG for that reaction will be a negative value.

Key Terms

- **exergonic reaction**: A chemical reaction where the change in the Gibbs free energy is negative, indicating a spontaneous reaction
- **endergonic reaction**: A chemical reaction in which the standard change in free energy is positive, and energy is absorbed
- **Gibbs free energy**: The difference between the enthalpy of a system and the product of its entropy and absolute temperature

Free Energy

Since chemical reactions release energy when energy-storing bonds are broken, how is the energy associated with chemical reactions quantified and expressed? How can the energy released from one reaction be compared to that of another reaction?

A measurement of free energy is used to quantitate these energy transfers. Free energy is called Gibbs free energy (G) after Josiah Willard Gibbs, the scientist who developed the measurement. Recall that according to the second law of thermodynamics, all energy transfers involve the loss of some amount of energy in an unusable form such as heat, resulting in entropy. Gibbs free energy specifically refers to the energy associated with a chemical reaction that is available after accounting for entropy. In other words, Gibbs free energy is usable energy or energy that is available to do work.

Calculating ΔG

Every chemical reaction involves a change in free energy, called delta G (ΔG). The change in free energy can be calculated for any system that undergoes a change, such as a chemical reaction. To calculate ΔG, subtract the amount of energy lost to entropy (denoted as ΔS) from the total energy change of the system. This total energy change in the system is called enthalpy and is denoted as ΔH. The formula for calculating ΔG is as follows, where the symbol T refers to absolute temperature in Kelvin (degrees Celsius + 273): $G = \Delta H - T\Delta S$.

The standard free energy change of a chemical reaction is expressed as an amount of energy per mole of the reaction product (either in kilojoules or kilocalories, kJ/mol or kcal/mol; 1 kJ = 0.239 kcal) under standard pH, temperature, and pressure conditions. Standard pH, temperature, and pressure conditions are generally calculated at pH 7.0 in biological systems, 25 degrees Celsius, and 100 kilopascals (1 atm pressure), respectively. It is important to note that cellular conditions vary considerably from these standard conditions; therefore, standard calculated ΔG values for biological reactions will be different inside the cell.

Endergonic and Exergonic Reactions

If energy is released during a chemical reaction, then the resulting value from the above equation will be a negative number. In other words, reactions that release energy have a $\Delta G < 0$. A negative ΔG also means that the products of the reaction have less free energy than the reactants because they gave off some free energy during the reaction. Reactions that have a negative ΔG and, consequently, release free energy, are called exergonic reactions. Exergonic means energy is

exiting the system. These reactions are also referred to as spontaneous reactions because they can occur without the addition of energy into the system. Understanding which chemical reactions are spontaneous and release free energy is extremely useful for biologists because these reactions can be harnessed to perform work inside the cell. An important distinction must be drawn between the term spontaneous and the idea of a chemical reaction that occurs immediately. Contrary to the everyday use of the term, a spontaneous reaction is not one that suddenly or quickly occurs. The rusting of iron is an example of a spontaneous reaction that occurs slowly, little by little, over time.

If a chemical reaction requires an input of energy rather than releasing energy, then the ΔG for that reaction will be a positive value. In this case, the products have more free energy than the reactants. Thus, the products of these reactions can be thought of as energy-storing molecules. These chemical reactions are called endergonic reactions; they are non-spontaneous. An endergonic reaction will not take place on its own without the addition of free energy.

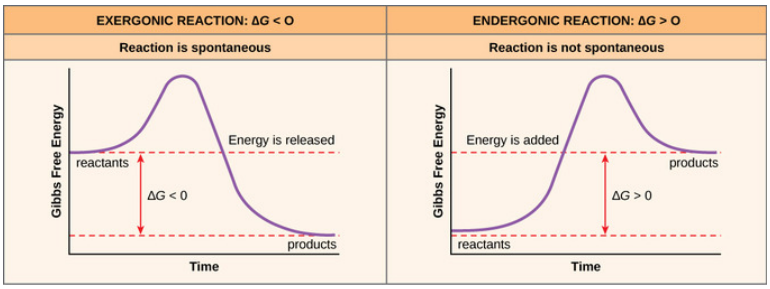

Figure 6.2A.1: **Exergonic and Endergonic Reactions**: Exergonic and endergonic reactions result in changes in Gibbs free energy. Exergonic reactions release energy; endergonic reactions require energy to proceed.

Free Energy and Biological Processes

In a living cell, chemical reactions are constantly moving towards equilibrium, but never reach it. A living cell is an open system: materials pass in and out, the cell recycles the products of certain chemical reactions into other reactions, and chemical equilibrium is never reached. In this way, living organisms are in a constant energy-requiring, uphill battle against equilibrium and entropy.

When complex molecules, such as starches, are built from simpler molecules, such as sugars, the anabolic process requires energy. Therefore, the chemical reactions involved in anabolic processes are endergonic reactions. On the other hand, the catabolic process of breaking sugar down into simpler molecules releases energy in a series of exergonic reactions. As in the example of rust above, the breakdown of sugar involves spontaneous reactions, but these reactions don't occur instantaneously. An important concept in the study of metabolism and energy is that of chemical equilibrium. Most chemical reactions are reversible. They can proceed in both directions, releasing energy into their environment in one direction, and absorbing it from the environment in the other direction.

Figure 6.2*A*.1: **Endergonic and Exergonic Processes**: Shown are some examples of endergonic processes (ones that require energy) and exergonic processes (ones that release energy). These include (a) a compost pile decomposing, (b) a chick hatching from a fertilized egg, (c) sand art being destroyed, and (d) a ball rolling down a hill.

6.2B: The First Law of Thermodynamics

The first law of thermodynamics states that energy can be transferred or transformed, but cannot be created or destroyed.

Learning Objectives

- Describe the first law of thermodynamics

Key Points

- According to the first law of thermodynamics, the total amount of energy in the universe is constant.
- Energy can be transferred from place to place or transformed into different forms, but it cannot be created or destroyed.
- Living organisms have evolved to obtain energy from their surroundings in forms that they can transfer or transform into usable energy to do work.

Key Terms

- **first law of thermodynamics**: A version of the law of conservation of energy, specialized for thermodynamical systems, that states that the energy of an isolated system is constant and can neither be created nor destroyed.
- **work**: A measure of energy expended by moving an object, usually considered to be force times distance. No work is done if the object does not move.

Thermodynamics is the study of heat energy and other types of energy, such as work, and the various ways energy is transferred within chemical systems. “Thermo-” refers to heat, while “dynamics” refers to motion.

The First Law of Thermodynamics

The first law of thermodynamics deals with the total amount of energy in the universe. The law states that this total amount of energy is constant. In other words, there has always been, and always will be, exactly the same amount of energy in the universe.

Energy exists in many different forms. According to the first law of thermodynamics, energy can be transferred from place to place or changed between different forms, but it cannot be created or destroyed. The transfers and transformations of energy take place around us all the time. For instance, light bulbs transform electrical energy into light energy, and gas stoves transform chemical energy from natural gas into heat energy. Plants perform one of the most biologically useful transformations of energy on Earth: they convert the energy of sunlight into the chemical energy stored within organic molecules.

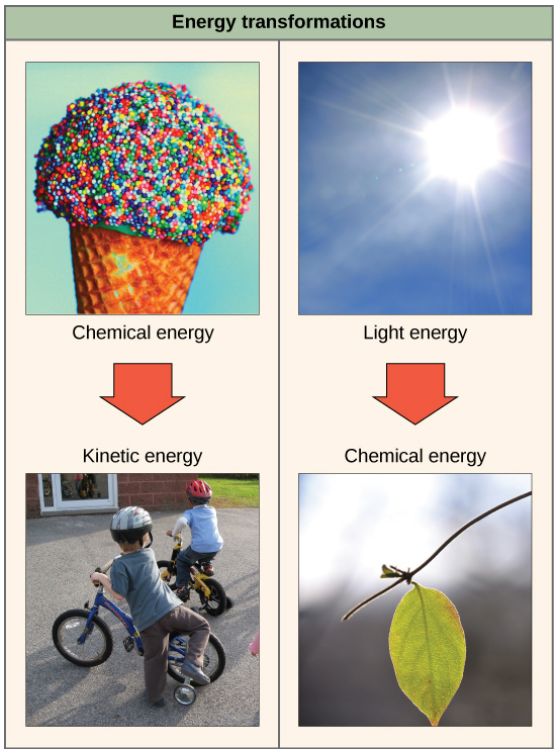

Figure 6.2B.1: **The first law of thermodynamics**: Shown are two examples of energy being transferred from one system to another and transformed from one form to another. Humans can convert the chemical energy in food, like this ice cream cone, into kinetic energy by riding a bicycle. Plants can convert electromagnetic radiation (light energy) from the sun into chemical energy.

The System and Surroundings

Thermodynamics often divides the universe into two categories: the system and its surroundings. In chemistry, the system almost always refers to a given chemical reaction and the container in which it takes place. The first law of thermodynamics tells us that energy can neither be created nor destroyed, so we know that the energy that is absorbed in an endothermic chemical reaction must have been lost from the surroundings. Conversely, in an exothermic reaction, the heat that is released in the reaction is given off and absorbed by the surroundings. Stated mathematically, we have:

$\Delta E = \Delta E_{sys} + \Delta E_{surr} = 0$

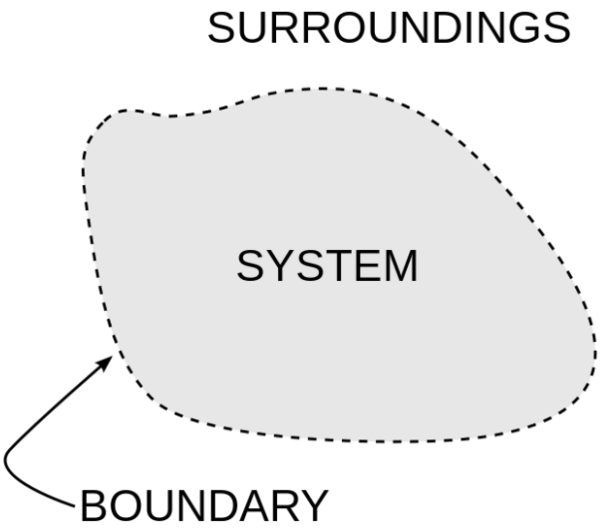

Figure 6.2B.1: **The system and surroundings**: A basic diagram showing the fundamental distinction between the system and its surroundings in thermodynamics.

Heat and Work

We know that chemical systems can either absorb heat from their surroundings, if the reaction is endothermic, or release heat to their surroundings, if the reaction is exothermic. However, chemical reactions are often used to do work instead of just exchanging heat. For instance, when rocket fuel burns and causes a space shuttle to lift off from the ground, the chemical reaction, by propelling the rocket, is doing work by applying a force over a distance.

If you've ever witnessed a video of a space shuttle lifting off, the chemical reaction that occurs also releases tremendous amounts of heat and light. Another useful form of the first law of thermodynamics relates heat and work for the change in energy of the internal system:

$$\Delta E_{sys}=Q+W$$

While this formulation is more commonly used in physics, it is still important to know for chemistry.

Figure 6.2B.1: **Rocket launch**: The powerful chemical reaction propelling the rocket lets off tremendous heat to the surroundings and does work on the surroundings (the rocket) as well.

6.2C: The Second Law of Thermodynamics

The second law of thermodynamics states that every energy transfer increases the entropy of the universe due to the loss of usable energy.

Learning Objectives

- Explain how living organisms can increase their order despite the second law of thermodynamics

Key Points

- During energy transfer, some amount of energy is lost in the form of unusable heat energy.
- Because energy is lost in an unusable form, no energy transfer is completely efficient.
- The more energy that is lost by a system to its surroundings, the less ordered and more random the system is.
- Entropy is a measure of randomness and disorder; high entropy means high disorder and low energy.
- As chemical reactions reach a state of equilibrium, entropy increases; and as molecules at a high concentration in one place diffuse and spread out, entropy also increases.

Key Terms

- **second law of thermodynamics**: Every energy transfer or transformation increases the entropy of the universe since all energy transfers result in the loss of some usable energy.
- **entropy**: A measure of randomness and disorder in a system.

The Second Law of Thermodynamics

A living cell ‘s primary tasks of obtaining, transforming, and using energy to do work may seem simple enough, but they are more problematic than they appear. The second law of thermodynamics explains why: No energy transfers or transformations in the universe are completely efficient. In every energy transfer, some amount of energy is lost in a form that is unusable. In most cases, this energy is in the form of heat. Thermodynamically, heat energy is defined as the energy transferred from one system to another that is not doing work. For example, when an airplane flies through the air, some of the energy of the flying plane is lost as heat energy due to friction with the surrounding air. This friction heats the air by temporarily increasing the speed of air molecules. Likewise, some energy is lost in the form of heat during cellular metabolic reactions. This is good for warm-blooded creatures like us because heat energy helps to maintain our body temperature. Strictly speaking, no energy transfer is completely efficient because some energy is lost in an unusable form.

Entropy

An important concept in physical systems is disorder (also known as randomness). The more energy that is lost by a system to its surroundings, the less ordered and more random the system is. Scientists define the measure of randomness or disorder within a system as entropy. High entropy means high disorder and low energy. To better understand entropy, remember that it requires energy to maintain structure. For example, think about an ice cube. It is made of water molecules bound together in an orderly lattice. This arrangement takes energy to maintain. When the ice cube melts and becomes water, its molecules are more disordered, in a random arrangement as opposed to a structure. Overall, there is less energy in the system inside the molecular bonds. Therefore, water can be said to have greater entropy than ice.

This holds true for solids, liquids, and gases in general. Solids have the highest internal energy holding them together and therefore the lowest entropy. Liquids are more disordered and it takes less energy to hold them together. Therefore they are higher in entropy than solids, but lower than gases, which are so disordered that they have the highest entropy and lowest amount of energy spent holding them together.

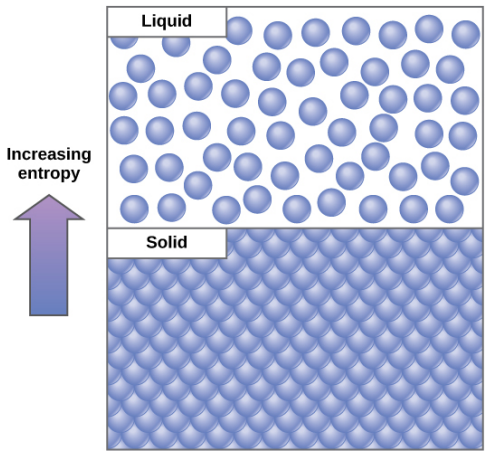

Figure $6.2C.1$: **Entropy**: Entropy is a measure of randomness or disorder in a system. Gases have higher entropy than liquids, and liquids have higher entropy than solids.

Entropy changes also occur in chemical reactions. In an exergonic chemical reaction where energy is released, entropy increases because the final products have less energy inside them holding their chemical bonds together. That energy has been lost to the environment, usually in the form of heat.

All physical systems can be thought of in this way. Living things are highly ordered, requiring constant energy input to be maintained in a state of low entropy. As living systems take in energy-storing molecules and transform them through chemical reactions, they lose some amount of usable energy in the process because no reaction is completely efficient. They also produce waste and by-products that are not useful energy sources. This process increases the entropy of the system's surroundings. Since all energy transfers result in the loss of some usable energy, the second law of thermodynamics states that every energy transfer or transformation increases the entropy of the universe. Even though living things are highly ordered and maintain a state of low entropy, the entropy of the universe in total is constantly increasing due to the loss of usable energy with each energy transfer that occurs. Essentially, living things are in a continuous uphill battle against this constant increase in universal entropy.

6.2D: Activation Energy

Activation energy is the energy required for a reaction to occur, and determines its rate.

Learning Objectives

- Discuss the concept of activation energy

Key Points

- Reactions require an input of energy to initiate the reaction; this is called the activation energy (E_A).
- Activation energy is the amount of energy required to reach the transition state.
- The source of the activation energy needed to push reactions forward is typically heat energy from the surroundings.
- For cellular reactions to occur fast enough over short time scales, their activation energies are lowered by molecules called catalysts.
- Enzymes are catalysts.

Key Terms

- **activation energy**: The minimum energy required for a reaction to occur.
- **catalysis**: The increase in the rate of a chemical reaction by lowering its activation energy.
- **transition state**: An intermediate state during a chemical reaction that has a higher energy than the reactants or the products.

Many chemical reactions, and almost all biochemical reactions do not occur spontaneously and must have an initial input of energy (called the activation energy) to get started. Activation energy must be considered when analyzing both endergonic and exergonic reactions. Exergonic reactions have a net release of energy, but they still require a small amount of energy input before they can proceed with their energy-releasing steps. This small amount of energy input necessary for all chemical reactions to occur is called the activation energy (or free energy of activation) and is abbreviated E_A.

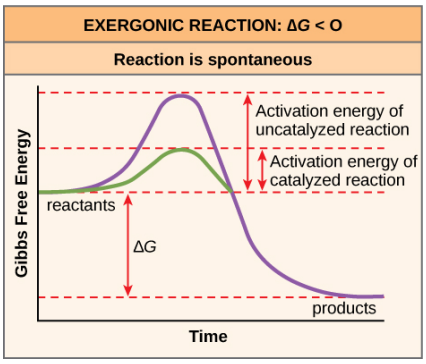

Figure 6.2D.1: **Activation energy**: Activation energy is the energy required for a reaction to proceed; it is lower if the reaction is catalyzed. The horizontal axis of this diagram describes the sequence of events in time.

Activation Energy in Chemical Reactions

Why would an energy-releasing, negative ΔG reaction actually require some energy to proceed? The reason lies in the steps that take place during a chemical reaction. During chemical reactions, certain chemical bonds are broken and new ones are formed. For example, when a glucose molecule is broken down, bonds between the carbon atoms of the molecule are broken. Since these are energy-storing bonds, they release energy when broken. However, to get them into a state that allows the bonds to break, the molecule must be somewhat contorted. A small energy input is required to achieve this contorted state, which is called the *transition state*: it is a high-energy, unstable state. For this reason, reactant molecules don't last long in their transition state, but very quickly proceed to the next steps of the chemical reaction.

Cells will at times couple an exergonic reaction (ΔG<0) with endergonic reactions (ΔG>0), allowing them to proceed. This spontaneous shift from one reaction to another is called energy coupling. The free energy released from the exergonic reaction is absorbed by the

endergonic reaction. One example of energy coupling using ATP involves a transmembrane ion pump that is extremely important for cellular function.

Free Energy Diagrams

Free energy diagrams illustrate the energy profiles for a given reaction. Whether the reaction is exergonic ($\Delta G<0$) or endergonic ($\Delta G>0$) determines whether the products in the diagram will exist at a lower or higher energy state than the reactants. However, the measure of the activation energy is *independent* of the reaction's ΔG. In other words, at a given temperature, the activation energy depends on the nature of the chemical transformation that takes place, but not on the relative energy state of the reactants and products.

Although the image above discusses the concept of activation energy within the context of the exergonic forward reaction, the same principles apply to the reverse reaction, which must be endergonic. Notice that the activation energy for the reverse reaction is larger than for the forward reaction.

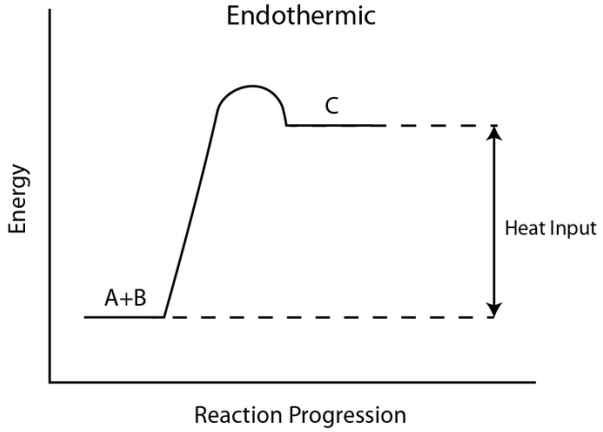

Figure 6.2*D*.1: **Activation energy in an endergonic reaction**: In this endergonic reaction, activation energy is still required to transform the reactants A + B into the product C. This figure implies that the activation energy is in the form of heat energy.

Heat Energy

The source of the activation energy needed to push reactions forward is typically heat energy from the surroundings. Heat energy (the total bond energy of reactants or products in a chemical reaction) speeds up the motion of molecules, increasing the frequency and force with which they collide. It also moves atoms and bonds within the molecule slightly, helping them reach their transition state. For this reason, heating up a system will cause chemical reactants within that system to react more frequently. Increasing the pressure on a system has the same effect. Once reactants have absorbed enough heat energy from their surroundings to reach the transition state, the reaction will proceed.

The activation energy of a particular reaction determines the rate at which it will proceed. The higher the activation energy, the slower the chemical reaction will be. The example of iron rusting illustrates an inherently slow reaction. This reaction occurs slowly over time because of its high E_A. Additionally, the burning of many fuels, which is strongly exergonic, will take place at a negligible rate unless their activation energy is overcome by sufficient heat from a spark. Once they begin to burn, however, the chemical reactions release enough heat to continue the burning process, supplying the activation energy for surrounding fuel molecules.

Like these reactions outside of cells, the activation energy for most cellular reactions is too high for heat energy to overcome at efficient rates. In other words, in order for important cellular reactions to occur at significant rates (number of reactions per unit time), their activation energies must be lowered; this is referred to as catalysis. This is a very good thing as far as living cells are concerned. Important macromolecules, such as proteins, DNA, and RNA, store considerable energy, and their breakdown is exergonic. If cellular temperatures alone provided enough heat energy for these exergonic reactions to overcome their activation barriers, the essential components of a cell would disintegrate.

The Arrhenius Equation

The Arrhenius equations relates the rate of a chemical reaction to the magnitude of the activation energy:

$$k=Ae^{E_a/RT}$$

where

- k is the reaction rate coefficient or constant
- A is the frequency factor of the reaction. It is determined experimentally.
- R is the Universal Gas constant
- T is the temperature in Kelvin

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6.3: ATP: Adenosine Triphosphate

Topic hierarchy

- 6.3A: ATP: Adenosine Triphosphate

6.3A: ATP: Adenosine Triphosphate

Cells couple the exergonic reaction of ATP hydrolysis with endergonic reactions to harness the energy within the bonds of ATP.

Learning Objectives

- Explain the role of ATP as the currency of cellular energy

Key Points

- Adenosine triphosphate is composed of the nitrogenous base adenine, the five-carbon sugar ribose, and three phosphate groups.
- ATP is hydrolyzed to ADP in the reaction $ATP+H2O \rightarrow ADP+Pi+$ free energy; the calculated ΔG for the hydrolysis of 1 mole of ATP is -57 kJ/mol.
- ADP is combined with a phosphate to form ATP in the reaction $ADP+Pi+\text{free energy} \rightarrow ATP+H2O$.
- The energy released from the hydrolysis of ATP into ADP is used to perform cellular work, usually by coupling the exergonic reaction of ATP hydrolysis with endergonic reactions.
- Sodium-potassium pumps use the energy derived from exergonic ATP hydrolysis to pump sodium and potassium ions across the cell membrane while phosphorylation drives the endergonic reaction.

Key Terms

- **energy coupling**: Energy coupling occurs when the energy produced by one reaction or system is used to drive another reaction or system.
- **endergonic**: Describing a reaction that absorbs (heat) energy from its environment.
- **exergonic**: Describing a reaction that releases energy (heat) into its environment.
- **free energy**: Gibbs free energy is a thermodynamic potential that measures the useful or process-initiating work obtainable from a thermodynamic system at a constant temperature and pressure (isothermal, isobaric).
- **hydrolysis**: A chemical process of decomposition involving the splitting of a bond by the addition of water.

ATP: Adenosine Triphosphate

Adenosine triphosphate (ATP) is the energy currency for cellular processes. ATP provides the energy for both energy-consuming endergonic reactions and energy-releasing exergonic reactions, which require a small input of activation energy. When the chemical bonds within ATP are broken, energy is released and can be harnessed for cellular work. The more bonds in a molecule, the more potential energy it contains. Because the bond in ATP is so easily broken and reformed, ATP is like a rechargeable battery that powers cellular process ranging from DNA replication to protein synthesis.

Molecular Structure

Adenosine triphosphate (ATP) is comprised of the molecule adenosine bound to three phosphate groups. Adenosine is a nucleoside consisting of the nitrogenous base adenine and the five-carbon sugar ribose. The three phosphate groups, in order of closest to furthest from the ribose sugar, are labeled alpha, beta, and gamma. Together, these chemical groups constitute an energy powerhouse. The two bonds between the phosphates are equal high-energy bonds (phosphoanhydride bonds) that, when broken, release sufficient energy to power a variety of cellular reactions and processes. The bond between the beta and gamma phosphate is considered “high-energy” because when the bond breaks, the products [adenosine diphosphate (ADP) and one inorganic phosphate group (P_i)] have a lower free energy than the reactants (ATP and a water molecule). ATP breakdown into ADP and P_i is called hydrolysis because it consumes a water molecule (hydro-, meaning “water”, and lysis, meaning “separation”).

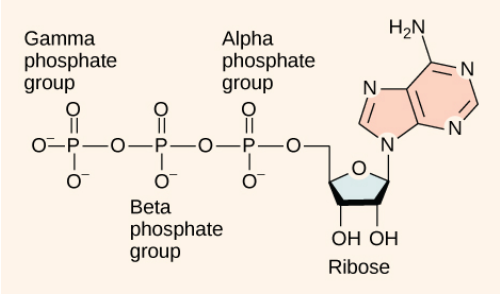

Figure 6.3A.1: **Adenosine Triphosphate (ATP)**: ATP is the primary energy currency of the cell. It has an adenosine backbone with three phosphate groups attached.

ATP Hydrolysis and Synthesis

ATP is hydrolyzed into ADP in the following reaction:

$ATP + H_2O \rightarrow ADP + P_i + \text{free energy}$

Like most chemical reactions, the hydrolysis of ATP to ADP is reversible. The reverse reaction combines ADP + P_i to regenerate ATP from ADP. Since ATP hydrolysis releases energy, ATP synthesis must require an input of free energy.

ADP is combined with a phosphate to form ATP in the following reaction:

$ADP + P_i + \text{free energy} \rightarrow ATP + H_2O$

ATP and Energy Coupling

Exactly how much free energy (ΔG) is released with the hydrolysis of ATP, and how is that free energy used to do cellular work? The calculated ΔG for the hydrolysis of one mole of ATP into ADP and P_i is −7.3 kcal/mole (−30.5 kJ/mol). However, this is only true under standard conditions, and the ΔG for the hydrolysis of one mole of ATP in a living cell is almost double the value at standard conditions: 14 kcal/mol (−57 kJ/mol).

ATP is a highly unstable molecule. Unless quickly used to perform work, ATP spontaneously dissociates into ADP + P_i, and the free energy released during this process is lost as heat. To harness the energy within the bonds of ATP, cells use a strategy called energy coupling.

Energy Coupling in Sodium-Potassium Pumps

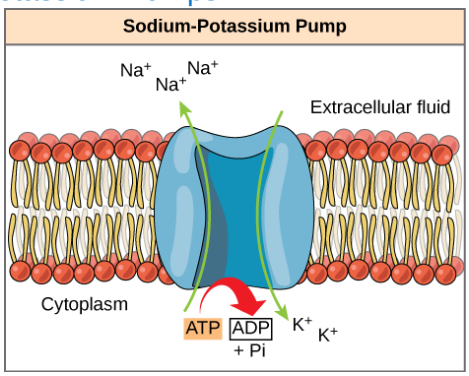

Figure 6.3A.1: **Energy Coupling**: Sodium-potassium pumps use the energy derived from exergonic ATP hydrolysis to pump sodium and potassium ions across the cell membrane.

Cells couple the exergonic reaction of ATP hydrolysis with the endergonic reactions of cellular processes. For example, transmembrane ion pumps in nerve cells use the energy from ATP to pump ions across the cell membrane and generate an action potential. The sodium-potassium pump (Na^+/K^+pump) drives sodium out of the cell and potassium into the cell. When ATP is hydrolyzed, it transfers its gamma phosphate to the pump protein in a process called phosphorylation. The Na^+/K^+ pump gains the free energy and undergoes a conformational change, allowing it to release three Na^+ to the outside of the cell. Two extracellular K^+ ions bind to the protein, causing the protein to change shape again and discharge the phosphate. By donating free energy to the Na^+/K^+ pump, phosphorylation drives the endergonic reaction.

Energy Coupling in Metabolism

During cellular metabolic reactions, or the synthesis and breakdown of nutrients, certain molecules must be altered slightly in their conformation to become substrates for the next step in the reaction series. In the very first steps of cellular respiration, glucose is broken down through the process of glycolysis. ATP is required for the phosphorylation of glucose, creating a high-energy but unstable intermediate. This phosphorylation reaction causes a conformational change that allows enzymes to convert the phosphorylated glucose molecule to the phosphorylated sugar fructose. Fructose is a necessary intermediate for glycolysis to move forward. In this example, the exergonic reaction of ATP hydrolysis is coupled with the endergonic reaction of converting glucose for use in the metabolic pathway.

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6.4: Enzymes

Topic hierarchy

- 6.4A: Enzyme Active Site and Substrate Specificity
- 6.4B: Control of Metabolism Through Enzyme Regulation

: Enzyme Active Site and Substrate Specificity

Enzymes catalyze chemical reactions by lowering activation energy barriers and converting substrate molecules to products.

Learning Objectives

- Describe models of substrate binding to an enzyme's active site.

Key Points

- The enzyme 's active site binds to the substrate.
- Increasing the temperature generally increases the rate of a reaction, but dramatic changes in temperature and pH can denature an enzyme, thereby abolishing its action as a catalyst.
- The induced fit model states an substrate binds to an active site and both change shape slightly, creating an ideal fit for catalysis.
- When an enzyme binds its substrate it forms an enzyme-substrate complex.
- Enzymes promote chemical reactions by bringing substrates together in an optimal orientation, thus creating an ideal chemical environment for the reaction to occur.
- The enzyme will always return to its original state at the completion of the reaction.

Key Terms

- **substrate**: A reactant in a chemical reaction is called a substrate when acted upon by an enzyme.
- **induced fit**: Proposes that the initial interaction between enzyme and substrate is relatively weak, but that these weak interactions rapidly induce conformational changes in the enzyme that strengthen binding.
- **active site**: The active site is the part of an enzyme to which substrates bind and where a reaction is catalyzed.

Enzyme Active Site and Substrate Specificity

Enzymes bind with chemical reactants called substrates. There may be one or more substrates for each type of enzyme, depending on the particular chemical reaction. In some reactions, a single-reactant substrate is broken down into multiple products. In others, two substrates may come together to create one larger molecule. Two reactants might also enter a reaction, both become modified, and leave the reaction as two products.

The enzyme's active site binds to the substrate. Since enzymes are proteins, this site is composed of a unique combination of amino acid residues (side chains or R groups). Each amino acid residue can be large or small; weakly acidic or basic; hydrophilic or hydrophobic; and positively-charged, negatively-charged, or neutral. The positions, sequences, structures, and properties of these residues create a very specific chemical environment within the active site. A specific chemical substrate matches this site like a jigsaw puzzle piece and makes the enzyme specific to its substrate.

Active Sites and Environmental Conditions

Environmental conditions can affect an enzyme's active site and, therefore, the rate at which a chemical reaction can proceed. Increasing the environmental temperature generally increases reaction rates because the molecules are moving more quickly and are more likely to come into contact with each other.

However, increasing or decreasing the temperature outside of an optimal range can affect chemical bonds within the enzyme and change its shape. If the enzyme changes shape, the active site may no longer bind to the appropriate substrate and the rate of reaction will decrease. Dramatic changes to the temperature and pH will eventually cause enzymes to denature.

Induced Fit and Enzyme Function

For many years, scientists thought that enzyme-substrate binding took place in a simple "lock-and-key" fashion. This model asserted that the enzyme and substrate fit together perfectly in one instantaneous step. However, current research supports a more refined view called induced fit. As the enzyme and substrate come together, their interaction causes a mild

n the enzyme's structure that confirms an ideal binding arrangement between the enzyme and the substrate. This ic binding maximizes the enzyme's ability to catalyze its reaction.

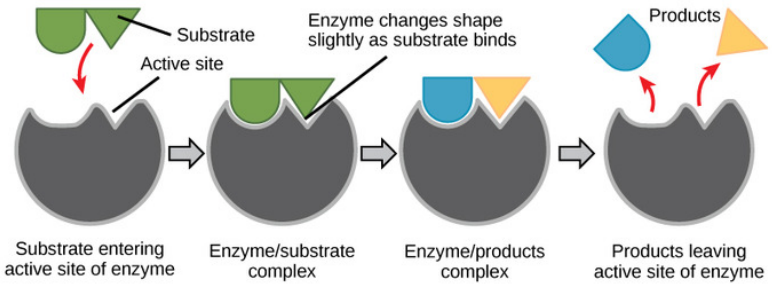

Figure 6.4A.1: **Induced Fit**: According to the induced fit model, both enzyme and substrate undergo dynamic conformational changes upon binding. The enzyme contorts the substrate into its transition state, thereby increasing the rate of the reaction.

Enzyme-Substrate Complex

When an enzyme binds its substrate, it forms an enzyme-substrate complex. This complex lowers the activation energy of the reaction and promotes its rapid progression by providing certain ions or chemical groups that actually form covalent bonds with molecules as a necessary step of the reaction process. Enzymes also promote chemical reactions by bringing substrates together in an optimal orientation, lining up the atoms and bonds of one molecule with the atoms and bonds of the other molecule. This can contort the substrate molecules and facilitate bond-breaking. The active site of an enzyme also creates an ideal environment, such as a slightly acidic or non-polar environment, for the reaction to occur. The enzyme will always return to its original state at the completion of the reaction. One of the important properties of enzymes is that they remain ultimately unchanged by the reactions they catalyze. After an enzyme is done catalyzing a reaction, it releases its products (substrates).

6.4B: Control of Metabolism Through Enzyme Regulation

Cells regulate their biochemical processes by inhibiting or activating enzymes.

Learning Objectives

- Explain the effect of an enzyme on chemical equilibrium

Key Points

- In competitive inhibition, an inhibitor molecule competes with a substrate by binding to the enzyme ‘s active site so the substrate is blocked.
- In noncompetitive inhibition (also known as allosteric inhibition), an inhibitor binds to an allosteric site; the substrate can still bind to the enzyme, but the enzyme is no longer in optimal position to catalyze the reaction.
- Allosteric inhibitors induce a conformational change that changes the shape of the active site and reduces the affinity of the enzyme’s active site for its substrate.
- Allosteric activators induce a conformational change that changes the shape of the active site and increases the affinity of the enzyme’s active site for its substrate.
- Feedback inhibition involves the use of a reaction product to regulate its own further production.
- Inorganic cofactors and organic coenzymes promote optimal enzyme orientation and function.
- Vitamins act as coenzymes (or precursors to coenzymes) and are necessary for enzymes to function.

Key Terms

- **coenzyme**: An organic molecule that is necessary for an enzyme to function.
- **allosteric site**: A site other than the active site on an enzyme.
- **cofactor**: An inorganic molecule that is necessary for an enzyme to function.

Control of Metabolism Through Enzyme Regulation

Cellular needs and conditions vary from cell to cell and change within individual cells over time. For example, a stomach cell requires a different amount of energy than a skin cell, fat storage cell, blood cell, or nerve cell. The same stomach cell may also need more energy immediately after a meal and less energy between meals.

A cell’s function is encapsulated by the chemical reactions it can carry out. Enzymes lower the activation energies of chemical reactions; in cells, they promote those reactions that are specific to the cell’s function. Because enzymes ultimately determine which chemical reactions a cell can carry out and the rate at which they can proceed, they are key to cell functionality.

Competitive and Noncompetitive Inhibition

The cell uses specific molecules to regulate enzymes in order to promote or inhibit certain chemical reactions. Sometimes it is necessary to inhibit an enzyme to reduce a reaction rate, and there is more than one way for this inhibition to occur. In competitive inhibition, an inhibitor molecule is similar enough to a substrate that it can bind to the enzyme’s active site to stop it from binding to the substrate. It “competes” with the substrate to bind to the enzyme.

In noncompetitive inhibition, an inhibitor molecule binds to the enzyme at a location other than the active site (an allosteric site). The substrate can still bind to the enzyme, but the inhibitor changes the shape of the enzyme so it is no longer in optimal position to catalyze the reaction.

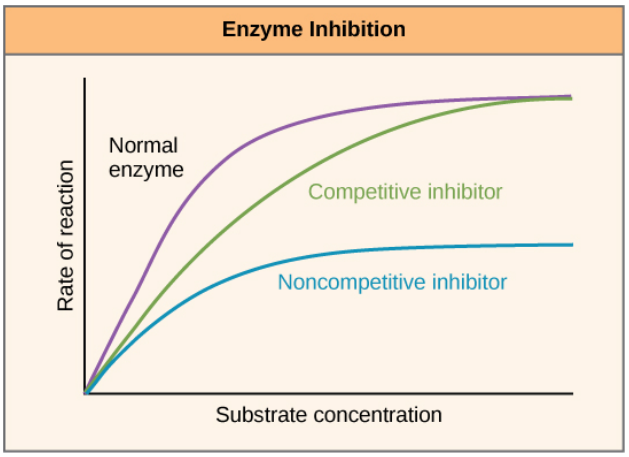

Figure 6.4*B*.1: **Enzyme inhibition**: Competitive and noncompetitive inhibition affect the rate of reaction differently. Competitive inhibitors affect the initial rate, but do not affect the maximal rate, whereas noncompetitive inhibitors affect the maximal rate.

Allosteric Inhibition and Activation

In noncompetitive allosteric inhibition, inhibitor molecules bind to an enzyme at the allosteric site. Their binding induces a conformational change that reduces the affinity of the enzyme's active site for its substrate. The binding of this allosteric inhibitor changes the conformation of the enzyme and its active site, so the substrate is not able to bind. This prevents the enzyme from lowering the activation energy of the reaction, and the reaction rate is reduced.

However, allosteric inhibitors are not the only molecules that bind to allosteric sites. Allosteric activators can increase reaction rates. They bind to an allosteric site which induces a conformational change that *increases* the affinity of the enzyme's active site for its substrate. This increases the reaction rate.

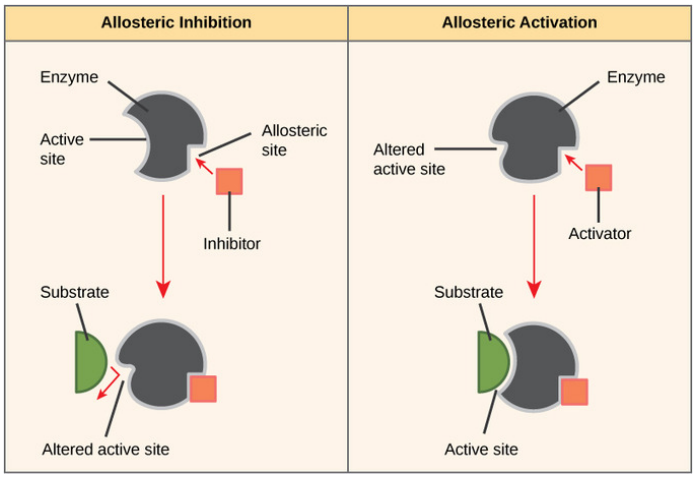

Figure 6.4*B*.1: **Allosteric inhibitors and activators**: Allosteric inhibitors modify the active site of the enzyme so that substrate binding is reduced or prevented. In contrast, allosteric activators modify the active site of the enzyme so that the affinity for the substrate increases.

Cofactors and Coenzymes

Many enzymes only work if bound to non-protein helper molecules called cofactors and coenzymes. Binding to these molecules promotes optimal conformation and function for their respective enzymes. These molecules bind temporarily through ionic or hydrogen bonds or permanently through stronger covalent bonds.

Cofactors are inorganic ions such as iron (Fe^{2+}) and magnesium (Mg^{2+}). For example, DNA polymerase requires a zinc ion (Zn^{2+}) to build DNA molecules. Coenzymes are organic helper molecules with a basic atomic structure made up of carbon and hydrogen. The most common coenzymes are dietary vitamins. Vitamin C is a coenzyme for multiple enzymes that take part in building collagen, an important component of connective tissue. Pyruvate dehydrogenase is a complex of several

enzymes that requires one cofactor and five different organic coenzymes to catalyze its chemical reaction. The availability of various cofactors and coenzymes regulates enzyme function.

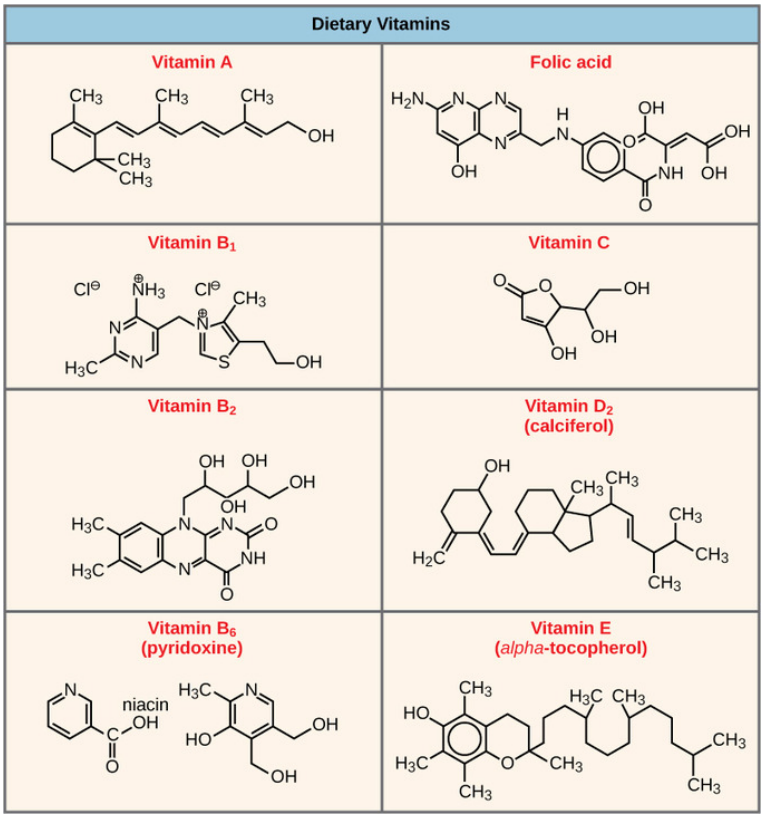

Figure 6.4B.1: **Vitamins**: Vitamins are important coenzymes or precursors of coenzymes and are required for enzymes to function properly. Multivitamin capsules usually contain mixtures of all the vitamins at different percentages.

Enzyme Compartmentalization

In eukaryotic cells, molecules such as enzymes are usually compartmentalized into different organelles. This organization contributes to enzyme regulation because certain cellular processes are contained in separate organelles. For example, the enzymes involved in the later stages of cellular respiration carry out reactions exclusively in the mitochondria. The enzymes involved in the digestion of cellular debris and foreign materials are located within lysosomes.

Feedback Inhibition in Metabolic Pathways

Feedback inhibition is when a reaction product is used to regulate its own further production. Cells have evolved to use feedback inhibition to regulate enzyme activity in metabolism, by using the products of the enzymatic reactions to inhibit further enzyme activity. Metabolic reactions, such as anabolic and catabolic processes, must proceed according to the demands of the cell. In order to maintain chemical equilibrium and meet the needs of the cell, some metabolic products inhibit the enzymes in the chemical pathway while some reactants activate them.

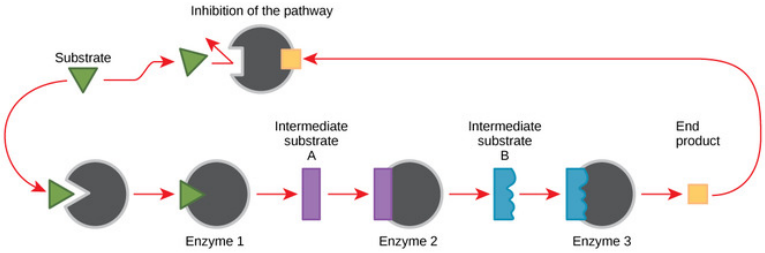

Figure 6.4B.1: **Feedback inhibition**: Metabolic pathways are a series of reactions catalyzed by multiple enzymes. Feedback inhibition, where the end product of the pathway inhibits an earlier step, is an important regulatory mechanism in cells.

The production of both amino acids and nucleotides is controlled through feedback inhibition. For an example of feedback inhibition, consider ATP. It is the product of the catabolic metabolism of sugar (cellular respiration), but it also acts as an allosteric regulator for the same enzymes that produced it. ATP is an unstable molecule that can spontaneously dissociate into ADP; if too much ATP were present, most of it would go to waste. This feedback inhibition prevents the production of additional ATP if it is already abundant. However, while ATP is an inhibitor, ADP is an allosteric activator. When levels of ADP are high compared to ATP levels, ADP triggers the catabolism of sugar to produce more ATP.

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CHAPTER OVERVIEW

7: CELLULAR RESPIRATION

Cellular respiration is a set of metabolic reactions and processes that take place in the cells of organisms to convert biochemical energy from nutrients into adenosine triphosphate (ATP), and then release waste products.

7.1: ENERGY IN LIVING SYSTEMS

7.1A: TRANSFORMING CHEMICAL ENERGY
7.1B: ELECTRONS AND ENERGY
7.1C: ATP IN METABOLISM
7.2: GLYCOLYSIS

7.2A: IMPORTANCE OF GLYCOLYSIS
7.2B: THE ENERGY-REQUIRING STEPS OF GLYCOLYSIS
7.2C: THE ENERGY-RELEASING STEPS OF GLYCOLYSIS
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7.4: OXIDATIVE PHOSPHORYLATION

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The electron transport chain uses the electrons from electron carriers to create a chemical gradient that can be used to power oxidative phosphorylation.

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7.1: Energy in Living Systems

Topic hierarchy

- 7.1A: Transforming Chemical Energy
- 7.1B: Electrons and Energy
- 7.1C: ATP in Metabolism

7.1A: Transforming Chemical Energy

Cellular respiration is the process of transforming chemical energy into forms usable by the cell or organism.

Learning Objectives

- Discuss the importance of cellular respiration

Key Points

- Organisms ingest organic molecules like the carbohydrate glucose to obtain the energy needed for cellular functions.
- The energy in glucose can be extracted in a series of chemical reactions known as cellular respiration.
- Cellular respiration produces energy in the form of ATP, which is the universal energy currency for cells.

Key Terms

- **aerobic respiration**: the process of converting the biochemical energy in nutrients to ATP in the presence of oxygen
- **adenosine triphosphate**: a multifunctional nucleoside triphosphate used in cells as a coenzyme, often called the "molecular unit of energy currency" in intracellular energy transfer
- **cellular respiration**: the set of the metabolic reactions and processes that take place in the cells of organisms to convert biochemical energy from nutrients into adenosine triphosphate (ATP)
- **catabolism**: the breakdown of large molecules into smaller ones usually accompanied by the release of energy

Introduction: Cellular Respiration

An electrical energy plant converts energy from one form to another form that can be more easily used. For example, geothermal energy plants start with underground thermal energy (heat) and transform it into electrical energy that will be transported to homes and factories.

Figure 7.1*A*.1: **Energy Plant**: This geothermal energy plant transforms thermal energy from deep in the ground into electrical energy, which can be easily used.

Like a generating plant, living organisms must take in energy from their environment and convert it into to a form their cells can use. Organisms ingest large molecules, like carbohydrates, proteins, and fats, and convert them into smaller molecules like carbon dioxide and water. This process is called cellular respiration, a form of catabolism, and makes energy available for the cell to use. The energy released by cellular respiration is temporarily captured by the formation of adenosine triphosphate (ATP) within the cell. ATP is the principle form of stored energy used for cellular functions and is frequently referred to as the energy currency of the cell.

The nutrients broken down through cellular respiration lose electrons throughout the process and are said to be oxidized. When oxygen is used to help drive the oxidation of nutrients the process is called aerobic respiration. Aerobic respiration is common among the eukaryotes, including humans, and takes place mostly within the mitochondria. Respiration occurs within the cytoplasm of prokaryotes. Several prokaryotes and a few eukaryotes use an inorganic molecule other than oxygen to drive the oxidation of their nutrients in a process called anaerobic respiration. Electron acceptors for anaerobic respiration include nitrate, sulfate, carbon dioxide, and several metal ions.

The energy released during cellular respiration is then used in other biological processes. These processes build larger molecules that are essential to an organism's survival, such as amino acids, DNA, and proteins. Because they synthesize new

molecules, these processes are examples of anabolism.

7.1B: Electrons and Energy

The transfer of electrons between molecules via oxidation and reduction allows the cell to transfer and use energy for cellular functions.

Learning Objectives

- Describe the role played by electrons in energy production and storage

Key Points

- When electrons are added to a compound, the compound is reduced; a compound that reduces another is called a reducing agent.
- When electrons are removed from a compound, the compound is considered oxidized; a compound that oxidizes another is called an oxidizing agent.
- The transfer of energy in the form of electrons allows the cell to transfer and use energy in an incremental fashion.
- The principle electron carriers are NAD+ and NADH because they can be easily oxidized and reduced, respectively.
- NAD+ is the oxidized form of the niacin and NADH is the reduced form after it has accepted two electrons and a proton.

Key Terms

- **oxidation**: A reaction in which the atoms of an element lose electrons and the valence of the element increases.
- **reduction**: A reaction in which electrons are gained and valence is reduced; often by the removal of oxygen or the addition of hydrogen.
- **nicotinamide adenine dinucleotide**: (NAD) An organic coenzyme involved in biological oxidation and reduction reactions.
- **electron shuttle**: molecules that bind and carry high-energy electrons between compounds in cellular pathways

Electrons and Energy

The removal of an electron from a molecule via a process called oxidation results in a decrease in the potential energy stored in the oxidized compound. When oxidation occurs in the cell, the electron (sometimes as part of a hydrogen atom) does not remain un-bonded in the cytoplasm. Instead, the electron shifts to a second compound, reducing the second compound (oxidation of one species always occurs in tandem with reduction of another).

The shift of an electron from one compound to another removes some potential energy from the first compound (the oxidized compound) and increases the potential energy of the second compound (the reduced compound). The transfer of electrons between molecules via oxidation and reduction is important because most of the energy stored in atoms is in the form of high-energy electrons; it is this energy that is used to fuel cellular functions. The transfer of energy in the form of electrons allows the cell to transfer and use energy in an incremental fashion: in small packages rather than as a single, destructive burst.

Electron carriers

In living systems, a small class of molecules functions as electron shuttles: they bind and carry high-energy electrons between compounds in cellular pathways. The principal electron carriers we will consider are derived from the vitamin B group, which are derivatives of nucleotides. These compounds can be easily reduced (that is, they accept electrons) or oxidized (they lose electrons). Nicotinamide adenine dinucleotide (NAD) is derived from vitamin B3, niacin. NAD^+ is the oxidized form of niacin; NADH is the reduced form after it has accepted two electrons and a proton (which together are the equivalent of a hydrogen atom with an extra electron). It is noteworthy that NAD^+*must* accept two electrons at once; it cannot serve as a one-electron carrier.

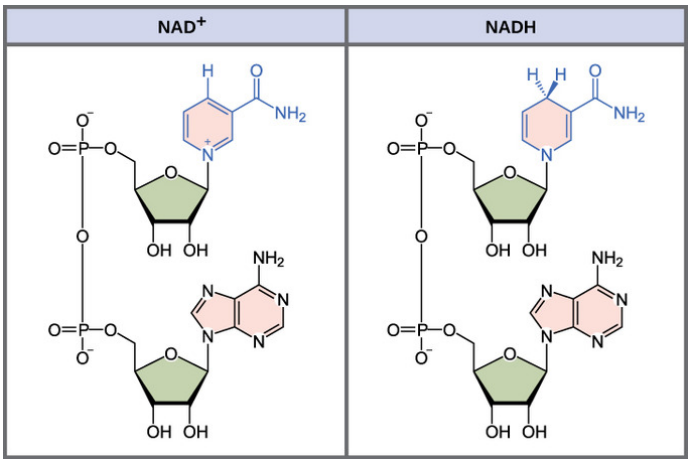

Figure 7.1*B*.1: **The structure of NADH and NAD+**: The oxidized form of the electron carrier (NAD+) is shown on the left and the reduced form (NADH) is shown on the right. The nitrogenous base in NADH has one more hydrogen ion and two more electrons than in NAD+.

NAD^+ can accept electrons from an organic molecule according to the general equation:

RH (Reducing agent) + NAD^+ (Oxidizing agent) → NADH (Reduced) + R (Oxidized)

When electrons are added to a compound, the compound is reduced. A compound that reduces another is called a reducing agent. In the above equation, RH is a reducing agent and NAD^+ is reduced to NADH. When electrons are removed from a compound, the compound is oxidized. In the above equation, NAD^+ is an oxidizing agent and RH is oxidized to R. The molecule NADH is critical for cellular respiration and other metabolic pathways.

Similarly, flavin adenine dinucleotide (FAD^+) is derived from vitamin B_2, also called riboflavin. Its reduced form is $FADH_2$. A second variation of NAD, NADP, contains an extra phosphate group. Both NAD^+ and FAD^+ are extensively used in energy extraction from sugars, and NADP plays an important role in anabolic reactions and photosynthesis.

7.1C: ATP in Metabolism

ATP, produced by glucose catabolized during cellular respiration, serves as the universal energy currency for all living organisms.

Learning Objectives

- Compare the two methods by which cells utilize ATP for energy.

Key Points

- Cells require a constant supply of energy to survive, but cannot store this energy as free energy as this would result in elevated temperatures and would destroy the cell.
- Cells store energy in the form of adenosine triphosphate, or ATP.
- Energy is released when the terminal phosphate group is removed from ATP.
- To utilize the energy stored as ATP, cells either couple ATP hydrolysis to an energetically unfavorable reaction to allow it to proceed or transfer one of the phosphate groups from ATP to a protein substrate, causing it to change conformations and hence energetic preference.

Key Terms

- **phosphorylation**: the addition of a phosphate group to a compound; often catalyzed by enzymes
- **adenosine triphosphate**: a multifunctional nucleoside triphosphate used in cells as a coenzyme, often called the "molecular unit of energy currency" in intracellular energy transfer
- **phosphate**: Any salt or ester of phosphoric acid

ATP in Living Systems

A living cell cannot store significant amounts of free energy. Excess free energy would result in an increase of heat in the cell, which would lead to excessive thermal motion that could damage and then destroy the cell. Rather, a cell must be able to handle that energy in a way that enables the cell to store energy safely and release it for use as needed. Living cells accomplish this by using the compound adenosine triphosphate (ATP). ATP is often called the "energy currency" of the cell and can be used to fill any energy need of the cell.

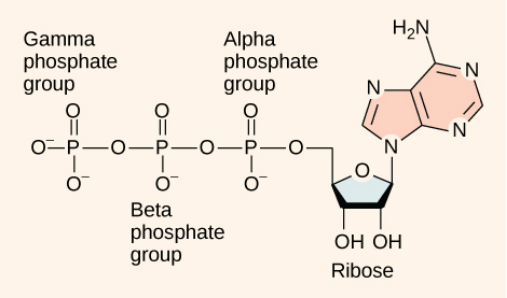

Figure 7.1C.1: **Adenosine triphosphate.**: ATP (adenosine triphosphate) has three phosphate groups that can be removed by hydrolysis to form ADP (adenosine diphosphate) or AMP (adenosine monophosphate).The negative charges on the phosphate group naturally repel each other, requiring energy to bond them together and releasing energy when these bonds are broken.

ATP Structure and Function

The core of ATP is a molecule of adenosine monophosphate (AMP), which is composed of an adenine molecule bonded to a ribose molecule and to a single phosphate group. Ribose is a five-carbon sugar found in RNA, and AMP is one of the nucleotides in RNA. The addition of a second phosphate group to this core molecule results in the formation of adenosine diphosphate (ADP); the addition of a third phosphate group forms adenosine triphosphate (ATP).

The addition of a phosphate group to a molecule requires energy. Phosphate groups are negatively charged and, thus, repel one another when they are arranged in a series, as they are in ADP and ATP. This repulsion makes the ADP and ATP

molecules inherently unstable. The release of one or two phosphate groups from ATP, a process called dephosphorylation, releases energy.

Energy from ATP

Hydrolysis is the process of breaking complex macromolecules apart. During hydrolysis, water is split, or lysed, and the resulting hydrogen atom (H^+) and a hydroxyl group (OH^-) are added to the larger molecule. The hydrolysis of ATP produces ADP, together with an inorganic phosphate ion (P_i), and the release of free energy. To carry out life processes, ATP is continuously broken down into ADP, and, like a rechargeable battery, ADP is continuously regenerated into ATP by the reattachment of a third phosphate group. Water, which was broken down into its hydrogen atom and hydroxyl group during ATP hydrolysis, is regenerated when a third phosphate is added to the ADP molecule, reforming ATP.

Obviously, energy must be infused into the system to regenerate ATP. In nearly every living thing on earth, the energy comes from the metabolism of glucose. In this way, ATP is a direct link between the limited set of exergonic pathways of glucose catabolism and the multitude of endergonic pathways that power living cells.

Phosphorylation

When ATP is broken down by the removal of its terminal phosphate group, energy is released and can be used to do work by the cell. Often the released phosphate is directly transferred to another molecule, such as a protein, activating it. For example, ATP supplies the energy to move the contractile muscle proteins during the mechanical work of muscle contraction. Recall the active transport work of the sodium-potassium pump in cell membranes. Phosphorylation by ATP alters the structure of the integral protein that functions as the pump, changing its affinity for sodium and potassium. In this way, the cell performs work, using energy from ATP to pump ions against their electrochemical gradients.

Sometimes phosphorylation of an enzyme leads to its inhibition. For example, the pyruvate dehydrogenase (PDH) complex could be phosphorylated by pyruvate dehydrogenase kinase (PDHK). This reaction leads to inhibition of PDH and its inability to convert pyruvate into acetyl-CoA.

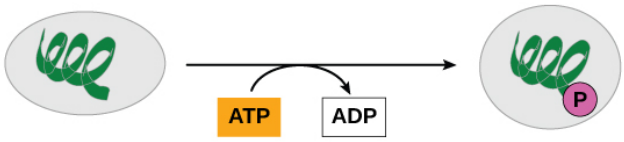

Figure 7.1C.1: **Protein phosphorylation**: In phosphorylation reactions, the gamma phosphate of ATP is attached to a protein.

Energy from ATP hydrolysis

The energy from ATP can also be used to drive chemical reactions by coupling ATP hydrolysis with another reaction process in an enzyme. In many cellular chemical reactions, enzymes bind to several substrates or reactants to form a temporary intermediate complex that allow the substrates and reactants to more readily react with each other. In reactions where ATP is involved, ATP is one of the substrates and ADP is a product. During an endergonic chemical reaction, ATP forms an intermediate complex with the substrate and enzyme in the reaction. This intermediate complex allows the ATP to transfer its third phosphate group, with its energy, to the substrate, a process called phosphorylation. Phosphorylation refers to the addition of the phosphate (~P). When the intermediate complex breaks apart, the energy is used to modify the substrate and convert it into a product of the reaction. The ADP molecule and a free phosphate ion are released into the medium and are available for recycling through cell metabolism. This is illustrated by the following generic reaction:

A + enzyme + ATP → [A enzyme −P] B + enzyme + ADP + phosphate ion

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7.2: Glycolysis

- 7.2A: Importance of Glycolysis
- 7.2B: The Energy-Requiring Steps of Glycolysis
- 7.2C: The Energy-Releasing Steps of Glycolysis
- 7.2D: Outcomes of Glycolysis

7.2A: Importance of Glycolysis

Glycolysis is the first step in the breakdown of glucose to extract energy for cellular metabolism.

Learning Objectives

- Explain the importance of glycolysis to cells

Key Points

- Glycolysis is present in nearly all living organisms.
- Glucose is the source of almost all energy used by cells.
- Overall, glycolysis produces two pyruvate molecules, a net gain of two ATP molecules, and two NADH molecules.

Key Terms

- **glycolysis**: the cellular metabolic pathway of the simple sugar glucose to yield pyruvic acid and ATP as an energy source
- **heterotroph**: an organism that requires an external supply of energy in the form of food, as it cannot synthesize its own

Nearly all of the energy used by living cells comes to them from the energy in the bonds of the sugar glucose. Glucose enters heterotrophic cells in two ways. One method is through secondary active transport in which the transport takes place against the glucose concentration gradient. The other mechanism uses a group of integral proteins called GLUT proteins, also known as glucose transporter proteins. These transporters assist in the facilitated diffusion of glucose. Glycolysis is the first pathway used in the breakdown of glucose to extract energy. It takes place in the cytoplasm of both prokaryotic and eukaryotic cells. It was probably one of the earliest metabolic pathways to evolve since it is used by nearly all of the organisms on earth. The process does not use oxygen and is, therefore, anaerobic.

Glycolysis is the first of the main metabolic pathways of cellular respiration to produce energy in the form of ATP. Through two distinct phases, the six-carbon ring of glucose is cleaved into two three-carbon sugars of pyruvate through a series of enzymatic reactions. The first phase of glycolysis requires energy, while the second phase completes the conversion to pyruvate and produces ATP and NADH for the cell to use for energy. Overall, the process of glycolysis produces a net gain of two pyruvate molecules, two ATP molecules, and two NADH molecules for the cell to use for energy. Following the conversion of glucose to pyruvate, the glycolytic pathway is linked to the Krebs Cycle, where further ATP will be produced for the cell's energy needs.

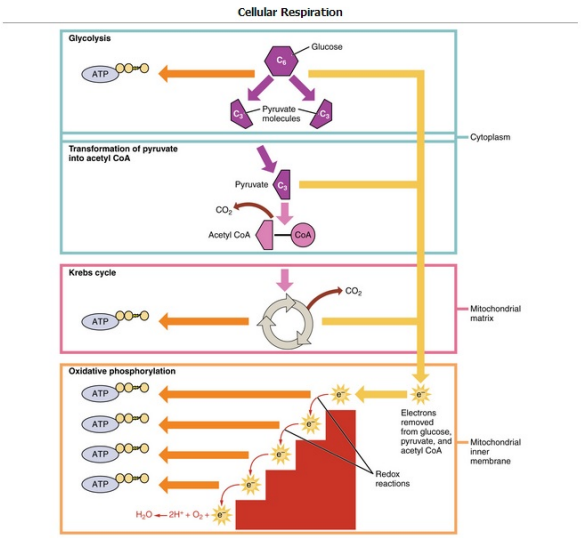

Figure 7.2A.1: **Cellular Respiration**: Glycolysis is the first pathway of cellular respiration that oxidizes glucose molecules. It is followed by the Krebs cycle and oxidative phosphorylation to produce ATP.

7.2B: The Energy-Requiring Steps of Glycolysis

In the first half of glycolysis, energy in the form of two ATP molecules is required to transform glucose into two three-carbon molecules.

Learning Objectives

- Outline the energy-requiring steps of glycolysis

Key Points

- ATP molecules donate high energy phosphate groups during the two phosphorylation steps, step 1 with hexokinase and step 3 with phosphofructokinase, in the first half of glycolysis.
- In steps 2 and 5, isomerases convert molecules into their isomers to allow glucose to be split eventually into two molecules of glyceraldehyde-3-phosphate, which continues into the second half of glycolysis.
- The enzyme aldolase in step 4 of glycolysis cleaves the six-carbon sugar 1,6-bisphosphate into two three-carbon sugar isomers, dihydroxyacetone-phosphate and glyceraldehyde-3-phosphate.

Key Terms

- **glucose**: a simple monosaccharide (sugar) with a molecular formula of C6H12O6; it is a principal source of energy for cellular metabolism
- **adenosine triphosphate**: a multifunctional nucleoside triphosphate used in cells as a coenzyme, often called the "molecular unit of energy currency" in intracellular energy transfer

First Half of Glycolysis (Energy-Requiring Steps)

In the first half of glycolysis, two adenosine triphosphate (ATP) molecules are used in the phosphorylation of glucose, which is then split into two three-carbon molecules as described in the following steps.

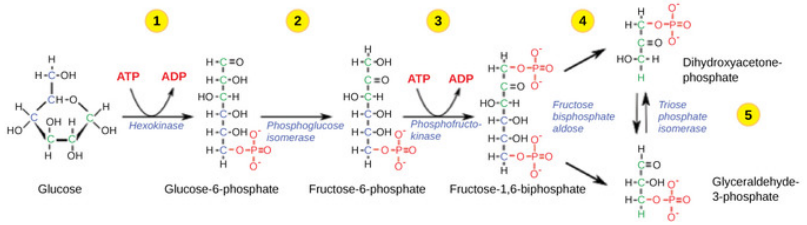

Figure 7.2*B*.1: **The first half of glycolysis: investment**: The first half of glycolysis uses two ATP molecules in the phosphorylation of glucose, which is then split into two three-carbon molecules.

Step 1. The first step in glycolysis is catalyzed by hexokinase, an enzyme with broad specificity that catalyzes the phosphorylation of six-carbon sugars. Hexokinase phosphorylates glucose using ATP as the source of the phosphate, producing glucose-6-phosphate, a more reactive form of glucose. This reaction prevents the phosphorylated glucose molecule from continuing to interact with the GLUT proteins. It can no longer leave the cell because the negatively-charged phosphate will not allow it to cross the hydrophobic interior of the plasma membrane.

Step 2. In the second step of glycolysis, an isomerase converts glucose-6-phosphate into one of its isomers, fructose-6-phosphate. An enzyme that catalyzes the conversion of a molecule into one of its isomers is an isomerase. (This change from phosphoglucose to phosphofructose allows the eventual split of the sugar into two three-carbon molecules).

Step 3. The third step is the phosphorylation of fructose-6-phosphate, catalyzed by the enzyme phosphofructokinase. A second ATP molecule donates a high-energy phosphate to fructose-6-phosphate, producing fructose-1,6-bisphosphate. In this pathway, phosphofructokinase is a rate-limiting enzyme. It is active when the concentration of ADP is high; it is less active when ADP levels are low and the concentration of ATP is high. Thus, if there is "sufficient" ATP in the system, the pathway slows down. This is a type of end-product inhibition, since ATP is the end product of glucose catabolism.

Step 4. The newly-added high-energy phosphates further destabilize fructose-1,6-bisphosphate. The fourth step in glycolysis employs an enzyme, aldolase, to cleave 1,6-bisphosphate into two three-carbon isomers: dihydroxyacetone-phosphate and glyceraldehyde-3-phosphate.

Step 5. In the fifth step, an isomerase transforms the dihydroxyacetone-phosphate into its isomer, glyceraldehyde-3-phosphate. Thus, the pathway will continue with two molecules of a single isomer. At this point in the pathway, there is a net investment of energy from two ATP molecules in the breakdown of one glucose molecule.

7.2C: The Energy-Releasing Steps of Glycolysis

In the second half of glycolysis, energy is released in the form of 4 ATP molecules and 2 NADH molecules.

Learning Objectives

- Outline the energy-releasing steps of glycolysis

Key Points

- The net energy release in glycolysis is a result of two molecules of glyceraldehyde-3- phosphate entering the second half of glycolysis where they are converted to pyruvic acid.
- Substrate -level phosphorylation, where a substrate of glycolysis donates a phosphate to ADP, occurs in two steps of the second-half of glycolysis to produce ATP.
- The availability of NAD+ is a limiting factor for the steps of glycolysis; when it is unavailable, the second half of glycolysis slows or shuts down.

Key Terms

- **NADH**: nicotinamide adenine dinucleotide (NAD) carrying two electrons and bonded with a hydrogen (H) ion; the reduced form of NAD

Second Half of Glycolysis (Energy-Releasing Steps)

So far, glycolysis has cost the cell two ATP molecules and produced two small, three-carbon sugar molecules. Both of these molecules will proceed through the second half of the pathway where sufficient energy will be extracted to pay back the two ATP molecules used as an initial investment while also producing a profit for the cell of two additional ATP molecules and two even higher-energy NADH molecules.

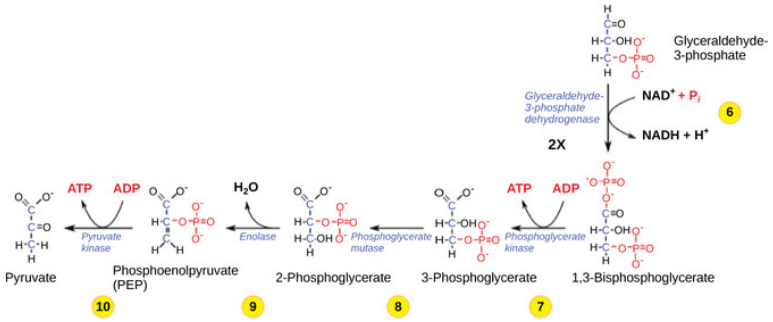

Figure 7.2C.1: **The second half of glycolysis: return on investment**: The second half of glycolysis involves phosphorylation without ATP investment (step 6) and produces two NADH and four ATP molecules per glucose.

Step 6. The sixth step in glycolysis oxidizes the sugar (glyceraldehyde-3-phosphate), extracting high-energy electrons, which are picked up by the electron carrier NAD^+, producing NADH. The sugar is then phosphorylated by the addition of a second phosphate group, producing 1,3-bisphosphoglycerate. Note that the second phosphate group does not require another ATP molecule.

Here, again, there is a potential limiting factor for this pathway. The continuation of the reaction depends upon the availability of the oxidized form of the electron carrier NAD^+. Thus, NADH must be continuously oxidized back into NAD^+ in order to keep this step going. If NAD^+ is not available, the second half of glycolysis slows down or stops. If oxygen is available in the system, the NADH will be oxidized readily, though indirectly, and the high-energy electrons from the hydrogen released in this process will be used to produce ATP. In an environment without oxygen, an alternate pathway (fermentation) can provide the oxidation of NADH to NAD^+.

Step 7. In the seventh step, catalyzed by phosphoglycerate kinase (an enzyme named for the reverse reaction), 1,3-bisphosphoglycerate donates a high-energy phosphate to ADP, forming one molecule of ATP. (This is an example of substrate-level phosphorylation.) A carbonyl group on the 1,3-bisphosphoglycerate is oxidized to a carboxyl group, and 3-phosphoglycerate is formed.

Step 8. In the eighth step, the remaining phosphate group in 3-phosphoglycerate moves from the third carbon to the second carbon, producing 2-phosphoglycerate (an isomer of 3-phosphoglycerate). The enzyme catalyzing this step is a mutase (isomerase).

Step 9. Enolase catalyzes the ninth step. This enzyme causes 2-phosphoglycerate to lose water from its structure; this is a dehydration reaction, resulting in the formation of a double bond that increases the potential energy in the remaining phosphate bond and produces phosphoenolpyruvate (PEP).

Step 10. The last step in glycolysis is catalyzed by the enzyme pyruvate kinase (the enzyme in this case is named for the reverse reaction of pyruvate's conversion into PEP) and results in the production of a second ATP molecule by substrate-level phosphorylation and the compound pyruvic acid (or its salt form, pyruvate). Many enzymes in enzymatic pathways are named for the reverse reactions since the enzyme can catalyze both forward and reverse reactions (these may have been described initially by the reverse reaction that takes place in vitro, under non-physiological conditions).

7.2D: Outcomes of Glycolysis

One glucose molecule produces four ATP, two NADH, and two pyruvate molecules during glycolysis.

Learning Objectives

- Describe the energy obtained from one molecule of glucose going through glycolysis

Key Points

- Although four ATP molecules are produced in the second half, the net gain of glycolysis is only two ATP because two ATP molecules are used in the first half of glycolysis.
- Enzymes that catalyze the reactions that produce ATP are rate-limiting steps of glycolysis and must be present in sufficient quantities for glycolysis to complete the production of four ATP, two NADH, and two pyruvate molecules for each glucose molecule that enters the pathway.
- Red blood cells require glycolysis as their sole source of ATP in order to survive, because they do not have mitochondria.
- Cancer cells and stem cells also use glycolysis as the main source of ATP (process known as aerobic glycolysis, or Warburg effect).

Key Terms

- **pyruvate**: any salt or ester of pyruvic acid; the end product of glycolysis before entering the TCA cycle

Outcomes of Glycolysis

Glycolysis starts with one molecule of glucose and ends with two pyruvate (pyruvic acid) molecules, a total of four ATP molecules, and two molecules of NADH. Two ATP molecules were used in the first half of the pathway to prepare the six-carbon ring for cleavage, so the cell has a net gain of two ATP molecules and 2 NADH molecules for its use. If the cell cannot catabolize the pyruvate molecules further (via the citric acid cycle or Krebs cycle), it will harvest only two ATP molecules from one molecule of glucose.

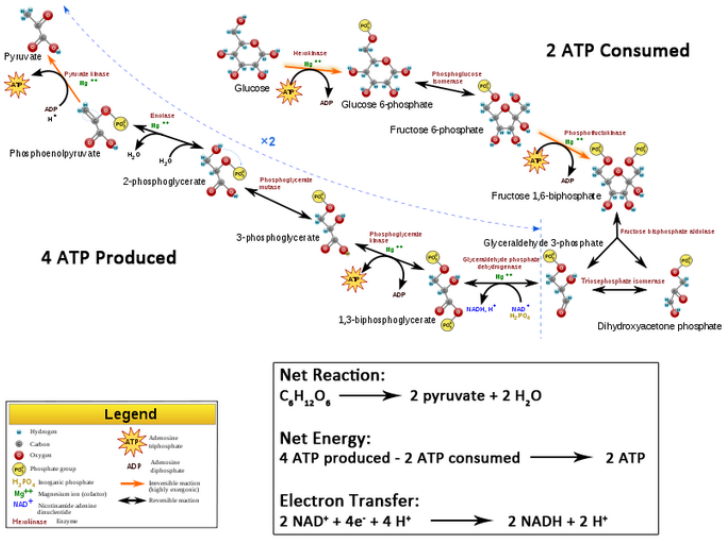

Figure 7.2*D*.1: **Glycolysis produces 2 ATP, 2 NADH, and 2 pyruvate molecules**: Glycolysis, or the aerobic catabolic breakdown of glucose, produces energy in the form of ATP, NADH, and pyruvate, which itself enters the citric acid cycle to produce more energy.

Mature mammalian red blood cells do not have mitochondria and are not capable of aerobic respiration, the process in which organisms convert energy in the presence of oxygen. Instead, glycolysis is their sole source of ATP. Therefore, if glycolysis is interrupted, the red blood cells lose their ability to maintain their sodium-potassium pumps, which require ATP to function, and eventually, they die. For example, since the second half of glycolysis (which produces the energy

molecules) slows or stops in the absence of NAD+, when NAD+ is unavailable, red blood cells will be unable to produce a sufficient amount of ATP in order to survive.

Additionally, the last step in glycolysis will not occur if pyruvate kinase, the enzyme that catalyzes the formation of pyruvate, is not available in sufficient quantities. In this situation, the entire glycolysis pathway will continue to proceed, but only two ATP molecules will be made in the second half (instead of the usual four ATP molecules). Thus, pyruvate kinase is a rate-limiting enzyme for glycolysis.

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Oxidation of Pyruvate and the Citric Acid Cycle

Topic hierarchy

- 7.3A: Breakdown of Pyruvate
- 7.3B: Acetyl CoA to CO2
- 7.3C: Citric Acid Cycle

The citric acid cycle is a series of reactions that produces two carbon dioxide molecules, one GTP/ATP, and reduced forms of NADH and FADH2.

7.3A: Breakdown of Pyruvate

After glycolysis, pyruvate is converted into acetyl CoA in order to enter the citric acid cycle.

Learning Objectives

- Explain why cells break down pyruvate

Key Points

- In the conversion of pyruvate to acetyl CoA, each pyruvate molecule loses one carbon atom with the release of carbon dioxide.
- During the breakdown of pyruvate, electrons are transferred to NAD+ to produce NADH, which will be used by the cell to produce ATP.
- In the final step of the breakdown of pyruvate, an acetyl group is transferred to Coenzyme A to produce acetyl CoA.

Key Terms

- **acetyl CoA**: a molecule that conveys the carbon atoms from glycolysis (pyruvate) to the citric acid cycle to be oxidized for energy production

Breakdown of Pyruvate

In order for pyruvate, the product of glycolysis, to enter the next pathway, it must undergo several changes to become acetyl Coenzyme A (acetyl CoA). Acetyl CoA is a molecule that is further converted to oxaloacetate, which enters the citric acid cycle (Krebs cycle). The conversion of pyruvate to acetyl CoA is a three-step process.

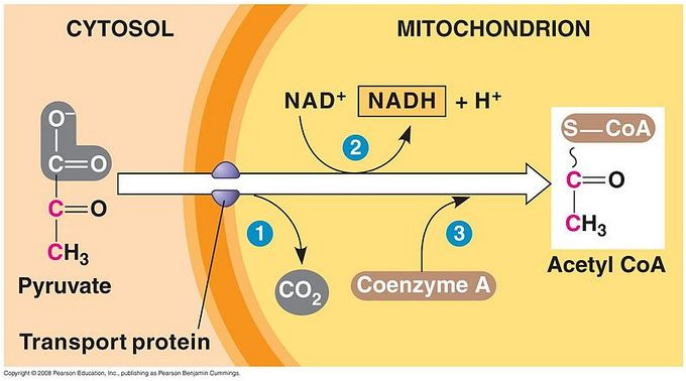

Figure 7.3A.1: **Breakdown of Pyruvate**: Each pyruvate molecule loses a carboxylic group in the form of carbon dioxide. The remaining two carbons are then transferred to the enzyme CoA to produce Acetyl CoA.

Step 1. A carboxyl group is removed from pyruvate, releasing a molecule of carbon dioxide into the surrounding medium. (Note: carbon dioxide is one carbon attached to two oxygen atoms and is one of the major end products of cellular respiration.) The result of this step is a two-carbon hydroxyethyl group bound to the enzyme pyruvate dehydrogenase; the lost carbon dioxide is the first of the six carbons from the original glucose molecule to be removed. This step proceeds twice for every molecule of glucose metabolized (remember: there are two pyruvate molecules produced at the end of glycolysis); thus, two of the six carbons will have been removed at the end of both of these steps.

Step 2. The hydroxyethyl group is oxidized to an acetyl group, and the electrons are picked up by NAD^+, forming NADH (the reduced form of NAD+). The high- energy electrons from NADH will be used later by the cell to generate ATP for energy.

Step 3. The enzyme-bound acetyl group is transferred to CoA, producing a molecule of acetyl CoA. This molecule of acetyl CoA is then further converted to be used in the next pathway of metabolism, the citric acid cycle.

7.3B: Acetyl CoA to CO2

The acetyl carbons of acetyl CoA are released as carbon dioxide in the citric acid cycle.

Learning Objectives

- Describe the fate of the acetyl CoA carbons in the citric acid cycle

Key Points

- The citric acid cycle is also known as the Krebs cycle or the TCA (tricarboxylic acid) cycle.
- Acetyl CoA transfers its acetyl group to oxaloacetate to form citrate and begin the citric acid cycle.
- The release of carbon dioxide is coupled with the reduction of NAD^+ to NADH in the citric acid cycle.

Key Terms

- **TCA cycle**: an alternative name for the Krebs cycle or citric acid cycle
- **Krebs cycle**: a series of enzymatic reactions that occurs in all aerobic organisms; it involves the oxidative metabolism of acetyl units and serves as the main source of cellular energy
- **oxaloacetate**: a four carbon molecule that receives an acetyl group from acetyl CoA to form citrate, which enters the citric acid cycle

Acetyl CoA to CO_2

Acetyl CoA links glycolysis and pyruvate oxidation with the citric acid cycle. In the presence of oxygen, acetyl CoA delivers its acetyl group to a four-carbon molecule, oxaloacetate, to form citrate, a six-carbon molecule with three carboxyl groups. During this first step of the citric acid cycle, the CoA enzyme, which contains a sulfhydryl group (-SH), is recycled and becomes available to attach another acetyl group. The citrate will then harvest the remainder of the extractable energy from what began as a glucose molecule and continue through the citric acid cycle.

In the citric acid cycle, the two carbons that were originally the acetyl group of acetyl CoA are released as carbon dioxide, one of the major products of cellular respiration, through a series of enzymatic reactions. For each acetyl CoA that enters the citric acid cycle, two carbon dioxide molecules are released in reactions that are coupled with the production of NADH molecules from the reduction of NAD^+ molecules.

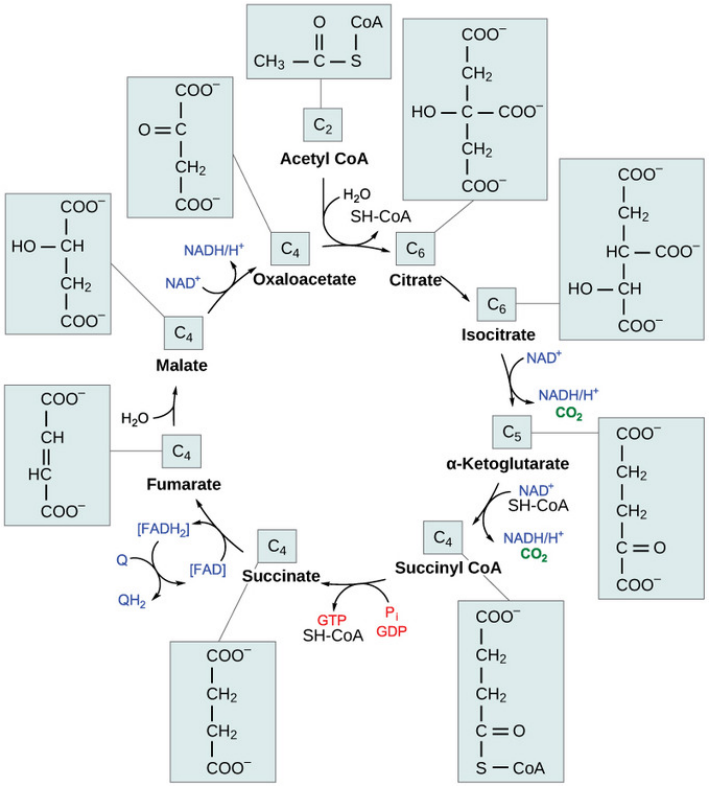

Figure 7.3B.1: **Acetyl CoA and the Citric Acid Cycle**: For each molecule of acetyl CoA that enters the citric acid cycle, two carbon dioxide molecules are released, removing the carbons from the acetyl group.

In addition to the citric acid cycle, named for the first intermediate formed, citric acid, or citrate, when acetate joins to the oxaloacetate, the cycle is also known by two other names. The TCA cycle is named for tricarboxylic acids (TCA) because citric acid (or citrate) and isocitrate, the first two intermediates that are formed, are tricarboxylic acids. Additionally, the cycle is known as the Krebs cycle, named after Hans Krebs, who first identified the steps in the pathway in the 1930s in pigeon flight muscle.

7.3C: Citric Acid Cycle

The citric acid cycle is a series of reactions that produces two carbon dioxide molecules, one GTP/ATP, and reduced forms of NADH and FADH2.

Learning Objectives

- List the steps of the Krebs (or citric acid) cycle

Key Points

- The four-carbon molecule, oxaloacetate, that began the cycle is regenerated after the eight steps of the citric acid cycle.
- The eight steps of the citric acid cycle are a series of redox, dehydration, hydration, and decarboxylation reactions.
- Each turn of the cycle forms one GTP or ATP as well as three NADH molecules and one FADH2 molecule, which will be used in further steps of cellular respiration to produce ATP for the cell.

Key Terms

- **citric acid cycle**: a series of chemical reactions used by all aerobic organisms to generate energy through the oxidization of acetate derived from carbohydrates, fats, and proteins into carbon dioxide
- **Krebs cycle**: a series of enzymatic reactions that occurs in all aerobic organisms; it involves the oxidative metabolism of acetyl units and serves as the main source of cellular energy
- **mitochondria**: in cell biology, a mitochondrion (plural mitochondria) is a membrane-enclosed organelle, often described as “cellular power plants” because they generate most of the ATP

Citric Acid Cycle (Krebs Cycle)

Like the conversion of pyruvate to acetyl CoA, the citric acid cycle takes place in the matrix of the mitochondria. Almost all of the enzymes of the citric acid cycle are soluble, with the single exception of the enzyme succinate dehydrogenase, which is embedded in the inner membrane of the mitochondrion. Unlike glycolysis, the citric acid cycle is a closed loop: the last part of the pathway regenerates the compound used in the first step. The eight steps of the cycle are a series of redox, dehydration, hydration, and decarboxylation reactions that produce two carbon dioxide molecules, one GTP/ATP, and reduced forms of NADH and FADH2. This is considered an aerobic pathway because the NADH and FADH2 produced must transfer their electrons to the next pathway in the system, which will use oxygen. If this transfer does not occur, the oxidation steps of the citric acid cycle also do not occur. Note that the citric acid cycle produces very little ATP directly and does not directly consume oxygen.

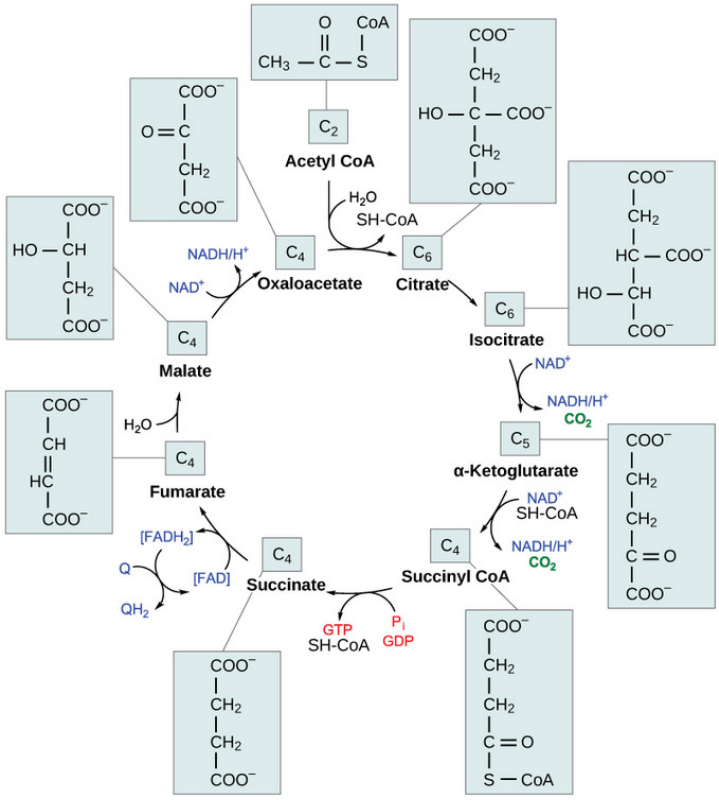

Figure 7.3C.1: **The citric acid cycle**: In the citric acid cycle, the acetyl group from acetyl CoA is attached to a four-carbon oxaloacetate molecule to form a six-carbon citrate molecule. Through a series of steps, citrate is oxidized, releasing two carbon dioxide molecules for each acetyl group fed into the cycle. In the process, three NAD+ molecules are reduced to NADH, one FAD molecule is reduced to FADH2, and one ATP or GTP (depending on the cell type) is produced (by substrate-level phosphorylation). Because the final product of the citric acid cycle is also the first reactant, the cycle runs continuously in the presence of sufficient reactants.

Steps in the Citric Acid Cycle

Step 1. The first step is a condensation step, combining the two-carbon acetyl group (from acetyl CoA) with a four-carbon oxaloacetate molecule to form a six-carbon molecule of citrate. CoA is bound to a sulfhydryl group (-SH) and diffuses away to eventually combine with another acetyl group. This step is irreversible because it is highly exergonic. The rate of this reaction is controlled by negative feedback and the amount of ATP available. If ATP levels increase, the rate of this reaction decreases. If ATP is in short supply, the rate increases.

Step 2. Citrate loses one water molecule and gains another as citrate is converted into its isomer, isocitrate.

Steps 3 and 4. In step three, isocitrate is oxidized, producing a five-carbon molecule, α-ketoglutarate, together with a molecule of $CO_2$and two electrons, which reduce NAD+ to NADH. This step is also regulated by negative feedback from ATP and NADH and by a positive effect of ADP. Steps three and four are both oxidation and decarboxylation steps, which release electrons that reduce NAD^+ to NADH and release carboxyl groups that form CO_2 molecules. α-Ketoglutarate is the product of step three, and a succinyl group is the product of step four. CoA binds the succinyl group to form succinyl CoA. The enzyme that catalyzes step four is regulated by feedback inhibition of ATP, succinyl CoA, and NADH.

Step 5. A phosphate group is substituted for coenzyme A, and a high- energy bond is formed. This energy is used in substrate-level phosphorylation (during the conversion of the succinyl group to succinate) to form either guanine triphosphate (GTP) or ATP. There are two forms of the enzyme, called isoenzymes, for this step, depending upon the type of animal tissue in which they are found. One form is found in tissues that use large amounts of ATP, such as heart and skeletal muscle. This form produces ATP. The second form of the enzyme is found in tissues that have a high number of anabolic pathways, such as liver. This form produces GTP. GTP is energetically equivalent to ATP; however, its use is more restricted. In particular, protein synthesis primarily uses GTP.

Step 6. Step six is a dehydration process that converts succinate into fumarate. Two hydrogen atoms are transferred to FAD, producing $FADH_2$. The energy contained in the electrons of these atoms is insufficient to reduce NAD^+ but adequate to reduce FAD. Unlike NADH, this carrier remains attached to the enzyme and transfers the electrons to the electron transport chain directly. This process is made possible by the localization of the enzyme catalyzing this step inside the inner membrane of the mitochondrion.

Step 7. Water is added to fumarate during step seven, and malate is produced. The last step in the citric acid cycle regenerates oxaloacetate by oxidizing malate. Another molecule of NADH is produced.

Products of the Citric Acid Cycle

Two carbon atoms come into the citric acid cycle from each acetyl group, representing four out of the six carbons of one glucose molecule. Two carbon dioxide molecules are released on each turn of the cycle; however, these do not necessarily contain the most recently-added carbon atoms. The two acetyl carbon atoms will eventually be released on later turns of the cycle; thus, all six carbon atoms from the original glucose molecule are eventually incorporated into carbon dioxide. Each turn of the cycle forms three NADH molecules and one $FADH_2$ molecule. These carriers will connect with the last portion of aerobic respiration to produce ATP molecules. One GTP or ATP is also made in each cycle. Several of the intermediate compounds in the citric acid cycle can be used in synthesizing non-essential amino acids; therefore, the cycle is amphibolic (both catabolic and anabolic).

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7.4: Oxidative Phosphorylation

7.4A: Electron Transport Chain

The electron transport chain uses the electrons from electron carriers to create a chemical gradient that can be used to power oxidative phosphorylation.

7.4B: Chemiosmosis and Oxidative Phosphorylation

Chemiosmosis is the movement of ions across a selectively permeable membrane, down their electrochemical gradient.

7.4C: ATP Yield

7.4A: Electron Transport Chain

The electron transport chain uses the electrons from electron carriers to create a chemical gradient that can be used to power oxidative phosphorylation.

Learning Objectives

- Describe how electrons move through the electron transport chain

Key Points

- Oxidative phosphorylation is the metabolic pathway in which electrons are transferred from electron donors to electron acceptors in redox reactions; this series of reactions releases energy which is used to form ATP.
- There are four protein complexes (labeled complex I-IV) in the electron transport chain, which are involved in moving electrons from NADH and $FADH_2$ to molecular oxygen.
- Complex I establishes the hydrogen ion gradient by pumping four hydrogen ions across the membrane from the matrix into the intermembrane space.
- Complex II receives $FADH_2$, which bypasses complex I, and delivers electrons directly to the electron transport chain.
- Ubiquinone (Q) accepts the electrons from both complex I and complex II and delivers them to complex III.
- Complex III pumps protons through the membrane and passes its electrons to cytochrome c for transport to the fourth complex of proteins and enzymes.
- Complex IV reduces oxygen; the reduced oxygen then picks up two hydrogen ions from the surrounding medium to make water.

Key Terms

- **prosthetic group**: The non-protein component of a conjugated protein.
- **complex**: A structure consisting of a central atom, molecule, or protein weakly connected to surrounding atoms, molecules, or proteins.
- **ubiquinone**: A lipid soluble substance that is a component of the electron transport chain and accepts electrons from complexes I and II.

Oxidative phosphorylation is a highly efficient method of producing large amounts of ATP, the basic unit of energy for metabolic processes. During this process electrons are exchanged between molecules, which creates a chemical gradient that allows for the production of ATP. The most vital part of this process is the electron transport chain, which produces more ATP than any other part of cellular respiration.

Electron Transport Chain

The electron transport chain is the final component of aerobic respiration and is the only part of glucose metabolism that uses atmospheric oxygen. Electron transport is a series of redox reactions that resemble a relay race. Electrons are passed rapidly from one component to the next to the endpoint of the chain, where the electrons reduce molecular oxygen, producing water. This requirement for oxygen in the final stages of the chain can be seen in the overall equation for cellular respiration, which requires both glucose and oxygen.

A complex is a structure consisting of a central atom, molecule, or protein weakly connected to surrounding atoms, molecules, or proteins. The electron transport chain is an aggregation of four of these complexes (labeled I through IV), together with associated mobile electron carriers. The electron transport chain is present in multiple copies in the inner mitochondrial membrane of eukaryotes and the plasma membrane of prokaryotes.

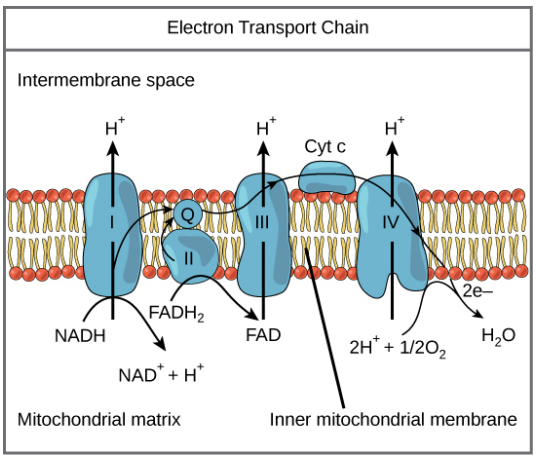

Figure 7.4*A*. 1: **The electron transport chain**: The electron transport chain is a series of electron transporters embedded in the inner mitochondrial membrane that shuttles electrons from NADH and $FADH_2$ to molecular oxygen. In the process, protons are pumped from the mitochondrial matrix to the intermembrane space, and oxygen is reduced to form water.

Complex I

To start, two electrons are carried to the first complex aboard NADH. Complex I is composed of flavin mononucleotide (FMN) and an enzyme containing iron-sulfur (Fe-S). FMN, which is derived from vitamin B_2 (also called riboflavin), is one of several prosthetic groups or co-factors in the electron transport chain. A prosthetic group is a non-protein molecule required for the activity of a protein. Prosthetic groups can be organic or inorganic and are non-peptide molecules bound to a protein that facilitate its function.

Prosthetic groups include co-enzymes, which are the prosthetic groups of enzymes. The enzyme in complex I is NADH dehydrogenase, a very large protein containing 45 amino acid chains. Complex I can pump four hydrogen ions across the membrane from the matrix into the intermembrane space; it is in this way that the hydrogen ion gradient is established and maintained between the two compartments separated by the inner mitochondrial membrane.

Q and Complex II

Complex II directly receives $FADH_2$, which does not pass through complex I. The compound connecting the first and second complexes to the third is ubiquinone (Q). The Q molecule is lipid soluble and freely moves through the hydrophobic core of the membrane. Once it is reduced to QH_2, ubiquinone delivers its electrons to the next complex in the electron transport chain. Q receives the electrons derived from NADH from complex I and the electrons derived from $FADH_2$ from complex II, including succinate dehydrogenase. This enzyme and $FADH_2$ form a small complex that delivers electrons directly to the electron transport chain, bypassing the first complex. Since these electrons bypass, and thus do not energize, the proton pump in the first complex, fewer ATP molecules are made from the $FADH_2$ electrons. The number of ATP molecules ultimately obtained is directly proportional to the number of protons pumped across the inner mitochondrial membrane.

Complex III

The third complex is composed of cytochrome b, another Fe-S protein, Rieske center (2Fe-2S center), and cytochrome c proteins; this complex is also called cytochrome oxidoreductase. Cytochrome proteins have a prosthetic heme group. The heme molecule is similar to the heme in hemoglobin, but it carries electrons, not oxygen. As a result, the iron ion at its core is reduced and oxidized as it passes the electrons, fluctuating between different oxidation states: Fe^{2+} (reduced) and Fe^{3+} (oxidized). The heme molecules in the cytochromes have slightly different characteristics due to the effects of the different proteins binding them, which makes each complex. Complex III pumps protons through the membrane and passes its electrons to cytochrome c for transport to the fourth complex of proteins and enzymes. Cytochrome c is the acceptor of electrons from Q; however, whereas Q carries pairs of electrons, cytochrome c can accept only one at a time.

Complex IV

The fourth complex is composed of cytochrome proteins c, a, and a_3. This complex contains two heme groups (one in each of the cytochromes a and a_3) and three copper ions (a pair of Cu_A and one Cu_B in cytochrome a_3). The cytochromes hold an oxygen molecule very tightly between the iron and copper ions until the oxygen is completely reduced. The reduced oxygen then picks up two hydrogen ions from the surrounding medium to produce water (H_2O). The removal of the hydrogen ions from the system also contributes to the ion gradient used in the process of chemiosmosis.

: Chemiosmosis and Oxidative Phosphorylation

Learning Objectives

- Describe how the energy obtained from the electron transport chain powers chemiosmosis and discuss the role of hydrogen ions in the synthesis of ATP

During chemiosmosis, electron carriers like NADH and FADH donate electrons to the electron transport chain. The electrons cause conformation changes in the shapes of the proteins to pump H+ across a selectively permeable cell membrane. The uneven distribution of H^+ ions across the membrane establishes both concentration and electrical gradients (thus, an electrochemical gradient) owing to the hydrogen ions' positive charge and their aggregation on one side of the membrane.

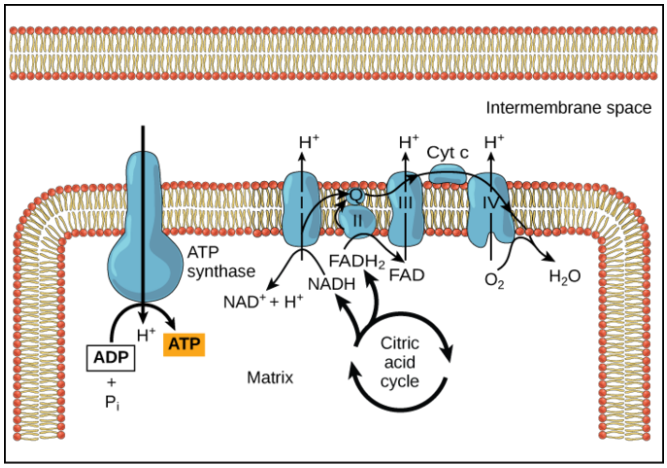

Figure 7.4B.1: **Chemiosmosis**: In oxidative phosphorylation, the hydrogen ion gradient formed by the electron transport chain is used by ATP synthase to form ATP.

If the membrane were open to diffusion by the hydrogen ions, the ions would tend to spontaneously diffuse back across into the matrix, driven by their electrochemical gradient. However, many ions cannot diffuse through the nonpolar regions of phospholipid membranes without the aid of ion channels. Similarly, hydrogen ions in the matrix space can only pass through the inner mitochondrial membrane through a membrane protein called ATP synthase. This protein acts as a tiny generator turned by the force of the hydrogen ions diffusing through it, down their electrochemical gradient. The turning of this molecular machine harnesses the potential energy stored in the hydrogen ion gradient to add a phosphate to ADP, forming ATP.

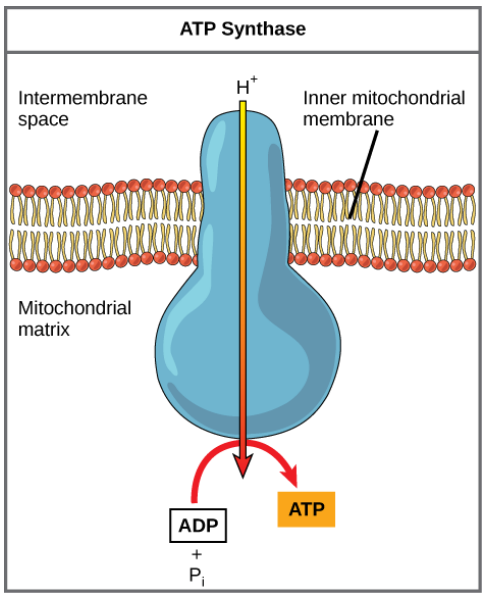

Figure 7.4*B*.1: **ATP Synthase**: ATP synthase is a complex, molecular machine that uses a proton (H+) gradient to form ATP from ADP and inorganic phosphate (Pi).

Chemiosmosis is used to generate 90 percent of the ATP made during aerobic glucose catabolism. The production of ATP using the process of chemiosmosis in mitochondria is called oxidative phosphorylation. It is also the method used in the light reactions of photosynthesis to harness the energy of sunlight in the process of photophosphorylation. The overall result of these reactions is the production of ATP from the energy of the electrons removed from hydrogen atoms. These atoms were originally part of a glucose molecule. At the end of the pathway, the electrons are used to reduce an oxygen molecule to oxygen ions. The extra electrons on the oxygen attract hydrogen ions (protons) from the surrounding medium and water is formed.

Key Points

- During chemiosmosis, the free energy from the series of reactions that make up the electron transport chain is used to pump hydrogen ions across the membrane, establishing an electrochemical gradient.
- Hydrogen ions in the matrix space can only pass through the inner mitochondrial membrane through a membrane protein called ATP synthase.
- As protons move through ATP synthase, ADP is turned into ATP.
- The production of ATP using the process of chemiosmosis in mitochondria is called oxidative phosphorylation.

Key Terms

- **ATP synthase**: An important enzyme that provides energy for the cell to use through the synthesis of adenosine triphosphate (ATP).
- **oxidative phosphorylation**: A metabolic pathway that uses energy released by the oxidation of nutrients to produce adenosine triphosphate (ATP).
- **chemiosmosis**: The movement of ions across a selectively permeable membrane, down their electrochemical gradient.

7.4C: ATP Yield

ese atoms were originally part of a glucose molecule. At the end of the pathway, the electrons are used to reduce an oxygen molecule to oxygen ions. The extra electrons on the oxygen attract hydrogen ions (protons) from the surrounding medium and water is formed.

ATP Yield

The amount of energy (as ATP) gained from glucose catabolism varies across species and depends on other related cellular processes.

LEARNING OBJECTIVES

Describe the origins of variability in the amount of ATP that is produced per molecule of glucose consumed

KEY TAKEAWAYS

Key Points

- While glucose catabolism always produces energy, the amount of energy (in terms of ATP equivalents) produced can vary, especially across different species.
- The number of hydrogen ions the electron transport chain complexes can pump through the membrane varies between species.
- NAD^+ provides more ATP than FAD^+ in the electron transport chain and can lead to variance in ATP production.
- The use of intermediates from glucose catabolism in other biosynthetic pathways, such as amino acid synthesis, can lower the yield of ATP.

Key Terms

- **catabolism**: Destructive metabolism, usually including the release of energy and breakdown of materials.

ATP Yield

In a eukaryotic cell, the process of cellular respiration can metabolize one molecule of glucose into 30 to 32 ATP. The process of glycolysis only produces two ATP, while all the rest are produced during the electron transport chain. Clearly, the electron transport chain is vastly more efficient, but it can only be carried out in the presence of oxygen.

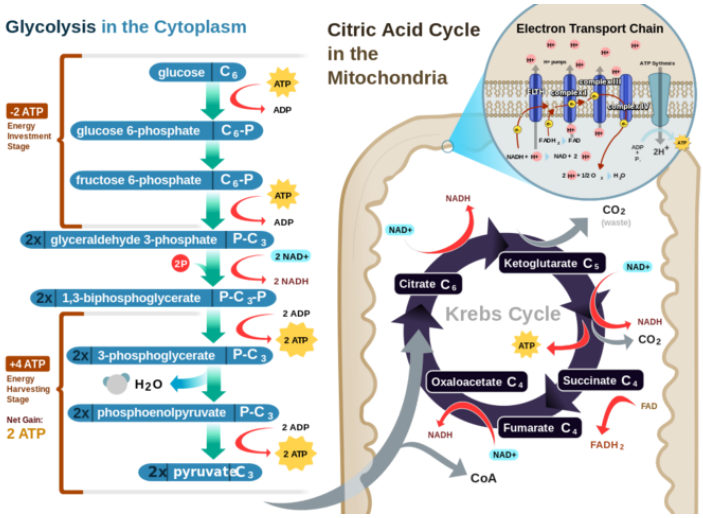

Figure 7.4C.1: **Cellular respiration in a eukaryotic cell**: Glycolysis on the left portion of this illustration can be seen to yield 2 ATP molecules, while the Electron Transport Chain portion at the upper right will yield the remaining 30-32 ATP molecules under the presence of oxygen.

The number of ATP molecules generated via the catabolism of glucose can vary substantially. For example, the number of hydrogen ions the electron transport chain complexes can pump through the membrane varies between species. Another source of variance occurs during the shuttle of electrons across the membranes of the mitochondria. The NADH generated

from glycolysis cannot easily enter mitochondria. Thus, electrons are picked up on the inside of mitochondria by either NAD^+ or FAD^+. These FAD^+ molecules can transport fewer ions; consequently, fewer ATP molecules are generated when FAD^+ acts as a carrier. NAD^+ is used as the electron transporter in the liver, and FAD^+ acts in the brain.

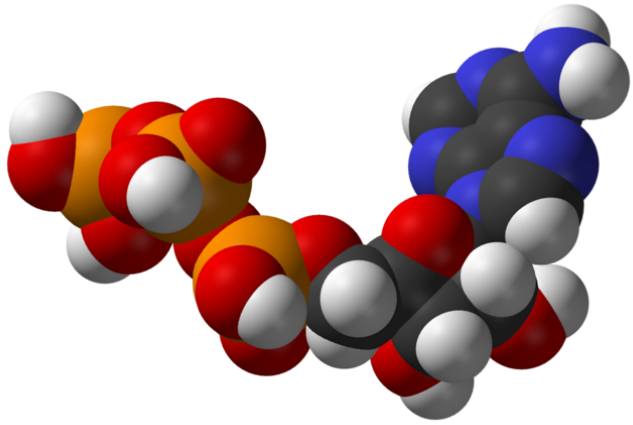

Figure 7.4C.1: **Adenosine triphosphate**: ATP is the main source of energy in many living organisms.

Another factor that affects the yield of ATP molecules generated from glucose is the fact that intermediate compounds in these pathways are used for other purposes. Glucose catabolism connects with the pathways that build or break down all other biochemical compounds in cells, but the result is not always ideal. For example, sugars other than glucose are fed into the glycolytic pathway for energy extraction. Moreover, the five-carbon sugars that form nucleic acids are made from intermediates in glycolysis. Certain nonessential amino acids can be made from intermediates of both glycolysis and the citric acid cycle. Lipids, such as cholesterol and triglycerides, are also made from intermediates in these pathways, and both amino acids and triglycerides are broken down for energy through these pathways. Overall, in living systems, these pathways of glucose catabolism extract about 34 percent of the energy contained in glucose.

Metabolism without Oxygen

Topic hierarchy

- 7.5A: Anaerobic Cellular Respiration

7.5A: Anaerobic Cellular Respiration

Some prokaryotes and eukaryotes use anaerobic respiration in which they can create energy for use in the absence of oxygen.

Learning Objectives

- Describe the process of anaerobic cellular respiration.

Key Points

- Anaerobic respiration is a type of respiration where oxygen is not used; instead, organic or inorganic molecules are used as final electron acceptors.
- Fermentation includes processes that use an organic molecule to regenerate NAD^+ from NADH.
- Types of fermentation include lactic acid fermentation and alcohol fermentation, in which ethanol is produced.
- All forms of fermentation except lactic acid fermentation produce gas, which plays a role in the laboratory identification of bacteria.
- Some types of prokaryotes are facultatively anaerobic, which means that they can switch between aerobic respiration and fermentation, depending on the availability of oxygen.

Key Terms

- **archaea**: A group of single-celled microorganisms. They have no cell nucleus or any other membrane-bound organelles within their cells.
- **anaerobic respiration**: A form of respiration using electron acceptors other than oxygen.
- **fermentation**: An anaerobic biochemical reaction. When this reaction occurs in yeast, enzymes catalyze the conversion of sugars to alcohol or acetic acid with the evolution of carbon dioxide.

Anaerobic Cellular Respiration

The production of energy requires oxygen. The electron transport chain, where the majority of ATP is formed, requires a large input of oxygen. However, many organisms have developed strategies to carry out metabolism without oxygen, or can switch from aerobic to anaerobic cell respiration when oxygen is scarce.

During cellular respiration, some living systems use an organic molecule as the final electron acceptor. Processes that use an organic molecule to regenerate NAD^+ from NADH are collectively referred to as fermentation. In contrast, some living systems use an inorganic molecule as a final electron acceptor. Both methods are called anaerobic cellular respiration, where organisms convert energy for their use in the absence of oxygen.

Certain prokaryotes, including some species of bacteria and archaea, use anaerobic respiration. For example, the group of archaea called methanogens reduces carbon dioxide to methane to oxidize NADH. These microorganisms are found in soil and in the digestive tracts of ruminants, such as cows and sheep. Similarly, sulfate-reducing bacteria and archaea, most of which are anaerobic, reduce sulfate to hydrogen sulfide to regenerate NAD^+ from NADH.

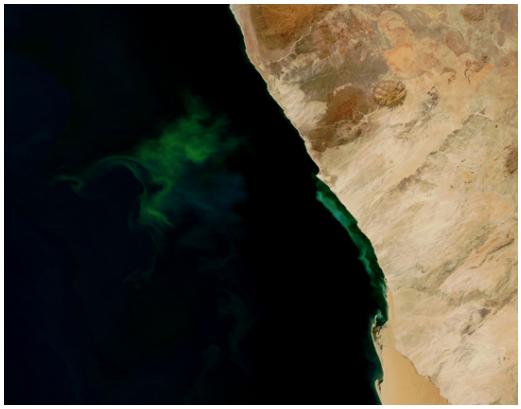

Figure 7.5A.1: **Anaerobic bacteria**: The green color seen in these coastal waters is from an eruption of hydrogen sulfide-producing bacteria. These anaerobic, sulfate-reducing bacteria release hydrogen sulfide gas as they decompose algae in the water.

Eukaryotes can also undergo anaerobic respiration. Some examples include alcohol fermentation in yeast and lactic acid fermentation in mammals.

Lactic Acid Fermentation

The fermentation method used by animals and certain bacteria (like those in yogurt) is called lactic acid fermentation. This type of fermentation is used routinely in mammalian red blood cells and in skeletal muscle that has an insufficient oxygen supply to allow aerobic respiration to continue (that is, in muscles used to the point of fatigue). The excess amount of lactate in those muscles is what causes the burning sensation in your legs while running. This pain is a signal to rest the overworked muscles so they can recover. In these muscles, lactic acid accumulation must be removed by the blood circulation and the lactate brought to the liver for further metabolism. The chemical reactions of lactic acid fermentation are the following:

Pyruvic acid + NADH ↔ lactic acid + NAD^+

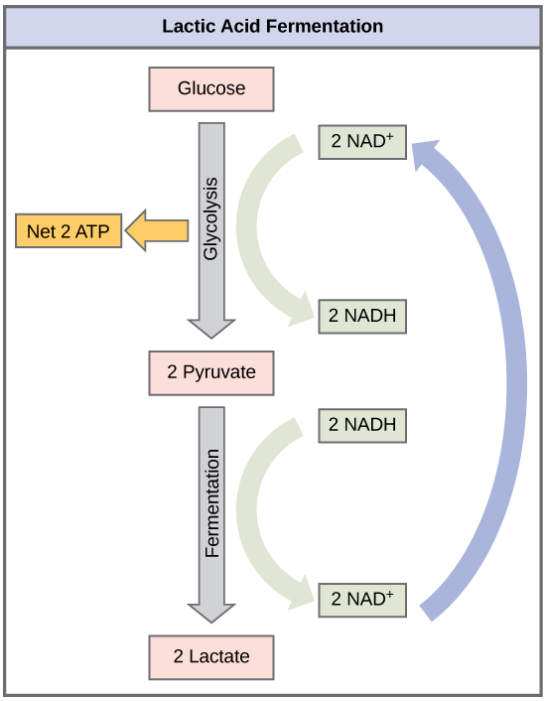

Figure 7.5A.1: **Lactic acid fermentation**: Lactic acid fermentation is common in muscle cells that have run out of oxygen.

The enzyme used in this reaction is lactate dehydrogenase (LDH). The reaction can proceed in either direction, but the reaction from left to right is inhibited by acidic conditions. Such lactic acid accumulation was once believed to cause

muscle stiffness, fatigue, and soreness, although more recent research disputes this hypothesis. Once the lactic acid has been removed from the muscle and circulated to the liver, it can be reconverted into pyruvic acid and further catabolized for energy.

Alcohol Fermentation

Another familiar fermentation process is alcohol fermentation, which produces ethanol, an alcohol. The use of alcohol fermentation can be traced back in history for thousands of years. The chemical reactions of alcoholic fermentation are the following (Note: CO_2 does not participate in the second reaction):

Pyruvic acid → CO2 + acetaldehyde + NADH → ethanol + NAD^+

Figure 7.5*A*. 1: **Alcohol Fermentation**: Fermentation of grape juice into wine produces CO2 as a byproduct. Fermentation tanks have valves so that the pressure inside the tanks created by the carbon dioxide produced can be released.

The first reaction is catalyzed by pyruvate decarboxylase, a cytoplasmic enzyme, with a coenzyme of thiamine pyrophosphate (TPP, derived from vitamin B_1 and also called thiamine). A carboxyl group is removed from pyruvic acid, releasing carbon dioxide as a gas. The loss of carbon dioxide reduces the size of the molecule by one carbon, making acetaldehyde. The second reaction is catalyzed by alcohol dehydrogenase to oxidize NADH to NAD^+ and reduce acetaldehyde to ethanol.

The fermentation of pyruvic acid by yeast produces the ethanol found in alcoholic beverages. Ethanol tolerance of yeast is variable, ranging from about 5 percent to 21 percent, depending on the yeast strain and environmental conditions.

Other Types of Fermentation

Various methods of fermentation are used by assorted organisms to ensure an adequate supply of NAD^+ for the sixth step in glycolysis. Without these pathways, that step would not occur and no ATP would be harvested from the breakdown of glucose.Other fermentation methods also occur in bacteria. Many prokaryotes are facultatively anaerobic. This means that they can switch between aerobic respiration and fermentation, depending on the availability of oxygen. Certain prokaryotes, like *Clostridia*, are obligate anaerobes. Obligate anaerobes live and grow in the absence of molecular oxygen. Oxygen is a poison to these microorganisms, killing them on exposure.

It should be noted that all forms of fermentation, except lactic acid fermentation, produce gas. The production of particular types of gas is used as an indicator of the fermentation of specific carbohydrates, which plays a role in the laboratory identification of the bacteria.

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7.6: Connections of Carbohydrate, Protein, and Lipid Metabolic Pathways

Topic hierarchy

- 7.6A: Connecting Other Sugars to Glucose Metabolism
- 7.6B: Connecting Proteins to Glucose Metabolism
- 7.6C: Connecting Lipids to Glucose Metabolism

: Connecting Other Sugars to Glucose Metabolism

Sugars, such as galactose, fructose, and glycogen, are catabolized into new products in order to enter the glycolytic pathway.

Learning Objectives

- Identify the types of sugars involved in glucose metabolism

Key Points

- When blood sugar levels drop, glycogen is broken down into glucose -1-phosphate, which is then converted to glucose-6-phosphate and enters glycolysis for ATP production.
- In the liver, galactose is converted to glucose-6-phosphate in order to enter the glycolytic pathway.
- Fructose is converted into glycogen in the liver and then follows the same pathway as glycogen to enter glycolysis.
- Sucrose is broken down into glucose and fructose; glucose enters the pathway directly while fructose is converted to glycogen.

Key Terms

- **disaccharide**: A sugar, such as sucrose, maltose, or lactose, consisting of two monosaccharides combined together.
- **glycogen**: A polysaccharide that is the main form of carbohydrate storage in animals; converted to glucose as needed.
- **monosaccharide**: A simple sugar such as glucose, fructose, or deoxyribose that has a single ring.

You have learned about the catabolism of glucose, which provides energy to living cells. But living things consume more than glucose for food. How does a turkey sandwich end up as ATP in your cells? This happens because all of the catabolic pathways for carbohydrates, proteins, and lipids eventually connect into glycolysis and the citric acid cycle pathways.

Metabolic pathways should be thought of as porous; that is, substances enter from other pathways, and intermediates leave for other pathways. These pathways are not closed systems. Many of the substrates, intermediates, and products in a particular pathway are reactants in other pathways. Like sugars and amino acids, the catabolic pathways of lipids are also connected to the glucose catabolism pathways.

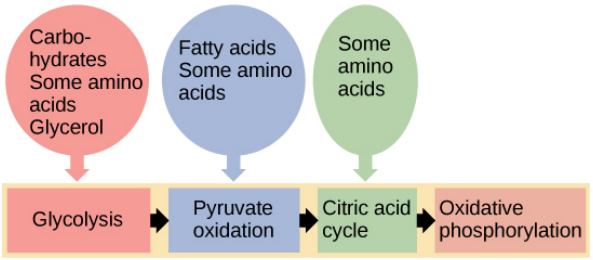

Figure 7.6A.1: **Glycogen Pathway**: Glycogen from the liver and muscles, hydrolyzed into glucose-1-phosphate, together with fats and proteins, can feed into the catabolic pathways for carbohydrates.

Glycogen, a polymer of glucose, is an energy-storage molecule in animals. When there is adequate ATP present, excess glucose is shunted into glycogen for storage. Glycogen is made and stored in both the liver and muscles. The glycogen is hydrolyzed into the glucose monomer, glucose-1-phosphate (G-1-P), if blood sugar levels drop. The presence of glycogen as a source of glucose allows ATP to be produced for a longer period of time during exercise. Glycogen is broken down into G-1-P and converted into glucose-6-phosphate (G-6-P) in both muscle and liver cells; this product enters the glycolytic pathway.

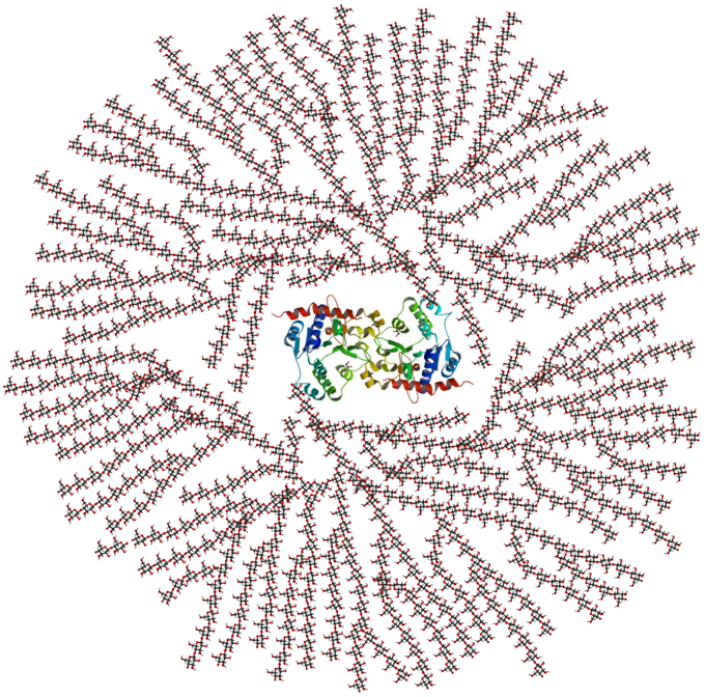

Figure 7.6A.1: **Glycogen Structure**: Schematic two-dimensional cross-sectional view of glycogen: A core protein of glycogenin is surrounded by branches of glucose units. The entire globular granule may contain around 30,000 glucose units.

Galactose is the sugar in milk. Infants have an enzyme in the small intestine that metabolizes lactose to galactose and glucose. In areas where milk products are regularly consumed, adults have also evolved this enzyme. Galactose is converted in the liver to G-6-P and can thus enter the glycolytic pathway.

Fructose is one of the three dietary monosaccharides (along with glucose and galactose) which are absorbed directly into the bloodstream during digestion. Fructose is absorbed from the small intestine and then passes to the liver to be metabolized, primarily to glycogen. The catabolism of both fructose and galactose produces the same number of ATP molecules as glucose.

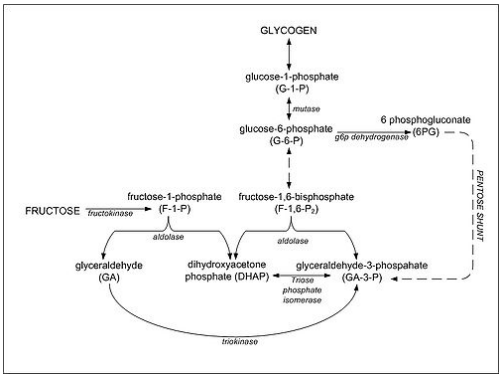

Figure 7.6A.1: **Fructose Metabolism**: Although the metabolism of fructose and glucose share many of the same intermediate structures, they have very different metabolic fates in human metabolism.

Sucrose is a disaccharide with a molecule of glucose and a molecule of fructose bonded together with a glycosidic linkage. The catabolism of sucrose breaks it down to monomers of glucose and fructose. The glucose can directly enter the glycolytic pathway while fructose must first be converted to glycogen, which can be broken down to G-1-P and enter the glycolytic pathway as described above.

7.6B: Connecting Proteins to Glucose Metabolism

Excess amino acids are converted into molecules that can enter the pathways of glucose catabolism.

Learning Objectives

- Describe the role played by proteins in glucose metabolism

Key Points

- Amino acids must be deaminated before entering any of the pathways of glucose catabolism: the amino group is converted to ammonia, which is used by the liver in the synthesis of urea.
- Deaminated amino acids can be converted into pyruvate, acetyl CoA, or some components of the citric acid cycle to enter the pathways of glucose catabolism.
- Several amino acids can enter the glucose catabolism pathways at multiple locations.

Key Terms

- **catabolism**: Destructive metabolism, usually including the release of energy and breakdown of materials.
- **keto acid**: Any carboxylic acid that also contains a ketone group.
- **deamination**: The removal of an amino group from a compound.

Metabolic pathways should be thought of as porous; that is, substances enter from other pathways and intermediates leave for other pathways. These pathways are not closed systems. Many of the substrates, intermediates, and products in a particular pathway are reactants in other pathways. Proteins are a good example of this phenomenon. They can be broken down into their constituent amino acids and used at various steps of the pathway of glucose catabolism.

Proteins are hydrolyzed by a variety of enzymes in cells. Most of the time, the amino acids are recycled into the synthesis of new proteins or are used as precursors in the synthesis of other important biological molecules, such as hormones, nucleotides, or neurotransmitters. However, if there are excess amino acids, or if the body is in a state of starvation, some amino acids will be shunted into the pathways of glucose catabolism.

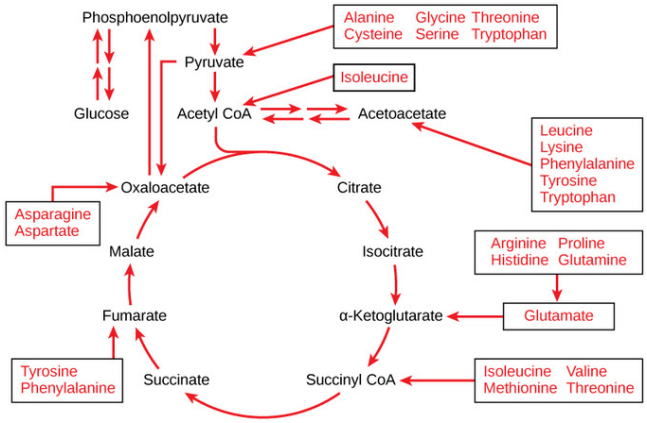

Figure 7.6$B.1$: **Connection of Amino Acids to Glucose Metabolism Pathways**: The carbon skeletons of certain amino acids (indicated in boxes) are derived from proteins and can feed into pyruvate, acetyl CoA, and the citric acid cycle.

Each amino acid must have its amino group removed (deamination) prior to the carbon chain's entry into these pathways. When the amino group is removed from an amino acid, it is converted into ammonia through the urea cycle. The remaining atoms of the amino acid result in a keto acid: a carbon chain with one ketone and one carboxylic acid group. In mammals, the liver synthesizes urea from two ammonia molecules and a carbon dioxide molecule. Thus, urea is the principal waste product in mammals produced from the nitrogen originating in amino acids; it leaves the body in urine. The keto acid can then enter the citric acid cycle.

When deaminated, amino acids can enter the pathways of glucose metabolism as pyruvate, acetyl CoA, or several components of the citric acid cycle. For example, deaminated asparagine and aspartate are converted into oxaloacetate and

enter glucose catabolism in the citric acid cycle. Deaminated amino acids can also be converted into another intermediate molecule before entering the pathways. Several amino acids can enter glucose catabolism at multiple locations.

7.6C: Connecting Lipids to Glucose Metabolism

Lipids can be both made and broken down through parts of the glucose catabolism pathways.

Learning Objectives

- Explain the connection of lipids to glucose metabolism

Key Points

- Many types of lipids exist, but cholesterol and triglycerides are the lipids that enter the pathways of glucose catabolism.
- Through the process of phosphorylation, glycerol can be converted to glycerol-3-phosphate during the glycolytic pathway.
- When fatty acids are broken down into acetyl groups through beta-oxidation, the acetyl groups are used by CoA to form acetyl-CoA, which enters the citric acid cycle to produce ATP.
- Beta-oxidation produces $FADH_2$ and NADH, which are used by the electron transport chain for ATP production.

Key Terms

- **beta-oxidation**: A process that takes place in the matrix of the mitochondria and catabolizes fatty acids by converting them to acetyl groups while producing NADH and FADH2.
- **lipid**: A group of organic compounds including fats, oils, waxes, sterols, and triglycerides; characterized by being insoluble in water; account for most of the fat present in the human body.

Like sugars and amino acids, the catabolic pathways of lipids are also connected to the glucose catabolism pathways. The lipids that are connected to the glucose pathways are cholesterol and triglycerides.

Cholesterol

Cholesterol contributes to cell membrane flexibility and is a precursor to steroid hormones. The synthesis of cholesterol starts with acetyl groups, which are transferred from acetyl CoA, and proceeds in only one direction; the process cannot be reversed. Thus, synthesis of cholesterol requires an intermediate of glucose metabolism.

Triglycerides

Triglycerides, a form of long-term energy storage in animals, are made of glycerol and three fatty acids. Animals can make most of the fatty acids they need. Triglycerides can be both made and broken down through parts of the glucose catabolism pathways. Glycerol can be phosphorylated to glycerol-3-phosphate, which continues through glycolysis.

Fatty acids are catabolized in a process called beta-oxidation that takes place in the matrix of the mitochondria and converts their fatty acid chains into two carbon units of acetyl groups, while producing NADH and $FADH_2$. The acetyl groups are picked up by CoA to form acetyl CoA that proceeds into the citric acid cycle as it combines with oxaloacetate. The NADH and $FADH_2$ are then used by the electron transport chain.

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7.7: Regulation of Cellular Respiration

Topic hierarchy

- 7.7A: Regulatory Mechanisms for Cellular Respiration
- 7.7B: Control of Catabolic Pathways

7.7A: Regulatory Mechanisms for Cellular Respiration

Cellular respiration can be controlled at each stage of glucose metabolism through various regulatory mechanisms.

Learning Objectives

- Explain the mechanisms that regulate cellular respiration.

Key Points

- Varying forms of the GLUT protein control the passage of glucose into the cells of specific tissues, thereby regulating cellular respiration.
- Reactions that are catalyzed by only one enzyme can go to equilibrium, which can cause the reaction to stall.
- If two different enzymes are necessary for a reversible reaction, there is greater opportunity to control the rate of the reaction and, as a result, equilibrium is reached less often.
- Enzymes are often controlled by binding of a molecule to an allosteric site on the protein.

Key Terms

- **enzyme**: a globular protein that catalyses a biological chemical reaction
- **allosteric**: a compound that binds to an inactive site, affecting the activity of an enzyme by changing the conformation of the protein (can activate or deactivate)
- **metabolism**: the complete set of chemical reactions that occur in living cells

Regulatory Mechanisms

Various mechanisms are used to control cellular respiration. As such, some type of control exists at each stage of glucose metabolism. Access of glucose to the cell can be regulated using the GLUT proteins that transport glucose. In addition, different forms of the GLUT protein control passage of glucose into the cells of specific tissues.

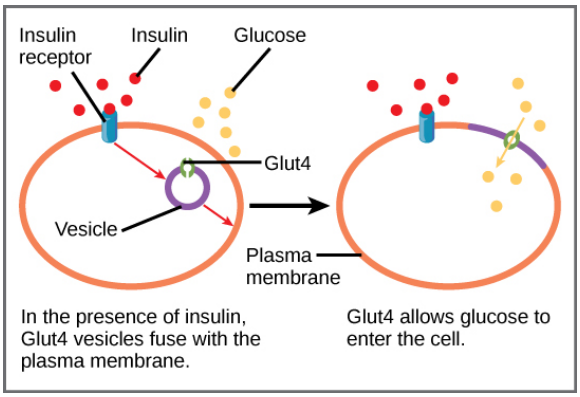

Figure 7.7A.1: **Glucose Transport**: GLUT4 is a glucose transporter that is stored in vesicles. A cascade of events that occurs upon insulin binding to a receptor in the plasma membrane causes GLUT4-containing vesicles to fuse with the plasma membrane so that glucose may be transported into the cell.

Some reactions are controlled by having two different enzymes: one each for the two directions of a reversible reaction. Reactions that are catalyzed by only one enzyme can go to equilibrium, stalling the reaction. In contrast, if two different enzymes (each specific for a given direction) are necessary for a reversible reaction, the opportunity to control the rate of the reaction increases and equilibrium is not reached.

A number of enzymes involved in each of the pathways (in particular, the enzyme catalyzing the first committed reaction of the pathway) are controlled by attachment of a molecule to an allosteric (non-active) site on the protein. This site has an effect on the enzyme's activity, often by changing the conformation of the protein. The molecules most commonly used in this capacity are the nucleotides ATP, ADP, AMP, NAD^+, and NADH. These regulators, known as allosteric effectors, may increase or decrease enzyme activity, depending on the prevailing conditions, altering the steric structure of the enzyme, usually affecting the configuration of the active site. This alteration of the protein's (the enzyme's) structure either increases or decreases its affinity for its substrate, with the effect of increasing or decreasing the rate of the reaction. The

attachment of a molecule to the allosteric site serves to send a signal to the enzyme, providing feedback. This feedback type of control is effective as long as the chemical affecting it is bound to the enzyme. Once the overall concentration of the chemical decreases, it will diffuse away from the protein, and the control is relaxed.

7.7B: Control of Catabolic Pathways

Catabolic pathways are controlled by enzymes, proteins, electron carriers, and pumps that ensure that the remaining reactions can proceed.

Learning Objectives

- Explain how catabolic pathways are controlled

Key Points

- Glycolysis, the citric acid cycle, and the electron transport chain are catabolic pathways that bring forth non-reversible reactions.
- Glycolysis control begins with hexokinase, which catalyzes the phosphorylation of glucose; its product is glucose-6-phosphate, which accumulates when phosphofructokinase is inhibited.
- The citric acid cycle is controlled through the enzymes that break down the reactions that make the first two molecules of NADH.
- The rate of electron transport through the electron transport chain is affected by the levels of ADP and ATP, whereas specific enzymes of the electron transport chain are unaffected by feedback inhibition.

Key Terms

- **phosphofructokinase**: any of a group of kinase enzymes that convert fructose phosphates to biphosphate
- **glycolysis**: the cellular metabolic pathway of the simple sugar glucose to yield pyruvic acid and ATP as an energy source
- **kinase**: any of a group of enzymes that transfers phosphate groups from high-energy donor molecules, such as ATP, to specific target molecules (substrates); the process is termed phosphorylation

Control of Catabolic Pathways

Enzymes, proteins, electron carriers, and pumps that play roles in glycolysis, the citric acid cycle, and the electron transport chain tend to catalyze non-reversible reactions. In other words, if the initial reaction takes place, the pathway is committed to proceeding with the remaining reactions. Whether a particular enzyme activity is released depends upon the energy needs of the cell (as reflected by the levels of ATP, ADP, and AMP).

Glycolysis

The control of glycolysis begins with the first enzyme in the pathway, hexokinase. This enzyme catalyzes the phosphorylation of glucose, which helps to prepare the compound for cleavage in a later step. The presence of the negatively-charged phosphate in the molecule also prevents the sugar from leaving the cell. When hexokinase is inhibited, glucose diffuses out of the cell and does not become a substrate for the respiration pathways in that tissue. The product of the hexokinase reaction is glucose-6-phosphate, which accumulates when a later enzyme, phosphofructokinase, is inhibited.

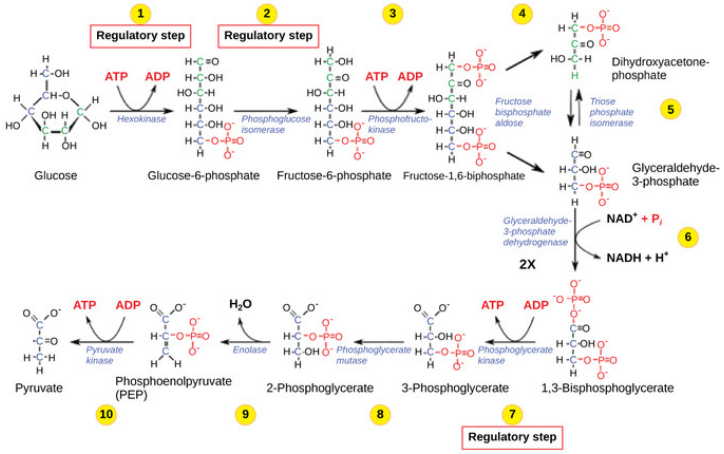

Figure 7.7*B*.1: **Glycolysis**: The glycolysis pathway is primarily regulated at the three key enzymatic steps (1, 2, and 7) as indicated. Note that the first two steps that are regulated occur early in the pathway and involve hydrolysis of ATP.

Phosphofructokinase is the main enzyme controlled in glycolysis. High levels of ATP, citrate, or a lower, more acidic pH decrease the enzyme's activity. An increase in citrate concentration can occur because of a blockage in the citric acid cycle. Fermentation, with its production of organic acids like lactic acid, frequently accounts for the increased acidity in a cell; however, the products of fermentation do not typically accumulate in cells.

The last step in glycolysis is catalyzed by pyruvate kinase. The pyruvate produced can proceed to be catabolized or converted into the amino acid alanine. If no more energy is needed and alanine is in adequate supply, the enzyme is inhibited. The enzyme's activity is increased when fructose-1,6-bisphosphate levels increase. (Recall that fructose-1,6-bisphosphate is an intermediate in the first half of glycolysis.) The regulation of pyruvate kinase involves phosphorylation, resulting in a less-active enzyme. Dephosphorylation by a phosphatase reactivates it. Pyruvate kinase is also regulated by ATP (a negative allosteric effect).

If more energy is needed, more pyruvate will be converted into acetyl CoA through the action of pyruvate dehydrogenase. If either acetyl groups or NADH accumulate, there is less need for the reaction and the rate decreases. Pyruvate dehydrogenase is also regulated by phosphorylation: a kinase phosphorylates it to form an inactive enzyme, and a phosphatase reactivates it. The kinase and the phosphatase are also regulated.

Citric Acid Cycle

The citric acid cycle is controlled through the enzymes that catalyze the reactions that make the first two molecules of NADH. These enzymes are isocitrate dehydrogenase and α-ketoglutarate dehydrogenase. When adequate ATP and NADH levels are available, the rates of these reactions decrease. When more ATP is needed, as reflected in rising ADP levels, the rate increases. α-Ketoglutarate dehydrogenase will also be affected by the levels of succinyl CoA, a subsequent intermediate in the cycle, causing a decrease in activity. A decrease in the rate of operation of the pathway at this point is not necessarily negative as the increased levels of the α-ketoglutarate not used by the citric acid cycle can be used by the cell for amino acid (glutamate) synthesis.

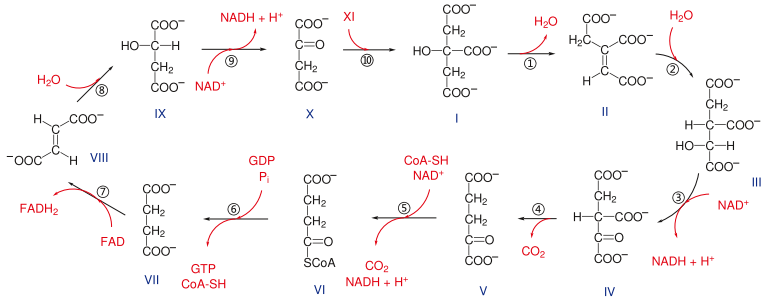

Figure 7.7*B*.1: **Citric Acid Cycle**: Enzymes, isocitrate dehydrogenase and α-ketoglutarate dehydrogenase, catalyze the reactions that make the first two molecules of NADH in the citric acid cycle. Rates of the reaction decrease when sufficient ATP and NADH levels are reached.

Electron Transport Chain

Specific enzymes of the electron transport chain are unaffected by feedback inhibition, but the rate of electron transport through the pathway is affected by the levels of ADP and ATP. Greater ATP consumption by a cell is indicated by a buildup of ADP. As ATP usage decreases, the concentration of ADP decreases: ATP begins to build up in the cell. This change in the relative concentration of ADP to ATP triggers the cell to slow down the electron transport chain.

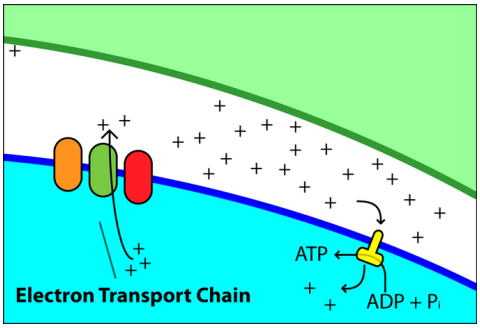

Figure 7.7*B*.1: **Electron Chain Transport**: Levels of ADP and ATP affect the rate of electron transport through this type of chain transport.

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CHAPTER OVERVIEW

8: PHOTOSYNTHESIS

8.1: OVERVIEW OF PHOTOSYNTHESIS

8.1A: THE PURPOSE AND PROCESS OF PHOTOSYNTHESIS
8.1B: MAIN STRUCTURES AND SUMMARY OF PHOTOSYNTHESIS
8.1C: THE TWO PARTS OF PHOTOSYNTHESIS
8.2: THE LIGHT-DEPENDENT REACTIONS OF PHOTOSYNTHESIS

8.2A: INTRODUCTION TO LIGHT ENERGY
8.2B: ABSORPTION OF LIGHT
8.2C: PROCESSES OF THE LIGHT-DEPENDENT REACTIONS
8.3: THE LIGHT-INDEPENDENT REACTIONS OF PHOTOSYNTHESIS

8.3A: CAM AND C4 PHOTOSYNTHESIS
8.3B: THE CALVIN CYCLE
8.3C: THE CARBON CYCLE

8.1: Overview of Photosynthesis

Topic hierarchy

- 8.1A: The Purpose and Process of Photosynthesis
- 8.1B: Main Structures and Summary of Photosynthesis
- 8.1C: The Two Parts of Photosynthesis

: The Purpose and Process of Photosynthesis

The process of photosynthesis converts light energy to chemical energy, which can be used by organisms for different metabolic processes.

Learning Objectives

- Describe the process of photosynthesis

Key Points

- Photosynthesis evolved as a way to store the energy in solar radiation as high-energy electrons in carbohydrate molecules.
- Plants, algae, and cyanobacteria, known as photoautotrophs, are the only organisms capable of performing photosynthesis.
- Heterotrophs, unable to produce their own food, rely on the carbohydrates produced by photosynthetic organisms for their energy needs.

Key Terms

- **photosynthesis**: the process by which plants and other photoautotrophs generate carbohydrates and oxygen from carbon dioxide, water, and light energy in chloroplasts
- **photoautotroph**: an organism that can synthesize its own food by using light as a source of energy
- **chemoautotroph**: a simple organism, such as a protozoan, that derives its energy from chemical processes rather than photosynthesis

The Importance of Photosynthesis

The processes of all organisms—from bacteria to humans—require energy. To get this energy, many organisms access stored energy by eating food. Carnivores eat other animals and herbivores eat plants. But where does the stored energy in food originate? All of this energy can be traced back to the process of photosynthesis and light energy from the sun.

Photosynthesis is essential to all life on earth. It is the only biological process that captures energy from outer space (sunlight) and converts it into chemical energy in the form of G3P (
Glyceraldehyde 3-phosphate) which in turn can be made into sugars and other molecular compounds. Plants use these compounds in all of their metabolic processes; plants do not need to consume other organisms for food because they build all the molecules they need. Unlike plants, animals need to consume other organisms to consume the molecules they need for their metabolic processes.

The Process of Photosynthesis

During photosynthesis, molecules in leaves capture sunlight and energize electrons, which are then stored in the covalent bonds of carbohydrate molecules. That energy within those covalent bonds will be released when they are broken during cell respiration. How long lasting and stable are those covalent bonds? The energy extracted today by the burning of coal and petroleum products represents sunlight energy captured and stored by photosynthesis almost 200 million years ago.

Plants, algae, and a group of bacteria called cyanobacteria are the only organisms capable of performing photosynthesis. Because they use light to manufacture their own food, they are called photoautotrophs (“self-feeders using light”). Other organisms, such as animals, fungi, and most other bacteria, are termed heterotrophs (“other feeders”) because they must rely on the sugars produced by photosynthetic organisms for their energy needs. A third very interesting group of bacteria synthesize sugars, not by using sunlight’s energy, but by extracting energy from inorganic chemical compounds; hence, they are referred to as chemoautotrophs.

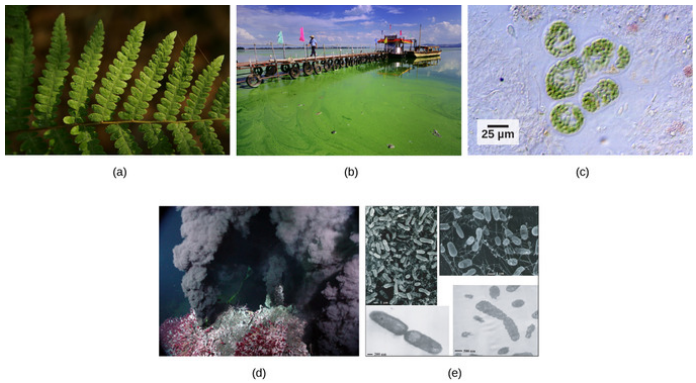

Figure 8.1*A*.1: **Photosynthetic and Chemosynthetic Organisms**: Photoautotrophs, including (a) plants, (b) algae, and (c) cyanobacteria, synthesize their organic compounds via photosynthesis using sunlight as an energy source. Cyanobacteria and planktonic algae can grow over enormous areas in water, at times completely covering the surface. In a (d) deep sea vent, chemoautotrophs, such as these (e) thermophilic bacteria, capture energy from inorganic compounds to produce organic compounds. The ecosystem surrounding the vents has a diverse array of animals, such as tubeworms, crustaceans, and octopi that derive energy from the bacteria.

The importance of photosynthesis is not just that it can capture sunlight's energy. A lizard sunning itself on a cold day can use the sun's energy to warm up. Photosynthesis is vital because it evolved as a way to store the energy in solar radiation (the "photo-" part) as high-energy electrons in the carbon-carbon bonds of carbohydrate molecules (the "-synthesis" part). Those carbohydrates are the energy source that heterotrophs use to power the synthesis of ATP via respiration. Therefore, photosynthesis powers 99 percent of Earth's ecosystems. When a top predator, such as a wolf, preys on a deer, the wolf is at the end of an energy path that went from nuclear reactions on the surface of the sun, to light, to photosynthesis, to vegetation, to deer, and finally to wolf.

8.1B: Main Structures and Summary of Photosynthesis

In multicellular autotrophs, the main cellular structures that allow photosynthesis to take place include chloroplasts, thylakoids, and chlorophyll.

Learning Objectives

- Describe the main structures involved in photosynthesis and recall the chemical equation that summarizes the process of photosynthesis

Key Points

- The chemical equation for photosynthesis is $6CO2+6H2O \rightarrow C6H12O6+6O2$.$6CO2+6H2O \rightarrow C6H12O6+6O2$.
- In plants, the process of photosynthesis takes place in the mesophyll of the leaves, inside the chloroplasts.
- Chloroplasts contain disc-shaped structures called thylakoids, which contain the pigment chlorophyll.
- Chlorophyll absorbs certain portions of the visible spectrum and captures energy from sunlight.

Key Terms

- **chloroplast**: An organelle found in the cells of green plants and photosynthetic algae where photosynthesis takes place.
- **mesophyll**: A layer of cells that comprises most of the interior of the leaf between the upper and lower layers of epidermis.
- **stoma**: A pore in the leaf and stem epidermis that is used for gaseous exchange.

Overview of Photosynthesis

Photosynthesis is a multi-step process that requires sunlight, carbon dioxide, and water as substrates. It produces oxygen and glyceraldehyde-3-phosphate (G3P or GA3P), simple carbohydrate molecules that are high in energy and can subsequently be converted into glucose, sucrose, or other sugar molecules. These sugar molecules contain covalent bonds that store energy. Organisms break down these molecules to release energy for use in cellular work.

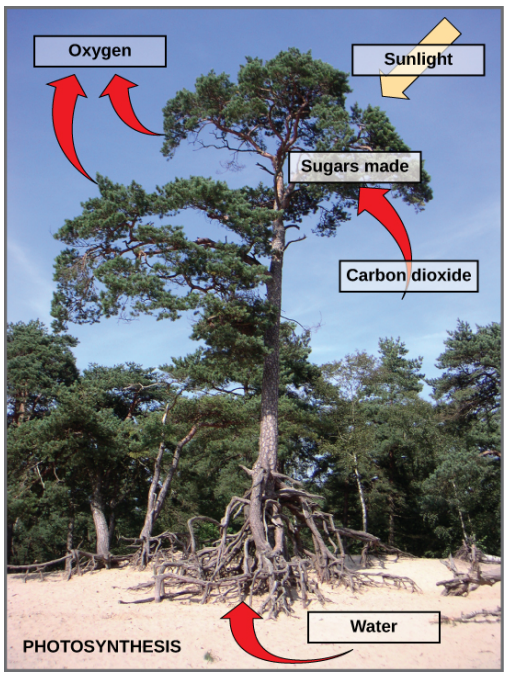

Figure 8.1B.1: **Photosynthesis**: Photosynthesis uses solar energy, carbon dioxide, and water to produce energy-storing carbohydrates. Oxygen is generated as a waste product of photosynthesis.

The energy from sunlight drives the reaction of carbon dioxide and water molecules to produce sugar and oxygen, as seen in the chemical equation for photosynthesis. Though the equation looks simple, it is carried out through many complex

steps. Before learning the details of how photoautotrophs convert light energy into chemical energy, it is important to become familiar with the structures involved.

Photosynthesis Equation							
Carbon dioxide	+	Water	SUNLIGHT →	Sugar	+	Oxygen	
$6CO_2$		$6H_2O$		$C_6H_{12}O_6$		$6O_2$	

Figure 8.1B.1: **Chemical equation for photosynthesis**: The basic equation for photosynthesis is deceptively simple. In reality, the process includes many steps involving intermediate reactants and products. Glucose, the primary energy source in cells, is made from two three-carbon GA3P molecules.

Photosynthesis and the Leaf

In plants, photosynthesis generally takes place in leaves, which consist of several layers of cells. The process of photosynthesis occurs in a middle layer called the mesophyll. The gas exchange of carbon dioxide and oxygen occurs through small, regulated openings called stomata (singular: stoma), which also play a role in the plant's regulation of water balance. The stomata are typically located on the underside of the leaf, which minimizes water loss. Each stoma is flanked by guard cells that regulate the opening and closing of the stomata by swelling or shrinking in response to osmotic changes.

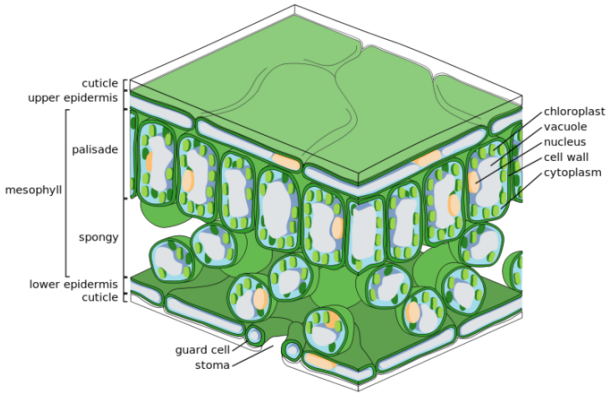

Figure 8.1B.1: **Structure of a leaf (cross-section)**: Photosynthesis takes place in the mesophyll. The palisade layer contains most of the chloroplast and principal region in which photosynthesis is carried out. The airy spongy layer is the region of storage and gas exchange. The stomata regulate carbon dioxide and water balance.

Photosynthesis within the Chloroplast

In all autotrophic eukaryotes, photosynthesis takes place inside an organelle called a chloroplast. For plants, chloroplast-containing cells exist in the mesophyll. Chloroplasts have a double membrane envelope composed of an outer membrane and an inner membrane. Within the double membrane are stacked, disc-shaped structures called thylakoids.

Embedded in the thylakoid membrane is chlorophyll, a pigment that absorbs certain portions of the visible spectrum and captures energy from sunlight. Chlorophyll gives plants their green color and is responsible for the initial interaction between light and plant material, as well as numerous proteins that make up the electron transport chain. The thylakoid membrane encloses an internal space called the thylakoid lumen. A stack of thylakoids is called a granum, and the liquid-filled space surrounding the granum is the stroma or "bed."

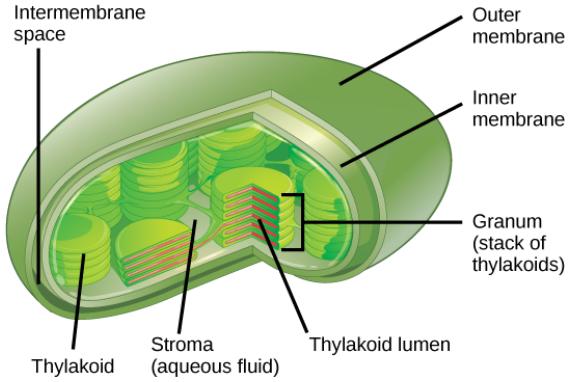

Figure 8.1B.1: **Structure of the Chloroplast**: Photosynthesis takes place in chloroplasts, which have an outer membrane and an inner membrane. Stacks of thylakoids called grana form a third membrane layer.

8.1C: The Two Parts of Photosynthesis

Light-dependent and light-independent reactions are two successive reactions that occur during photosynthesis.

Learning Objectives

- Distinguish between the two parts of photosynthesis

Key Points

- In light-dependent reactions, the energy from sunlight is absorbed by chlorophyll and converted into chemical energy in the form of electron carrier molecules like ATP and NADPH.
- Light energy is harnessed in Photosystems I and II, both of which are present in the thylakoid membranes of chloroplasts.
- In light-independent reactions (the Calvin cycle), carbohydrate molecules are assembled from carbon dioxide using the chemical energy harvested during the light-dependent reactions.

Key Terms

- **photosystem**: Either of two biochemical systems active in chloroplasts that are part of photosynthesis.

Photosynthesis takes place in two sequential stages:

1. The light-dependent reactions;
2. The light-independent reactions, or Calvin Cycle.

Light-Dependent Reactions

Just as the name implies, light-dependent reactions require sunlight. In the light-dependent reactions, energy from sunlight is absorbed by chlorophyll and converted into stored chemical energy, in the form of the electron carrier molecule NADPH (nicotinamide adenine dinucleotide phosphate) and the energy currency molecule ATP (adenosine triphosphate). The light-dependent reactions take place in the thylakoid membranes in the granum (stack of thylakoids), within the chloroplast.

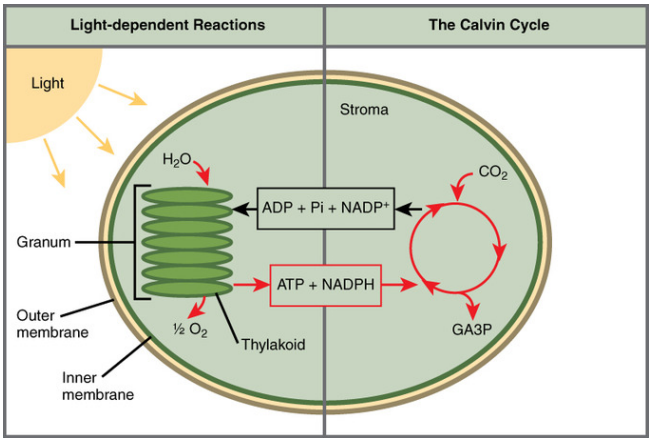

Figure 8.1C.1: **The two stages of photosynthesis**: Photosynthesis takes place in two stages: light-dependent reactions and the Calvin cycle (light-independent reactions). Light-dependent reactions, which take place in the thylakoid membrane, use light energy to make ATP and NADPH. The Calvin cycle, which takes place in the stroma, uses energy derived from these compounds to make GA3P from CO2.

Photosystems

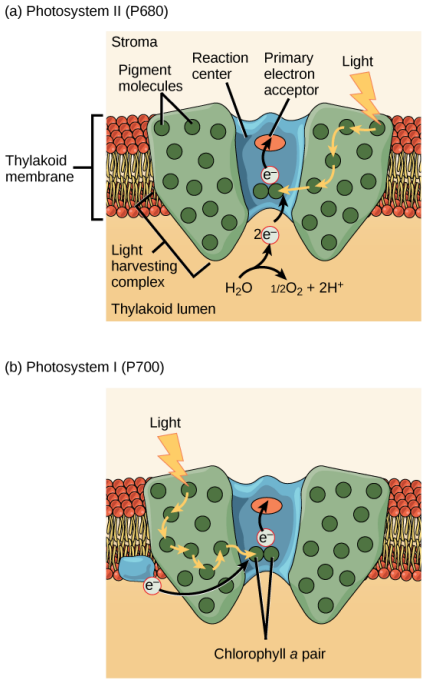

Figure 8.1C.1: **Photosystems I & II**: As explained above, the photosystems manipulate electrons with energy harvested from light.

The process that converts light energy into chemical energy takes place in a multi-protein complex called a photosystem. Two types of photosystems are embedded in the thylakoid membrane: photosystem II (PSII) and photosystem I (PSI). Each photosystem plays a key role in capturing the energy from sunlight by exciting electrons. These energized electrons are transported by “energy carrier” molecules, which power the light-independent reactions.

Photosystems consist of a light-harvesting complex and a reaction center. Pigments in the light-harvesting complex pass light energy to two special chlorophyll *a* molecules in the reaction center. The light excites an electron from the chlorophyll *a* pair, which passes to the primary electron acceptor. The excited electron must then be replaced. In photosystem II, the electron comes from the splitting of water, which releases oxygen as a waste product. In photosystem I, the electron comes from the chloroplast electron transport chain.

The two photosystems oxidize different sources of the low-energy electron supply, deliver their energized electrons to different places, and respond to different wavelengths of light.

Light-Independent Reactions

In the light-independent reactions or Calvin cycle, the energized electrons from the light-dependent reactions provide the energy to form carbohydrates from carbon dioxide molecules. The light-independent reactions are sometimes called the Calvin cycle because of the cyclical nature of the process.

Although the light-independent reactions do not use light as a reactant (and as a result can take place at day or night), they require the products of the light-dependent reactions to function. The light-independent molecules depend on the energy carrier molecules, ATP and NADPH, to drive the construction of new carbohydrate molecules. After the energy is transferred, the energy carrier molecules return to the light-dependent reactions to obtain more energized electrons. In addition, several enzymes of the light-independent reactions are activated by light.

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8.2: The Light-Dependent Reactions of Photosynthesis

Topic hierarchy

- 8.2A: Introduction to Light Energy
- 8.2B: Absorption of Light
- 8.2C: Processes of the Light-Dependent Reactions

: Introduction to Light Energy

All electromagnetic radiation, or light energy, travels at a particular wavelength and carries a certain amount of energy.

Learning Objectives

- Explain the difference between short and long wavelengths.

Key Points

- The amount of energy of a wave can be determined by measuring its wavelength, the distance between consecutive points of a wave.
- Visible light is a type of radiant energy within the electromagnetic spectrum; other types of electromagnetic radiation include UV, infrared, gamma, and radio rays as well as X-rays.
- The difference between wavelengths relates to the amount of energy carried by them; short, tight waves carry more energy than long, wide waves.

Key Terms

- **electromagnetic spectrum**: the entire range of wavelengths of all known radiations consisting of oscillating electric and magnetic fields, including gamma rays, visible light, infrared, radio waves, and X-rays
- **wavelength**: the length of a single cycle of a wave, as measured by the distance between one peak or trough of a wave and the next; it corresponds to the velocity of the wave divided by its frequency
- **visible light**: the part of the electromagnetic spectrum, between infrared and ultraviolet, that is visible to the human eye

What Is Light Energy?

The sun emits an enormous amount of electromagnetic radiation (solar or light energy). Humans can see only a fraction of this energy, which is referred to as “visible light.” The manner in which solar energy travels is described as waves. Scientists can determine the amount of energy of a wave by measuring its wavelength, the distance between consecutive points of a wave, such as from crest to crest or from trough to trough.

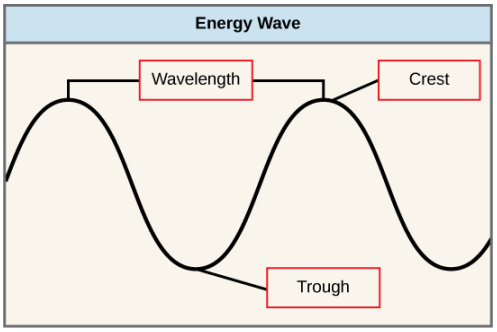

Figure 8.2A.1: **Wavelengths**: The wavelength of a single wave is the distance between two consecutive points of similar position (two crests or two troughs) along the wave.

Visible light constitutes only one of many types of electromagnetic radiation emitted from the sun and other stars. The electromagnetic spectrum is the range of all possible frequencies of radiation. The electromagnetic spectrum shows several types of electromagnetic radiation originating from the sun, including X-rays and ultraviolet (UV) rays. The higher-energy waves can penetrate tissues and damage cells and DNA, which explains why both X-rays and UV rays can be harmful to living organisms. Scientists differentiate the various types of radiant energy from the sun within the electromagnetic spectrum.The difference between wavelengths relates to the amount of energy carried by them.

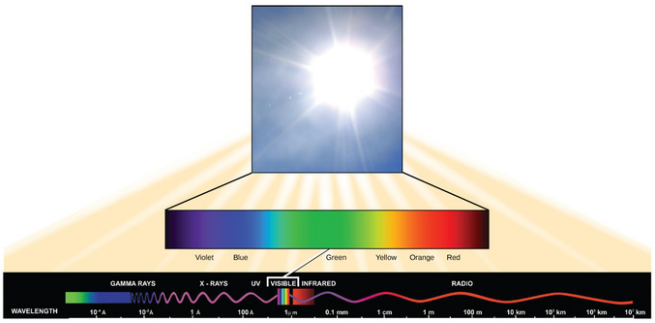

Figure 8.2*A*.1: The Electromagnetic Spectrum: The sun emits energy in the form of electromagnetic radiation. This radiation exists at different wavelengths, each of which has its own characteristic energy. All electromagnetic radiation, including visible light, is characterized by its wavelength.

Each type of electromagnetic radiation travels at a particular wavelength. The longer the wavelength, the less energy is carried. Short, tight waves carry the most energy. This may seem illogical, but think of it in terms of a person moving a heavy rope. It takes little effort by a person to move a rope in long, wide waves. To make a rope move in short, tight waves, a person would need to apply significantly more energy.

8.2B: Absorption of Light

Pigments, like chlorophyll and carotenoids, absorb and reflect light at a certain region of the electromagnetic spectrum.

Learning Objectives

- Differentiate between chlorophyll and carotenoids.

Key Points

- Plant pigment molecules absorb only light in the wavelength range of 700 nm to 400 nm; this range is referred to as photosynthetically-active radiation.
- Violet and blue have the shortest wavelengths and the most energy, whereas red has the longest wavelengths and carries the least amount of energy.
- Pigments reflect or transmit the wavelengths they cannot absorb, making them appear in the corresponding color.
- Chorophylls and carotenoids are the major pigments in plants; while there are dozens of carotenoids, there are only five important chorophylls: *a*, *b*, *c*, *d*, and bacteriochlorophyll.
- Chlorophyll *a* absorbs light in the blue-violet region, chlorophyll *b* absorbs red-blue light, and both *a* and *b* reflect green light (which is why chlorophyll appears green).
- Carotenoids absorb light in the blue-green and violet region and reflect the longer yellow, red, and orange wavelengths; these pigments also dispose excess energy out of the cell.

Key Terms

- **chlorophyll**: Any of a group of green pigments that are found in the chloroplasts of plants and in other photosynthetic organisms such as cyanobacteria.
- **carotenoid**: Any of a class of yellow to red plant pigments including the carotenes and xanthophylls.
- **spectrophotometer**: An instrument used to measure the intensity of electromagnetic radiation at different wavelengths.

Absorption of Light

Light energy initiates the process of photosynthesis when pigments absorb the light. Organic pigments have a narrow range of energy levels that they can absorb. Energy levels lower than those represented by red light are insufficient to raise an orbital electron to an excited, or quantum, state. Energy levels higher than those in blue light will physically tear the molecules apart, a process called bleaching. For example, retinal pigments can only “see” (absorb) 700 nm to 400 nm light; this is visible light. For the same reasons, plant pigment molecules absorb only light in the wavelength range of 700 nm to 400 nm; plant physiologists refer to this range for plants as photosynthetically-active radiation.

The visible light seen by humans as the color white light actually exists in a rainbow of colors in the electromagnetic spectrum, with violet and blue having shorter wavelengths and, thus, higher energy. At the other end of the spectrum, toward red, the wavelengths are longer and have lower energy.

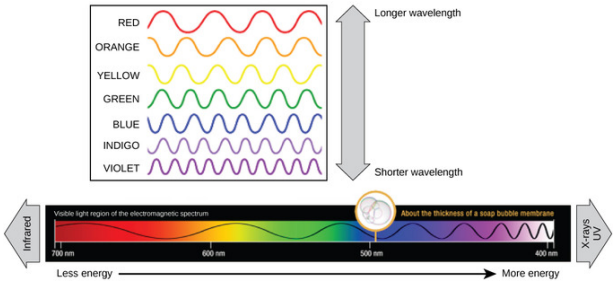

Figure 8.2B.1: **Visible Light**: The colors of visible light do not carry the same amount of energy. Violet has the shortest wavelength and, therefore, carries the most energy, whereas red has the longest wavelength and carries the least amount of energy.

Understanding Pigments

Different kinds of pigments exist, each of which has evolved to absorb only certain wavelengths or colors of visible light. Pigments reflect or transmit the wavelengths they cannot absorb, making them appear in the corresponding color.

Chlorophylls and carotenoids are the two major classes of photosynthetic pigments found in plants and algae; each class has multiple types of pigment molecules. There are five major chlorophylls: *a*, *b*, *c* and *d*, along with a related molecule found in prokaryotes called bacteriochlorophyll.

With dozens of different forms, carotenoids are a much larger group of pigments. The carotenoids found in fruit, such as the red of tomato (lycopene), the yellow of corn seeds (zeaxanthin), or the orange of an orange peel (β-carotene), are used to attract seed-dispersing organisms. In photosynthesis, carotenoids function as photosynthetic pigments that are very efficient molecules for the disposal of excess energy. When a leaf is exposed to full sun, the light-dependent reactions are required to process an enormous amount of energy; if that energy is not handled properly, it can do significant damage. Therefore, many carotenoids are stored in the thylakoid membrane to absorb excess energy and safely release that energy as heat.

Each type of pigment can be identified by the specific pattern of wavelengths it absorbs from visible light, which is the absorption spectrum. Chlorophyll *a* absorbs light in the blue-violet region, while chlorophyll *b* absorbs red-blue light. Neither *a* or *b* absorb green light; because green is reflected or transmitted, chlorophyll appears green. Carotenoids absorb light in the blue-green and violet region and reflect the longer yellow, red, and orange wavelengths.

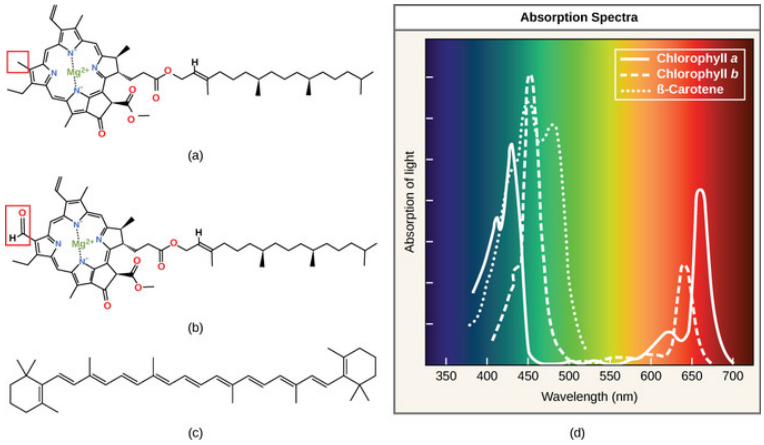

Figure 8.2$B.1$: **Chlorophyll a and b**: (a) Chlorophyll *a*, (b) chlorophyll *b*, and (c) β-carotene are hydrophobic organic pigments found in the thylakoid membrane. Chlorophyll *a* and *b*, which are identical except for the part indicated in the red box, are responsible for the green color of leaves. β-carotene is responsible for the orange color in carrots. Each pigment has (d) a unique absorbance spectrum.

Many photosynthetic organisms have a mixture of pigments. In this way organisms can absorb energy from a wider range of wavelengths. Not all photosynthetic organisms have full access to sunlight. Some organisms grow underwater where light intensity and quality decrease and change with depth. Other organisms grow in competition for light. Plants on the rainforest floor must be able to absorb any light that comes through because the taller trees absorb most of the sunlight and scatter the remaining solar radiation

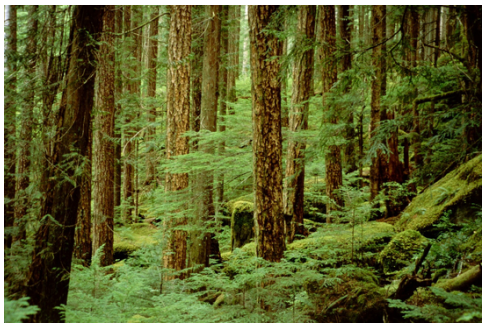

Figure 8.2$B.1$: **Pigments in Plants**: Plants that commonly grow in the shade have adapted to low levels of light by changing the relative concentrations of their chlorophyll pigments.

When studying a photosynthetic organism, scientists can determine the types of pigments present by using a spectrophotometer. These instruments can differentiate which wavelengths of light a substance can absorb. Spectrophotometers measure transmitted light and compute its absorption. By extracting pigments from leaves and placing these samples into a spectrophotometer, scientists can identify which wavelengths of light an organism can absorb.

8.2C: Processes of the Light-Dependent Reactions

Light-dependent reactions, which take place in photosystem I and II, convert solar energy into NADPH and ATP.

Learning Objectives

- Describe how light energy is converted into ATP and NADPH.

Key Points

- Light energy splits water and extracts electrons in photosystem II (PSII); then electrons are moved from PSII to cytochrome b6f to photosystem I (PSI) and reduce in energy.
- Electrons are re-energized in PSI and those high energy electrons reduce $NADP^+$ to NADPH.
- In non-cyclic photophosphorylation, cytochrome b6f uses the energy of electrons from PSII to pump hydrogen ions from the lumen to the stroma; this energy allows ATP synthase to attach a third phosphate group to ADP, which forms ATP.
- In cyclic photophosphorylation, cytochrome b6f uses the energy of electrons from both PSII and PSI to create more ATP and to stop the production of NADPH, maintaining the right proportions of NADPH and ATP.

Key Terms

- **photosystem**: Either of two biochemical systems, active in chloroplasts, that are part of photosynthesis.
- **photophosphorylation**: The addition of a phosphate (PO43-) group to a protein or other organic molecule by photosynthesis.
- **chemiosmosis**: The movement of ions across a selectively permeable membrane, down their electrochemical gradient.

How Light-Dependent Reactions Work

The overall function of light-dependent reactions, the first stage of photosynthesis, is to convert solar energy into chemical energy in the form of NADPH and ATP, which are used in light-independent reactions and fuel the assembly of sugar molecules. Protein complexes and pigment molecules work together to produce NADPH and ATP.

Producing Chemical Energy

Light energy is converted into chemical energy in a multiprotein complex called a photosystem. Two types of photosystems, photosystem I (PSI) and photosystem II (PSII), are found in the thylakoid membrane inside the chloroplast. Each photosystem consists of multiple antenna proteins that contain a mixture of 300–400 chlorophyll *a* and *b* molecules, as well as other pigments like carotenoids. Cytochrome b6f complex and ATP synthase are also major protein complexes in the thylakoid membrane that work with the photosystems to create ATP and NADPH.

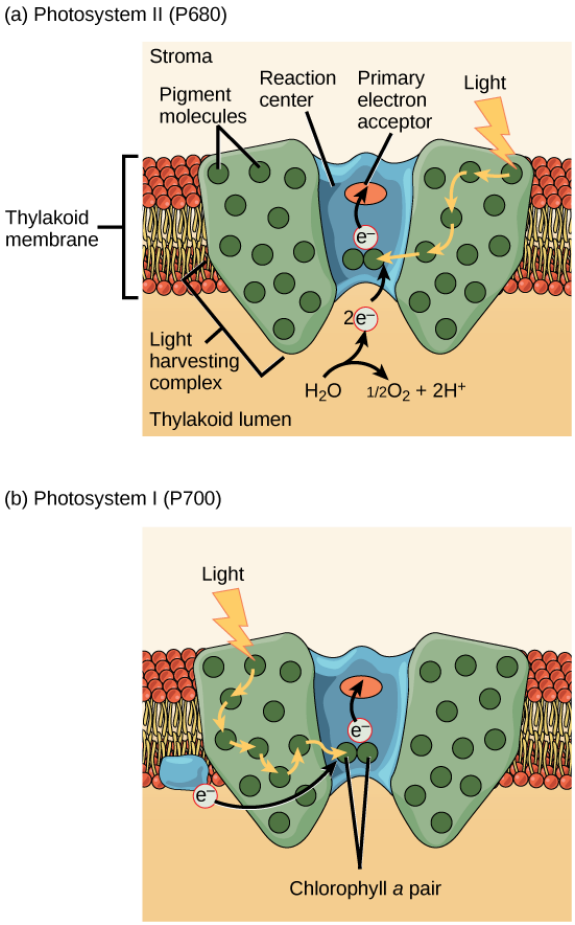

Figure 8.2C.1: **Photosystems I & II**: A photosystem consists of a light-harvesting complex and a reaction center. Pigments in the light-harvesting complex pass light energy to two special chlorophyll a molecules in the reaction center. The light excites an electron from the chlorophyll a pair, which passes to the primary electron acceptor. The excited electron must then be replaced. In (a) photosystem II, the electron comes from the splitting of water, which releases oxygen as a waste product. In (b) photosystem I, the electron comes from the chloroplast electron transport chain.

The two photosystems absorb light energy through proteins containing pigments, such as chlorophyll. The light-dependent reactions begin in photosystem II. In PSII, energy from sunlight is used to split water, which releases two electrons, two hydrogen atoms, and one oxygen atom. When a chlorophyll *a* molecule within the reaction center of PSII absorbs a photon, the electron in this molecule attains a higher energy level. Because this state of an electron is very unstable, the electron is transferred to another molecule creating a chain of redox reactions called an electron transport chain (ETC). The electron flow goes from PSII to cytochrome b6f to PSI; as electrons move between these two photosystems, they lose energy. Because the electrons have lost energy prior to their arrival at PSI, they must be re-energized by PSI. Therefore, another photon is absorbed by the PSI antenna. That energy is transmitted to the PSI reaction center. This reaction center, known as P700, is oxidized and sends a high-energy electron to reduce NADP+ to NADPH. This process illustrates oxygenic photosynthesis, wherein the first electron donor is water and oxygen is created as a waste product.

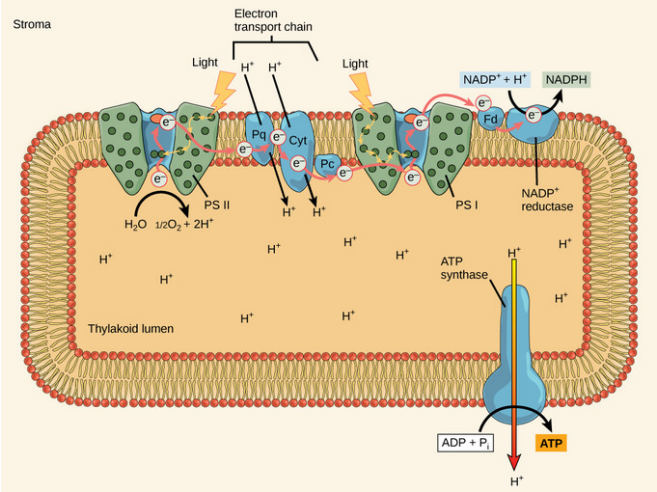

Figure 8.2C.1: **Photosystem II**: In the photosystem II (PSII) reaction center, energy from sunlight is used to extract electrons from water. The electrons travel through the chloroplast electron transport chain to photosystem I (PSI), which reduces NADP+ to NADPH. The electron transport chain moves protons across the thylakoid membrane into the lumen. At the same time, splitting of water adds protons to the lumen while reduction of NADPH removes protons from the stroma. The net result is a low pH in the thylakoid lumen and a high pH in the stroma. ATP synthase uses this electrochemical gradient to make ATP.

Cytochrome b6f and ATP synthase work together to create ATP. This process, called photophosphorylation, occurs in two different ways. In non-cyclic photophosphorylation, cytochrome b6f uses the energy of electrons from PSII to pump hydrogen ions from the lumen (an area of high concentration) to the stroma (an area of low concentration). The energy released by the hydrogen ion stream allows ATP synthase to attach a third phosphate group to ADP, which forms ATP. This flow of hydrogen ions through ATP synthase is called chemiosmosis because the ions move from an area of high to an area of low concentration through a semi-permeable structure. In cyclic photophosphorylation, cytochrome b6f uses the energy of electrons from both PSII and PSI to create more ATP and to stop the production of NADPH. Cyclic phosphorylation is important to maintain the right proportions of NADPH and ATP, which will carry out light-independent reactions later on.

The net-reaction of all light-dependent reactions in oxygenic photosynthesis is: $2H_2O$ + 2NADP+ + 3ADP + 3Pi → O_2 + 2NADPH + 3ATP

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8.3: The Light-Independent Reactions of Photosynthesis

Topic hierarchy

- 8.3A: CAM and C4 Photosynthesis
- 8.3B: The Calvin Cycle
- 8.3C: The Carbon Cycle

8.3A: CAM and C4 Photosynthesis

Some plants have evolved mechanisms to increase the CO_2 concentration in their leaves under hot and dry conditions.

Learning Objectives

- Compare C4 and CAM photosynthesis

Key Points

- The process of photosynthesis in desert plants has evolved mechanisms to conserve water.
- Plants that use crassulacean acid metabolism (CAM) photosynthesis fix CO_2 at night, when their stomata are open.
- Plants that use C4 carbon fixation concentrate carbon dioxide spatially, using “bundle sheath cells” which are inundated with CO_2.

Key Terms

- **crassulacean acid metabolism**: A carbon fixation pathway that evolved in some plants as an adaptation to arid conditions, in which the stomata in the leaves remain shut during the day to reduce evapotranspiration, but open at night to collect carbon dioxide (CO2).
- **C4 carbon fixation**: A form of photosynthesis in which plants concentrate CO2 spatially, with a RuBisCO reaction centre in a “bundle sheath cell” that is inundated with CO2

Photosynthesis in desert plants has evolved adaptations that conserve water. In harsh, dry heat, every drop of water must be used to survive. Because stomata must open to allow for the uptake of CO_2, water escapes from the leaf during active photosynthesis. Desert plants have evolved processes to conserve water and deal with harsh conditions. A more efficient use of CO_2 allows plants to adapt to living with less water.

Some plants such as cacti can prepare materials for photosynthesis during the night by a temporary carbon fixation and storage process, because opening the stomata at this time conserves water due to cooler temperatures. In addition, cacti have evolved the ability to carry out low levels of photosynthesis without opening stomata at all, a mechanism for surviving extremely dry periods.

Figure 8.3A.1: **Cactus**: The harsh conditions of the desert have led plants like these cacti to evolve variations of the light-independent reactions of photosynthesis. These variations increase the efficiency of water usage, helping to conserve water and energy.

CAM Photosynthesis

Xerophytes, such as cacti and most succulents, also use
phosphoenolpyruvate (PEP) carboxylase to capture carbon dioxide in a process called crassulacean acid metabolism (CAM). In contrast to C_4 metabolism, which *physically* separates the CO_2 fixation to PEP from the Calvin cycle, CAM *temporally* separates these two processes.

CAM plants have a different leaf anatomy from C_3 plants, and fix the CO_2 at night, when their stomata are open. CAM plants store the CO_2 mostly in the form of malic acid via carboxylation of phosphoenolpyruvate to oxaloacetate, which is then reduced to malate. Decarboxylation of malate during the day releases CO_2 inside the leaves, thus allowing carbon fixation to 3-phosphoglycerate by RuBisCO. Sixteen thousand species of plants use CAM.

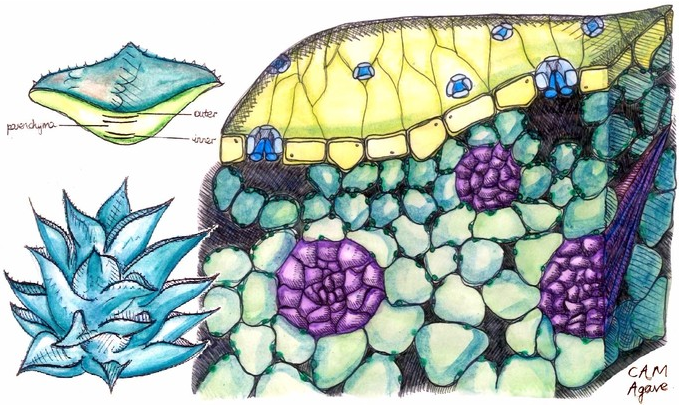
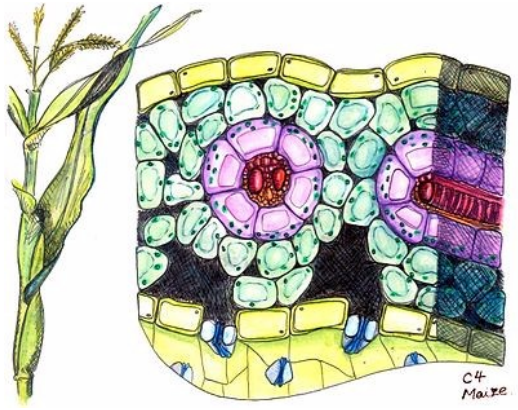


Figure 8.3A.1: **Cross section of agave, a CAM plant**: Cross section of a CAM (crassulacean acid metabolism) plant, specifically of an agave leaf. Vascular bundles shown. Drawing based on microscopic images courtesy of Cambridge University Plant Sciences Department.

C_4 Carbon Fixation

The C4 pathway bears resemblance to CAM; both act to concentrate CO_2 around RuBisCO, thereby increasing its efficiency. CAM concentrates it temporally, providing CO_2 during the day and not at night, when respiration is the dominant reaction.

C_4 plants, in contrast, concentrate CO_2 spatially, with a RuBisCO reaction centre in a "bundle sheath cell" that is inundated with CO_2. Due to the inactivity required by the CAM mechanism, C_4 carbon fixation has a greater efficiency in terms of PGA synthesis.



Cross section of maize, a C_4 plant

Cross section of a C4 plant, specifically of a maize leaf. Drawing based on microscopic images courtesy of Cambridge University Plant Sciences Department.

C_4 plants can produce more sugar than C_3 plants in conditions of high light and temperature. Many important crop plants are C_4 plants, including maize, sorghum, sugarcane, and millet. Plants that do not use PEP-carboxylase in carbon fixation are called C3 plants because the primary carboxylation reaction, catalyzed by RuBisCO, produces the three-carbon 3-phosphoglyceric acids directly in the Calvin-Benson cycle. Over 90% of plants use C_3 carbon fixation, compared to 3% that use C_4 carbon fixation; however, the evolution of C_4 in over 60 plant lineages makes it a striking example of convergent evolution.

8.3B: The Calvin Cycle

The Calvin cycle is organized into three basic stages: fixation, reduction, and regeneration.

Learning Objectives

- Describe the Calvin Cycle

Key Points

- The Calvin cycle refers to the light-independent reactions in photosynthesis that take place in three key steps.
- Although the Calvin Cycle is not directly dependent on light, it is indirectly dependent on light since the necessary energy carriers (ATP and NADPH) are products of light-dependent reactions.
- In fixation, the first stage of the Calvin cycle, light-independent reactions are initiated; CO2 is fixed from an inorganic to an organic molecule.
- In the second stage, ATP and NADPH are used to reduce 3-PGA into G3P; then ATP and NADPH are converted to ADP and $NADP^+$, respectively.
- In the last stage of the Calvin Cycle, RuBP is regenerated, which enables the system to prepare for more CO_2 to be fixed.

Key Terms

- **light-independent reaction**: chemical reactions during photosynthesis that convert carbon dioxide and other compounds into glucose, taking place in the stroma
- **rubisco**: (ribulose bisphosphate carboxylase) a plant enzyme which catalyzes the fixing of atmospheric carbon dioxide during photosynthesis by catalyzing the reaction between carbon dioxide and RuBP
- **ribulose bisphosphate**: an organic substance that is involved in photosynthesis, reacts with carbon dioxide to form 3-PGA

The Calvin Cycle

In plants, carbon dioxide (CO_2) enters the leaves through stomata, where it diffuses over short distances through intercellular spaces until it reaches the mesophyll cells. Once in the mesophyll cells, CO_2 diffuses into the stroma of the chloroplast, the site of light-independent reactions of photosynthesis. These reactions actually have several names associated with them. Other names for light-independent reactions include the Calvin cycle, the Calvin-Benson cycle, and dark reactions. The most outdated name is dark reactions, which can be misleading because it implies incorrectly that the reaction only occurs at night or is independent of light, which is why most scientists and instructors no longer use it.

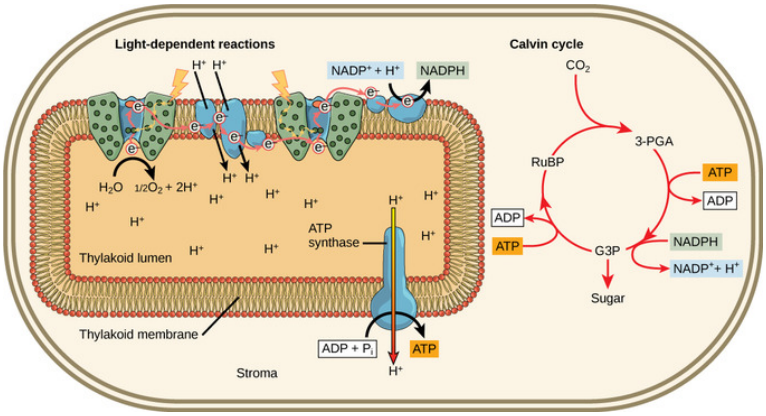


Figure 8.3B.1: **Light Reactions**: Light-dependent reactions harness energy from the sun to produce chemical bonds, ATP, and NADPH. These energy-carrying molecules are made in the stroma where the Calvin cycle takes place. The Calvin cycle is not totally independent of light since it relies on ATP and NADH, which are products of the light-dependent reactions.

The light-independent reactions of the Calvin cycle can be organized into three basic stages: fixation, reduction, and regeneration.

Stage 1: Fixation

In the stroma, in addition to CO_2,two other components are present to initiate the light-independent reactions: an enzyme called ribulose bisphosphate carboxylase (RuBisCO) and three molecules of ribulose bisphosphate (RuBP). RuBP has five atoms of carbon, flanked by two phosphates. RuBisCO catalyzes a reaction between CO_2 and RuBP. For each CO_2 molecule that reacts with one RuBP, two molecules of 3-phosphoglyceric acid (3-PGA) form. 3-PGA has three carbons and one phosphate. Each turn of the cycle involves only one RuBP and one carbon dioxide and forms two molecules of 3-PGA. The number of carbon atoms remains the same, as the atoms move to form new bonds during the reactions (3 atoms from $3CO_2$ + 15 atoms from 3RuBP = 18 atoms in 3 atoms of 3-PGA). This process is called carbon fixation because CO_2 is "fixed" from an inorganic form into organic molecules.

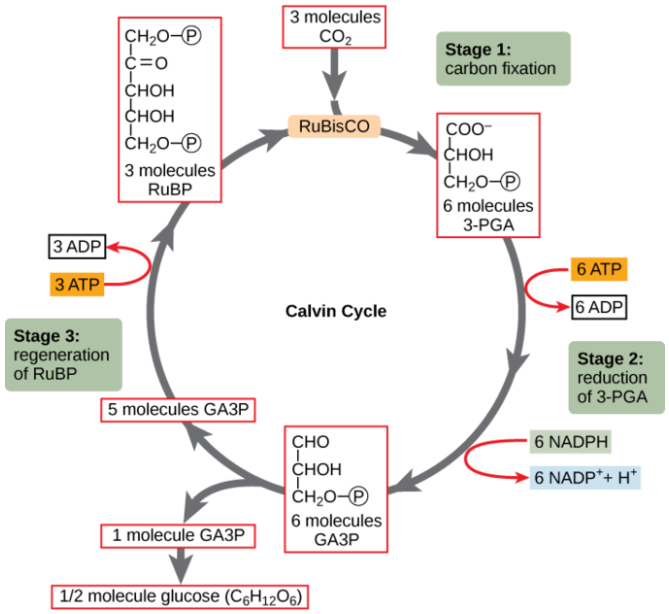


Figure 8.3*B*.1: **The Calvin Cycle**: The Calvin cycle has three stages. In stage 1, the enzyme RuBisCO incorporates carbon dioxide into an organic molecule, 3-PGA. In stage 2, the organic molecule is reduced using electrons supplied by NADPH. In stage 3, RuBP, the molecule that starts the cycle, is regenerated so that the cycle can continue. Only one carbon dioxide molecule is incorporated at a time, so the cycle must be completed three times to produce a single three-carbon GA3P molecule, and six times to produce a six-carbon glucose molecule.

Stage 2: Reduction

ATP and NADPH are used to convert the six molecules of 3-PGA into six molecules of a chemical called glyceraldehyde 3-phosphate (G3P). This is a reduction reaction because it involves the gain of electrons by 3-PGA. Recall that a reduction is the gain of an electron by an atom or molecule. Six molecules of both ATP and NADPH are used. For ATP, energy is released with the loss of the terminal phosphate atom, converting it to ADP; for NADPH, both energy and a hydrogen atom are lost, converting it into $NADP^+$. Both of these molecules return to the nearby light-dependent reactions to be reused and reenergized.

Stage 3: Regeneration

At this point, only one of the G3P molecules leaves the Calvin cycle and is sent to the cytoplasm to contribute to the formation of other compounds needed by the plant. Because the G3P exported from the chloroplast has three carbon atoms, it takes three "turns" of the Calvin cycle to fix enough net carbon to export one G3P. But each turn makes two G3Ps, thus three turns make six G3Ps. One is exported while the remaining five G3P molecules remain in the cycle and are used to regenerate RuBP, which enables the system to prepare for more CO_2 to be fixed. Three more molecules of ATP are used in these regeneration reactions.

: The Carbon Cycle

All organisms need energy to perform life functions, and energy that is released is reused in other ways.

Learning Objectives

- Describe the importance of the carbon cycle

Key Points

- Every single atom of energy is conserved by changing form or moving from one type of energy to another, so waste does not exist in nature.
- Photosynthesis absorbs light energy to build carbohydrates, and aerobic cellular respiration releases energy by using oxygen to metabolize carbohydrates.
- Photosynthesis consumes carbon dioxide and produces oxygen, and aerobic respiration consumes oxygen and produces carbon dioxide.
- Both photosynthesis and cellular respiration use electron transport chains to capture the energy necessary to drive other reactions.

Key Terms

- **heterotroph**: an organism that requires an external supply of energy in the form of food, as it cannot synthesize its own
- **cellular respiration**: the set of the metabolic reactions and processes that take place in the cells of organisms to convert biochemical energy from nutrients into adenosine triphosphate (ATP)
- **aerobic**: living or occurring only in the presence of oxygen

The Carbon Cycle

Figure $8.3C.1$: **Photosynthesis and Aerobic Respiration**: Photosynthesis consumes carbon dioxide and produces oxygen. Aerobic respiration consumes oxygen and produces carbon dioxide. These two processes play an important role in the carbon cycle.

Whether the organism is a bacterium, plant, or animal, all living things access energy by breaking down carbohydrate molecules. But if plants make carbohydrate molecules, why would they need to break them down, especially when it has been shown that the gas organisms release as a "waste product" (CO_2) acts as a substrate for the formation of more food in photosynthesis? Living things need energy to perform life functions. In addition, an organism can either make its own food or eat another organism; either way, the food still needs to be broken down. Finally, in the process of breaking down food, called cellular respiration, heterotrophs release needed energy and produce "waste" in the form of CO_2 gas.

In nature, there is no such thing as waste. Every single atom of matter and energy is conserved, recycling over and over, infinitely. Substances change form or move from one type of molecule to another, but their constituent atoms never disappear.

CO_2 is no more a form of waste than oxygen is wasteful to photosynthesis. Both are byproducts of reactions that move on to other reactions. Photosynthesis absorbs light energy to build carbohydrates in chloroplasts, and aerobic cellular respiration releases energy by using oxygen to metabolize carbohydrates in the cytoplasm and mitochondria.

ynthesis consumes carbon dioxide and produces oxygen. Aerobic respiration consumes oxygen and produces carbon e. Both processes use electron transport chains to capture the energy necessary to drive other reactions. These two powerhouse processes, photosynthesis and cellular respiration, function in biological, cyclical harmony to allow organisms to access life-sustaining energy that originates millions of miles away in the sun.

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CHAPTER OVERVIEW

9: CELL COMMUNICATION

9.1: SIGNALING MOLECULES AND CELLULAR RECEPTORS

9.1A: SIGNALING MOLECULES AND CELLULAR RECEPTORS
9.1B: FORMS OF SIGNALING
9.1C: TYPES OF RECEPTORS
9.1D: SIGNALING MOLECULES
9.2: PROPAGATION OF THE CELLULAR SIGNAL

9.2A: BINDING INITIATES A SIGNALING PATHWAY
9.2B: METHODS OF INTRACELLULAR SIGNALING
9.3: RESPONSE TO THE CELLULAR SIGNAL

9.3A: TERMINATION OF THE SIGNAL CASCADE
9.3B: CELL SIGNALING AND GENE EXPRESSION
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9.3D: CELL SIGNALING AND CELL GROWTH
Cell growth is promoted by ligands known as growth factors.

9.3E: CELL SIGNALING AND CELL DEATH
9.4: SIGNALING IN SINGLE-CELLED ORGANISMS

9.4A: SIGNALING IN YEAST
9.4B: SIGNALING IN BACTERIA

Signaling Molecules and Cellular Receptors

Topic hierarchy

- 9.1A: Signaling Molecules and Cellular Receptors
- 9.1B: Forms of Signaling
- 9.1C: Types of Receptors
- 9.1D: Signaling Molecules

9.1A: Signaling Molecules and Cellular Receptors

Cellular communication ensures regulation of biological processes within various environments from single-celled to multicellular organisms.

Learning Objectives

- Explain the importance of cell communication

Key Points

- The ability of cells to communicate through chemical signals originated in single cells and was essential for the evolution of multicellular organisms.
- In multicellular organisms, cells send and receive chemical messages constantly to coordinate the actions of distant organs, tissues, and cells.
- Cells can receive a message, transfer the information across the plasma membrane, and then produce changes within the cell in response to the message.
- Single-celled organisms, like yeast and bacteria, communicate with each other to aid in mating and coordination.
- Cellular communication has developed as a means to communicate with the environment, produce biological changes, and, if necessary, ensure survival.

Key Terms

- **biofilm**: a thin film of mucus created by and containing a colony of bacteria and other microorganisms

Introduction: Signaling Molecules and Cellular Receptors

Imagine what life would be like if you and the people around you could not communicate. You would not be able to express your wishes to others, nor could you ask questions to find out more about your environment. Social organization is dependent on communication between the individuals that comprise that society; without communication, society would fall apart.

Figure 9.1*A*.1: **Communication is Key**: Have you ever become separated from a friend while in a crowd? If so, you know the challenge of searching for someone when surrounded by thousands of other people. If you and your friend have cell phones, your chances of finding each other are good. A cell phone's ability to send and receive messages makes it an ideal communication device.

As with people, it is vital for individual cells to be able to interact with their environment. This is true whether a cell is growing by itself in a pond or is one of many cells that form a larger organism. In order to properly respond to external stimuli, cells have developed complex mechanisms of communication that can receive a message, transfer the information across the plasma membrane, and then produce changes within the cell in response to the message.

In multicellular organisms, cells send and receive chemical messages constantly to coordinate the actions of distant organs, tissues, and cells. The ability to send messages quickly and efficiently enables cells to coordinate and fine-tune their

functions.

While the necessity for cellular communication in larger organisms seems obvious, even single-celled organisms communicate with each other. Yeast cells signal each other to aid mating. Some forms of bacteria coordinate their actions in order to form large complexes called biofilms or to organize the production of toxins to remove competing organisms. The ability of cells to communicate through chemical signals originated in single cells and was essential for the evolution of multicellular organisms. The efficient and error-free function of communication systems is vital for all forms of life.

9.1B: Forms of Signaling

The major types of signaling mechanisms that occur in multicellular organisms are paracrine, endocrine, autocrine, and direct signaling.

Learning Objectives

- Describe four types of signaling found in multicellular organisms

Key Points

- Cells communicate via various types of signaling that allow chemicals to travel to target sites in order to elicit a response.
- Paracrine signaling occurs between local cells where the signals elicit quick responses and last only a short amount of time due to the degradation of the paracrine ligands.
- Endocrine signaling occurs between distant cells and is mediated by hormones released from specific endocrine cells that travel to target cells, producing a slower, long-lasting response.
- Autocrine signals are produced by signaling cells that can also bind to the ligand that is released, which means the signaling cell and the target cell can be the same or a similar cell.
- Direct signaling can occur by transferring signaling molecules across gap junctions between neighboring cells.

Key Terms

- **endocrine signaling**: signals from distant cells that originate from endocrine cells, usually producing a slow response, but having a long-lasting effect
- **autocrine signaling**: produced by signaling cells that can also bind to the ligand that is released: the signaling cell and the target cell can be the same or a similar cell (prefix auto- means self)
- **paracrine signaling**: a form of cell signaling in which the target cell is near (para = near) the signal-releasing cell

Forms of Signaling

There are four categories of chemical signaling found in multicellular organisms: paracrine signaling, endocrine signaling, autocrine signaling, and direct signaling across gap junctions. The main difference between the different categories of signaling is the distance that the signal travels through the organism to reach the target cell. It is also important to note that not all cells are affected by the same signals.

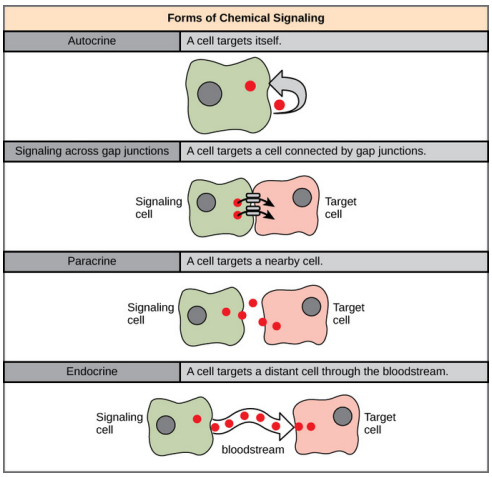

Figure 9.1*B*.1: **Forms of Chemical Signaling**: In chemical signaling, a cell may target itself (autocrine signaling), a cell connected by gap junctions, a nearby cell (paracrine signaling), or a distant cell (endocrine signaling). Paracrine signaling acts on nearby cells, endocrine signaling uses the circulatory system to transport ligands, and autocrine signaling acts on the signaling cell. Signaling via gap junctions involves signaling molecules moving directly between adjacent cells.

Paracrine Signaling

Signals that act locally between cells that are close together are called paracrine signals. Paracrine signals move by diffusion through the extracellular matrix. These types of signals usually elicit quick responses that last only a short amount of time. In order to keep the response localized, paracrine ligand molecules are normally quickly degraded by enzymes or removed by neighboring cells. Removing the signals will reestablish the concentration gradient for the signal, allowing them to quickly diffuse through the intracellular space if released again.

One example of paracrine signaling is the transfer of signals across synapses between nerve cells. A nerve cell consists of a cell body, several short, branched extensions called dendrites that receive stimuli, and a long extension called an axon, which transmits signals to other nerve cells or muscle cells. The junction between nerve cells where signal transmission occurs is called a synapse. A synaptic signal is a chemical signal that travels between nerve cells. Signals within the nerve cells are propagated by fast-moving electrical impulses. When these impulses reach the end of the axon, the signal continues on to a dendrite of the next cell by the release of chemical ligands called neurotransmitters by the presynaptic cell (the cell emitting the signal). The neurotransmitters are transported across the very small distances between nerve cells, which are called chemical synapses. The small distance between nerve cells allows the signal to travel quickly; this enables an immediate response.

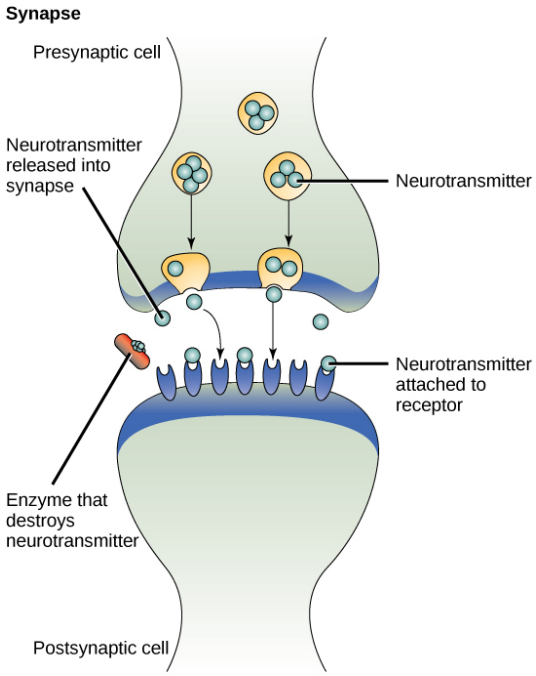

Figure 9.1B.1: **Synapsis**: The distance between the presynaptic cell and the postsynaptic cell—called the synaptic gap—is very small and allows for rapid diffusion of the neurotransmitter. Enzymes in the synapatic cleft degrade some types of neurotransmitters to terminate the signal.

Endocrine Signaling

Signals from distant cells are called endocrine signals; they originate from endocrine cells. In the body, many endocrine cells are located in endocrine glands, such as the thyroid gland, the hypothalamus, and the pituitary gland. These types of signals usually produce a slower response, but have a longer-lasting effect. The ligands released in endocrine signaling are called hormones, signaling molecules that are produced in one part of the body, but affect other body regions some distance away.

Hormones travel the large distances between endocrine cells and their target cells via the bloodstream, which is a relatively slow way to move throughout the body. Because of their form of transport, hormones get diluted and are present in low concentrations when they act on their target cells. This is different from paracrine signaling in which local concentrations of ligands can be very high.

Autocrine Signaling

Autocrine signals are produced by signaling cells that can also bind to the ligand that is released. This means the signaling cell and the target cell can be the same or a similar cell (the prefix auto- means self, a reminder that the signaling cell sends a signal to itself). This type of signaling often occurs during the early development of an organism to ensure that cells develop into the correct tissues and take on the proper function. Autocrine signaling also regulates pain sensation and inflammatory responses. Further, if a cell is infected with a virus, the cell can signal itself to undergo programmed cell death, killing the virus in the process. In some cases, neighboring cells of the same type are also influenced by the released ligand. In embryological development, this process of stimulating a group of neighboring cells may help to direct the differentiation of identical cells into the same cell type, thus ensuring the proper developmental outcome.

Direct Signaling Across Gap Junctions

Gap junctions in animals and plasmodesmata in plants are connections between the plasma membranes of neighboring cells. These water-filled channels allow small signaling molecules, called intracellular mediators, to diffuse between the two cells. Small molecules, such as calcium ions (Ca^{2+}), are able to move between cells, but large molecules, like proteins and DNA, cannot fit through the channels. The specificity of the channels ensures that the cells remain independent, but can quickly and easily transmit signals. The transfer of signaling molecules communicates the current state of the cell that is directly next to the target cell; this allows a group of cells to coordinate their response to a signal that only one of them may have received. In plants, plasmodesmata are ubiquitous, making the entire plant into a giant communication network.

9.1C: Types of Receptors

Receptors, either intracellular or cell-surface, bind to specific ligands, which activate numerous cellular processes.

Learning Objectives

- Compare internal receptors with cell-surface receptors

Key Points

- Intracellular receptors are located in the cytoplasm of the cell and are activated by hydrophobic ligand molecules that can pass through the plasma membrane.
- Cell-surface receptors bind to an external ligand molecule and convert an extracellular signal into an intracellular signal.
- Three general categories of cell-surface receptors include: ion -channel, G- protein, and enzyme -linked protein receptors.
- Ion channel -linked receptors bind a ligand and open a channel through the membrane that allows specific ions to pass through.
- G-protein-linked receptors bind a ligand and activate a membrane protein called a G-protein, which then interacts with either an ion channel or an enzyme in the membrane.
- Enzyme-linked receptors are cell-surface receptors with intracellular domains that are associated with an enzyme.

Key Terms

- **integral protein**: a protein molecule (or assembly of proteins) that is permanently attached to the biological membrane
- **transcription**: the synthesis of RNA under the direction of DNA

Types of Receptors

Receptors are protein molecules in the target cell or on its surface that bind ligands. There are two types of receptors: internal receptors and cell-surface receptors.

Internal receptors

Internal receptors, also known as intracellular or cytoplasmic receptors, are found in the cytoplasm of the cell and respond to hydrophobic ligand molecules that are able to travel across the plasma membrane. Once inside the cell, many of these molecules bind to proteins that act as regulators of mRNA synthesis to mediate gene expression. Gene expression is the cellular process of transforming the information in a cell's DNA into a sequence of amino acids that ultimately forms a protein. When the ligand binds to the internal receptor, a conformational change exposes a DNA-binding site on the protein. The ligand-receptor complex moves into the nucleus, binds to specific regulatory regions of the chromosomal DNA, and promotes the initiation of transcription. Internal receptors can directly influence gene expression without having to pass the signal on to other receptors or messengers.

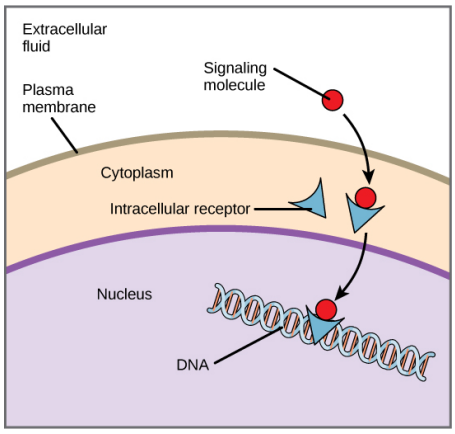

Figure 9.1C.1: **Intracellular Receptors**: Hydrophobic signaling molecules typically diffuse across the plasma membrane and interact with intracellular receptors in the cytoplasm. Many intracellular receptors are transcription factors that interact with DNA in the nucleus and regulate gene expression.

Cell-Surface Receptors

Cell-surface receptors, also known as transmembrane receptors, are cell surface, membrane-anchored, or integral proteins that bind to external ligand molecules. This type of receptor spans the plasma membrane and performs signal transduction, converting an extracellular signal into an intracellular signal. Ligands that interact with cell-surface receptors do not have to enter the cell that they affect. Cell-surface receptors are also called cell-specific proteins or markers because they are specific to individual cell types.

Each cell-surface receptor has three main components: an external ligand-binding domain (extracellular domain), a hydrophobic membrane-spanning region, and an intracellular domain inside the cell. The size and extent of each of these domains vary widely, depending on the type of receptor.

Cell-surface receptors are involved in most of the signaling in multicellular organisms. There are three general categories of cell-surface receptors: ion channel-linked receptors, G-protein-linked receptors, and enzyme-linked receptors.

Ion Channel-Linked Receptors

Ion channel-linked receptors bind a ligand and open a channel through the membrane that allows specific ions to pass through. To form a channel, this type of cell-surface receptor has an extensive membrane-spanning region. In order to interact with the phospholipid fatty acid tails that form the center of the plasma membrane, many of the amino acids in the membrane-spanning region are hydrophobic in nature. Conversely, the amino acids that line the inside of the channel are hydrophilic to allow for the passage of water or ions. When a ligand binds to the extracellular region of the channel, there is a conformational change in the protein's structure that allows ions such as sodium, calcium, magnesium, and hydrogen to pass through.

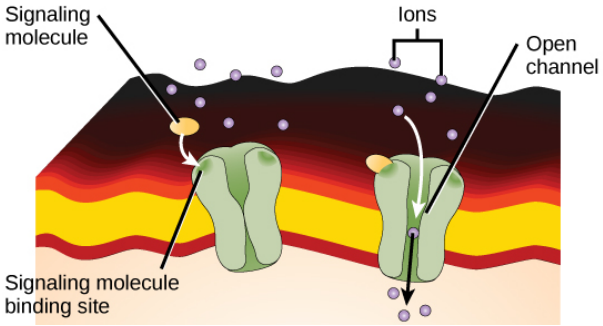

Figure 9.1C.1: **Gated-Ion Channels**: Gated ion channels form a pore through the plasma membrane that opens when the signaling molecule binds. The open pore then allows ions to flow into or out of the cell.

G-Protein Linked Receptors

G-protein-linked receptors bind a ligand and activate a membrane protein called a G-protein. The activated G-protein then interacts with either an ion channel or an enzyme in the membrane. All G-protein-linked receptors have seven transmembrane domains, but each receptor has its own specific extracellular domain and G-protein-binding site.

Cell signaling using G-protein-linked receptors occurs as a cyclic series of events. Before the ligand binds, the inactive G-protein can bind to a newly-revealed site on the receptor specific for its binding. Once the G-protein binds to the receptor, the resultant shape change activates the G-protein, which releases GDP and picks up GTP. The subunits of the G-protein then split into the α subunit and the β subunit. One or both of these G-protein fragments may be able to activate other proteins as a result. Later, the GTP on the active α subunit of the G-protein is hydrolyzed to GDP and the β subunit is deactivated. The subunits reassociate to form the inactive G-protein, and the cycle starts over.

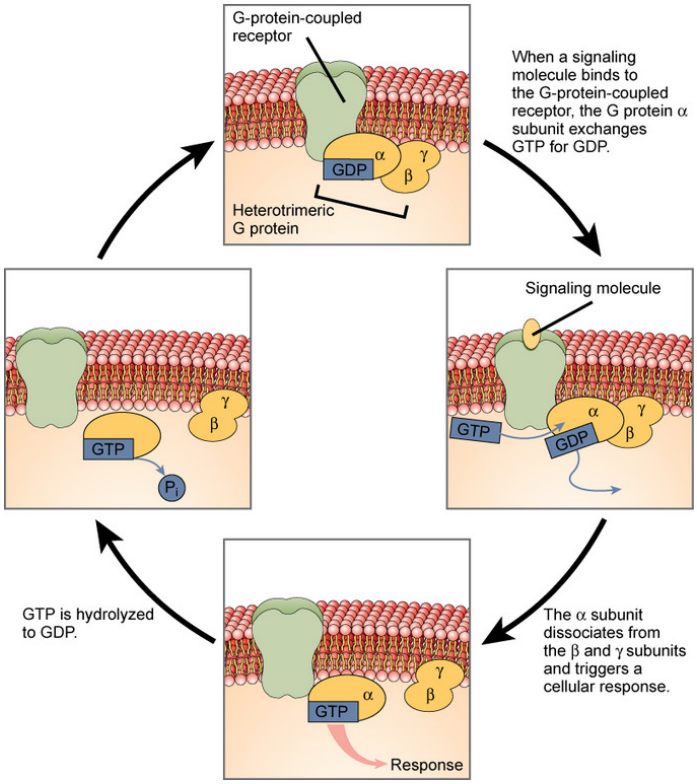

Figure 9.1C.1: **G-proteins**: Heterotrimeric G proteins have three subunits: α, β, and γ. When a signaling molecule binds to a G-protein-coupled receptor in the plasma membrane, a GDP molecule associated with the α subunit is exchanged for GTP. The β and γ subunits dissociate from the α subunit, and a cellular response is triggered either by the α subunit or the dissociated β pair. Hydrolysis of GTP to GDP terminates the signal.

Enzyme-Linked Receptors

Enzyme-linked receptors are cell-surface receptors with intracellular domains that are associated with an enzyme. In some cases, the intracellular domain of the receptor itself is an enzyme or the enzyme-linked receptor has an intracellular domain that interacts directly with an enzyme. The enzyme-linked receptors normally have large extracellular and intracellular domains, but the membrane-spanning region consists of a single alpha-helical region of the peptide strand. When a ligand binds to the extracellular domain, a signal is transferred through the membrane and activates the enzyme, which sets off a chain of events within the cell that eventually leads to a response. An example of this type of enzyme-linked receptor is the tyrosine kinase receptor. The tyrosine kinase receptor transfers phosphate groups to tyrosine molecules. Signaling molecules bind to the extracellular domain of two nearby tyrosine kinase receptors, which then dimerize. Phosphates are then added to tyrosine residues on the intracellular domain of the receptors and can then transmit the signal to the next messenger within the cytoplasm.

9.1D: Signaling Molecules

Signaling molecules are necessary for the coordination of cellular responses by serving as ligands and binding to cell receptors.

Learning Objectives

- Compare and contrast the different types of signaling molecules: hydrophobic, water-soluble, and gas ligands

Key Points

- Signaling molecules can range from small proteins to small ions and can be hydrophobic, water-soluble, or even a gas.
- Hydrophobic signaling molecules (ligands) can diffuse through the plasma membrane and bind to internal receptors.
- Water-soluble ligands are unable to pass freely through the plasma membrane due to their polarity and must bind to an extracellular domain of a cell -surface receptor.
- Other types of ligands can include gases, such as nitric oxide, which can freely diffuse through the plasma membrane and bind to internal receptors.

Key Terms

- **ligand**: an ion, molecule, or functional group that binds to another chemical entity to form a larger complex
- **hydrophobic**: lacking an affinity for water; unable to absorb, or be wetted by water

Signaling Molecules

Produced by signaling cells and the subsequent binding to receptors in target cells, ligands act as chemical signals that travel to the target cells to coordinate responses. The types of molecules that serve as ligands are incredibly varied and range from small proteins to small ions like calcium (Ca^{2+}).

Small Hydrophobic Ligands

Small hydrophobic ligands can directly diffuse through the plasma membrane and interact with internal receptors. Important members of this class of ligands are the steroid hormones. Steroids are lipids that have a hydrocarbon skeleton with four fused rings; different steroids have different functional groups attached to the carbon skeleton. Steroid hormones include the female sex hormone, estradiol, which is a type of estrogen; the male sex hormone, testosterone; and cholesterol, which is an important structural component of biological membranes and a precursor of steriod hormones. Other hydrophobic hormones include thyroid hormones and vitamin D. In order to be soluble in blood, hydrophobic ligands must bind to carrier proteins while they are being transported through the bloodstream.

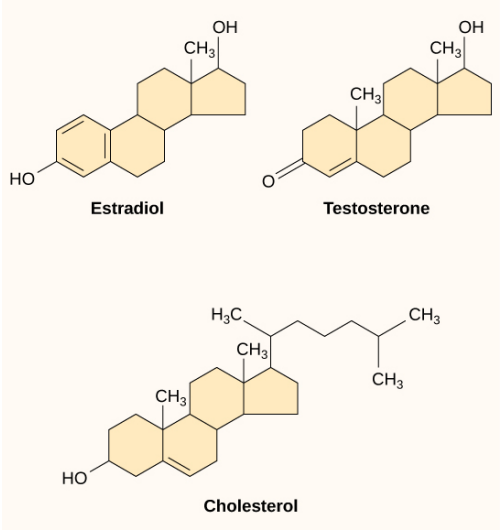

Figure 9.1D.1: **Steroid Hormones**: Steroid hormones have similar chemical structures to their precursor, cholesterol. Because these molecules are small and hydrophobic, they can diffuse directly across the plasma membrane into the cell, where they interact with internal receptors.

Water-Soluble Ligands

Water-soluble ligands are polar and, therefore, cannot pass through the plasma membrane unaided; sometimes, they are too large to pass through the membrane at all. Instead, most water-soluble ligands bind to the extracellular domain of cell-surface receptors. Cell-surface receptors include: ion-channel, G-protein, and enzyme-linked protein receptors. The binding of these ligands to these receptors results in a series of cellular changes. These water soluble ligands are quite diverse and include small molecules, peptides, and proteins.

Other Ligands

Nitric oxide (NO) is a gas that also acts as a ligand. It is able to diffuse directly across the plasma membrane; one of its roles is to interact with receptors in smooth muscle and induce relaxation of the tissue. NO has a very short half-life; therefore, it only functions over short distances. Nitroglycerin, a treatment for heart disease, acts by triggering the release of NO, which causes blood vessels to dilate (expand), thus restoring blood flow to the heart.

Propagation of the Cellular Signal

Topic hierarchy

- 9.2A: Binding Initiates a Signaling Pathway
- 9.2B: Methods of Intracellular Signaling

: Binding Initiates a Signaling Pathway

Ligand binding to cell-surface receptors activates the receptor's intracellular components setting off a signaling pathway or cascade.

Learning Objectives

- Recognize the relationship between a ligand's structure and its mechanism of action.

Key Points

- Signaling pathways can be complicated since most cellular proteins can affect different downstream events.
- Cell -surface receptors are integral in signaling pathways.
- Ion channel -linked receptors open a channel once a ligand binds allowing specific ions to pass through the membrane.
- G-protein-linked receptors activate a membrane protein called a G-protein once a ligand binds.
- Enzyme -linked receptors are cell-surface receptors with intracellular domains.

Key Terms

- **ligand**: an ion, molecule, or functional group that binds to another chemical entity to form a larger complex
- **receptor**: a protein on a cell wall that binds with specific molecules so that they can be absorbed into the cell in order to control certain functions

Cell-surface receptors, also known as transmembrane receptors, are membrane-anchored (integral) proteins that bind to external ligand molecules. This type of receptor spans the plasma membrane and performs signal transduction in which an extracellular signal is converted into an intracellular signal. Ligands that interact with cell-surface receptors do not have to enter the cell that they affect. Cell-surface receptors are also called cell-specific proteins or markers because they are specific to individual cell types. Each cell-surface receptor has three main components: an external ligand-binding domain, a hydrophobic membrane-spanning region, and an intracellular domain inside the cell. The ligand-binding domain is also called the extracellular domain. The size and extent of each of these domains vary widely, depending on the type of receptor. Cell-surface receptors are involved in most of the signaling in multicellular organisms.

There are three general categories of cell-surface receptors: ion channel-linked receptors, G-protein-linked receptors, and enzyme-linked receptors.

1. Ion channel-linked receptors bind a ligand and open a channel through the membrane that allows specific ions to pass through. To form a channel, this type of cell-surface receptor has an extensive membrane-spanning region. When a ligand binds to the extracellular region of the channel, there is a conformational change in the protein's structure that allows ions such as sodium, calcium, magnesium, and hydrogen to pass through.
2. G-protein-linked receptors bind a ligand and activate a membrane protein called a G-protein. The activated G-protein then interacts with either an ion channel or an enzyme in the membrane. All G-protein-linked receptors have seven transmembrane domains, but each receptor has its own specific extracellular domain and G-protein-binding site.
3. Enzyme-linked receptors are cell-surface receptors with intracellular domains that are associated with an enzyme. In some cases, the intracellular domain of the receptor itself is an enzyme. Other enzyme-linked receptors have a small intracellular domain that interacts directly with an enzyme. When a ligand binds to the extracellular domain, a signal is transferred through the membrane, activating the enzyme. Activation of the enzyme sets off a chain of events within the cell that eventually leads to a response.

After the ligand binds to the cell-surface receptor, the activation of the receptor's intracellular components sets off a chain of events that is called a signaling pathway or a signaling cascade. In a signaling pathway, second messengers, enzymes, and activated proteins interact with specific proteins, which are in turn activated in a chain reaction that eventually leads to a change in the cell's environment. The events in the cascade occur in a series, much like a current flows in a river. Interactions that occur before a certain point are defined as upstream events; events after that point are called downstream events.

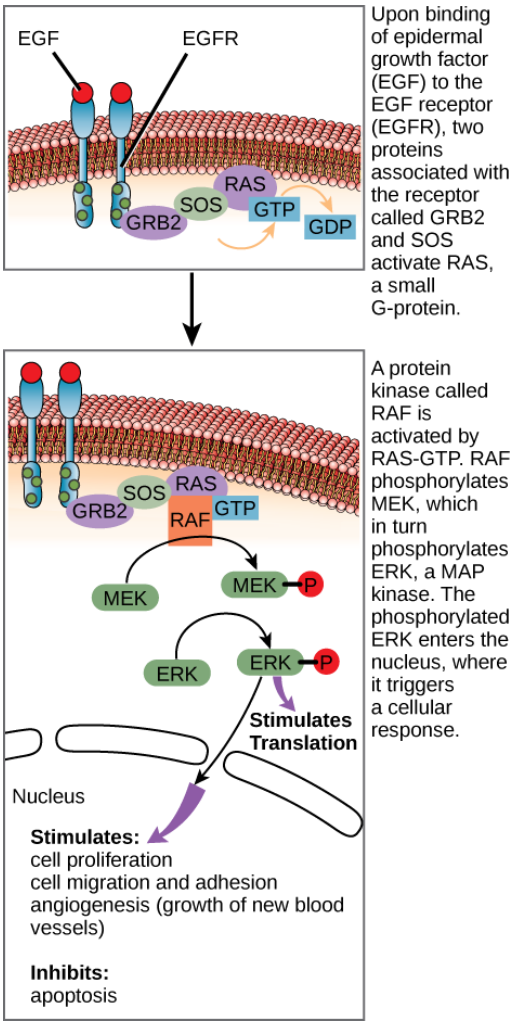

Figure 9.2*A*.1: **Ligand Initiated Signaling Pathway**: An example of ligand initiated signaling pathways is when epidermal growth factor (EGF) binds to its receptor. A complex cascade of downstream events causes the cell to grow and divide.

Signaling pathways can get very complicated very quickly because most cellular proteins can affect different downstream events, depending on the conditions within the cell. A single pathway can branch off toward different endpoints based on the interplay between two or more signaling pathways. The same ligands are often used to initiate different signals in different cell types. This variation in response is due to differences in protein expression in different cell types. Another complicating element is signal integration of the pathways in which signals from two or more different cell-surface receptors merge to activate the same response in the cell. This process can ensure that multiple external requirements are met before a cell commits to a specific response.

The effects of extracellular signals can also be amplified by enzymatic cascades. At the initiation of the signal, a single ligand binds to a single receptor. However, activation of a receptor-linked enzyme can activate many copies of a component of the signaling cascade, which amplifies the signal.

9.2B: Methods of Intracellular Signaling

Signaling pathway induction activates a sequence of enzymatic modifications that are recognized in turn by the next component downstream.

Learning Objectives

- Explain how the binding of a ligand initiates signal transduction throughout a cell

Key Points

- Phosphorylation, the addition of a phosphate group to a molecule such as a protein, is one of the most common chemical modifications that occurs in signaling pathways.
- The activation of second messengers, small molecules that propagate a signal, is a common event after the induction of a signaling pathway.
- Calcium ion, cyclic AMP, and inositol phospholipids are examples of widely-used second messengers.

Key Terms

- **second messenger**: any substance used to transmit a signal within a cell, especially one which triggers a cascade of events by activating cellular components
- **phosphorylation**: the addition of a phosphate group to a compound; often catalyzed by enzymes

The induction of a signaling pathway depends on the modification of a cellular component by an enzyme. There are numerous enzymatic modifications that can occur which are recognized in turn by the next component downstream.

One of the most common chemical modifications that occurs in signaling pathways is the addition of a phosphate group (PO_4^{-3}) to a molecule such as a protein in a process called phosphorylation. The phosphate can be added to a nucleotide such as GMP to form GDP or GTP. Phosphates are also often added to serine, threonine, and tyrosine residues of proteins where they replace the hydroxyl group of the amino acid. The transfer of the phosphate is catalyzed by an enzyme called a kinase. Various kinases are named for the substrate they phosphorylate. Phosphorylation of serine and threonine residues often activates enzymes. Phosphorylation of tyrosine residues can either affect the activity of an enzyme or create a binding site that interacts with downstream components in the signaling cascade. Phosphorylation may activate or inactivate enzymes; the reversal of phosphorylation, dephosphorylation by a phosphatase, will reverse the effect.

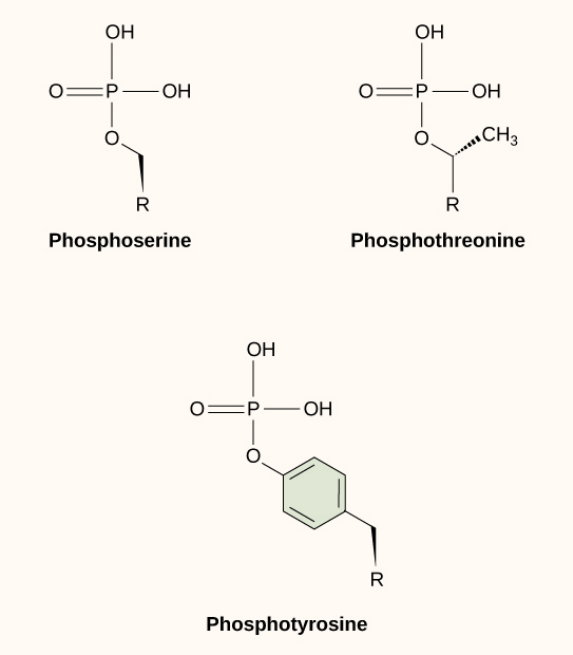

Figure 9.2*B*.1: **Example of phosphorylation**: In protein phosphorylation, a phosphate group (PO4-3) is added to residues of the amino acids serine, threonine, and tyrosine.

The activation of second messengers is also a common event after the induction of a signaling pathway. They are small molecules that propagate a signal after it has been initiated by the binding of the signaling molecule to the receptor. These molecules help to spread a signal through the cytoplasm by altering the behavior of certain cellular proteins.

Calcium ion is a widely-used second messenger. The free concentration of calcium ions (Ca^{2+}) within a cell is very low because ion pumps in the plasma membrane continuously use adenosine-5′-triphosphate (ATP) to remove it. For signaling purposes, Ca^{2+} is stored in cytoplasmic vesicles, such as the endoplasmic reticulum, or accessed from outside the cell. When signaling occurs, ligand-gated calcium ion channels allow the higher levels of Ca^{2+} that are present outside the cell (or in intracellular storage compartments) to flow into the cytoplasm, which raises the concentration of cytoplasmic Ca^{2+}. The response to the increase in Ca^{2+} varies, depending on the cell type involved. For example, in the β-cells of the pancreas, Ca^{2+} signaling leads to the release of insulin, whereas in muscle cells, an increase in Ca^{2+} leads to muscle contractions.

Another second messenger utilized in many different cell types is cyclic AMP (cAMP). Cyclic AMP is synthesized by the enzyme adenylyl cyclase from ATP. The main role of cAMP in cells is to bind to and activate an enzyme called cAMP-dependent kinase (A-kinase). A-kinase regulates many vital metabolic pathways. It phosphorylates serine and threonine residues of its target proteins, activating them in the process. A-kinase is found in many different types of cells; the target proteins in each kind of cell are different. Differences give rise to the variation of the responses to cAMP in different cells.

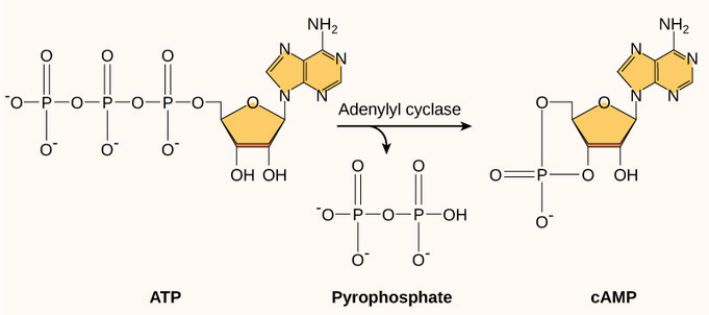

Figure 9.2B.1: **Example of cAMP as a second messenger**: This diagram shows the mechanism for the formation of cyclic AMP (cAMP). cAMP serves as a second messenger to activate or inactivate proteins within the cell. Termination of the signal occurs when an enzyme called phosphodiesterase converts cAMP into AMP.

Present in small concentrations in the plasma membrane, inositol phospholipids are lipids that can also be converted into second messengers. Because these molecules are membrane components, they are located near membrane-bound receptors and can easily interact with them. Phosphatidylinositol (PI) is the main phospholipid that plays a role in cellular signaling. Enzymes known as kinases phosphorylate PI to form PI-phosphate (PIP) and PI-bisphosphate (PIP_2).

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9.3: Response to the Cellular Signal

Topic hierarchy

- 9.3A: Termination of the Signal Cascade
- 9.3B: Cell Signaling and Gene Expression
- 9.3C: Cell Signaling and Cellular Metabolism
- 9.3D: Cell Signaling and Cell Growth

Cell growth is promoted by ligands known as growth factors.

- 9.3E: Cell Signaling and Cell Death

9.3A: Termination of the Signal Cascade

Signal cascades convey signals to the cell through the phosphorylation of molecules by kinases.

Learning Objectives

- Describe the process by which the signal cascade in cell communication is terminated

Key Points

- The chain of events that conveys the signal through the cell is called a signaling pathway or cascade.
- Phosphorylation, a major component of signal cascades, adds a phosphate group to proteins, thereby changing their shapes and activating or inactivating the protein.
- Degrading or removing the ligand so it can no longer access its receptor terminates the signal.
- Enzymes like phosphotases can remove phosphate groups on proteins during dephosphorylation and reverse the cellular modifications produced by signaling cascades.

Key Terms

- **signaling cascade**: the chain of events that conveys the signal through the cell
- **phosphorylation**: the addition of a phosphate group to a compound; often catalyzed by enzymes
- **dephosphorylation**: the removal of phosphate groups from a compound; often catalyzed by enzymes

Termination of the Signal Cascade

Ligand binding to the receptor allows for signal transduction through the cell. The chain of events that conveys the signal through the cell is called a signaling pathway or cascade. Signaling pathways are often very complex because of the interplay between different proteins. A major component of cell signaling cascades is the phosphorylation of molecules by enzymes known as kinases. Phosphorylation adds a phosphate group to serine, threonine, and tyrosine residues in a protein, changing their shapes, and activating or inactivating the protein.

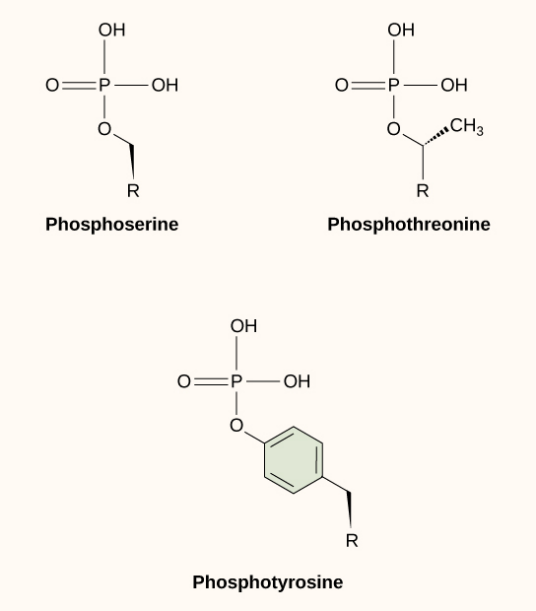

Figure 9.3A.1: **Phosphorylation**: In protein phosphorylation, a phosphate group is added to residues of the amino acids serine, threonine, and tyrosine.

The aberrant signaling often seen in tumor cells is proof that the termination of a signal at the appropriate time can be just as important as the initiation of a signal. One method of terminating or stopping a specific signal is to degrade or remove the ligand so that it can no longer access its receptor. One reason that hydrophobic hormones like estrogen and testosterone trigger

long-lasting events is because they bind carrier proteins. These proteins allow the insoluble molecules to be soluble in blood, but they also protect the hormones from degradation by circulating enzymes.

Inside the cell, many different enzymes reverse the cellular modifications that result from signaling cascades. For example, phosphatases are enzymes that remove the phosphate group attached to proteins by kinases in a process called dephosphorylation. Cyclic AMP (cAMP) is degraded into AMP by phosphodiesterase, and the release of calcium stores is reversed by the Ca^{2+} pumps that are located in the external and internal membranes of the cell.

3: Cell Signaling and Gene Expression

Gene expression, vital for cells to function properly, is the process of turning on a gene to produce RNA and protein.

Learning Objectives

- Describe the regulation of gene expression

Key Points

- Each cell controls when and how its genes are expressed.
- Malfunctions in the control of gene expression are detrimental to the cell and can lead to the development of many diseases, such as cancer.
- In prokaryotic cells, the control of gene expression is mostly at the transcriptional level.
- In eukaryotic cells, the control of gene expression is at the epigenetic, transcriptional, post-transcriptional, translational, and post-translational levels.

Key Terms

- **translation**: a process occurring in the ribosome in which a strand of messenger RNA (mRNA) guides assembly of a sequence of amino acids to make a protein
- **gene expression**: the transcription and translation of a gene into messenger RNA and, thus, into a protein
- **transcription**: the synthesis of RNA under the direction of DNA

Gene Expression

For a cell to function properly, necessary proteins must be synthesized at the proper time. All cells control or regulate the synthesis of proteins from information encoded in their DNA. The process of turning on a gene to produce RNA and protein is called gene expression.

Whether in a simple unicellular organism or a complex multi-cellular organism, each cell controls when and how its genes are expressed. For this to occur, there must be a mechanism to control when a gene is expressed to make RNA and protein; how much of the protein is made; and when it is time to stop making that protein because it is no longer needed. The regulation of gene expression conserves energy and space. It would require a significant amount of energy for an organism to express every gene at all times, so it is more energy efficient to turn on the genes only when they are required. In addition, only expressing a subset of genes in each cell saves space because DNA must be unwound from its tightly-coiled structure to transcribe and translate the DNA. Cells would have to be enormous if every protein were expressed in every cell all the time. The control of gene expression is extremely complex. Malfunctions in this process are detrimental to the cell and can lead to the development of many diseases, including cancer.

Prokaryotic versus Eukaryotic Gene Expression

To understand how gene expression is regulated, we must first understand how a gene codes for a functional protein in a cell. The process occurs in both prokaryotic and eukaryotic cells, just in slightly different manners. Prokaryotic organisms are single-celled organisms that lack a cell nucleus; their DNA floats freely in the cell cytoplasm. To synthesize a protein, the processes of transcription and translation occur almost simultaneously. When the resulting protein is no longer needed, transcription stops. As a result, the primary method to control what type of protein and how much of each protein is expressed in a prokaryotic cell is the regulation of DNA transcription. All of the subsequent steps occur automatically. When more protein is required, more transcription occurs. Therefore, in prokaryotic cells, the control of gene expression is mostly at the transcriptional level.

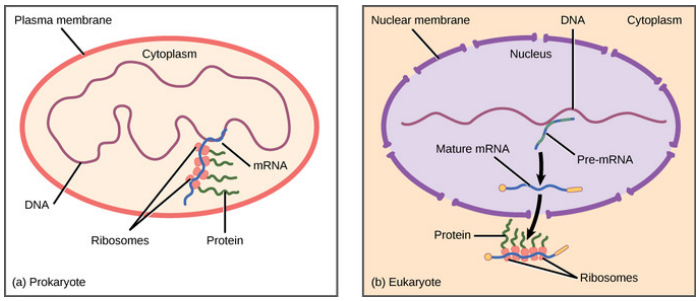

Figure $9.3B.1$: **Prokaryotic & Eukaryotic Gene Expression**: Prokaryotic transcription and translation occur simultaneously in the cytoplasm; regulation occurs at the transcriptional level. Eukaryotic gene expression is regulated during transcription and RNA processing, which take place in the nucleus, and during protein translation, which takes place in the cytoplasm. Further regulation may occur through post-translational modifications of proteins.

Eukaryotic cells, in contrast, have intracellular organelles that add to their complexity. In eukaryotic cells, the DNA is contained inside the cell's nucleus where it is transcribed into RNA. The newly-synthesized RNA is then transported out of the nucleus into the cytoplasm where ribosomes translate the RNA into protein. The processes of transcription and translation are physically separated by the nuclear membrane: transcription occurs only within the nucleus, and translation occurs only outside the nucleus in the cytoplasm. The regulation of gene expression can occur at all stages of the process. Regulation may occur when the DNA is uncoiled and loosened from nucleosomes to bind transcription factors (epigenetic level); when the RNA is transcribed (transcriptional level); when the RNA is processed and exported to the cytoplasm after it is transcribed (post-transcriptional level); when the RNA is translated into protein (translational level); or after the protein has been made (post-translational level).

The rush of adrenaline that leads to greater glucose availability is an example of an increase in metabolism.

Learning Objectives

- Explain how cellular metabolism can be altered

Key Points

- The activation of β-adrenergic receptors in muscle cells by adrenaline leads to an increase in cyclic AMP.
- Cyclic AMP activates PKA (protein kinase A), which phosphorylates two enzymes.
- Phophorylation of the first enzyme promotes the degradation of glycogen by activating intermediate GPK that in turn activates GP, which catabolizes glycogen into glucose.
- Phosphorylation of the second enzyme, glycogen synthase (GS), inhibits its ability to form glycogen from glucose.
- The inhibition of glucose to form glycogen prevents a futile cycle of glycogen degradation and synthesis, so glucose is then available for use by the muscle cell.

Key Terms

- **cyclic adenosine monophosphate**: cAMP, a second messenger derived from ATP that is involved in the activation of protein kinases and regulates the effects of adrenaline
- **epinephrine**: (adrenaline) an amino acid-derived hormone secreted by the adrenal gland in response to stress
- **protein kinase A**: a family of enzymes whose activity is dependent on cellular levels of cyclic AMP (cAMP)

Increase in Cellular Metabolism

As the environments of most organisms are constantly changing, the reactions of metabolism must be finely regulated to maintain a constant set of conditions within cells. Metabolic regulation also allows organisms to respond to signals and interact actively with their environments. Two closely-linked concepts are important for understanding how metabolic pathways are controlled. Firstly, the *regulation* of an enzyme in a pathway is how its activity is increased and decreased in response to signals. Secondly, the *control*exerted by this enzyme is the effect that these changes in its activity have on the overall rate of the pathway. For example, an enzyme may show large changes in activity (*i.e.* it is highly regulated), but if these changes have little effect on the rate of a metabolic pathway, then this enzyme is not involved in the control of the pathway.

The result of one such signaling pathway affects muscle cells and is a good example of an increase in cellular metabolism. The activation of β-adrenergic receptors in muscle cells by adrenaline leads to an increase in cyclic adenosine monophosphate (also known as cyclic AMP or cAMP) inside the cell. Also known as epinephrine, adrenaline is a hormone (produced by the adrenal gland attached to the kidney) that prepares the body for short-term emergencies. Cyclic AMP activates PKA (protein kinase A), which in turn phosphorylates two enzymes. The first enzyme promotes the degradation of glycogen by activating intermediate glycogen phosphorylase kinase (GPK) that in turn activates glycogen phosphorylase (GP), which catabolizes glycogen into glucose. (Recall that your body converts excess glucose to glycogen for short-term storage. When energy is needed, glycogen is quickly reconverted to glucose.) Phosphorylation of the second enzyme, glycogen synthase (GS), inhibits its ability to form glycogen from glucose. In this manner, a muscle cell obtains a ready pool of glucose by activating its formation via glycogen degradation and by inhibiting the use of glucose to form glycogen, thus preventing a futile cycle of glycogen degradation and synthesis. The glucose is then available for use by the muscle cell in response to a sudden surge of adrenaline—the “fight or flight” reflex.

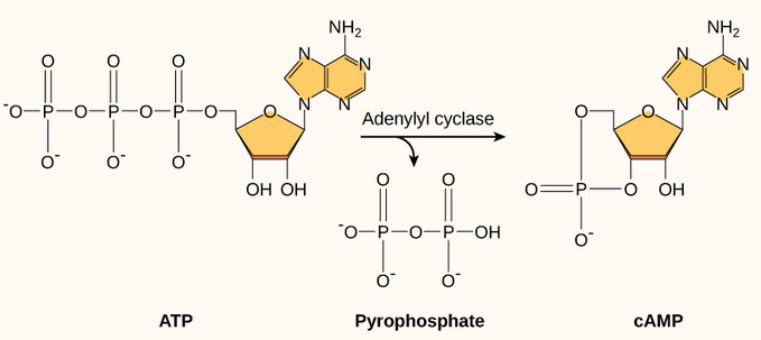

Figure $9.3C.1$: **Formation of Cyclic AMP**: This diagram shows the mechanism for the formation of cyclic AMP (cAMP). cAMP serves as a second messenger to activate or inactivate proteins within the cell.

9.3D: Cell Signaling and Cell Growth

Learning Objectives

- Explain how cell growth is affected by growth factors.

Cell Growth

Cell signaling pathways play a major role in cell division. Cells do not normally divide unless they are stimulated by signals from other cells. The ligands that promote cell growth are called growth factors. Most growth factors bind to cell-surface receptors that are linked to tyrosine kinases. These cell-surface receptors are called receptor tyrosine kinases (RTKs). Activation of RTKs initiates a signaling pathway that includes a G-protein called RAS, which activates the MAP kinase pathway described earlier. The enzyme MAP kinase then stimulates the expression of proteins that interact with other cellular components to initiate cell division. In addition, uncontrolled cell growth leads to cancer; mutations in the genes encoding protein components of signaling pathways are often found in tumor cells.

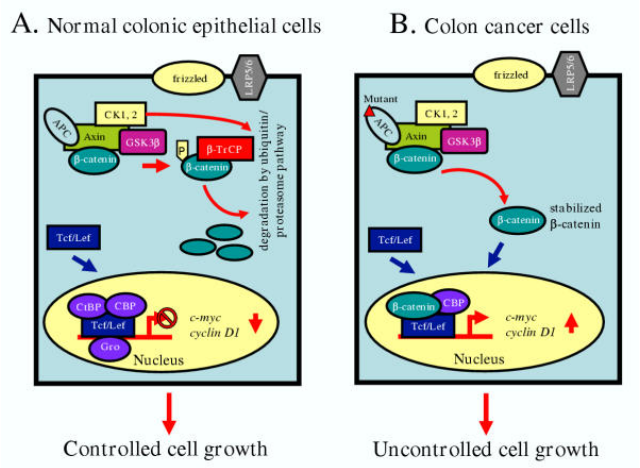

Figure 9.3D.1: **Uncontrolled Cell Growth**: Colorectal cancer occurs after numerous mutations to a normal cell.

Cancer Biologists & Uncontrolled Cell Growth

Cancer biologists study the molecular origins of cancer with the goal of developing new prevention methods and treatment strategies that will inhibit the growth of tumors without harming the normal cells of the body. Signaling pathways control cell growth. These pathways are controlled by signaling proteins, which are, in turn, expressed by genes. Mutations in these genes can result in malfunctioning signaling proteins. This prevents the cell from regulating its cell cycle, triggering unrestricted cell division and cancer. The genes that regulate the signaling proteins are one type of oncogene: a gene that has the potential to cause cancer. The gene encoding RAS is an oncogene that was originally discovered when mutations in the RAS protein were linked to cancer. Further studies have indicated that 30 percent of cancer cells have a mutation in the RAS gene that leads to uncontrolled growth. If left unchecked, uncontrolled cell division can lead tumor formation and metastasis, the growth of cancer cells in new locations in the body.

Cancer biologists have been able to identify many other oncogenes that contribute to the development of cancer. For example, HER2 is a cell-surface receptor that is present in excessive amounts in 20 percent of human breast cancers. Cancer biologists realized that gene duplication led to HER2 overexpression in 25 percent of breast cancer patients and developed a drug called Herceptin (trastuzumab), a monoclonal antibody that targets HER2 for removal by the immune system. Herceptin therapy helps to control signaling through HER2. Its use, in combination with chemotherapy, has helped to increase the overall survival rate of patients with metastatic breast cancer.

Key Points

- Normally, cells do not divide unless they are stimulated by signals from other cells.
- Most growth factors, which promote cell growth, bind to cell-surface receptors that are linked to tyrosine kinases.
- MAP kinase stimulates the expression of proteins that interact with other cellular components to initiate cell division.

- Uncontrolled cell growth leads to cancer.

Key Terms

- **receptor**: a protein on a cell wall that binds with specific molecules so that they can be absorbed into the cell in order to control certain functions
- **growth factor**: a naturally-occurring substance capable of stimulating cellular growth, proliferation, and cellular differentiation
- **oncogene**: any gene that contributes to the conversion of a normal cell into a cancerous cell when mutated or expressed at high levels

9.3E: Cell Signaling and Cell Death

When a cell is damaged, unnecessary, or dangerous to an organism, a cell can initiate the mechanism for cell death known as apoptosis.

Learning Objectives

- Describe how apoptosis is initiated

Key Points

- Apoptosis allows a cell to die in a controlled manner by preventing the release of damaging molecules from inside the cell.
- Internal checkpoints to monitor a cell's health exist; if abnormalities are observed, a cell can also spontaneously initiate the process of apoptosis.
- In some cases, such as a viral infection or cancer, the cell's normal checks and balances fail.
- External signaling can also initiate apoptosis.
- Apoptosis is also essential for normal embryological development; unnecessary cells that appear during the early stages of development will eventually be eliminated through cell signaling.

Key Terms

- **apoptosis**: a process of programmed cell death
- **glycoprotein**: a protein with covalently-bonded carbohydrates

Apoptosis

When a cell is damaged, superfluous, or potentially dangerous to an organism, a cell can initiate a mechanism to trigger programmed cell death, or apoptosis. Apoptosis allows a cell to die in a controlled manner that prevents the release of potentially damaging molecules from inside the cell.

Internal Signaling

There are many internal checkpoints that monitor a cell's health; if abnormalities are observed, a cell can spontaneously initiate the process of apoptosis. However, in some cases such as a viral infection or uncontrolled cell division due to cancer, the cell's normal checks and balances fail.

External Signaling

External signaling can also initiate apoptosis. For example, most normal animal cells have receptors that interact with the extracellular matrix, a network of glycoproteins that provides structural support for cells in an organism. The binding of cellular receptors to the extracellular matrix initiates a signaling cascade within the cell. However, if the cell moves away from the extracellular matrix, the signaling ceases, and the cell undergoes apoptosis. This system keeps cells from traveling through the body and proliferating out of control, as happens with tumor cells that metastasize.

Another example of external signaling that leads to apoptosis occurs in T-cell development. T-cells are immune cells that bind to foreign macromolecules and particles, targeting them for destruction by the immune system. Normally, T-cells do not target "self" proteins (those of their own organism), a process that can lead to autoimmune diseases. In order to develop the ability to discriminate between self and non-self, immature T-cells undergo screening to determine whether they bind to so-called self proteins. If the T-cell receptor binds to self proteins, the cell initiates apoptosis to remove the potentially dangerous cell.

Apoptosis and Embryos

Apoptosis is also essential for normal embryological development. In vertebrates, for example, early stages of development include the formation of web-like tissue between individual fingers and toes. During the course of normal development, these unnecessary cells must be eliminated, enabling fully separated fingers and toes to form. A cell signaling mechanism triggers apoptosis, which destroys the cells between the developing digits.

Figure 9.3E.1: **Apoptosis**: The histological section of a foot of a 15-day-old mouse embryo, visualized using light microscopy, reveals areas of tissue between the toes which apoptosis will eliminate before the mouse reaches its full gestational age at 27 days.

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Signaling in Single-Celled Organisms

Topic hierarchy

- 9.4A: Signaling in Yeast
- 9.4B: Signaling in Bacteria

9.4A: Signaling in Yeast

Yeasts utilize cell-surface receptors, mating factors, and signaling cascades in order to communicate.

Learning Objectives

- Describe how cell signaling occurs in single-celled organisms such as yeast

Key Points

- Budding yeasts participate in a process that is similar to sexual reproduction that entails two haploid cells combining to form a diploid cell.
- Budding yeasts secrete a signaling molecule called mating factor when trying to find another haploid yeast cell that is ready to mate.
- In yeast, a cell signaling cascade is initiated when a mating factor binds to cell-surface receptors in other yeast cells.
- A cell signaling cascade includes protein kinases and GTP-binding proteins that are similar to G-proteins.
- Yeasts have 130 types of kinases, but they do not contain tyrosine kinases, which are utilized by multicellular organisms to control complex forms of development and communication.

Key Terms

- **kinase**: any of a group of enzymes that transfers phosphate groups from high-energy donor molecules, such as ATP, to specific target molecules (substrates); the process is termed phosphorylation
- **GTP-binding protein**: a protein which binds GTP and catalyzes its conversion to GDP
- **G protein**: any of a class of proteins, found in cell membranes, that pass signals between hormone receptors and effector enzymes

Signaling in Yeast

Yeasts are single-celled eukaryotes; therefore, they have a nucleus and organelles characteristic of more complex life forms. Comparisons of the genomes of yeasts, nematode worms, fruit flies, and humans illustrate the evolution of increasingly-complex signaling systems that allow for the efficient inner workings that keep humans and other complex life forms functioning correctly.

The components and processes found in yeast signals are similar to those of cell-surface receptor signals in multicellular organisms. Budding yeasts are able to participate in a process that is similar to sexual reproduction that entails two haploid cells combining to form a diploid cell. In order to find another haploid yeast cell that is prepared to mate, budding yeasts secrete a signaling molecule called mating factor. When mating factor binds to cell-surface receptors in other yeast cells that are nearby, they stop their normal growth cycles and initiate a cell signaling cascade that includes protein kinases and GTP-binding proteins that are similar to G-proteins.

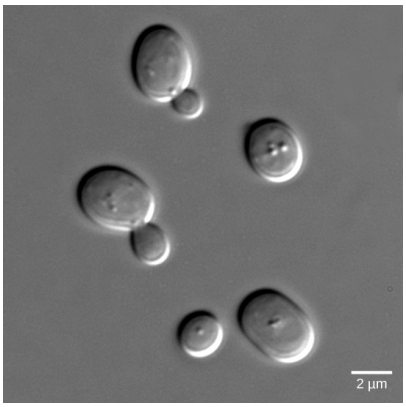

Figure 9.4A.1: **Budding Yeasts**: Budding Saccharomyces cerevisiae yeast cells can communicate by releasing a signaling molecule called mating factor. In this micrograph, they are visualized using differential interference contrast microscopy, a light microscopy technique that enhances the contrast of the sample.

Cellular Communication in Yeasts

Kinases are a major component of cellular communication. Studies of these enzymes illustrate the evolutionary connectivity of different species. Yeasts have 130 types of kinases. More complex organisms such as nematode worms and fruit flies have 454 and 239 kinases, respectively. Of the 130 kinase types in yeast, 97 belong to the 55 subfamilies of kinases that are found in other eukaryotic organisms. The only obvious deficiency seen in yeasts is the complete absence of tyrosine kinases. It is hypothesized that phosphorylation of tyrosine residues is needed to control the more sophisticated functions of development, differentiation, and cellular communication used in multicellular organisms.

Because yeasts contain many of the same classes of signaling proteins as humans, these organisms are ideal for studying signaling cascades. Yeasts multiply quickly and are much simpler organisms than humans or other multicellular animals. Therefore, the signaling cascades are also simpler and easier to study, although they contain similar counterparts to human signaling

9.4B: Signaling in Bacteria

Bacterial signaling allows bacteria to monitor cellular conditions and communicate with each other.

Learning Objectives

- Describe how cell signaling occurs in single-celled organisms such as bacteria

Key Points

- Gene expression in bacteria is initiated when the population density of the bacteria reaches a certain level.
- Bacterial signaling is called quorum sensing because cell density is the determining factor for signaling.
- Quorum sensing uses autoinducers, which are secreted by bacteria to communicate with other bacteria of the same kind, as signaling molecules.
- Autoinducers may be small, hydrophobic molecules, or they can be larger peptide-based molecules; regardless, each type of molecule has a different mode of action.
- Some bacteria form biofilms, which are complex colonies of bacteria that exchange chemical signals to coordinate the release of toxins that attack the host.

Key Terms

- **quorum sensing**: a method of communication between bacterial cells by the release and sensing of small diffusible signal molecules
- **autoinducer**: any of several compounds, synthesized by bacteria, that have signalling functions in quorum sensing
- **biofilm**: a thin film of mucus created by and containing a colony of bacteria and other microorganisms

Signaling in Bacteria

Signaling in bacteria, known as quorum sensing, enables bacteria to monitor extracellular conditions, ensure sufficient amounts of nutrients are present, and avoid hazardous situations. There are circumstances, however, when bacteria communicate with each other.

The first evidence of bacterial communication was observed in a bacterium that has a symbiotic relationship with Hawaiian bobtail squid. When the population density of the bacteria reached a certain level, specific gene expression was initiated: the bacteria produced bioluminescent proteins that emitted light. Because the number of cells present in the environment (the cell density) is the determining factor for signaling, bacterial signaling was named quorum sensing. Interestingly, in politics and business, a quorum is the minimum number of members required to be present to vote on an issue.

Quorum sensing uses autoinducers as signaling molecules. Autoinducers are signaling molecules secreted by bacteria to communicate with other bacteria of the same kind. The secreted autoinducers can be small, hydrophobic molecules, such as acyl-homoserine lactone (AHL), or larger peptide-based molecules. Each type of molecule has a different mode of action. When AHL enters target bacteria, it binds to transcription factors, which then switch gene expression on or off. The peptide autoinducers stimulate more complicated signaling pathways that include bacterial kinases. The changes in bacteria following exposure to autoinducers can be quite extensive. The pathogenic bacterium *Pseudomonas aeruginosa* has 616 different genes that respond to autoinducers.

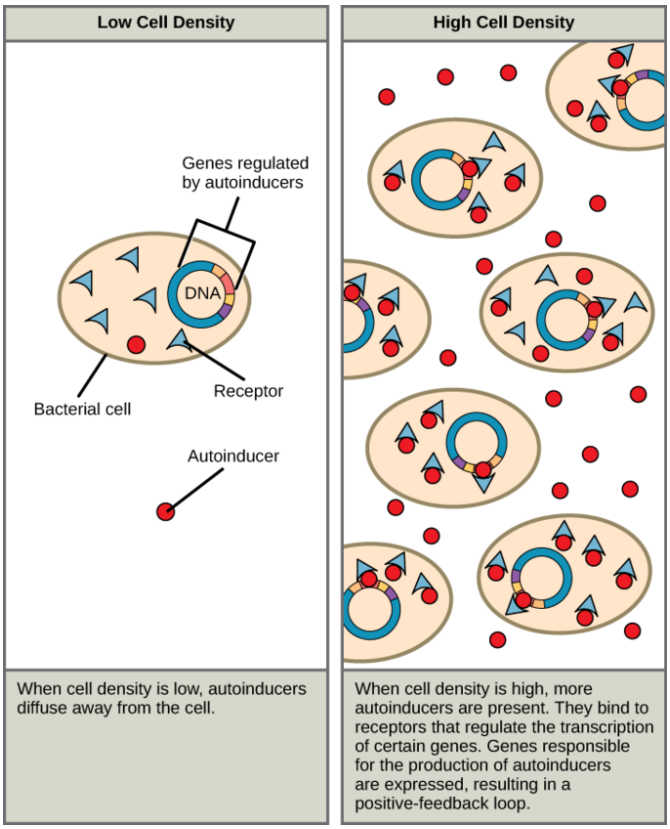

Figure $9.4B.1$: **Autoinducers**: Autoinducers are small molecules or proteins produced by bacteria that regulate gene expression.

Some species of bacteria that use quorum sensing form biofilms, which are complex colonies of bacteria (often containing several species) that exchange chemical signals to coordinate the release of toxins that attack the host. Bacterial biofilms can sometimes be found on medical equipment. When biofilms invade implants, such as hip or knee replacements or heart pacemakers, they can cause life-threatening infections.

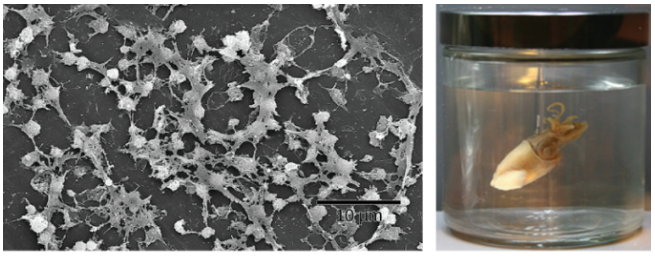

Figure $9.4B.1$: **Bacterial Biofilms**: Cell-cell communication enables these (a) Staphylococcus aureus bacteria to work together to form a biofilm inside a hospital patient's catheter, seen here via scanning electron microscopy. S. aureus is the main cause of hospital-acquired infections. (b) Hawaiian bobtail squid have a symbiotic relationship with the bioluminescent bacteria Vibrio fischeri. The luminescence makes it difficult to see the squid from below because it effectively eliminates its shadow. In return for camouflage, the squid provides food for the bacteria. Free-living V. fischeri do not produce luciferase, the enzyme responsible for luminescence, but V. fischeri living in a symbiotic relationship with the squid do. Quorum sensing determines whether the bacteria should produce the luciferase enzyme.

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22.4D: Bacterial Foodborne Diseases
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23.3D: Rhizaria
Fungi
24: Fungi

G

Glomeromycota
24.3F: Glomeromycota
glycolysis
7.2: Glycolysis
7.2A: Importance of Glycolysis
Gnathostomes
29.2B: Gnathostomes - Jawed Fishes
Golgi apparatus
4.4C: The Golgi Apparatus
gravitropism
30.7D: Plant Responses to Gravity
growing tip
30.5A: Meristems

H

heat of vaporization of water
2.2C: Water's Heat of Vaporization
hormone
37.2A: How Hormones Work
hypertonic
5.2F: Tonicity
hypothalamus
35.4B: Brain - Midbrain and Brain Stem
hypotonic
5.2F: Tonicity

I

In vitro fertilization
43.7D: Infertility
infertility
43.7D: Infertility
isotonic
5.2F: Tonicity

K

karyotype
13.1C: Identification of Chromosomes and Karyotypes

L

Logistic Model
45.2B: Logistic Population Growth
lysosomes
4.4D: Lysosomes

M

Macromolecules
3: Biological Macromolecules
Malpighian Tubules
41.3C: Malpighian Tubules of Insects
Mechanoreceptors
36.2A: Somatosensory Receptors
Mental Illness
35.6C: Neurodevelopmental Disorders - Mental Illnesses
meristems
30.5A: Meristems
Metagenomics
17.4B: Pharmacogenomics, Toxicogenomics, and Metagenomics
MRSA
22.4C: Antibiotics: Are We Facing a Crisis?

N

Nephridia
41.3B: Flame Cells of Planaria and Nephridia of Worms
nervous system
35: The Nervous System
Neurodegenerative Disorders
35.6A: Neurodegenerative Disorders

O

oogenesis
43.3C: Gametogenesis (Spermatogenesis and Oogenesis)
osmolarity
5.2F: Tonicity
osmosis
5.2E: Osmosis
ossification
38.2C: Bone Development

P

Parkinson's disease
35.6A: Neurodegenerative Disorders
passive transport
5.2A: The Role of Passive Transport
Peroxisomes
4.4E: Peroxisomes

phagocytosis
5.4A: Endocytosis
Pharmacogenomics
17.4B: Pharmacogenomics, Toxicogenomics, and Metagenomics
photoperiodism
30.7B: The Phytochrome System and Red Light Response
phototropins
30.7C: Blue Light Response
phototropism
30.7C: Blue Light Response
Phylogenetic Trees
20.1A: Phylogenetic Trees
20.1B: Limitations of Phylogenetic Trees
phytochrome
30.7B: The Phytochrome System and Red Light Response
pinocytosis
5.4A: Endocytosis
Plague of Athens
22.4A: History of Bacterial Diseases
programmed cell death
16.4G: Programmed Cell Death
Properties of Life
1.2A: Properties of Life
Prosimians
29.7A: Characteristics and Evolution of Primates
Protists
23: Protists

R
Radiolarians
23.3D: Rhizaria
Reversing Denaturation
3.3D: Denaturation and Protein Folding
Rhizaria
23.3D: Rhizaria
Root Apical Meristem
30.5A: Meristems

S
schizophrenia
35.6C: Neurodevelopmental Disorders - Mental Illnesses
Selective Permeability
5.2B: Selective Permeability
Signal Summation
35.2D: Signal Summation
skototropism
30.7C: Blue Light Response
somatosensation
36.2: Somatosensation
spermatogenesis
43.3C: Gametogenesis (Spermatogenesis and Oogenesis)
spinal cord
35.4C: Spinal Cord
stroke
35.6D: Other Neurological Disorders
superbug
22.4C: Antibiotics: Are We Facing a Crisis?
Synaptic Plasticity
35.2E: Synaptic Plasticity

T
thalamus
35.4B: Brain - Midbrain and Brain Stem
Thermoreception
36.2C: Thermoreception
thermoregulation
33.3C: Homeostasis - Thermoregulation
tonicity
5.2F: Tonicity
Toxicogenomics
17.4B: Pharmacogenomics, Toxicogenomics, and Metagenomics

V
Vacuoles
4.4A: Vesicles and Vacuoles
Vesicles
4.4A: Vesicles and Vacuoles

W
wax
3.2B: Waxes

Z
zygomycota
24.3B: Zygomycota - The Conjugated Fungi

Glossary

Sample Word 1 | Sample Definition 1